KIRLOSKAR 3GK550ETA 4SR1 · GK3.8703 · 25 KVA

Replacement Engine Control Unit

Complete documentation: the machine, the circuits, the specification, and the research all of it rests on.

This replaces the **Engine Control Unit** block on page 1 of the Kirloskar troubleshooting wiring diagram, manual 08-GP3-60-001. Every pin, signal and colour in these pages was read out of that document or measured in simulation; where something is inferred rather than confirmed, it says so on the line.

The engine has not been inspected yet. Of twelve items on the unknowns register, two are answered by inference and none is confirmed — every confirmation needs the machine. The CPCB IV+ certification position in Memo 06 is a decision for the machine's owners, not an engineering conclusion. **Read that memo before fitting anything.**

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Build Status — What Is Done and What Is Not

As of 20 September 2026, end of session. Counts in this document are generated from the repository, not maintained by hand: the block results come from `sim/run_sim.py`, the unknowns from the specification's own register.

At a glance

	Count
Circuit blocks simulated	**24**
Blocks passing every check	**24**
Blocks failing a check	**0**
Research memos	**12**
Documents in the bible	**18**
Unknowns closed	**4**
Unknowns partly answered	**4**
Unknowns still open	**9**
Phase	**2** — schematic capture **complete**: 10 sheets, root wired, project netlist checked

All 24 blocks pass. On 19 September it was 16, after a temperature-corner pass found six blocks that fail cold or hot and turned prose caveats into enforced checks. All six are now answered, and the answers are part specifications rather than widened checks — every one of them is written up in `docs/bom_requirements.md` with the block that demands it, and every failing corner is kept as pinned evidence of *why* the requirement exists.

One of the six was not a part-selection problem at all. The metering unit's coil is copper, copper moves 0.39 %/°C, and no better resistor exists to buy. It needed closed-loop current control, which put a 50 mΩ shunt and a sense amplifier on ECU pin 88's return — new schematic content, not a BOM line. See §1.7.

Schematic complete; the PCB is prepared for hand routing, with the housing and connector as placeholders (14 October). Firmware runs on the PC against a simulated engine. The KiCad project is in `hw/`, with all ten hierarchical sheets created and the first — `power_input` — drawn and netlist-checked. See §7.

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1. Circuit blocks — simulated and passing

Each block is a SPICE netlist plus a set of falsifiable checks. A block "passes" when every claim it makes about itself holds. These are design verification, not layout.

1.1 Power chain — 9 blocks

Block	What it covers	Status
`emi_filter`	Two-stage LC, CISPR 25 conducted, differential mode	Passing
`feed_emi`	VBAT_INJ / VBAT_EGR feed filters, CISPR 25 Class 5 against modelled load harmonics	Passing
`reverse_battery`	Ideal-diode P-FET vs series Schottky	Passing
`transient_clamp`	ISO 7637-2 pulse 2a, +112 V / 2 Ω / 50 µs	Passing
`load_dump`	ISO 7637-2 pulse 5b, 40 V / 0.5 Ω / 400 ms	**FAILS — deliberately**
`buck_preregulator`	6–40 V to 5 V at 400 kHz	Passing
`sensor_rail`	5 V sensor distribution, one PTC per group	**FAILS — cold corner**
`mcu_pdn`	3V3 decoupling impedance, 100 mΩ target	**FAILS — cold corner**
`supervisor`	Watchdog, brownout, fail-safe gate kill path	Passing
`negative_pulses`	ISO 7637-2 pulses 1 and 3a, negative excursions	**FAILS — real gap**

Why `load_dump` fails, and why that is correct. The TVS chosen against pulse 2a absorbs 46 J at 116 W under pulse 5b — roughly 100× both its single-pulse energy rating and its continuous dissipation rating. The failure is the deliverable: it says the part selection is wrong, not that the simulation is. The fix is a higher standoff voltage so the TVS stays off during a normal clamped load dump, rather than a larger part to absorb it. ~~Blocked on U16~~ — closed 9 Oct 2026: the owner chose to design for the suppressed (clamped) load dump, so pulse 5b applies.

1.2 Sensor front-ends — 7 blocks

Block	What it covers	Status
`sensor_ratiometric`	Pins 41/35/80/37, 0.5–4.5 V into a 3.3 V ADC	Passing
`sensor_differential`	Shared sensor ground on pin 34, single-ended vs differential	Passing
`ntc_frontend`	Pin 79 charge-air temperature, divider vs current source	Passing
`battery_sense`	Pins 04/06, 6–40 V rail into a 3.3 V ADC	Passing
`discrete_input`	Pins 20/24/71, active-high switched battery	Passing
`trip_module_sense`	Trip-module state, supervised three-state loop	Passing
`cam_frontend`	Pin 46 cam Hall, 5 V swing divided into the 3.3 V domain	Passing

`ntc_frontend` simulates **both** candidate topologies because **U2** is unresolved. One resistance measurement at the machine picks between them; the front-end is designed so that measurement selects a populate option rather than unblocking a redesign.

1.3 Speed and position — 1 block

Block	What it covers	Status
`vr_conditioner`	Crank VR sensor, pins 52/74/30	Passing

1.4 Injection — 3 blocks

Block	What it covers	Status
`injector_boost`	Why the boost stage exists, pins 03/05	**FAILS — both corners**
`injector_turnoff`	Turn-off recirculation into the boost rail	Passing
`boost_converter`	Injector rail reservoir sizing	Passing

`boost_converter` passes but carries a recorded gap: it has **no bleed path.** Its charge source regulates the rail from below only. Harmless while current flowed outward; `injector_turnoff` now pushes energy back in, and a hold-current cutoff would climb the rail about 2 V per event with no way down.

1.5 Actuators — 3 blocks

Block	What it covers	Status
`metering_unit_pwm`	Pin 88, low-side PWM into the fuel metering solenoid	**FAILS — both corners**
`relay_driver`	Pins 50/69, low-side FET with flyback	Passing
`egr_hbridge`	Pins 59/81, positional actuator, feedback on 37	**FAILS — shoot-through**

1.6 Communications — 1 block

Block	What it covers	Status
`can_termination`	Split vs single termination, J1939 250 kbit/s	**FAILS — tempco**

1.7 The six corners that were failing, and what each one bought

All six are closed. None was closed by widening a check: each is answered by specifying a part, with the failing corner kept as pinned evidence. The requirements live in `docs/bom_requirements.md` under the tags below.

Block	Was	Answer	Tag
`can_termination`	123.95 Ω at 200 ppm/°C vs J1939's own 120 Ω	thin-film, 0.1 %, 50 ppm/°C → 121.10 Ω at stacked worst case	`CAN-TERM`
`sensor_rail`	4.15 A at cold + −30 % vs a 3.0 A polyfuse ceiling	PTC R25 2.0 → 3.0 Ω, solved against the ceiling → 2.88 A	`SENSOR-PTC`

Block	Was	Answer	Tag
`mcu_pdn`	197 mΩ at a wet electrolytic's cold ESR vs 100 mΩ	polymer bulk, ESR **band** 20–50 mΩ → 60 / 92 mΩ	`PDN-BULK`
`metering_unit_pwm`	0.862 A cold / 0.462 A hot vs a 0.675 A setpoint	closed-loop current control → 0.679 / 0.683 A	`MU-ISENSE`
`negative_pulses`	VIN −0.528 V vs the LM5164's −0.3 V	active clamp → −0.068 V	`NEG-CLAMP`
`injector_boost`	ratio missed 11.6× at both corners	11.6× was this block's *nominal*, never a requirement	—

Three of those were more than part swaps, and they are the ones worth reading.

The metering unit needed an architecture change. Over −40…+125 °C the coil's resistance goes 7.39 → 13.82 Ω and at fixed duty the current follows it, ±30 %. A hysteretic current regulator holds 0.679 / 0.679 / 0.683 A at −40 / 25 / +125 °C. Closed loop only works while the regulator has duty left: at the hot corner the coil can draw at most 0.977 A at 100 % duty, so the setpoint sits at 69 % of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop.

The negative clamp needed a FET, for a structural reason. A Schottky's drop at the 15 A pulse 1 delivers decomposes as 0.367 V junction + 0.180 V bulk, and only the bulk term falls when parts are paralleled — four of them still leave 0.375 V. The rating is really a demand for ≤ 20 mΩ effective clamp impedance *including the junction*, which no junction device meets and a 5 mΩ FET clears four times over. Still open but now bounded: the comparator's propagation delay is unmodelled, so the question went from *"is a 2 ms excursion acceptable"* to *"is a sub-microsecond one"*.

The PDN requirement had to be a band, not a ceiling. Bulk ESR is also what damps the regulator-against-bulk resonance, so at 20 mΩ the peak climbs back to 92 mΩ. "ESR ≤ 50 mΩ" would have been satisfied by a 5 mΩ part that fails the block.

A near-miss worth recording. The active-clamp branch first shared the battery node with the unprotected one, and at 13 mΩ to ground it shunted the fault for that branch too — reporting −0.177 V where the true Schottky-only figure is −0.528 V, i.e. reporting the problem as already solved. Caught because the clamp diode's own ammeter read 13 mA where 15 A was expected.

Two stale checks were also found. `negative_pulses` still demanded 0.707 J from the TVS, from before the clamp Schottky existed — the TVS no longer conducts on negative pulses at all, and the requirement

moved onto the part that now carries it (13.58 A peak, 14.5 mJ). And a burst-accumulation check conflated first-pulse settling with accumulation; repeats 2–5 are identical to the printed precision.

One block passes in a way worth reading. `emi_filter` was not failed at its corner, because its −40 dB gate is documented in its own final check as being of unknown adequacy pending measured hardware emissions. Failing a corner against an admittedly arbitrary number manufactures a verdict. The X7R −15 % corner is recorded as regression pins with a note that the value sits below the nominal gate.

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2. Circuits not yet built at all

These have no netlist, no schematic and no checks. They are the real gaps.

Missing circuit	Pins	Why it is not built
Negative-input protection for the buck	pin 21 chain	**Found 19 Sep.** The LM5164's VIN is rated −0.3 V to 100 V; the negative limit is not a mirror of the positive one and nothing was sized against it. The TVS clamps correctly to −38 V, which is still two orders of magnitude outside what the buck tolerates. A missing component, not a mis-sized one
~~Intake throttle driver~~	—	**Closed 10 Nov 2026 (U8).** The owner's "throttle" is fuelling to reach and hold 1500 rpm: the firmware's speed governor (`govern.c`), through the injectors and the metering unit. Bosch's drawing gives a throttle valve no pin, so the ECU drives none
~~Pre/post-heat output~~	—	**Closed 10 Nov 2026 (U14), by the owner's rule: follow Bosch.** The MD1CS162 terminal diagram has no heater pin, so the ECU has no heater output. The DSE4522 switches pre/post-heat itself (enabled at 50 °C)

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3. Verification not yet done

The blocks above are each simulated in isolation, at one temperature, against the stimuli named. These are the gaps in the *verification*, not in the circuits.

Gap	Consequence
~~No temperature corner anywhere~~	**Done 19 Sep**, −40/+125 °C across all blocks. Six fail at a corner — `emi_filter` and `can_termination` from *ordinary specified* component behaviour, not stacked worst cases. Being converted from prose into enforced checks
~~ISO 7637-2 negative pulses unsimulated~~	**Done 19 Sep.** The TVS is now modelled as the bidirectional part it is, which is what made the question answerable. Result: the buck's VIN sees −40.7 V against a sourced −0.3 V absolute maximum — see the new gap below
Common-mode EMI unmodelled	`emi_filter` covers differential mode only. Most real automotive failures are common mode, and modelling it honestly needs a layout to give the return path
Reverse-battery transient — now load-bearing	The switch model turns off instantly, so it always looks perfect. That was a caveat; since `negative_pulses` it is a design dependency. If the controller opens fast enough the buck is isolated; if not it sees the full clamped negative voltage. No controller part is chosen and no turn-off delay is sourced, so both bookends are simulated and neither is claimed

Gap	Consequence
Watchdog fault-detection time	`supervisor` bounds only what happens *after* reset asserts, not how long detection takes

4. Documents

Document	State
Reconciled Specification	**Binding.** Rewritten §3 on the fitted controller, §4 gained a hardware fail-safe requirement, register extended to U16
Field Handbook	Current
Sensor Reference	Current
Circuit Reference	Generated from the simulations at build time, so it cannot drift
Memos 01–12	Dated records. Superseded conclusions carry banners rather than being rewritten

Memo 01 and Memo 02 are substantially wrong and say so at the top. Memo 01 misidentified the engine and advised discarding the correct search term. Memo 02 analysed a controller that is not fitted. They are kept because the reasoning that produced them is the record of how the error happened.

5. Unknowns

Closed — 4

#	Question	Answer
U11	Engine identity	`3GK550ETA 4SR1`, app code `GK3.8703`, CPCB IV+ under certificate `P25/2825/24`
U1	ECU connector	Bosch `1 928 405 192 / 194`, **code C** keying
U13	Which controller is fitted	**DSE4522 MKII AMF**, not the KG640C the diagram shows
U12	Does the fuel relay remove power?	**No — it only signals**

Three of these four closed against what this project had reasoned. That is the single most useful thing to know about the desk work: it is productive, and it is not reliable without the machine.

Partly answered — 4

U4 CAN expectations (addresses 234 and 44 confirmed) · **U5** injector drive profile (part confirmed 10 Nov: Bosch 0 445 B76 1X3, CRI-2-16; its electrical data is unpublished, so the drive's numbers are confirmed on the bench) · ~~U8~~ throttle (closed 10 Nov: the governor) · ~~U14~~ pre/post-heat (closed 10 Nov: not on Bosch's drawing; the DSE switches it)

Open — 8

U2 charge-air sensor topology · **U6** pin 09 function · **U7** pins 04/06 sense or feed (no longer blocking: the board uses them as load feeds since 8 Oct) · **U9** catalyst sensor routing · **U10** SENT signal identity · **U15** what `CRS-878` denotes · **U16** alternator rectifier suppressed or not · plus U12's contact-type confirmation, which gates nothing

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6. Decisions open, and blocking

Decision	Blocks	Why it cannot wait
~~Integrated driver IC vs discrete FETs~~	—	**DECIDED 19 Sep 2026, memo 12.** Discrete FETs, keeping the 100 V rail. MC33816 caps at 72 V, L9781 at 80 V (ST data brief, 8 Oct), and both are pre-drivers that need external FETs anyway
~~Passive pulldown vs active clamp~~	—	**DECIDED 19 Sep 2026, memo 12.** Active clamp, built discretely, at every gate. The integrated route relocates this problem rather than removing it
MC33816 last-time-buy 06/08/2027	Fallback path only	Needs a distributor conversation, not another search
~~L9781 datasheet unretrievable~~	—	**CLOSED 8 Oct 2026.** The owner supplied ST's data brief DB3720: `VTank` up to 80 V, a pre-driver for 11 external FETs. The runner-up is confirmed out against the 100 V rail (memo 12 §2.4)
Ideal-diode controller for Q1	`power_input` sheet, `reverse_battery`, `negative_pulses`	Its turn-off time is load-bearing and unsourced. `VBAT_REV_GATE` is undriven on the sheet until a part is chosen
Negative-clamp comparator for Q2	`power_input` sheet, `negative_pulses`	Its propagation delay is the last unmodelled term in the negative-pulse chain. `NCLAMP_GATE` is undriven until a part is chosen
Air-intake shutoff	Nothing electrical	Owners' decision. Nothing in this design can stop an oil-fuelled runaway
Trip-module pickup target	Module ordering	Whether it shares our flywheel target. Sharing is cheaper; separate preserves the independence that is the reason for fitting it

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7. Phases

Phase	Output	State
0	Reconciled specification	Done
1	Research memos	Done — twelve
1.5	Block-level simulation	**Done — 28 blocks, 28 passing** (stop_kill 10 Nov, boost_ovp 12 Nov 2026)
2	**KiCad hierarchical schematic capture**	**Complete — 380 components, 378 with footprints; connectivity OK with 2 named open items (power-ground pin, trip-loop pin)**

Phase	Output	State
3	4-layer PCB layout, DRC, fab outputs	**Prepared for hand routing (14 Oct)**: board with all 379 parts placed by block, classes and rules set, DRC clean, 499 connections to route (decisions log §13). Housing and J1001 are placeholders until measured (brief Group 6)
4	Firmware skeleton	**Core done on the PC, with production logging — 16/16 engine scenarios, 101 unit tests; calibrated from the old ECU's recording (§14). Service link (20 Oct): bootloader, end-of-line record, underspeed warning, laptop tool over USB and CAN — 7 bootloader tests, the tool tested end to end** (decisions log §11, §12, §16). **Simulator (20 Oct): `ecusim`, one program for Windows, macOS and Linux, runs the firmware live or from scripts in a browser; all 16 scenarios pass as scripts** (§17). Next: S32K148 dev kit and the chip layers

Phase 2 — what exists

The project is `hw/ecu25kva.kicad_pro`. Everything in it is generated, and the generators refuse to overwrite a sheet that has content, so Eeschema edits are safe against a re-run.

Sheet	State	Blocks it realises
`power_input`	**Drawn, netlist-checked**	`emi_filter`, `transient_clamp`, `load_dump`, `negative_pulses`, `reverse_battery`
`rails`	**Drawn, netlist-checked**	`buck_preregulator`, `mcu_pdn`, `sensor_rail`
`mcu`	**Drawn, netlist-checked**	`supervisor`, `mcu_pdn`
`injector`	**Drawn, netlist-checked**	`boost_converter`, `injector_boost`, `injector_turnoff`
`metering_egr`	**Half drawn** — metering unit done; EGR blocked on a rail rating	`metering_unit_pwm`, `egr_hbridge`
`sensors_analog`	**Drawn, netlist-checked**	`sensor_ratiometric`, `sensor_differential`, `ntc_frontend`, `battery_sense`
`speed_inputs`	**Drawn, netlist-checked**	`vr_conditioner`, `cam_frontend`
`discrete_io`	**Drawn, netlist-checked**	`discrete_input`, `trip_module_sense`, `relay_driver`
`can`	**Drawn, netlist-checked**	`can_termination`
`connector`	**Drawn, netlist-checked**	— (not a circuit: the harness boundary)

The S32K148 symbol is generated from NXP's own pin table, and no pin number is retyped anywhere in the chain. KiCad 7 ships no S32K symbol. The package pin numbers live only in `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx`, which is not a download but one of eleven workbooks embedded as attachments inside the Reference Manual PDF. That file had been extracted once and not kept, so `docs/pinmap.md` cited a source the repo no longer held. It is now in `refs/`, with checksums, and `hw/extract_pinmux.py` makes the extraction reproducible. Re-extracting confirmed three of pinmap.md's claims independently: 128 port + 16 supply = 144 pins with every package pin accounted for; the six driver-gate pins are all PE=0 PS=0; and exactly four pins on the whole part carry a reset pull, which are precisely the four JTAG/RESET pins pinmap.md reserved.

Three parts have moved from a class to a part number, each against a retrieved datasheet, and all three are in `docs/bom_requirements.md`: `LM5164DDAR` (buck), `TLV76733QWDRBRQ1` (3V3), `TPS3850G33DRCT` (supervisor). Two of them supersede rows in research memo 07, which is now annotated in place.

The buck row is worth reading as a process failure, not just a part change. Memo 07 chose a 60 V part for load-dump margin, correctly, at the time. The input-protection work then raised the TVS standoff and put the pulse 2a clamp at 73.3 V — and `transient_clamp.cir` has carried a *passing* check reading "60 V-class parts are NO LONGER viable downstream" ever since. The contradiction sat between an executable file and a prose table for days, and only the executable one knew. Nothing compares them. It surfaced when someone had to draw the buck and needed a pinout.

Check the netlist, not the plot. Three bugs on the first sheet were invisible on a correctly-plotting page and obvious in the exported netlist: a rotation helper with 90 and 270 swapped, which put the clamp Schottky's cathode on ground; custom part fields starting at property id 2, which KiCad reserves, silently renaming the first one "Footprint"; and labels offset from their wires for looks, which detaches them silently while the page still reads as connected.

A wire endpoint landing mid-span on another wire does not connect — not in KiCad 7's netlister, and not even with an explicit junction element in the file. Interactively, Eeschema *splits* the underlying wire when you drop a junction on it; a generated file has no such split, so the two cross without meeting. The page plots exactly as intended and the netlist reports every tapped part as unconnected. Found on the rails sheet, where the input capacitor, the UVLO divider and the RON resistor all tapped the middle of one wire and all three came back unconnected. `schlib.Rail` now emits one segment per interval and is the only thing allowed to draw a multi-tap rail.

Drawing the sheets is finding things nothing else was going to find. Six so far, all of the same shape — a document and an executable file disagreeing, with nothing that compares them:

- **No watchdog pin.** The pin map assigned 37 signals and none serviced the external watchdog memo 11's supervisor topology requires. A window watchdog nobody feeds resets the board on a timer. `PTE0` / `PTE1` added.
- **The buck was a 60 V part on a 73.3 V rail.** `transient_clamp.cir` had carried a *passing* check saying 60 V-class parts were no longer viable for days, while memo 07's BOM still named one.

- **The trip sense was a GPIO on a three-state loop.** `trip_module_sense.cir` designs and checks three bands with real margin; a GPIO reads two, which discards exactly the distinction the bleed resistor exists to create. Moved `PTD6` → `PTB0` (ADC0_SE4).
- **The cam front-end put 5 V on a 3.3 V pin.** Spec §4 and pinmap §1.3 both gave pin 46's front-end as "pull-up to `5V_SENSOR` → timer capture", and no `sim/blocks/*.cir` file simulated a cam front-end at all — `vr_conditioner.cir` covers the crank and there was no cam equivalent, so the swing was never checked. `cam_frontend.cir` was written, and one 16 kΩ resistor turns the pull-up into the same 10 k/16 k divider the ratiometric channels already use.
- **A ground-referenced kill clamp cannot service a high-side gate.** Spec §4's fail-safe list names pins 03/05 alongside the four low-side gates, and `supervisor.cir` sized a ground-referenced clamp; pinmap §2 confirmed the reset state of all six pins against NXP's own table. All three agree, and all three are silent on the fact that the source of a high-side N-FET swings to the boost rail. The injector sheet puts the 470 Ω across gate and *source*, and moves the kill path into the driver's shutdown input.
- **The power ground has no connector pin.** `GP3.8703.C4.pdf` shows pin 21 as the battery positive feed and no battery negative anywhere, because it is a field troubleshooting diagram and says so: it omits "power grounds, unused pins and internal-only pins". Every other sheet returns to `GND` — the injector low sides put 18 A into it — and at the harness boundary that net has nowhere to go. It cannot be solved by choosing one of the 50 unknown pins: a wrong ground pin is a harness that has to be remade, not a rework.

A seventh finding came out of the parts, not the pins. Every clamp earlier in this project was chosen by forward drop. The crank input's four clamp diodes have to be chosen by *reverse leakage* instead: Schottky leakage at 125 °C puts ~0.90 V of offset across the comparator input's 4.489 kΩ Thévenin, which is 4.5× the whole ±198 mV hysteresis window — the comparator latches and a running engine has no speed signal. `vr_conditioner.cir` cannot see this: its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Tagged `VR-CLAMP` in `docs/bom_requirements.md`.

The battery rail's rating is now enforced, not narrated. VBAT_PROT reaches 40.0 V for 400 ms on a load dump and 73.3 V for ~50 µs on pulse 2a, and `transient_clamp.cir` checks what that excludes: a 40 V-abs-max integrated driver by 1.8×, with the dump alone leaving it no margin. The practical consequence is that integrated automotive H-

bridges and smart switches — a 40–45 V class — are excluded from everything battery-connected here, which is what blocked the EGR half of `metering_egr`.

Two open items were closed by drawing, not by deciding. The injector current-sense placement (`pinmap` §1.6, INFERRED per-bank, "settle when the sheet is drawn") is settled: low-side, ground-referenced, one per bank, because cylinders 1 and 3 fire 240 crank degrees apart and never overlap. And the boost reservoir's missing bleed path — which `injector_turnoff.cir` asked for by name — is now modelled in `boost_converter.cir` as arrangement C and drawn as a 22 kΩ resistor: one hold-cutoff event clears in 21.4 ms against the 26.67 ms between cylinders, for 455 mW burned continuously.

The root sheet is wired and the project has a netlist for the first time. Until 21 September 2026 `hw/ecu25kva.kicad_sch`'s ten sheet symbols carried **no hierarchical pins**, which meant every sheet's own netlist was correct and the project had none: a name on two sheets was two nets and nothing compared them. Each symbol now carries one pin per net its child exports — 162 pins — with a stub and a label, and two labels with the same name are one net.

Exporting it immediately found the second half of the same problem: **`kicad-cli` reported annotation errors**, because each sheet generator allocated `R1`, `C1`, `U1` from its own counter. Ten sheets, ten parts called `R1`. `schlib.REF_BASE` now gives each sheet its own hundred — KiCad's own sheet-number × 100 convention, applied at generation time rather than by a hand pass in Eeschema.

components	**201**, every reference unique
named nets with ≥2 pins	**108** (the four `5V_MAIN` islands merged into one)
named nets with 1 pin	**18**, each a recorded open item
sheet paths	10
`GND` / `3V3_MCU` / `5V_MAIN` / `VBAT_PROT`	139 / 36 / 28 / 12 pins, all spanning the project

Exporting it for the first time found three real defects that ten clean per-sheet netlists could not.

1. **`5V_MAIN` was four nets.** The buck's 5 V output is a board rail — the CAN transceivers' VCC, the five difference amplifiers and the discrete inputs' pull-ups all run off it — and it was a *local* label on every one of those sheets. A local label stops at its sheet's edge, so three of the four copies had no regulator on them, and each sheet's own netlist was perfectly correct. `rails` now exports it.
2. **`3V3_MCU` was three.** Same mechanism, on `can` (both transceivers' VIO) and `speed_inputs` (the comparator's supply, both upper clamp diodes and the bias divider — five pins and no rail).

3. **Five supply pins had no decoupling capacitor.** The difference amplifiers on `sensors_analog` were the only undecoupled supply pins on the board. No simulation block was ever going to say so: `sensor_differential.cir` models an ideal amplifier with no supply pin at all.

All three are now permanent checks rather than one-off observations: one walks every `power_in` pin and asks whether a capacitor sits on its net, and one fails any sheet-local net whose name shadows a project-wide one — the exact shape of defects 1 and 2.

`hw/check_netlist.py` is the gate, and it is the counterpart to `sim/run_sim.py`: that one guards the component *values*, this one guards the *connectivity*. It enforces the same rule — **an exemption is not a pass**. Every single-pin net must be listed with its reason, and a net that stops being single-pin must be removed from the list or the check fails. An open item that quietly closes is as much a drift as one that quietly opens.

The full schematic set plots to `docs/pdf/ECU-Schematics.pdf`, 11 pages.

Phase 2 is done. What remains under these sheets is not schematic work — it is eighteen named open ends, and every one waits on a datasheet, a part number, or the machine:

- **nine** driver / current-sense-amplifier interfaces (requirements drawn, parts not chosen)
- **two** controller gates on `power_input`
- **five** for the EGR bridge, blocked on a gate driver above 73.3 V
- **two** with no part and no pin: `BARO`, and `TRIP_LOOP` — which needs one of the 50 unconfirmed connector pins, as does `GND`

Correction, 21 September 2026 — the connector's part number was in the repo and the sheet said it was not. `hw/connector.kicad_sch` was drawn from `refs/ecu-pinout-extracted.md`, which predates the field visit and still says "a complete pinout requires the connector part number and the OEM pin list". `refs/field-evidence-2026-09-18.md` §2b closed U1 three days earlier, off a photograph of the housing angled to the light: **Bosch `1 928 405 192` / `1 928 405 194`, code C**, `>PA66-GF50<`, EDC17 family — read off the physical part, which beats a catalogue match because it cannot be a mis-identification. **Code C is load-bearing:** Bosch codes these housings mechanically so differently-coded plugs cannot mate. What is still missing is the OEM *pin list*, a different document, and the board-side land pattern — neither number returns a catalogue hit, so the footprint has to come through a Bosch distributor. Same shape as the six findings above, and this one was mine: reading the older reference and not the newer one.

The last two placeholder symbols are gone, 22 September 2026. The crank comparator is TLV3201-Q1 — the clamp's comparator, which meets every line of `VR-COMP`. The five difference amplifiers are **INA592** at their native G = ½: the requirement was ≥ 60 dB CMRR and the drawn discrete network could only reach ~48 dB; the on-die network gives 88 dB minimum. Putting that real figure into `sensor_differential.cir` in place of an inferred 80 dB raised the differential front-end's advantage from 90× to **394×**, referred to the sensor. INA592 is not AEC-Q100; no qualified G = ½ difference amplifier was found.

Baro drawn, 22 September 2026 — 2 → 1 open end. Spec §7 deviation 2 adds it deliberately; NXP MPXHZ6115A from a retrieved datasheet, on 5V_MAIN so the existing `SENSOR_5V_MON` channel makes it ratiometric. Its datasheet does not state AEC-Q100; Infineon's KP236 is listed qualified but its datasheet would not resolve, and it is a footprint change — so that is recorded, not guessed.

The one open end left is `TRIP_LOOP`, and no part closes it. It needs a connector pin *and* a harness wire — it is a new signal, not on the OEM diagram — which is field-brief task 5.2.

Connector decided, 22 September 2026: the OEM Bosch 94-way. Spec §7 deviation 1 had said the ECU would use functionally-split sealed connectors with an adapter harness; `hw/connector.kicad_sch` drew the OEM 94-way on the board. The owner decided for the connector described in the field evidence — Bosch `1 928 405 192` / `194`, **code C** — so the sheet stands as drawn and the spec's deviation is struck through and marked reversed. No adapter harness. What that makes load-bearing: the board-side land pattern (field-brief task 5.3, via a Bosch distributor), code-C keying, and the loom's own answers to task 5.1 (power ground) and 5.2 (`TRIP_LOOP`).

The EGR bridge is drawn, 22 September 2026 — 7 → 2 open ends. It had been blocked since 20 September on one datasheet: DRV8873-Q1's 40 V VM against a 73.3 V rail. It went discrete — four 100 V FETs, two **AUIRS2184S** — and the decode `EGR-DRIVER` demands comes from the driver's structure, not a logic gate: one input per leg, so each leg's high and low are exclusive by construction and the 270 A state becomes a high-side brake. The same family as the injector drivers, chosen the opposite way for the opposite reason. **Coast is asserted by copper** on a shared, pulled-down shutdown, which costs one MCU pin, `PTB10`.

The negative clamp latched, and a passing check hid it — 22 September 2026, 8 → 7 open ends. Choosing its comparator meant setting real thresholds, and the model's were wrong: engage near +0.08 V, release above +0.12 V. At the end of every negative pulse the returning source drove 1.35 A back through the still-on clamp — 6.7 mV across 5 mΩ, never reaching release — so it **latched on and shorted the source.** The

check measured only the clamp's minimum and said RESOLVED; a recovery check failed on both pulses. It is a low-side ideal diode and now releases above −6.8 mV, below zero even with the comparator's full offset.

Two more numbers on that block were artifacts. With the ~100 ns comparator-and-driver delay in, the default timestep let ngspice commit the hysteretic switch on a rejected trial step: the "−0.068 V" minimum was really **−0.177 V** — still inside the −0.3 V rating by 1.70×, which answers the question this block had carried since 20 September. And the pinned 4.74 A clamp-diode peak and a 15.8 mV "settling" difference on pulse 3a were coarse-step overshoot, confirmed on the unchanged original netlist: 4.114 A fine, equal to the flat top and to (206.5 − 0.4) / 50 Ω.

Choosing the clamp comparator found two under-rated parts and a drawn FET of the wrong polarity, 22 September 2026. `NEG-CLAMP` specified its FET at **40 V** and the backstop Schottky was **60 V** — both with a terminal on `VBAT_PROT`, both sized for the negative pulses they catch, neither checked against pulse 2a driving the *same* node to **+73.3 V**. `transient_clamp.cir` carried "every part on VBAT_PROT must clear 73.3 V" as a principle, not a list; it is now a **table**, every part on the rail with its rating and margin, and a part below the line fails. Both are 100 V class now.

And the reverse-battery FET: the sheet drew an **N-FET** with an undriven controller gate, while `reverse_battery.cir` has always modelled a **P-FET**. The node swings +73.3 V to −47.8 V and no controller IC checked spans it — LM74700-Q1 stops at +65 V, LM5050-1-Q1 at −0.3 V. A P-FET with a passive gate network *is* the block's switch model and has no IC ratings to violate. It gives up ideal-diode hold-up on a *shorted*-input dropout, which the spec does not require; that is on the page.

Current sense chosen, 22 September 2026: INA181A1-Q1 on all three channels, 12 → 9 open ends. One part and one gain — 20 — set for the injector, so the 18 A peak reads 1.80 V and a fault to 32.8 A is still measured rather than clipped. Supplied from 3V3_MCU, which makes its output ADC-safe by construction and removes the need for a clamp. The metering FET was also redrawn: it had been placed at `rot=270` since 20 September, which gives diagonal drain and source wires — the same drawing bug the injector sheet had, fixed there first.

Gate drivers chosen, 21 September 2026: AUIRS2181S on all eight gates, and the open-end count fell from 20 to 12. Its first listed application is common-rail injection, and the deciding property is one a parametric search would never filter for: it has **no cross-conduction interlock**. In a half bridge that is the less safe choice; here the high and low switches are in *series* with the coil and must both be on, so UCC27282-Q1 and

UCC27712-Q1 — which have interlock — could not fire an injector at all. It also clears the two ratings the obvious 120 V part (UCC27211A-Q1) fails: 4.1 V of headroom on its HB absolute maximum, and a −1 V HS DC minimum that the freewheel diode violates on every hold off-time. The kill path moves to `HIN`, the one ground-referenced point in the high-side path.

The drivers needed a rail the board did not have — 10–20 V — and **it is made from the boost rail** so it survives cranking. The LM5164 already on the board was the obvious candidate and is ruled out by its own VIN absolute maximum of 100 V against a ~102 V rail. A discrete follower with a current limit carries the ~3 mA; if the boost stops, every driver drops into UVLO and every gate turns off.

Choosing the second part found an error in the first sheet it touched. The injector high side was drawn on 21 September with the battery feeding the bank node through a plain **diode-OR**, on the argument that it needed no extra MCU pin. Looking for a gate driver for it showed that it is wrong three ways, and only the third is visible from a datasheet:

1. **It cannot regulate hold.** With the battery on a diode the only switch in the loop is the low side, and opening it sends the coil's current into the 100 V rail — a fast decay against −87 V, which is turn-off, not chopping.
2. **It leaves connector pins 03 and 05 permanently live**, so a harness short to ground draws current whenever the battery is connected.
3. **The gate driver cannot be bootstrapped.** A bootstrap capacitor charges when the switch node goes low; a battery diode holds it at 12.8 V, so it never does.

The replacement is what `injector_turnoff.cir` already modelled and spends a paragraph of its header justifying: **two high-side switches per bank** onto a common node, plus `D_fw` from **ground**. Hold chopping moves to the high side, the node swings to −0.7 V each off-time, and the bootstrap charges. It costs two MCU pins — `PTB5` and `PTA17`, both FTM0 channels, so the whole injector stage stays on one timer and its edges stay phase-locked. The pin map's "six gate pins" section is now eight.

Phase 3 started 21 September 2026 — the first part is chosen. `TPS40210QDGQRQ1`, the injector boost controller, and it was chosen on **one spec that runs backwards from the usual**: tOFF(min) = 200 ns max. A boost needs 94.1% duty at the 6 V cranking corner; at 150 kHz that off-time floor allows 97.0%, and at the 2.2 MHz most wide-input boost controllers in this class use, the same 200 ns caps duty at 56% — below even the 86.6% the rail needs at a *nominal* battery. Switching slower is the requirement, and the part that looks less modern is the one that works.

And it hit the 73.3 V rail rating for the third time. VDD absolute maximum is 52 V. Same shape as DRV8873-Q1's 40 V `VM`, which took that part off the EGR bridge — solvable here only because this pin draws 2.5 mA plus gate charge rather than a motor's current, so a series resistor and a zener hold it inside the rating. All of it is checked as executable arithmetic under `boost_converter`, not as prose. *(Corrected 22 September: the clamp as first drawn — 47 Ω and a 43 V zener checked as an ideal 43.0 V source — gives 61.9 V with a real part. It is now 220 Ω and a 36 V BZG03C36-HM3, 46.6 V worst case. See §7's parts entry.)*

The converter stage is drawn — `injector` moved to A1 to hold it — and the loop compensation is the one thing on it marked as a starting point rather than a result, because a current-mode boost's loop depends on the inductor's real DCR and the capacitor's real ESR and is a bench measurement.

Small-signal parts chosen, 22 September 2026. Choosing them found two errors, both in gate drive.

1. **Every kill-FET gate could see 73.3 V.** `GATE_KILL` is pulled to VBAT_PROT on purpose, so the kill path has a supply during a 3V3 brownout. Nothing on the net draws DC, so the net follows the rail up to the pulse-2a peak. That is 3.7× a small-signal FET's ±20 V Vgs rating. A BZX84-C12-Q now clamps the net at 12.7 V. See `GATE-KILL-CLAMP`.
2. **The relay FETs' gates were driven from a 3.3 V pin.** The schematic noted them "fully enhanced at 3.0 V", and no 100 V automotive FET checked is specified below Vgs = 4.5 V. An SN74AHCT1G125-Q1 on 5V_MAIN now drives each gate. See `RELAY-GATE`.

Parts chosen: 2N7002BK for the nine kill FETs and the reset inverter, PMV280ENEA for the relay FETs. The same pass fixed a diagonal wire from VBAT_PROT to each relay flyback diode, a drawing bug present since the sheet was first drawn.

Power FETs chosen, 22 September 2026: four part numbers for all sixteen. SQD50P08-25L is the reverse-battery P-FET. IPD90N10S4-06 ×8 covers the negative clamp, metering, the EGR bridge and the injector battery switches. SQM85N15-19 ×5 covers the injector boost-side and low-side switches. DMN15H310SK3 is the boost switch, the one part that needs its Rds(on) specified at a 4 V gate. Two things the choice changed:

- **The symbols.** Every FET had been drawn as `Q_NMOS_GSD`, which is the SOT-23 pin order. DPAK and D²PAK are G-D-S with the drain on the tab. With the TO-252 footprint on the old symbol, the source would have gone to the tab. The power FETs now use `Q_NMOS_GDS` and `Q_PMOS_GDS`, which have the same pin positions and different numbers. The netlist shows every drain on pin 2.

- **Checks that had used placeholders.** The bootstrap droop assumed 60 nC; the chosen part's maximum is 120 nC, which doubles the droop to 126 mV (1.1% of the bootstrap supply). The boost's cranking duty had ignored the switch; with its drop at a hot junction it needs 94.8% of the 97.0% available. The gate pulldowns were sized for Crss = 500 pF; the worst chosen part is 300 pF max.

Diodes chosen, 22 September 2026, with one design error found and one class left open.

- **The boost controller's VDD clamp did not clamp.** It had been checked as an ideal 43.0 V source behind 47 Ω. The real 43 V part is 40–46 V with +0.12 %/K and 45 Ω of impedance, which puts **61.9 V** on a 52 V pin at 125 °C. Its 40 V minimum also conducts on the load dump it was chosen to avoid. It is now 220 Ω and a BZG03C36-HM3: 46.6 V worst case, and 33 mA on the dump. The chosen boost switch's small gate charge is what makes 220 Ω possible. A 39 V zener was tried first, and the `transient_clamp` rail table rejected it at 1.05×. See `BOOST-VDD-CLAMP`.
- **Three stock footprints number their pads against the symbol.** TO-277A (the backstop Schottky) and SOT-23 zeners have the cathode on pad 3. `hw/gen_footprints.py` now derives renumbered copies into `hw/lib/ecu25kva.pretty`. The GATE_KILL zener, footprinted in the previous commit with the stock SOT-23, had been reversed.
- **LOWV-CLAMP is open**, covering the 15 input clamps at 3.0 V and 3.3 V. A sub-5 V zener at tens of µA leaks enough to matter, and the model's sharp-knee zener hides it.
- **Phase-3 notes.** The seven VS-16EDH02HM3 are SMPD, which has no KiCad 7 footprint. The BAV199 pairs are drawn as separate half-parts and need one package per pair.

Transistors and magnetics, 22 September 2026: three more open items, and two of them are design decisions, not part choices.

- **12V_GATE pass transistor.** The DPAK part could not be retrieved. The verified PZTA42-Q in SOT223 would have run 1.01× over its hot rating, so two 3.3 kΩ collector resistors now take half the power. The **105 °C board ambient is an assumption**; the spec states none.
- **POWER-PATH.** The injector and EGR current all flows through pin 21, the EMI filter and a reverse FET sized for 5 A. Pins 04/06 are separate 10 A battery feeds that are used only as senses. Architecture decision needed.
- **BOOST-SLOPE.** By TI's Equation 9, the drawn boost goes subharmonic while cranking (82 mΩ against a 44.9 mΩ ceiling). The 330 µH it needs does not exist at a 2.2 A saturation rating, and a smaller inductor makes the ceiling lower.

- **SENSOR-PTC.** A real PTC is specified from 0.40 to 5.00 Ω, not "3.0 Ω ±30%", and its hold current halves at 85 °C.
- The netlist gate now counts the **power ground** as an open item instead of passing it silently: GND reaches the connector only through sensor-ground pins.

Every chip resistor and capacitor is packaged, 22 September 2026. 188 of them, sized by `hw/assign_passives.py` from the voltage across each part and its continuous power at a 105 °C board. Ratings come from Vishay's CRCW e3 and KEMET's automotive X7R tables, and the netlist gate now fails on any R or C left without a footprint. See `PASSIVE-PACKAGES`. The pass found that 4.7 µF at 100 V does not exist in automotive X7R, so three input capacitors became 2 × 2.2 µF. The EMI model was updated to match.

The 38 parts still without a footprint are all either blocked or chosen separately:

- Blocked on POWER-PATH: L101, L102, F101.
- Blocked on BOOST-SLOPE: L401.
- Blocked on SENSOR-PTC: F201–F203.
- Blocked on LOWV-CLAMP: the 15 zeners at 3.0 and 3.3 V.
- Land pattern needed in phase 3: the seven SMPD diodes, L201 and J1001.
- BAV199 pairs, drawn as separate halves: 5.
- Not yet chosen: C313 (polymer) and C401 (electrolytic). Both chosen 8 October, decisions log §9.

Footprints are the phase-3 input, not a phase-2 gate: 264 of 302 components carry one, and they are exactly the parts with a retrieved datasheet behind them.

POWER-PATH built, 8 October 2026. The injector and EGR loads moved off pin 21. Pin 04 now feeds `VBAT_INJ` (the hold switches and the boost input) and pin 06 feeds `VBAT_EGR` (the EGR bridge). Each has a TVS, an SQM120P10-10m1L reverse P-FET, a Schottky backstop and 4.4 µF of bulk. New block `power_path.cir` (18 checks): EGR stall gives Tj 139 °C at a 105 °C board (155 °C while cranking), and the injector gives Tj 117 °C. The netlist gate checks the split. The result is 333 components, 314 with footprints, and 208 passives packaged. L101, L102 and F101 are unblocked and now need pin 21's reduced load stated before they are sized. Conducted emissions on the two new rails are open. This was done on macOS with KiCad 10 for `kicad-cli` and KiCad 7.0.11's symbol libraries for generation, and ngspice 42.

Pin 21's filter and fuse chosen, 8 October 2026. Pin 21's load is now stated as an upper bound of 2.95 A: the LM5164's 1.75 A peak limit caps the buck's input current, plus the relays, the metering freewheel and the boost VDD feed. The parts are L101 Würth 7447709220 (22 µH, Isat 6.5 A), L102 Würth 744314047 (0.47 µH), and F101 Littelfuse 0437005.WRA (5 A, 63 V, 3.16 A at 105 °C). F101 used to be 20 A. The EMI model now carries the inductors' RDC and self-resonance, and its 150 kHz margin fell to 0.56 dB. The fuse's pulse I^2t is now computed from the simulated waveforms: 16 % of melting on pulse 1, against a 20 % rule of thumb whose pulse-cycling data is still open. 317 of 333 components now have footprints.

SENSOR-PTC built, 8 October 2026. The three PTCs (F201–F203) are gone. Each sensor group now has a TPS7B4253-Q1 tracker following 5V_MAIN, all behind one TPS48110-Q1 overvoltage cut-off. The controller was chosen over the LTC4368 for AEC-Q100: decisions log §7, asked and explained in plain terms. `sensor_rail.cir` was rewritten with pulse 2a through the real pin-21 filter. The trackers' input peaks at 22.3 V (limit 45 V), because the controller's own gate network makes the FET a source follower. The block also found that the clamp node reaches 76.6 V with an unsaturated filter inductor, against 73.3 V in `transient_clamp.cir`; that is now a check. The result is 353 components, 335 with footprints, and 225 passives packaged. Open: the 47 µF hold-up part, and two land patterns.

BOOST-SLOPE built, 8 October 2026: all three approved redesigns are now in. The TPS40210-Q1 runs from its own output through a 12V_GATE-style follower (34.3–45.0 V); the battery only starts it, through 180 Ω and a Schottky. L401 is 150 µH, Würth 7447709151, not the approved 100 µH. Two questions to the owner settled that (decisions log §8): the owner's 10 × 10 × 4 mm suggestion saturates at 1.4 A, and the 12 × 12 mm 150 µH beats 100 µH on stability and saturation. The sense resistor is 62 mΩ. Equation 9 holds at the worst corner by 1.12×. Building it found that the oscillator's −20 % frequency corner would have failed the 100 µH design outright. The result is 362 components, 345 with footprints, and 230 passives packaged.

The three non-ceramic capacitors chosen, 8 October 2026 (decisions log §9). Each is a polymer type, because a polymer's ESR barely moves in the cold. C313 is a KEMET T599 150 µF with a 22 mΩ resistor in series. No datasheet gives an ESR minimum, so the resistor is `PDN-BULK`'s floor. C211 is a Chemi-Con HXC 47 µF 63 V. C401 is now two HXC 82 µF 80 V cans in series, because no part rated above 100 V publishes a cold ESR; the owner chose that over one 160 V wet can. The injector still reaches 18 A in 45 µs at −40 °C, aged. The sims run the 41 µF stack, and every

check uses its worst-case low point, 85.1 V. The 22 kΩ bleed is split 2 × 11 kΩ to balance the cans. The result is 365 components, 351 with footprints, and 232 passives packaged.

Four land patterns drawn, 8 October 2026. Every chosen part with no KiCad land pattern now has one, drawn in `hw/gen_footprints.py` from the maker's recommended layout: - the SMPD for the seven VS-16EDH02HM3 (Vishay 95604; the tab is the cathode, so it is pad 1, matching the symbol's K); - WE-PD 1050 for L201; - TI DGX0019A for the TPS48110-Q1, position 16 empty as on the part; - TI DDA0008J for the three TPS7B4253-Q1. Its thermal pad has a 2.6 × 3.2 solder-mask-defined opening, which no stock variant has.

All four parse and plot in kicad-cli. They are written in KiCad 7 format, because the installed KiCad 10 library is newer than the project files. 363 of 365 components now have a footprint. The two left are J1001, which needs the Bosch drawing, and D806, the trip-loop clamp.

A filter on each new supply feed, 8 October 2026 (decisions log §10). New block `feed_emi.cir` measures conducted emissions the way CISPR 25 does, through a 5 µH / 50 Ω LISN. As drawn, the injector feed was 33 dB over Class 5 and the EGR feed 48 dB over, because POWER-PATH had taken both off pin 21's filter. Each feed now has a Coilcraft XAL1510 10 µH inductor, two 82 µF HXC bulk cans and two R-C dampers (0.68 Ω + HXC). Margins are 17.7 dB (injector) and 9.3 dB (EGR) under the limit. The dampers keep the filter stable against the constant-power loads, 1.47× with aged capacitors. The filter also cuts pulse 2a at the loads to 28.7 V. It sets a firmware requirement: hold chopping and EGR PWM at or below 25 kHz. The result is 379 components, 377 with footprints, 236 passives packaged, and 26 of 26 blocks passing.

Firmware, proven on the PC, 11 October 2026 (decisions log §11). `firmware/core/` is portable C. It covers crank/cam decoding, injection timing, the speed governor with droop, rail pressure, sensor faults, latched shutdown and J1939 to the DSE4522. `firmware/sim/` runs it against a Python model of the engine, generator and rail. All eleven spec scenarios pass (`.venv/bin/python firmware/sim/run_sil.py`, 36 s), and so do 46 unit tests (`make -C firmware/sim test`). A mid-build expert review found 2 critical and 7 important defects, among them a start that peaked 3 rpm under the overspeed trip. All nine are fixed, each with a test that failed first. The watchdog kick period is now fixed at 10 ms, so the TPS3850 window straps (SET0/SET1, CWD) can be chosen to accept 5–20 ms. A final review found that the firmware spec had assumed a normally-open metering valve, which contradicts the trip module's design. The firmware now assumes normally closed. **The real valve must be identified**: if it is normally open, the trip module's contacts have to move.

Production logging, 11 October 2026 (decisions log §12). Fault memory with snapshots, a 64-entry event log, lifetime counters (engine hours, starts, trips, watchdog resets, time at load) and a trip recorder now survive power-off. They live in a 3 KB image saved to two alternating copies with CRC-32. DM2, DM3, DM4 and engine hours are read out over J1939. The watchdog now pulses only while the main loop is alive. The ms counter wraps hourly, a lost tooth logs as a crank fault, and a too-long fault message is never half-sent. There are now 15 scenarios and 78 unit tests, all passing.

Field data, 13 October 2026 (`refs/field-evidence-2026-10-13.md`). Jeevan sent: - **Firing order 1-2-3**, which confirms the firmware assumption. - **Governor class ISO 8528-5 G2**, which becomes a requirement. The simulated governor meets every G2 limit. On the real engine, though, the load-rejection peak must stay under our own 1650 rpm trip, which is tighter than G2's +12 %. - **The KOEL injector number** `4H.508.70.0.PR.REV-1` (U5). - **Photos of the old ECU**: Bosch `0 281 021 133` , type `MD1CS162` , KOEL `T4.435.01.0` . Its connector face shows **6 large pins (1–6) and 4 rows of 22**, so the J1001 placeholder is corrected. A second, unidentified ECU is also recorded. - **An engine-off CAN trace.** It holds only UDS diagnostics (0x7E0/0x7E8), so **U4 stays open.** Its ECU names itself `KOEL_TC_82.5KVA` / `4R1190ETA4G1` , a 4-cylinder 82.5 kVA set, not ours; logged as **U17**.

The September DSE record also shows that the DSE shuts down only on the DM1 **Protect** lamp, which our firmware never sets. Fixed on 14 October with the owner's go-ahead: a shutdown fault (overspeed, crank loss) now sets **red and Protect** in DM1; a sensor fault sets amber only. Decisions log §12, addendum.

PCB prepared for hand routing, 14 October 2026 (decisions log §13). `hw/ecu25kva.kicad_pcb` holds all 379 parts placed by block, with: - the 4-layer stack-up - ten net classes, with widths computed from current - a custom rule for pads inside a part

`hw/check_pcb.py` passes: footprints, nets, classes, widths, and DRC with no clearance or courtyard errors. 499 connections are left to route.

The outline, holes, cooling band and J1001 are **placeholders**; the inspection brief's new Group 6 asks for the measurements. **The parts don't fit one side of the spec's 160 × 120 mm guess**, so the placeholder is 180 × 130 mm. If the old board is smaller, parts go on both sides. The routing guide is `docs/layout-guide.md` .

The old ECU's own measurements, engine running, 14 October 2026 (`refs/field-evidence-2026-10-14.md`). This is a 35-minute calibration recording (XCP, MDF 3.0) from a 3-cylinder engine at 1500 rpm on a 12 V system, with no faults stored. It is not a CAN bus trace, so **U4 stays open**. It contradicts five firmware assumptions: - **The metering valve**

behaves as normally open. Current falls as pressure rises, and rises to stop the engine. Firmware assumes normally closed. This is a safety change: the trip-module contacts have to move. Confirm with the valve's part number. - **Injection timing** is 11.5–15° before TDC. Firmware uses 5°. - **Rail pressure** follows load, 827–1057 bar. Firmware uses a fixed 1000 bar. - **Injector constants:** dead time about 330 µs and flow constant about 0.0020. Firmware uses 300 µs and 0.0009, so it would hold each injector open about twice as long as needed. - **The governor is isochronous,** with no droop. Firmware applies 4 %.

Two more readings: - The OEM governor dips **9.8 %** on a load step, at G2's limit. Our simulation is likely optimistic. - The intake-air-temperature input reads 0 mV, so no sensor is fitted (U2).

Applied the same day with the owner's go-ahead (decisions log §14): normally-open valve, measured injector law, and timing and rail tables. 80 unit tests and 15/15 scenarios pass. **Still open: the trip module has to cut the injector feed, pin 04**, not the valve. That is written up in spec §3 item 6, with the field survey in brief Group 7. It is external wiring, and not yet done.

The Bosch pinout becomes the master, 7–8 November 2026 (decisions log §18). The owner sent Bosch's terminal diagram for the MD1CS162 (`refs/T4.435.04.0.PR-terminal-diagram-MD1CS162.pdf`, pin list `refs/bosch-pinout-md1cs162.md`). The owner's ruling: we replicate this ECU, the drawing applies to our engine, and we use only Bosch pins. It disagrees with KOEL's harness diagram on power: - **No permanent battery feed.** Pin 71 (ignition) closes the main relay in hardware, through a new wake FET on pin 50. The relay feeds pins 04/06, the board's only supply. The electronics chain moved from pin 21 to pin 04. - **Pin 21 is the water-in-fuel input.** Pins 01/02 are the power ground, which closes the gap open since 19 September. Pin 09 is the GCU's analog sync input. Pin 14 is the diagnosis request and pin 25 the stop switch. - **A latch bug found and fixed.** Pin 50's flyback diode to the board's supply would have fed the board through the always-live relay coil and held the relay in for good (22.8 mA, 2.7 times its release current). It is now a TVS to ground. `sim/blocks/main_relay.cir` shows both. - **The trip loop is gone.** It had no Bosch pin. The trip module now opens the main relay's coil feed (spec §3 item 7). - **Firmware, ABI 2:** - After-run: the relay is held until the engine has stopped and the fault memory is written, at most 60 s. - Pin 25 latches SHUTDOWN, like the ignition. **Since 10 Nov it also kills the gates in hardware**, for when the firmware is what's misbehaving (decisions log §18). - Water in fuel raises an amber SPN 97. - The service link runs at 1000 kbit/s, Bosch's rate for 87/86. - **The netlist check had been passing its decoupling check on nothing** under KiCad 10. It now parses KiCad 10's netlist and checks 50 supply pins.

Every check passes on this Windows machine: 26/26 circuit blocks; the netlist with **0 open items**; the firmware unit tests, the boot tests, 16/16 scenarios and the service tool; and the board check, schematic parity included. KiCad 10.0.6 drops every pin on an auto-named net from parity, so since 9 Nov `check_pcb.py` checks those pins against the exported netlist (HANDOFF §2). **The Rust simulator is updated but unbuilt**: there was no Rust toolchain here.

The prototype board was brought up to the Bosch pinout on 9 November (decisions log §18, "The prototype, brought up to it"): - 30 new parts, each input's front end placed at J1007. - The three new inputs on PD8–PD10. - 0 DRC errors, nothing unconnected, and the fab files regenerated.

The boost cannot run away, 12 November 2026 (decisions log §19). The owner asked for fixes for the 100 V boost and for heat: - **A hardware overvoltage cutoff.** A 110 V zener (D417) drives the boost controller's DIS/EN pin, so an open feedback resistor can no longer push the rail into its 150 V parts. It trips at 105.0–119.3 V over both tolerance corners (`sim/blocks/boost_ovp.cir`). A 5.1 V clamp protects the pin, and `BOOST_DIS` (PTD15 on production, PD14 on the prototype) lets the MCU stop the boost too. Five new parts on each board; `check_netlist.py` checks the circuit. - **Firmware, ABI 3:** above 106 V on the rail, it stops the boost and shuts down (red DTC, SPN 520192). It also stops the boost while shut down. An **EGR stall guard** lets the bridge coast after 1.5 s over 5 A (amber DTC, SPN 2791 FMI 6), and holds it off after the third stall. - **Coating and heat:** the board is conformal-coated before it goes on the engine. Bring-up step 7 adds a 10-minute full-load heat check with a 100 °C limit.

The application in Simulink and Stateflow, 13 November 2026 (decisions log §20, `model/README.md`). The owner asked for a state-flow model that generates the code the ECU runs: - **Twelve models**, built by scripts in `model/` : six Stateflow charts (the engine state machine, and the boost, EGR stall, water-in-fuel, underspeed and main-relay guards) and six Simulink models (governor, rail control, the two tables, injector law, trip decision). Constants are read from the firmware's C. - **Embedded Coder generates the C** into `firmware/gen/` . Building with `ECU_MODEL` swaps it in behind the same functions. - **Proven equal:** all unit tests and 16/16 scenarios pass on the generated build; the back-to-back test agrees exactly on millions of random inputs (the tables to a unit in the last place). - `ecu_app` : the whole application closed loop with an engine, simulated in Simulink. - **Flashing:** the code goes in the firmware image like any other. The STM32 chip layer is still the missing piece, for either build. - **14 Nov:** boost-rail warnings (held high: the

zener clamping; low while fuelling: a dead boost), the state machine owning the ignition cycle, and `./verify.sh` to run every check at once (decisions log §20).

---

8. Work at the machine

Ordered by what it unblocks.

Re-ordered 21 September 2026, after the schematics were drawn. Two items moved to the top that were not on this list before, because they came out of capture rather than out of a document. The full tasks are `docs/research/08-engine-inspection-brief.md` **Group 5**, written for whoever is at the machine.

Task	Closes	Unblocks
~~Continuity from every populated cavity to chassis — where is the power ground?~~	—	**Closed 8 Nov 2026 by Bosch's drawing: pins 01/02.** A continuity check still confirms it
~~Cavity census~~	—	~~A pin for `TRIP_L00P`~~ The loop is gone (no Bosch pin)
~~Whether the main relay has its own suppressor~~	—	**Closed 10 Nov:** pin 50 drives the main relay and 69 the buzzer relay (owner). Pin 50's TVS, and pin 69's flyback diode to the board's supply (fed, like the buzzer coil, from the main-relay line), are safe with or without a suppressor in the relay. Settled by the owner on 10 Nov: pin 25's polarity as Bosch draws it; pins 09 and 14 as analog and digital inputs, functions configured later
~~Pin 21 with the ignition off~~	—	**Closed 10 Nov 2026 by the owner:** pin 21 is the water-in-fuel sensor, as Bosch draws it
~~Trace who switches the heater feed~~	~~U14~~	Closed 10 Nov 2026: not on Bosch's drawing, so not the ECU's
~~Follow the throttle's 6-pin cable to where it ends~~	~~U8~~	Closed 10 Nov 2026: speed is governed by fuel quantity; no throttle driver
~~Injector part number from the body~~	**U5**	**Part confirmed 10 Nov (owner):** Bosch 0 445 B76 1X3 (CRI-2-16), as on the drawing. Its coil data is not published, so the 100 V boost and 18 A peak stay the CRI-2 family's and are measured on the bench with a real injector (bring-up step 4); the setpoints are firmware values
~~Alternator rectifier part number~~	~~U16~~	Closed 9 Oct 2026: designed for the suppressed load dump, no field check needed
Resistance across the charge-air element, cold	**U2**	The `ntc_frontend` populate choice
Crank VR output while cranking, cold *(supervised)*	—	Confirms the ±198 mV hysteresis window against the real sensor, not a 2 V estimate
Contact pitch and row spacing, with calipers	—	Confirms which Bosch family drawing is the right one for the board-side footprint
Relay contact-type check	U12 confirmation	Nothing — recorded for completeness

Nothing on this list can grow the board any more: U8 (throttle) and U14 (heater) closed on 10 Nov, because neither is on Bosch's drawing. Everything left changes a value or closes a register entry.

---

9. Resume here — next session

Updated 8 November 2026, after the Bosch pinout (decisions log §18). Every check passes: - 26/26 circuit simulations - the netlist gate, with **0 open items** - the firmware unit tests, 7 bootloader tests, 16/16 engine scenarios and the service tool end to end - the PCB rules and generator self-tests - the board check, including the schematic links and parity (on Windows, pins on auto-named nets are checked against the exported netlist: HANDOFF §2) - all 26 prototype tests (9 Nov)

On Windows these all run with KiCad 10's bundled Python and ngspice, plus `zig cc` (pip `ziglang`) for the firmware. See HANDOFF §2.

Where it stands

- **Schematic:** complete, 405 parts, **on Bosch's pinout** since 8 Nov (decisions log §18). Regenerating it is deterministic.
- **PCB:** prepared for hand routing, not routed (decisions log §13).
- **Prototype board (17 October; Bosch pinout 9 November):** routed and fab-ready in `prototype/` : the full production circuit with an STM32G474, screw terminals, 300 × 200 mm, 4 layers. 0 DRC errors, nothing unconnected, schematic parity clean. Gerbers, drill, positions and BOM in `prototype/fab/` ; bring-up guide `prototype/docs/bring-up.md` (decisions log §15). **Reviewed 19 October:** the review found the watchdog kill path lost by the 15 October supervisor fix. D302 (BAT54Q) restores it, in production too. The 5 V buck's output and switch node now run on the outer layers, sized for its 1 A. The pour checks were made able to fail.
- **Firmware:** runs on the PC, calibrated from the old ECU's own recording (§14). **Service link (20 October):** a bootloader with the image header written last, the end-of-line record with its analog trims, an underspeed warning (SPN 190 FMI 18, amber, no shutdown), and a laptop tool (`firmware/tools/ecu_service.py`) over USB or CAN channel 2, all proven on the PC (decisions log §16, firmware guide §16). The chip layers (flash, CAN, USB, reset) are still to write.
- **Simulator (20 October):** `simulator/` , mode 1 (firmware only). `ecusim` is a single Rust program. It runs the unchanged firmware against the engine model, live or from `.seq` scripts, at 1× to about 100×, and traces everything in a browser. Firmware builds load from the UI. The 16 SIL scenarios pass as scripts. Guide `simulator/README.md` ; decisions log §17. Modes 2 (board fed over USB) and 3 (bench rig) are next.

- **Safety change written up, not yet wired:** the overspeed trip opens the main relay's coil feed (spec §3 item 7, which supersedes item 6's pin 04).
- **Stale since 8 Nov:** the Rust simulator's build (its source is updated to ABI 3).
- **The prototype is on the Bosch pinout as of 9 Nov.** Placement and routing were updated in place; decisions log §18.

Waiting on the machine (inspection brief)

1. ~~7.1, the metering valve's part number.~~ **Confirmed 10 Nov (owner):** Bosch 0 928 400 820 (CB18), as on the drawing. Bosch's 0 928 400 metering units are normally open, as the firmware and the stop design assume.
2. **Group 6, the old ECU's measurements:** housing, board, and the J1001 header. **Deferred (owner, 10 Nov 2026):** the prototype is tested physically first, and the production board is changed and fitted to the housing only after that. It blocks the production board, not the prototype.
3. ~~5.1 / 6.4, which pins are battery ground.~~ Bosch's drawing: 01/02. Pin 25 follows Bosch's drawing (owner, 10 Nov). Pins 09 and 14 are settled as inputs; their functions are configured later.
4. **A running-engine J1939 trace on the DSE's CAN** (U4). The 14 Oct file was an XCP recording, not a bus trace.
5. **7.2–7.4, any existing trip module and the main relay's coil feed:** needed when the trip is fitted.
6. **Emissions certification (CPCB IV+, certificate `P25/2825/24`).** It was granted with Bosch's ECU and calibration, so a replacement ECU will probably need it re-checked. **Deferred (owner, 12 Nov 2026):** the ECU is proven working on the prototype first. The certification is taken up before any unit goes into a customer's genset. It doesn't block the prototype.

Needs a person

- **Routing the PCB:** `docs/layout-guide.md`, once Group 6 replaces the placeholders.
- **MC33816AE last-time-buy:** a distributor question (fallback path only).

Next that needs neither

- **Build the Rust simulator** at ABI 3 (`cargo test`), and rebuild the committed Windows `.exe`.
- **Quote and build the prototype:** 2 oz outer copper, 300 × 200 mm, 95 pin necks at 0.15 mm (decisions log §15, "The cost to know before quoting").
- **The S32K148 port:** running the firmware core on NXP's evaluation board, with the engine signals simulated on the bench. It needs the board, a debugger and a toolchain on this machine.

Handoff — state of the project, 28 September 2026

Written for the next session (human or Claude) picking this up on another machine. It says what exists, what is verified, what is open and what to do next. `docs/status.md` remains the long-form record; this is the short one.

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1. Standing rules — these override defaults

1. **Safety and reliability come before cost, size and convenience.** Check at worst-case corners (datasheet min/max, −40 °C and +125 °C, 105 °C board ambient, reverse battery, ISO 7637-2 pulse 2a), never at typical. Prefer a topology that removes a failure mechanism over one that manages it. State failure modes and which direction is safe.
2. **No AI attribution in commits.** No `Co-Authored-By`, no "Generated with" line. (Commits `1332322`..`7a189fe` carry such trailers from before the rule; they are left alone deliberately — do not rewrite that history.)
3. **Deliverables are files in this repository.** No externally hosted pages.
4. **Verify with the project venv and `set -o pipefail`.** Never a bare `python3`, and never `| tail` in a chain that gates a commit — a piped tail once let a 22/24 suite into a commit claiming 24/24.
5. **Nothing is claimed without a check that runs.** A caveat in prose gets believed; a corner that is written as an enforced check is the one that gets caught. Every finding in this project was found this way.

---

2. Setting up on a new machine

```
git clone https://github.com/mohan4220/ecu25kva.git
cd ecu25kva
python3 -m venv .venv                # .venv is git-ignored
.venv/bin/pip install numpy markdown matplotlib pypdf schemdraw
```

System packages (not pip):

Need	Why	Notes
`kicad` 7.x, with `kicad-cli`	netlist export, SVG/PDF plots	schematics are generated, never hand-edited
KiCad stock symbol + footprint libraries	`/usr/share/kicad/{symbols,footprints}`	`hw/gen_footprints.py` derives from them
`libngspice.so.0`	the simulator, driven by ctypes	override the path with `NGSPICE_LIB=` if it lives elsewhere

Need	Why	Notes
`inkscape` , `imagemagick`	only for eyeballing rendered sheets	optional

On macOS (set up 8 October 2026, KiCad 10.0.7, ngspice 42): KiCad from kicad.org or `brew install --cask kicad`, and Chrome for the PDF build. **Use ngspice 42, not Homebrew's 47** — on 47, three blocks fail (`injector_turnoff` and `boost_converter` never finish, and `metering_unit_pwm` misses by 3 %), and the suite stalls at random. Build 42 from the release tarball (clang needs one warning disabled):

```
curl -fsSL -o ngspice-42.tar.gz \
  https://sourceforge.net/projects/ngspice/files/ng-spice-rework/old-releases/42/ngspice-42.tar
tar xzf ngspice-42.tar.gz && cd ngspice-42
./configure --prefix=$HOME/opt/ngspice-42 --with-ngshared --enable-xspice \
  --enable-cider --disable-debug --without-x CFLAGS=-O2 \
  CXXFLAGS="-O2 -Wno-invalid-specialization"
make -j8 && make install
```

Nothing lives at the Linux paths, so the scripts read these instead — put them in `~/.zshrc`:

```
export NGSPICE_LIB=$HOME/opt/ngspice-42/lib/libngspice.0.dylib
export SPICE_SCRIPTS=$HOME/opt/ngspice-42/share/ngspice/scripts   # else no spinit
export PATH="/Applications/KiCad/KiCad.app/Contents/MacOS:$PATH"
export KICAD_SYMBOL_DIR=$HOME/opt/kicad7-libs/kicad-symbols-7.0.11   # see below
export KICAD_FOOTPRINT_DIR=/Applications/KiCad/KiCad.app/Contents/SharedSupport/footprints
export CHROME="/Applications/Google Chrome.app/Contents/MacOS/Google Chrome"
```

Generate against KiCad 7.0.11's symbols, not KiCad 10's. The generators copy each library symbol into the sheet, so KiCad 10's library would change every symbol on every regenerated sheet (and `schlib` cannot read its tab-indented format anyway). With the 7.0.11 library an unchanged sheet regenerates identically, apart from UUIDs. KiCad 10 is used only for `kicad-cli`.

```
mkdir -p ~/opt/kicad7-libs && cd ~/opt/kicad7-libs
curl -fsSL https://gitlab.com/kicad/libraries/kicad-symbols/-/archive/7.0.11/kicad-symbols-7.0.1
```

The PDFs come out two pages shorter on this Chrome (326, not 328) with no source change — font metrics, not content. Don't commit a rebuild for that alone.

On Windows (set up 8 November 2026, KiCad 10.0.6), nothing beyond KiCad is installed. KiCad's bundled Python makes the venv (`…/KiCad/10.0/bin/python.exe -m venv .venv`, then `.venv/Scripts/python.exe`). It also runs `gen_pcb.py`, because it has `pcbnew`. Its `ngspice.dll` runs the simulations, and `pip install`

`ziglang` into the venv gives a C compiler (`.venv/Scripts/python.exe -m ziglang cc`) for the firmware, which has no `make` here: `run_sil.py` then uses the libraries already in `firmware/sim/build/` . Set:

```
export PATH="/c/Users/<you>/AppData/Local/Programs/KiCad/10.0/bin:$PATH"
export NGSPICE_LIB="C:/Users/<you>/AppData/Local/Programs/KiCad/10.0/bin/ngspice.dll"
export KICAD_SYMBOL_DIR="C:/Users/<you>/opt/kicad7-libs/kicad-symbols-7.0.11"
export PYTHONUTF8=1       # several files are UTF-8; Windows defaults to cp1252
```

Two traps: - **Write Python patches with `chr(92)` for a backslash.** The shell wrapper halves doubled backslashes before Python sees them, so a `"\\n"` written into a heredoc arrives as `"\n"` , a real newline. - **KiCad 10.0.6's schematic parity drops every pin on an auto-named net**, `Net-(...)` or `unconnected-(...)` , and reports "No corresponding pin found in schematic". It drops those pins and no others: 198 pads on the production board, 98 on the prototype. The netlist this same KiCad exports lists those pins on the right nets. 10.0.7 on macOS passed. Since 9 Nov, `check_pcb.py` checks exactly those reports against the exported netlist and passes a pad only if the schematic puts that pin on that same net. It prints how many it checked that way. Swapping two such pads' nets on a copy of the board still fails.

There is no Rust toolchain on that machine, so the simulator was not rebuilt.

Verify the whole project in one go: `./verify.sh` (since 14 Nov; `--boards` adds the KiCad board checks, `--matlab` the Simulink closed loop). It names every step that failed. By hand, the core of it is:

```
set -o pipefail
.venv/bin/python sim/run_sim.py | tail -3        # expect 28/28 blocks pass
.venv/bin/python hw/check_netlist.py | tail -6   # expect "schematic connectivity OK", 0 open i
.venv/bin/python hw/assign_passives.py           # expect 256 packaged, 0 unresolved
.venv/bin/python docs/build_pdf.py | tail -1     # rebuilds docs/pdf/ECU-Bible.pdf
```

The application models (Simulink/Stateflow) are generated too (since 13 Nov, `model/README.md`). Edit `model/mdl_<name>.m` , then in MATLAB `build_all({'<name>'})` , `gen_code({'<name>'})` and `sim_ecu_app` (in `model/test`). Then rebuild both firmware builds and run the unit tests, the scenarios (`ECU_LIB=libecu_model`) and `model/test/run_b2b.sh` . On Windows call `.venv/Lib/site-packages/ziglang/zig.exe cc` directly: `python -m ziglang cc` hides compiler errors.

The schematics are generated. Edit `hw/gen_sheet_*.py` , then `.venv/bin/python hw/gen_sheet_<name>.py --force` . Without `--force` the generator refuses to overwrite a non-empty file. After adding or re-valuing any R or C, re-run `assign_passives.py` and then regenerate the sheets again so the footprints land.

Schematic UUIDs are derived, not random (since 15 Oct). A symbol's UUID comes from its sheet title and reference, and a sheet's from its name, so regenerating an unchanged sheet is byte-identical (`hw/test_schlib.py`). The PCB links every footprint to its symbol by these. Renaming a sheet, retitling it, or changing a reference re-links those parts. Before routing, re-run `gen_pcb.py`; after routing, use KiCad's Update PCB from Schematic with "re-link by reference". `check_pcb.py` fails on any stale link.

---

3. Where things stand

Phase 2 (schematic capture) is complete and checked. Ten hierarchical sheets, root wired, project netlist verified. Phase 3 (part selection) is most of the way through. The PCB is prepared for hand routing (`hw/ecu25kva.kicad_pcb`, decisions log §13): every part placed by block, every rule set, with the housing and J1001 as placeholders until the old ECU is measured (brief Group 6). The firmware core runs on the PC against a simulated engine, with production logging (fault memory, events, counters, trip recorder) and all 16 scenarios passing (`firmware/`, decisions log §11–12). Since 20 October it also has a service link (decisions log §16, firmware guide §16): a bootloader, the end-of-line record with analog trims, an underspeed warning, and a laptop tool over USB or CAN channel 2. Only the chip layers are missing; the prototype is first flashed over SWD. **A simulator anyone can run** (`simulator/`, decisions log §17): `ecusim` is one Rust program that runs that firmware live or from scripts, traced in a browser, with firmware builds loadable from the page. It is built for a Windows user with nothing installed. **A prototype board is routed and fab-ready** (`prototype/`, decisions log §15): the full production circuit with an STM32G474 and screw terminals, 300 × 200 mm, 0 DRC errors, nothing unconnected. Fab files in `prototype/fab/`, bring-up guide `prototype/docs/bring-up.md`.

Gate	State
Simulation blocks	**26 of 26 pass**
Netlist connectivity	**passes**, with **0** open items (Bosch pinout, 8 Nov)
Footprints	**404 of 405** components
Chip R and C packaged	**253 of 253**

Verified part numbers are in `docs/bom_requirements.md` under "Chosen parts" — every one against a datasheet that was actually retrieved and read. That now covers the MCU, regulators, supervisor, gate drivers, current-sense and difference amplifiers, comparator, the

boost controller, all sixteen power FETs, the small-signal FETs, 27 diodes, the gate-rail transistors, the buck inductor, and the packaging of every chip resistor and capacitor.

Design decisions taken with the owner are in `docs/decisions-log.md` — the four approved findings, the discrete-input redesign and the sensor-supply redesign, with the options and reasoning for each.

---

4. What is open, in the order I would do it

4.1 Three approved redesigns: all three built

1. **POWER-PATH — built 8 October 2026.** Pin 04 now feeds `VBAT_INJ` and pin 06 feeds `VBAT_EGR`, each through a TVS, an SQM120P10-10m1L reverse P-FET and a Schottky (`power_path.cir`, and the netlist gate checks the split). Pin 21's L101, L102 and F101 are chosen for its new 2.95 A bound (`docs/bom_requirements.md` § POWER-PATH, with the thin margins stated). Left over: F101's pulse-cycling data. The two new rails' conducted emissions were closed 8 October with a filter on each feed (decisions log §10). It puts a firmware limit on hold chopping and EGR PWM: ≤ 25 kHz. The original item read: move the injector and EGR supply onto connector pins 04/06 (10 A fused each in the loom) with their own protection, leaving pin 21 feeding the electronics through the EMI filter and reverse FET. Today every high-current load runs through a path whose reverse FET was sized for 5 A and whose filter never stated a DC current. Blocks: L101, L102, F101. See `docs/bom_requirements.md` § POWER-PATH.
2. **SENSOR-PTC — built 8 October 2026.** The cut-off is a TPS48110-Q1 (chosen over the LTC4368 for AEC-Q100, decisions log §7). `sensor_rail.cir` is rewritten and the trackers' input peaks at 22.3 V. C211 and both land patterns were done 8 October. The original item read: one TPS7B4253-Q1 per sensor group, ADJ tracking 5V_MAIN, behind a 100 V-class overvoltage cut-off (an LTC4368-class controller was the candidate; **its datasheet was not retrieved yet — do that first**) with a 47 µF hold-up capacitor. The cut-off exists because the tracker's IN pin is rated 45 V and the rail reaches 73.3 V for 50 µs. Replaces F201–F203 and the whole SENSOR-PTC requirement. `sensor_rail.cir` needs rewriting with it, including the fault-containment check (one group shorted, the other two and the MCU unaffected).
3. **BOOST-SLOPE — built 8 October 2026.** VDD comes from a follower off the output (34.3–45.0 V), with the battery for start-up only. L401 is **150 µH**, Würth 7447709151 (decisions log §8 revised the approved 100 µH), and the sense resistor is 62 mΩ. Equation 9 holds by 1.12× at the worst corner, oscillator −20 % included. The original item read: feed the TPS40210's VDD from the clamped output side to steepen its internal ramp, and drop L401 to 100 µH (Würth 7447709101, Isat 3.1 A typ, AEC-

Q200). Re-check TI Equation 9 at the cranking corner after the change; the current design is 1.83× over its ceiling there. The check that pins the finding is in `run_sim.py` under `boost_converter`.

4.2 Parts still unchosen

The three non-ceramic capacitors were chosen 8 October 2026 (decisions log §9): C313 is a KEMET T599 polymer tantalum with a 22 mΩ series resistor (R306) as its ESR floor. C211 is a Chemi-Con HXC 47 µF 63 V. C401 + C420 are two HXC 82 µF 80 V cans in series, balanced by the 2 × 11 kΩ bleed. What is left:

- **The trip-loop clamp** — the LOWV-CLAMP residual. It is the one input that still needs a continuous clamp, because its healthy band *is* the clamp level. It is also blocked on a connector pin.

4.3 Blocked on the machine or a supplier

- ~~**The power ground.**~~ Closed 8 Nov 2026: Bosch's terminal diagram puts it on pins 01/02, and the netlist check now requires it.
- ~~`IRIP_LOOP` **has no connector pin**~~ Removed 8 Nov 2026: no Bosch pin. The trip module opens the main relay's coil feed (spec §3 item 7).
- **Pin 25's polarity: as Bosch draws it** (owner, 10 Nov): pressed connects it to terminal 31. Its hardware kill depends on this; first power-up shows it (bring-up step 6).
- **Pins 09 and 14 (owner, 10 Nov):** an analog input and a digital switch input, as built. Their functions are configured in firmware later; until then they are read and unused.
- **The Bosch 94-way land pattern** — needs a distributor drawing for 1 928 405 192 / 194, code C. The board carries the OEM connector (spec §7 deviation 1 reversed), so this is the board's own header.
- ~~**U16, the alternator rectifier**~~ — closed 9 October by the owner. The board is designed for the suppressed (clamped, about 40 V) load dump, as most modern alternators are.
- **MC33816AE last-time-buy 06/08/2027** — needs a distributor, not another search. It only matters for the fallback path. (The L9781 item is closed: the owner supplied ST's data brief on 8 October, confirming its 80 V tank ceiling.)

4.4 The component still without a footprint

Parts	Why
J1001	the Bosch 94-way header: no public drawing. The harness mate is Bosch 1 928 405 063 (code C); the likely header is TE 936524-2 (154 = 60 + 94, 2.5 mm, restricted by TE). Ask the distributor, or buy and measure (memo 03 addendum). On the board as `Bosch_94_PLACEHOLDER` (6 large + 4 × 22, read off the old ECU); measure it from the old ECU (brief 6.3)
~~D806~~	the trip-loop clamp. Gone with the loop, 8 Nov 2026

`hw/gen_footprints.py` holds the project footprints. It derives two from stock by renumbering pads: a TO-277A and a SOT-23 zener, whose stock pad numbering would otherwise have reversed the part. It also **draws** four from the makers' recommended layouts (8 October): SMPD for D408–D414 (tab = cathode = pad 1), WE-PD 1050 for L201, TI DGX0019A for U203 (position 16 empty) and TI DDA0008J for U204–U206 (thermal pad 2.95 × 4.9 copper, 2.6 × 3.2 solder-mask-defined opening). They are written in KiCad 7 format. The installed KiCad 10 library is newer than KiCad 7 can open, so the script skips re-deriving the two stock copies from it. TI's optional thermal vias under the DDA pad are left to layout.

4.5 Then

Firmware on the S32K148 evaluation board, with an Arduino generating the engine signals (the core is already proven on the PC, §11 of the decisions log). Routing the PCB by hand (`docs/layout-guide.md`), once Group 6's measurements replace the outline and J1001 placeholders. Rebuild and check the board with the commands in the README ("Preparing the PCB"). The generator needs KiCad 10's own Python; one footprint KiCad 10 dropped (the TPS40210's HVSSOP-10) is kept from KiCad 7.0.11 in `hw/lib/kicad7.pretty` .

---

5. Traps worth knowing before editing

These each cost a debugging session already:

- **KiCad's schematic-parity check doesn't notice a stale footprint link:** it falls back to matching by reference. `check_pcb.py` compares the paths itself.
- **kicad-cli 10 silently ignores a custom-rules file it can't parse.** Every number needs units (`0mm`). `check_pcb.py` confirms the file is the generated one.
- **A wire endpoint landing mid-span does not connect** in KiCad 7, even with a junction in the file. The page plots correctly and the netlist says "unconnected". Use `schlib.Rail` , which splits every segment at its taps.
- **A label's justification is read in its own rotated frame.** At 180°, "left" puts the text back across its own wire. Both label types are fixed in `schlib` ; keep it that way.
- **Pin numbers are not pad numbers.** A SOT-23 FET symbol (G-S-D) on a DPAK footprint puts the source on the tab. Power FETs use the G-D-S symbols; BJTs in SOT-23 are B-E-C; BZX84 zeners and TO-277A diodes need the renumbered footprints from `gen_footprints.py` .

- **Auto-generated net names move.** `Net-(R407-Pad2)` changes the moment a part is added ahead of it, and `assign_passives.py` keys voltages on net names — so give any node whose voltage matters a real label.
- `--force`, or the generator keeps the existing sheet.
- **The ngspice version matters.** The suite passes on ngspice 42 and fails on 47 (see §2). Upgrading the simulator is a change that needs the whole suite re-run, not a routine update.
- **kicad-cli's DRC report stops listing unconnected items at about 500.** A test that reads that list for one net passes blind on a badly broken board. Read connectivity in pcbnew instead (`gen_board.py clusters`).
- **pcbnew SWIG objects never match with `is`.** Each access hands back a new wrapper; compare `m_Uuid`. `f is not fp` silently let a new part collide with its own courtyard.
- **A symbol added ahead of others on a sheet renumbers them,** and the symbol UUIDs follow the references. On a routed prototype board, run `gen_board.py sync`: it matches footprints to parts by value and pad nets, renames them, refreshes their fields and places only what is new. A circuit change that renames nets or moves pins goes through `migrate` first (`BOSCH_MIGRATE` is the worked example). Then come `sync`, `finish`, `gnd_stitch` (ground vias for the new parts) and `tidy`. To move an MCU pin, `rip` its nets before `migrate` refits the pads.
- **The router walls in a pin between two busy neighbours.** Bosch's three new inputs on PE7–PE9 (bottom row, the CAN corridor) blocked PE8 and three CAN lines through two passes and a rip-up. On PD8–PD10, free pins facing J1007, they routed in one pass. Pick a pin facing its circuit before rip-up rounds.
- **The simulator's `EcuIo` copies `ecu_io` by hand.** Change the struct or an entry point, bump `ECU_ABI_VERSION` (`ecu.h`) and `ABI_VERSION` (`simulator/crates/sim/src/fw.rs`); `io_size_matches` catches a missed field.
- **A dynamic library loaded twice from one path is the first one again.** `Firmware::load` copies every upload to a fresh name. The C core's state is global, so a firmware is one ECU: a `Sim` claims it, and a second waits.
- **Simulation models hide things.** An idealised zener hid both the leakage that started LOWV-CLAMP and an 84 mV loading error that a passing check had pinned as the expected value. When a model's part gets chosen, re-fit the model to the datasheet.

---

6. Repository map

Path	What it is
`hw/gen_sheet_*.py`	the ten sheet generators — the schematic source of truth

Path	What it is
`hw/schlib.py`	drawing primitives, pin geometry, the chosen-parts table
`hw/assign_passives.py`	packages every chip R and C from the voltage across it
`hw/gen_symbols.py` , `hw/gen_footprints.py`	project symbols and renumbered footprints
`hw/check_netlist.py`	the connectivity gate: shadowed nets, decoupling, rails, open items
`sim/blocks/*.cir`	25 self-checking SPICE blocks
`sim/run_sim.py`	runs them and asserts every claim as arithmetic
`simulator/`	`ecusim` : the firmware against the engine model, in a browser (guide `simulator/README.md`)
`docs/bom_requirements.md`	requirements, chosen parts, and the open design items
`docs/status.md`	the long-form running record
`docs/decisions-log.md`	decisions put to the owner, with options and reasoning
`docs/pinmap.md`	pin assignments, including the inverted discrete inputs
`docs/research/`	twelve research memos
`refs/`	OEM wiring diagram, machine photographs, extracted pinout, field evidence

`README.md` is stale in its counts (it said 163 pages, 17 blocks and nine memos; it is now 328 pages, 24 blocks and twelve memos, and it has been corrected). Worth a pass.

ENGINE CONTROL UNIT · KIRLOSKAR 3GK550ETA 4SR1

Genset ECU Field Handbook

Everything known about replacing the engine control unit on a 25 kVA diesel genset — the machine, the 94-pin interface, every circuit with its simulated result, and the two risks that decide whether this ever runs.

Rev 1 13 Sep 2026 17 circuits verified 10 open unknowns Source: 08-GP3-60-001

Circuit reference → Sensor Reference →

01 The machine

02 The 94-pin interface

03 The circuits

04 Stopping a runaway

05 The two real risks

06 What we don't know

07 Running the simulations

01 The machine

The engine is a **Kirloskar 3GK550ETA 4SR1** — three cylinders, turbocharged and aftercooled, common-rail injection, cooled EGR with a diesel oxidation catalyst and no SCR. **26.5 kW / 36 hp at 1500 rpm**, governing at constant speed to make 50 Hz. Application code `GK3.8703`, built 12 September 2025.

That constant speed matters more than it first appears, and it comes up twice in this handbook: once because it makes the timing problem easier than an automotive engine's, and once because the genset's own output frequency is rigidly tied to shaft speed, which turns out to be what is left of the runaway protection.

The panel controller is a **Deep Sea Electronics DSE4522 MKII AMF**, India special — *not* the Kirloskar KG640C the OEM wiring diagram shows. That distinction is not academic; §04 explains what it cost.

PARAMETER	VALUE	STATUS
Engine	`3GK550ETA 4SR1 · app code GK3.8703`	Confirmed — rating plate
Rated output	`26.5 kW / 36 hp @ 1500 rpm · ISO 3046`	Confirmed — rating plate
Type approval	`ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24`	Confirmed — rating plate
Genset	`25 kVA, 415 V 3-phase, 50 Hz, 4-pole alternator`	Confirmed — controller config
Fuel system	`Common rail, metering unit + 3 injectors`	Confirmed
Aftertreatment	`Cooled EGR + DOC — no SCR, no DEF, no DPF`	Confirmed
Emissions tier	`CPCB IV+ (IGES4) · GSR 804(E), 3 Nov 2022`	Confirmed
Electrical	`12 V, 75 Ah`	Confirmed
Panel controller	`DSE4522 MKII AMF (India SP) · 4522-001-01`	Confirmed — photographed
ECU connector	`Bosch 1 928 405 192 / 194 · code C · 94-way`	Confirmed — markings
Our MCU	`NXP FS32K148HAT0MLQT, 144-LQFP`	Selected

ON THE NAME "GK550" — WE GOT THIS WRONG

The engine was first described to us as a **GK550**. An earlier revision of this handbook said no Kirloskar document used that string, that it was a garbled run-together of the genset prefix and the engine family number, and that it "will find nothing" when ordering parts. The rating plate reads `3GK550ETA 4SR1`. **The original description was correct and our inference was wrong.** The reasoning that produced it was sound — five hardware constraints read off the wiring diagram, all of which match the real engine — but it converged on a different Kirloskar engine that happens to satisfy the same five. Use `3GK550ETA 4SR1` and `GK3.8703`.

02 The 94-pin interface

The OEM ECU presents one 94-pin connector. The troubleshooting diagram traces **44 of those pins**; the other 50 — power grounds, unused positions, whatever drives the intake throttle — are simply not on a document written for field technicians. A genuine drop-in replacement needs the connector part number to get the rest.

Every wire in the diagram is colour-coded by signal class, and that code is load-bearing rather than cosmetic: it is how we established that the discrete inputs are switched-battery rather than switched-ground, and that the fuel metering unit is driven low-side. The same colours are used below.

- DC positive · sensor supply · injector high side
- DC negative · sensor ground · injector low side
- Analog in · PWM out
- Digital in · frequency
- CAN H
- CAN L

PIN	SIGNAL	CLASS	FRONT-END
21	**Battery + feed**	supply	EMI filter → fuse → TVS → reverse-battery FET
04 / 06	**V_V_BAT_1R / 2R**	supply	120 k / 10 k divider → ADC · 10 A fused each · 04 feeds the injector, 06 the EGR bridge (POWER-PATH, 8 Oct 2026)
09	**Synchronization ground**	?	Function unresolved
11 / 41 / 79 / 34	**Boost pressure + air temp**	analog	Combo sensor, 4-pin, shared supply and ground
33	**Coolant temperature**	analog	NTC, pull-up divider · ground splices to 34
32 / 35 / 08	**Rail pressure**	analog	Ratiometric · dedicated sensor ground
39 / 80	**Oil pressure**	analog	Ratiometric · ground splices to 36

PIN	SIGNAL	CLASS	FRONT-END
15 / 37 / 36	**EGR position**	analog	Potentiometer, shares EGR connector
52 / 74 / 30	**Crankshaft speed**	frequency	Variable reluctance, differential, twisted pair
45 / 46 / 44	**Camshaft position**	frequency	Hall effect, already digital
43 / 23	**Coolant switch · droop switch**	digital	Switch-to-return + debounce
20 / 24 / 71	**Audio abort · override · ignition**	switched batt	**Active high** — zener clamp, see §03
01 / 02	**G_G_B AT1 / AT2**	ground	Switch returns, or main power grounds — unresolved
88	**Fuel metering unit**	PWM low side	Low-side driver + current sense
59 / 81	**EGR high / low**	PWM	H-bridge, position feedback on pin 37
03	**Injector high side — bank A**	high side	Cylinders **1 and 3**, joined at splice V5
05	**Injector high side — bank B**	high side	Cylinder 2
73 / 07 / 29	**Injector low side — cyl 1 / 3 / 2**	low side	Current-controlled peak-and-hold
50 / 69	**Main relay · buzzer relay**	low side	N-FET + flyback diode
55 / 77	**CAN channel 1**	CAN H CAN L	To the KG640C
87 / 86	**CAN channel 2**	CAN H CAN L	Also wired in the OEM system

> **INJECTORS SHARE THEIR HIGH SIDES**
>
> Pin 03 feeds cylinders 1 *and* 3. Only the low sides are individual. That is five pins, not six, and two boost-fed bank switches rather than three — which matters, because the boost stage is the expensive part of the board. It is safe because on a three-cylinder four-stroke any two cylinders are 240 crank degrees apart, or 26.7 ms at 1500 rpm, against an injection lasting perhaps 3 ms. That separation holds under either candidate firing order, so the unknown firing order stops mattering.

Signals that are not on this connector

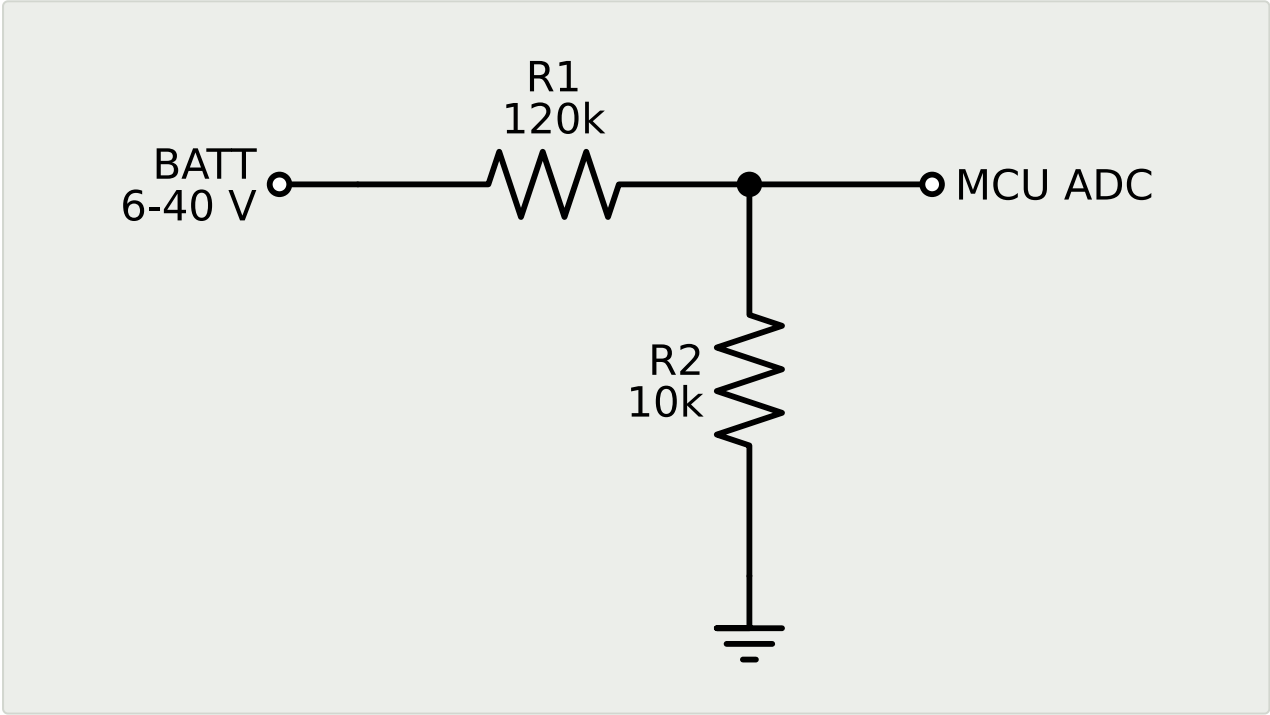
The fuel level sender and the water-in-fuel switch both appear in the wiring diagram, and neither reaches the ECU — they route through the XC11/XC1 harness to the panel. They belong to the GCU. Earlier design passes had built front-ends for both.

Also absent from the entire system: any magnetic pickup, any DEF or urea sensor, any LIN bus, any SCR, and any separate overspeed switch. What the diagram labels `OVERSIDE SWITCH` on pin 24 traces to `-S7 Override SS SW` — one component, not two.

03 The circuits

Every circuit below exists as a SPICE netlist that runs, with checks that are claims about the circuit rather than about the simulator. A failing check means the design is wrong. All six currently pass — but two of them only after simulation caught errors that had survived four rounds of document review.

Battery voltage sense pins 04, 06

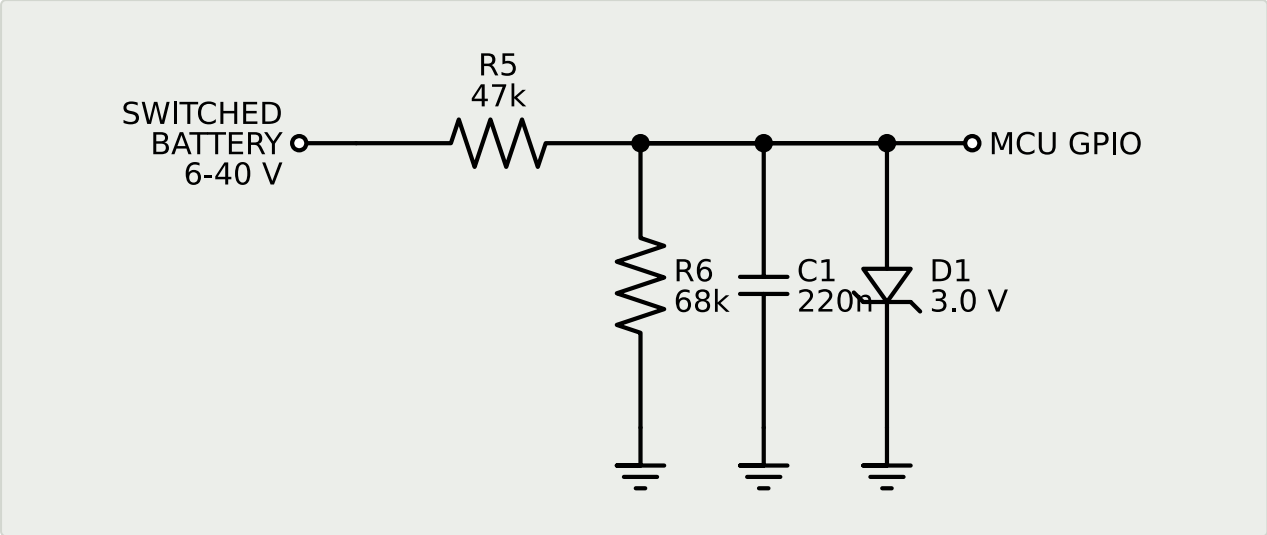


A plain resistive divider scaling the 6–40 V rail into the ADC. Two channels, one per fused rail, which doubles as fuse-integrity monitoring: voltage present downstream means the fuse is intact.

The inherited value was **75 k / 10 k**, and it would have destroyed an ADC pin. That ratio was sized for a 5 V ADC; this MCU's analog supply is 3.3 V. At the top of the input range it delivers 4.69 V into a 3.3 V pin. Corrected to **120 k / 10 k**.

PASS	Peak output at 40 V in	3.069 V
PASS	Headroom below the 3.3 V rail	0.231 V
PASS	Reading at the 6 V cranking dip	0.462 V · 573 LSB
PASS	Linearity residual	< 1 µV

Discrete switch input `pins 20, 24, 71 · active high`



These are switched-battery inputs: the switch connects the pin to battery, not to ground. Earlier documents had them as pull-up-to-5 V with the switch closing to ground — the opposite sense, with a front-end that cannot survive battery voltage.

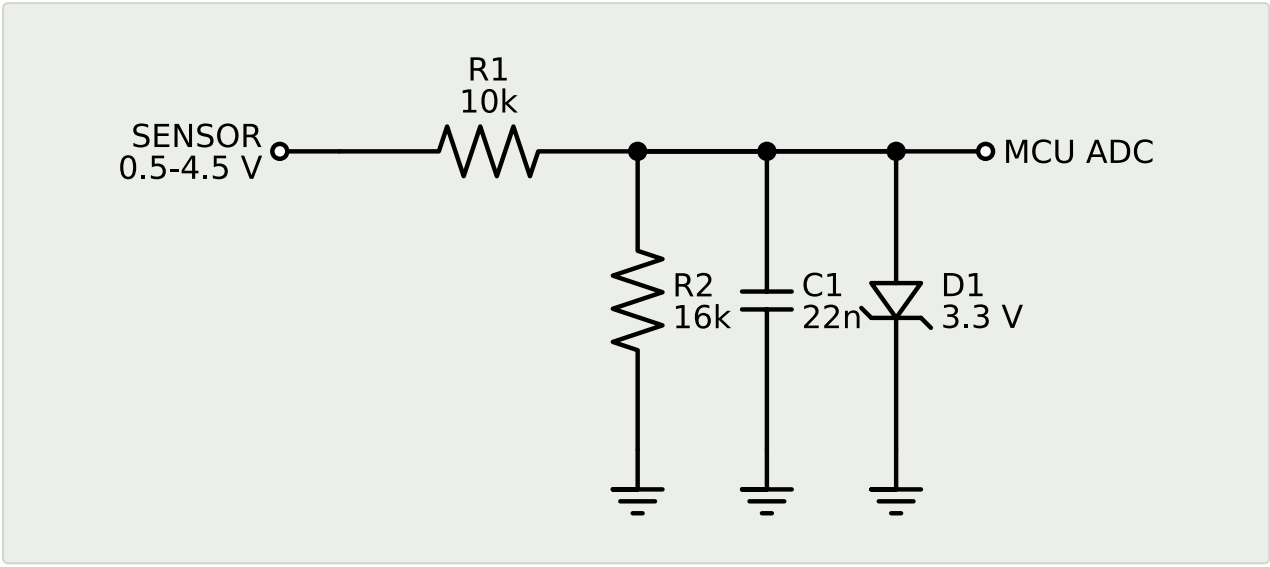
This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 2.31/6 = 0.385. Staying under 3.3 V at 40 V needs it below 3.3/40 = 0.0825. Those do not overlap, so no fixed ratio works at both ends — the original was impossible, not merely mis-valued.

A 3.0 V zener sets the clamp level and the divider only biases into it, so the logic level stops tracking the battery and goes flat.

PASS	Level at 6 V cranking dip	2.863 V
PASS	Level at 40 V load dump	2.954 V
PASS	Spread across the whole range	92 mV
PASS	Threshold crossings during a 5-bounce contact	0
PASS	Debounce time constant	6.11 ms

Debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two button presses.

Ratiometric sensor front-end pins 41, 35, 80, 37

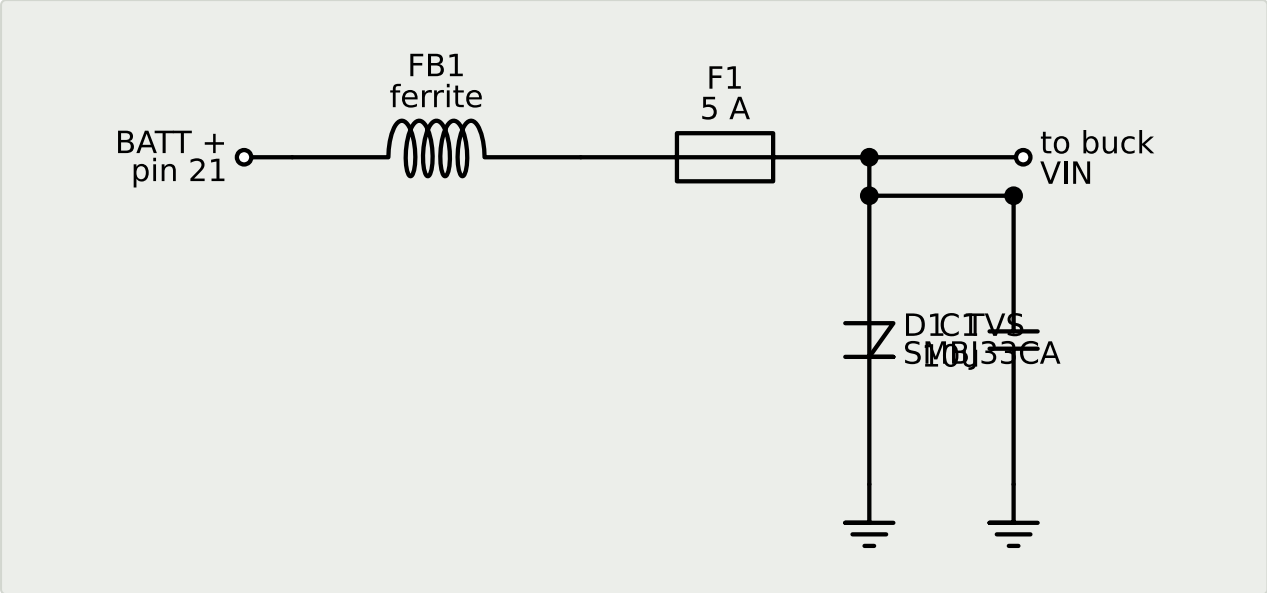


Boost pressure, rail pressure, oil pressure and EGR position are all 0.5–4.5 V ratiometric sensors — their output is a fraction of their own 5 V supply, so the ADC reference has to track that same rail or a drifting supply reads as a pressure change.

Full scale is deliberately placed at 2.77 V, below the 3.3 V clamp rather than hard against the ADC ceiling. That costs about 4% of span and buys a front-end that a 40 V harness short cannot damage.

PASS	Sensor zero (0.5 V) maps to	0.308 V
PASS	Sensor full scale (4.5 V) maps to	2.769 V
PASS	ADC span used	74.6 %
PASS	Low-pass corner	1175 Hz
PASS	Loss at the sensor's own 100 Hz	0.03 dB
PASS	Attenuation at 100 kHz switching noise	38.6 dB
PASS	Pin voltage on a 40 V harness short	3.31 V · 3.7 mA

Transient clamp `battery input · ISO 7637-2 pulse 2a`

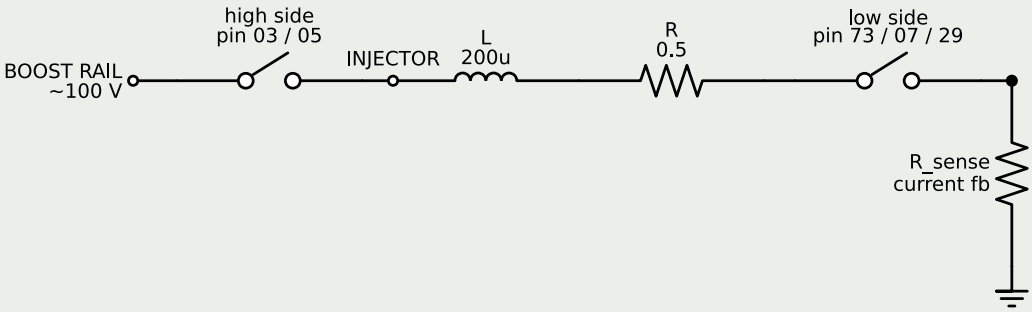


This is the circuit that decides whether the board survives its electrical environment, so it gets simulated rather than assumed. Pulse 2a models a sudden current interruption in another device sharing the harness: **+112 V through 2 Ω for 50 µs**, at Level IV severity.

PASS	Clamped voltage at the buck input	50.0 V
PASS	Margin inside the LM5164's 100 V rating	2.0 ×
PASS	Recovery after the pulse	12.98 V

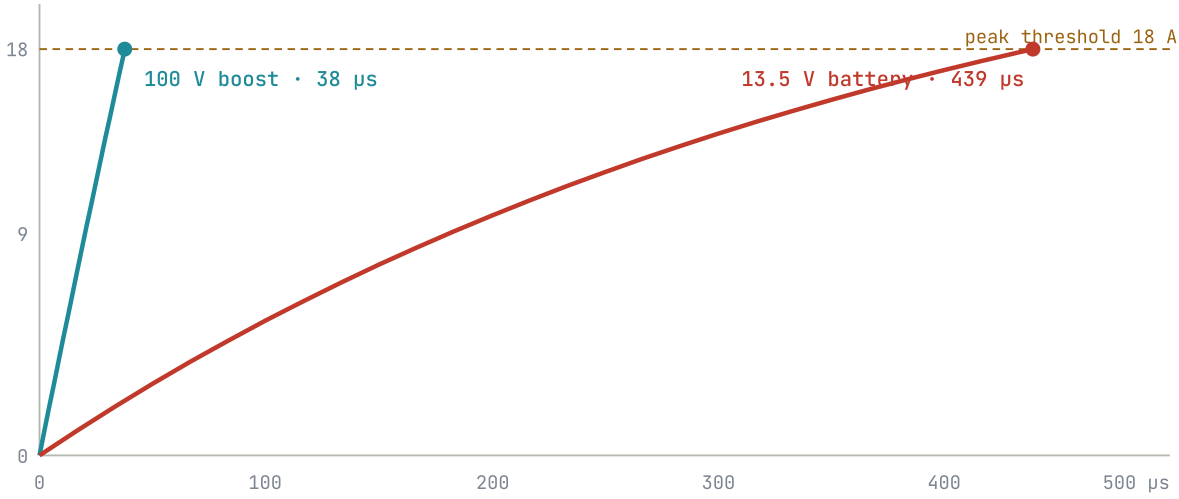
> **A SPEC TARGET THAT SHOULD BE DROPPED**
>
> Earlier documents asked for a clamp "under 42 V". An SMBJ33CA cannot do that — the real part is specified at 53.3 V clamping at rated pulse current, so 50 V here is the component behaving correctly. The 42 V figure was never derivable from a datasheet. Chasing it with a lower-voltage TVS would make the clamp conduct during a legitimate suppressed load dump and force it to absorb that energy. Drop the target; keep the part.

Injector drive — why the boost stage exists pins 03, 05 · 73, 07, 29



An injector is an inductor with resistance, so current through it cannot change instantly — `di/dt = V/L`. The needle does not lift until current reaches its peak threshold, so every microsecond spent getting there is dead time in series with every injection event.

This is the single most expensive stage on the board. Here is its justification as a measurement rather than an assertion.

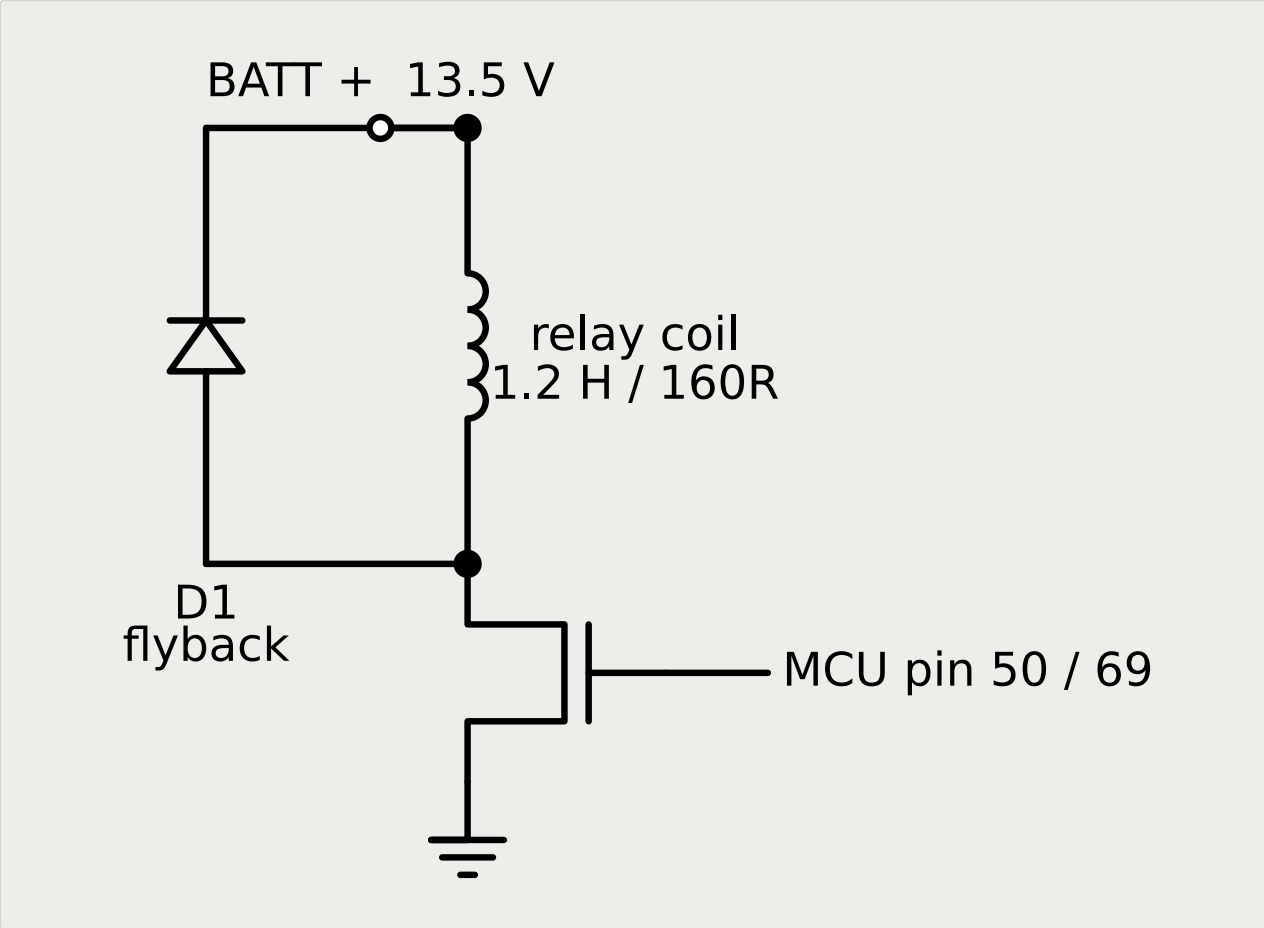


Injector current reaching its 18 A peak threshold. Modelled at 200 µH / 0.5 Ω, the mid-range of the envelope for solenoid common-rail injectors of this class.

PASS	Time to 18 A from a 100 V boost rail	37.8 µs
PASS	Time to 18 A from the 13.5 V battery	439.5 µs
PASS	Boost is faster by	11.6 ×
PASS	Boost drive, as a share of a 1 ms injection	3.8 %
PASS	Battery drive, as a share of a 1 ms injection	44.0 %

Without the boost stage, nearly half of the shortest injection event is spent merely lifting the needle. That is why no production common-rail ECU drives injectors from the battery rail, and why the boost converter is not optional.

Relay driver `pins 50, 69`



A low-side N-FET switching a relay coil. The flyback diode is the entire point: a coil is an inductor, and interrupting its current generates `-L·di/dt` across it. The diode gives the collapsing field somewhere to circulate.

PASS	Drain peak with the flyback diode	14.3 V
PASS	Drain peak without it	unbounded

The simulator reports hundreds of kilovolts for the undiode case. Read that as a model artifact — an ideal switch opening instantly across an inductor implies an unbounded voltage. A real MOSFET avalanches at its breakdown rating and turns the coil's stored energy into heat in the die, once per relay operation, until it fails. The magnitude is fictional; the conclusion is not.

04 Stopping a runaway

A 25 kVA set in runaway is a mechanical hazard to anyone near it. The failure that matters is a controller hanging with fuel still flowing — and a controller cannot be its own protection against that.

This section has now been wrong twice, in opposite directions, and the second time is worth understanding before trusting the third.

We first read the system as sending engine speed to the panel over CAN only, which would leave overspeed protection depending on the very microcontroller and bus most likely to have failed. We then found the panel controller had an independent alternator-frequency speed path, concluded the pessimism was unwarranted, and relaxed the requirement for a standalone trip module. **That second reading was about the wrong device.**

The OEM wiring diagram shows a Kirloskar `KG640C`. The controller actually fitted, photographed front and back, is a **Deep Sea Electronics DSE4522 MKII AMF**. And DSE's own manual for its siblings says plainly that with a CAN engine, *"compatible ECU's receive the start command via CAN and transmit the engine speed to the DSE controller"*.

So engine speed reaches this controller from us. Its configured 1710 rpm overspeed shutdown reads a number our ECU sends, and a hung ECU transmitting a stale, plausible speed defeats it completely. The first, pessimistic reading was closer to the truth than the second.

> WHAT THIS CHANGED
>
> The standalone magnetic-pickup trip module — a GAC SSW675 or a Murphy HD9063, both with Indian distributors — goes back to **mandatory**. The relaxation to "recommended" was bought entirely by the independent-speed claim, and that claim described a controller that is not on this machine.

One independent path does survive, and it is worth having — but it is a different mechanism and a more conditional one. The DSE4522 measures **generator frequency** directly at its terminals 21–24, straight off the alternator's three phases, without touching CAN at all.

Generator frequency	Over-frequency shutdown	Fuel relay, terminal 3	Fuel system stays powered

Measured directly at terminals 21–24. On a 4-pole alternator with no slip, 1500 rpm *is* 50 Hz.	→ Configured to trip at `56.0 Hz` — that is **1680 rpm**, below the CAN-dependent 1710 rpm trip.	→ DC Output A. 5 A continuous — a coil driver, consistent with operating the 70 A `-13RB1` contactor.	→ **Measured 18 Sep 2026: the relay signals, it does not remove power.** The chain ends here, and it ends broken.

No link before the last one touches this ECU, so a hung ECU does not defeat them. But note what is being relied on: a *generator* protection doing the work of an *engine* protection. It inherits that mechanism's dependencies — the alternator must be excited and producing measurable output. An excitation failure removes it. That is a narrower guarantee than a magnetic pickup watching the flywheel, and it should not be described as equivalent to one.

THE GATE CLOSED, AND IT CLOSED BADLY

Every protective action this controller can take ends at the same place: it de-energises the fuel relay on terminal 3. The question was whether those contacts remove supply, or merely signal. **Measured at the machine on 18 September 2026: they signal.**

So the chain terminates in the failure it was meant to catch. Read together with the CAN finding above, the two compound into one sentence worth being blunt about: **the controller either cannot see the fault or cannot act on it.** Overspeed it reads from our CAN, so a hung ECU hides it. Over-frequency it measures for itself — and then acts through a relay that only signals.

Nothing on this machine currently interrupts fuelling without the cooperation of the ECU being protected against. That is what the standalone trip module now exists to fix, and it is why its contacts have to sit in series with the fuel metering unit's own battery feed through fuse `8F1` — not in the GCU relay chain, which has just failed this exact test.

One caveat kept for honesty: the contact-type check that disambiguates a normally-closed relay is not confirmed to have been done first. It is still wanted. Nothing above changes if it comes back the other way, because the unsafe reading is already the one being designed to.

AND THE ONE NOBODY CONSIDERED

A diesel that starts burning its own lubricating oil — through a failed turbo seal, or crankcase fumes drawn into the intake — has a fuel supply that **none of the protections above can touch**. Every one of them cuts diesel or electrical power. The standard countermeasure is a mechanical **air-intake shutoff valve**, and no such device appears anywhere in this design.

It was never considered and rejected. It was never considered. Whether to fit one is a decision for the machine's owners — the part is mechanical, not electronic — but it should be a decision rather than an omission.

05 The two real risks

Neither is a hardware problem, and neither is solved by better circuit design. They are the two things that decide whether this project ends with a working board on a bench or a running genset.

You cannot buy the injector calibration

The drive stage can be designed, built, and fully verified against a known inductive load. That is real engineering progress and it should proceed.

What it cannot do is deliver the *right amount of fuel*. Producing an electrically correct current pulse is not the same as knowing how much fuel that pulse injected — and that mapping, fuel mass against pulse width across the rail pressure range, is proprietary flow-bench data tied to a specific injector's hydraulic geometry. Bosch and Denso ship it with their own ECUs and not otherwise.

It gets harder in the **ballistic region**, where pulses are too short for the needle to reach full lift and delivery becomes steeply non-linear — which is exactly where a genset at light load spends its time. And empirical tuning on a running engine is not a fallback: at 19.7:1 compression, wrong fuelling risks combustion damage and turbocharger overspeed long before trial and error converges.

ROUTE TO THE DATA	COST	LEGALITY
Supplier licence	Structured around OEM production volumes. Realistically out of reach at this scale.	The sanctioned route.
Measure it yourself	Buy injectors, measure delivery directly on a rate-of-injection rig — Bosch tube or Zeuch method. Bounded and controllable with access to a fuel-injection lab.	Entirely legal. First-party measurement of purchased hardware.
Extract the OEM calibration	Cheapest in engineering time if a donor ECU exists, but the binary is undocumented and likely protected.	Contested. Implicates the manufacturer's copyright. Not recommended without legal clearance.

THE ONE WORTH PURSUING

Route two. It is legal, the cost is bounded, and a university with a fuel-injection lab is a far more reachable thing than a Bosch development agreement.

The regulatory position

This is an emissions-certified genset in service. CPCB IV+ type approval attaches to an engine family at the manufacturer level — it is not a per-unit certificate that travels with the machine regardless of what is done to it.

READ THIS BEFORE PLANNING DEPLOYMENT

CPCB's own RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy **must not be altered** — and that is under CPCB's own sanctioned retrofit scheme, which separately excludes modifications limited to engine control. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not touch the engineering. Designing the board, building it, and bench-testing it against simulated crank and cam signals are all unaffected. It bears on putting it into service on a genset that is in use.

And it compounds the calibration problem in a way neither risk shows alone: this engine meets its emissions limits *partly through calibration* — EGR scheduling and injection timing — not through hardware alone. A replacement ECU can therefore move it out of compliance without a single component changing.

This is a decision for the project's owners to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and had not previously been examined — not to make the decision for anyone.

06 What we don't know

Nothing on this list may be guessed into copper. Most of it needs somebody at the machine with a multimeter and a camera rather than more research — which is worth knowing, because it means one site visit closes most of it.

Four closed on 18 September, from photographs and one manufacturer's manual. Three of the four closed *against* something this project had inferred, which is a fair measure of how much the desk research was carrying.

#	CLOSED	ANSWER
U11	**Engine identity**	3GK550ETA 4SR1, app code GK3.8703 — rating plate. **Our inferred model number was wrong.**
U13	**Which controller is fitted**	DSE4522 MKII AMF (India SP), serial 11021794 — **not the KG640C** the diagram shows. Cost us §04's argument.
U1	**The connector**	Bosch 1 928 405 192 / 194, **code C** keying. 94 ways: power 1–8, then 9–28, 29–50, 51–72, 73–94.
U3	**Does the controller have a speed pickup?**	No — DSE's terminal list runs 1–35 with no pickup input.

#	STILL OPEN	BLOCKS	HOW IT CLOSES
U12	**Does the fuel relay remove power, or only signal?**	**CLOSED 18 Sep 2026 — it only signals.** §04's safety gate, resolved against us	Closed by measurement. The trip module becomes the only ECU-independent shutdown.
U4	**Exact J1939 PGN set**	CAN firmware	Mostly answered — addresses 234 / 44, and the controller wants oil pressure, coolant temp, coolant level and speed from us. Log the bus for the rest.
U2	**Boost air temp — NTC or ratiometric?**	One sensor front-end	One resistance reading across the sensor, engine cold

#	STILL OPEN	BLOCKS	HOW IT CLOSES
U5	**Injector part number**	Boost rail target, not the driver itself	Read the injector body
U6	**What pin 09 "synchronization ground" does**	Ground plan	Continuity, battery disconnected
U7	**Are pins 04/06 sense-only or load feeds?**	No longer blocking: the board now uses them as load feeds (POWER-PATH, 8 Oct 2026)	Measure current at the fuse
U8	**Is the 6-pin intake throttle on the ECU?**	Whether a throttle driver is in scope	Follow the cable by hand
U9	**Where the catalyst temp sensor goes**	Whether an EGT front-end is needed	Trace on the engine
U10	**Which signal uses SENT**	Possible digital sensor front-end	Inspect the harness
U14	**Pre-heat / post-heat — controller or ECU?**	Whether a heater output is in scope. We have none.	Trace the heater supply. The controller config enables both at 50 °C.
U15	**What "CRS-878" denotes**	Possibly bears on the injector	Ask KOEL, or find DSE's engine-profile list
—	**Are pins 01/02 switch returns or power grounds?**	Ground plan	Continuity to battery negative

U12 NOW STANDS ALONE

With the controller identified, every protective action it can take ends at one relay. U12 asks what that relay's contacts actually carry, and the whole of §04's second line of defence rests on the answer. It is a multimeter and ten minutes, and nothing else on this list substitutes for it.

07 Running the simulations

You already have a SPICE engine. KiCad ships `libngspice`, and the harness in the repository drives it directly, so the circuits run and check themselves without installing anything further.

```
.venv/bin/python sim/run_sim.py              # every block
.venv/bin/python sim/run_sim.py injector     # blocks matching "injector"
```

Output is one line per claim, green or red. Because the claims are about the circuit — "reads logic high at a 6 V cranking dip", "never exceeds the MCU's absolute maximum" — a red line means the design is wrong, not that the simulator is unhappy.

The netlists are plain SPICE and are not tied to this harness. They open directly in **KiCad**'s built-in simulator (Tools → Simulator), in **LTspice**, or in a standalone **ngspice**. Each carries its own `.control` block, so it runs without further setup.

Read the comments inside each netlist before the netlist itself. They carry the reasoning — why the discrete input is not a divider, why the sensor full scale sits below the clamp, why the 42 V clamp target should be abandoned — which the schematic alone cannot tell you.

COMPANION TO THE FIELD HANDBOOK

Sensor Reference

Every sensor in the system — what it measures, how its signal reaches the processor, and what the engine actually does with the number. Including the ones that belong to the panel rather than the ECU, and the ones we would add.

14 on the ECU 6 panel-side 7 proposed Rev 1 · 13 Sep 2026

Circuit reference → ← Field Handbook

00 How to read an entry
01 Position & speed
02 Fuel system
03 Air path
04 Engine health
05 Electrical
06 Operator inputs
07 Panel-side
08 What we would add
09 Channel budget

00 How to read an entry

Every sensor below has the same five rows, because those are the five things you need to know before you can either build its front-end or write its firmware:

- **Measures** — the physical quantity and its useful range.
- **Signal** — the electrical form it arrives in, its ECU pins, and its wire class from the Kirloskar diagram's own colour legend.
- **Acquisition** — the front-end circuit, which MCU peripheral reads it, and how often.
- **Used for** — what the engine does with the number. This is the row that matters and the one most sensor documentation leaves out.
- **Fault detection** — how firmware knows the reading is lying.

Sampling rates are proposals, not measurements — they follow from each signal's own bandwidth and from what the control loop using it needs. Marked **inferred** where that is the case.

> **A NOTE ON SAMPLE RATE**
>
> The instinct is to sample everything fast. Resist it. Coolant temperature has a thermal time constant measured in tens of seconds, so sampling it at 1 kHz buys nothing and costs ADC bandwidth that rail pressure genuinely needs. Rate should follow the physics, and the physics here spans four orders of magnitude — from crank edges every 667 µs to a fuel level that moves over hours.

01 Position & speed

Everything angle-related derives from these two. Get them wrong and nothing else in the system can be right, because injection timing is meaningless without knowing where the crankshaft is.

Crankshaft speed

pins 52 / 74 / 30

ON ECU

MEASURES	Crank angular position and speed, from a toothed wheel. Nominal running speed **1500 rpm**; must also resolve cranking speed, roughly 100–250 rpm.
SIGNAL	**Variable reluctance**, differential. 🔵 `Frequency I/P High` on pin 52, ⚫ `Frequency I/P Low` on pin 74, ⚫ `sensor ground` on pin 30. Twisted pair. Output is AC whose amplitude scales with speed — near-zero while cranking, several volts at idle.
ACQUISITION	Zero-cross adaptive-threshold conditioner producing one clean digital edge per tooth, into a FlexTimer input capture channel. Every edge is timestamped; at 1500 rpm with a 60-tooth wheel that is an edge every **667 µs**. The missing-tooth gap gives the angular reference.
USED FOR	**Speed governing** — the primary feedback for the metering unit loop. **Injection timing** — every injection event is scheduled against crank angle derived from here. **Overspeed shutdown** — the ECU's own first line of defence. **Crank sequencing** — the starter is disengaged the moment speed crosses the catch threshold. **Reported over CAN** in EEC1, PGN 61444.
FAULT DETECTION	Missing edges, implausible tooth intervals, or loss of the sync gap. Loss of crank signal on a running engine is a shutdown, not an alarm — the ECU can no longer time injection.

Camshaft position `pins 45 / 46 / 44` ON ECU

MEASURES	Which half of the four-stroke cycle the crank angle refers to. One pulse per two crank revolutions.
SIGNAL	**Hall effect**, already a clean digital square wave — which is why it needs no conditioning IC and has no Low pin. 🔴 `Switch Battery 5 V` on pin 45, 🔵 `Frequency I/P` on pin 46, ⚫ `sensor ground` on pin 44.
ACQUISITION	Pull-up to the protected sensor rail, straight into a second FlexTimer capture channel. One event per 80 ms at running speed.
USED FOR	**Cylinder phase**, and only that. The crank sensor says the engine is at 15° before top dead centre; the cam sensor says whether that is cylinder 1's compression stroke or its exhaust stroke. Without it you can measure speed perfectly and still inject into the wrong cylinder.
FAULT DETECTION	Absent pulse, or a pulse at the wrong crank angle relative to the sync gap. Loss after sync is survivable for a while — the engine can run on the last known phase — but it must not be allowed to re-sync blind.

02 Fuel system

Rail pressure `pins 32 / 35 / 08` ON ECU

MEASURES	Common-rail fuel pressure. Typically **250–1800 bar** on an engine of this class — **inferred**, the exact range needs the sensor part number.
SIGNAL	Ratiometric **0.5–4.5 V**. 🔴 `Switch Battery 5 V` on pin 32, 🟣 `Analog I/P` on pin 35, ⚫ `sensor ground` on pin 08 — a **dedicated** ground, not spliced with any other sensor. That is deliberate: this is the most accuracy-critical analog channel on the engine, and a shared return would inject other sensors' currents into its reference.
ACQUISITION	Ratiometric front-end into the ADC. Sampled fast — **1 kHz** or better, **inferred** — because it is the feedback for a closed loop that has to reject the pressure ripple every injection event causes.
USED FOR	**Closed-loop rail pressure control**: the error between measured and target pressure drives the fuel metering unit's PWM duty. Also sets the **injection quantity calculation** — the same pulse width delivers more fuel at higher rail pressure, so quantity is always a function of both. **Over-pressure protection** is a hard shutdown.
FAULT DETECTION	Out-of-range reading, or pressure that fails to follow a metering unit command — which distinguishes a dead sensor from a stuck metering unit. Loss of rail pressure feedback means injection quantity becomes unknown; treat as shutdown.

Fuel metering unit current `pin 88, sensed on-board` ON ECU

MEASURES	Current through the metering unit solenoid — the actuator's own feedback, not an external sensor.
SIGNAL	Shunt and amplifier on the ECU's low-side driver. The metering unit itself is only 2-pin: 🔴 `battery` via the 20 A fuse, and ⚫ `PWM low side` to pin 88.
ACQUISITION	ADC channel synchronised to the PWM cycle, so current is sampled at a consistent point in the switching period rather than at random phase.
USED FOR	**Inner loop of the rail pressure cascade.** Rail pressure sets a target current; current control sets PWM duty. Two loops because solenoid force follows current, and current follows duty only loosely once coil temperature and battery voltage move. Also gives **open and short circuit detection** on the actuator.
FAULT DETECTION	Zero current with duty commanded means an open coil or blown fuse. Full current at zero duty means a shorted driver — which is dangerous, since it drives the rail to maximum pressure.

Injector current, ×3

pins 73 / 07 / 29, sensed on-board

ON ECU

MEASURES	Current through each injector solenoid during its firing event. Peak **12–24 A**, hold **8–13 A**.
SIGNAL	Sense resistor in each low-side leg, read by the driver IC's own current comparators.
ACQUISITION	The MC33816 regulates current in hardware against programmed thresholds — the MCU sets the profile, the driver enforces it microsecond by microsecond. Far too fast for firmware to close.
USED FOR	**Peak-and-hold profile enforcement.** A high current opens the needle, then a lower current holds it, because holding at peak would cook the solenoid. Also **injector diagnosis**: a coil that fails open or short shows up immediately in the current trace.
FAULT DETECTION	Failure to reach peak within the expected time means an open coil or a collapsed boost rail. The handbook's injector chart shows why that timing is diagnostic — the rise time is set by the rail voltage and the coil, so a slow rise means one of them is wrong.

03 Air path

Boost pressure `pins 11 / 41 / 34` ON ECU

MEASURES	Absolute pressure in the intake manifold, downstream of the turbocharger and aftercooler. Typically **1–3 bar absolute** on a turbocharged engine of this size — **inferred**.
SIGNAL	Ratiometric 0.5–4.5 V. One half of a **four-pin combination sensor** shared with manifold air temperature: 🔴 `Switch Battery 5 V` on pin 11, 🟣 `Analog I/P Pressure` on pin 41, ⚫ `sensor ground` on pin 34 shared with both legs and with the coolant sensor.
ACQUISITION	Ratiometric front-end into the ADC. **100 Hz** is ample — manifold pressure changes with turbo speed, which has real inertia. **Inferred.**
USED FOR	**Air mass estimation**, with air temperature, via the ideal gas law — and air mass is what sets the **smoke-limit fuelling cap**. A diesel that injects more fuel than its air charge can burn produces visible smoke and heat rather than torque, so the fuelling limit tracks boost. Also **EGR control**, since EGR displaces fresh air, and **turbo overspeed protection**.
FAULT DETECTION	Range check, plus plausibility against barometric pressure at rest — with the engine stopped, manifold pressure must read close to ambient. A sensor that fails high would silently allow over-fuelling, so this one defaults conservative on fault.

Manifold air temperature `pin 79`

`TOPOLOGY OPEN — U2`

MEASURES	Intake charge temperature after the aftercooler. Roughly **−20 to +100 °C**.
SIGNAL	`Analog I/P Temperature` on pin 79, sharing the combination sensor's supply and ground. **Whether it is an NTC or a conditioned ratiometric output is unresolved.** The wire colour class cannot distinguish them — the coolant NTC uses the same magenta class. Industry convention for combination pressure-and-temperature sensors is an NTC element, so that is the working assumption.
ACQUISITION	Pull-up divider if NTC, ratiometric front-end if not. The board carries a **DNP pull-up footprint** so either works without a respin — the cheapest possible insurance against a question we cannot currently answer.
USED FOR	**Air density correction.** Boost pressure alone does not give air mass; the same pressure at 20 °C and at 80 °C is meaningfully different charge. Feeds the same smoke-limit calculation. Also **intercooler diagnostics** — a rising delta over ambient means the aftercooler is fouling.
FAULT DETECTION	Range check, and plausibility against coolant temperature at cold start: on a machine that has stood overnight, intake air, coolant and ambient should all agree within a few degrees. That check is free and catches a surprising number of wiring errors.

EGR valve position

`pins 15 / 37 / 36` — ON ECU

MEASURES	Actual position of the EGR valve, 0–100% travel.
SIGNAL	Ratiometric potentiometer. 🔴 `Switch Battery 5 V` on pin 15, 🟣 `Analog I/P` wiper on pin 37, ⚫ `sensor ground` on pin 36. Shares its five-pin connector with the actuator's own 🟣 `EGR High` and ⚫ `EGR Low` drive pins.
ACQUISITION	Ratiometric front-end into the ADC, **100 Hz**, matched to the valve's mechanical response. **Inferred.**
USED FOR	**Closed-loop EGR position control** — the H-bridge drives until measured position matches demand. EGR rate is one of the two things this engine uses to meet its NOx limit, the other being injection timing, which is exactly why the certification question in the handbook bites so hard. Also **stuck-valve detection**.
FAULT DETECTION	Position failing to follow demand within a timeout means a stuck valve, a failed motor, or a broken position sensor — distinguishable by whether current is flowing. A stuck-open EGR valve at load is a derate condition.

Intake throttle position

`6-pin connector, routing unknown` — UNRESOLVED — U8

MEASURES	Intake throttle plate position. On a CPCB IV+ diesel the throttle is not a power control — it creates intake depression to drive EGR flow when the pressure differential is otherwise insufficient.
SIGNAL	**Six pins**, which is two more than a motor pair plus a three-wire position sensor accounts for. The extra two are unexplained. It does not appear on any ECU pin traced in the wiring diagram — it may belong to a variant, or route somewhere we have not found.
ACQUISITION	Not designed. Blocked until the connector is traced on the engine.
USED FOR	If present: **EGR flow authority at low load**, and **a clean shutdown** — closing the throttle starves the engine of air and stops it smoothly.
FAULT DETECTION	Not designed.

04 Engine health

These are the sensors that stop the engine before it destroys itself. They earn their place in an ECU that never optimises anything.

Coolant temperature `pin 33` ON ECU

MEASURES	Engine coolant temperature, **−40 to +130 °C**. Normal running is roughly 80–95 °C.
SIGNAL	**NTC thermistor, two wires only** — no excitation pin, unlike the ratiometric sensors. 🟣 `Analog I/P` on pin 33; its ⚫ `sensor ground` does not get its own ECU pin but splices into the boost sensor's return at pin 34.
ACQUISITION	Pull-up divider into the ADC, linearised in firmware against the sensor's resistance-temperature table. **2 Hz is plenty** — an engine block's thermal time constant is tens of seconds. The pull-up is chosen to put maximum sensitivity around 85 °C, where the decisions are.
USED FOR	**Cold-start fuelling enrichment** — a cold diesel needs more fuel and earlier injection to light off. **Overheat derate then shutdown. EGR inhibit when cold**, since EGR on a cold engine fouls the intake. **Reported over CAN** in ET1, PGN 65262.
FAULT DETECTION	Open circuit reads full pull-up voltage and short reads zero — both outside any physical temperature, so both are unambiguous. Critically, **an open circuit must not read as "very cold"**, which would trigger enrichment on a hot engine. The independent coolant switch below exists precisely because this channel can fail.

Coolant temperature switch `pin 43, return on 01 / 02` ON ECU

MEASURES	A single over-temperature threshold. Not a measurement — a trip.
SIGNAL	🔵 `Digital I/P` on pin 43, switch-to-return. Its return leg is one of the `G_G_B` pins.
ACQUISITION	Pull-up and RC debounce into a GPIO, polled at **10 ms**.
USED FOR	**A second, independent path to overheat shutdown.** This is the same design instinct as the overspeed discussion in the handbook: the analog coolant channel and this switch fail in different ways, so requiring only one of them to trip means a single sensor failure cannot leave the engine cooking. It is cheap redundancy on the failure that quietly destroys engines.
FAULT DETECTION	Wired normally-closed where possible, so an open harness reads as tripped rather than as healthy.

Lubricating oil pressure `pins 39 / 80` ON ECU

MEASURES	Engine oil gallery pressure, typically **0–10 bar**. Normal running above roughly 2 bar; idle above roughly 1 bar. **Inferred.**
SIGNAL	Ratiometric 0.5–4.5 V, three pins: 🔴 `Switch Battery 5 V` on pin 39, 🟣 `Analog I/P` on pin 80, and a ⚫ `sensor ground` that splices with the EGR sensor's into pin 36.
ACQUISITION	Ratiometric front-end into the ADC at **10 Hz**. **Inferred** — oil pressure responds within a revolution, but no decision needs faster.
USED FOR	**Low oil pressure shutdown**, the single most important protection on any engine — bearing damage from oil starvation happens in seconds. The threshold is **speed-dependent**: pressure that is normal at idle is a fault at rated speed, so the limit is a function of the crank sensor's reading. Also **start inhibit** until pressure builds, and reported over CAN in EFL/P1, PGN 65263.
FAULT DETECTION	Range check. The dangerous failure is reading high when pressure is actually low, so a plausibility check against engine state helps: with the engine stopped, oil pressure must read near zero. A sensor that reads 3 bar on a stopped engine is lying.

05 Electrical

Battery voltage, ×2 pins 04 / 06 ON ECU

MEASURES	Voltage on two separately fused battery rails, **6–40 V** covering cranking sag through clamped load dump. Nominal 12 V, charging 13.5–14.4 V.
SIGNAL	Two 🔴 `DC positive` feeds, `V_V_BAT_1R` and `V_V_BAT_2R`, each behind its own 10 A fuse.
ACQUISITION	**120 kΩ / 10 kΩ** divider into the ADC — no active conditioning, this is a slow rail rather than a delicate signal. **10 Hz.** The divider value was corrected from an inherited 75 kΩ / 10 kΩ that would have put 4.69 V on a 3.3 V pin.
USED FOR	**Low and high battery alarms. Charging system diagnosis** — voltage that never rises above resting means the alternator is not charging. **Injector timing compensation**, since boost converter recharge time depends on supply voltage. And because the two rails are separately fused, **comparing them infers a blown fuse** without a dedicated fuse-status circuit. Reported over CAN in VEP1, PGN 65271.
FAULT DETECTION	The two channels check each other: they should track within a volt. A persistent divergence means a fuse, a corroded connection, or a failed divider.

Ignition / run request `pin 71` ON ECU

MEASURES	Whether the panel is commanding the engine to run.
SIGNAL	Switched battery, active high. Downstream of the GCU's 70 A ignition relay.
ACQUISITION	Zener-clamped bias network into a GPIO — not a divider, for the reason given in the handbook. Polled at **10 ms**.
USED FOR	**Run enable and orderly shutdown.** Losing this input is the normal stop request: the ECU stops injection, closes the throttle if fitted, and reports the stop over CAN. It is also, per the handbook's safety section, the **signalling half of a path whose other half is physical** — the same relay that drops this input is believed to remove power from the fuel system entirely, which is what makes a hung ECU survivable.
FAULT DETECTION	Not detectable as an open circuit — an open wire and a genuine stop request are electrically identical. That is an accepted limitation of matching the OEM harness, and it is recorded in the FMEA rather than engineered around, because changing the sense would break compatibility.

06 Operator inputs

Not sensors in the strict sense, but they arrive on the same connector and firmware treats them identically — debounced digital state with a defined meaning.

INPUT	PIN	SENSE	WHAT IT DOES	STATUS
Audio abort	20	🔴 `active high`	Silences the buzzer without clearing the underlying fault. The fault stays active, the annunciator stays lit, and only the noise stops — so an operator can think without losing the alarm.	Confirmed
Override SS	24	🔴 `active high`	Function genuinely unconfirmed. Labelled `OVERSIDE SWITCH` at the ECU and `-S7 Override SS SW` at the panel — **one switch, not two**, despite earlier documents listing an overspeed switch and an override switch separately.	Open
Droop select	23	🔵 `switch to return`	Selects droop versus isochronous governing. In droop, speed falls slightly as load rises, which lets multiple sets share load without fighting. Isochronous holds speed exactly, which is right for a set running alone.	Confirmed

> **DO NOT WIRE CONTROL LOGIC TO PIN 24 YET**
>
> The override switch's actual function is a guess. Read its state, log it, report it over CAN — but do not let it change what the engine does until somebody has confirmed what the panel intends by it. A switch whose meaning you have assumed is worse than a switch you ignore.

WHAT THE CONTROLLER NOW DEMANDS FROM US — CONFIRMED 18 SEP 2026

The fitted **DSE4522**'s own configuration takes **oil pressure**, **coolant temperature**, **coolant level** and **engine speed** from the ECU over CAN rather than from its own analogue inputs, on source addresses 234 (engine) and 44 (instrumentation). Its analogue inputs A and B are configured *Not Used* for exactly that reason.

Two consequences for this design. Those four values are no longer optional instrumentation — the controller's protections act on them, at 0.99 bar oil pressure and 200 °C coolant. And **ECU Data Fail is configured to shut the set down**, so our CAN transmission is itself a watchdog held by another device. Engine speed is the uncomfortable one: the controller's overspeed trip reads the number we send, which is why handbook §04 now calls a standalone trip module mandatory.

07 Panel-side sensors

These exist in the machine and appear in the wiring diagram, but they do not reach the ECU. They belong to the panel controller — a Deep Sea **DSE4522 MKII AMF**, identified 18 Sep 2026; earlier revisions of this page said KG640C, which is what the OEM wiring diagram shows but not what is fitted. They are documented here because earlier design passes built ECU front-ends for two of them, and because knowing what the panel already measures tells you what the ECU does not need to.

SENSOR	GOES TO	SIGNAL	USED FOR
Fuel level	Panel via XC pin 22	Resistive float sender, 2-pin	Tank level display and low-fuel alarm. **Not an ECU signal** — earlier documents wrongly gave it an ECU front-end.
Water in fuel	Panel via XC pin 16/1	Conductivity probe, 3-pin with built-in electronics	Water-in-fuel warning. Water in a common-rail system destroys injectors, so this is a genuine alarm. **Not an ECU signal.**
Catalyst temperature	Unresolved — U9	2-pin, one sensor only	DOC inlet or outlet temperature. Earlier documents specified three EGT channels; the diagram shows one.

SENSOR	GOES TO	SIGNAL	USED FOR
Alternator voltage & frequency	GCU terminals 27–30	3-phase AC sense, sampled at 5 kHz	Output metering, and — per the handbook's safety section — **the GCU's only engine speed source**. On a 4-pole alternator 1500 rpm is rigidly 50 Hz, which is what makes overspeed detection possible without an ECU.
Load current	GCU terminals 35–42	Current transformers	Load metering, overload protection, load-dependent start/stop.
Charging alternator D+	GCU	Excitation sense	Charge-failure alarm, and a secondary indication the engine is turning.

08 What we would add

The OEM sensor set is the minimum to run the engine and meet emissions. These are the additions worth designing space for now, since a connector position or an unused ADC channel costs almost nothing at design time and cannot be retrofitted cheaply.

Ordered by how much each earns its place.

ADDITION	INTERFACE	WHY IT EARNS A PLACE	COST TO DESIGN IN NOW
Barometric pressure	On-PCB MEMS, I²C	Altitude and weather change air density, and therefore the smoke-limit fuelling cap. Without it the engine either smokes at altitude or is derated everywhere to be safe. Needs no harness at all — barometric pressure is the same inside the enclosure as outside.	One IC, two I²C pins, a vented enclosure port. **Already in the design.**
Magnetic pickup	VR, to a standalone trip module	The independent overspeed path discussed in the handbook. Currently recommended rather than mandatory, because the GCU's alternator-frequency path already bypasses the ECU — but it is the only option that also survives a GCU firmware hang.	A flywheel sensor, a trip module, and panel wiring. No ECU pins at all, which is the point.
Exhaust gas temperature, ×2	Type K thermocouple, CJC amplifier	DOC inlet and outlet. The difference across a catalyst tells you whether it is working; a single sensor tells you much less. Also protects the turbine from over-temperature.	Two amplifier ICs, two ADC channels, four connector positions.
Fuel temperature	NTC, 2-wire	Diesel density varies with temperature, so the same injected volume is a different mass. Matters most for fuel consumption reporting and for hot-restart behaviour, where return fuel has been heated.	One ADC channel, one pull-up, two connector positions.

ADDITION	INTERFACE	WHY IT EARNS A PLACE	COST TO DESIGN IN NOW
Crankcase pressure	Low-range ratiometric	The best single indicator of engine wear available. Rising blow-by past the rings shows here long before anything else notices, which turns an unplanned failure into scheduled maintenance.	One ADC channel and a sensor. Genuinely the highest-value addition for a machine expected to run for years.
Ambient air temperature	NTC, 2-wire	Separates engine-related temperature rise from a hot day. Makes the derate strategy fair, and makes the aftercooler diagnostic in §03 meaningful.	One ADC channel.
Accelerometer	MEMS, SPI or I²C, on-PCB	Vibration signature as a condition-monitoring input — bearing wear, misfire, and mounting problems all have signatures. Speculative for a genset, but nearly free if the board is being laid out anyway.	One IC and a few pins. Firmware effort is the real cost, not hardware.

WHAT FUTURE-PROOFING ACTUALLY MEANS HERE

Not fitting every sensor on the list. It means **reserving the channels and connector positions** so that adding one later is a firmware change and a harness leg, not a board respin. Concretely: leave four ADC channels unrouted but broken out, keep two spare positions on the sensor connector, put the I²C bus on a header even if only the barometric sensor uses it now, and leave the second CAN channel populated.

The MCU has room for all of it — the channel budget below shows the design using well under half of what the part provides. Spending some of that headroom on unpopulated options is the cheapest decision available at this stage.

09 Channel budget

What the sensor set costs in MCU resources, against what the S32K148 provides. The headroom is the argument for designing the additions in now.

RESOURCE	USED NOW	WITH ALL ADDITIONS	S32K148 PROVIDES	HEADROOM
ADC channels	14	19	up to 64	ample
Timer capture	2	2	64 across 8 FlexTimers	ample
Timer output / PWM	8	8	shared with above	ample
Digital in	5	5	abundant GPIO	ample
CAN	2	2	3 FlexCAN	1 spare
I²C	1	1	2 LPI2C	1 spare
SPI	1 (injector driver)	2	3 LPSPI	1 spare

The ADC count is the one to watch, and it is not close to a limit. The caveat from the MCU memo stands: how many of those 64 channels are actually broken out on the 144-pin package needs the reference manual's pin-mux table, which is a Phase 2 item.

ENGINE CONTROL UNIT · KIRLOSKAR 3GK550ETA 4SR1

Circuit Reference

Every circuit block that has been built and verified, with the schematic, the reasoning, the simulated result, and what each model leaves out. A dashed part is on the board but not in the netlist below it — its impedance is either folded into a lumped value elsewhere in the same circuit, or left out entirely. The checks are claims about the solid parts only.

24 blocks all checks **passing** generated from **sim/blocks/** ← Field handbook

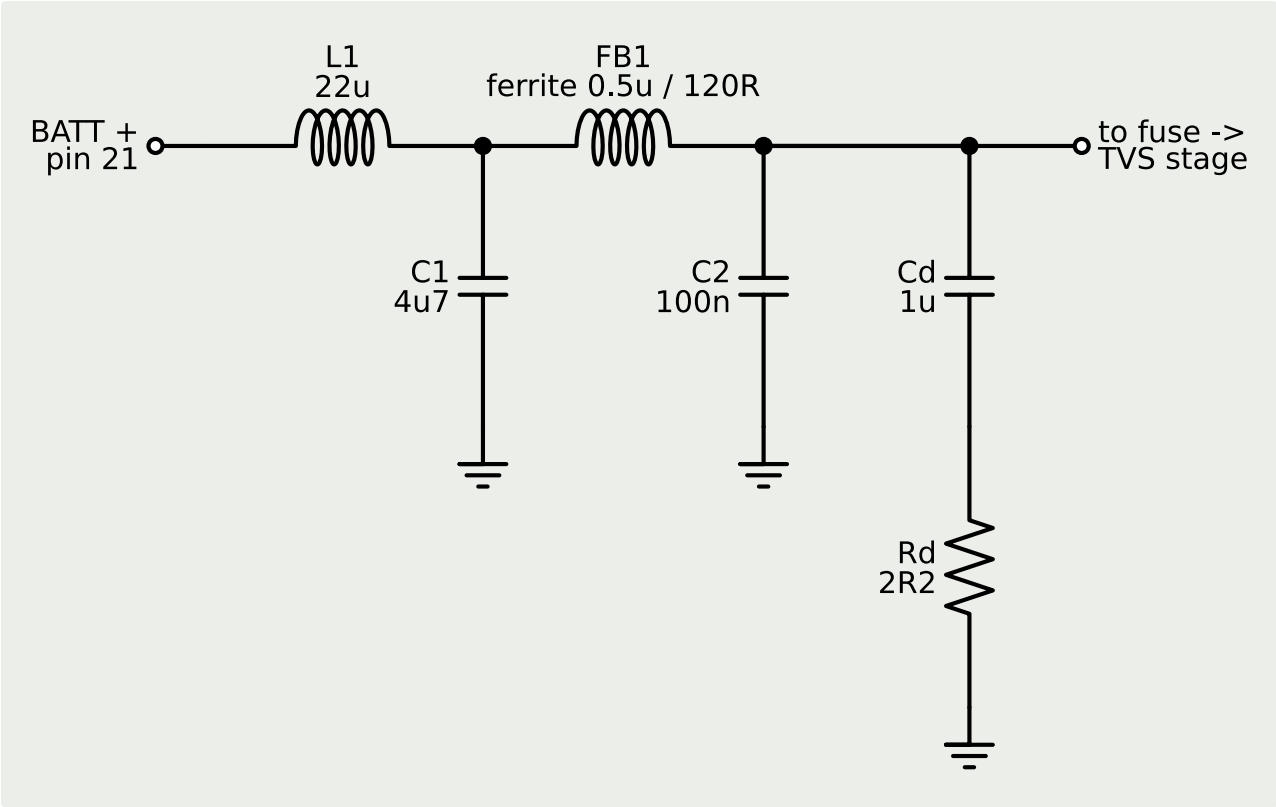
Sensor reference →

01 Power chain
02 Sensor front-ends
03 Speed and position
04 Injection
05 Actuators
06 Communications

01 Power chain

From the battery terminal to the MCU rail. Six stages, each one a different failure it has to survive.

EMI input filter `pin 21` `18/18 checks`



Two LC stages between the battery and everything else, measured 50 ohm to 50 ohm the way a LISN would see it.

Why it looks like that. One stage is not enough, and the reason is the capacitor: C1's own lead inductance puts its self-resonance near 1.9 MHz, and above that it stops being a capacitor. Stage two picks up there. The damper across C2 is not optional either — without it the ferrite and C2 ring at 717 kHz and hand back 12 dB right where the switching frequency's second harmonic sits.

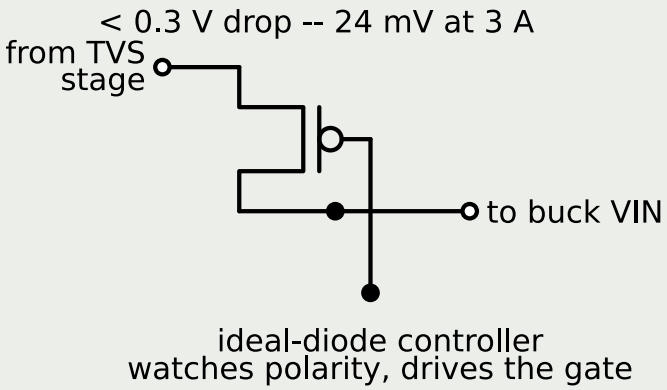
> **What the model does not cover.** Differential mode only. Common-mode noise — both wires swinging together against chassis, which is most real automotive failures — needs the common-mode choke, and that cannot be modelled honestly without a layout to give it a return path.

PASS insertion loss at 150 kHz (band start)		-40.564 dB
PASS insertion loss at 400 kHz (switching f)		-50.734 dB
PASS insertion loss at 1 MHz		-72.103 dB
PASS insertion loss at 30 MHz		-129.540 dB
PASS insertion loss at 100 MHz (FM band)		-95.679 dB
PASS worst insertion loss anywhere above 150 kHz		-40.663 dB
PASS ... worst point sits at		151.5 kHz
PASS is 40 dB enough for Class 5?	unknown -- needs measured unfiltered emissions on hardware	
PASS ... IL@150kHz, C1 at X7R's own -15% temp derate (C1=3.74uF)		-39.242 dB
PASS ... IL@400kHz, same corner		-49.291 dB
PASS pin 21 DC current, upper bound (buck at its peak limit)		2.949 A
PASS L101 7447709220: saturation margin over the bound (Isat 6.5 A typ)		2.204 x
PASS ... self-heating at the bound, on a 105 C board		117.384 C
PASS ... under its 125 C operating maximum		117.384 C
PASS L102 744314047: saturation margin (Isat 9 A typ)		3.052 x

PASS F101 0437005.WRA: continuous rating at 105 C (80% x 79%)	3.160 A
PASS ... over the pin 21 bound by	1.072 x
PASS ... the largest 437A rating still rated 63 V (7 A is 35 V, under the 40 V load dump)	63.000 V

Netlist: sim/blocks/emi_filter.cir

Reverse-battery protection – 10/10 checks



A P-FET held on by a controller that watches polarity, compared against the obvious Schottky.

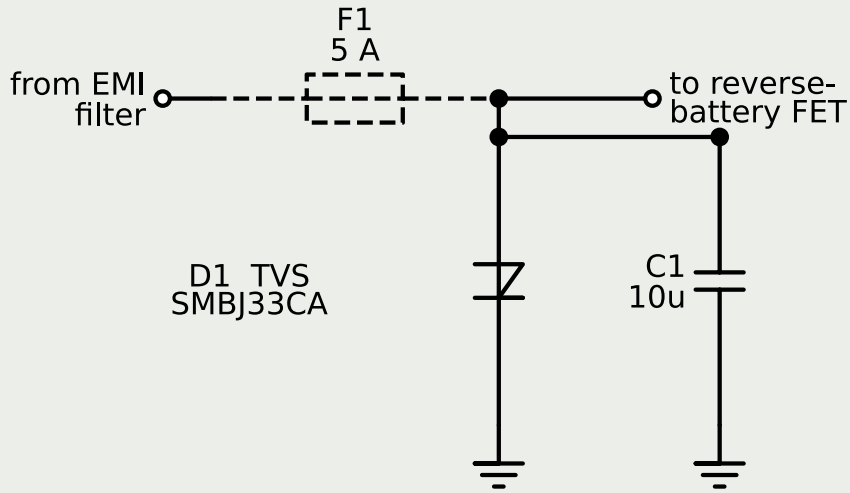
Why it looks like that. Somebody will reverse the jump leads eventually. A Schottky survives it and charges you 0.46 V on every amp forever; the ideal-diode arrangement costs a controller IC and drops 24 mV. At 3 A that is 1.4 W of heat that simply never happens.

> **What the model does not cover.** Steady state only. A real controller takes about a microsecond to turn the FET off, and current flows backwards through the body diode during that window. The switch model here turns off instantly, so it will always look perfect in reverse — that number is a datasheet question.

PASS Schottky drop at 3 A		0.465 V
PASS ideal-diode FET drop at 3 A		0.024 V
PASS FET drop stays under 0.3 V spec to 5 A		0.040 V
PASS ... Schottky does NOT meet it	0.52 V at 5 A -- fails the 0.3 V spec	
PASS ... at the sheet's specified ceiling, 40 mOhm at Vgs = -4.5 V		0.200 V at 5 A
PASS ... still inside the 0.3 V requirement by		1.500 x
PASS power burnt in the Schottky at 3 A		1.395 W
PASS ... and in the FET instead		0.072 W
PASS Schottky blocks -14 V (leak into 1k)		1.000 mV
PASS ideal-diode FET blocks -14 V		0.014 mV

Netlist: sim/blocks/reverse_battery.cir

Transient clamp `pin 21` `30/30 checks`



F1 dashed: on the board per spec sec.4, but this netlist folds all series impedance into Rsrc -- F1 is not modelled, and its survival under pulse 2a's repeated ~38 A stress is untested

A bidirectional TVS behind a 5 A fuse, against ISO 7637-2 pulse 2a: +112 V through 2 ohms for 50 us. Second and third stages of spec sec.4's pin 21 order — EMI filter, then fuse, then this TVS, then the reverse-battery FET.

Why it looks like that. Clamps to 50 V, which leaves 2x margin inside the buck controller's 100 V rating and keeps 60 V-class parts viable everywhere downstream.

> **What the model does not cover.** It misses the 42 V figure the earlier documents asked for, and that figure should be dropped: it was arbitrary, with no standard behind it. The real constraints are the parts' ratings. The fuse (dashed in the schematic) is not modelled either — this netlist folds all series impedance into Rsrc — and its survival under pulse 2a's repeated ~38 A instantaneous stress has not been tested, unlike load_dump, which did check this for its own pulse.

PASS pulse actually applied (source peak)	
PASS clamped voltage at buck input	
PASS ... margin inside LM5164's 100 V rating	
PASS ... 60 V-class parts are NO LONGER viable downstream	
PASS ... so the reverse-battery FET needs 100 V class, not 60	
PASS ... EVERY part on VBAT_PROT must clear this, not just the FET	
PASS ... so a 40 V-abs-max integrated driver is EXCLUDED from this rail	
PASS ... and the load dump alone already leaves it no margin	
PASS ... minimum class for anything battery-connected	80 V or better: above the 73.3 V pulse 2a clamp with margin. 60 V
PASS rail table: LM5164 buck, VIN -- VIN abs max	
PASS rail table: Q1 reverse FET, P-ch -- |Vds| class	

```
PASS rail
table: Q2
negative-clamp
FET -- Vds
class
```

```
PASS rail
table: D103
negative-clamp
Schottky -- Vr
class
```

```
PASS rail
table:
TPS40210 boost
ctrl, VDD
(220R + 36 V
clamp, 46.6 V
worst) --
clamped VDD
```

```
PASS rail
table: Boost
inductor / FET
(150 V class)
-- Vds class
```

```
PASS rail
table:
Injector
battery-side
blocking
diodes -- Vr
class
```

```
PASS rail
table: Relay
flyback diodes
(discrete_io)
-- Vr class
```

```
PASS rail
table: Relay
low-side FETs
(discrete_io)
-- Vds class
```

```
PASS rail
table:
Metering FET
and freewheel
(metering_egr)
-- Vds / Vr
class
```

```
PASS rail
table:
Injector low-
side FETs
(drain on
boost rail) --
Vds class
```

```
PASS rail
table: EGR
bridge FETs,
all four
(metering_egr)
-- Vds class
```

```
PASS ... what
the two
corrected
parts WERE
```

```
PASS GATE_KILL
unclamped vs a
20 V Vgs
rating (the
finding)
```

```
PASS GATE_KILL
with BZX84-
C12-Q (12.7 V
max at 5 mA) -
- margin
```

```
PASS ... hot:
BZX84-C12-Q
+10 mV/K max,
100 K above 25
C
```

```
PASS ... zener
current at the
pulse peak
```

```
PASS ... 47k
pullup
dissipation at
the pulse peak
```

```
PASS recovers
to nominal
after pulse
```

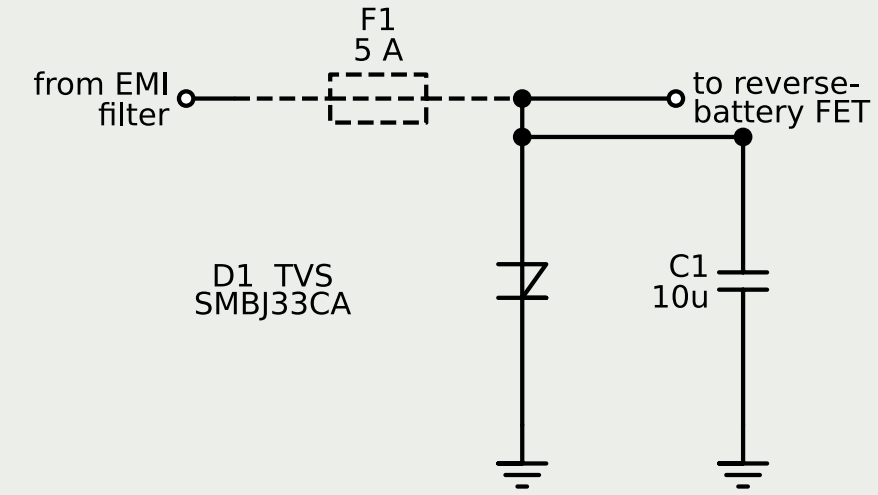
```
PASS F101:
pulse 2a I2t
through the
fuse
```

```
PASS ... as a
fraction of
0437005.WRA's
2.30 A2s
melting I2t
```

Netlist: sim/blocks/transient_clamp.cir

Load dump pin 21

11/11 checks



pulse 5b: Us*=40 V, 0.5R, 400 ms
D1 absorbs ~46 J at ~116 W -- FAILS an SMBJ-class part's rating
F1 dashed (drawn, not in this netlist) does not rescue it either:
fault current is 3.04 A, 0.6x F1's rating, and won't clear in 400 ms

The same TVS and capacitor as transient_clamp, against ISO 7637-2 pulse 5b: a suppressed load dump, 40 V through 0.5 ohms for 400 ms — four orders of magnitude longer than pulse 2a.

Why it looks like that. Clamps to 38 V, comfortably inside every voltage rating in the chain — the same story as pulse 2a. But 400 ms is long enough that the limit stops being clamp voltage and becomes dissipated energy: the TVS absorbs roughly 46 J at ~116 W average, which is on the order of 100x an SMBJ-class part's single-pulse energy rating and its continuous power rating both. This block FAILS on purpose — a TVS sized against pulse 2a's peak voltage was never sized against pulse 5b's energy, because nothing before this block asked the energy question. But the whole failure turns on one inequality: re-run at Us* = 35 V, the figure usually quoted for a 12 V suppressed dump, and the TVS never reaches its 36.7 V breakdown — 0.000 A, 0 J, no problem at all. So the finding is not that an SMBJ33CA cannot survive a load dump. It is that a TVS whose standoff sits BELOW the dump level gets forced to conduct during a normal event and then has to absorb it. The fix is a higher standoff (SMBJ40CA/45CA class), not a bigger energy rating: the TVS then stays off and the buck, rated 100 V, rides 40 V without complaint. Energy absorption only becomes unavoidable under pulse 5a.

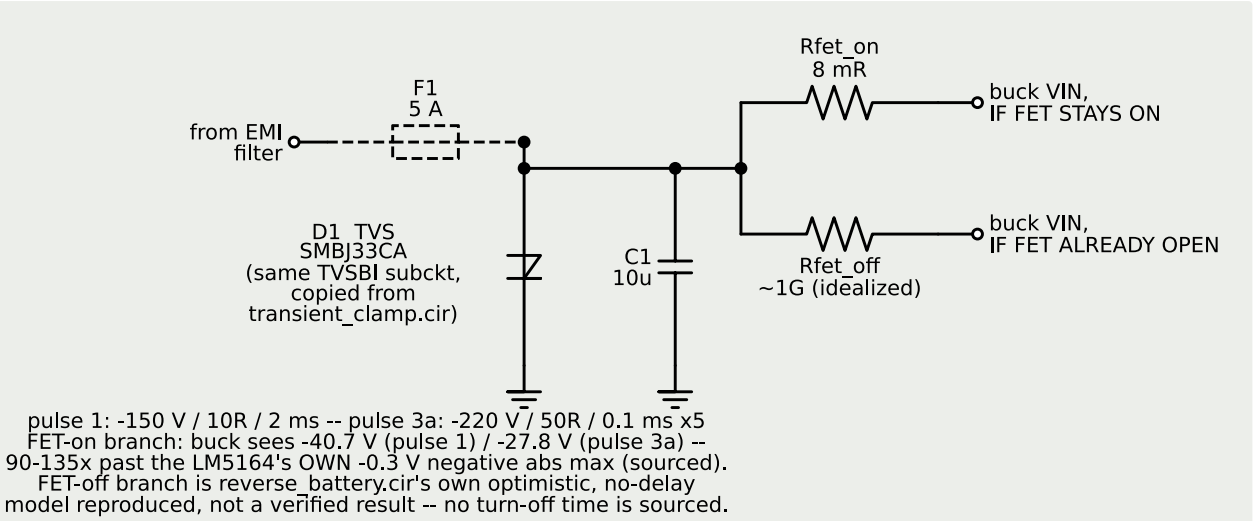
> **What the model does not cover.** Whether this pulse even applies: 5b assumes the alternator's rectifier is suppressed/clamping. That has not been confirmed for this genset — if it is not, the real event is pulse 5a (65–87 V, unclamped), a harder problem this block does not simulate. Also not modelled: the real pulse's exponential-decay shape (approximated here as a trapezoid, which is pessimistic in the right direction), and the TVS's actual junction thermal transient. The 5 A fuse (dashed in the schematic, not a netlist element) does not rescue this failing case either: the fault current is 3.04 A, 0.6x the fuse's rating, and it will not clear in 400 ms.

PASS pulse actually applied (source peak)	40.000 V
PASS clamped voltage through the 400 ms plateau	39.604 V
PASS ... inside LM5164's 100 V rating	2.525 x
PASS ... 60 V-class parts still viable	39.604 V
PASS TVS current through the plateau -- now ZERO, the TVS stays off	0.000 A
PASS average power dissipated in the TVS	0.000 W
PASS total energy absorbed by the TVS over the pulse	0.000 J
PASS energy stays within a single-pulse (~0.5 J) rating	0.000 J
PASS average power stays within continuous (~3 W) rating	0.000 W
PASS ... fraction of the continuous rating now used	0.000 x
PASS recovers to nominal after pulse	13.366 V

Netlist: `sim/blocks/load_dump.cir`

Negative transients `pin 21`

41/41 checks



The same TVS and bulk capacitor as transient_clamp/load_dump, against ISO 7637-2's two worst NEGATIVE pulses: pulse 1 (−150 V through 10 Ω for 2 ms, a supply disconnection from an inductive load) and pulse 3a (−220 V through 50 Ω for 0.1 ms, simulated as a 5-pulse burst at the standard's own 90 ms minimum repeat). Both Level IV, both sourced from research memo 06's own severity table. This became answerable only once the TVS stopped being modelled as a single diode, which conducts forward — and silently wrong — in reverse polarity.

Why it looks like that. The buck's own datasheet (LM5164, Absolute Maximum Ratings, fetched directly) rates VIN at −0.3 V to 100 V — the negative limit is not a mirror of the positive one. With the reverse-battery FET assumed to still be conducting — the honest assumption, since no turn-off delay is sourced anywhere and pulse 3a is only 0.1 ms wide — the buck's VIN pin sees −40.7 V on pulse 1 and −27.8 V on pulse 3a: 90–135× past that −0.3 V rating, not a thin margin. The reverse-battery stage does nothing to prevent this on the timescale these pulses occupy; it would help enormously if the controller reacts fast enough, and nothing here — or in reverse_battery.cir — says whether it does. Pulse 1's TVS energy (0.707 J single event) exceeds load_dump's own ~0.5 J class bound by 41%, a second genuine failure, not a manufactured one — TOLERANCE shows it flips to a PASS at Level III instead of IV, the same one-inequality shape load_dump's own Us* finding has. Pulse 3a's bulk capacitor (C1) does something transient_clamp never needed to consider: its own RC (500 µs) is five times the pulse's 90 µs flat top, so C1 arrests the excursion before the TVS ever reaches its 36.7 V breakdown at nominal value — the TVS

carries zero current, and nothing accumulates across the 5-pulse burst. TOLERANCE shows this is fragile: the same MLCC DC-bias derating emi_filter.cir already documents (up to 60% loss) is enough to flip the TVS into conducting at −38.4 V instead of −27.8 V — still a small energy (4.2 mJ/pulse, far under any rating), but a different regime, and the buck sees more negative voltage either way.

> **What the model does not cover.** The FET's actual turn-off delay is not modelled anywhere in this repo — reverse_battery.cir's own RESULT NOTE says its switch model 'will always look perfect in reverse' because it has none, and no controller part is chosen to pull one from a datasheet. Both bookends (FET fully on, FET already open) are simulated directly rather than guessing at a number in between. The burst is 5 repeats, not the standard's ~40000 (1 hr at 90 ms minimum repeat); nothing here rules out a slow thermal drift that would only show up over the full count, and TVS junction temperature is not modelled at all, the same gap load_dump's own RESULT NOTE states for its much larger energy question. Pulse 1's real ISO waveform is a decaying transient, not the flat top used here — pessimistic in the right direction, same simplification load_dump uses for pulse 5b.

PASS pulse 1 actually applied (source peak)
PASS pulse 3a actually applied (source peak)
PASS battery node, pulse 1 -- now set by the clamp diode, not the TVS
PASS buck VIN pin, pulse 1, WITH the negative clamp fitted
PASS ... evidence: a Schottky ALONE still misses LM5164's own -0.3 V abs max (sourced, not INFERRED), by
PASS battery node, pulse 3a -- now set by the clamp diode, not the TVS
PASS buck VIN pin, pulse 3a, WITH the negative clamp fitted
PASS ... evidence: same, pulse 3a -- a Schottky alone misses by
PASS RESOLVED: buck VIN, pulse 1, with the active clamp
PASS ... margin inside the rating
PASS ... the dip past the -100 mV engage threshold is slew x delay
PASS RESOLVED: buck VIN, pulse 3a, with the active clamp
PASS ... margin inside the rating
PASS ... and the improvement over the Schottky alone
PASS pulse 1: clamped VIN RECOVERS after the pulse

PASS ... it does not latch into a short on the returning source	
PASS pulse 3a: clamped VIN recovers after the fifth burst	
PASS drawn network: ENGAGE threshold at VIN	
PASS drawn network: RELEASE threshold at VIN	
PASS ... with TLV3201-Q1's full 5 mV offset against it, still below zero -- the clamp cannot hold on against reverse current	
PASS ... and the model's ACLAMP thresholds are these two numbers	
PASS ... the question this block carried since 20 Sep: is the sub-microsecond window before the clamp engages acceptable?	YES, and not marginally: VIN's fall rate is set by the 10 o
PASS ... comparison only: IF the FET had already opened (idealized, reverse_battery.cir's own optimistic model, not verified against any real turn-off delay)	
PASS ... same comparison, pulse 3a	
PASS ... so the reverse-battery stage helps only if it reacts in time, and nothing sources that time	FET-on branch tracks the ba
PASS bat1 recovers to its own Rsrc1/Rload divider baseline (11.25 V, not 13.5 V)	

```
PASS bat3a recovers
to its own divider
baseline (6.75 V)
between every burst
pulse
```

```
PASS burst: peak-to-
peak spread across
repeats 2-5
(accumulation would
show up here)
```

```
PASS ... and repeat 1
matches them: nothing
is still settling at
5 ms
```

```
PASS ... no monotonic
drift across repeats
2-5
```

```
PASS TVS energy,
pulse 1 -- zero now
the clamp diode
conducts first
```

```
PASS ... so the CLAMP
DIODE now carries it:
peak current, pulse 1
```

```
PASS ... vs the 20 A
class already
specified for it
```

```
PASS ... clamp diode
energy, pulse 1
```

```
PASS ... peak clamp-
diode current, pulse
3a (50 ohm source, so
milder)
```

```
PASS ... which is the
flat-top current,
(206.5 - 0.4) / 50
ohm
```

```
PASS ... vs
load_dump's own ~0.5
J single-pulse class
bound (the TVS now
clears it by not
conducting, not by
absorbing less)
```

```
PASS TVS energy,
pulse 3a (nominal C1
-- TVS stays below
its own breakdown,
does not conduct)
```

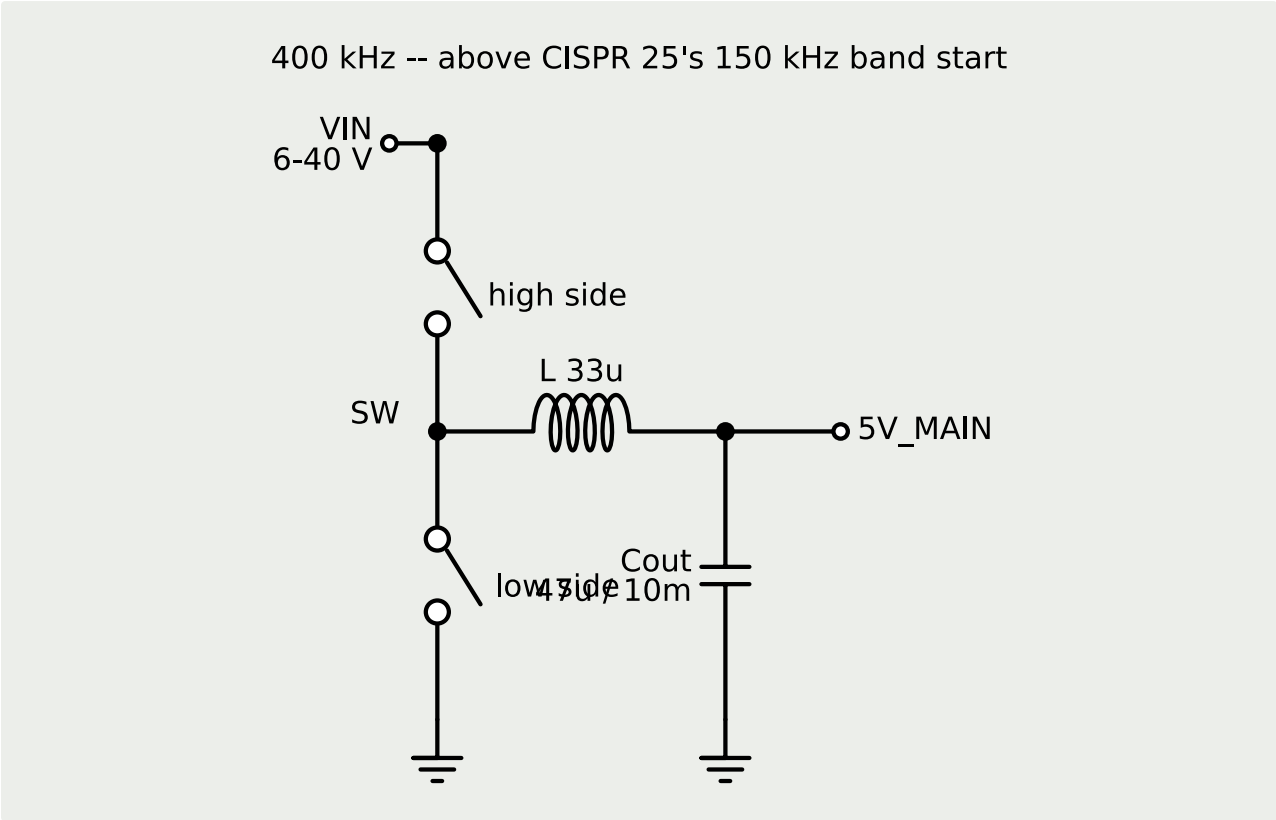
```
PASS ... so nothing
here can accumulate
at nominal C1 -- see                                TVS current is numerically zero thro
TOLERANCE for the
corner where it would
```

```
PASS F101: pulse 1
I2t through the fuse
```

```
PASS ... as a
fraction of
0437005.WRA's 2.30
A2s melting I2t
(repeated pulse: 20%
rule of thumb, pulse-
cycling data open)
```

Netlist: sim/blocks/negative_pulses.cir

Buck pre-regulator – 10/10 checks



6–40 V down to 5 V at 400 kHz, run open-loop at fixed duty so the output filter is visible.

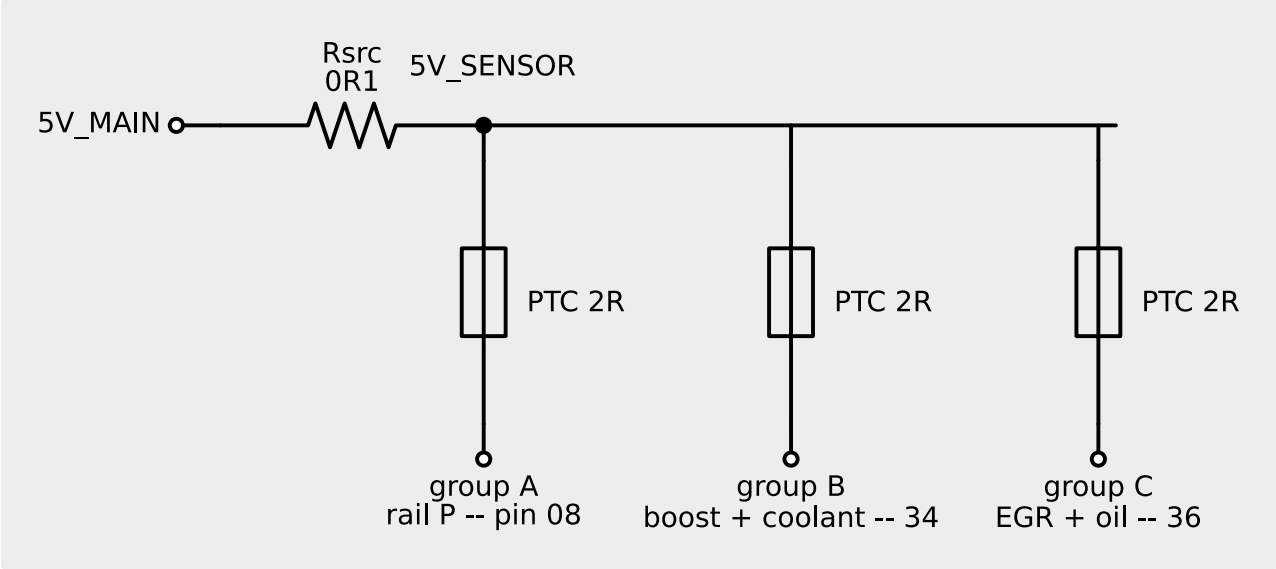
Why it looks like that. 400 kHz is chosen against CISPR 25, not for size: the conducted band starts at 150 kHz, so a fundamental below it would put every harmonic inside the measurement from the first one up. The inductor is sized at HIGH line, not low — ripple current grows with input voltage even though duty cycle falls.

> **What the model does not cover.** Ideal switches with a fixed on-resistance. Good enough for ripple, which is set by L and C; useless for efficiency, which is set by switching overlap, gate charge and reverse recovery, none of which this model has.

PASS output at 13.5 V in (open loop, D=0.370)	4.926 V
PASS output at 40 V in (open loop, D=0.125)	4.945 V
PASS inductor ripple at 13.5 V	0.256 A
PASS inductor ripple at 40 V	0.348 A
PASS ... worst case is HIGH line, not low	1.362 x
PASS worst-case ripple as fraction of 1 A load	34.840 %
PASS output ripple at 13.5 V	13.845 mV
PASS output ripple at 40 V	21.423 mV
PASS L1 (7447714330) Isat hot against the 1.18 A worst peak	1.720 x
PASS ... DCR loss at 1 A	0.079 W

Netlist: sim/blocks/buck_preregulator.cir

5V_SENSOR distribution – 24/24 checks



One tracking regulator per sensor group, behind one overvoltage cut-off, with pulse 2a applied through the real pin-21 filter.

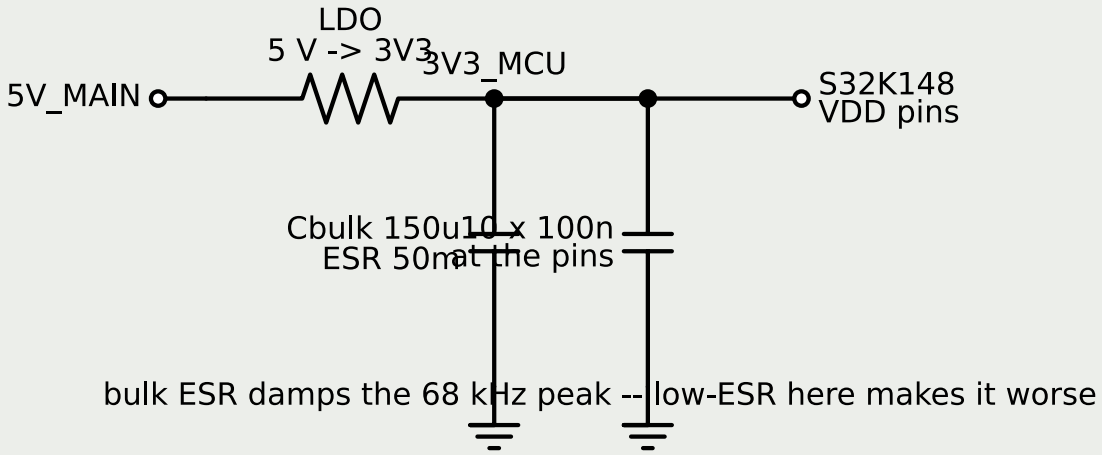
Why it looks like that. A shorted sensor harness is a routine fault. Each group's TPS7B4253-Q1 limits its own short at 301–520 mA, so the other two groups keep tracking 5V_MAIN and the MCU never sees the fault. The PTCs this replaced could not do that against a 1 A buck, and a battery short back-fed 5V_MAIN through them. The trackers survive 45 V and the rail reaches 73 V on pulse 2a, so a TPS48110-Q1 cut-off opens above 40.5–42.1 V and stays closed through the 40 V load dump. Its own inrush network (22 nF from gate to ground) also makes the FET a source follower for fast edges, so the trackers' input peaks near 22 V either way. The cut-off was chosen over an LTC4368 because it is AEC-Q100: decisions log section 7.

> **What the model does not cover.** The controller is modelled by function: a divider, a hysteretic comparator, a 2.6–4 µs delay, and a square-law FET with the datasheet's gate network. The trackers are datasheet arithmetic, not a transistor model. A sensor wire shorted to battery during pulse 2a is a double fault that this design does not cover; it is recorded.

PASS pulse 2a, worst corner: trackers' input peak (VSNS)	22.468 V
PASS ... against their 45 V absolute maximum	22.468 V
PASS ... nominal corner	22.073 V
PASS ... the OV comparator did open the FET (rail above VSNS)	79.950 us
PASS all trackers at their limit, hold-up at 26.3 uF (C211 tol + aged): VSNS low	9.231 V
PASS ... headroom over 5 V + 0.52 V dropout	3.711 V
PASS TPS48110-Q1 VS/CS slew through 100 ohm + 22 nF, top corner	4.880 V/us
PASS TPS48110-Q1 VS/CS slew through 100 ohm + 22 nF, droop corner	4.933 V/us
PASS clamp node with L1 NOT saturated (transient_clamp says 73.3)	76.487 V
PASS ... still inside the 100 V class every part there is	76.487 V
PASS OV trip, lowest tolerance	40.521 V
PASS ... above the 40 V suppressed load dump by	0.521 V
PASS OV trip, highest tolerance	42.082 V
PASS load dump passes through: 40 V on the trackers' 45 V IN	5.000 V
PASS inrush into 47 uF at turn-on (SLUSEE5E Eq. 3)	0.162 A
PASS C211 against the 40 V load dump, which the cut-off passes	1.575 x
PASS ... ESR drop, -55 C, end of life, all trackers at limit	0.250 V
PASS cranking at 6 V: headroom over 5V_MAIN + dropout	0.460 V
PASS shorted group: its tracker limits at (max)	0.520 A
PASS ... the other two still track 5V_MAIN to	4.000 mV
PASS ... load on 5V_MAIN from three ADJ pins	16.500 uA
PASS ... shorted tracker's dissipation at 13.5 V (thermal shutdown cycles it; the datasheet calls the output short-circuit proof)	7.020 W
PASS output shorted to battery: 40 V load dump on OUT's -1..45 V	40.000 V
PASS ... coincident with pulse 2a is a double fault -- recorded, not covered	decisions log section 7

Netlist: sim/blocks/sensor_rail.cir

3V3_MCU decoupling – 13/13 checks



The impedance the S32K148 sees looking into its own supply, from 1 kHz to 100 MHz.

Why it looks like that. The bulk capacitor here is sized by a resonance, not by charge. The first version used 10 uF with low ESR and peaked at 3.4 ohms around 48 kHz — the regulator's own output inductance ringing against the bulk capacitance, decades away from the bulk/ceramic anti-resonance the design was drawn around. Bigger AND lossier fixes it, which is why a plain aluminium part beats a ceramic in that position. It was pulled a second time on 18 September, from 47 uF to 150 uF: at 47 uF the block cleared its target only at the nominal value of the LDO's output inductance, a number no chosen part pins down, and a 26% margin that evaporates when an unmeasured parameter is restated is not a margin. At 150 uF that resonance has dropped and damped enough that a different peak is now the tallest — the bulk/ceramic anti-resonance at 2.6 MHz, floored near 60 mOhm — and that one does not move with the LDO at all. The verdict has stopped depending on the unpinned number. More bulk beyond 150 uF buys nothing, because it does not touch the peak that now dominates.

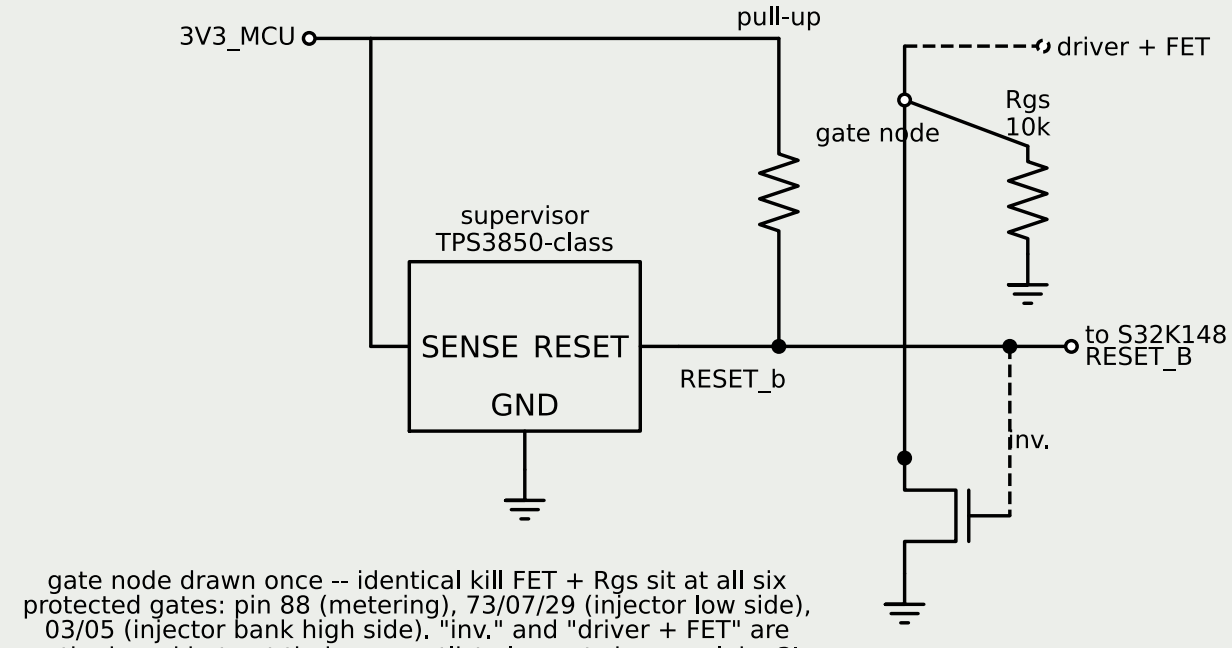
> **What the model does not cover.** Lumped: no planes, no via inductance, no spreading inductance, so it is optimistic above about 100 MHz — which is exactly where the sweep stops. An earlier version swept to 1 GHz and obediently reported ESL times omega at the last point as a 'peak'.

PASS peak impedance, 1 kHz to 100 MHz		60.355 mOhm
PASS ... and it has moved to the MHz anti-resonance		2.630 MHz
PASS ... i.e. bulk against ceramic, no longer regulator L against bulk	2.63 MHz -- the LDO resonance is no longer the limiter	
PASS ten 100 nF parts at 100 MHz		46.923 mOhm
PASS one 1 uF part at 100 MHz		357.421 mOhm
PASS ... same capacitance, this much worse		7.617 x
PASS ten-part net at the 3.5 MHz anti-resonance		55.862 mOhm
PASS ... one-part net is LOWER there (its ESR damps it)	34 mOhm vs 56 -- more ESR, better damping	
PASS ... evidence: peak with a WET electrolytic at its cold-ESR corner (Rb=200 mOhm, 4x) -- the corner that forced the spec		196.700 mOhm
PASS ... so a wet bulk part misses the 100 mOhm target by		96.700 mOhm
PASS RESOLVED: peak with the specified polymer bulk at its cold-ESR limit (Rb=72.2 mOhm, 96 uF: T599 + 22 mOhm, aged)		71.915 mOhm
PASS ... and at the low-ESR end the same part reaches warm (Rb=21.78 mOhm: the 22 mOhm resistor alone) -- the under-damped side		86.931 mOhm
PASS ... worst of the two ends, margin under target		13.069 mOhm

Netlist: sim/blocks/mcu_pdn.cir

Supervisor / fail-safe kill path –

34/34 checks



gate node drawn once -- identical kill FET + Rgs sit at all six protected gates: pin 88 (metering), 73/07/29 (injector low side), 03/05 (injector bank high side). "inv." and "driver + FET" are on the board but not their own netlist elements here -- claim 3's FAIL (see RESULT NOTE) is about Rgs, drawn and modelled as 10k.

A TPS3850-class windowed supervisor watching 3V3_MCU — the same rail mcu_pdn.cir characterizes — whose RESET drives, through one inverting stage, a kill FET at every protected driver gate (pin 88 metering, 73/07/29 injector low side, 03/05 injector bank high side), in parallel with a standing 10 kΩ gate-source pulldown.

Why it looks like that. Research memo 11 found this board has no external watchdog, no brownout supervisor, and nothing defining output state during MCU reset — and that spec §3's other lines of defence both collapsed on 18 Sep 2026 (the fitted DSE4522 takes engine speed from this ECU over CAN, and its fuel relay signals rather than removes power), making this the first line, not defence-in-depth. Five falsifiable claims came out of that memo; two mattered most. Claim 4, the central one, PASSES with a wide margin: across six brownout ramps spanning 293 µs to 1 s, the kill path always finishes (gate below Vgs(th), fought by an adversarial driver the whole time) before 3V3_MCU crosses 2.97 V — the S32K148's own PLL-guarantee floor — with margin that GROWS at slower ramps, so the memo's flagged-open 'slow sag' case does not threaten it. Claim 3 FAILS, and not narrowly: swept across the memo's own illustrative Crss range under this design's own 2.8 V/µs boost-rail turn-off edge, the 10 kΩ pulldown alone lets the coupled gate voltage exceed a representative Vgs(th) starting at Crss = 50 pF — the LOW end of that range, not an extended one — and the memo's own hand arithmetic claiming '~5.6 mV' at 200 pF is off by three orders of

magnitude (560 µA through 10 kΩ is 5.6 V). A 300 Ω–1 kΩ pulldown clears it with real margin; the fix is a component VALUE, not a new part.

> **What the model does not cover.** The pin-mux gap is real and not papered over: which physical PTxx pins 88/73/07/29/03/05 land on is still a Phase-2 item, so their individual reset pull-state defaults cannot be stated — this model treats every protected gate generically, with no GPIO drive assumed at all except as a deliberately adversarial stimulus. The supervisor IC's own comparator propagation delay (assert side) is not sourced anywhere and is modelled as zero; claim 4's tightest margin (18 µs, at the fastest ramp) is the one that delay would eat into first. The inverting stage and kill FET's own gate-drive dynamics are collapsed into one switch with an INFERRED Ron. Ciss=2 nF and Vgs(th)=1.0 V are class figures, not a chosen part's datasheet numbers — no injector/metering FET has been selected yet.

PASS claim 1: gate below Vgsth, kill engaged (bound: 10-40 us)
PASS ... actual kill-path discharge time
PASS claim 1b: gate stays below Vgsth despite an adversarial driver actively holding it high (GPIO-independence)
PASS claim 2: final V(gate) with the 470 ohm pulldown fitted
PASS ... vs the same driver with no pulldown
PASS ... driver output impedance needed to still reach 9.5 V
PASS ... the 100 ohm driver this block assumed is too weak by
PASS claim 3 (evidence): peak V(gate), Crss=20 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=50 pF, Rpd=10k

PASS claim 3 (evidence): peak V(gate), Crss=100 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=200 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=500 pF, Rpd=10k
PASS claim 3 (evidence): peak V(gate), Crss=1000 pF, Rpd=10k
PASS ... memo 11's own hand arithmetic at Crss=200 pF claimed ~5.6 mV -- off by 1000x (560 uA * 10 kOhm = 5.6 V, not 5.6 mV); thi
PASS ... so the 10k pulldown fails even at the memo's OWN low end
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=293 us
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=585 us
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=1.17 ms

PASS claim 4: gate safe before 2.97V guarantee floor, ramp=10 ms
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=100 ms
PASS claim 4: gate safe before 2.97V guarantee floor, ramp=1000 ms
PASS ... tightest margin is at 18 us at 293 us -- the FASTEST ramp tried
PASS claim 5: kill-path latency vs one 26.67 ms cylinder interval
PASS ... caveat: excludes the watchdog's not own fault-DETECTION time
PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=220 ohm
PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=470 ohm
PASS claim 6 (sweep): peak V(gate) at Crss=500 pF, Rpd=1000 ohm

```
PASS claim 6
(sweep): peak
V(gate) at
Crss=500 pF,
Rpd=2200 ohm
```

```
PASS claim 6
(sweep): peak
V(gate) at
Crss=500 pF,
Rpd=4700 ohm
```

```
PASS ... same
sweep read at
Vgsth=2.5 V,
the
permissive
end
```

```
PASS ...
largest
pulldown that
holds the
gate below
Vgsth
```

```
PASS ... cost
of 470 ohm:
drive current
per gate held
at 10 V
```

```
PASS ... and
across all
six driver
gates
```

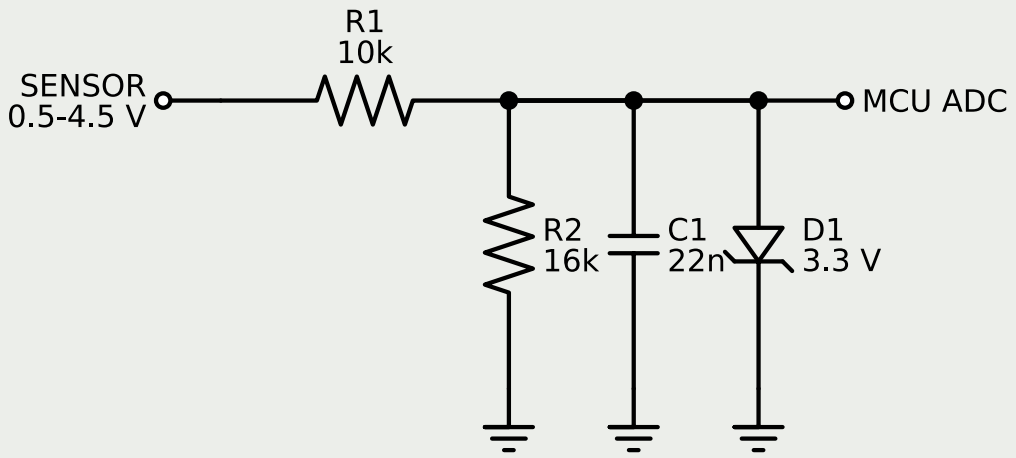
```
PASS chosen
FETs' worst
Crss (max)
against the
500 pF sizing
```

Netlist: `sim/blocks/supervisor.cir`

02 Sensor front-ends

Everything that turns something on the engine into a number in the ADC.

Ratiometric sensor input pins 41, 35, 80, 37 16/16 checks



The 0.5–4.5 V pressure and position channels, scaled into a 3.3 V ADC.

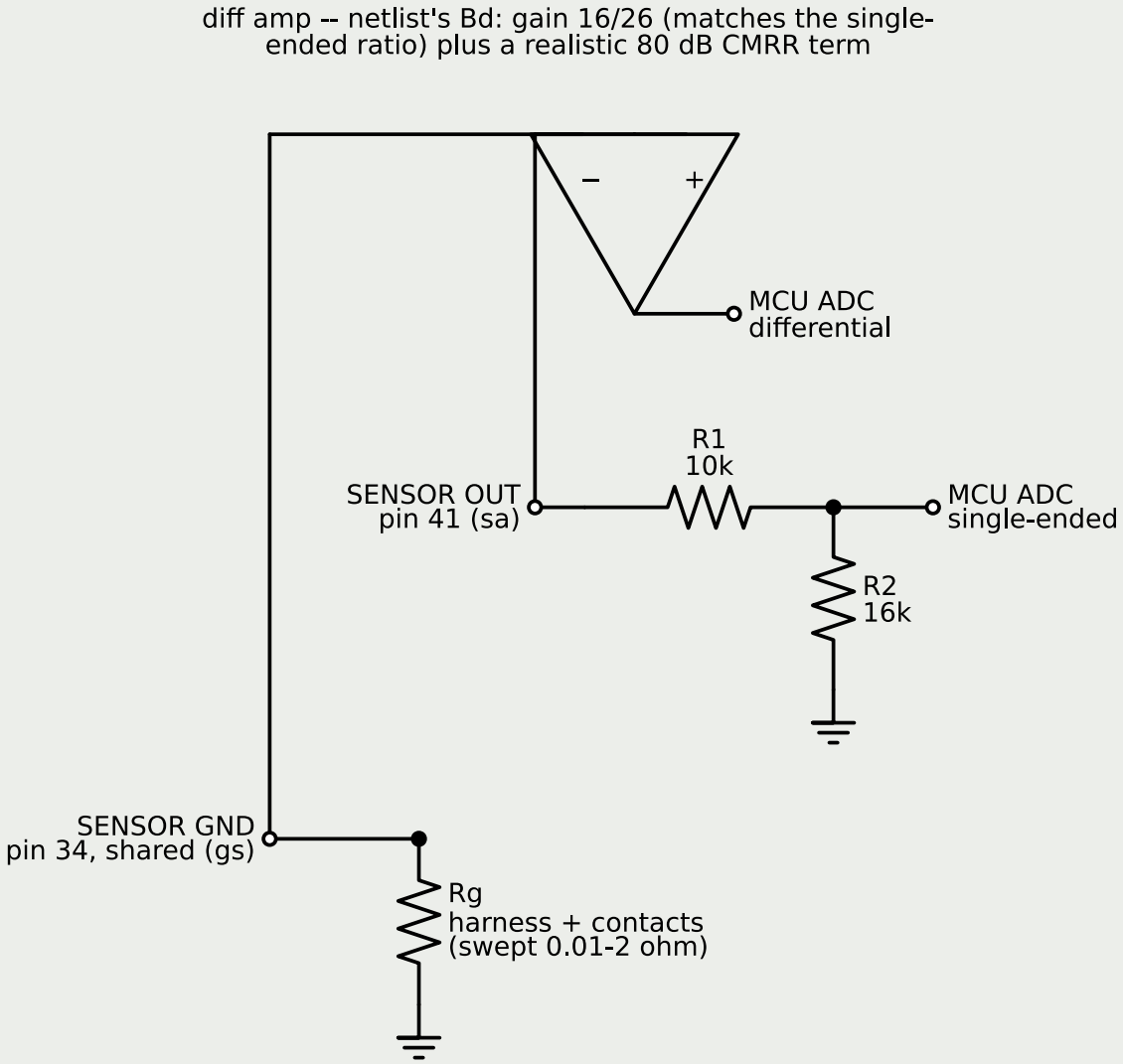
Why it looks like that. Full scale deliberately lands at 2.77 V rather than hard against the ceiling, so the protection clamp can do its job on a harness fault without ever touching the signal in normal running. That costs a little span and buys a front-end 40 V cannot damage. The last four checks cover the ratiometric property itself, which this block was named for and did not model until the spec review caught it: a sensor's output is a fraction of its own supply, so a reading taken against an assumed 5.000 V carries the rail's whole error band — 10% here. Giving the rail its own ADC channel through an identical divider removes it, and the divider ratio cancels algebraically, so its tolerance drops out too.

> **What the model does not cover.** Ground offset. Every network here measures against a perfect ground, and pins 34 and 36 are shared returns — which is why spec §4 now specifies a differential front-end for those channels and leaves this one for pin 08, the dedicated return. Also not modelled: divider tolerance matching, on which the rail correction depends, and ADC sampling time against the source impedance.

PASS sensor zero (0.5 V) maps to	0.308 V
PASS sensor full scale (4.5 V) maps to	2.769 V
PASS ... clears the 3.3 V clamp	0.531 V
PASS span used of ADC range	74.592 %
PASS low-pass corner	1,174.898 Hz
PASS loss at sensor's own 100 Hz	-0.031 dB
PASS attenuation at 100 kHz switching	-38.596 dB
PASS 40 V harness short: clamp node (one BAV199 above 3.3 V)	4.326 V
PASS ... pad behind 1k	3.931 V
PASS ... current injected into the pad, against +/-3 mA	0.395 mA
PASS ... fault current limited to	3.567 mA
PASS raw reading moves with a +/-5% rail drift	10.000 %
PASS ... so an assumed 5.000 V rail is a 10% error band	10.0% of reading, and nothing in the raw value reveals it
PASS corrected reading is flat	0.000 %
PASS correction recovers the true fraction	0.500
PASS ... divider ratio cancels, so its tolerance drops out too	measurand = Vsig/Vrail -- k appears in both and divides away

Netlist: sim/blocks/sensor_ratiometric.cir

Differential sensor input `pin 34 group` `9/9 checks`



Not drawn: Ib (netlist, 20 mA into gs) -- the coolant sensor's own return current, sharing this ground via splice V3. It is what moves gs; Rg is what decides how far it moves.

The same channel measured sensor-to-sensor-ground instead of sensor-to-ECU-ground, swept against the harness return's resistance.

Why it looks like that. Pin 34 is spliced between the boost sensor and the coolant sensor, so it carries both sensors' return current. A single-ended input reads the resulting offset as signal. Research memo 09 found this is exactly what Deep Sea Electronics specifies against (+/-2 V common mode on every sender channel) and what SEDEMAC's dedicated sensor common point exists for. Note what the OEM did with the channel that could not tolerate it: rail pressure gets pin 08, spliced with nothing. The schematic below draws one divider (R1/R2, the single-ended path) plus a generic amplifier for the differential path,

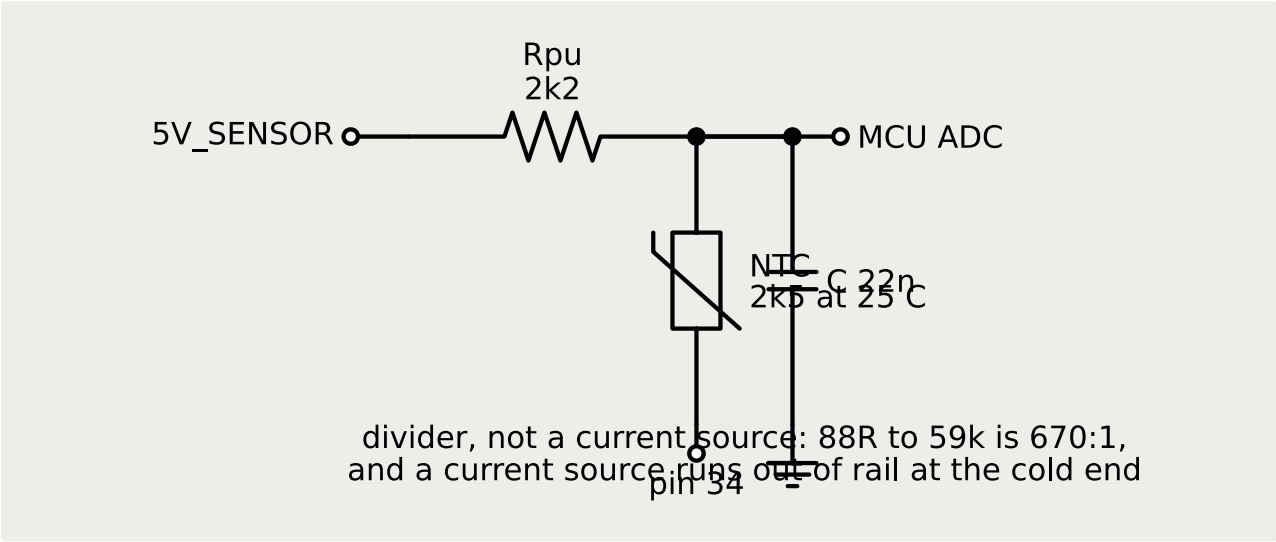
because that is what the netlist models — a behavioral gain block (Bd: 16/26 gain, matched to R1/R2's own ratio, plus an 80 dB CMRR term), not a second resistor divider. An earlier version of this drawing showed two dividers joined by a plain wire, which matched neither the netlist nor any circuit that would actually reject a ground offset.

> **What the model does not cover.** The error is about half a percent of span at 1 ohm of return resistance, and that understates it. It is a bias rather than noise, so it never averages out; it grows as contacts corrode; and it is indistinguishable from a real reading — a boost pressure biased high permits more fuel, and the first symptom is smoke, not a fault code. Rg, the swept harness + connector resistance, is drawn; Ib, the 20 mA coolant-sensor current that actually moves gs, is not — it is a stimulus, not a part of this front end, and it is captioned rather than drawn for that reason.

PASS clean harness (50 mOhm), single-ended error	
PASS 1 ohm return, ground offset present	
PASS ... single-ended reading error	
PASS ... differential reading error, referred to the sensor	
PASS ... single-ended, referred to the sensor: the whole offset	
PASS differential is better by	
PASS ... it was 90x at the inferred 80 dB and 16/26 gain; the chosen part's MINIMUM 88 dB is what moved it	the argument for differential got stronger, not weaker, when a real
PASS single-ended error at 2 ohm (bad contact)	
PASS ... single-ended needs contact resistance below	~0.1 ohm forever, to hol

Netlist: sim/blocks/sensor_differential.cir

NTC temperature input `pin 79` `11/11 checks`



The thermistor branch of U2, as a pull-up divider and as a constant-current source, across the full –40 to 150 C range.

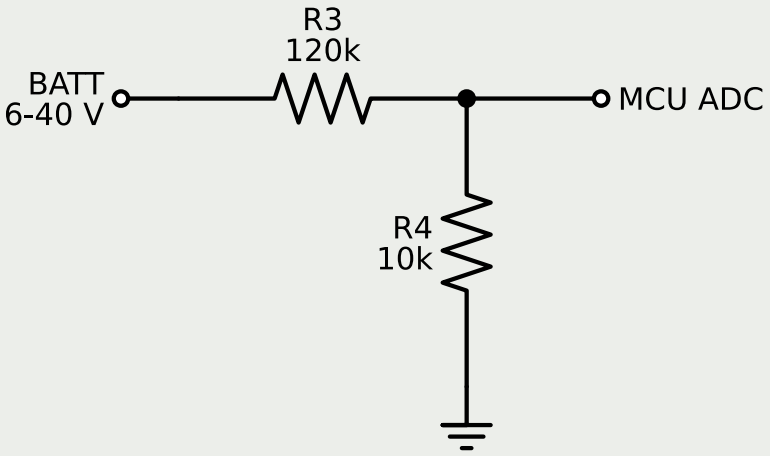
Why it looks like that. Memo 09 found DSE drives its temperature senders with a fixed 15 mA and measures differentially, and the obvious move is to copy it. Checked against an actual charge-air NTC it does not transfer: 88 ohms to 59 kilohms is a 670:1 span against the 50:1 a genset sender covers, and a current source sized to give useful signal at 150 C demands 5.9 V from a 5 V rail at –40 C. The divider's non-linearity is what saves it — it is bounded by the supply at both ends.

> **What the model does not cover.** Which branch this engine actually needs is still U2, and one resistance measurement at the machine settles it. The self-heating figure should be revisited once the real sensor's dissipation constant is known.

PASS divider at 150 C (88 ohm)		0.192 V
PASS divider at 25 C (2.5k)		2.664 V
PASS divider at -40 C (59k)		4.820 V
PASS ... never leaves the 5 V rail		4.820 V
PASS 100 uA source needs this at -40 C		5.899 V
PASS ... but the rail is 5 V	5.9 V demanded -- out of compliance, reading clips	
PASS 100 uA source at 150 C gives only		8.800 mV
PASS current that WOULD stay in compliance		84.746 uA
PASS ... leaving this at 150 C		7.458 mV
PASS divider self-heating power at 25 C		2.829 mW
PASS ... error at ~2 mW/C dissipation constant		1.414 degC

Netlist: sim/blocks/ntc_frontend.cir

Battery voltage sense pins 04, 06 9/9 checks



R3/R4: the adopted divider (spec sec.4, pins 04/06). battery_sense.cir also carries a rejected R1/R2 75k/10k divider, sized for a 5 V ADC -- not drawn here.

The 6–40 V system rail scaled into the ADC.

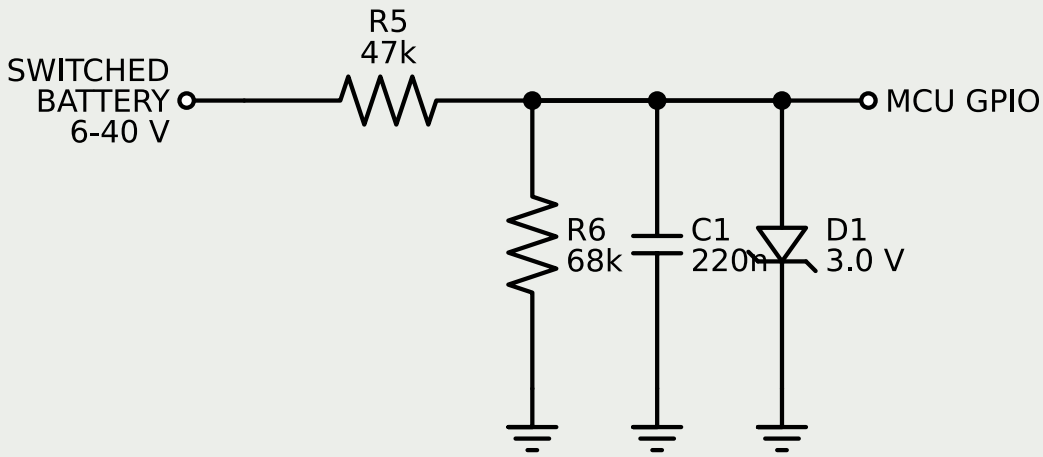
Why it looks like that. The inherited 75k/10k divider was sized for a 5 V ADC and reaches 4.69 V at 40 V — it would have damaged the input pin. 120k/10k peaks at 3.07 V and still resolves 573 counts at the 6 V cranking dip. The schematic below draws only the adopted 120k/10k divider, and it now carries the netlist's own designators for it, R3/R4 — spec §4 gives the same 120k/10k for pins 04/06. The rejected 75k/10k comparison is R1/R2 in battery_sense.cir; it is not drawn, since it is not the circuit on the board, and using its designators for the drawn divider (an earlier version of this drawing did) pointed a reader at the wrong resistors.

> **What the model does not cover.** Nothing subtle. A resistive divider is exactly linear and the check confirms it, which is there to catch a wrong topology rather than a wrong value.

PASS	inherited 75k/10k peak at 40 V	4.694 V
PASS	... and so exceeds a 3.3 V ADC	4.69 V > 3.3 V -- would damage input
PASS	corrected 120k/10k peak at 40 V	3.069 V
PASS	... stays under the 3.3 V rail	3.069
PASS	headroom below rail	0.231 V
PASS	reading at 6 V cranking dip	0.462 V
PASS	... resolves to usable ADC counts	572.867 LSB
PASS	divider is linear (max residual)	0.000 V
PASS	slope matches 10/130	0.077

Netlist: sim/blocks/battery_sense.cir

Discrete switch input `pins 20, 24, 71` `15/15 checks`



The switched-battery inputs, active high.

Why it looks like that. This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 0.385; staying under 3.3 V at 40 V needs it below 0.0825. No fixed ratio satisfies both, so the original was impossible rather than merely mis-valued. A 3.0 V zener sets the level and the divider only biases into it, which holds the output flat within 92 mV across the whole range.

> **What the model does not cover.** The debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two separate presses.

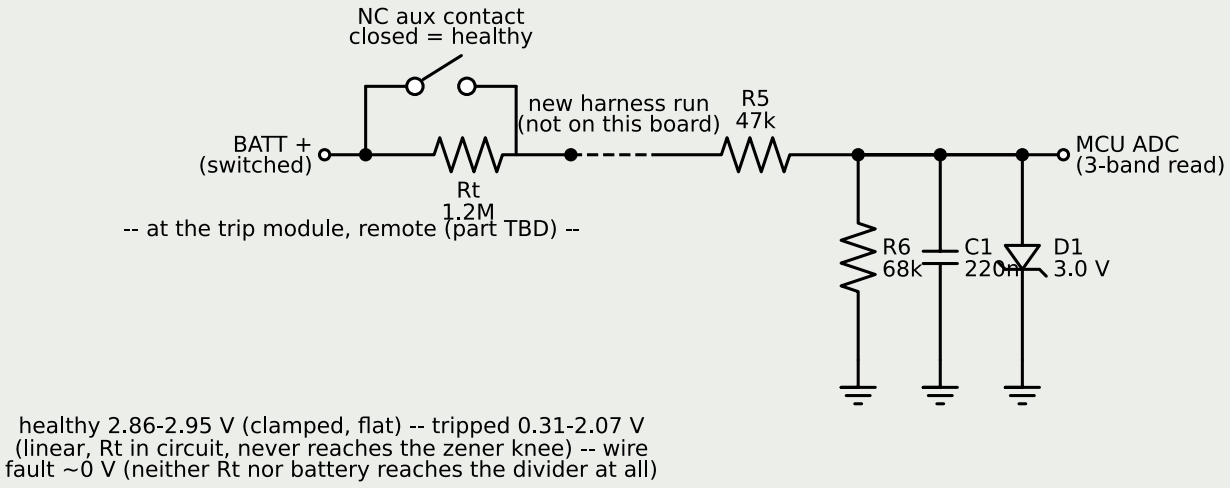
PASS	pad max over -24..40 V in (weak, -40 C)	3.300 V
PASS	pad min over -24..40 V in (weak, -40 C)	0.020 V
PASS	pad max over -24..40 V in (typical, 125 C)	3.300 V
PASS	pad min over -24..40 V in (typical, 125 C)	0.026 V
PASS	base at -24 V reverse battery, against VEBO 6 V	-1.047 V
PASS	6 V cranking dip, hFE 30, -40 C: pad reads LOW (active)	0.027 V
PASS	... input at which the weak cold part reaches VIL	2.500 V
PASS	typical part, 125 C: input that first pulls the pad below VIH	1.300 V
PASS	... so a harness floating at 1 V still reads inactive	3.230 V
PASS	pull-up channel, contact OPEN, hFE 30, -40 C: pad LOW	0.055 V
PASS	pull-up channel, contact CLOSED, 125 C: pad HIGH	3.300 V
PASS	pad edges through a 5-chatter bounce	1.000
PASS	settled LOW after the bounce	0.028 V
PASS	debounce time constant (10k || 4.7k) x 4.7 uF	15.027 ms
PASS	contact wetting current at 12 V (≥ 1 mA for non-gold contacts)	1.120 mA

Netlist: sim/blocks/discrete_input.cir

Trip-module sense

new pin, no OEM harness

14/14 checks



healthy 2.86-2.95 V (clamped, flat) -- tripped 0.31-2.07 V (linear, Rt in circuit, never reaches the zener knee) -- wire fault ~0 V (neither Rt nor battery reaches the divider at all)

A three-state supervised loop reading whether the mandatory standalone overspeed trip module's contacts are open or closed — not a copy of the input above, on purpose.

Why it looks like that. Spec §3: U12 closed badly on 18 Sep 2026 — the fitted DSE4522's fuel relay only signals, it does not remove power, so a standalone trip module in series with the fuel metering unit's own battery feed (fuse 8F1) is mandatory again, and this ECU must sense its state rather than read a trip as a fuelling anomaly. discrete_input.cir's own accepted limitation — an open wire reads identically to 'not asserted', accepted there because changing the sense would break OEM harness compatibility — does not carry over: this pin has no OEM harness to stay compatible with, and an undetectable broken wire here would read 'not tripped' at the exact moment a trip occurs, which is the one failure spec §3 exists to catch. So this block adds an end-of-line resistor (Rt, 1.2 MΩ) in parallel with the module's own auxiliary contact, physically at the module end of the new harness run, giving three distinguishable ADC bands on the same R5/R6/zener front end discrete_input.cir already validates: healthy (contact closed, Rt shorted) clamps to 2.86–2.95 V exactly as that file's input does; tripped (contact open, Rt in circuit) is a plain, unclamped linear divider that stays 0.31–2.07 V across 6–40 V, never reaching the zener's knee; wire fault (a break anywhere in the run removes both the contact and Rt the same way) reads ~0 V. A contact-bounce transient, reusing discrete_input.cir's own nasty bounce profile inverted, shows the reading never leaves the healthy band during the bounce itself — C1's

time constant on this branch is long enough to filter chatter shorter than a few hundred microseconds — and settles cleanly into the tripped band only once the contact stays open.

> **What the model does not cover.** No trip module is chosen (memo 02's costed GAC SSW675 / Murphy HD9063 candidates survive costing but neither's aux-contact rating is pulled), so Rt=1.2 MΩ is this file's own INFERRED choice with no datasheet or industry EOL-loop convention behind it. The loop is high impedance (the same few-hundred-microamp class discrete_input.cir's divider runs at), not the low-Z, milliamp-class loop industrial supervised-contact practice normally uses — chosen to reuse the existing front-end rather than build a new one, and the cost is a genuinely tight margin: at the 6 V cranking dip with Rt at +5% tolerance, the tripped band's floor sits only 0.147 V above the wire-fault threshold, about half the nominal-case margin, before any harness leakage is stacked on top. This pin also needs to land on an ADC-capable channel, not a bare GPIO — a pin-mux consideration for whoever assigns the physical PTxx map.

PASS healthy (contact closed) band floor
PASS healthy band ceiling
PASS tripped (Rt in circuit) band floor, at 6V cranking dip
PASS tripped band ceiling, at 40V
PASS wire fault reads near zero
PASS tripped band never reaches the zener's 3.0V knee
PASS margin: wire-fault threshold to tripped floor (nominal Rt)
PASS margin: tripped ceiling to healthy floor (nominal Rt)

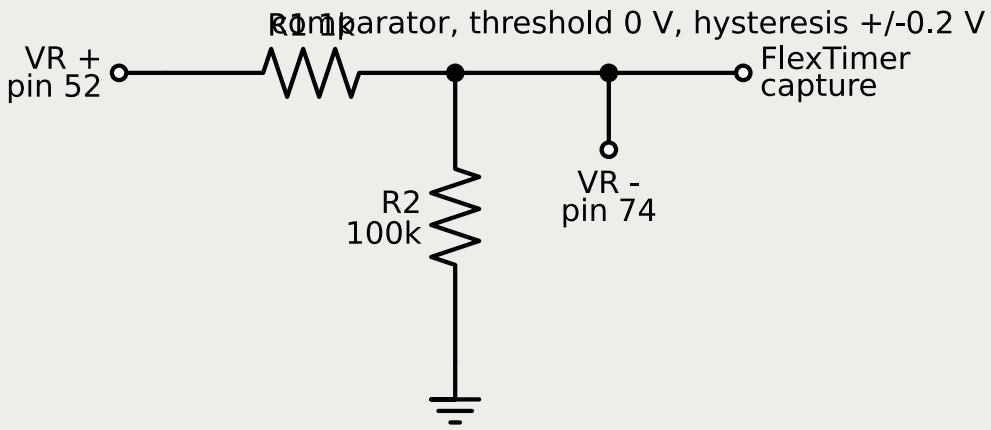
PASS ... both bands separated with real margin, not touching	wire-fault margi
PASS during bounce (t<4.5ms), reading never leaves the healthy band	
PASS settled value (t=100ms) approaches the DC asymptote (hand: 0.6206 V)	
PASS ... which sits inside the tripped band, not wire-fault or healthy	
PASS ... global minimum over the whole run stays clear of wire-fault	
PASS open-wire detection: needed here, unlike pins 20/24/71	no OEM harness to stay compatible with, and an undetectable open reads

Netlist: sim/blocks/trip_module_sense.cir

03 Speed and position

The most safety-critical input on the board.

Crank speed conditioner `pins 52, 74, 30` `7/7 checks`



A variable-reluctance sensor into a zero-crossing comparator, run at both running speed and cranking speed.

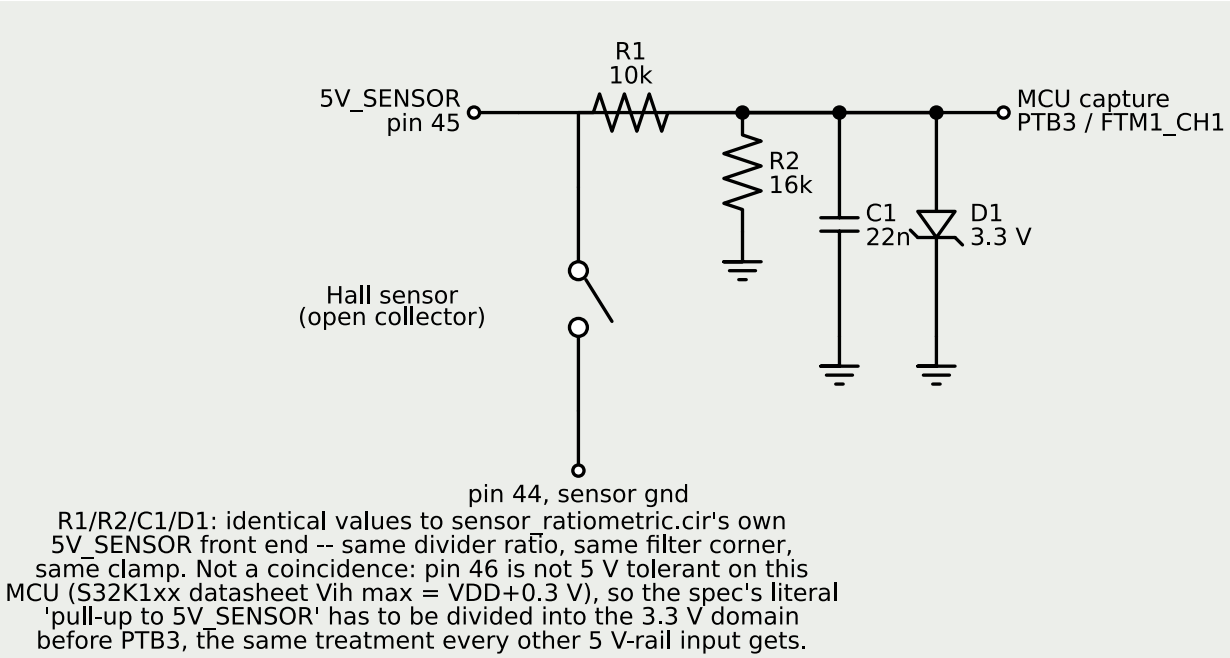
Why it looks like that. A VR sensor generates voltage by induction, so its amplitude is proportional to speed: roughly 20 V peak at 1500 rpm and 2 V at cranking. A fixed 5 V threshold, which looks perfectly sensible against a 20 V signal, produces all 30 teeth at running speed and NOTHING at cranking. That is the worst failure shape there is — it works on the bench and the engine never starts. Zero-crossing detection does not care about amplitude, and the crossing is also where the waveform is steepest and timing jitter is smallest.

> **What the model does not cover.** A real VR signal is not a clean sine — tooth edges distort it and the missing-tooth sync gap produces one large excursion per revolution. Amplitude at true cranking speed on a cold engine with a weak battery can be under a volt, which is when starting matters most, and the hysteresis window needs checking against the real sensor rather than this 2 V estimate.

PASS zero-cross, running (1500 rpm, 30 teeth)		30.000 edges
PASS zero-cross, cranking (150 rpm, 3 teeth)		3.000 edges
PASS fixed 5 V threshold, running		30.000 edges
PASS fixed 5 V threshold, cranking		0.000 edges
PASS ... so a fixed threshold never starts the engine	works at 1500 rpm, blind at 150 rpm -- passes on the bench	
PASS running tooth period		666.667 us
PASS ... jitter across the window		0.000 us

Netlist: sim/blocks/vr_conditioner.cir

Cam signal front-end `pins 45, 46, 44` `12/12 checks`



A cam Hall sensor's open-collector output, pulled up and divided into the 3.3 V domain before PTB3/FTM1_CH1, instead of the spec's literal pull-up straight to the 5 V sensor rail.

Why it looks like that. PTB3 is not 5 V tolerant for anything this design should rely on — the S32K148's own datasheet puts guaranteed valid-high recognition at VDD+0.3 V, about 3.6 V, and treats a higher DC injection allowance as a fault margin, not a design basis for a pin that sees 5 V every cam revolution for the machine's whole life. Nothing in sim/blocks/ ever modelled this pin before — there is a conditioner for the crank sensor and no cam equivalent — so the overvoltage sat uncaught. The divider (10k/16k) and filter (22 nF, 1176 Hz corner) reuse sensor_ratiometric.cir's own values exactly, since both share the same 5V_SENSOR rail and the same target domain: 94x above the cam's 12.5 Hz signal (half crank speed at 1500 rpm, one lobe per revolution assumed) and still well below switching-frequency noise.

> **What the model does not cover.** The real Hall sensor's saturation voltage, sink current, and internal pull-up (if any) are not modelled — no part chosen. Lobe count is INFERRED at 1 per cam revolution; the TOLERANCE section shows the filter margin survives a 3-lobe alternative just as well. A real cam signal's edge shape and any duty-cycle asymmetry from target

> geometry are not modelled, the same limitation vr_conditioner.cir already states for the crank signal's own non-ideal waveform.

PASS sensor off, 5V_SENSOR nominal (5.0 V) → pad	3.077 V
PASS ... clears the 3.6 V abs max	0.523 V
PASS sensor_rail.cir's own fault floor (4.5 V) → pad	2.769 V
PASS ... still reads logic high (⩾2.31 V, 0.7*VDD)	2.769 V
PASS above-nominal rail (5.25 V) → pad	3.231 V
PASS ... still clears the 3.6 V abs max	0.369 V
PASS ... WAS: the zener's loading of the signal at this point	0.084 V
PASS never exceeds 3.6 V abs max in transient	3.077 V
PASS rising edges, 170 ms window at 12.5 Hz (80 ms period)	2.000 edges
PASS 40 V harness short: clamp node (BAV199-Q to 3.3 V)	4.326 V
PASS ... current injected into the pad behind 1k, vs +/-3 mA	0.395 mA
PASS ... fault current limited to	3.567 mA

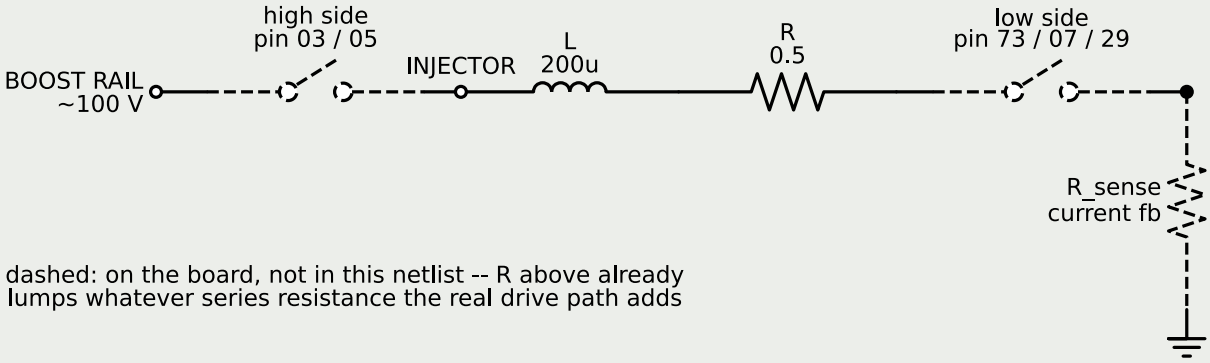
Netlist: sim/blocks/cam_frontend.cir

04 Injection

The most expensive stage on the board and the reason it exists.

Why the boost rail exists `pins 03, 05`

13/13 checks



The same injector driven from the battery and from a 100 V rail, compared on time to reach the 18 A peak threshold.

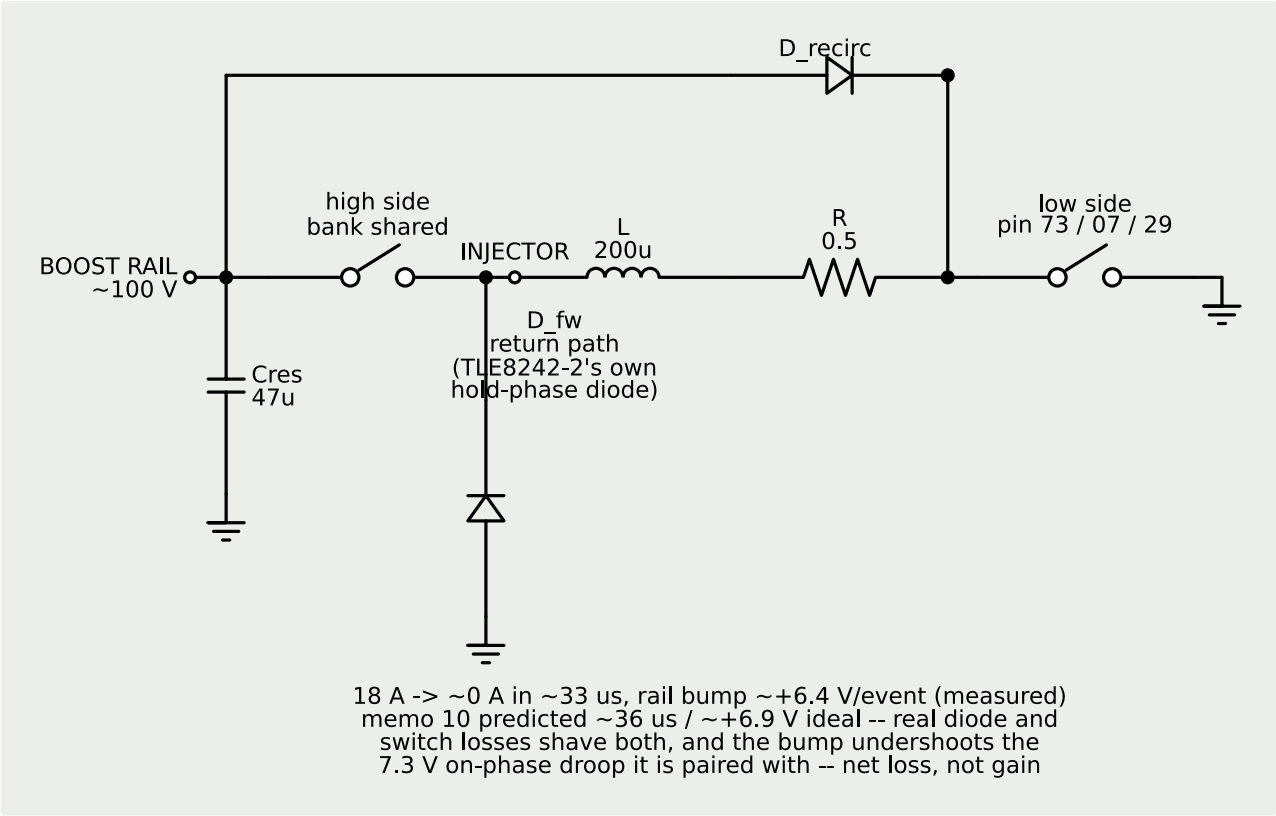
Why it looks like that. 439 us from the battery, 38 us from the boost rail. On a 1 ms injection the battery drive spends 44% of the event just lifting the needle; the boost rail spends 3.8%. That ratio is the entire argument for the stage.

> **What the model does not cover.** An RL model of the injector with no magnetic saturation and no moving needle. The envelope it sits in — 65–115 V, 12–24 A peak — is family-wide from memo 04, not this injector's measured values. That is U5. Also not modelled: the high/low-side switches and the sense resistor, dashed in the schematic — this is a turn-on-only step response straight onto L/R, and R already stands in for whatever series resistance the real drive path adds.

PASS time to 18 A peak from 13.5 V battery	439.496 us
PASS time to 18 A peak from 100 V boost	37.796 us
PASS boost is faster by	11.628 x
PASS battery drive eats this much of a 1 ms injection	43.950 %
PASS boost drive eats this much of a 1 ms injection	3.780 %
PASS ... evidence: ratio at -40C ambient, vs the 11.6x NOMINAL (not a requirement)	9.856 x
PASS ... REQUIREMENT: boost still decisively faster at -40C (>5x)	9.856 x
PASS ... and battery drive still eats this much of a 1 ms injection at -40C	36.760 %
PASS ... time to 18 A from 100 V boost, -40C ambient	37.296 us
PASS ... evidence: ratio at +125C ambient, vs the 11.6x NOMINAL (not a requirement)	19.017 x
PASS ... REQUIREMENT: boost still decisively faster at +125C (>5x)	19.017 x
PASS ... and battery drive still eats this much of a 1 ms injection at +125C	73.210 %
PASS ... time to 18 A from 100 V boost, +125C ambient	38.496 us

Netlist: sim/blocks/injector_boost.cir

Turn-off / recirculation `pins 73, 07, 29` `15/15 checks`



The low-side switch opening at 18 A, with a recirculation diode routing the coil's current back into the same boost cap (Cres, 47 uF) instead of a clamp that just burns it — plus a second diode (D_fw) the loop needs to close, and a comparison against freewheeling to the battery instead.

Why it looks like that. Research memo 10 surveyed real injector-driver topologies and ruled out a Zener/active clamp (Nexperia AN50003 flags repetitive active clamp as reliability-"Questionable") and a bare battery freewheel (too slow) in favor of recirculation to the existing boost rail — confirmed production diesel-injector practice via a 1988 Marelli Autronica patent, not a textbook flourish. Simulated: 33 us to collapse 18 A to zero (memo predicted ~36 us), 25x faster than freewheeling to the battery (830 us — slower than the memo's own simplified estimate, because a real RL circuit's exponential tail matters more at 13.5 V than at 100 V). The rail bump per event is +6.4 V, under the memo's +6.9 V ideal because real diode and switch losses are modelled here, and it undershoots the 7.3 V on-phase droop it is paired with — net loss, not gain, so three back-to-back events never push the rail over its 100 V setpoint. But that verdict hinges on which current is being cut off: boost_converter.cir's own arrangement A has the battery supply HOLD current, not the boost rail, and recirculating a hold-current cutoff through the same diode has no offsetting boost-side draw to net against

— modelled separately here, it overshoots to 102 V on a single event. boost_converter.cir's charge regulation (Bchga) has no bleed path either way; it just happens not to need one for the peak-only case.

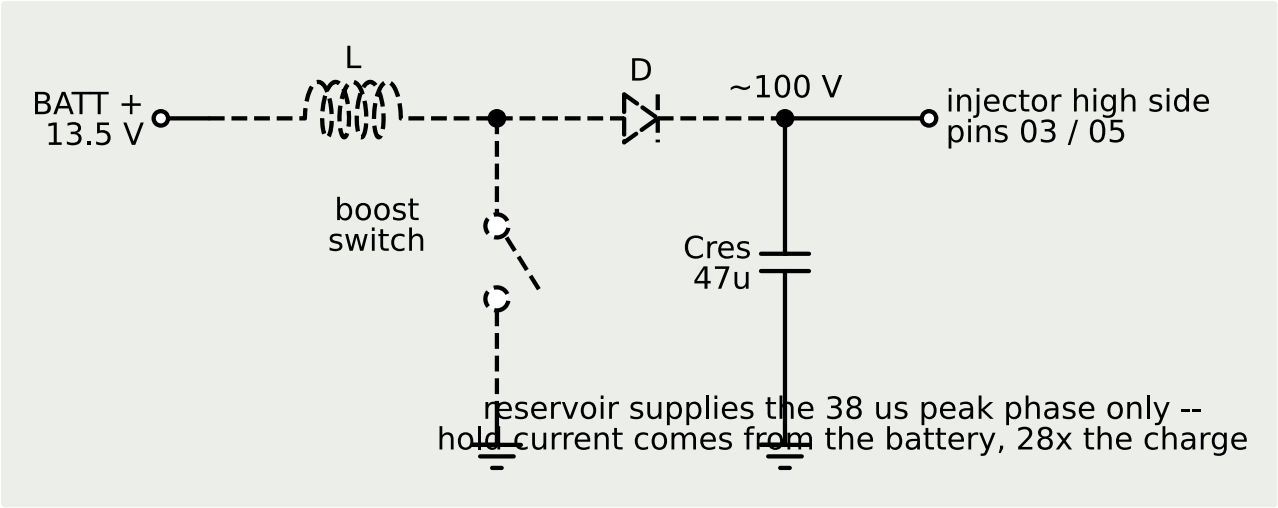
> **What the model does not cover.** The differential-mode EMI edge (memo's own claim 5) is not modelled — the switches here are ideal with zero transition time, so any dV/dt read off this simulation would be a solver artifact, not a FET's real slew rate. Also not modelled: diode reverse recovery (Qrr), the diode's thermal transient, and — inherited from injector_boost.cir — magnetic saturation and needle motion.

PASS turn-off time, 18 A → ~0 A on the boost rail	
PASS ... vs freewheel-to-battery baseline (comparison only)	
PASS ... boost recirculation is faster by	
PASS on-phase droop, pilot event	
PASS ... matches boost_converter.cir's own droop figure	
PASS recovery bump, pilot event (isolated)	
PASS ... undershoots the droop it is paired with (real losses)	
PASS rail never exceeds 100 V setpoint (3-event circuit)	
PASS rail recovered by the next cylinder (26.67 ms after post)	
PASS hold-current cutoff into the SAME cap, no offsetting draw	
PASS ... THIS is the scenario that overshoots 100 V	
PASS ... so the verdict hinges on whether hold-current cutoffs share the boost-recirculation diode	peak-only recirculation never overshoots; hold-current reci
PASS peak instantaneous power in D_recirc during turn-off	
PASS ... exceeds the ~3.6 W worst-case AVERAGE by	

```
PASS differential-
mode edge vs                                  not modelled here (ideal switches) --
emi_filter.cir
```

Netlist: sim/blocks/injector_turnoff.cir

Boost rail reservoir `pins 03, 05` `105/105 checks`



The reservoir capacitor under a real injection event, with and without the hold phase drawn from the same rail.

Why it looks like that. The converter cannot deliver 18 A — it delivers a trickle and the capacitor delivers the pulse, so the capacitor is the component that matters. The design question is where the HOLD current comes from, and the answer is not subtle: hold costs 29 times the peak phase's charge, so a rail supplying both collapses completely. Hold comes from the battery. 50 mA of average charging recovers the rail with room to spare in the 26.67 ms before the next cylinder fires.

> **What the model does not cover.** Charge balance only. L, D and the switch are dashed in the schematic because the converter physically has them, but there is no inductor, no switch and no control loop in the netlist — just a current source standing in for the converter's average output. It says nothing about the converter's stability or its inrush, and the collapsed voltage in the second case should be read as 'far below usable' rather than as a number.

PASS droop, peak phase only (41 uF)

PASS ... rail stays usefully above battery

PASS droop, peak AND hold from the rail

PASS ... so the rail collapses

PASS hold phase costs this much more charge

PASS recovered by the next injection (26.67 ms)

PASS ... 50 mA average charging is enough

PASS one hold-cutoff event puts this on the rail

PASS without a bleed, three events stack to

PASS ... and it STAYS there -- nothing can discharge it

PASS with a 22k bleed, same three events reach

PASS ... but the rail is back at setpoint before the next cylinder

PASS time to clear one event		
PASS ... against the 240-degree budget at 1500 rpm		
PASS ... cost, burned continuously whenever the rail is up		
PASS TPS40210-Q1 VDD absolute maximum		
PASS ... against the pulse 2a rail it would otherwise sit on		
PASS ... so it CANNOT go straight on VBAT_PROT	same shape as DRV8873-Q1's 40 V VM.	
PASS WAS: 47 ohm + BZG03C43-M, VDD at pulse 2a, hot		
PASS ... over the 52 V absolute maximum by		
PASS ... and its 40 V minimum conducts on the load dump it was chosen to stay out of		43

PASS VDD_BAT at pulse 2a: 180 ohm + BZG03C36, VZ max, 125 C, ZZ bound (reaches VDD through the Schottky)
PASS ... inside the absolute maximum by
PASS ... zener current during the 50 us pulse
PASS ... series resistor, peak dissipation for 50 us
PASS load dump, VZ min at -40 C: zener current for 400 ms
PASS ... zener dissipation for 400 ms (1.25 W is the STEADY rating at 25 C; this leans on thermal mass, not read off a Zth curve)
PASS start-up VDD at the 6 V cranking dip (Qg typ, through the Schottky)
PASS ... above the UVLO turn-on ceiling by

```
PASS ... at
twice the
typical gate
charge
(DMN15H310SK3
gives no
maximum) --
THIN, stated
```

```
PASS duty
needed at the
6 V cranking
corner
```

```
PASS ... at
13.5 V, for
contrast
```

```
PASS duty the
part allows
at 150 kHz
```

```
PASS ...
margin at the
cranking
corner
```

```
PASS ... so
the cranking
corner is
reachable
```

```
PASS WAS: 330
uH, VDD 5.16
V on the
battery: 100%
RISNS(max) at
6 V
```

```
PASS ... the
drawn 82 mOhm
exceeded it -
- subharmonic
while
cranking
```

```
PASS VDD from
the output
follower,
lowest (VZ
min, -40 C,
low current,
cold Vbe, 56
ohm drop)
```

```
PASS ...
highest (VZ
max, 125 C,
hot Vbe)
```

```
PASS ...
inside the 52
V absolute
maximum by
```

```
PASS ...
still
regulating at
the rail's
worst droop
(RAIL_MIN),
headroom
```

```
PASS ... pass
transistor,
normal (104 V
rail, lowest
VDD)
```

```
PASS ...
junction at a
105 C board
(104 K/W)
```

```
PASS ... VDD
shorted:
limited to
one Vbe
across 56 ohm
```

```
PASS ... pass
transistor
into the
short
```

```
PASS ... each
3.3k 2512
into the
short (0.59 W
allowed at
105 C)
```

```
PASS IS: 80%
RISNS(max),
worst corner
(VDD 34.3 V,
120 uH, 120
kHz, 6 V)
```

```
PASS ... 62
mOhm inside
it by
```

```
PASS Eq. 49:
peak inductor
current at
the cranking
corner
```

```
PASS ...
sense ceiling
so the limit
never trips
in normal
running
```

```
PASS ... 62
mOhm inside
it by
```

```
PASS current
limit,
highest (180
mV)
```

```
PASS ... past
L401's 10%
saturation
(2.7 A typ)
by -- stated
in decisions
log 8
```

```
PASS ...
under its 30%
saturation
(3.3 A typ)
by
```

```
PASS ...
start-up is
the
exception:
until the
output passes                                          subharmonic is possible for
about 36 V,
VDD is the
battery feed
and the ramp
is shallow
```

```
PASS Q401
gate charge
(typ) under
the 25 nC
ceiling by
```

```
PASS input
current at
the cranking
corner (50 mA
out, 85%)
```

```
PASS duty
needed with
switch, sense
and L401
drops, hot
```

PASS ... inside what the part allows at the +7% frequency corner by	
PASS the same 200 ns off-time at 2.2 MHz allows only	
PASS ... which does not reach even the 13.5 V duty	
PASS ... so switching SLOWER is the requirement here	200 ns of off-time is a floor in time, no
PASS bank node worst case (reference +2%, one recirculation bump)	
PASS ... so VB reaches	
PASS AUIRS2181S: VB absolute maximum 625 V, margin	
PASS UCC27211A-Q1, the obvious 120 V candidate: HB margin	
PASS ... 4 V of headroom on an ABSOLUTE maximum -- rejected	
PASS bank node's lowest point, D_fw conducting 18 A	

PASS ... AUIRS2181S logic operational down to -5 V; clear by	
PASS ... UCC27211A-Q1's HS DC minimum is -1 V -- violated in NORMAL operation, not in a fault	
PASS cross-conduction interlock on AUIRS2181S	none (datasheet feature table: 2181 'no', 2183/2184 'yes') -- UCC27
PASS bootstrap droop over the 38 us peak phase (SQM85N15-19 120 nC max, 150 uA IQBS, 1 uF)	
PASS ... as a fraction of the ~11.5 V bootstrap supply	
PASS why not the LM5164 already on the board: its VIN abs max	
PASS ... against the rail it would sit on	
PASS 12V_GATE above the driver's UVLO+ ceiling (9.8 V) by	

```
PASS 12V_GATE
low corner
(BZX84-C13-Q
min, low
current, hot
Vbe)
PASS ...
above the
driver UVLO+
ceiling (9.8
V) by
PASS 12V_GATE
high corner
(C13-Q max,
cold Vbe)
PASS ...
under the
chosen FETs'
20 V Vgs
rating by
PASS current
limit: one
Vbe across 56
ohm
PASS ...
against the 5
mA design
load
PASS WAS:
pass
transistor
alone, normal
(88 V at 5
mA)
PASS ... over
what PZTA42-Q
may dissipate
at a 105 C
board
PASS ... and
shorted, held
by the limit
PASS IS:
2x3.3k
collector
resistors --
transistor,
normal
PASS ...
junction at a
105 C board
```

PASS ... transistor into a shorted output

PASS ... junction, shorted, 105 C board

PASS ... each 2512 resistor, shorted (0.59 W allowed at 105 C on a 1 W part derated linearly 70-155 C)

PASS ... lowest boost rail that still regulates 12V_GATE

PASS ... against the rail's worst droop (RAIL_MIN)

PASS standing load on the boost: bleed + gate rail + divider + its own VDD follower and zener bias

PASS ... so the peak-phase draw now recovers in

PASS ... still inside the 26.67 ms between cylinders by

PASS reservoir ESR, worst: two cans, -55 C, end of life
PASS time to 18 A at -40 C, lowest rail, worst ESR
PASS ... against 38 us nominal: still a 4.5% slice of a 1 ms injection
PASS rejected 1x 160 V wet can at an estimated 8 ohm aged and cold: highest current it can push
PASS one can, first charge, worst capacitance split
PASS ... under its 80 V rating by
PASS one can, settled on the 2 x 11k bleed, worst leakage
PASS ... each 11k 2512 dissipates
PASS reservoir ripple: injection pulses + charger, continuous

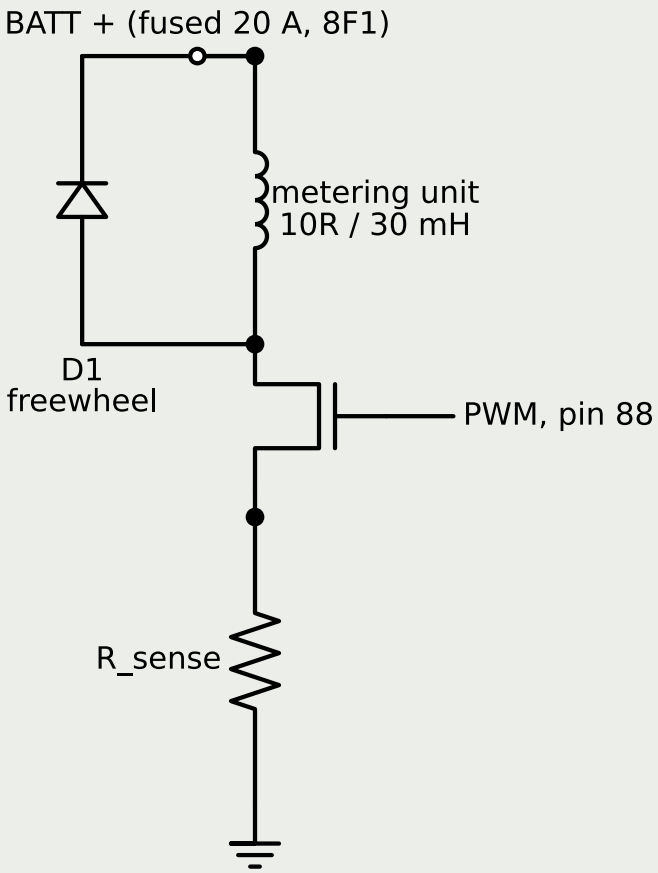
```
PASS ...
under the 2.1
A rated at
125 C, 100
kHz, by
```

Netlist: `sim/blocks/boost_converter.cir`

05 Actuators

Outputs that move something.

Fuel metering unit driver `pin 88` `25/25 checks`



Low-side PWM into the metering solenoid at three frequencies, all at the same duty.

Why it looks like that. Rail pressure is controlled by the AVERAGE current through this coil, and all three frequencies produce the same average — which is what makes the control law independent of the PWM constant. What changes is ripple, and ripple is not purely a nuisance here: a proportional solenoid has stiction, and deliberate dither keeps it moving. At 100 Hz the period is comparable to the coil's own 3 ms time constant and the result is 152% ripple — that is not dither, that is chopping, and the valve follows the PWM rather than its average.

> **What the model does not cover.** The correct dither amplitude is a property of the specific metering unit's mechanics and is manufacturer data tied to a part number — the same gap as U5, in a smaller form. Keeping the frequency a firmware constant

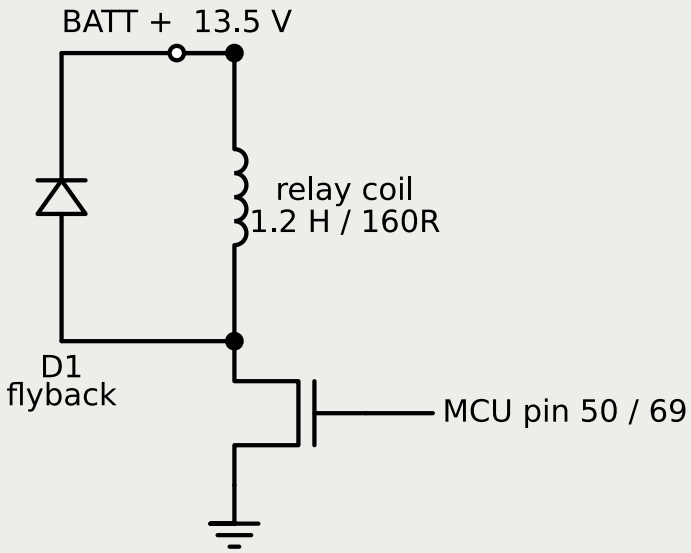
> rather than a hardware one means it can be tuned on the machine. Coil inductance is also not constant: it falls as the armature moves and as the core saturates.

PASS mean current at 100 Hz		0.642 A
PASS mean current at 1 kHz		0.637 A
PASS mean current at 10 kHz		0.635 A
PASS ripple at 100 Hz		0.969 A
PASS ... as fraction of mean		150.919 %
PASS ripple at 1 kHz		0.118 A
PASS ripple at 10 kHz		0.012 A
PASS ripple scales as 1/f		9.965 x
PASS 100 Hz is not dither, it is chopping	151% ripple -- valve follows the PWM	
PASS ... evidence, OPEN LOOP: mean current at 100 Hz, -40C ambient		0.854 A
PASS ... evidence, OPEN LOOP: mean current at 1 kHz, -40C ambient		0.862 A
PASS ... evidence, OPEN LOOP: mean current at 10 kHz, -40C ambient		0.858 A
PASS ... so open-loop duty misses the 0.675 A setpoint at -40C by		27.132 %
PASS ... control law: frequencies still agree, -40C		0.992 %
PASS ... RESOLVED, CLOSED LOOP: mean current at -40C		0.679 A
PASS ... evidence, OPEN LOOP: mean current at 100 Hz, +125C ambient		0.458 A
PASS ... evidence, OPEN LOOP: mean current at 1 kHz, +125C ambient		0.462 A
PASS ... evidence, OPEN LOOP: mean current at 10 kHz, +125C ambient		0.459 A
PASS ... so open-loop duty misses the 0.675 A setpoint at +125C by		-31.883 %
PASS ... control law: frequencies still agree, +125C		0.727 %
PASS ... RESOLVED, CLOSED LOOP: mean current at +125C		0.682 A
PASS closed loop needs duty headroom: max coil current at +125C		0.977 A
PASS ... setpoint as a fraction of that ceiling		69.100 %
PASS closed loop at nominal 25C		0.679 A
PASS ... hysteresis band it actually regulates within		0.133 A

Netlist: sim/blocks/metering_unit_pwm.cir

Relay driver pins 50, 69

6/6 checks



A low-side FET switching a relay coil, run with and without the flyback diode.

Why it looks like that. With the diode the drain stays at 14.3 V on turn-off. Without it the coil's stored energy has nowhere to go except through the FET.

> **What the model does not cover.** The simulated peak without the diode is hundreds of kilovolts, which is the ideal-switch model taken literally. A real MOSFET avalanches at its breakdown voltage and dissipates that energy in the die instead — once per relay operation, until it fails. The magnitude is a model artifact; the conclusion is not.

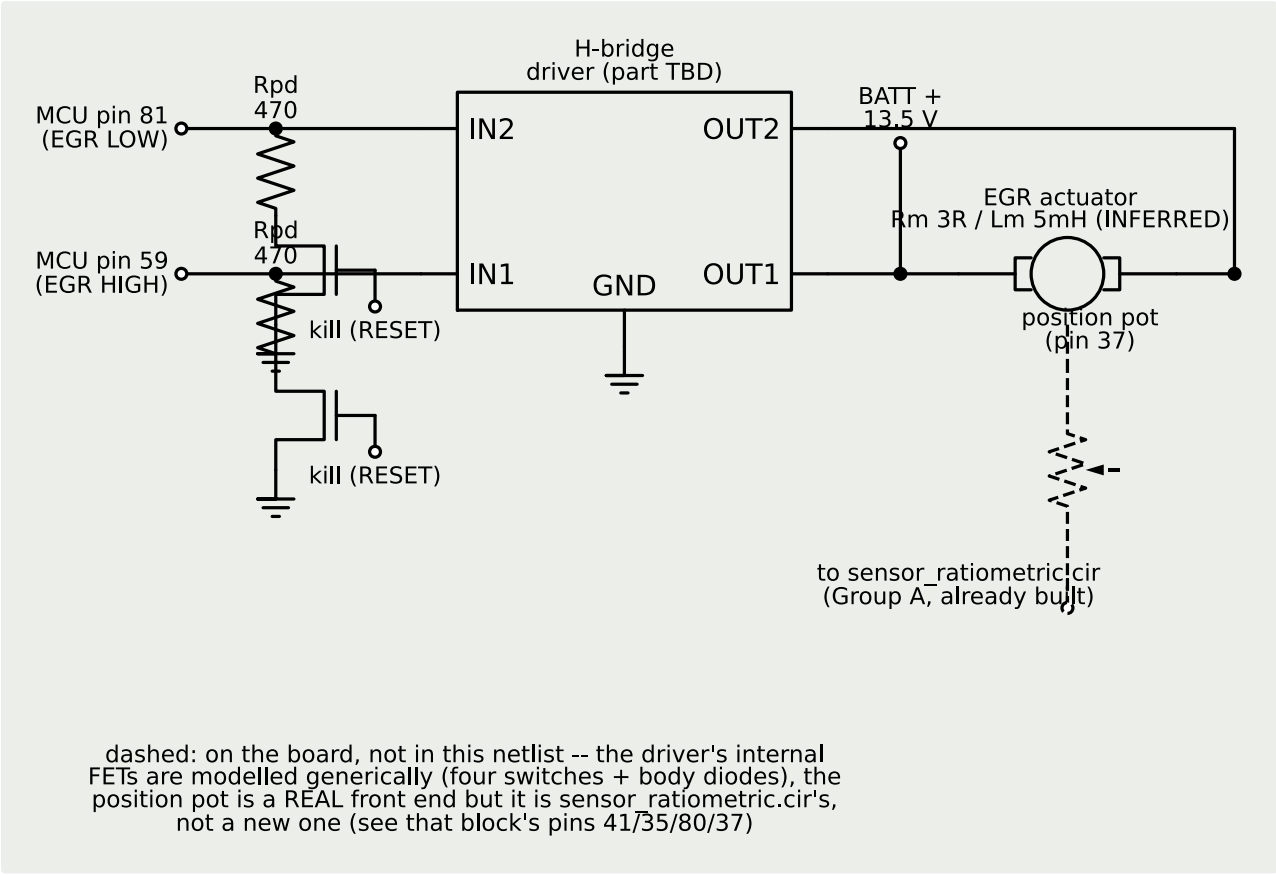
PASS drain peak WITH flyback diode	14.274 V
PASS drain peak WITHOUT diode	402,899.522 V
PASS ... diode suppresses the kick by	28,226.510 x
PASS gate overdrive from a bare 3.3 V GPIO (was)	0.600 V
PASS gate overdrive from the AHCT buffer at VCC 4.5 V (is)	1.700 V
PASS ... FET drop at coil current, Rds(on) at the 4.5 V point	36.450 mV

Netlist: sim/blocks/relay_driver.cir

EGR actuator H-bridge

pins 59, 81, position feedback pin 37

19/19 checks



Two logic/PWM lines into an external H-bridge driving a positional DC-motor actuator both ways — the one output on this board that is not a low-side switch — modelled as four generic switch positions with anti-parallel body diodes, plus the same active kill-clamp (Rpd 470 Ω, keyed to the supervisor's RESET) spec §4 requires at every driver gate.

Why it looks like that. This is a positional actuator, not an on/off solenoid: it is held at a commanded position, which is why it needs a bridge. Spec §4's outputs table lists pins 59/81 in the same row set as the six gates supervisor.cir already protects, but that file's own header enumerates only 88/73/07/29/03/05 — EGR was never added because this block did not exist. Reusing supervisor supervisor.cir's validated kill topology closes that gap rather than re-deriving it. The two-input "sign" scheme this design assumes (IN1 drives the S1+S4 diagonal, IN2 drives S3+S2) means no combination the ECU can present ever commands one leg's own high and low side together — real shoot-through is a property of the eventual driver IC's internal interlock, a part not yet chosen, not something these two lines can cause on their own. Stall current (jammed, or simply sitting at its own end of travel — normal operation for a positional actuator, not a fault) comes out to 4.36 A at an assumed 3 Ω armature resistance, comfortably past a class-typical 2–3 A small-

actuator driver rating: the eventual part needs active current limiting, not a fixed series resistor. Commanding coast with current already flowing is checked with and without the body diodes, the same comparison relay_driver.cir runs for its flyback diode: with them the bridge nodes stay within a diode drop of the rail; without them the ideal-switch model produces a multi-megavolt artifact, because the motor's stored energy has nowhere else to go.

> **What the model does not cover.** No H-bridge driver IC or FET is chosen — Ron, the body-diode figures, and the motor's Rm/Lm are all INFERRED class figures, the same gap as every other unselected FET on this board. Whether the eventual driver is integrated or discrete FETs behind a gate driver is undecided (memo 12 settled this for the injector and metering stages but not for EGR), so the 470 Ω kill-clamp value here is a uniform-standard choice, not a fresh Miller-coupling derivation for the 13.5 V domain EGR actually runs in. No mechanical model of the actuator exists — no return-spring constant, no gear ratio, no end-stop switches — so "coast lets the spring close the valve" is stated as the assumed fail-safe direction, not simulated. Position feedback on pin 37 is not modelled in this file at all: it is Group A ratiometric, already built and checked in sensor_ratiometric.cir, and building a second front end here would duplicate a proven circuit rather than reuse it.

PASS kill-clamp: gate below Vgsth (hand estimate 22.8 ns)		22.
PASS ... adversarial driver (100R/10V) still held below Vgsth		0
PASS ... adversarial settle voltage (hand: 10*4.95/104.95=0.471 V)		0
PASS stall current (Vemf=0, hand: 13.5/3.1)		4
PASS running current (Vemf=9V, hand: 4.5/3.1)		1
PASS coast current (IN1=IN2=0)		0
PASS stall current exceeds a class-typical 2-3 A driver rating		4
PASS ... so the eventual driver needs active current limiting, not a fixed series resistor	4.35 A at Rm=3 ohm alone exceeds a 2-3 A class	
PASS forward steady current at t=19.9ms matches section 2's stall figure		4
PASS WITH diodes: low side stays within a diode drop of ground (hand: -1.09 V)		-1
PASS WITH diodes: high side stays within a diode drop of the rail (hand: 14.59 V)		14
PASS WITHOUT diodes: low side swings far past ground		9,038,255
PASS WITHOUT diodes: high side swings far past the rail		9,038,269
PASS ... recirculation diodes are load-bearing, not decorative	bounded to 14.6 V with diodes vs 9.04e+06 V w	

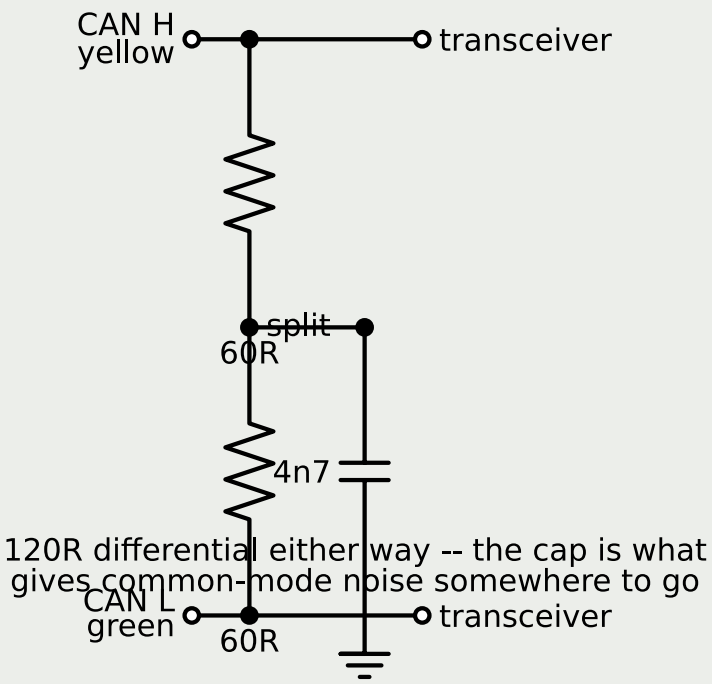
PASS position feedback (pin 37) is Group A ratiometric, proven in sensor_ratiometric.cir	0.308-2.769 V into the 3.3 V ADC, see check_sensor_ratiome
PASS rejected naive wiring: rail current with IN1=IN2 asserted	270
PASS ... matching the two-legs-of-2x50mOhm hand figure	270
PASS ADOPTED: decoded DIR/PWM driver, both inputs asserted	4
PASS ... so decode removes the fault state entirely	270 A naive vs 4.35 A decoded

Netlist: sim/blocks/egr_hbridge.cir

06 Communications

The bus to the genset controller.

CAN split termination `CAN H, CAN L` 11/11 checks



A split 60+60 termination with a capacitor to ground at the midpoint, against a single 120 ohm resistor.

Why it looks like that. Both are 120 ohms differential and a schematic label cannot tell them apart. The difference is common mode: a single resistor gives noise on both wires together nowhere to go, while the split version shunts it to ground without touching the differential signal, because the midpoint is a virtual ground for differential drive. That matters more here than on a truck. The fitted controller is a Deep Sea DSE4522, and DSE's manual is explicit that with a CAN engine the ECU transmits engine speed to it — so this bus carries a safety-relevant value, and it runs beside 415 V three-phase wiring and contactor coils in the same panel.

> **What the model does not cover.** Termination is not the whole common-mode story; the choke is the other half and needs a layout. And this does not decide whether to fit termination at all: a 120 ohm terminator belongs at each END of the backbone, and if this ECU is not an end node, fitting one makes the bus worse. That is U4 — so the board should carry the parts with a link and let the machine decide. The DSE4522 terminates CAN on its terminals 18/19/20 and DSE specify 120 ohm cable, which settles the cable but not which nodes carry the terminators.

PASS split termination, differential at 250 kbit/s	119.993 ohm
PASS single 120R, differential at 250 kbit/s	119.993 ohm
PASS ... identical differentially	0.000 ohm
PASS split termination, common-mode at 10 MHz	30.189 ohm
PASS single 120R, common-mode at 10 MHz	500.000 kohm
PASS ... split shunts common mode better by	16,562.490 x
PASS split common-mode at the bit rate stays high	138.099 ohm
PASS ... evidence: at 200 ppm/C thick-film tempco (matched direction, -40/125C) -- the corner that forced the requirement	123.952 ohm
PASS ... so ordinary thick-film misses J1939's own 120 ohm by	3.952 ohm
PASS RESOLVED: specified 0.1%/50 ppm thin-film, worst-case stacked	121.104 ohm
PASS ... margin remaining inside the 120 +/-2 ohm window	0.896 ohm

Netlist: sim/blocks/can_termination.cir

Genset ECU — Reconciled Specification

Date: 2026-09-13 **Status:** Phase 0 — supersedes all four prior artifacts
Target: Functional replacement for the `ENGINE CONTROL UNIT` block of Kirloskar wiring diagram `08-GP3-60-001`, on a 25 kVA diesel genset with a 12 V / 75 Ah battery.

---

1. Why this document exists

Four artifacts precede this one:

Artifact	Rev	Role
Genset ECU Design Brief	J	Spec
ECU Circuit Guide	H	Per-block why/how/prove
ECU Circuit Schematics	H	15 topology sheets
PCB Design Roadmap	A	Brief → fab sequence

They were written by reverse-engineering the Kirloskar wiring diagram from memory of it, not from the document itself. Now that `GP3.8703.C4.pdf` is in hand, roughly a third of their content is wrong — not stylistically, but in ways that change the BOM, the driver topology and the pin count.

This document is the single source of truth from here. Where it contradicts an artifact, this document wins. The artifacts remain useful for the reasoning behind blocks that survived unchanged, and the Circuit Guide's "prove it" bench procedures remain valid for those blocks.

Raw extracted evidence: `refs/ecu-pinout-extracted.md`.

---

2. What changed, and why it matters

2.1 Injector drive is bank-shared

The diagram traces injector high-side pin `03` to **both** cylinder 1 and cylinder 3, joined at splice V5. Pin `05` serves cylinder 2 alone. Low-side selects are individual: `73` (cyl 1), `07` (cyl 3), `29` (cyl 2).

Five pins, two banks. The artifacts specify an independent high/low pair per cylinder — six pins, three banks — which would put three boost stages on a board that needs two.

This is the conventional arrangement: the boost converter and high-side switch are the expensive, thermally-significant parts, so they are shared across cylinders that never fire simultaneously. Cylinder selection happens on the low side, which is cheap.

Consequence: two high-side bank switches, one shared boost rail, three low-side current-controlled selects.

2.2 Fuel metering unit is low-side PWM, not an H-bridge

Metering unit pin 2 sits on fused battery (20 A, `8F1`). Pin 1 goes to ECU pin `88` on a black wire — the low-side class per the page-2 legend. The ECU sinks current; it does not source it, and it never reverses polarity.

The Design Brief specifies a TLE9201SG H-bridge with one leg tied off. That works electrically but buys a reversing bridge for a solenoid that only pulls one way.

Consequence: one high-current low-side driver with current sense, closed-loop in firmware. One fewer part family to source.

2.3 Fuel level and water-in-fuel are not ECU signals

Both route through the XC11/XC1 36-pin harness to the panel — fuel level to XC pin 22 (wire `1C2`), water-in-fuel to XC pin 16/1 (wire `4B`). Neither reaches an ECU pin anywhere in the diagram. They belong to the GCU.

Consequence: two analog front-ends and two connector positions removed.

REVERSED for water-in-fuel, 8 Nov 2026 (decisions log §18). Bosch's terminal diagram wires the water-in-fuel sensor (F 002 H23 583) to ECU pin 21, `I_S_WFLS`, active high from the main-relay line. It is an ECU input again: an NPN-buffered switch input, and an amber DTC (SPN 97). Fuel level stays with the GCU.

2.4 Discrete inputs are active-high, not active-low

`AUDIO ABORT` (20), `OVERSIDE SWITCH` (24) and `IGNITION` (71) are all **red** wires — switched battery. The switch connects the pin *to battery*, not to ground.

The artifacts specify pull-up to 5 V with the switch closing to ground. That is the opposite sense, and the front-end is wrong for it: these pins see battery voltage, not 5 V, so they need a divider rated for the full 6–40 V input range plus a clamp.

The fail-safe reasoning also inverts. Circuit Guide Group D argues that normally-closed wiring makes an open harness read as a fault. With switched-battery inputs an open wire reads "not asserted" — indistinguishable from a switch that simply isn't pressed. That

argument does not hold here and should not be repeated in the new design. Detecting an open wire on these inputs requires either a pull-down and a diagnostic current source, or accepting that it is not detectable. **We accept it is not detectable** and note it in the FMEA, because changing the sense would break harness compatibility.

Pins `01` (`G_G_B AT1`) and `02` (`G_G_B AT2`) are ECU-provided return/ground legs for the switch circuits. The coolant switch (`43`) and droop switch (`23`) are cyan — digital-input class — and are conventional switch-to-return inputs.

2.5 Both CAN channels are real

Pins `55` / `77` and `87` / `86` are both drawn as twisted pairs with bidirectional markers. The artifacts treat CAN1 as a spare, DNP on the harness. In the OEM system both are wired.

Consequence: populate both transceivers and both connector positions.

2.6 Scope reductions confirmed

None of the following exist anywhere in this system: magnetic pickup unit, DEF/urea sensor, LIN bus, SCR, separate overspeed switch, barometric sensor in the harness. There is **one** catalyst temperature sensor connector (2-pin), not three EGT channels.

`OVERSIDE SWITCH` (pin 24) and `Override SS SW -S7` are one component. The Design Brief lists them as two.

---

3. Safety position

> **Revised 18 Sep 2026 — second revision, and it moves in the pessimistic direction.** The genset controller has been photographed and identified. It is a **Deep Sea Electronics DSE4522 MKII AMF (India SP)**, part `4522-001-01`, serial 11021794 — **not the KG640C** that the OEM wiring diagram shows and that research memo 02 analysed. This section was built on memo 02. Its central claim is now known to be false for the device actually fitted, and the relaxation that claim bought is withdrawn.

What changed

The previous revision rested on this: *the controller senses speed from alternator frequency, not from CAN, so its overspeed trip cannot be defeated by a hung ECU.*

That is true of the KG640C. **It is not true of the DSE4522.** From DSE's own operator manual for the DSE4510/4520 MKII (document `057-260`, the 4522's siblings):

> "If the unit has been configured for CAN, compatible ECU's receive the start command via CAN and **transmit the engine speed to the DSE controller**."

and, in the engine-at-rest detection logic:

> "Engine speed is zero **as detected by the CAN ECU**"

Engine speed reaches this controller from our ECU, over CAN. The configured overspeed shutdown at 1710 rpm therefore reads a number we send. An ECU that hangs while transmitting a stale, plausible speed defeats it completely — which is precisely the failure mode this section exists to guard against.

What is confirmed, and what replaces it

No magnetic pickup — now confirmed rather than inferred. The DSE4522's full terminal list runs 1–35 and contains no speed-pickup input of any kind: 1–2 DC supply, 3–4 outputs A (FUEL) and B (START), 5 charge fail/excite, 6–9 outputs C–F, 10 sensor common, 11–13 analogue sensors, 14–17 digital inputs, 18–20 CAN, 21–24 generator voltage sensing, 25–28 mains, 29–35 CTs, charge alternator and comms. Read from the manufacturer's own terminal table, not from a diagram's silence.

An independent path does still exist — but it is a different mechanism, and a more conditional one. The module measures **generator frequency directly** at terminals 21–24, straight off the alternator's three phases. It does not use CAN to do this. The configuration sets:

Protection	Trip	Source	Equivalent speed (4 poles)
Generator **over-frequency** shutdown	**56.0 Hz** (112 %)	Alternator terminals 21–24 — **independent of CAN**	**1680 rpm**
Engine **over-speed** shutdown	1710 rpm	**CAN, from our ECU**	1710 rpm

On a 4-pole alternator at no slip, speed and output frequency are rigidly linked. The over-frequency trip fires at **1680 rpm, before** the CAN-dependent overspeed trip at 1710 rpm, and it cannot be defeated by anything our ECU does or fails to do.

Its conditions, stated plainly. It is a *generator* protection being relied on as an *engine* protection, and it inherits that mechanism's dependencies: the alternator must be excited and producing measurable output, and the AVR must be working. An excitation failure removes the protection. That is a narrower guarantee than a magnetic pickup watching the flywheel, and it should not be described as equivalent to one.

Where the chain terminates — U12 CLOSED 18 Sep 2026, and it closed badly

> **The measurement came back: the fuel relay cuts a signal. It does not remove power.**
>
> Reported from the machine alongside two other findings — the intake throttle and the pre-heat/post-heat heaters are all fed **from the battery**, not from the ECU.
>
> This is the outcome the inspection brief told the reader to flag immediately rather than save for a writeup, and the reasoning below was written before the answer was known. It stands as written; what follows is what it now means.
>
> **Both independent lines of defence are gone, for two separate reasons.**
>
> The controller has no independent speed path — it takes engine speed from this ECU over CAN (U13, closed 18 Sep). And its every protective action terminates in a relay that only signals. So a hung ECU keeps the fuel metering unit energised, and the device that is supposed to catch that is asking the hung ECU how fast the engine is turning.
>
> What survives: generator over-frequency at 56 Hz, measured from the alternator, which does not depend on this ECU for its *measurement* — but which acts through the same signalling relay. It can detect the runaway. It cannot stop it.
>
> **Confidence, stated honestly.** Brief task 1.4 exists because a normally-closed relay produces the same voltage reading with the opposite meaning, and it is not confirmed that 1.4 was performed before this result. The wording reported ("cuts signal") is a statement about function rather than a raw voltage, which suggests tracing rather than a single measurement — but that is an inference about how the work was done, not evidence. Adopting the unsafe reading costs nothing if it is wrong; adopting the safe one and being wrong costs the machine. Confirmation is still wanted, and it does not gate anything, because nothing below changes if it comes back the other way.

Every protective action this controller can take ends at the same place: it de-energises **DC Output A, terminal 3, the FUEL relay** (rated 10 A for 10 s, 5 A continuous — a relay coil driver, consistent with it operating the 70 A `-13RB1` contactor in the OEM diagram).

So the independent over-frequency path is only as good as what that relay's contacts carry. If they remove supply from the ECU and the fuel metering unit, the path works. If they merely signal a hung ECU, **the path terminates in the failure it was meant to catch**, and there is no independent protection at all.

That is U12, unchanged in substance and now the single load-bearing measurement in this section.

The position for this design

1. The ECU implements overspeed shutdown in firmware as a **primary** function, with the crank sensor as its input, at the highest scheduling priority. First line, and it is ours.
2. The controller's **generator over-frequency shutdown at 56 Hz**, measured directly from the alternator, is the **second line**. It is independent of this ECU but conditional on alternator excitation. We must not weaken it: the ECU must not latch the fuel metering unit on in a way that survives its supply being removed, and must fail safe when the ignition input at pin 71 drops.
3. A standalone MPU-driven overspeed trip module is **mandatory again**, not recommended. The relaxation in the previous revision was bought entirely by the claim that the controller had an independent *speed* path; that claim is false for the fitted device. What remains is a generator-frequency protection with an excitation dependency, terminating in a relay whose function is unverified. That is not enough to carry a 25 kVA set on its own. Two candidates are costed in memo 02 (GAC SSW675, Murphy HD9063) — that costing survives even though the memo's device analysis does not.
4. ~~**The gate: U12 must be closed by measurement before any build takes injection authority.**~~ **Closed 18 Sep 2026, in the failing direction.** The gate was written to admit the possibility that U12 came back clean. It did not. The relay signals, so line 2 detects a runaway without being able to act on it, and item 3's "mandatory" is no longer a precaution against an unverified relay — it is the only shutdown path on this machine that does not run through the ECU being protected against.
5. ~~**The trip module must break the fuel metering unit's supply directly.**~~ **SUPERSEDED 14 Oct 2026: see item 6.** The valve turned out normally open, and then breaking its supply *opens* it. The original text is kept below because the reasoning about the GCU relay chain still holds; only the contact's position changes.

This follows from U12 and is now a requirement, not a preference. A trip module that signals the same GCU relay chain inherits exactly the defect U12 just exposed. Its contacts have to sit in series with the metering unit's own battery feed — the one through fuse `8F1` — so that tripping removes fuel authority regardless of what the ECU, the controller, or the GCU relay board is doing. Both costed candidates in memo 02 are contact-output devices and can be wired this way; the costing survives, the wiring assumption behind it does not.

6. **CORRECTED 14 Oct 2026: the trip module must break the injector supply, ECU pin 04.** The old ECU's own recording (`refs/field-evidence-2026-10-14.md` §3a; decisions log §14) shows a **normally-open** metering valve: its current falls as rail pressure rises, and rises to 1684 mA to stop the engine. With such a valve, removing its supply gives *full* flow. Item 5's arrangement would leave the pump delivering to the rail, with the rail limited only by its relief valve, and **would not stop a runaway at all.**

On this board, **all injector energy enters on pin 04** (`V_BAT_1R` → `VBAT_INJ`): - it feeds the boost converter that charges the 100 V reservoir - it feeds the two battery high sides that carry the hold current

Nothing else powers an injector. So the trip module's contacts go **in series with the pin 04 feed, after its 10 A loom fuse.** When it trips: - **Hold current stops at once,** because the battery path is gone. - **The boost reservoir runs down within a few injections.** It is 41 µF and loses about 8.3 V per peak phase (`gen_sheet_injector.py`, from `boost_converter.cir`), so roughly a dozen peak phases empty it from 100 V. That is about 0.3 s at 1500 rpm, and it is an estimate from those two numbers, not a simulation. - **Pin 21 (logic), pin 06 (EGR) and the metering valve's own feed are untouched.** The ECU stays alive to log the trip and report it over CAN.

The contact carries real current. The pin 04 feed is 10.98 A peak and 3.8 A RMS, behind a 10 A loom fuse (`gen_sheet_power_input.py`). If the module's own contact isn't rated for that, it drives a **run relay** instead: - a 12 V automotive relay with a normally-open contact rated ≥ 30 A, and coil suppression - its coil is fed through the module's "healthy" contact, so the relay is **energised to run** - then a trip, a module power loss, or a broken coil wire all stop the engine

What the ECU sees: pin 04's voltage is already measured on the harness side (`V_BAT_1R_SNS`, ADC). Pin 04 dropping while pin 21 stays up *is* a trip, so the firmware can recognise and report one without a new input. Whether that makes the supervised `TRIP_LOOP` input unnecessary is **open**, for the owner. The loop also reports a cut wire to the module, which the voltage alone doesn't.

7. **SUPERSEDES ITEM 6, 8 Nov 2026 (Bosch pinout, decisions log §18): the trip module opens the main relay's coil feed.** Under Bosch's wiring pins 04/06 come from the main relay's contact and are the board's *only* supply, so breaking pin 04 now drops the ECU as well. Item 6's "the ECU stays alive to log the trip" no longer holds. The trip loop had no Bosch pin and is gone.

The module's contact goes **in series with the relay coil's Term. 30 feed**, between the coil's fuse and relay terminal 86. When it trips, the relay opens and takes everything with it: - **The injectors**, on 04, both their hold path and the boost converter's input. - **The EGR**, on 06. - **The ECU**, also on 04. - **The metering unit and the buzzer relay**, on the main-relay line.

No injection is possible with 04 dead. The normally-open metering valve going to full flow therefore feeds a rail with no outlet but its relief valve, which is item 6's reasoning applied one step earlier.

- **The contact carries only the coil current**, about 84 mA, so the module's own contact is enough and item 6's run relay is not needed.
- **The ECU cannot defeat it.** The ECU switches only the coil's low side (pin 50, and the wake FET from pin 71). With the coil's feed open, neither can close the relay.
- **What the ECU sees:** nothing. It loses power. The GCU sees the J1939 traffic stop and alarms on it. A logged trip is what this gives up. The module's own indication, and the GCU's, record it instead.

The owners may of course revisit item 3 with evidence. What they should not do is inherit the previous revision's conclusion, because the fact it was built on turned out to describe a controller that is not on this machine.

What this ECU must do about the trip module

Item 6 puts the trip module's contacts in series with the injector feed, pin 04 (item 5, the metering unit's feed, is superseded). That interface is the module's, not ours, but it constrains this design in three ways.

We must not be able to defeat it. The trip module removes pin 04. Nothing on this board may power the injector stage from anywhere else: not pin 21's `VBAT_PROT`, not pin 06's `VBAT_EGR`, and not through a diode-OR or a protection part's leakage path. Today the schematic has none. `VBAT_INJ` is fed only from `V_BAT_1R` through Q103. The boost controller's own supply (`VDD_BAT` from `VBAT_PROT` through R405) and the high-side bootstraps (from `12V_GATE`) come from pin 21. They carry milliamps of control current, not injector current. With pin 04 open they keep the drivers alive but give them nothing to switch. That property must survive every revision. **Any path around the trip contact silently restores the failure the module exists to remove.** Any such path would be a wiring error that silently restores the failure mode the module exists to remove.

We should sense its state, and treat it as an input rather than a surprise. A trip that removes fuel while the ECU is still commanding injection produces a fault the ECU should recognise and report over CAN, rather than reading it as a fuelling anomaly and compensating.

The cheapest form is one discrete input sensing whether the module's contacts are closed. That is a new connector position and a new front-end, and it does not exist in the OEM harness — a deliberate addition, listed under §7.

We must fail safe into it, not against it. The module acts on overspeed sensed by its own magnetic pickup. The ECU's own overspeed shutdown (item 1) acts on the crank sensor. These are independent measurements of the same quantity, and they will disagree near the threshold. The ECU must not treat its own reading as authoritative to the point of trying to keep running through a trip.

Not yet decided: whether the module's pickup shares the flywheel target with our crank sensor or gets its own. Sharing is cheaper and physically simpler; separate targets preserve the independence that is the entire point of fitting the module. This is a mounting question for whoever fits it, and it should be answered before the module is ordered rather than after.

What this section still does not address

A diesel that begins burning its own lubricating oil — through a failed turbocharger seal or crankcase fumes drawn into the intake — has a fuel supply that **none of the three lines above can interrupt**. Every one of them cuts diesel or electrical power. The standard countermeasure is a mechanical or pneumatic **air-intake shutoff valve**, and no such device appears anywhere in this design, this register, or §9's out-of-scope list. It was never considered rather than considered and rejected.

This is recorded here, unresolved, because it is a decision for the machine's owners: the part is mechanical, not electronic, and fitting it is not within this PCB's scope. It should not be allowed to stay invisible.

---

4. ECU pin map

Since 8 Nov 2026 the pin map is Bosch's terminal diagram for this ECU (`refs/bosch-pinout-md1cs162.md`, 46 pins, decisions log §18). It replaced the KOEL wiring diagram below where they disagree, and they disagree on power. The tables in this section are kept as written, with the Bosch changes in this table:

Pin	Bosch	What the board does
21	`I_S_WFLS`, water in fuel	Switch input, active high. **Not** a battery feed: there is none
04 / 06	`V_V_BAT_1R/2R`, main relay contact	The board's only supply. 04 feeds the electronics chain (5 A fuse, EMI filter, TVS, reverse FET, buck) and the injectors; 06 the EGR
50	`O_S_MRLY`	Closes the board's own supply relay; held through the after-run

Pin	Bosch	What the board does
71	`I_S_T15`	Ignition, and wakes the board: a second FET on pin 50
01 / 02	`G_G_BAT1/2`	Power ground
09	`I_A_SYNC`	GCU synchronisation input, 0–10 V divider to ADC
14	`I_S_DIA`	Diagnosis request, contact to ground
25	`I_S_STST`	Stop / emergency switch, contact to ground: SHUTDOWN

Confirmed pins from the wiring diagram. **44 of 94.** See §8 for the remaining 50.

Power

Pin	Signal	Front-end
21	~~Battery + feed~~ **Water in fuel (Bosch)**	~~Main input: EMI filter → fuse → TVS → reverse-battery FET~~ That chain now takes pin 04
04	`V_V_BAT_1R` (10 A fused)	**120 k / 10 k** divider → ADC; **injector feed** `VBAT_INJ` since 8 Oct 2026 (POWER-PATH, decisions log §2a, §5); **the electronics feed** since 8 Nov 2026
06	`V_V_BAT_2R` (10 A fused)	**120 k / 10 k** divider → ADC; **EGR feed** `VBAT_EGR` since 8 Oct 2026 (POWER-PATH, decisions log §2a, §5)
09	~~`SYNCHRONIZATION GROUND`~~ `I_A_SYNC` (Bosch)	Analog input from the GCU

The battery-sense divider is **120 k / 10 k, not the 75 k / 10 k** inherited from the earlier artifacts. That value was sized for a 5 V ADC and delivers 4.69 V at a 40 V input — above the S32K148's 3.3 V analog supply. The corrected divider peaks at 3.07 V and still resolves 573 ADC counts at the 6 V cranking dip. Verified in `sim/blocks/battery_sense.cir`.

Ratiometric analog — 10 k / 16 k divider, 22 nF shunt, 3.3 V clamp, rail measured and corrected in firmware

Revised 14 Sep 2026. This section previously specified *1 kΩ series, 100 nF shunt, ADC ref tracks* `5V_SENSOR`, which contradicted the simulated front-end and cannot be built on this MCU. Both halves were wrong, and for the same reason the 75 k / 10 k battery divider was wrong — they were written for a 5 V part:

- `VREFH` **cannot track** `5V_SENSOR`**.** The S32K148's ADC reference may not exceed its 3.3 V analog supply. A reference at 5 V is not an option on any 3.3 V device.
- **1 kΩ in series scales nothing.** With no division, a sensor at 4.5 V arrives at a 3.3 V pin. The series resistor is a fault-current limiter, not an attenuator.

So the signal must be divided, and the question becomes how to keep the *ratiometric* property after dividing it.

Decision: divide, then measure the rail and correct in firmware. `5V_SENSOR` gets its own ADC channel. Every ratiometric reading is computed as a fraction of the measured rail rather than of an assumed 5.000 V, which is what the sensor's output actually represents. Cost: one ADC channel, zero analog parts. The channel budget has room — see the sensor reference.

Rejected alternative: dividing `5V_SENSOR` into `VREFH` by the same 16/26 ratio. It fits numerically (5 V × 16/26 = 3.08 V, inside the 3.3 V limit) and gives true ratiometric conversion with no firmware arithmetic. It is rejected because `VREFH` is shared by every channel on that ADC, so it would silently make **battery sense and the NTC channels ratiometric to the sensor rail as well** — both of which need an absolute reference. It would also need a buffer, since a resistive divider is too noisy and too high-impedance to drive a reference input.

Single-ended or differential is decided by the ground, not by the channel. Where the sensor's return is shared, the other sensor's current develops an offset across the harness resistance that a single-ended input reads as signal: `sim/blocks/sensor_differential.cir` measures 12 mV of bias at 1 Ω of return resistance against 0.13 mV differential. Research memo 09 found both Deep Sea Electronics and SEDEMAC treat this as a first-class problem. So:

- **Shared ground (pins 34, 36) → differential front-end.**
- **Dedicated ground (pin 08) → single-ended is sufficient**, because there is no other sensor's current in that return. This mirrors the OEM's own ranking: they gave rail pressure a dedicated return precisely so it would not need one.

Pin	Signal	Group ground	Front-end
11 / 41	Boost: 5 V excitation / pressure signal	34	Differential
79	Boost: temperature signal — **topology unresolved, see §8**	34	Differential
34	Sensor ground — boost + coolant	—	—
33	Coolant temperature (NTC, pull-up divider)	34 (spliced)	Differential
15 / 37	EGR position: 5 V / wiper	36	Differential
36	Sensor ground — EGR + oil pressure	—	—
39 / 80	Oil pressure: 5 V / signal	36 (spliced)	Differential
32 / 35	Rail pressure: 5 V / signal	08	Single-ended
08	Sensor ground — rail pressure, dedicated	—	—

Pin	Signal	Group ground	Front-end
—	`5V_SENSOR` rail monitor	—	Divider → ADC, for the correction above

Sensor-ground grouping mirrors the OEM harness exactly: `34` boost+coolant, `36` EGR+oil, `08` rail alone, `30` crank alone, `44` cam alone. Rail pressure gets a dedicated return because it is the most accuracy-critical analog channel on the engine.

Speed

Pin	Signal	Front-end
52 / 74	Crank VR: Frequency I/P High / Low	Differential adaptive-threshold conditioner → timer capture
30	Crank sensor ground	Shield terminates here, ECU end only
45 / 46	Cam Hall: 5 V / Frequency I/P	**This row was wrong — see the correction below**
44	Cam sensor ground	—

Discrete inputs

Pin	Signal	Sense	Front-end
43	Coolant switch	Switch-to-return	Pull-up + RC debounce
23	Droop switch	Switch-to-return	Pull-up + RC debounce
01 / 02	`G_G_B AT1` / `AT2`	—	Switch return legs — but see note below, these may be power grounds
20	Audio abort	**Active-high**	47 k / 68 k bias + 3.0 V zener clamp + 220 nF
24	Override SS	**Active-high**	same
71	Ignition	**Active-high**	same

The active-high front-end is not a divider, and cannot be. Reading logic high at a 6 V cranking dip needs a divider ratio above 2.31/6 = 0.385; staying under 3.3 V at 40 V needs it below 3.3/40 = 0.0825. Those constraints do not overlap, so no fixed ratio works at both ends. The zener sets the clamp level and the divider only biases into it, which makes the logic level flat to within 92 mV across the whole range instead of tracking the battery. Verified in `sim/blocks/discrete_input.cir`.

Pins 01/02 need checking. Memo 03 infers from the connector drawing that pins 1, 2, 5 and 6 are large power contacts. Pins 05 and 06 fit that — injector high side and a battery rail. Switch return legs would not. The likelier reading is that `G_G_B` denotes battery ground and these are the ECU's main power grounds, which would also explain why no power-ground pin appears anywhere in the extraction. Queued as a continuity measurement.

Outputs

Pin	Signal	Driver
03	Injector high side — bank A (cyl 1 + 3)	Boosted high-side switch
05	Injector high side — bank B (cyl 2)	Boosted high-side switch
73 / 07 / 29	Injector low side — cyl 1 / 3 / 2	Current-controlled low-side, peak-and-hold
88	Fuel metering unit PWM	High-current low-side + current sense
59 / 81	EGR High / EGR Low	H-bridge + position feedback on pin 37
50	Main relay	Low-side FET + flyback
69	Buzzer relay	Low-side FET + flyback

`Cam front-end (pin 46) — this row was wrong, CORRECTED 19 Sep 2026`

The outputs table said "pull-up to `5V_SENSOR` → timer capture" for the cam frequency input. A Hall sensor pulled up to the 5 V sensor rail swings to 5 V, and the S32K148 is a 3.3 V part. Taken literally, that drives an MCU pin above its absolute maximum.

It survived because **nothing simulated it.** There is a `vr_conditioner` block for the crank sensor and no cam equivalent, so the mechanism that catches this class of error never looked. Found by the pin-map work, on the way to something else.

This is the second front-end here specified in a way that would have destroyed a pin. The first was a divider on the battery sense — caught by `battery_sense.cir`, precisely because a block existed.

Now built: `sim/blocks/cam_frontend.cir` divides the 5 V swing into the 3.3 V domain and verifies it across the rail's own range — 3.072 V nominal, 3.147 V at 5.25 V high line, 2.769 V at the 4.5 V fault floor `sensor_rail.cir` established, where it still reads a valid logic high. All clear the 3.6 V absolute maximum. It also checks a 40 V harness short, which clamps the pad to 3.308 V with fault current held to 3.7 mA.

`Nothing on this board protects the buck against a NEGATIVE input — GAP, found 19 Sep 2026`

`sim/blocks/negative_pulses.cir` simulated ISO 7637-2's negative transients for the first time, which became possible only once the TVS stopped being modelled as a single diode.

The LM5164's VIN pin is rated −0.3 V to 100 V. The negative limit is not a mirror of the positive one, and nothing in this design was ever sized against it. Read from TI's own absolute-maximum table.

With the reverse-battery FET assumed still conducting — the honest assumption, since no turn-off delay is sourced anywhere and pulse 3a is only 0.1 ms wide:

Pulse	Buck VIN	Past the −0.3 V rating by
1 (−150 V / 10 Ω / 2 ms)	**−40.7 V**	135×
3a (−220 V / 50 Ω / 0.1 ms)	**−27.8 V**	93×

The bidirectional TVS clamps correctly. It clamps to its own breakdown, and −38 V is a perfectly good clamp that is still two orders of magnitude outside what the buck tolerates. **This is a missing component, not a mis-sized one** — the chain has no series element or shunt diode addressing a negative excursion.

And the reverse-battery stage cannot be assumed to cover it. If its controller opens fast enough the buck is isolated almost perfectly; if not, the FET conducts and the buck sees the full clamped negative voltage. `reverse_battery.cir` models instant turn-off and its own RESULT NOTE warns it "will always look perfect in reverse". **No controller part is chosen and no turn-off delay is sourced**, so both bookends are simulated and neither is claimed. That delay is now a load-bearing number, and this decision should be taken with the controller selection rather than separately.

`Every driver gate is held off by hardware, not by firmware — REQUIRED 18 Sep 2026`

Research memo 11 establishes the hazard with the MCU's own datasheet numbers rather than by argument. Two facts combine badly:

- **S32K148 GPIOs are high-impedance out of reset**, until firmware configures them. High-impedance on a gate driver is not "off" — it is undefined, and the gate can be brought up by leakage or by capacitive coupling from a switching node.
- **There is a band where nothing internal resets the part.** Low Voltage Reset always forces a reset at 2.50–2.7 V, but Low Voltage Detect at 2.8–3.0 V only does so if firmware has set `LVDRE`. Below the datasheet's own 2.97 V correctness guarantee and above LVR, the core may execute wrong instructions **with its outputs still driven**.

The pin that makes this concrete is **88**. The fuel metering unit is fed from battery through fuse `8F1` and switched low-side by this ECU (§2.2). A gate floating high during reset energises the metering unit with no firmware in control of it — and, since U12, with nothing downstream able to remove its power either.

Therefore, as a binding requirement on every driver gate in the table above:

1. A **gate–source pulldown** at each gate node, sized by the FET actually chosen rather than fixed here:

> `Rpd < Vgs(th) / (Crss · dV/dt)`

The dV/dt in that formula is the weakest number in this requirement, and this document previously overstated its standing. It read "2.8 V/µs from `injector_turnoff.cir`", which is a citation that file explicitly refuses: its own comments say reporting a dV/dt from that model "would be exactly the kind of promotion-to-sourced the project's own discipline rules out." The figure is memo 10's **inferred** estimate — 100 V in ~36 µs — and `supervisor.cir` imports it as an assumed input waveform, labelled INFERRED there too. Every file closer to the number than this one hedges it; only the spec stated it flatly, beside genuinely simulated figures, where it read as one of them.

It is also an average, not a peak, and the error runs in the unsafe direction. 100 V over 36 µs is the mean slope of the whole turn-off. Miller coupling happens during the plateau, where a real FET slews far faster. The requirement's sensitivity to that, at Crss = 500 pF:

dV/dt	gate through 470 Ω	Rpd actually needed
2.8 V/µs (the assumed average)	0.66 V — ok	< 714 Ω
10 V/µs	2.35 V — over threshold	< 200 Ω
28 V/µs	6.58 V — over threshold	< 71 Ω

At 200 Ω the gate driver would need an output impedance near 10 Ω; at 71 Ω a passive pulldown cannot be driven at all. **So the 470 Ω answer survives only at the assumed average**, and a modest factor on the real slew rate removes the passive-pulldown topology entirely rather than merely resizing it.

A second assumption pulls the other way. The 470 Ω also assumes Vgs(th) = 1.0 V, the conservative end of a 1.0–2.5 V class range with no part chosen. Read at 2.5 V the same sweep gives **1000 Ω and a ≤52.6 Ω driver**. So:

Assumption	Direction	Effect on 470 Ω
Vgs(th) 1.0 V vs 2.5 V	permissive if wrong	470 Ω is **pessimistic** — costs drive current, not safety
dV/dt 2.8 V/µs (an average, not a plateau slew)	**unsafe if wrong**	470 Ω is **optimistic** — a real slew of 10 V/µs demands 200 Ω

The slew-rate assumption is the larger lever and the dangerous direction. **Treat 470 Ω as a demonstration, not a value to build to:** what it actually shows is that a passive pulldown alone lives in a narrow window bounded by two numbers nobody has measured. That conclusion holds whichever way either assumption lands, and it is the argument for the active clamp — which `sim/blocks/supervisor.cir` has already validated, and which research memo 12 recommends building discretely regardless of the driver-architecture decision.

At a 1.0 V threshold that is **under 1.79 kΩ for Crss = 200 pF, 714 Ω at 500 pF, 357 Ω at 1000 pF**. `sim/blocks/supervisor.cir` claim 6 sweeps this directly and lands on **470 Ω** at Crss = 500 pF — twenty times smaller than the figure this requirement first carried.

That value constrains the gate driver, and the two cannot be specified apart. At 10 kΩ the pulldown was a perturbation on the driver. At 470 Ω it is the bottom half of a divider with the driver's own output impedance:

> `Vgs = Vdrv · Rpd/(Rpd + Rdrv)` — holding 9.5 V across 470 Ω needs **Rdrv ≤ 24.7 Ω**

It also costs **21.3 mA per gate and 1.28 W across all six**, continuously, whenever the FETs conduct. Stating the pulldown as a single number hid both the coupling and the power.

This reopens a topology question memo 11 closed on the bad arithmetic. The memo considered an active clamp and dismissed it. A passive pulldown strong enough to reject Miller coupling at 500 pF is strong enough to fight its own driver and burn over a watt doing it — which is the trade an active clamp exists to avoid, and which an integrated driver in the MC33814 class handles internally. That decision is now coupled to the driver-architecture decision below and should be taken with it.

This requirement said "10 kΩ" when first written on 18 Sep 2026, and that was wrong. Memo 11 §4 computed the Miller-coupled current correctly at 140–560 µA and then mis-multiplied, reporting 5.6 mV across 10 kΩ where the answer is 5.6 V. `sim/blocks/supervisor.cir` was built to check that paragraph and fails it at 50 pF, the low end of the memo's own range. The memo now carries the correction inline. The error is recorded rather than quietly patched because it survived a review, a spec commit, and was caught only by simulating the claim. 2. A **windowed supervisor** monitoring `3V3_MCU`, tripping at **~3.05–3.10 V**, above LVD's 3.0 V maximum, so the trip does not depend on whether firmware ever configured `LVDRE`. Its watchdog window must close well inside one cylinder interval (26.67 ms at 1500 rpm). 3. The supervisor's reset, inverted, driving a **kill transistor at each protected gate**, so the safe state is enforced independently of GPIO configuration. A plain timeout watchdog is defeated by a stuck loop that still kicks it; the window is what closes that.

With the normally-open valve (14 Oct 2026), the kill still stops the engine. It also holds the metering driver's gate off, which leaves the valve fully *open*, but the same kill holds every injector gate off, so nothing is injected and rail pressure goes to the relief valve. The safe state is "no injection", not "no rail pressure".

This is the ECU's own fail-safe and it is **separate from the standalone trip module** of §3. The trip module removes fuel authority from outside the board; this keeps the board's own outputs defined while it still has power. Neither substitutes for the other, and since U12 neither is optional.

Open, and blocking the detail rather than the requirement: the physical `PTxx` assignment for pins 88, 73/07/29 and 03/05 is a phase-2 task (§5), so the per-pin reset pull defaults cannot yet be looked up. The requirement above does not depend on them — it exists precisely so that the answer does not matter — but the analysis cannot be finished until the pin map is fixed.

`Input protection — RESOLVED 20 Sep 2026, and it moved three other parts`

Two decisions, taken together because they are coupled.

1. TVS: SMBJ33CA → SMCJ43CA. The standoff had to clear the 40 V clamped load dump so the part stops conducting during a normal event — that is what `load_dump`'s failure was actually saying. 43 V is the smallest standard bin above 40 V, and smallest matters because clamping voltage rises with the bin and everything downstream sees it.

The package changed too, and that was forced by simulation rather than chosen. At the 600 W SMBJ size, pulse 2a clamped at **88.6 V** against the buck's 100 V absolute maximum — 11 % margin, too thin to design to. The 1500 W SMCJ part has half the dynamic resistance at the same standoff and clamps at **73.3 V**, restoring 1.37× margin.

Result: `load_dump` now draws **0.000 A and absorbs 0.000 J** — the TVS simply stays off through the dump, which was the predicted fix and is now the simulated one.

2. A negative clamp diode at the buck input. The TVS holds the node at its own −47.8 V breakdown, which is a correct clamp and still two orders of magnitude outside the buck's **−0.3 V** rating. A 60 V / 20 A-class Schottky, anode to ground and cathode to VIN, conducts long before the TVS does:

Pulse	Buck VIN before	After
1	−40.7 V	**−0.53 V**
3a	−27.8 V	**−0.39 V**

The residual is honest and still fails its check. 0.53 V is 1.76× the −0.3 V rating, not inside it. No silicon diode reaches 0.3 V at the ~15 A this fault delivers. Closing the last 0.23 V needs series impedance — the EMI filter's ferrite sits upstream and is not modelled here — or a judgement that a sub-microsecond 0.2 V excursion past a DC absolute maximum is acceptable. **That is a question for the part's manufacturer, and it is now a bounded one rather than a missing component.**

What this moved elsewhere: the reverse-battery P-FET sits downstream of the clamp, so it now needs **100 V class, not 60 V** — its third revision, each time because the clamp moved. `transient_clamp.cir` now carries an explicit check stating the downstream requirement, so the next move does not have to be noticed by a human reading three files.

`The EGR driver must decode IN1/IN2 internally — REQUIRED 19 Sep 2026`

`sim/blocks/egr_hbridge.cir` argued that no IN1/IN2 combination could short a leg, because each input line drives a *diagonal* pair. A review falsified it against the block's own wiring: IN1 commands leg-A-high and leg-B-low, IN2 commands leg-B-high and leg-A-low, so asserting **both** turns all four switches on and shorts **both legs** rail-to-ground at once. Simulated at **270 A**, matching the hand figure of two legs of 2 × 50 mΩ across 13.5 V.

The reasoning failed by checking each input line alone and never asking what the two do together.

Nothing on this board currently prevents that state. The gate pulldowns hold both lines low at reset, which is correct and not the issue — the exposure is any fault or firmware error that drives both high.

The requirement: the EGR driver must decode IN1/IN2 so that the 11 state means brake or coast and never shoot-through. Integrated H-bridge drivers do this internally, and that now counts as a reason to prefer one here — note that memo 12's driver decision covered the *injectors* and explicitly did not address EGR. If the driver is discrete, the interlock has to be built, and **a per-leg dead time is not sufficient on its own**: this fault is a logic state, not a timing overlap.

RESOLVED 20 Sep 2026. The same bridge driven through a decoded DIR/PWM interface, with **both** inputs asserted — the combination that shorts the naive arrangement — draws **4.35 A**, which is the motor's own stall current, not a short. 270 A against 4.35 A. The fault state does not exist under decode; it is not merely made less likely.

`egr_hbridge.cir` now simulates both, and the naive wiring is kept as **evidence of why the requirement exists** rather than as a live failure — the same treatment `battery_sense.cir` gives the divider it rejected.

What to draw: an integrated H-bridge with a decoded DIR/PWM (or equivalent non-shoot-through) input interface, current limiting, and headroom over the actuator's stall. `egr_hbridge.cir` puts stall at **2.6–6.4 A** across a plausible armature resistance range, with no actuator chosen — so specify a part with comfortable margin over 6.4 A and confirm against the actuator once U8-class information arrives.

Candidate parts are not named here: none has been checked against a retrieved datasheet, and this project's rule is that an unread figure does not become a design number.

`Driver architecture — DECIDED 19 Sep 2026, research memo 12`

Discrete power MOSFETs with a gate-driver IC for the injector banks, keeping the 100 V boost rail. An integrated smart low-side switch for the metering unit. An active kill-clamp at every gate of both, built discretely.

The decision was forced by one confirmed number. The two integrated injector-driver ICs that fit this design's channel count and current class both cap their boost rail **below the 100 V already chosen**: MC33816's `VBOOST` absolute maximum is **72 V** (NXP datasheet Rev 10.0), and L9781's tank voltage is up to 80 V (ST data brief DB3720, supplied by the owner 8 Oct 2026, after the datasheet failed retrieval five times).

Adopting either would force the boost rail down and re-derive three already-passing blocks. Memo 12 re-did that arithmetic: at 72 V the peak-phase ramp stretches from 38 µs to **53 µs**, which still fits a pilot injection but consumes half of it.

And neither part removes the work. Both are pre-drivers, not integrated power stages — external MOSFETs, external boost inductor and diode, and external sense shunt are required either way. MC33816 additionally brings its own microcode toolchain. MC33814 is the wrong class outright: 1.3 A continuous against an 18 A requirement, and no boost stage at all.

On the coupled clamp question, the integrated route relocates the problem rather than removing it — and does so in a form worth copying. MC33816's high-side pre-driver is *actively forced low* on reset, falling back to a weak 500 kΩ–2 MΩ resistor only once its bootstrap capacitor is exhausted. That is the same two-tier structure `supervisor.cir` already validated, on a different bias rail, and with no datasheet statement about how fast that bootstrap decays under brownout — where `supervisor.cir` claim 4 gives a timed guarantee. **So build the active clamp discretely regardless.** This was never a question the IC-versus-discrete decision could have answered differently.

What would change this: a retrieved L9781 datasheet showing a tank voltage at or above 100 V, or a decision to accept a ≤72 V rail on other grounds.

One sourcing item escalated and unresolved: MC33816AE carries a distributor last-time-buy date of 06/08/2027, which needs a human with distributor access. The L9781 item is closed: the owner supplied ST's data brief on 8 Oct 2026, confirming the 80 V tank ceiling.

CAN

Pin	Signal
55 / 77	CAN channel 1 — H / L
87 / 86	CAN channel 2 — H / L

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5. MCU selection

Recommendation: NXP S32K148. This resolves the contradiction in the artifacts — Design Brief §03 specifies MPC5744P while its own §02 block diagram and §06 sensor table still say S32K148 and FTM.

The argument for MPC5744P was angle-synchronous injection timing needing the eTPU2.

> **Corrected 2026-09-13** after research memo 05 challenged the original reasoning and was right to. This section first argued that constant 1500 rpm operation relaxes the timing problem because a crank degree is 111 µs and a 60-tooth wheel gives an edge every 667 µs. That checks the wrong variable. Crank-edge *capture* rate was never the binding constraint in either case — at 6000 rpm a 60-tooth wheel produces an edge every **83 µs**, which is about **6,700 CPU cycles** at 80 MHz against an NVIC latency of roughly 12. The comparison was never close, so constant speed cannot be what makes it comfortable. The reasoning below replaces it.
>
> **Re-corrected 18 Sep 2026.** This box said "an edge every 28 µs, which is over 2000 CPU cycles". 28 µs is the **one-degree-of-crank** period at 6000 rpm (10,000 µs ÷ 360), not a tooth-edge period — a unit conflation, off by a factor of six, sitting inside a box that was itself written to fix an earlier reasoning error. A 60-tooth wheel at 6000 rpm gives 166.7 µs per tooth and 83.3 µs per edge counting both polarities.
>
> The conclusion is unaffected and in fact strengthened: the correct figure gives *more* margin, not less. It is corrected because a wrong number that happens to support the right answer is still a wrong number, and this one would have been inherited by anyone reasoning forward from it.

What the eTPU2 actually buys. It is a co-processor that autonomously schedules and re-schedules many angle-domain output-compare events across a wide and *rapidly changing* speed range, without consuming the main CPU's interrupt budget and without the core's scheduling jitter — cache effects, higher-priority ISRs, RTOS latency — reaching the injection timing chain. That matters when rpm sweeps 800–6000 with tight transient response and pilot/main/post multi-pulse strategies all landing within microseconds of a moving target angle.

Why we don't need it. This set governs to a fixed 1500 rpm, and the pin map in §4 shows a single main injection pulse per cylinder per cycle — one bank high-side plus one low-side select, with no multi-pulse rate shaping in scope. The constraint that does bind is **injector pulse-width resolution**, which sets fuel-quantity metering accuracy and is independent of crank speed. FlexTimer output-compare runs off the bus clock: one tick is 12.5 ns at 80 MHz, 8.9 ns at 112 MHz HSRUN. Against a single-pulse metering requirement on the order of 1–4 µs, that is roughly 100–450× headroom.

The S32K148 also carries 2× Programmable Delay Blocks, which trigger ADC conversions or timer events at a defined delay from a capture event in hardware. That recovers part of what the eTPU2 offers — jitter isolation on the timing chain — without the eTPU2.

Caveat carried forward: the 1–4 µs resolution figure is a general diesel-injector engineering figure, not this injector's datasheet. Revisit if U5 closes with a part demanding materially finer control, or if multi-pulse injection ever enters scope.

What the S32K148 buys in exchange:

- Free S32 Design Studio; GCC ARM toolchain; cheap CMSIS-DAP/J-Link probes instead of a Power Architecture Nexus pod
- Stock KiCad symbol availability, or trivially built from NXP's own EVB files
- Real availability in small quantities in India
- 3× FlexCAN (we need 2); dual 12-bit ADC against a demand of ~14 channels — 8 sensor inputs, 2 battery-rail senses, and 4 current-sense returns from the injector, metering and EGR drivers; FlexTimer instances well beyond the ~10 timing channels required (crank capture, cam capture, 3 injector low-side, 2 injector high-side, metering PWM, 2 EGR bridge PWM)
- No eTPU2 microcode, which is a specialist skill and a hiring risk on a small team

Confirmed in Phase 1 — see `docs/research/05-mcu-selection.md`. Orderable part **FS32K148HAT0MLQT**, 144-pin LQFP, 80 MHz RUN / 112 MHz HSRUN grade, −40…+125 °C. In stock at DigiKey India at roughly ₹1,900–2,000/unit. The S32K146 fallback was checked and is in worse supply shape (backordered, 12–52 week spread), so it is not needed.

Two items deferred to Phase 2: confirming how many ADC channels the 144-pin package actually breaks out (needs the reference manual's pin-mux table, not the datasheet), and pad-by-pad verification of the SOT486-2 package drawing before the footprint is locked.

Pin map done, 19 Sep 2026 — see `docs/pinmap.md`. The first of the two deferred items above is closed: the Reference Manual's own embedded `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx` (not the datasheet) gives 128 GPIO-capable pins on the 144-pin package, of which this design commits 37, including all 94-way connector signals with a confirmed identity (spec §4). It also settles §4's own deferred fail-safe detail: all six gate pins (88, 73/07/29, 03/05) are confirmed high-impedance with no pull at reset, exactly the condition §4's hardware kill-clamp was already designed against. One new finding surfaced in the process: pin 46's cam front-end, as specified above, would put a 5 V swing on a 3.3 V MCU pin — flagged there, not fixed here. Package pad-pitch verification remains open.

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6. Power architecture

Unchanged from the artifacts in topology; the reasoning there is sound.

Stage	Function
EMI input filter	Common-mode choke, CISPR 25 conducted margin
Reverse-battery	Ideal-diode controller + P-FET, < 0.3 V drop
Transient clamp	Bidirectional TVS, ISO 7637-2 pulses incl. load dump 5b
Buck pre-regulator	6–40 V → 5 V, wide-input synchronous
`5V_SENSOR`	PTC-protected per sensor group, isolated from MCU rails
`3V3_MCU`	LDO off 5 V
Supervisor	Windowed watchdog + brown-out reset

Input range 6–40 V: 6 V covers the cranking dip on a 12 V system, 40 V bounds the clamped load-dump transient.

Bosch wiring, 8 Nov 2026: there is no permanent battery feed. The chain above starts at pin 04, the main relay's contact. Pin 71 (ignition) closes the relay through a wake FET on pin 50, and the MCU holds it through the after-run: it stops the engine, writes the fault memory, then releases it. Pin 50's turn-off clamp is a TVS to ground. A diode to the board's supply would feed the board through the always-live coil (`sim/blocks/main_relay.cir`).

Injector boost rail is new relative to the power chain as drawn: a separate boost stage generating the injector opening voltage, with its own reservoir bank sized for the pulsed draw of two banks. Target voltage is set by the injector part — see §8.

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7. Deliberate deviations from the OEM design

Recorded so they are choices, not drift:

1. ~~**Connector split.** The OEM uses one 94-pin connector. We use functionally-split sealed connectors. This simplifies harness build and field diagnosis, at the cost of needing an adapter harness to mate with an unmodified Kirloskar loom. That adapter is a deliverable, not an afterthought.~~

REVERSED 22 September 2026, by the owner. The ECU carries the **OEM connector described in the field evidence** — Bosch `1 928 405 192` / `1 928 405 194`, **code C**, 94-way, EDC17 family (`refs/field-evidence-2026-09-18.md` §2b) — and plugs straight into the unmodified Kirloskar loom. **No adapter harness.** This is no longer a deviation from the OEM design.

Consequences, all already reflected elsewhere: the board-side land pattern is now load-bearing (field brief task 5.3, and a Bosch distributor quoting both numbers, since neither returns a catalogue hit); **code C** keying must match exactly; and the harness-side questions — which cavity is the power ground (task 5.1), and a pin and wire for the new `TRIP_LOOP` (task 5.2) — are answered in the OEM loom itself rather than designed into an adapter. 2. **Onboard barometric sensor.** Not present in the OEM harness. Added on-PCB behind a vented port; costs nothing in harness terms and improves fuelling correction. 3. **Independent overspeed path.** Added, per §3. The OEM does not have one.

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8. Unknowns register

Nothing in this list may be guessed into copper. Each needs an answer before the sheet that depends on it is drawn.

Status updated at phase 1 exit, 14 Sep 2026. Three states, and the middle one carries weight: **Open** means no answer yet; **Answered** means phase-1 research produced an answer that is good enough to design against but is *inferred*, not confirmed, and still carries the upgrade path in its memo; **Closed** means confirmed outright. Nothing here is Closed yet — every confirmation on this list needs the machine.

#	Unknown	Status	Blocks	How to close
U1	ECU connector identity	**CLOSED 18 Sep 2026 — Bosch**, part numbers `1 928 405 192` and `1 928 405 194`, `code C` mechanical keying. 94 cavities: power pins 1–8 in their own chamber, then rows 9–28, 29–50, 51–72, 73–94	—	Closed by markings on the part. Note: an adapter housing **must be code C** or it will not mate

#	Unknown	Status	Blocks	How to close
U2	Boost air-temp (pin 79): NTC or ratiometric?	Open	Sheet 3b front-end	One resistance reading, sensor cold — brief task 1.3, closes it outright. **Mitigation: a front-end that works either way 13 Oct 2026:** the Bosch catalogue's T-MAP boost sensors carry a 2.5 kΩ NTC — likely an NTC; the resistance reading still decides it
U3	Controller MPU input — does it exist?	**CLOSED 18 Sep 2026 — No.** The fitted controller is a **DSE4522 MKII**; DSE manual `057-260` 's terminal table runs 1–35 with no speed-pickup terminal	§3	Closed from the manufacturer's own terminal list
U4	Controller J1939 expectations	**Largely answered** — CAN source addresses **234** (engine) / **44** (instrumentation); J1939-75 instrumentation on, alarms off; the controller takes **oil pressure, coolant temperature, coolant level and ENGINE SPEED** from our ECU over CAN; `ECU Data Fail → Shutdown`	Firmware, CAN sheet	Remaining: the exact PGN set. Log the live bus, or read DSE publication 057-004 **13 Oct 2026:** an engine-off trace arrived but holds only UDS diagnostics (0x7E0/0x7E8), no J1939; still needs a running-engine trace on the DSE CAN
U5	Injector part number and drive profile	**Partly answered** — memo 04 gives a family envelope (65–115 V, 12–24 A peak, 8–13 A hold, 40–100 A/ms) sufficient to design the driver	Boost rail *target* voltage, not the driver sheet itself	Photograph the injector body for a Bosch `0445 1xx xxx` or Kirloskar `F6.xxx.xx.x.pr`. **Separately: the calibration data is a different problem — see §9 and memo 04 13 Oct 2026:** KOEL injector number `4H.508.70.0.PR.REV-1` ; the Bosch number on the injector body still needed (`refs/field-evidence-2026-10-13.md`)
U6	Pin 09 `SYNCHRONIZATION GROUND` function	Open	Power/ground plan	OEM pin list; drawn red despite the name
U7	Pins 04/06 — sense only, or load feeds?	Open — no longer blocking	The board now uses them as load feeds (POWER-PATH, 8 Oct 2026), relying on the 10 A loom fuse sizing the wire. Measuring still confirms the OEM's own use	Measure on the live engine, or OEM pin list

#	Unknown	Status	Blocks	How to close
U8	6-pin intake throttle — on the ECU at all?	**Partially answered 18 Sep 2026** — reported **battery-fed**, so the ECU does not supply it	Whether a throttle *driver* is in scope. Supply and control are separate questions and only supply is answered	Still needs brief task 3.11: follow the cable and see whether it terminates at the 94-way connector. A battery-fed actuator can still be ECU-commanded
U9	Catalyst temp sensor — connected where?	Open	Whether an EGT front-end is in scope	Trace on the engine
U10	`SENT` in the colour legend — which signal?	Open	Possible digital sensor front-end	Inspect harness; SENT is SAE J2716
U11	Engine identity	**CLOSED 18 Sep 2026** — rating plate reads **3GK550ETA 4SR1**, app code **GK3.8703**, 26.5 kW at 1500 rpm, type approval `ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24`. Memo 01's inferred `3R550ETA 4G1` was **wrong**; the original "GK550" was right	—	Closed by plate
U13	Which controller is fitted	**CLOSED 18 Sep 2026 — DSE4522 MKII AMF (India SP)**, part `4522-001-01`, serial 11021794. Photographed front and rear. **Not a KG640C.** Memo 02 superseded; spec §3 rewritten	—	Closed by photograph
U14	Pre-heat / post-heat: driven by the controller or by the ECU?	**Partially answered 18 Sep 2026** — reported **battery-fed**	Whether a heater *output* is in scope. As with U8, supply is not control	Who switches that battery feed is still open. The DSE4522 config enables both at 50 °C, so the controller is the likely switch — photograph the heater relay and trace its coil
U15	What does `CRS-878` denote in the controller's engine profile?	**NEW 18 Sep 2026**	Possibly bears on U5 — "CRS" plausibly identifies the common-rail system	Ask KOEL, or find DSE's engine-profile list
U17	Which machine are the old ECU (`MD1CS162`, KOEL `T4.435.01.0`) and the 1 Oct CAN trace from?	**NEW 13 Oct 2026** — the trace's ECU reports `KOEL_TC_82.5KVA`, `4R1190ETA4G1` (4-cylinder), not our 25 kVA `3GK550ETA4SR1`	Whether the trace and that ECU describe our engine at all	Ask Jeevan

#	Unknown	Status	Blocks	How to close
U16	**Is the battery-charging alternator's rectifier suppressed (avalanche/clamping), or unsuppressed?**	**NEW 18 Sep 2026. CLOSED 9 Oct 2026 by the owner: design for the suppressed case** (clamped load dump, ISO 7637-2 pulse 5b, about 40 V), which the board already does. The ECU is built from scratch, not matched to this machine's alternator, and modern alternators clamp.	**The input-protection topology.** Suppressed ⇒ ISO 7637-2 pulse 5b, ~40 V clamped, and the fix is a TVS standoff above the dump level — a part number. Unsuppressed ⇒ pulse 5a at 65–87 V unclamped, tens of joules that nothing downstream can ride out, and a different circuit	The alternator's rectifier part number (look for a suppression/avalanche bin), or a scope on the battery terminal during a live disconnect under load. The scope test needs the set running — a supervised activity, not a solo checklist item **Photo 9 Oct 2026** (`refs/charging-alternator-2026-10-09.jpg`): a compact belt-driven unit with the regulator and rectifier inside a black rear cover, a stud B+ terminal and a one-wire plug (green wire). Only production stamps are visible (`G90 518 13`, `30.11.24 21:07`, `09.11.24`, so it was built Nov 2024); no catalogue match. Still needed: the maker's label or part number, usually on the rear cover or a sticker
U12	**Does the 70 A ignition relay `-13RB1` remove power from the ECU and fuel metering unit, or only signal the ECU?**	**CLOSED 18 Sep 2026 — it only signals.** Reported from the machine. The relay does not remove power; a hung ECU keeps the fuel metering unit energised	**§3's safety gate, resolved against us.** The trip module is mandatory, and §3 item 5 now requires its contacts to break the metering unit's own battery feed rather than signal the same relay chain	Closed. Task 1.4's normally-closed check is still wanted as confirmation, but nothing depends on it — the unsafe reading is already adopted

~~**U12 now stands alone as the load-bearing unknown.**~~ **U12 closed 18 Sep 2026, and the answer was the bad one.** U13 cost this project its safety argument: the fitted DSE4522 takes engine speed from our ECU over CAN, so the controller's overspeed trip is not independent of the thing it is meant to protect against. What independence remained was the generator over-frequency shutdown at 56 Hz — and every protective action the controller can take terminates in the same fuel relay on terminal 3.

That relay signals. It does not remove power. So the second line of defence can detect a runaway and cannot stop one. The two findings compound: the controller either cannot see the fault (overspeed, read from our CAN) or cannot act on it (over-frequency, acting through a signalling relay).

Nothing on this machine currently interrupts fuelling without the cooperation of the ECU being protected against. The trip module in §3 item 3 is what closes that, and §3 item 5 now specifies where its contacts have to sit — in series with the metering unit's own battery feed, not in the GCU relay chain that just failed this test.

Worth stating plainly, because it is the pattern of this whole phase: **four of the five unknowns closed by field evidence closed against what this project had reasoned.** U11 (engine identity), U13 (which controller), U1 (connector — this one closed *for* us), and now U12. Desk inference has a measured track record here, and it is poor.

U1, U3, U11 and U13 all closed on 18 September from photographs and one manufacturer's manual. Three of the four closed *against* an inference this project had made, which is the honest measure of how much desk research was carrying.

U3 and U11 were the other two on that list and are now answered, both by inference from primary documents rather than by confirmation. Neither answer is fragile — but both are the kind of finding that a single photograph at the machine would settle permanently, which is why they stay on the register rather than leaving it.

U5 has moved off the critical list for a reason worth stating plainly: memo 04's family envelope means the **injector driver can be designed now, without the part number**. What the part number buys is a boost-rail target instead of a range. The thing that genuinely gates running this engine is not on this register at all — it is the injector calibration data, recorded in §9 and memo 04.

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9. Out of scope

- Injection calibration: timing maps, quantity maps, rail pressure setpoint schedules. The hardware will be capable; the maps are a dyno programme, not a PCB deliverable.
- Emissions certification. **But read this before planning deployment** — research memo 06 found, in primary sources, that the regulatory position is worse than "we simply won't certify it." CPCB's RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy must not be altered even under CPCB's own sanctioned retrofit scheme, and excludes changes limited to engine control from that scheme entirely. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not block the engineering: designing the board, building it, and bench-testing it against simulated signals are unaffected. It bears on **putting it into service on a genset that is in use**, and it compounds the calibration risk in memo 04 — this engine meets its limits partly through calibration, so a replacement ECU can move it out of compliance without any hardware changing at all.

This is the project owners' decision to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and was not previously examined, not to make the decision for them. - The panel controller. We talk to the **DSE4522 MKII AMF**; we do not replace it. (Earlier revisions of this line said KG640C, which is what the OEM wiring diagram shows and not what is fitted — see §3.) - Panel wiring, ATS, alternator metering.

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10. Phase sequence

Phase	Output	Status
0	This document	Done
1	Research memos: engine ID, controller, MCU confirmation, injector and connector sourcing	**Done** — nine memos
1.5	**Block-level simulation: topology and component values verified before capture**	**In progress — 17 blocks, all passing**
2	KiCad hierarchical schematic capture	Not started
3	4-layer PCB layout, DRC, fab outputs	Not started
4	Firmware skeleton: HAL, J1939, crank sequencing, fault manager	Not started

Phase 1.5 was added on 18 Sep 2026 to name work that was already happening. The simulated blocks are neither research memos nor schematic capture, and leaving them unnamed meant the roadmap did not describe the most productive activity on the project — the one that caught a divider that would have destroyed an ADC pin, a discrete input that was impossible as specified, and a decoupling network thirty times over its impedance target. It also gives the two gaps the circuit review found — load dump, and injector turn-off energy — a phase to belong to instead of floating as review findings.

Phase 2's blocked sheets are now fewer: **U1 is closed** (Bosch `1 928 405 192` / `194`, code C), so the connector and adapter-harness sheets are unblocked. U2 blocks one sensor front-end, U8 blocks whether a throttle driver exists at all, and U5 sets the boost rail's target voltage but not the driver itself. Power, CAN, MCU support, speed inputs, the relay outputs and the analog front-ends are all fully specified above.

MCU Pin Assignment — S32K148 to the 94-way Bosch Connector

Date: 2026-09-19 **Status:** Phase 2, first artifact. Unblocks the per-pin fail-safe detail spec §4 deferred and the two Phase-2 items memo 05 §7 left open (how many ADC/FTM channels the 144-pin package actually breaks out; per-pin reset defaults for the six gate pins).

> **Bosch pinout, 8 November 2026** (decisions log §18). The connector pins below are now Bosch's terminal diagram for this ECU (`refs/bosch-pinout-md1cs162.md`), which replaced KOEL's harness diagram where they disagreed. The changes are marked **BOSCH** in the tables: - Pin 21 is the water-in-fuel input, not battery. - Pins 04/06 are the board's only supply, from the main relay, which pin 71 wakes. - Pins 01/02 are the power ground. - Pins 09, 14 and 25 are new inputs. - The trip loop is gone.
>
> 45 MCU pins are committed, against 42 before; 46 since the boost stop on `PTD15` (12 Nov 2026).

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0. Preamble — package, sources, method

Package assumed: 144-pin LQFP, orderable part `FS32K148HAT0MLQT`. This is the package already selected in spec §5 and confirmed in `docs/research/05-mcu-selection.md` — chosen as headroom against the 50 still-unresolved OEM connector pins (U1's register), not because the *confirmed* 44-pin budget needs it. This document does not revisit that choice; it uses the `S32K148_144lqfp` column of NXP's own pin table throughout.

Documents retrieved this session, fresh, not from memory:

- **NXP, *S32K1xx Data Sheet*, Rev. 15, 5 March 2026** — `nxp.com/docs/en/data-sheet/S32K1xx.pdf` , the same revision `docs/research/05-mcu-selection.md` cites. Used for the LVR/LVD electrical thresholds (§3 below) and to confirm the datasheet itself defers the pinout table to the Reference Manual (§10.1, "For package pinouts and signal descriptions, refer to the Reference Manual" — this is a direct quote, not a paraphrase, and it is why memo 05 could not close its own two Phase-2 items from the datasheet alone).
- **NXP, *S32K1xx Series Reference Manual*, Rev. 13, 04/2020** — retrieved from `community.nxp.com` (the same mirror memo 05 found but did not open). **This PDF has eleven Excel workbooks embedded as file**

attachments, including `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx` — the actual chip-specific pin-mux table the Reference Manual's own Chapter 4 says exists but does not inline ("IO Signal Description Input Multiplexing sheet(s) attached to the Reference Manual"). Extracted with `pdfdetach` and read directly (893 rows, `IO Signal Table` tab). **This is the primary source for every `PTxx` ↔ peripheral ↔ package-pin-number claim in this document, all tagged CONFIRMED below**, and it is the single fact this whole exercise turned on: the manual's *narrative* text was retrievable by search all along, but the actual per-pin table was not, because it lives in an attachment nobody had opened.

- Sensor front-end topology and signal counts: `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` §4/§5, `refs/ecu-pinout-extracted.md`, `sim/blocks/*.cir` (read, not modified), `docs/research/05-mcu-selection.md`, `docs/research/12-driver-architecture.md`.

One revision gap, disclosed rather than smoothed over. The Reference Manual pulled this session is Rev. 13 (04/2020); the datasheet memo 05 cites is Rev. 15 (2026). No higher-revision Reference Manual was located or checked. Pin multiplexing is a silicon mask property, not a document-editing one, and no revision-history entry in either document (datasheet's own changelog was searched) mentions a pinout or signal-mux change for the S32K148 specifically — but this is **INFERRED**, not independently confirmed against a Rev.-15-class Reference Manual, because none was found. If a newer RM surfaces before schematic capture, diff its `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx` against the one this document used before trusting a pin unchanged.

Package pin budget, checked rather than assumed: the embedded table lists 157 `PTxx` port pins across the whole S32K1xx family tree; **128 of them are bonded out on the 144-pin LQFP package** (the rest exist only on the 176-pin part or the BGA). This document commits **39** of those 128 to a signal (37 as first written, plus the two watchdog pins §1.8 adds). That leaves 89 GPIO-capable pins unused — enough margin that the 50 still-unknown OEM connector pins (spec §8) will not run the package out of pins once their function is known, which was an open question memo 05 flagged and this answers: **the 144-pin package was the right headroom call.**

Reserved and excluded from this map, deliberately: `PTA4` / `PTC4` / `PTC5` (JTAG TMS/TCK/TDI) and `PTA5` (dedicated `RESET_b`) are the only four pins on the whole part with a reset default other than high-impedance-disabled (Reference Manual §4.4, Table 4-3 — see §2 below). None of the 94 connector signals are placed on them. `PTB6` / `PTB7` (`XTAL` / `EXTAL`)

are not used by any signal here either, but whether this design needs an external crystal at all is not decided anywhere in the repo — flagged in §5.

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1. The mapping table

Columns: **ECU pin** (Bosch 94-way, per spec §4 / `refs/ecu-pinout-extracted.md`), **Signal**, **Front-end** (from spec §4 / `sim/blocks/`, not re-derived here), `PTxx`, **Peripheral**, **Reset pull** (CONFIRMED from the embedded IO Signal Table — every non-JTAG/RESET pin reads `PE=0, PS=0`, i.e. **disabled, high-impedance, no pull**, matching RM §4.4 Table 4-3's blanket "Others: Disabled, High impedance" row), and a **Note** giving the one-line reason for anything not a free choice.

A pin bearing "—" in the `PTxx` column is not an MCU signal at all — it is a power, ground, or excitation-supply net that terminates at the power tree or the sensor-ground plane, not at a GPIO. Those are listed so that every one of spec §4's 44 confirmed connector pins is accounted for, not silently dropped.

1.1 Power

ECU pin	Signal	Front-end	PTxx	Peripheral	Reset pull	Note
04	`V_V_BAT_1R`, **BOSCH: the board's supply**	5 A fuse → EMI filter → TVS → reverse FET → buck; also the injector feed; and a 120k/10k sense divider	`PTA0`	ADC0_SE0	Hi-Z, no pull	The main relay's contact. Paired with pin 06 on the *other* ADC instance so both rails sample simultaneously
06	`V_V_BAT_2R`, the EGR feed	120k/10k divider	`PTA2`	ADC1_SE0	Hi-Z, no pull	Paired with 04 for simultaneity — a real S32K148 property (two independent SAR ADCs), not decorative
01 / 02	`G_G_BAT1/2`, **BOSCH: power ground**	GND	—	—	—	Closes the "no power ground" item open since 19 Sep
21	**BOSCH:** `I_S_WFLS`, water in fuel	see §1.4				Was "Battery +" from KOEL's diagram. There is no permanent feed

1.2 Analog sensors

ECU pin	Signal	Front-end	PTxx	Peripheral	Reset pull	Note
11	Boost sensor 5V excitation	`5V_SENSOR` distribution	—	—	—	Power net, not an MCU input

ECU pin	Signal	Front-end	`PTxx`	Peripheral	Reset pull	Note
41	Boost pressure	Differential (shared gnd 34)	`PTA1`	ADC0_SE1	Hi-Z, no pull	
79	Boost temperature	Differential or NTC — **U2 open, topology TBD**	`PTA3`	ADC1_SE1	Hi-Z, no pull	One ADC pin serves either candidate front-end in `ntc_frontend.cir`; the pin choice does not depend on U2's answer
34	Sensor ground (boost+coolant)	Instrumentation-amp reference	—	—	—	Analog ground-plane net, not a GPIO
33	Coolant temperature	Differential (shared gnd 34)	`PTA6`	ADC0_SE2	Hi-Z, no pull	
15	EGR 5V excitation	`5V_SENSOR` distribution	—	—	—	Power net
37	EGR position	Differential (shared gnd 36)	`PTA15`	ADC1_SE12	Hi-Z, no pull	
36	Sensor ground (EGR+oil)	Instrumentation-amp reference	—	—	—	Analog ground-plane net
39	Oil pressure 5V excitation	`5V_SENSOR` distribution	—	—	—	Power net
80	Oil pressure	Differential (shared gnd 36)	`PTA7`	ADC0_SE3	Hi-Z, no pull	
32	Rail pressure 5V excitation	`5V_SENSOR` distribution	—	—	—	Power net
35	Rail pressure	Single-ended (dedicated gnd 08)	`PTA16`	ADC1_SE13	Hi-Z, no pull	
08	Sensor ground (rail, dedicated)	—	—	—	—	Analog ground-plane net
09	**BOSCH:** `I_A_SYNC`, GCU synchronisation input	22k/10k divider, 0–10 V → 0–3.125 V, ADC tail	`PTB0`	ADC0_SE4	Hi-Z, no pull	The trip sense's old channel. The GCU's range and meaning are not on the drawing; read, no function yet
—	`5V_SENSOR` rail monitor	Divider → ADC (spec §4's ratiometric-correction channel)	`PTC14`	ADC0_SE12	Hi-Z, no pull	No connector pin — internal PCB net, required by the firmware ratiometric-correction scheme spec §4 already specifies
—	Onboard barometric sensor	New, per spec §7 deviation #2	`PTD18`	ADC1_SE16	Hi-Z, no pull	No connector pin — this is the 8th "sensor input" in memo 05's 14-channel ADC budget

1.3 Speed

ECU pin	Signal	Front-end	PTxx	Peripheral	Reset pull	Note
52 / 74	Crank VR High/Low	Differential adaptive-threshold conditioner (vr_conditioner.cir)	*(conditioner input, no direct MCU pin)*	—	—	The conditioner's zero-cross comparator already collapses this to one digital edge stream — see next row
—	Crank conditioner output	Zero-cross comparator output, ±198 mV hysteresis (hw/speed_inputs.kicad_sch)	PTB2	FTM1_CH0 (input capture)	Hi-Z, no pull	**Forced in kind, not in pin:** whatever pin is chosen must be FTM/capture-capable; PTB2 is a free choice among many valid options. FTM configured for both-edge capture, per RM's CnSC[ELSA:ELSB]
30	Crank sensor ground	Shield termination, ECU end	—	—	—	Ground/shield net
45	Cam Hall 5V excitation	5V_SENSOR distribution	—	—	—	Power net
46	Cam frequency I/P	10 k pull-up to 5V_SENSOR + **16 k to ground**, 22 nF, 3.3 V zener (cam_frontend.cir)	PTB3	FTM1_CH1 (input capture)	Hi-Z, no pull	**CORRECTED 20 Sep 2026** — spec §4's literal "pull-up to 5V_SENSOR " overvoltages the pad. See §4
44	Cam sensor ground	—	—	—	—	Ground net

1.4 Discrete inputs

ECU pin	Signal	Front-end	PTxx	Peripheral	Reset pull	Note
43	Coolant switch	10k pull-up to 5 V + NPN buffer, **inverted**	PTD5	GPIO input	Hi-Z, no pull	Pad LOW = contact OPEN. See discrete_input.cir
23	Droop switch	10k pull-up to 5 V + NPN buffer, **inverted**	PTD7	GPIO input	Hi-Z, no pull	Pad LOW = contact OPEN
21	**BOSCH:** water in fuel, active high	Same NPN buffer, **inverted**, no pull-up (open = no water)	PTD6	GPIO input	Hi-Z, no pull	From the main-relay line through the sensor's switch
14	**BOSCH:** diagnosis request, contact to ground	10k pull-up to 5 V + NPN buffer, **inverted**	PTD8	GPIO input	Hi-Z, no pull	GCU relay contact

ECU pin	Signal	Front-end	PTxx	Peripheral	Reset pull	Note
25	**BOSCH:** stop / emergency switch, contact to ground	10k pull-up to 5 V + NPN buffer, **inverted**	PTD9	GPIO input	Hi-Z, no pull	GCU relay contact; stop on closed, as Bosch draws it (owner, 10 Nov). Also kills the gates in hardware, through Q302 on the kill inverter's gate (decisions log §18)
20	Audio abort (active-high)	10k/4.7k/4.7 µF + NPN buffer, **inverted**	PTD11	GPIO input	Hi-Z, no pull	Pad LOW = input active. Was a 3.0 V zener clamp until 22 Sep 2026
24	Override SS (active-high)	Same NPN buffer, **inverted**	PTD12	GPIO input	Hi-Z, no pull	Pad LOW = input active
71	Ignition (active-high)	Same NPN buffer, **inverted**; **BOSCH:** also gates the wake FET on pin 50	PTD16	GPIO input	Hi-Z, no pull	Pad LOW = input active
—	~~Trip-module sense~~	**Removed 8 Nov 2026:** no Bosch pin. `PTB0` now reads pin 09				The history below is kept

CORRECTION, 20 September 2026 — the trip sense needs an ADC, not a GPIO.

This row originally read `PTD6`, GPIO input, "placeholder pin only — front-end is undesigned". The front-end has since been designed, and designing it invalidated the pin class. `sim/blocks/trip_module_sense.cir` builds a **three-state supervised loop** and checks all three bands with real margin:

State	Node voltage, swept 6–40 V
healthy — contact closed, bleed resistor shorted	2.863 – 2.954 V
tripped — contact open, 1.2 MΩ bleed in circuit	0.310 – 2.068 V
wire fault — nothing reaches the divider	0.000 V

A GPIO reads two states. It separates healthy from not-healthy and nothing else, which throws away exactly the distinction the bleed resistor exists to create. A supervised loop that cannot tell a trip from a cut wire is an unsupervised loop with extra parts in it.

`PTD6` is not ADC-capable — checked against `refs/s32k148-144lqfp-altfn.csv`, and neither is any other pin in §1.4, which is correct for the channels there and wrong for this one. The channel moves to **`PTB0` (ADC0_SE4 / ADC1_SE14)**: free, bonded out on the 144-LQFP at pin 78, `PE=0 / PS=0` at reset. `PTD6` returns to the spare pool.

Committed count is unchanged at 39; the spare GPIO count is unchanged at 89.

Updated 21 September 2026: committed count is now **41** and the spare GPIO count **87**, after `PTB5` and `PTA17` were added for the battery-side high switches (§1.5). **Updated 22 September 2026: 42** committed, **86** spare, after `PTB10` was added for the EGR bridge's shared shutdown (§1.5).

1.5 Outputs — the six gate pins are in §2

ECU pin	Signal	Front-end	`PTxx`	Peripheral	Reset pull	Note
03	Injector HS bank A (cyl 1+3)	**Boost-rail** high-side switch, peak phase	`PTC0`	FTM0_CH0	Hi-Z, no pull	See §2
05	Injector HS bank B (cyl 2)	**Boost-rail** high-side switch, peak phase	`PTC1`	FTM0_CH1	Hi-Z, no pull	See §2
03	Injector HS bank A — **battery side**	**Battery** high-side switch, hold chopping	`PTB5`	FTM0_CH5	Hi-Z, no pull	**ADDED 21 Sep 2026** — see below
05	Injector HS bank B — **battery side**	**Battery** high-side switch, hold chopping	`PTA17`	FTM0_CH6	Hi-Z, no pull	**ADDED 21 Sep 2026** — see below
73	Injector LS cyl 1	Current-controlled low-side	`PTC2`	FTM0_CH2	Hi-Z, no pull	See §2
07	Injector LS cyl 3	Current-controlled low-side	`PTC3`	FTM0_CH3	Hi-Z, no pull	See §2
29	Injector LS cyl 2	Current-controlled low-side	`PTB4`	FTM0_CH4	Hi-Z, no pull	See §2
88	Fuel metering PWM	High-current low-side + current sense	`PTD10`	FTM2_CH0	Hi-Z, no pull	See §2
59	EGR High (bridge IN1)	H-bridge	`PTB8`	FTM3_CH0	Hi-Z, no pull	Not one of the six — see §2's scope note
81	EGR Low (bridge IN2)	H-bridge	`PTB9`	FTM3_CH1	Hi-Z, no pull	Not one of the six
—	EGR bridge enable (shared $\overline{SD}$)	Pulled LOW on the sheet — **coast** at reset	`PTB10`	FTM3_CH2	Hi-Z, no pull	**ADDED 22 Sep 2026.** The bridge is two AUIRS2184S, one input per leg; coast needs both shutdowns asserted, and they default asserted by copper. Same timer as IN1/IN2

ECU pin	Signal	Front-end	`PTx x`	Peripheral	Reset pull	Note
50	Main relay, **BOSCH: the board's own supply**	Low-side FET, a second FET woken from pin 71, **SMAJ48CA to ground, no flyback diode**	`PTD13`	GPIO output	Hi-Z, no pull	Held through the after-run. A diode to VBAT_PROT fed the board through the coil (decisions log §18)
69	Buzzer relay	Low-side FET + flyback	`PTD14`	GPIO output	Hi-Z, no pull	
—	`BOOST_DIS` , the boost converter's stop	100 Ω + BAT54Q into the TPS40210's DIS/EN, beside the 110 V overvoltage zener	`PTD15`	GPIO output	Hi-Z, no pull	**ADDED 12 Nov 2026** (decisions log §19). Hi-Z at reset leaves the boost running, as before. GPIO-HD, so the 4.5 mA it sources is within its drive

ADDED 21 September 2026 — a peak-and-hold stage needs two high-side switches per bank, and this map had allocated one.

The row above originally read "Boosted high-side switch," one pin per bank, with hold current arriving from the battery through a diode-OR. Drawing the sheet accepted that; **choosing a gate driver for it did not.** Three things are wrong with a diode:

1. **It cannot regulate hold.** With the battery on a diode the only switch in the loop is the low side, and when the low side opens the coil's current goes to the recirculation diode and into the 100 V boost rail — a *fast* decay against −87 V, which is turn-off, not chopping. Hold needs a slow freewheel around the coil.
2. **It leaves connector pins 03 and 05 permanently live.** A diode from the battery to a connector pin means a harness short to ground draws current whenever the battery is connected, with no switch anywhere to stop it.
3. **The gate driver cannot be bootstrapped.** A bootstrap capacitor charges when the switch node goes low. A battery diode holds that node at 12.8 V, so it never does.

What replaces it is what `injector_turnoff.cir` already models: two high-side switches onto a common bank node, plus `D_fw` from **ground** to that node — the freewheel path whose necessity that block's header spends a paragraph on. Hold chopping then happens on the high side, the node swings to −0.7 V every off-time, and the bootstrap charges.

Both new pins are FTM0 channels, like the five injector pins already allocated, so the whole stage stays on one timer and its edges stay phase-locked to each other. `PTB5` is a GPIO-HD (high-drive) pad; both are `PE = 0, PS = 0` at reset, confirmed against the same IO Signal Table §2 uses — so they inherit §2's finding exactly.

1.6 Current-sense channels (internal, no connector pin)

Memo 05's ADC budget names "4 current-sense returns from the injector, metering and EGR drivers." These are shunts on the driver stage, not connector signals, so they have no ECU pin — but they need an MCU `PTxx` and are placed here for completeness.

Associated signal	`PTxx`	Peripheral	Reset pull	Note
Injector bank A current sense (pins 03/73)	`PTC15`	ADC0_SE13	Hi-Z, no pull	**INFERRED split**: assumes one shunt per high-side bank, not per low-side cylinder. Memo 12 says "per channel or per bank" without deciding; this map picks per-bank because that is what makes the 4-channel figure in memo 05/§5 add up (2 banks + metering + EGR = 4). Three-channel per-cylinder sensing would need a 5th ADC pin — trivially available, but not what the existing budget assumes. **SETTLED 21 Sep 2026, when the injector sheet was drawn** — `hw/injector.kicad_sch` puts the shunts **low-side and ground-referenced**, one per bank, in the shared source return. Cylinders 1 and 3 fire 240 crank degrees apart and never overlap, so one shunt reads whichever is conducting — the same fact that lets the high side be bank-shared. A high-side shunt would read the same current and need an amplifier whose common-mode range reaches 100 V. Per-bank confirmed, 5 mΩ.
Injector bank B current sense (pin 05, low side 29)	`PTD19`	ADC1_SE17	Hi-Z, no pull	Paired with bank A on the other ADC instance for simultaneous sampling
Metering unit current sense (pin 88)	`PTC16`	ADC0_SE14	Hi-Z, no pull	
EGR bridge current sense (pins 59/81)	`PTD22`	ADC1_SE18	Hi-Z, no pull	

1.7 CAN

BOSCH (8 Nov 2026): 87/86 is Bosch's CAN 1, the application line at 1000 kbit/s for flashing and UDS (the service link); 55/77 is CAN 2, the PT line at 250 kbit/s for J1939 and OBD. The channel numbers below are this project's, kept to avoid renaming nets.

Both channels are real (spec §2.5). The S32K148 has three FlexCAN instances; this design needs two, exactly as memo 05 notes ("1 spare instance"). CAN0/CAN1 are used, CAN2 is left spare — a free choice among three equally-valid instances, not a forced one.

ECU pin	Signal	`PTxx`	Peripheral	Reset pull	Note
55	CAN channel 1, H	*(via transceiver)*	—	—	Bus-side signal; transceiver CANH pin, not an MCU pin
77	CAN channel 1, L	*(via transceiver)*	—	—	Transceiver CANL pin
—	CAN channel 1, MCU side	`PTE5`	FlexCAN0_TX	Hi-Z, no pull	To transceiver 1 TXD
—	CAN channel 1, MCU side	`PTE4`	FlexCAN0_RX	Hi-Z, no pull	From transceiver 1 RXD
87	CAN channel 2, H	*(via transceiver)*	—	—	Transceiver 2 CANH
86	CAN channel 2, L	*(via transceiver)*	—	—	Transceiver 2 CANL
—	CAN channel 2, MCU side	`PTC7`	FlexCAN1_TX	Hi-Z, no pull	To transceiver 2 TXD
—	CAN channel 2, MCU side	`PTC6`	FlexCAN1_RX	Hi-Z, no pull	From transceiver 2 RXD
—	CAN channel 3 (spare)	*(unassigned)*	FlexCAN2	—	Instance exists, no pins committed

1.8 SPI / supervisor interface

No SPI-dependent part is currently decided anywhere in the repo. The handbook's speculative peripheral table (`docs/handbook/sensors.html`) once counted "1 SPI channel for the injector driver," but memo 12 (19 Sep 2026) decided discrete FETs with a conventional gate-driver IC for the injectors and a smart low-side switch for the metering unit — neither is a microcode/SPI-programmed part in memo 12's recommendation. So: **no SPI pin is committed in this map.** `LPSPI1` (`PTB14` SCK / `PTB15` SIN / `PTB16` SOUT / `PTB17` PCS3) is entirely free and is the natural reservation if a future EEPROM, external diagnostic port, or SPI-configured driver IC is added — but forcing a pin assignment ahead of a part choice would be exactly the kind of guess this project's own discipline rules out.

The supervisor interface is not a data bus at all — that is the point of it. `sim/blocks/supervisor.cir` and `docs/research/11-supervisor-and-fail-safe-outputs.md` specify a TPS3850-class windowed supervisor whose `RESET` output drives the S32K148's own `RESET_b` pin **and**, through one inverting stage, a dedicated kill transistor at each of the six gate nodes — entirely in analog/discrete hardware, with no MCU involvement in that path by design (that is what lets it protect against a *hung* MCU).

CORRECTED 20 Sep 2026, during schematic capture. This section previously said "the only MCU pin in this interface is `RESET_b`", and that was wrong in a way that mattered: the *kill* path needs no MCU pin,

but the **watchdog** does. A window watchdog is not a passive monitor — the MCU has to present a falling edge on `WDI` inside the window, every window, or the part asserts and resets the board on a timer. A supervisor topology with no pin to kick it is a supervisor that guarantees a reset loop.

Two pins are therefore added here, both taken from the 91 spare GPIO the map leaves free, both `GPIO-HD`, both `PE=0 / PS=0` at reset (confirmed from the same embedded IO Signal Table as every other row in this document):

Signal	`PTxx`	Package pin	Peripheral	Reset pull	Note
`RESET_b`	`PTA5`	141	dedicated	**weak pull-up**	**Forced — this is the only `RESET_b` pin on the part.** Dedicated function, not muxed. Reset default per RM Table 4-3: weak pull-up enabled (the one deliberate exception among the pins this document touches)
`WDT_KICK`	`PTE0`	138	GPIO output	Hi-Z, no pull	MCU → TPS3850 `WDI`. Free choice of pin; what is *not* free is that some pin must do this
`WDT_FAULT`	`PTE1`	137	GPIO input	Hi-Z, no pull	TPS3850 `WDO` → MCU, a live input. Since 19 Oct 2026 `WDO` also pulls `RESET` low through D302 (BAT54Q), so a watchdog timeout resets the MCU and kills the gates. The MCU is held in reset for as long as `WDO` is low, so by the time firmware runs it reads high again: it no longer tells a watchdog timeout from a rail excursion

That takes the committed count from 37 signals to **39**, and the spare GPIO from 91 to **89**.

The supervisor part is also now chosen, which §1.8 previously left as a class: **TPS3850G33** (`TPS3850G33DRCT`, VSON-10). The variant matters and the reasoning is in `hw/gen_symbols.py` beside the symbol — briefly, the datasheet's own nomenclature table gives `G` = thresholds at ±4 % of nominal and `H` = ±7 %, and with the part's ±0.8 % accuracy that puts G33's worst-case undervoltage trip at **3.143 V** against H33's **3.044 V**. The supervisor has to assert *before* the S32K148's own LVD (3.0 V maximum), so G33 clears it by 143 mV and H33 by 44 mV — close enough that the two could fire in either order, and a supervisor that might lose the race to the thing it supervises is not doing its job. The cost is a tighter window the 3V3 rail has to live inside (3.143–3.459 V), which is a constraint this map now places on the rails design.

---

2. The gate pins and their reset behaviour — six, now eight

Spec §4's binding requirement (added 18 Sep 2026) says every driver gate on pins **88, 73/07/29, 03/05** must be held off by hardware during reset, because S32K148 GPIOs are high-impedance out of reset. The requirement explicitly left its own per-pin detail open:

> "the physical `PTxx` assignment for pins 88, 73/07/29 and 03/05 is a phase-2 task (§5), so the per-pin reset pull defaults cannot yet be looked up."

That is now answered, pin by pin, from the manufacturer's own IO Signal Table (not inferred):

ECU pin	`PTxx`	`PE` (pull enable)	`PS` (pull select)	Reset state
88	`PTD10`	0	0	Disabled, high-impedance
73	`PTC2`	0	0	Disabled, high-impedance
07	`PTC3`	0	0	Disabled, high-impedance
29	`PTB4`	0	0	Disabled, high-impedance
03	`PTC0`	0	0	Disabled, high-impedance
05	`PTC1`	0	0	Disabled, high-impedance
03 (battery side)	`PTB5`	0	0	Disabled, high-impedance — **added 21 Sep 2026**
05 (battery side)	`PTA17`	0	0	Disabled, high-impedance — **added 21 Sep 2026**

CONFIRMED against both the Reference Manual's own prose (§4.4, Table 4-3: "Others: Disabled — High impedance — `ibe=0, obe=0, pue=0, pus=0`") and the per-pin `PE` / `PS` / `Reset` fields in the embedded `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx`, read directly for each of these six pins.

The two added on 21 September 2026 read the same way, from the same table — and they matter differently, because they are *high-side* gates. `hw/injector.kicad_sch` found that a ground-referenced kill clamp cannot service a high-side gate at all: the source swings to the boost rail, so a pulldown to ground would hold `Vgs` at minus the bank voltage and a kill FET referenced to ground cannot short a gate floating 100 V up. Those four gates get 470 Ω across **gate and source**, and the kill path moves into the driver's own shutdown input. Their reset state is still worth recording, and it is still high-impedance.

None of the six defaults to a pull that works against the fail-safe requirement — and none defaults to a pull that helps it either. There is no pull at all. The only four pins on the entire part with any reset-time pull are `PTA4` / `PTC4` / `PTC5` (JTAG) and `PTA5` (`RESET_b`) — none of which are used for gate duty. So the exposure spec §4 and memo 11 already reasoned about from first principles (a floating gate, brought up by leakage or Miller coupling, with nothing internal to the MCU resisting it) is **exactly the confirmed condition for all six pins**, not a worst-case assumption that turned out pessimistic or optimistic. This closes the open item cleanly: the hardware kill-clamp `supervisor.cir`

already validated does not change, because it was designed to not depend on this answer, and the answer it did not depend on turned out to match the design's own worst case precisely.

Scope note on EGR (pins 59/81). Spec §4's fail-safe list names only the six pins above — not the EGR bridge. A concurrently-updated file, `sim/blocks/egr_hbridge.cir`, argues EGR's own safe state is different in kind (coast, not off, so a spring return can close the valve) and is drafting its own gate-kill treatment. That file is outside this task's scope and was not modified here; it is noted so the two documents are not read as disagreeing by omission.

---

3. Datasheet numbers reconfirmed, not just cited

Spec §4 and memo 11 state the S32K148's brownout thresholds as VLVR 2.50–2.7 V, VLVD 2.8–3.0 V, and a 2.97 V PLL-engaged correctness-guarantee floor. Re-read directly from the Rev. 15 datasheet's own electrical characteristics tables this session (not taken on trust from the earlier memo):

- `VLVR` (LVR falling threshold, RUN/HSRUN/STOP): **2.50 / 2.58 / 2.7 V** (min/typ/max)
- `VLVD` (falling low-voltage detect threshold): **2.8 / 2.875 / 3.0 V**
- "S32K148 will operate from 2.7 V when executing from internal FIRC. When the PLL is engaged S32K148 is guaranteed to operate from **2.97 V**." — quoted directly, twice, at different points in the datasheet (electrical characteristics footnote and I/O operating-voltage table, both consistent).

CONFIRMED, exact match to spec/memo 11's figures. No arithmetic correction needed here — recorded because the task asked for datasheet numbers to be checked, not carried forward on trust a second time.

---

4. What did not fit, and what is INFERRED

Cam signal front-end (pin 46) — a finding, not a fix. Spec §4 describes pin 46's front-end as "Pull-up to `5V_SENSOR` → timer capture," i.e. the Hall sensor's open-drain output pulled directly to the 5 V sensor rail, into an MCU timer-capture pin. Read literally, that puts a 5 V logic swing on a 3.3 V-rail MCU pin — over the S32K148's absolute maximum (`VDD + 0.3 V`, roughly 3.6 V at `VDD = 3.3 V`, per the datasheet's own operating-voltage figures in §3 above). **No `sim/blocks/*.cir` file simulates a cam front-end at all** (the built-block list in `docs/status.md` §1.3 has `vr_conditioner.cir` for crank; there is no cam equivalent) —

so this was never checked, and the pin map surfaces it rather than papering over it with a pin choice. `PTB3` is assigned as the *capture pin*, but the front-end feeding it needs a level clamp or divider before the pad, matching the discipline already applied to pins 04/06 (battery sense) and 20/24/71 (discrete inputs). **This should become a `sim/blocks/cam_frontend.cir`-class check before the cam sheet is drawn**, not solved here.

RESOLVED 20 September 2026 — the check was written and the sheet now carries the fix. `sim/blocks/cam_frontend.cir` exists and runs in the suite, and `hw/speed_inputs.kicad_sch` draws the front-end it validates: the 10 k pull-up to `5V_SENSOR` stays (the open-collector sensor needs it), and a 16 k to ground turns it into the same 10 k/16 k divider the ratiometric sensor channels already use — 5.00 V becomes 3.077 V, and 4.50 V (`sensor_rail.cir`'s own fault floor for a healthy group) becomes 2.769 V, both inside `[2.31 V logic-high floor, 3.6 V Vih abs-max]`. It costs one resistor. The §1.3 row for pin 46 is updated accordingly.

Injector current-sense split (§1.6) is INFERRED, not confirmed. Memo 12 leaves "per channel or per bank" open; this map assumes per-bank (2 channels) because that is the only split consistent with memo 05's stated "4 current-sense returns" total once metering (1) and EGR (1) are subtracted. A per-cylinder scheme (3 channels) is equally buildable and would just need one more free ADC1 pin — **settle when the injector driver sheet's shunt placement is actually drawn.**

RM revision currency (§0) is INFERRED. Rev. 13 (04/2020) was the newest S32K148 IO signal table located; the datasheet elsewhere cited is Rev. 15 (2026). No pin-mux errata were found in either document's revision history, but a newer Reference Manual was not checked because none was found. Low risk (pinout is a silicon property), not zero.

Crystal reservation (`PTB6` / `PTB7`) is an open question, not a finding. Nothing in the repo states whether this design runs off the internal FIRC or an external crystal. Neither pin is used by any signal in this map, so there is no conflict either way — but if CAN bit-timing accuracy (a real automotive concern, not evaluated here) ends up requiring a crystal reference, those two pins should be reserved before anything else claims them.

SPI (§1.8): no pin committed, for the reason stated there — no SPI-dependent part is decided. `LPSPI1` is free and is the natural landing spot.

---

5. Open questions

1. ~~Cam front-end overvoltage (§4)~~ — **CLOSED 20 September 2026.** `cam_frontend.cir` was written, the divider it validates is drawn on `hw/speed_inputs.kicad_sch`, and the §1.3 row now states the real front-end rather than spec §4's literal one.
2. ~~Injector current-sense placement (§1.6)~~ — **CLOSED 21 September 2026.** Per-bank, low-side, ground-referenced, 5 mΩ. Drawn on `hw/injector.kicad_sch`.
3. **Crystal vs FIRC (§4)** — decides whether `PTB6` / `PTB7` are reserved.
4. **U6 (pin 09) and pins 01/02** — still open per spec §8/§4; no MCU pin assigned, consistent with "nothing guessed into copper."
5. **The 50 unconfirmed OEM connector pins (spec §8)** — cannot be placed on `PTxx` without a function to place. 91 of the 128 GPIO-capable pins on the 144-pin package remain uncommitted, which is comfortable headroom once they are known. **Sharpened 21 September 2026, by drawing the connector sheet: two of those 50 are not headroom, they are blockers.** The wiring diagram shows pin 21 as the battery positive feed and shows *no battery negative anywhere* — its own scope note says it omits "power grounds, unused pins and internal-only pins" — so `GND`, the net every other sheet returns to and the one the injector low sides put 18 A into, has no pin. And `TRIP_LOOP`, the supervised trip-module loop `trip_module_sense.cir` designs, is a **new** signal that is not on the OEM diagram at all, so it needs one of the 50 as well. Neither can be chosen: a wrong ground pin is not a rework, it is a harness that has to be remade. Both wait on the connector part number and the OEM pin list, or on continuity measurement at the machine.
6. **RM revision currency (§0)** — worth a diff against a newer Reference Manual if one surfaces before layout.

BOM requirements

What the schematic has to buy, and the simulation that says so.

Every row here exists because a block in `sim/blocks/` failed without it. The rule this file follows is the same one the netlists follow: **a figure nobody has read in a datasheet does not become a design number.** So these are requirements stated as specifiable, purchasable properties — tempco bands, ESR bands, on-resistance ceilings — not manufacturer part numbers. Naming a part is a separate step, and it needs a retrieved datasheet.

Where a part number *does* appear below, it is because it was checked.

Input protection

`TVS` — transient suppressor, battery input

SMCJ43CA. Bidirectional, 43 V standoff, 1500 W package.

Both halves of that were forced by simulation, not chosen:

- **43 V standoff** is the smallest standard bin above the 40 V clamped load dump. Below it the part conducts during a *normal* ISO 7637-2 pulse 5b event rather than a fault — which is what `load_dump.cir` was failing on, not "the TVS is too small."
- **1500 W (SMC) package, not 600 W (SMB).** At the SMB size, pulse 2a clamps at 88.6 V against the buck's 100 V absolute maximum — 11% margin. The SMC part has roughly half the dynamic resistance at the same standoff and clamps at 73.3 V, 1.37×.

Checked by `load_dump.cir` (0.000 A, 0.000 J through the TVS) and `transient_clamp.cir`.

`NEG-CLAMP` — negative active clamp, buck VIN

N-channel MOSFET, **Vds ≥ 80 V (100 V class), Rds(on) ≤ 10 mΩ at Vgs = 4.5 V**, source to ground, drain to VIN, driven by a comparator sensing VIN against a −0.1 V threshold. **In parallel with** the 100 V / 20 A clamp Schottky, which stays as the instant passive backstop.

> **CORRECTED 22 September 2026 — both parts were under-rated for the rail they sit on.** This tag originally read **Vds ≥ 40 V** and the backstop Schottky was **60 V**. Both have a terminal on `VBAT_PROT`, and both were sized for the *negative* pulses they exist to catch — but pulse 2a drives the **same node to +73.3 V**, where a 40 V FET is 0.55× rated and a 60 V Schottky 0.82×. The 40 V FET had zero margin on the 40 V load dump

> alone. `transient_clamp.cir` already carried the rule "every part on VBAT_PROT must clear 73.3 V" — as a principle, not a list, so nothing compared these two against it. It is now a table in `run_sim.py` : every part with a terminal on this rail, its rating, and the margin, and a part below the line fails.

The requirement is really an *effective clamp impedance*, and it is worth stating as one: the LM5164's VIN absolute maximum is −0.3 V, pulse 1 delivers 150 V through 10 Ω, so

```
Rclamp ≤ 0.3 V / 15 A = 20 mΩ, junction drop included
```

No junction device meets that, and the reason is structural rather than a sizing miss. Decomposing the Schottky's own drop at 15 A:

term	value
junction, `N·Vt·ln(I/Is)`	0.367 V
bulk, `I·Rs`	0.180 V
total	**0.547 V**

Only the bulk term falls when parts are paralleled; the junction term is logarithmic in current and barely moves. Four Schottkys in parallel still leave 0.375 V. A FET in its ohmic region has no junction offset at all — 15 A through 5 mΩ is 75 mV.

Checked by `negative_pulses.cir` : buck VIN reaches −0.068 V on pulse 1 (4.4× inside the rating) and −0.027 V on pulse 3a (11×), against −0.528 V and −0.393 V with the Schottky alone.

Closed 22 September 2026 — and closing it found a latch the block had been reporting as a pass.

The control is **TLV3201-Q1** (50 ns max, push-pull, no phase inversion beyond the rails) into **UCC27517A-Q1** (4 A, 23 ns max at 12 V, output held low in UVLO). The comparator's 52 mA would take about half a microsecond to charge the clamp FET's gate; the driver takes nanoseconds.

The clamp is a low-side ideal diode, and it has to let go. The model engaged it near VIN = +0.08 V and released it only above +0.12 V. At the end of every negative pulse the source returns to +13.5 V and drives current back through the still-on clamp — 1.35 A through 5 mΩ is 6.7 mV, never reaching the release threshold — so **the clamp latched on and shorted the source for the rest of the run.** `run_sim.py` checked only the clamp branch's *minimum* and reported it RESOLVED. A recovery check added the same day failed against it on both pulse 1 and pulse 3a. With a real battery instead of the test generator's 10 Ω, that is a dead short after the first negative pulse.

The thresholds are now **engage below −100 mV, release above −6.8 mV**, from the drawn network and checked as arithmetic against it:

sense node	10 kΩ from VIN, 499 kΩ from 5V_MAIN: 0.9804·VIN + 98.2 mV, so −100 mV arrives as 0 V and no negative reference is needed
reference	10 kΩ to ground, **536 kΩ** from the output: 0 V low, 91.6 mV high
release, worst case	−6.8 mV nominal, **−1.8 mV** with the comparator's full 5 mV offset — still below zero, so it cannot hold on against reverse current (523 kΩ, the first value, sat at +0.5 mV)

The window before it engages is now a number, and it is inside the rating. With the ~100 ns delay (50 + 23 ns max plus gate charge) and a 10 ns timestep cap, VIN crosses −100 mV, keeps falling ~90 ns, and bottoms at **−0.177 V** — inside the LM5164's −0.3 V by **1.70×**. The Schottky backstop barely conducts. The previous −0.068 V / 4.4× was optimistic by 2.6×: at the default step, ngspice committed the hysteretic switch on a rejected trial step and hid the dip.

Supplies are the power-up argument. The comparator runs from 5V_MAIN, which is dead whenever VIN is too low to run the buck — so it cannot hold the clamp on and short an incoming battery. The driver runs from 12V_GATE, held up ~400 ms by the boost reservoir, so the clamp keeps its gate through a whole pulse.

Clamp Schottky

100 V / 20 A class — it was 60 V; see the correction above. It carries what the TVS used to: **13.58 A peak, 14.5 mJ** on pulse 1, measured at its own ammeter in `negative_pulses.cir` rather than inferred from 150 V / 10 Ω.

A 100 V Schottky has a higher forward drop than the 60 V part `negative_pulses.cir` models (`SCHN`, BV = 60 V). That drop only matters for the comparator's propagation delay — the window before the active clamp engages — and is re-measured when the comparator is chosen.

Reverse-battery FET

P-channel, |Vds| ≥ 80 V, Rds(on) ≤ 40 mΩ at Vgs = −4.5 V, drain to the battery side, source to the load, with a **passive gate network**: 47 kΩ gate-to-ground and a 12 V zener gate-to-source.

> **CORRECTED 22 September 2026 — the sheet had drawn an N-FET, and `reverse_battery.cir` has always modelled a P-FET.** Its header says "a P-FET held on by a controller that watches the polarity", and its switch model closes on positive polarity. Found while choosing the controller the N-FET would have needed.

No controller IC checked can sit on this node. It swings from **+73.3 V** (pulse 2a, after the TVS) to **−47.8 V** (pulse 1, the bidirectional TVS's negative clamp — bidirectional precisely so a reversed battery does not forward-bias it). That is a 121 V span:

Controller	Rated	Fails by
LM74700-Q1	ANODE −65 V to **+65 V**	8.3 V on the positive side
LM5050-1-Q1	IN **−0.3 V** to +100 V	47.5 V on the negative side — and IN draws 320 µA, so a series resistor to protect it corrupts its 20 mV sense

A P-FET needs no controller: `Vgs = -V(load)`, so it is on whenever the load side is positive and off when it is pulled to zero. That **is** the block's switch model, with no IC ratings on a 121 V swing to violate.

What it gives up, recorded rather than hidden. An ideal-diode controller blocks reverse *current*; this blocks reverse *polarity*. On a dropout that *shorts* the input, bulk capacitance discharges backward through the FET. The spec carries no hold-up requirement, and the negative clamp is already sized for this FET staying on through a whole pulse — its turn-off via 47 kΩ and `Ciss` takes tens of microseconds.

The 40 mΩ ceiling gives 0.2 V at 5 A against the block's 0.3 V requirement. The model's 8 mΩ is its header's own "mid-range for a 100 V P-channel" and is optimistic; the ceiling is what `run_sim.py` checks.

Power

`PDN-BULK` — 3V3_MCU bulk capacitor

> **Chosen, 8 October 2026 (decisions log §9):** KEMET T599D157M006ATE025 (150 µF polymer tantalum, 25 mΩ max) **in series with R306, 22 mΩ 1 %.** A datasheet gives an ESR maximum and never a minimum, so the resistor is the band's floor: the pair is 21.8–72.2 mΩ across temperature and end of life. A sweep shows the target holds from about 17 to 95 mΩ, so the 50 mΩ ceiling below was conservative. `mcu_pdn.cir` checks both ends at 96 µF: 71.9 and 86.9 mΩ.

Polymer aluminium electrolytic or polymer tantalum, 150 µF, specified by an **ESR band of 20–50 mΩ across −40…+125 °C** — not by a 25 °C typical figure.

Three things about that are load-bearing:

- **The part class changes, not the value.** A wet aluminium or tantalum electrolyte loses conductivity as it freezes, roughly 4× ESR at −40 °C, which puts the PDN peak at 197 mΩ against a 100 mΩ target. A solid polymer electrolyte has no such mechanism.

- **The cold number is what gets specified.** A supplier's 25 °C ESR does not answer this block's question, and asking for it is how the original corner got in.
- **A band, not a ceiling.** Bulk ESR is also what damps the regulator-against-bulk resonance. At 20 mΩ the peak climbs back to 92 mΩ, 8 mΩ under target. "ESR ≤ 50 mΩ" would be satisfied by a 5 mΩ part that fails this block.

The 100 mΩ target is derived — 50 mV allowed ripple over a 0.5 A transient step — not a round convention. Checked by `mcu_pdn.cir`.

`SENSOR-PTC` — 5 V sensor rail, one per sensor group

> **Superseded and built, 8 October 2026 (decisions log §4, §7).** `sensor_rail.cir` checks it: VSNS peaks at 22.3 V on pulse 2a against the trackers' 45 V. The OV window is 40.52–42.08 V with 0.1 % resistors; 1 % would put the bottom at 39.8 V, inside the 40 V dump. Inrush is 0.16 A into 47 µF. Cranking headroom is 0.46 V. The controller's VS and CS pins see 4.9 V/µs through 100 Ω + 22 nF, against their 60 V/µs limit. The hold-up, C211, was chosen 8 October (Chemi-Con HXC, decisions log §9). The TPS48110-Q1 (DGX0019A) and TPS7B4253-Q1 (DDA0008J) land patterns are drawn in `hw/gen_footprints.py` from TI's own layouts. The 10 µF 50 V output capacitors are 2220, the only case KEMET makes at 50 V; they face the harness, so buy them with flexible termination.
>
> The PTCs are replaced by one **TPS7B4253-Q1** tracking regulator per group (TI SLVSCP3D, rev. Nov 2025), following 5V_MAIN to ±4 mV and limiting at 301–520 mA. The three sit behind one **TPS48110-Q1** overvoltage cut-off (TI SLUSEE5E, rev. Apr 2026, AEC-Q100 Grade 1) switching an IPD90N10S4-06, with 47 µF of hold-up after it. The cut-off opens for pulse 2a (73.3 V; the trackers are rated 45 V) and stays closed through the 40 V load dump. The text below is the PTC requirement it replaces, kept for the record.

Resettable PTC, hold current ≥ 200 mA, **R25 = 3.0 Ω nominal**.

Sized against the ceiling rather than guessed. The polyfuse has to be what clears a shorted harness, not the 5 V regulator's own current limit, so fault current must stay under 3.0 A:

```
I < 3.0 A  ⟹  Rsrc + Rptc > 5/3 = 1.667 Ω
corner = R25 × 0.8 (cold) × 0.7 (-30% spread) = 0.56 × R25
⟹  R25 > 2.78 Ω
```

At the same stacked −40 °C + −30% corner the specified part draws 2.88 A, 0.12 A under the ceiling. The larger series resistance was re-checked against the rest of the block rather than assumed harmless: healthy-

group voltage holds at 4.69 V against a 4.5 V floor, and nominal fault current stays inside the 0.3–3.0 A window the polyfuse needs to trip on.

Checked by `sensor_rail.cir`.

Drivers

`MU-ISENSE` — metering unit current sense, ECU pin 88 return

50 mΩ shunt on the low side, into **either** a current-sense amplifier feeding an ADC channel **or** a gate driver with an integrated current-regulation comparator.

The amplifier is not optional: 50 mΩ at the 0.675 A setpoint is 33.75 mV, too small to take straight to the S32K148's ADC at useful resolution.

This one is not a part-selection fix at all — it is an architecture change, and it is the only failing corner in the suite that no component choice could have answered. The coil is copper, copper moves 0.39%/°C, and over −40…+125 °C its resistance goes 7.39 → 13.82 Ω. At a fixed 50% duty the mean current follows it: **0.862 A cold against 0.462 A hot**, +27% and −32% around the setpoint. There is no better resistor to buy.

A hysteretic current regulator holds 0.679 / 0.679 / 0.683 A at −40 / 25 / +125 °C. **Closed loop only works while the regulator has duty left** — at the hot corner the coil can draw at most 13.5/13.82 = 0.977 A at 100% duty, so the setpoint sits at 69% of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop behaviour rather than reporting anything.

Checked by `metering_unit_pwm.cir`.

`EGR-DRIVER` — EGR H-bridge

A bridge driver with **internal DIR/PWM decode** — one logic input pair that cannot express both legs of a diagonal conducting at once.

Two logic lines wired straight to two diagonal switch pairs short the bridge: with both inputs asserted it draws **270 A**. Under a decoded interface the same input combination draws **4.35 A**, the motor's own stall current. The fault state stops existing rather than being avoided by firmware convention.

Current capability: headroom over a 2.6–6.4 A stall.

> **Drawn 22 September 2026 — discrete, two `AUIRS2184STR`.** No integrated bridge clears the rail: DRV8873-Q1, the candidate, has a 40 V VM absolute maximum against 73.3 V, and the class caps at 40–45 V. So four **100 V** N-FETs (≤ 20 mΩ, headroom over the 6.4 A stall), and the decode this tag requires comes from the **driver's structure** rather than a logic

> gate: AUIRS2184S has **one input per leg** — IN high puts HO on, IN low puts LO on — with cross-conduction prevention and ~400 ns deadtime. Each leg's high and low are exclusive by construction, so the 270 A state, both inputs high, becomes a **high-side brake**.
>
> | EGR_EN | IN1 IN2 | bridge |
> |---|---|---|
> | high | 1 0 / 0 1 | forward / reverse |
> | high | 1 1 / 0 0 | high-side / low-side brake |
> | **low** | any | **COAST** |
>
> **Coast is asserted by copper.** Both drivers' SD̄ share `EGR_SD`, pulled down 10 kΩ, so with `PTB10` Hi-Z at reset both legs are off and the return spring closes the valve. A kill FET on the same node gives coast on `GATE_KILL` without the MCU. One 20 mΩ shunt in the shared low-side return feeds an INA181A1-Q1: 2.56 V at a 6.4 A stall.
>
> Pinout and logic are from the IR2184 datasheet (same die) and the AUIRS2181(4)S family table; **the automotive part's own datasheet was not retrieved** — confirm against it at sourcing.

Checked by `egr_hbridge.cir`, which keeps the naive wiring as pinned evidence of why the requirement exists.

`GATE-PULLDOWN` — fail-safe gate pulldowns, six driver gates

470 Ω, gate to source, on every driver gate.

The largest standard value that still holds a gate below Vgs(th) = 1.0 V against Miller coupling at Crss = 500 pF. The commonly-reached-for 10 kΩ fails even at the *low* end of the memo's own Crss envelope — 1.155 V at 50 pF already exceeds the threshold.

Cost, stated because it is real: 21.3 mA per gate held at 10 V, 1.28 W across all six.

Checked by `supervisor.cir`.

Signal

`CAN-TERM` — CAN split termination, Rah / Ral

60.0 Ω **thin-film**, initial tolerance **±0.1%** or better, tempco **±50 ppm/°C** or better.

Ordinary standard-grade thick-film runs 100–200 ppm/°C. At 200 ppm/°C over the full −40/+125 °C span, both resistors shifting together, the differential impedance reaches 123.95 Ω against J1939's own 120 Ω — and 120 Ω here is what the standard specifies, not a round convention, so missing it is a real finding.

The contrast is a part class, with no meaningful cost or availability penalty at this quantity. With both mechanisms stacked the adverse way (60 × 1.00825 × 1.001 = 60.5555 Ω) the specified part lands at 121.10 Ω, inside the ±2 Ω window with 0.90 Ω to spare. Checked at the **worst case**, not the nominal — so passing means the requirement is sufficient, not merely that it helps.

Checked by `can_termination.cir`.

`VR-CLAMP` — crank input clamp diodes, four off

Reverse leakage **≤1 µA at 125 °C**. Forward drop is explicitly *not* a requirement.

This inverts the rule every earlier clamp on this board was chosen by. The negative-clamp Schottky's 0.547 V at 15 A decomposes into a junction term and a bulk term, and that decomposition is what ruled out paralleling junction devices for the power path. These four diodes conduct only on a fault, into a 4.7 kΩ series resistor that is already doing the limiting — what matters is what they do when they are **off**.

A reverse-biased diode's leakage flows out of the 4.489 kΩ Thévenin at the comparator input and appears there as an offset voltage, against a hysteresis window of ±198 mV:

Class	Leakage at 125 °C	Offset	Against the window
BAT54S-class Schottky	~200 µA (≈2 µA at 25 °C, ~100× hotter)	0.90 V	**4.5× the whole window** — latches
BAV199-class low-leakage silicon	<1 µA (5 nA max at 25 °C)	<4.5 mV	2% of the window

The two diodes on a leg are reverse-biased symmetrically by `VR_BIAS`, and the legs are matched, so leakages partly cancel and what survives is partly common-mode. That is not a thing to defend a speed input on, so the requirement is stated instead.

`vr_conditioner.cir` does **not** model this — its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Found while drawing `hw/speed_inputs.kicad_sch`.

The same tag covers the **differential TVS across the pair**: bidirectional, standoff ≥48 V. The standoff has to clear the *signal*, not just the rail — VR amplitude is proportional to speed, about 20 V peak at 1500 rpm, so a 24 V part would clip a runaway engine's own crank signal at the one moment the ECU most needs to keep counting teeth. 48 V clears 3600 rpm.

`VR-COMP` — crank zero-cross comparator

Single **3.3 V** supply, **push-pull** rail-to-rail output, propagation delay **≤1 µs**. Input common-mode range: **mid-rail only**.

> **Chosen 22 September 2026:** `TLV3201AQDCKRQ1` — the same part as the negative clamp's comparator, so one part number serves both. 2.7–5.5 V, push-pull, input range (VEE) − 0.2 V to (VCC) + 0.2 V, **50 ns max** against the 1 µs requirement, 5 mV offset against a ±198 mV window.

Three of those four requirements are unusually easy, deliberately:

- The output swing *is* the hysteresis window (3.3 V × 4489/(4489+33000) = 0.395 V), so an open-drain part would make the window depend on its pull-up value and its V_{OL}.
- Biasing the floating VR coil to `VR_BIAS` means the inputs never leave mid-rail, so **no common-mode range including ground is needed** — the usual hard constraint on a zero-cross detector, avoided by biasing the sensor rather than level-shifting it.
- 1 µs is 0.009 crank degrees at 1500 rpm on a 60-tooth wheel.
- Input offset voltage and its tempco are non-issues: 10 mV is 5% of the window.

Checked by `vr_conditioner.cir`, which establishes the topology (a fixed 5 V threshold produces edges at 1500 rpm and **none** at cranking speed) and the ±0.2 V window the E24 network above lands on.

`BOOST-BLEED` — injector boost reservoir bleed

22 kΩ, 1 W, across the reservoir. Since 8 October 2026 the reservoir is two 82 µF / 80 V cans in series (41 µF, decisions log §9), and the bleed is **2 × 11 kΩ 1 W**, one across each can (R401, R435). Same total current and power; the split also balances the stack, at 52.4 V per can worst case.

The charger regulates from **one side only**: `boost_converter.cir`'s `Bchg` adds 50 mA while the rail is below setpoint and adds nothing above it. Nothing else in the circuit can remove charge, so any excursion above 100 V is permanent. `injector_turnoff.cir` measured the mechanism that produces one — a hold-current cutoff, whose energy came from the **battery** through the high side's diode-OR and so has no boost-side draw to net against — and its RESULT NOTE asked for this resistor by name.

Sizing is not a duty-cycle argument; it is "one event is gone before the next cylinder fires":

event charge	10 A decaying to 0 over 20 µs = 100 µC
on 41 µF	2.45 V of rail rise
above setpoint	charger is **off**, so the whole 100 V / 22 kΩ = 4.55 mA discharges it
time to clear	**21.4 ms**, against **26.67 ms** between cylinders at 1500 rpm
cost	**455 mW**, burned continuously whenever the rail is up

Checked by `boost_converter.cir` arrangement C, which runs two identical reservoirs — one bare, one bled — through three such events at the real cylinder spacing. Bare, the rail stacks to 107.4 V and stays there.

A comparator-gated shunt costs nothing at idle and is the alternative. It is more parts, and it is a part nobody has chosen.

Over a complete injection event there is no pumping at all — the peak phase draws 342 µC from the reservoir and end-of-injection recirculation returns roughly 100 µC of it. This resistor is not there for normal running. It is there because one-sided regulation has no answer for an excursion from *any* cause.

`INJ-RECIRC` — injector recirculation diodes, five off

150 V, 20 A, ultrafast. Three on the low sides (one per cylinder, into the boost reservoir) and two on the high sides (the battery diode-OR).

Research memo 10 names the class. Not a Zener active clamp across the FET — Nexperia AN50003 rates repetitive active clamp reliability- "Questionable" for gate-oxide wear-out. Not a bare freewheel to battery, which is too slow: the same problem the boost rail exists to fix on the opening side.

Sized by the pulse, not the average. `injector_turnoff.cir` measures 28.7 W instantaneous during turn-off against roughly 3.6 W averaged over all nine events per second — a factor of eight. The average would never have sized this part.

`FEED-EMI` — a filter on each high-current feed

Each of VBAT_INJ (pin 04) and VBAT_EGR (pin 06) gets the same filter (decisions log §10, `feed_emi.cir`): - the existing 2 × 2.2 µF ceramics on the FET side; - a **10 µH** inductor, Coilcraft XAL1510-103 (L103, L104); - on the load side, **two 82 µF / 80 V HXC** bulk cans; - **two dampers**, each an HXC in series with **0.68 Ω 1206**.

Check	Result
Class 5, worst band, 25 kHz chopping/PWM	INJ 17.7 dB under (12.2 aged), EGR 9.3 dB under
As drawn before	INJ 33 dB over, EGR 48 dB over
Filter output impedance peak, aged	0.64 Ω, against −0.94 Ω of constant-power load at 9 V: 1.47×
Pulse 2a across the inductor	48.7 V, against its 55 V rating
Pulse 2a at the loads	28.7 V (VBAT_PROT: 73.3 V)
EGR stall ripple per bulk can	1.6 A, against 1.68 A at 25 kHz: 1.05× at the 125 °C rating

Firmware requirement: the injector hold chopping and the EGR PWM run at **≤ 25 kHz**. At 50 kHz the EGR feed is 1 dB over.

Not covered: common mode, which needs a layout, as for `emi_filter`.
The limits are research memo 06's secondary sources.

Chosen parts

These are the entries that have moved from "a class" to "a part number", each against a datasheet that was actually retrieved and read. They are listed separately from the requirements above because they are answers, not constraints.

Function	Part	Why this one
Buck pre-regulator	**LM5164DDAR**	100 V input, 1 A, 6–100 V range. See below — it replaces a 60 V part that can no longer survive this input.
3V3 regulator	**TLV76733QWDRBRQ1**	1% over load *and* temperature. The supervisor makes that load-bearing.
Supervisor	**TPS3850G33DRCT**	±4% window variant, so its undervoltage trip clears the MCU's own LVD by 143 mV rather than 44 mV.
Injector boost controller	**TPS40210QDGQRQ1**	Chosen on its **200 ns maximum off-time** — the one spec that makes 94% duty reachable at the 6 V cranking corner. See below.
Gate drivers, all eight gates	**AUIRS2181STR** ×5	**No cross-conduction interlock** — the injector's high and low switches are in series with the coil and must both be on. See below.
Difference amplifiers	**INA592IDR** ×5	Native G = ½, 88 dB CMRR min against a 60 dB floor the discrete network could not reach. **Not AEC-Q100** — see below.
Barometric sensor	**MPXHZ6115A6T1**	Spec §7 deviation 2, on-PCB behind a vented port. AEC-Q100 **not stated** in its datasheet — see below.
EGR bridge drivers	**AUIRS2184STR** ×2	One input per leg: each leg's high and low exclusive by construction. See `EGR-DRIVER`.
Negative-clamp control	**TLV3201AQDCKRQ1** + **UCC27517AQDBVRQ1**	A low-side ideal diode: engage below −100 mV, release above −6.8 mV. See `NEG-CLAMP`.
Current-sense amplifiers	**INA181A1QDBVRQ1** ×3	VCM reaches −0.2 V, so it sits across a ground-referenced shunt; supplied from 3V3_MCU so its output is ADC-safe by construction.
Kill FETs and reset inverter	**2N7002BK** ×10	60 V, ±20 V Vgs, Vth 1.1–2.1 V, 2 Ω max at Vgs 5 V against the 5 Ω `supervisor.cir` assumes. AEC-Q101. Needs `GATE-KILL-CLAMP`.
GATE_KILL clamp	**BZX84-C12-Q**	11.4–12.7 V at 5 mA, +10 mV/K max. AEC-Q101. The plain BZX84 series is no longer automotive-qualified; the `-Q` suffix is.
Relay FETs	**PMV280ENEA** ×3	100 V, ±20 V Vgs, 432 mΩ max at Vgs 4.5 V. AEC-Q101. Driven through a buffer — see `RELAY-GATE`. The third is the main relay's **wake** FET, gated from pin 71 through 10k / 22k and a BZX84-C12-Q (Bosch pinout, 8 Nov).
Relay gate buffers	**CAHCT1G125QDBVRQ1** ×2	SN74AHCT1G125-Q1 on 5V_MAIN: VIH 2 V reads a 3.3 V GPIO, VOH 4.4 V min drives the FET. AEC-Q100.
Reverse-battery FET	**SQD50P08-25L**	−80 V P-channel, ±20 V Vgs, 31 mΩ max at −4.5 V against a 40 mΩ ceiling. AEC-Q101, TO-252.

Function	Part	Why this one
Pin 21 filter inductor L101	**7447709220**	Würth WE-PD 1210, 22 µH ±20%, IR 5.3 A, Isat 6.5 A typ, RDC 28 mΩ max, SRF 10 MHz typ, 120 V, AEC-Q200 Grade 1, −40 to +125 °C. Datasheet rev 003.001 (2024-05-08).
Pin 21 second stage L102	**744314047**	Würth WE-HCI 7050, 0.47 µH ±20%, IR 18 A, Isat 9 A typ, RDC 1.35 mΩ, SRF 235 MHz typ, AEC-Q200 Grade 1, −40 to +150 °C. Datasheet rev 004.001 (2025-02-06).
Pin 21 fuse F101	**0437005.WRA**	Littelfuse 437A, 1206 fast-acting, 5 A, 63 V DC, 50 A interrupting, melting I^2t 2.30 A^2s, 15.9 mΩ, AEC-Q200. Datasheet revised 09/08/21. 79 % re-rating at 105 °C read off its curve.
Sensor trackers, ×3	**TPS7B4253QDDARQ1**	TI TPS7B4253-Q1, SLVSCP3D (rev. Nov 2025): tracks ADJ to ±4 mV, 300 mA, limit 301–520 mA, dropout 520 mV max at 200 mA, IN −40…45 V, OUT −1…45 V. AEC-Q100. DDA PowerPAD-8; land pattern drawn from TI DDA0008J (`gen_footprints.py`).
Sensor-supply OV cut-off	**TPS48110AQDGXRQ1**	TI TPS4811-Q1, SLUSEE5E (rev. Apr 2026): 100 V abs max, OV 1.16–1.20 V rising, 2.6–4 µs to gate off, 4 A pull-down, 60 V/µs limit on drain-side pins. AEC-Q100 Grade 1. DGX VSSOP-19; land pattern drawn from TI DGX0019A (`gen_footprints.py`). Chosen over LTC4368 (not AEC-Q100): decisions log §7.
Boost inductor L401	**7447709151**	Würth WE-PD 1210, 150 µH ±20%, IR 2.1 A, Isat 2.7 A (10 %) / 3.3 A (30 %) typ, RDC 200 mΩ max, 120 V, AEC-Q200 Grade 1. Datasheet rev 003.001 (2024-05-08). Decisions log §8.
Boost VDD reference	**BZG03C39-HM3**	Vishay BZG03C-M, 37–41 V at 10 mA, ZZ 40 Ω, +0.06–0.11 %/K, AEC-Q101, SMA. Doc 86574 Rev. 1.0.
Reverse-battery FETs, pins 04/06	**SQM120P10-10m1L** ×2	−100 V P-channel, ±20 V Vgs, 10.1 mΩ max at −10 V, 20.5 mΩ at 175 °C, 15.0 mΩ at −4.5 V, RthJA 40 °C/W. AEC-Q101, TO-263. Vishay doc 76943 Rev. A, retrieved 8 Oct 2026. `POWER-PATH` .
100 V N-channel	**IPD90N10S4-06** ×8	Negative clamp, metering, four EGR, two injector battery switches (60 V class needed; one part number instead of two). 6.7 mΩ max at 10 V, ±20 V Vgs, AEC-Q101, TO-252. The logic-level sibling `-S4L-06` was rejected for its ±16 V Vgs: every one of these gates is driven from 12V_GATE.
Injector 150 V FETs	**SQM85N15-19** ×5	Two boost-side high sides, three low sides. 19 mΩ max at 10 V — the model's 20 mΩ. ±20 V Vgs, AEC-Q101, TO-263. Qg 120 nC max sets the bootstrap droop: 126 mV.
Boost switch	**DMN15H310SK3**	150 V, Vth 3 V max, 350 mΩ max at Vgs **4 V** — specified below the 5.7 V it gets while cranking. Qg 8.7 nC typ. AEC-Q101, TO-252. Cranking duty with its drop, hot: 94.8% against 97.0% available.
Injector recirculation, blocking, freewheel	**VS-16EDH02HM3** ×7	200 V, 16 A avg, 250 A surge, trr 32 ns, AEC-Q101. SMPD (TO-263AC): land pattern drawn from Vishay doc 95604, tab = cathode (`gen_footprints.py`). Average current per diode is tens of mA; the rating is for the 18 A peaks.
Boost diode	**VS-2EMH02HM3**	200 V, 2 A, trr 25 ns, AEC-Q101, SMA.
Bootstrap and relay flyback diodes	**VS-1EMH02HM3** ×8	200 V, 1 A, 50 A surge, AEC-Q101, SMA. The relay flybacks needed only 100 V; one part number instead of two.
Metering freewheel	**VS-3ECH02HM3**	200 V, 3 A, 130 A surge, AEC-Q101, SMC.

Function	Part	Why this one
Negative-clamp backstop Schottky	**SS20PH102HM3**	100 V, 20 A, 310 A surge, AEC-Q101, TO-277A. Stock footprint puts the cathode on pad 3; the project's `TO-277A_K1_A2` renumbers it (`hw/gen_footprints.py`).
Battery TVS	**SMCJ43CA-HM3**	43 V standoff, 47.8–52.8 V breakdown, 69.4 V clamping at 21.6 A, 1500 W, AEC-Q101.
Crank-pair TVS	**SMAJ48CA-HE3**	48 V standoff, 53.3–58.9 V breakdown, 400 W, AEC-Q101. One more clamps pin 50's drain in place of its flyback diode (`main_relay.cir` , 8 Nov).
12 V clamps	**BZX84-C12-Q** ×3, **BZX84-C13-Q**	GATE_KILL, reverse-FET gate, 12V_GATE reference. SOT-23 with pin 3 as the cathode, so the project's `SOT-23_Zener_K1_A2` footprint is used. The GATE_KILL zener's first footprint, the stock SOT-23, had its anode where the symbol puts the cathode.
VDD clamp	**BZG03C36-HM3**	See `BOOST-VDD-CLAMP` : it replaced a 43 V clamp that did not clamp.
Buck inductor	**Würth 7447714330**	WE-PD 1050, 33 µH, Isat 2.9 A typ against a 1.18 A worst peak (1.72× after a −30% hot derate), 79 mΩ, AEC-Q200 Grade 1. This closes the Isat gap `buck_preregulator.cir` left open. Land pattern drawn from the Würth drawing (`gen_footprints.py`).
12V_GATE pass / limit	**PZTA42-Q**, **PMBT3904-Q**	See `GATE-RAIL` . PMBT3904-Q is SOT-23 B-E-C, so it uses the `Q_NPN_BEC` symbol.
3V3_MCU bulk (C313)	**T599D157M006ATE025** + 22 mΩ 1 % (R306)	KEMET polymer tantalum, 150 µF, 6.3 V (5.4 V at 125 °C), 25 mΩ max, −55/+150 °C, AEC-Q200, 7343-31. The resistor is `PDN-BULK` 's ESR floor; see decisions log §9.
Sensor hold-up (C211)	**HHXC630ARA470MHA0G**	Chemi-Con HXC hybrid polymer, 47 µF 63 V, 8 × 10 mm SMD, 40 mΩ max, Z(−55)/Z(20) ≤ 2 at 100 kHz, 4000 h at 125 °C, AEC-Q200. 1.58× over the 40 V load dump; 26.3 µF tolerance-and-aged corner in `sensor_rail.cir` .
Feed filter inductors (L103, L104)	**XAL1510-103MED** ×2	Coilcraft XAL1510, 10 µH, 9.0 mΩ max, Isat 26.3 A, Irms 16 A (20 °C rise), SRF 10.1 MHz, 55 V across the winding, AEC-Q200 Grade 1, 165 °C part. Doc 947. Stock KiCad footprint. See `FEED-EMI` .
Feed filter bulk and dampers (C111–C118)	**HHXC800ARA820MJC5G** ×8	The boost reservoir's part, reused; with R110–R113, 0.68 Ω 1206. See `FEED-EMI` .
Boost reservoir (C401, C420)	**HHXC800ARA820MJC5G** ×2, in series	Chemi-Con HXC hybrid polymer, 82 µF 80 V, 10 × 12.5 mm SMD, 28 mΩ max, cold ratio ≤ 2, 2.1 A rms at 125 °C, AEC-Q200. 41 µF / 160 V stack, balanced by the 2 × 11 kΩ bleed. Chosen over one 160 V wet can because no part above 100 V publishes a cold ESR (decisions log §9).

`PASSIVE-PACKAGES` — 188 resistors and capacitors, sized by what is across them

`hw/assign_passives.py` gives every chip resistor and capacitor its package, and `hw/check_netlist.py` fails if a part is added or re-valued without re-running it.

The voltage comes first. Each net carries a continuous and a peak voltage: battery nets are 24 V jump-start and 73.3 V pulse 2a, the boost rail is 104 V. The voltage across a part is the worse of its two nets, with

two exceptions. Inside a bootstrap domain, a part sees only the 13.6 V gate swing. And `PAIR_V` overrides the few series strings and AC-only parts where "the worse net" over-states what is across them. Both are keyed by net names, which is why several internal nodes were given labels: the auto-names `Net-(R407-Pad2)` move whenever a part is added ahead of them.

- **Resistors: Vishay D/CRCW e3 ratings** (AEC-Q200, standard mode). The peak voltage must clear Umax by the rail table's 1.09×. The continuous V^2/R must fit the power rating derated to 0.588× at a 105 °C board. Nothing is smaller than 0603.
- **Capacitors: KEMET C1023 X7R AUTO** (AEC-Q200). The rating must clear the peak voltage and 1.5× the continuous one, and nothing up to 1 µF is rated below 25 V. The part picks the smallest case that KEMET's waterfall table actually lists at that rating. The table is parsed by x-coordinate into `hw/lib/kemet_x7r_auto.json` (`hw/tools/parse_kemet_waterfall.py`), because the PDF's text layout compresses the data columns and misreads them.

Result: 121 × 0603, 36 × 0805, 5 × 1206, 7 × 1210 and 12 × 2010, plus the stated 2512 parts and the shunts.

Things this pass found:

- **4.7 µF at 100 V is not in the automotive X7R range**; 1210 stops at 50 V. Three places needed it: the EMI filter's C1, the buck input and the boost input. Each is now two 2.2 µF, 100 V, 1210 parts. `emi_filter.cir` models the pair as 4.4 µF with ESR and ESL halved. Its cold corner is 0.6 dB worse, and the check was re-pinned.
- **The twelve 470 Ω gate pulldowns land in 2010.** At 100% duty (metering, EGR) 12 V across 470 Ω is 0.31 W per gate, as well as 25 mA of driver current. They were sized for Crss = 500 pF, and the worst chosen FET is 300 pF, so 1 kΩ may now serve. That would need a re-run of `supervisor.cir` claim 6 before it could be claimed.
- **DC bias on bulk MLCCs is not modelled.** Two parts are affected: the 47 µF 10 V buck output and the 10 µF 6.3 V on 3V3_MCU. Both lose a large part of their capacitance at their working voltage. The 1.5× rule does not capture that.

`POWER-PATH` — what the input filter and reverse FET actually carry (CLOSED 8 October 2026)

Built. Pin 04 now feeds `VBAT_INJ` and pin 06 feeds `VBAT_EGR`. Each feed runs through its own SMCJ43CA TVS, an **SQM120P10-10m1L** reverse P-FET (passive gate, as Q1), an SS20PH102HM3 Schottky backstop and 2 × 2.2 µF 100 V. The reasons for the split are in `docs/decisions-log.md` §2a. `power_path.cir` checks these figures:

	EGR, pin 06	Injector, pin 04
Load	6.4 A stall, continuous	10 A hold 3 ms / 26.67 ms + 0.98 A boost
FET drop at 175 °C RDS(on)	0.131 V	0.225 V peak
FET dissipation	0.84 W	0.30 W (3.8 A RMS)
Tj at a 105 °C board	139 °C	117 °C
Tj while cranking (Vgs −6 V)	155 °C (RDS(on) inferred)	—

The netlist gate checks that each rail leaves its own FET, reaches its own loads, and that VBAT_PROT no longer reaches either power stage.

What this leaves open:

- **L101, L102 and F101 are chosen (8 October 2026)** for what pin 21 now carries, stated as an upper bound of **2.95 A**. The bound is 1.75 A for the buck, from the LM5164's IPEAK1 maximum, since its average input current cannot exceed its peak limit. Then 0.30 A for two relay coils at 24 V, 0.86 A for the metering freewheel, and 0.04 A for the boost VDD feed. `run_sim.py` checks all of it under `emi_filter`, `transient_clamp` and `negative_pulses`:

Part	Choice	Against the 2.95 A bound
L101	Würth **7447709220**, 22 µH WE-PD	Isat 6.5 A typ (2.2×); 117 °C on a 105 °C board, under its 125 °C maximum
L102	Würth **744314047**, 0.47 µH WE-HCI 7050	Isat 9 A typ (3.05×)
F101	Littelfuse **0437005.WRA**, 5 A 437A	3.16 A continuous at 105 °C (1.07×); pulse 1 I^2t 16 % of melting, pulse 2a 2.5 %

Thin margins, stated:

- **F101's 1.07×** is against a bound the buck cannot actually sustain; the estimated real load is 0.25–0.5 A at the input. 5 A is also the largest 437A rating still rated for 63 V, since 7 A is 35 V, under the 40 V load dump. If the fuse opens on overload, the board loses power, which is the fail-safe direction.
- **The 20 % pulse ceiling** for the repeated pulse 1 is a fuse-fatigue rule of thumb, not a retrieved figure. Littelfuse pulse-cycling data is open.
- **The EMI filter's margins at 150 kHz and 400 kHz fell** to 0.56 dB and 0.73 dB once the model carried the real inductors' RDC and self-resonance (they were 1.17 and 1.55 dB with ideal inductors). Lf is 0.47 µH, not the 0.5 µH it was modelled at. The gates are still the unvalidated placeholders `emi_filter` already flags.
- **Conducted emissions from the EGR PWM and the injector chopping** no longer pass through the pin-21 filter. **Closed 8 October 2026, see `FEED-EMI`:** as drawn the feeds were 33 and 48 dB over Class 5, and each now has its own filter.

- Anything placed on VBAT_INJ or VBAT_EGR has to tolerate about −0.5 V on pulse 1. The Schottky holds the rail there, and there is no active clamp on these rails. The present loads are switch and bridge FETs, a diode and an inductor, and all of them do.

Bosch pinout, 8 November 2026 (decisions log §18): pin 21 is the water-in-fuel input, not a battery feed. The electronics chain (F101, L101, L102, the TVS, Q101) now takes **pin 04**, beside the injector feed. Its load is unchanged, so the parts are too.

The original requirement follows.

Every high-current load on the board draws from VBAT_PROT: the injector battery switches (10 A hold), the EGR bridge (up to 6.4 A stall) and the relay coils. VBAT_PROT is fed through connector pin 21, then the 20 A fuse, the 22 µH + 0.5 µH EMI filter, the TVS and the reverse FET. Two blocks sized parts on that path, and neither was written with this load in mind:

- `reverse_battery.cir` states its claim at **5 A** (0.2 V).
- `emi_filter.cir` was written for the buck's 400 kHz noise and never states a DC current.

A 22 µH inductor that does not saturate at 15–20 A is a large part.

The connector has two more battery feeds, pins **04** and **06** (`V_BAT_1R/2R` , each fused 10 A in the loom). The spec says they "feed high-current loads [?]", and the board uses them only as ADC senses. Moving the injector and EGR supply onto 04/06, with its own protection, would leave pin 21 feeding the electronics: a few amps through the filter. Keeping everything on pin 21 means resizing the filter, the fuse and the reverse FET for the full load, and relying on one contact to carry it.

This is an architecture decision. L101, L102 and F101 stay unchosen until it is made.

`BOOST-SLOPE` — current-mode stability constrains the boost inductor (CLOSED 8 October 2026)

Built (decisions log §2b, §8). The TPS40210-Q1 runs from its own output through a follower: 34.3–45.0 V, with a 39 V zener, PZTA42-Q, 2 × 3.3 kΩ and a PMBT3904-Q limiter, as 12V_GATE. The battery feeds it only at start-up, through 180 Ω, the 36 V clamp and a Schottky. **L401 is Würth 7447709151, 150 µH**, and the sense resistor is **62 mΩ**. `run_sim boost_converter` checks, at the worst corner (VDD 34.3 V, 120 µH, oscillator −20 %, 6 V in):

Check	Result
Equation 9, TI's 80 % ceiling	69.4 mΩ: 62 mΩ inside by 1.12×
Equation 49, no false trip	65.1 mΩ: inside by 1.05×
Current limit	1.94–2.90 A: 7 % past L401's 10 % saturation, 14 % under its 30 % point, in a fault only
Start-up VDD at 6 V	4.77 V against the 4.5 V threshold
Standing load on the boost	17.6 mA; peak-phase recovery 10.6 ms against 26.7 ms

Stated rather than hidden: at twice the switch FET's typical gate charge (no maximum is given), start-up margin is 17 mV. Before the output passes about 36 V, the ramp is still the battery's.

The original finding follows.

TPS40210-Q1 applies a fixed internal ramp of fSW × VDD / 20. TI's Equation 9 (SLVS861F §7.3.3) therefore caps the sense resistor at `VDD·L·fSW / (60·(VOUT + VD - VIN))`, and TI recommends 80% of that.

	330 µH, VDD ≈ battery
13.5 V in	ceiling 99.7 mΩ (at 80%): 82 mΩ **passes**
6 V cranking, VDD 5.16 V	ceiling 44.9 mΩ (at 100%): 82 mΩ **fails**, 1.83× over

The failure is subharmonic oscillation while cranking. TI says the voltage loop still regulates, but ripple grows. The inductor makes it worse. A 330 µH part rated to saturate above the 2.2 A current limit is not a part that exists in SMD: the search found 330 µH parts at 0.2–0.4 A. The buildable direction is a smaller inductor, which lowers the ceiling further: 100 µH allows 30 mΩ.

TI's own remedy is to feed VDD from the output side for a steeper ramp. VDD's 52 V absolute maximum rules out connecting it to the 100 V rail directly; it would need a clamped feed, such as the 36 V zener that is already there. Either that, or accept subharmonic operation while cranking. **L401 stays unchosen** until one of those is decided. The candidate at 100 µH is Würth **7447709101**: WE-PD 1260, Isat 3.1 A typ, 110 mΩ max, AEC-Q200 Grade 1.

`SENSOR-PTC` — real PTCs are not specified the way this was written (OPEN)

The requirement below asks for R25 = 3.0 Ω nominal with a −30% spread. PTC datasheets do not work that way. Bourns MF-USMF020 (AEC-Q200) is specified as a **range, 0.40–5.00 Ω**. At its 0.40 Ω minimum it contributes 0.14× of the 2.78 Ω the fault-current ceiling needs. Its hold current also derates from 0.20 A at 23 °C to **0.10 A at 85 °C**, half the requirement.

The fault-current ceiling has to come from a fixed series resistor, or from the regulator's own limit, with the PTC only doing the tripping. Its hold current then has to be chosen at the hot corner.

`LOWV-CLAMP` — the 3.0 V and 3.3 V input clamps

Analog and cam inputs: closed 22 September 2026. The ten 3.3 V zeners became **BAV199-Q**. That part is a low-leakage double diode, 80 nA max at 150 °C, AEC-Q101. Its common pin sits on the signal, K2 goes to 3V3_MCU and A1 to ground. It is drawn as one 3-pin part, so a pair is one package; the speed input's split halves were redrawn the same way.

These inputs clamp only on a fault: a channel's normal maximum is 3.08 V, below the rail. A rail clamp in that position brings a problem the zener had hidden. The BAV199-Q's upper diode has a high Vf, which is the price of low leakage: 0.9 V at 1 mA. On a 40 V harness short, the MCU pad's own ESD diode sits in parallel with it and would carry most of the 3.6 mA fault current. That is over the S32K148's ±3 mA injection limit. So every clamped channel now has **1 kΩ between the clamp node and the pad**. The fault node rides at 4.33 V, the pad at 3.93 V, and **0.40 mA** is injected. That is checked in `sensor_ratiometric` and `cam_frontend`, which now model the diode pair, the 1 kΩ resistor and the pad's ESD diode.

The same models exposed what the zener had been doing in normal operation. At a 5.25 V rail the cam input sits at 3.23 V, and the 3.3 V zener model was loading it by **84 mV**. The old check had pinned that loading as the expected value.

Discrete inputs: closed 22 September 2026, by removing the clamp. The five switch inputs are now NPN buffers.

- **Circuit:** 10 kΩ input, 4.7 kΩ and 4.7 µF from base to ground, a BAV199-Q lower diode on the base, and a PMBT3904-Q whose collector is the MCU pad. The pad has a 10 kΩ pull-up to 3V3_MCU.
- **Why:** a 6–40 V input cannot be divided into logic range without clamping continuously. That rules out a zener (soft knee), a rail clamp (the pad parks at 3.9 V) and a shunt reference (its ~35 µA bias, TI SNOS633N Fig. 5-4). The buffer needs no clamp: the pad sees only 0–3.3 V.
- **Checked at the worst corner**, per the owner's rule that safety and reliability come first:
- hFE 30 (the datasheet minimum of 60, halved for −40 °C) still pulls the pad to 0.03 V from a 6 V input.
- A typical part at 125 °C does not turn on below a 1.3 V input.
- On −24 V reverse battery the base is held at −1.05 V, against a 6 V VEBO.
- One pad edge for a 5-chatter bounce.

- **Wetting current:** 1.1 mA through a closed contact at 12 V. The 100 kΩ first drawn gave 0.11 mA, at which non-gold contacts oxidise open in the field.
- **Logic is inverted**: pad LOW = input active, or contact open on the two contact-to-ground channels. `docs/pinmap.md` §1 records it.
- **Failure modes are for the firmware spec.** A transistor that fails open leaves the pad stuck high, and a C-E short leaves it stuck low. One polarity cannot make every channel fail safe.

Residual: the trip loop keeps its clamped analog front end, because its healthy band *is* the 3.0 V clamp level. It is also still blocked on a connector pin.

`GATE-KILL-CLAMP` — 12 V zener on GATE_KILL

`GATE_KILL` is pulled up by 47 kΩ to **VBAT_PROT**, not 3V3_MCU, and for a good reason: a pullup on the rail being supervised has no supply during the brownout it exists to catch. But nothing on the net draws DC: it is nine FET gates and a reset inverter's drain. So with the inverter off, the net follows the battery rail, and that rail reaches **73.3 V** on pulse 2a. That is 3.7× the ±20 V Vgs rating of any small-signal FET.

This was not caught while the kill path was drawn. It surfaced on 22 Sep 2026, when the first kill FET was being chosen and its Vgs rating had to be checked against what drives its gate.

A BZX84-C12-Q from GATE_KILL to ground holds the net at 12.7 V max, or 13.7 V at 125 °C. Its current at the pulse peak is 1.3 mA, and the 47k dissipates 78 mW for the pulse's duration. Below 12 V, which includes cranking, the zener is out of circuit and GATE_KILL is the battery, which is what the pullup is there for. The numbers are checked in `run_sim` `transient_clamp`.

`RELAY-GATE` — the relay FETs' gates at 5 V, through a buffer

The relay sheet drove each FET's gate straight from a 3.3 V GPIO, and its field note said "fully enhanced at 3.0 V". No part backed that claim. The 100 V automotive small-signal FETs checked were PMV280ENEA, DMN10H220LQ and BSS123N. Every one of them specifies Rds(on) at Vgs = 4.5 V and no lower, and their Vth runs up to 2.5–2.8 V. That leaves 0.5–0.8 V of overdrive, less when cold, and the datasheets guarantee nothing there. The 100 V class is not optional here, because the drain is on VBAT_PROT through the coil.

So an SN74AHCT1G125-Q1 on 5V_MAIN sits between the pin and the gate. Its TTL inputs read 3.3 V logic (VIH 2 V at VCC 4.5–5.5 V), and its output is 4.4 V minimum. OE is tied low. A 100 k pulldown on the input

keeps the relay off while the GPIO is Hi-Z from reset. The existing 10 k pulldown on the gate covers the case where 5V_MAIN is absent. At 84 mA of coil current, the PMV280ENEA drops 36 mV.

`INA592IDR` — the difference amplifiers, and a stronger argument

TI SBOS914F, October 2018, revised April 2021. Retrieved and text-extracted. Five, one per differential channel on `sensors_analog`.

Gain	native **G = ½** (±IN inputs, SENSE tied to OUT)
CMRR	**88 dB minimum**
Gain error	±0.03 % max
Supply	4.5–36 V single; runs from 5V_MAIN
Output	within 220 mV of each rail
Input	common mode to 3·(V+) − 2·REF = 15 V at G = ½; only absolute maximum is 10 mA, and the ~12 kΩ internal network holds a 40 V harness short to about 3.3 mA

Why a difference amplifier and not an op-amp. The requirement was ≥ 60 dB CMRR, the floor `sensor_differential.cir` tried. The drawn design — an op-amp with a discrete 26k/16k network — could not reach it: 0.1 % discrete resistors give about 48 dB. The network has to be matched on-die. G = ½ also puts a 0.5–4.5 V sensor at 0.25–2.25 V, straight into the ADC.

And the argument got stronger with a real number. The block had modelled 80 dB, marked INFERRED, at gain 16/26. With the chosen part's **minimum** 88 dB at G = ½ — and both paths' errors referred to the sensor, since they no longer share a gain — differential now beats single-ended by **394×** on a 1 Ω corroded return, up from 90×.

A **1 kΩ** now sits between each amplifier and its ADC tail: the amplifier runs from 5 V and can rail at 4.78 V on a harness fault, where the 3.3 V zener at the pin would otherwise take the amplifier's full short-circuit current. With 1 kΩ it takes 1.5 mA.

Not AEC-Q100, and nothing better was found. INA2132 was the dual alternative, but its "G = ½" (datasheet Figure 12) is a single-ended attenuator; its matched network only gives a *difference* G = 1, which pushes a 4.5 V sensor past its 4 V swing on 5 V. No qualified difference amplifier with a differential G = ½ turned up.

`MPXHZ6115A6T1` — the onboard barometric sensor

NXP (Freescale) MPXA6115A series data sheet, Rev. 7.3, April 2015, retrieved and text-extracted. Spec §7 deviation 2 adds it on purpose: "on-PCB behind a vented port; costs nothing in harness terms and improves fuelling correction."

Range	15–115 kPa absolute
Transfer	`VOUT = VS × (0.009·P - 0.095)` , P in kPa
Supply	4.75–5.25 V, 10 mA max
Accuracy	±1.5 % of span, 0–85 °C; compensated −40…125 °C
Package	super small outline, case 98ARH99066A — a footprint KiCad ships

Supplied from **5V_MAIN**, which `SENSOR_5V_MON` already measures — so the reading is ratiometric-correctable with no new channel. Through the same 10k/16k divider every 5 V channel on the sheet uses: **2.51 V at sea level** (101.3 kPa), 1.92 V at 2000 m. No zener at the MCU pin: every other channel on that sheet arrives from the harness; this never leaves the board.

AEC-Q100 is not stated in the retrieved datasheet. Infineon's KP236 is listed automotive-qualified with a 40–115 kPa range and would be the choice, but its datasheet did not resolve by any route this session — Mouser timed out twice, Infineon's own host redirected. It is a **footprint change**, not a drop-in, so the choice is recorded rather than guessed: confirm qualification at sourcing, or retrieve the KP236 sheet and swap.

The **vented port** is a mechanical requirement the schematic cannot carry.

`INA181A1QDBVRQ1` — one part, one gain, three channels

TI SLYS018F, April 2018, revised October 2024. Retrieved and text-extracted.

Common mode	**−0.2 V to 26 V** — covers a ground-referenced shunt
Gain	20 V/V (A1), ±1 % max over temperature
Offset	**±150 µV max at VCM = 0 V**, 1 µV/°C max drift
Bandwidth	350 kHz (A1)
Supply	2.7–5.5 V; output swings to VS − 0.02 V
CMRR	84 dB min

Channel	Shunt	Full scale	Where it sits
Injector bank A → `PTC15`	5 mΩ	1.80 V at 18 A peak, 1.00 V at 10 A hold	ADC0
Injector bank B → `PTD19`	5 mΩ	same	ADC1 — simultaneous with bank A
Metering → `PTC16`	50 mΩ	675 mV at the 0.675 A setpoint, 844 counts	ADC0

The gain is chosen for the injector. 1.80 V at the peak leaves headroom to 3.28 V, so a fault current up to **32.8 A is measured rather than clipped** — which is what an overcurrent check needs. Offset is worth 30 mA of

injector current (0.17 % of peak) and 3 mA of metering current (0.44 % of setpoint). Bandwidth resolves the 38 µs ramp to peak with about 0.45 µs of lag.

Supplied from 3V3_MCU, which makes the output ADC-safe by construction. It cannot swing above VS − 0.02 V, so these three channels need no 3.3 V zener at the MCU pin — unlike every channel on `sensors_analog`, which arrives from the harness.

This closes `MU-ISENSE`'s open half. That tag specified the 50 mΩ shunt and said "the amplifier below is not optional"; this is the amplifier. `IN-` goes to GND, and the tag's layout requirement stands: a Kelvin tap at the shunt's own ground pad, not a via into the pour.

`AUIRS2181STR` — chosen because it has *no* interlock

Infineon (International Rectifier) AUIRS2181(4)S, datasheet dated 10 January 2014, still hosted by Infineon. Retrieved and text-extracted this session. **Its first listed typical application is "Piezo / common rail Injection."** Confirm lifecycle status at sourcing — the datasheet is old even though the part is still listed.

Offset	VB **625 V** absolute maximum; VS operational **−5 V to +600 V**
Supply	VCC 10–20 V; UVLO+ 8.0 / 8.9 / **9.8 V** max
Logic	VIH 2.5 V min, VIL 0.8 V max — direct from a 3.3 V MCU pin
Output	1.9 A / 2.3 A typ, **1.4 A / 1.8 A min** short-circuit at 15 V
Delay	160 / 200 ns typ, 270 / 330 ns max
Interlock	**None** on 2181/21814 (the 2183/2184 have it)
Package	SOIC-8, AEC-Q100

Three properties chose it, each against a part that failed on it:

Property	This part	Why it matters	Failed it
No cross-conduction interlock	none	In a half bridge, HO and LO together is shoot-through and every driver prevents it. Here the "high" and "low" switches are in **series with the coil** and must **both** be on for it to conduct at all	UCC27282-Q1, UCC27712-Q1 — could not fire an injector
Offset rating	625 V abs max, **5.4×** margin	Bank node ~104.1 V worst case (41 µF reservoir), VB 12 V above it = 116.2 V	UCC27211A-Q1 — HB 120 V abs max, **3.8 V** of headroom
Negative VS	operational to −5 V, clears by **3.7 V**	`D_fw` drops the bank node 1.0–1.3 V below ground on **every** hold off-time	UCC27211A-Q1 — HS DC min −1 V, **violated in normal operation**

The first is the one that is not a rating, and the one nobody searching a parametric table would filter for: the familiar reflex is that a half-bridge driver *without* interlock is the less safe choice. In this circuit interlock is not a safety feature, it is a failure to function.

Five in use. Four on `injector` — each package pairs one high side with one low side, by package and not by circuit; UB2's low side is spare. One on `metering_egr`, low side only, so the board carries one gate-driver part number instead of two.

The kill path moves to `HIN`. The part has no shutdown pin, and a ground-referenced FET cannot short a gate floating 100 V up. `HIN` *is* ground-referenced, so `supervisor.cir`'s kill FET goes there behind a 1 kΩ from the MCU. It is slower than the low-side gate clamp — 330 ns maximum driver turn-off against 23 ns — and that is acceptable because it is not the path the fail-safe argument rests on: the low sides are in **series** with every coil, and killing any one stops that cylinder.

Checked by `boost_converter`'s rating checks in `run_sim.py`.

`GATE-RAIL` — 12V_GATE, a discrete follower off the boost rail

The board had no rail between 5 V and 40 V, and the drivers need 10–20 V. **Its source is the boost rail, and the reason is cranking:** a regulator from `VBAT_PROT` cannot make 12 V from a 6–9 V cranking battery, and a gate rail that collapses during cranking means no injection exactly when the engine needs it. `BOOST_100V` is the one rail already designed to hold up from 6 V.

Not the LM5164 already on the board — its VIN absolute maximum is **100 V** (TI datasheet §5.1), and this rail sits at ~102 V.

Part	Value	Why
Bias	2 × 47 kΩ	87 V split so neither resistor carries all of it; 0.87 mA
Reference	**BZX84-C13-Q**	13 V − Vbe − limit drop = **12.07 V** at 5 mA; 11.35–13.55 V across tolerance and temperature
Collector	**2 × 3.3 kΩ, 2512**	Takes 33 V of the drop at 5 mA; 0.45 W each into a short
Pass	**PZTA42-Q**, SOT223	300 V, AEC-Q101. **0.27 W** normal, 0.26 W shorted → Tj 133 °C at a 105 °C board
Limit	**PMBT3904-Q + 56 Ω**	0.65 V / 56 Ω = **11.6 mA**
Output	10 µF 25 V	Holds through each driver's gate-charge pulse

Load is about 3 mA (five drivers' quiescent current plus gate charge), designed for 5 mA. **12V_GATE clears the drivers' UVLO+ ceiling by 2.27 V.** Standing load on the boost is now 9.65 mA — bleed, gate rail and divider — of its 50 mA, and the peak-phase draw still recovers in 8.48 ms against 26.67 ms between cylinders.

Why the collector resistors. The design first called for a DPAK NPN rated ≥160 V. No such datasheet could be retrieved: onsemi and Diodes both refused the download. The part that was retrieved, PZTA42-Q, is 104 K/W in SOT223, so it may dissipate only 0.43 W at a 105 °C board. The pass transistor alone would carry 0.44 W normally and 1.16 W into a short. Putting 6.6 kΩ in the collector moves half the power into two

2512 resistors. The cost is headroom: 12V_GATE now needs the boost rail above 46.5 V, which is half of its worst droop. **The 105 °C board ambient is an assumption** — no ambient is stated anywhere in the spec.

Its failure runs the safe way. If the boost stops, 12V_GATE collapses, every driver drops into UVLO and holds its outputs low, and every gate on the board turns off.

`TPS40210QDGQRQ1` — chosen on an off-time, not on an input range

TI SLVS861F, August 2008, revised June 2020. Retrieved and text-extracted this session.

Input	4.5 V–52 V, **absolute maximum 52 V** (Table 6.1)
Reference	700 mV, 686–714 mV over −40/+125 °C (Table 6.5)
Off-time floor	**tOFF(min) = 170 ns typ, 200 ns max** (Table 6.6)
UVLO	4.25 V typ, **4.5 V max** turn-on (Table 6.5)
Supply current	1.5 mA typ, **2.5 mA max**, not switching
Internal regulator	BP = 8 V typ (7–9 V), 0–15 mA
Gate driver	400 mA source / sink
Package	HVSSOP-10 PowerPAD (DGQ), 3 × 3 mm, AEC-Q100 grade 1

The number that chose it is the off-time, and the reasoning runs backwards from the usual. A boost from a 6 V cranking dip to 100 V needs

```
D = 1 - 6 / (100 + 0.9) = 94.05 %
```

and a controller's minimum off-time is a floor **in time**, not in duty. At 150 kHz, 200 ns is 3 % of the period, so `Dmax` = 97.0 % and the cranking corner clears by **2.95 points**. At 2.2 MHz — where most wide-input boost controllers in this class sit, and where the magnetics would be far smaller — the same 200 ns is 44 % of the period and `Dmax` falls to **56 %**, below even the 86.6 % the rail needs at a *nominal* 13.5 V battery.

So "higher switching frequency, smaller inductor" is the wrong optimisation here, and the part that looks less modern is the one that works.

THE RATING THAT DOES NOT FIT — the third time on this board. VDD's absolute maximum is **52 V**; `VBAT_PROT` reaches **73.3 V** for about 50 µs on ISO 7637-2 pulse 2a. Over by **1.41×**.

This is the same shape as DRV8873-Q1's 40 V `VM`, which took that part off the EGR bridge — and it is solvable here for a reason that did not apply there. A motor bridge's `VM` carries the motor current; this pin carries 2.5 mA plus gate charge. So it gets its own clamp:

`BOOST-VDD-CLAMP` — 220 Ω, 1 W, and a BZG03C36-HM3

Pulse 2a, 73.3 V	121 mA through the zener for 50 µs; **VDD 46.6 V** worst case against 52 V abs max (1.12×)
Load dump, 40 V for 400 ms	33 mA at the zener's cold minimum (32.7 V), **1.09 W** for 400 ms. The 1.25 W rating is steady-state at 25 °C, so this relies on thermal mass; not yet checked against a Zth curve
Cranking, 6 V in	2.5 mA + 8.7 nC × 150 kHz = 3.8 mA, dropping 0.84 V → **VDD = 5.16 V**
against UVLO turn-on max 4.5 V	**0.66 V of margin**; 0.125 V even at the 25 nC gate-charge ceiling

The worst case is taken as VZ max (38 V at 10 mA), heated to 125 °C at the +0.11 %/K maximum tempco. On top of that sits the 40 Ω maximum dynamic impedance, applied from 10 mA up to the pulse current. That last figure is a small-signal value at 10 mA, and dynamic resistance falls with current, so it is a bound, not an estimate.

What was drawn first, and why it was wrong. The first design was 47 Ω and "a 43 V zener", and it was checked as an ideal 43.0 V source. The real part in that class, BZG03C43-M, is 40–46 V at 10 mA, up to +0.12 %/K, with a 45 Ω impedance. With 47 Ω in series, a quarter of an amp flows during the pulse. At 125 °C that puts **61.9 V** on a 52 V pin. The value was also chosen so the clamp would stay out of conduction during the 40 V load dump, but its tolerance band starts at 40.0 V. Found on 22 September 2026, when the zener was chosen and its datasheet read. The old design is kept as a pinned check.

The fix is in the resistor. The chosen boost switch, DMN15H310SK3, has a small gate charge, which lowers the pin's current enough to allow 220 Ω. That cuts the pulse current by a factor of five and limits load-dump conduction to 33 mA. Once the dump current is limited, a lower zener is safe, and it gives back the margin the tempco takes. 39 V was tried first and gave 49.4 V. That is inside 52 V but only 1.05×, below the 1.09× the `transient_clamp` rail table requires of every part on this rail, and the table failed it.

The power stage is untouched by any of this — the inductor and FET see the rail directly, and are 150 V class for it, the same class the injector switches already carry.

Checked by `boost_converter.cir`'s rating checks in `run_sim.py` — not with a switching model, which that block deliberately does not have, but as executable arithmetic on the datasheet's own numbers. Same treatment `transient_clamp.cir` gives the 73.3 V rail.

What is NOT settled by a datasheet: the loop compensation. `Rcomp` 10 kΩ and `Ccomp` 6.8 nF put the error amplifier's zero at 2.3 kHz, below the right-half-plane zero a boost puts at `(1-D)² · Rload / (2π·L)` = 17.3 kHz at 13.5 V in — and that zero **moves with duty cycle**, so the cranking corner is the hard one. These are a starting point. A current-mode

boost's loop depends on the inductor's real DCR, the capacitor's real ESR and the FET's real switching behaviour; it gets measured on the bench.

`LM5164DDAR` — and the 60 V part it replaces

Research memo 07 §4 selected **TPS54360B-Q1**, reasoning that a 60 V part beat a 42 V LM5175-Q1 "for load-dump margin". That reasoning was correct when it was written. It is not correct now, and the change came from the input-protection work above:

- The TVS standoff had to rise to 43 V so the part stops conducting during a *normal* clamped load dump.
- A higher standoff clamps higher. Pulse 2a now clamps at **73.3 V**.
- `transient_clamp.cir` carries a **passing** check that says so in as many words: *"60 V-class parts are NO LONGER viable downstream."*

A 60 V buck on this input is a part that fails the first ISO 7637-2 pulse 2a event. The LM5164 was already the part every rating check in the suite was written against — its −0.3 V / 100 V VIN limits are what `negative_pulses` and `transient_clamp` check — but nothing had ever reconciled the BOM table with it. **This is what a contradiction between two documents looks like when only one of them is executable.**

Operating range 6–100 V covers the spec's own 6–40 V requirement, with the 6 V end matching the cranking dip the input range was built around. 1 A against an estimated 400–500 mA board load.

Configuration values, all derived from the datasheet rather than picked:

Part	Value	From
`RON`	31.6 kΩ	The tON table in §5.5 is linear in RON/VIN: at 12 V, 25 kΩ gives 830 ns, so K = 398 ns·V/kΩ. For 400 kHz at 5 V out, RON = tON·VIN/K is 31.4 kΩ at **both** 13.5 V and 40 V — the VIN feedforward is what holds the frequency fixed. E96 value 31.6 kΩ.
FB divider	38.3 kΩ / 12.1 kΩ	VREF is 1.2 V (1.181–1.218, ±1.5%), so 5 V needs 1 + R1/R2 = 4.167.
EN/UVLO divider	75 kΩ / 24.9 kΩ	Enable rises at 1.5 V typ; 0.249 ratio puts turn-on at 6.02 V, the spec's cranking-dip floor. The pin is rated to 100 V, so the divider programs the threshold rather than protecting the pin.
BST cap	2.2 nF 50 V X7R	Not a range. The datasheet specifies the part: *"a high-quality 2.2nF 50V X7R ceramic capacitor between BST and SW"*.
`L` / `Cout`	33 µH / 47 µF, 10 mΩ ESR	`buck_preregulator.cir`, sized at the 40 V worst case rather than nominal. The ESR is part of the value — it sets the ripple.

The RON figure is worth one more line: it independently reproduces `buck_preregulator.cir`'s own gate timings, 925 ns at 13.5 V and 312.5 ns at 40 V. The netlist got those from D = Vout/Vin; the datasheet gets them from RON. They agree.

`TLV76733QWDRBRQ1` — accuracy as a requirement

Memo 07 §6 named `TLV1117-33QDCYRQ1`, explicitly *"class pricing, not fetched"*. Choosing the supervisor turned 3V3 accuracy into a hard constraint: outside **3.143–3.459 V** the board resets itself. TLV767-Q1 is **1% over load and temperature** — 3.267–3.333 V, about 125 mV of margin at each end. 1 A, 2.5–16 V in, AEC-Q100, VSON-8.

The fixed-output version is the one specified. Its pin 3 is `SNS` where the adjustable version has `FB`, and they are not interchangeable. `SNS` must not float; it is tied to `OUT`.

`TPS3850G33DRCT`

See `docs/pinmap.md` §1.8 and `hw/gen_symbols.py`. The variant letter is the decision: `G` sets thresholds at ±4% of nominal and `H` at ±7%, and with the part's ±0.8% accuracy that is a worst-case undervoltage trip of **3.143 V** against **3.044 V**. The supervisor has to assert before the S32K148's own LVD (3.0 V max), so G33 clears it by 143 mV and H33 by 44 mV — close enough that the two could fire in either order, and a supervisor that might lose the race to the thing it supervises is not doing its job. The cost is the tighter rail window above.

MCU

NXP S32K148, `FS32K148HAT0MLQT` — 144-pin LQFP, 80 MHz, −40…+125 °C.

Confirmed in research memo 05 against the S32K1xx data sheet Rev. 15 with page-level citations. The full reasoning is there; what matters for the BOM:

- The **144-pin** package is headroom against unknown U1 (50 of the OEM ECU's 94 pins are still unresolved), not a requirement of the confirmed channel budget, which fits inside a 100-pin part.
- **LQFP over MAPBGA**: identical instance counts — CAN/ADC/FTM/I²C are silicon-level, not package-level — and 0.5 mm pitch is inspectable by eye or AOI, where a BGA needs X-ray or destructive inspection. A real cost for a small team doing early assembly.
- **80 MHz `H` grade, not 112 MHz `U`**: the `U` speed grade is *not valid* at the 125 °C `M` temperature grade, per the datasheet's own ordering-table footnote. Memo 05 §6 shows the extra clock is not needed to close the timing budget, so there is no reason to trade thermal margin for headroom the design does not use. This corrects the spec §5 framing, which cites 112 MHz.
- In stock at DigiKey's India storefront at single-unit quantities. Not a lead-time part. The S32K146 fallback was checked and is *not* needed — it is the one that is out of stock, with 12–52 week lead times.

Still to check before footprint lock: package drawing 98ASS23177W (SOT486-2) has not been independently fetched. The PCB roadmap's warning stands — a hand-sourced footprint is the most likely place to introduce a silent error.

Design decisions put to the owner

Every decision here was a fork the evidence could not settle on its own: either it changed the architecture, or it traded cost against reliability. Each entry gives the question as it was asked, the options offered, what was chosen, and why the question existed at all.

Dates are 2026. The standing rule from 22 September — **safety and reliability before cost, size and convenience** — is what several of these answers turn on.

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1. The harness connector

Asked: the specification (§7, deviation 1) called for the OEM 94-way connector to be replaced by functionally-split sealed connectors plus an adapter harness. The physical part had since been identified from photographs of the machine. Keep the deviation, or put the OEM connector on the board?

Chosen: the OEM Bosch 94-way on the board, no adapter. Spec deviation 1 is recorded as reversed.

Consequence: the board-side land pattern is now load-bearing and has to come from a Bosch distributor. Neither part number returns a public catalogue hit.

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2. Four part-selection findings, approved together

These four were presented as recommendations and approved with one "ok" on 22 September. Each one came out of choosing a part and discovering the drawn circuit could not accept it.

2a. Where the injector and EGR current comes from — POWER-PATH

The problem: every high-current load (injector hold ≈10 A, EGR up to 6.4 A stall, relay coils) draws through connector pin 21, a 20 A fuse, the EMI filter and a reverse-battery FET. The reverse-battery block sized that FET for **5 A**, and the EMI filter block never stated a DC current at all. A 22 µH inductor that does not saturate at 15–20 A is a large part. Meanwhile pins 04 and 06 are separate battery feeds, fused 10 A each in the loom, that this board uses only to measure voltage.

Options: move the injector and EGR supply onto pins 04/06 with their own protection, leaving pin 21 for the electronics; or keep everything on pin 21 and resize the fuse, filter and reverse FET for the full load,

with one contact carrying it.

Chosen: move the high-current loads to pins 04/06. Status: built 8 October 2026 (`power_path.cir` , power_input sheet).

How it was built took three more questions, recorded in §5.

2b. Boost converter stability while cranking — BOOST-SLOPE

The problem: the injector boost controller (TPS40210-Q1) adds a fixed internal slope-compensation ramp. Its Equation 9 caps the current-sense resistor at `VDD·L·fSW / (60·(VOUT + VD - VIN))` , and 80% of that is the recommended value. The drawn 82 mΩ passes at 13.5 V (ceiling 99.7 mΩ) and **fails at the 6 V cranking corner** (ceiling 44.9 mΩ, so 1.83× over): the converter goes subharmonic exactly when the engine is being started. Worse, a 330 µH inductor that saturates above the 2.2 A current limit is not a part that exists in surface mount — the search found 330 µH parts rated 0.2–0.4 A — and the buildable direction, a smaller inductor, lowers the ceiling further (100 µH allows 30 mΩ).

Options: feed the controller's supply pin from the output side, which is TI's own remedy and steepens the ramp, and use 100 µH; or accept subharmonic operation while cranking.

Chosen: feed VDD from the clamped output side, 100 µH. Revised 8 October: 150 µH (Würth 7447709151), see §8. Status: built 8 October 2026.

2c. Sensor-supply protection — SENSOR-PTC

The problem: the requirement asked for a resettable fuse of "R25 = 3.0 Ω nominal, −30% spread". Real PTCs are not specified that way: Bourns MF-USMF020 gives a **range, 0.40–5.00 Ω**, and at its 0.40 Ω minimum it contributes 14% of the series resistance the fault-current ceiling assumed. Its hold current also halves from 0.20 A at 23 °C to 0.10 A at 85 °C, against a ≥200 mA requirement.

Chosen (first pass): a fixed series resistor sets the fault ceiling and the fuse only trips. This was later superseded — see decision 4.

2d. The 3.0 V and 3.3 V input clamps — LOWV-CLAMP

The problem: fifteen zeners clamp the analog, speed and discrete inputs. They were specified at 5 mA but sit at tens of microamps, where a sub-5 V zener's knee is soft. The datasheet allows **10 µA of leakage at only 1 V.** On a discrete input, 10 µA through the ~28 kΩ source is 0.28 V against a logic-high margin of 0.36–0.41 V while cranking; on an analog input it is a signal-dependent error on the ADC. The simulation's idealised sharp-knee zener hid all of it.

Chosen: replace them with a low-leakage clamp to the 3.3 V rail (BAV199). Status: done for the analog and cam inputs; the discrete inputs needed a different answer — see decision 3.

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3. The discrete switch inputs

Why it came back: BAV199 works where a clamp only conducts on a fault. The five switch inputs clamp **continuously**, because a 6–40 V input cannot be divided into logic range otherwise: reading a logic high at 6 V needs a divider ratio above 0.385, and staying under 3.3 V at 40 V needs one below 0.0825. Those do not overlap. Three candidate clamps were checked and all three fail:

Candidate	Why it fails
3.0 V zener (BZX84-C3V0-Q)	soft knee at µA; the leakage that started this
Rail clamp (BAV199-Q to 3V3)	parks the pad at 3.9 V continuously — beyond the pin's rating, as an operating point rather than a fault
Shunt reference (LM4040-Q1 3.0 V)	draws ~35 µA of bias below breakdown (TI SNOS633N, Fig. 5-4); the cranking-dip input can supply only 20 µA, so the node sags to ~2.6 V with no guaranteed bound, and it fails outright on the pull-up channels

Options offered: an NPN buffer per input; a stronger drive plus the LM4040, with special cases for the two contact-to-ground channels; or a dedicated automotive switch-detection IC on SPI.

Chosen: the NPN buffer. 10 kΩ into the base, 4.7 kΩ and 4.7 µF to ground, a BAV199-Q diode holding the base against reverse battery, and the transistor's collector as the MCU pin with a 10 kΩ pull-up to 3.3 V. The pad now only ever sees 0–3.3 V, and nothing on it has to conduct continuously.

What the safety rule changed inside this decision: the input resistor was first drawn at 100 kΩ, which draws 0.11 mA through a closed contact at 12 V. Non-gold switch and relay contacts at that current oxidise into an intermittent open — a field failure, not a bench one. 10 kΩ gives 1.1 mA. Everything is checked at the worst corner rather than typical: hFE 30 (datasheet minimum 60, halved for −40 °C) still pulls the pad to 0.03 V from a 6 V input, and a typical part at 125 °C stays off below a 1.3 V input.

Cost: the logic is inverted — pad low means input active — which is recorded in the pin map, and the failure modes (transistor open reads inactive, collector-emitter short reads active) are handed to the firmware plausibility spec, because one polarity cannot make every channel fail safe.

Status: done.

4. The 5 V sensor supply

Why it came back: the fix approved in 2c does not survive the real source impedance. The sensor rail block modelled the 5 V supply as ideal behind 0.1 Ω, but it comes from a **1 A buck**. So:

- a shorted sensor harness hits the buck's current limit long before any fuse trips; 5V_MAIN collapses, 3V3_MCU follows it, and the supervisor shuts the engine down. One chafed wire stops the machine.
- a sensor supply wire shorted to **battery** back-feeds 24 V into 5V_MAIN through the fuse. No series resistor or PTC prevents that.

Options offered:

Option	What it buys	What it costs
Tracker per group + overvoltage cut-off	each group survives a short to ground *and* to battery; other groups and the MCU are unaffected	about 8 more parts
Series resistor + PTC (as approved in 2c)	cheapest	a harness short browns out the MCU; a battery short back-feeds the rail
Trackers fed directly, with a lower main TVS	fewer parts	rewrites the whole board's transient design, and needs the alternator question (U16) answered first

Chosen: a tracking regulator per sensor group (TPS7B4253-Q1, whose output survives −1 to +45 V and limits at 301–520 mA, tracking 5V_MAIN to ±4 mV) behind a 100 V-class overvoltage cut-off.

Why the cut-off is needed: the tracker's input pin is rated 45 V and this board's battery rail reaches 73.3 V for 50 µs on ISO 7637-2 pulse 2a. The cheap alternatives were worked through and discarded: a series resistor with a TVS must clamp below 45 V, so it would conduct through the entire 400 ms load dump (about 7 J), and any series resistance breaks the 0.52 V of dropout headroom at a 6 V cranking input; an emitter or source follower loses its own Vbe/Vgs at that same corner. A cut-off instead disconnects for the 50 µs pulse, while a 47 µF capacitor holds the sensors up (0.16 V of droop), and stays connected through the 40 V dump.

Status: built 8 October 2026. The cut-off controller is the **TPS48110-Q1** (see §7). `sensor_rail.cir` was rewritten around it.

5. How the POWER-PATH feeds are built (8 October)

§2a decided that the loads move to pins 04/06. These three questions decided how. Each was asked with a recommended option, and each was answered with it.

5a. Which loads go on which pin

Asked: how should the high-current loads be split across pins 04 and 06?

Options: - **04 injector, 06 EGR** (recommended). Pin 04 feeds the injector hold switches and the boost converter input; pin 06 feeds the EGR H-bridge. Separate rails, separate protection: an EGR short blows only its own fuse, and no two fuses are paralleled. The injector averages well under its fuse rating (10 A only during holds of about 1–3 ms), and the EGR's 6.4 A stall is inside its 10 A fuse. - **04 + 06 paralleled** onto one high-current rail. Two fuses in parallel never share evenly: one blows first and dumps the whole load on the other.

Chosen: 04 injector (`VBAT_INJ`), 06 EGR (`VBAT_EGR`).

Why it was a question: U7, whether the OEM uses pins 04/06 as load feeds at all, is still open. What makes them usable is the harness itself: each pin has its own 10 A loom fuse, and a fuse is sized to protect its wire.

5b. Reverse-battery protection on the new rails

Asked: what should provide reverse-battery protection on the two new rails?

Options: - **A P-FET per rail** (recommended). Same passive topology as Q1 on pin 21: a gate resistor to ground plus a zener, and no controller IC, so nothing has to be rated for the +73 / −48 V swing. It needs a lower-RDS(on) part, chosen from a retrieved datasheet, for 6.4 A continuous at a 105 °C board. The reversed-battery fault is removed rather than managed. - **Fuse only.** No series element: a reversed battery drives current through the bridge FETs' body diodes until the 10 A loom fuse blows. No drop or heat in normal running, but it relies on the FETs surviving the fuse's clearing time — a managed failure, which the standing rule argues against.

Chosen: a P-FET per rail — SQM120P10-10m1L (Vishay doc 76943, retrieved 8 October).

Why it was a question: pin 21's SQD50P08-25L could not simply be copied. At the EGR's 6.4 A stall it would dissipate 1.3–1.8 W, which takes a DPAK on a 105 °C board past its junction limit. The chosen part runs at 139 °C (155 °C while cranking), checked in `power_path.cir`.

5c. What stays on pin 21

Asked: which loads stay on pin 21, the filtered electronics feed?

Options: - **Buck, relays, metering** (recommended). Pin 21 keeps the buck (all the electronics), the relay coils (~84 mA) and the metering valve's freewheel diode (0.86 A max; the valve itself is powered by the

OEM's 20 A fuse 8F1). The boost controller's VDD stays on VBAT_PROT until BOOST-SLOPE moves it. Pin 21 then carries a couple of amps. - **Buck only.** Move the relay coils and the metering freewheel onto the new rails as well. Pin 21 becomes electronics-only, but the relays and the metering valve then depend on the 04/06 fuses too.

Chosen: buck, relays and metering stay on pin 21.

Consequence: L101, L102 and F101 are unblocked and are now sized for pin 21's reduced load, which still has to be stated as a number.

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6. The toolchain on a new machine (7–8 October)

Setting the project up on macOS raised three questions. They are about tools, not the circuit, but two of them decide which simulator and which libraries the circuits are checked against.

6a. Which KiCad

Asked: the project was generated and verified with kicad-cli 7.0.11, but Homebrew offers only KiCad 10. Install 7.0.11 from kicad.org, KiCad 10 from Homebrew, or skip KiCad for now?

Options: 7.0.11, which matches the verified netlists (recommended at the time); KiCad 10, which is easier to keep updated, but its netlist export and libraries may differ; or no KiCad.

Chosen: KiCad 10 (10.0.7, installed by the owner).

Consequence: two things had to change for KiCad 10. `kicad-cli` 10 writes its netlist one token per line, so the parsers in `check_netlist.py` and `assign_passives.py` were changed to accept either layout. The results were identical to KiCad 7's, and `passive_fp.json` came out byte for byte the same. The generators copy library symbols into each sheet, so they are pointed at **KiCad 7.0.11's symbol library**, downloaded separately. With it, an unchanged sheet regenerates identically apart from UUIDs. KiCad 10's library would have changed every symbol on every regenerated sheet. That library choice was not put to the owner; it follows from this answer.

6b. Which ngspice

What happened: Homebrew's ngspice 47 failed three blocks that pass on the original machine. `injector_turnoff` and `boost_converter` never finished, and `metering_unit_pwm` missed by 3 %. The suite also stalled at random. Fixes for all three were found: a ramp on an on/off charger model, a step-size cap and a time-weighted average. But the random stall remained.

The owner's position: first, that the circuits must work on the latest version too. Then, after the random stall: build ngspice 42 from source rather than spend more time on 47.

Chosen: ngspice 42, built from source into `~/opt/ngspice-42`. The committed suite passes 24/24 on it unchanged (25/25 with `power_path`).

Still open: the suite does not pass on ngspice 47 for the three causes above. Upgrading the simulator is a change that needs the whole suite re-run (`docs/HANDOFF.md` §2 and §5).

6c. Keep the ngspice 47 fixes or not

Asked: with 42 passing the committed code, revert the 47-specific edits to the netlists, `run_sim.py` and `ngrun.py`, and commit only the setup changes?

Chosen: revert. The 47 fixes are not in the repository; §6b records what they were.

7. The overvoltage cut-off in front of the sensor trackers (8 October)

§4 chose a tracking regulator per sensor group "behind a 100 V-class overvoltage cut-off", and named an LTC4368-class controller as the candidate. Its datasheet had never been retrieved. Retrieving it, and looking for an alternative, turned the candidate into a choice.

What the part is for

The three sensor regulators (TPS7B4253-Q1, one per group) survive at most **45 V** on their input. This board's battery rail reaches **73.3 V** for about 50 µs on ISO 7637-2 pulse 2a, a standard automotive surge test. So the regulators need a fast automatic switch in front of them:

- normally **on**, passing battery power through;
- above about 45 V it switches **off** within a few microseconds, and a 47 µF capacitor after it keeps the sensors powered meanwhile;
- when the surge has passed, it switches back **on**.

It is a small controller chip driving one power transistor. It must stay **on** through the 40 V load dump, which lasts 400 ms, far longer than the capacitor could bridge.

The question

Asked: which overvoltage cut-off controller should feed the three sensor trackers?

The first time it was asked, the options were given only as datasheet figures, and the owner replied that they could not decide between two part numbers. The question was then explained as above, with this

comparison:

	TPS48110-Q1 (Texas Instruments)	LTC4368 (Analog Devices)
Survives	100 V	100 V
Switches off in	2.6–4 µs after the threshold	under 8 µs
Automotive-qualified (AEC-Q100)	**yes**, Grade 1	**no**: C, I and H grades only (H = −40 to +125 °C)
Around it	a few more parts: charge-pump capacitor, unused current-sense pins tied off	simpler, 10 pins
Datasheet	SLUSEE5E (April 2026), from ti.com	Rev A, read through a mirror because analog.com timed out
Extra check needed	a 60 V/µs limit on how fast its input may rise	—

Recommended: TPS48110-Q1. Electrically, either would do the job. The deciding point is qualification. AEC-Q100 means the chip has passed the automotive industry's stress tests for temperature cycling, vibration and lifetime. Every other IC on this board has it, and the LTC4368 would have been the one exception, on the part that protects the sensor supply. The standing rule puts reliability before cost and convenience.

Chosen: TPS48110-Q1 ("ok use the recommended one").

Consequences:

- **Slew limit:** the input slew rate on pulse 2a's edge has to be checked against the 60 V/µs limit. The `sensor_rail` rewrite does that.
- **Threshold window:** the OV threshold must sit above the 40 V load dump at its lowest tolerance, and trip early enough that the regulators' input stays under 45 V. That includes the rise during the controller's 2.6–4 µs reaction time. The window is narrow, and the `sensor_rail` rewrite pins it.
- **The switch transistor** is the board's existing 100 V N-FET, IPD90N10S4-06, so there is one part number fewer to buy.

What the simulation then showed. Pulse 2a was applied through the real pin-21 filter. The trackers' input peaks at about **22 V**, nowhere near 45 V, at every corner. TI's own inrush network does most of that: the 22 nF from gate to ground holds the FET's gate, so the FET can only follow the surge up to its gate voltage. The OV comparator then opens it fully once the rail passes 40.5–42.1 V, and that timing turned out not to be critical. The 60 V/µs slew limit is met by feeding the controller's supply pins through 100 Ω and 22 nF, which gives 4.9 V/µs. The 47 µF hold-up is enough even aged to 33 µF with all three trackers at their limit. One thing it showed about the rest of the board: with the filter inductor not saturated, the clamp node reaches **76.6 V**, not the 73.3 V `transient_clamp.cir` reports. Every part there is 100 V class, and the figure is now a check.

A limit this design does not remove

The trackers' outputs survive −1 V to +45 V, so a sensor supply wire shorted to battery is survived at 24 V (jump start) and at 40 V (load dump). A short to battery **at the same time as** a pulse 2a surge (73.3 V) would exceed it. That is a double fault, and it is recorded rather than designed out.

8. The boost inductor, revisited (8 October)

§2b approved two changes to the injector boost converter: feed the controller's supply (VDD) from the clamped output side, and use a 100 µH inductor (Würth 7447709101). Building it showed the 100 µH part does not fit every rule.

What the inductor and its sense resistor have to do

The TPS40210-Q1 measures inductor current through a small sense resistor, and that one resistor sets two things:

- **Stability.** TI's Equation 9 caps the resistor at 80 % of VDD × L × fSW / (60 × (VOUT + VD − VIN)). Below it the converter is stable; above it, it goes "subharmonic" and oscillates at half its switching frequency. The worst corner is the 6 V cranking dip. Feeding VDD from the boost output raises VDD from about 5 V to about 32 V while cranking, which is the whole remedy.
- **The current limit.** The controller cuts each cycle when the resistor's voltage reaches 120–180 mV, a 1.5 : 1 spread over tolerance and temperature. The limit has to sit high enough never to trip in normal running (TI's Equation 49), and low enough that in a fault the inductor does not saturate badly.

The first question

Asked: with the approved 100 µH inductor, all three rules cannot be met strictly. Allow the fault current limit to reach the inductor's softer 30 % saturation point, or find a bigger inductor?

The owner's answer: "150 µH inductor comes in size 10x10x4 SMD package."

Checked against Würth's own datasheets:

- **7447714151, 10 × 10 × 4 mm:** 150 µH, Isat 1.4 A typ. It cannot work. Normal cranking peaks at about 1.14 A, so TI's rule puts the current limit's minimum near 1.8 A and its maximum near 2.7 A. The part would saturate in normal running, not just in a fault.
- **7447709151, 12 × 12 mm:** 150 µH, Isat 2.7 A (10 %) and 3.3 A (30 %) typ, in the **same footprint** as the approved 100 µH part.

The second question

Asked: which boost inductor?

	100 µH 7447709101 (as approved)	150 µH 7447709151
Sense resistor	50 mΩ	62 mΩ
Current limit, min–max	2.4–3.6 A	1.9–2.9 A
Limit max vs 10 % saturation	16 % over	7 % over
Limit max vs 30 % saturation	5 % under	14 % under
Stability margin (Eq. 9, 80 % rule)	1.07×	1.30×
No false trip while cranking (Eq. 49)	1.27×	1.07×

Recommended and chosen: 150 µH, Würth 7447709151, with a 62 mΩ sense resistor. It has more stability margin and less fault-current saturation, in the same footprint. This revises the 100 µH in §2b; the VDD-from-output remedy stands.

Why neither choice is strict: the 1.5 : 1 spread of the current- limit threshold means the limit's maximum is always 1.5 × its minimum. The minimum has to clear normal cranking current, so the maximum lands above the first softening point of any inductor this size. Partial saturation during a fault is not damaging: the controller still cuts the current every cycle.

How it was built

- **VDD from the output.** A follower off the 100 V rail gives VDD 34.3–45.0 V. It is 12V_GATE's circuit with a 39 V zener (BZG03C39, same Vishay family as the 36 V clamp): a PZTA42-Q pass transistor, 2 × 3.3 kΩ collector resistors, and a PMBT3904-Q limiting at 11.6 mA. A follower off a 100 V rail needs that protection whatever it feeds. If VDD shorted, the follower would otherwise burn.
- **Start-up from the battery.** The old 220 Ω feed became a 180 Ω start-up path, with its 36 V clamp, into a Schottky (SS20PH102, already on the board). The Schottky stops the follower pushing about 0.1 A back into the battery rail. 180 Ω, not 220 Ω, buys back the Schottky's drop: VDD is 4.77 V at the 6 V cranking dip, against the 4.5 V start threshold.
- **Found while building it:** the oscillator's timing resistor returns to VDD, and TI specifies the frequency −20 % / +7 % across VDD. Equation 9 scales with frequency, so the stability check has to use −20 %. At that corner the 150 µH design keeps 1.12×. The 100 µH design would have failed outright (42.9 mΩ against its 50 mΩ). The choice in this section was the right one for a reason nobody had seen yet.
- **Stated, not hidden:** the switch FET's datasheet gives only a typical gate charge. At twice that, start-up margin falls to 17 mV. Until the output passes about 36 V, the controller runs from the battery feed with a

shallow ramp, so subharmonic operation is possible for the first milliseconds of a cranking start, not in running.

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9. The three capacitors that are not ceramic (8 October)

Every chip capacitor on the board is a multilayer ceramic, packaged automatically by `assign_passives.py`. Three are not, because each needs something a ceramic cannot give: a lot of capacitance at high voltage, or a controlled internal resistance. They were left open until now.

Ref	Job	Chosen
C313	bulk on the MCU's 3.3 V supply	KEMET T599D157M006ATE025, 150 µF polymer tantalum, **plus R306, 22 mΩ 1 %** in series
C211	holds the sensor supply up while the cut-off is open	Chemi-Con HHXC630ARA470MHA0G, 47 µF 63 V hybrid polymer, 8 × 10 mm
C401 + C420	the injector's 100 V energy tank	**two** Chemi-Con HHXC800ARA820MJC5G, 82 µF 80 V hybrid polymer, 10 × 12.5 mm, in series = 41 µF, 160 V

All three are AEC-Q200, surface-mount, and rated to 125 °C or higher. Chemi-Con figures are from its 2026 catalogue (CAT. No. E1001A); KEMET's from the part's own specification sheet.

The idea behind all three: ESR

Every real capacitor has a small resistance in series with it, its **ESR**. In wet aluminium electrolytics it rises steeply in the cold, because the liquid inside thickens: typically 4–10× at −40 °C. A **polymer** or **hybrid polymer** capacitor replaces most or all of that liquid with a conductive solid, so its ESR barely moves. Chemi-Con guarantees the HXC series' impedance at −55 °C is **at most twice** its 20 °C value. That one property is why all three choices are polymer types.

C313 — the bulk capacitor needs a *minimum* resistance

`mcu_pdn.cir` showed the bulk capacitor's ESR is not just a loss but the damping for a resonance between the regulator and the capacitor. Too little ESR and the supply rings; too much and it stops filtering. The requirement (`PDN-BULK`) was therefore a band: 20–50 mΩ.

Two findings changed how it was met:

- **A capacitor datasheet gives a maximum ESR, never a minimum.** No part can be bought "at least 20 mΩ". So the floor is a separate resistor: 22 mΩ, 1 %, in series with a capacitor whose own ESR is 25 mΩ at most. Even if the capacitor's ESR were zero, the pair is 21.8 mΩ.

- **The ceiling was conservative.** A sweep of the same model shows the 100 mΩ target holds from about 17 mΩ up to about 95 mΩ. The pair's worst high end is 72.2 mΩ: resistor +1 % plus the capacitor at KEMET's 2× end-of-life limit. Both ends pass at the smallest capacitance the datasheet allows (96 µF): 71.9 and 86.9 mΩ.

No question was put to the owner: the resistor is the only way to meet a requirement both ends of which are needed.

C211 — the sensor hold-up

When the TPS48110-Q1 opens for a voltage spike (§7), this capacitor keeps the three sensor regulators running. The only new demands are a voltage rating above the 40 V load dump the switch lets through, and a capacitance that survives aging. The HXC part gives 63 V (1.58× over 40 V). At −20 % tolerance and −30 % end of life it is 26.3 µF. `sensor_rail.cir` now runs its two worst corners at 26.3 µF, not the 33 µF it used before, and still passes: the rail's lowest point is 9.2 V. Its ESR drop at −55 °C, aged, with all three regulators at their limit, is 0.25 V. No question was needed.

C401 — the injector tank, and the question

What it does. The boost converter can only trickle-charge this capacitor. The capacitor alone delivers the injector's opening pulse: 18 A within about 40 µs, every injection. Its ESR is in series with the injector. At 18 A, every ohm of ESR costs 18 V of push and slows the opening.

What the search found. The tank sits at 100 V and reaches 104 V, so it needs a part rated above 100 V. In the catalogues searched, **no capacitor rated above 100 V publishes its ESR at all**, cold or warm. Chemi-Con's 160 V parts give only a 120 Hz loss figure. Vishay's specified-cold series stops at 80 V. Parts rated up to 80 V do publish it, including HXC.

Asked: "C401 is the injector's 100 V energy tank … No single capacitor rated above 100 V publishes its cold ESR. Which way?"

	2 × 80 V hybrid in series (recommended)	1 × 160 V wet electrolytic
Part	2 × Chemi-Con HXC 82 µF 80 V, SMD	1 × Chemi-Con GXF 51 µF 160 V, radial
ESR at −55 °C, aged	224 mΩ, from the datasheet	not specified; estimated ~8 Ω
Time to 18 A at −40 °C, lowest rail	45 µs (38 µs nominal)	never reaches 18 A at the estimate: 10.2 A at most
Cost and space	one extra can, 10 × 12.5 mm	cheapest, one part

The owner's answer: "2× 80 V in series (Recommended)."

Why it matters: the injector controller ends the opening phase when the current reaches its peak threshold. A tank that cannot push 18 A through its own resistance never ends that phase, and the engine misfires. That would happen on a cold start, late in the part's life, which is the hardest fault to find in the field.

How two in series work:

- **Capacitance halves, voltage rating doubles.** 41 µF, 160 V. The sims now run at 41 µF. The droop per injection is 8.3 V (it was 7.3 V). On the smallest stack the datasheet allows (−20 %, then −30 % aged) it is 14.9 V, so the rail never falls below 85.1 V. Every check that used the old 92.7 V low point now uses that figure. The gate-driver bank node rises to 104.1 V, since each hold-cutoff bump is 2.45 V on 41 µF. Recovery time does not change; it depends on charge, not capacitance.
- **The bleed balances the stack.** Two capacitors in series only share the voltage evenly if something forces them to. The 22 kΩ bleed that `BOOST-BLEED` already required is now 2 × 11 kΩ, one across each can (R401, R435). With the worst leakage mismatch the datasheet allows, each can settles at 52.4 V on an 80 V part. The first charge from 0 V is the worst moment: it splits by capacitance alone, so 71.0 V can land on one can. That is 1.13× under its rating. Total bleed current and power are unchanged.
- **Ripple:** the injection pulses plus the charger's switching current, counted as if it never stopped, are 1.06 A rms against 2.1 A rated at 125 °C.

All of it is executable arithmetic in `run_sim.py` under `boost_converter`, beside the converter's own datasheet checks.

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10. A filter on each new supply feed (8 October)

POWER-PATH (§2a, §5) moved the injector and EGR currents onto connector pins 04 and 06. That was right for current and heat, but it took the board's two noisiest loads away from pin 21's EMI filter. Each new feed had only its reverse FET and 4.4–8.8 µF of ceramic capacitors. The question was whether that is enough to meet the radio-interference limit this project designs to. The limit is CISPR 25 Class 5, the strictest automotive class (research memo 06), the same target pin 21's filter is held to.

What makes the noise

Both loads switch large currents on and off many thousands of times a second, straight from the battery wire:

- **Injector hold.** After the 100 V opening pulse, each injector is held open at about 10 A by switching the battery on and off ("chopping").
- **EGR valve motor.** Driven by PWM: full battery on and off, up to 6.4 A at stall.
- **Boost converter input.** A smaller, smoother 150 kHz ripple.

Each on-off edge sends a spray of harmonics down the harness. A radio receiver near the wiring would hear it, and the standard limits how much is allowed.

What the model found

New block `feed_emi.cir` reproduces CISPR 25's measurement: the standard 5 µH / 50 Ω test network (LISN), 200 mm of harness, then the board. `run_sim.py` adds up each load's harmonics against the Class 5 limits, band by band, with the peak and average detectors.

	Worst point against Class 5
VBAT_INJ as drawn	**33 dB over**
VBAT_EGR as drawn	**48 dB over**
VBAT_INJ with the filter	17.7 dB under (12.2 dB under with aged capacitors)
VBAT_EGR with the filter	9.3 dB under

48 dB is about 250 times too much voltage. No layout trick closes a gap that size; it needs a filter.

The filter, per feed

- **Coilcraft XAL1510-103**, 10 µH, rated 16 A for a 20 °C rise. Between the reverse FET and the loads.
- **Two 82 µF / 80 V hybrid capacitors** (the same Chemi-Con HXC part as C401) on the load side. These supply the chopping current, so the inductor only carries the average.
- **Two dampers:** another HXC each, in series with a 0.68 Ω resistor.
- The existing 2 × 2.2 µF ceramics now sit on the connector side, which makes it a two-capacitor filter.

Both feeds get the identical circuit: one parts list, one layout block.

Why the dampers. An inductor and capacitor together form a resonant circuit, here near 3 kHz. Two of the loads on the injector feed, the boost converter and the hold-current loop, try to draw *constant power*. To the filter that looks like a negative resistance: about 0.94 Ω at a 9 V cranking dip with a hot injector coil. If the filter's own impedance peak is higher than that, the supply can oscillate. Undamped it peaks at 1.24 Ω. With the dampers it peaks at 0.64 Ω, even with end-of-life capacitors, a

margin of 1.47×. The damping is a plain resistor, not a wet capacitor's internal resistance, because that rises about 8× at −40 °C (§9), exactly when cranking makes it matter.

Why this inductor. The Würth 10 µH alternative saturates at 8.8 A typical and is rated only to 75 °C ambient. The Coilcraft part saturates at 26 A and is rated to 165 °C. Its one limit is 55 V across the winding. The model puts ISO pulse 2a through the filter: at most 48.7 V across it, 1.12× inside. In return the filter swallows most of pulse 2a, so the feed's loads see at most 28.7 V, where VBAT_PROT sees 73.3 V.

A requirement this puts on the firmware

The hold chopping and the EGR PWM frequencies have not been chosen; they are firmware. Harmonic strength in the radio bands grows with switching frequency, so the filter was sized at **25 kHz**. At 50 kHz the EGR feed would be 1 dB over. **Both must stay at or below 25 kHz.** That is above hearing, which is the usual reason for choosing about 20 kHz anyway.

What it costs, and what would change it

Two inductors (15 × 16 mm), eight hybrid capacitors (10 × 12.5 mm) and four resistors: 14 parts, roughly US$15–20 a board. No question was put to the owner. Without the filter the board fails Class 5 by 33–48 dB, and the only alternative is a looser emissions class. That is the owner's call. A stationary genset has no car radio, so a lower class may be defensible, but it is not this log's decision to make. If the owner relaxes the class, these filters can shrink.

Decided 12 October 2026: the owner keeps CISPR 25 Class 5. The filters stay as designed. The PCB layout is held to Class 5 practice too: tight switching loops, an unbroken ground plane, and filters placed at the connector. The layout guide carries this.

Limits of the model. Like `emi_filter.cir`, it covers differential mode only. Common-mode noise, both wires moving together against the chassis, needs a layout to model. The 100 ns switching edges and the in-phase sum of harmonics are deliberate worst cases.

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11. Firmware on the PC first (10–11 October)

The hardware was complete to schematic, but no firmware existed. A competing team already runs firmware on their board, with an Arduino faking the engine signals. The question was how to catch up without waiting for our own PCB.

The choice

The control software is written as **plain C that touches no hardware** (`firmware/core/`). It runs on the PC against a **simulated engine** (`firmware/sim/engine.py`). The same C files later run unchanged on an NXP S32K148 evaluation board, and after that on our own board. Every control and safety function is proven first on the PC, where a mistake costs nothing.

All the firmware sees is one table of numbers (`ecu_io` in `ecu.h`). The inputs are sensor voltages and switch states. The outputs are injection commands, the metering-valve duty, the watchdog pin and the CAN messages. The firmware has four entry points: start up, a crank tooth arrived, a cam pulse arrived, and one millisecond passed. The simulator fills the table and reads it back, and the S32K148 port will do the same from its pins. Injection is given as **"start at time T, for D microseconds"** per cylinder, which is what the chip's timer executes. That keeps the simulator's time step coarse without making the fuel amounts coarse.

What the firmware does

- **Reads the crank wheel.** A 60-2 wheel has 58 teeth and a gap where two are missing. The gap shows up as a tooth period more than 1.5× the one before. Two gaps 58 teeth apart mean "synchronised". The cam pulse, once per two revolutions, says which revolution has cylinder 1 firing.
- **Injects** once per cylinder per cycle, 5° before top dead centre. It schedules at least 1° ahead, so the start time is never already past.
- **Governs speed.** It holds 1500 rpm, or falls 4 % from no load to full load when the droop switch is closed. On reaching 400 rpm the setpoint **ramps up at 500 rpm/s** rather than jumping, explained below.
- **Holds rail pressure** at 1000 bar by driving the metering valve, treated as normally closed: more duty, more fuel flow. See the final review below.
- **Checks every sensor.** One that stays out of range for 100 ms raises a fault code. From the first moment it goes out of range, its last good value is held.
- **Shuts down and latches.** Above 1650 rpm (checked on every tooth), on loss of the crank signal, or on ignition off. It stays shut down until the ignition is cycled. The trip cancels any injection not yet started.
- **Talks J1939 to the DSE4522:**
- engine speed every 20 ms
- coolant temperature every 1 s
- oil pressure every 500 ms
- active fault codes every 1 s

What the simulator proved

Eleven scenarios, run with one command (`.venv/bin/python firmware/sim/run_sil.py`), all pass in 36 s:

#	Scenario	Result
1	Start	synchronised in 116 teeth; 1500 rpm by 5 s; peak 1534 rpm
2	0 → 20 kW load step	dips to 1449 rpm, back within ±15 rpm
3	20 → 0 kW load rejection	peaks at 1547 rpm, well under the 1650 trip
4	Droop switch	speed falls 3.8 % (want 4 ± 1 %)
5	Runaway (stuck injector)	trips at 1650.6 rpm; no new injection after
6	Crank signal lost	shuts down 50 ms later, fault SPN 636
7	Ignition off	shuts down with no fault code
8	Coolant sensor unplugged	fault SPN 110 FMI 3 after 1.0 s; engine keeps running
9	CAN timing and content	every message on time to the ms; speed within 0.1 %
10	Rail pressure	within 1.6 % of 1000 bar at full load
11	Watchdog	WDI falls every 10 ms

There are also 46 unit tests on the separate pieces (`make -C firmware/sim test`).

What an expert review caught

Halfway through, a separate reviewer acting as a diesel-ECU firmware specialist read the code. Every finding was reproduced as a failing test before it was fixed. The ones worth knowing:

- **The start nearly tripped overspeed.** On reaching 400 rpm the speed loop asked for 1500 at once, fuel hit its limit, and the engine peaked at 1647 rpm, 3 rpm short of the trip. The ramping setpoint brought the peak down to 1534.
- **A crank-sensor noise spike read as overspeed.** One extra edge in mid-tooth made the speed read 1700 rpm. An edge arriving in less than half a tooth's time is now ignored, because no engine accelerates that fast.
- **A cancelled injection could still fire.** The cancel did not change the sequence number a timer driver watches. Now it does.
- **A hot engine looked like a shorted sensor.** With the placeholder coolant sensor, 0.2 V is a real 129 °C. That would have been flagged as a short and replaced by a harmless 90 °C. The coolant short threshold is now 0.1 V.
- **A stray cam pulse shifted everything by a revolution.** The engine would have injected on the wrong stroke. A cam pulse in the wrong revolution now stops injection until the next one agrees.
- Smaller fixes:
- a timestamp 1 µs out of order read as signal loss

- an out-of-range sensor's raw value was used during the 100 ms debounce

What the final review caught

After all ten steps, a second reviewer read the whole branch. Again, each finding became a failing test before it was fixed.

- **The metering valve's polarity contradicted the safety design.** This firmware spec had assumed a normally-open valve. The parent spec's trip module stops fuel by *cutting the valve's supply*, and that only works if the valve is **normally closed**: unpowered, no fuel. With a normally-open valve, cutting its supply opens it fully. The runaway run showed the rail climbing to 1695 bar after the trip. The firmware now assumes normally closed, to agree with the parent spec. Polarity is a single constant (`METER_NC`), and shutdown always commands the closed position. **The real valve still has to be identified.** If it is normally open, the constant flips *and* the trip module's contacts must move to the injector supply.
- **A stray edge before cranking blinded the decoder for 30 s.** The noise filter compared every tooth with a stale 30-second "period". After any pause longer than 50 ms, the decoder now starts afresh.
- **A 1 ms ignition flicker cleared a latched trip** while the engine was still turning, and the ECU restarted it. The latch now clears only after the ignition has been off for 500 ms with the engine stopped.
- **The speed reading could drop to 0 for one tick** if a tooth arrived just after the tick read the clock. That doubled the fuel demand for a moment. The time comparison is now signed, as it already was in the decoder.
- **Scenario 5 could not tell an edge-checked trip from a late one.** With the per-edge overspeed check deleted, it still passed. It now checks that the ECU is shut down on the very edge it first reads above 1650 rpm, and that rail pressure falls after the trip. With the check deleted, the scenario now fails.

Choices made while building

- **Signal-loss timeout of 50 ms, not 100.** The scenario requires shutdown within 100 ms. At 400 rpm the longest normal gap between edges is 7.5 ms.
- **Watchdog kicked every 10 ms.** This settles the open item on the MCU sheet. The TPS3850's window (its SET0/SET1 straps and CWD capacitor) has to accept a 5–20 ms kick period. That is well inside the 26.67 ms cylinder interval the spec requires.
- **Tuned gains.** These are placeholders until real engine tests:
- speed: 0.5 mg/rpm proportional, 2.0 mg/(rpm·s) integral
- rail: 0.002 duty/bar proportional, 0.01 duty/(bar·s) integral

The gentler of the two passing sets was kept. A real engine has delays this model lacks. - **A stray cam pulse or a noise edge** does not stop the engine. A single spike should not shut down a genset.

What is assumed, and how to change it

Each assumption is a single named constant in a header, marked `assumed` or `calibration`: - the 60-2 wheel and cylinder 1 TDC at 90° after the gap (`CRANK_TDC1_DEG`) - firing order 1-2-3 - start of injection, 5° before TDC - the injector flow law - fuel at full load, 34 mg (sets the droop) - every sensor's volts-to-units table (`sensors.c`)

The engine model's constants are guesses for a 3-cylinder genset diesel, also marked. When the real engine's figures arrive, each is a one-line change, and the scenarios say at once whether the control still holds.

Out of scope, and next

Not done yet: - injector current regulation (that lives in the driver chip) - pilot and post injections - real calibration maps - EGR control

The next step is the S32K148 evaluation board with an Arduino generating the engine signals: the competitor's setup, but with control logic that is already proven.

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12. Logs a production ECU keeps (11 October)

This ECU goes into a production genset that runs for months at a time. Until now, everything the firmware knew disappeared at power-off. A technician arriving at a stopped set would have found nothing.

Four records

All four live in one **persistent image** of about 3 KB. That fits the S32K148's 4 KB emulated EEPROM.

- **Fault memory.** Up to 16 faults, kept after they clear. Each has an occurrence count, the first and last time it was seen, and a **snapshot taken the first time**:
- speed
- load
- fuel
- rail pressure
- coolant
- oil
- battery
- state

It answers "what was the engine doing when this happened?" - **Event log.** The last 64 of: - power-ups and their cause (a watchdog reset is a firmware hang caught in the field) - ignition on and off - crank, run and shutdown, with the reason - faults set and cleared - stored-fault clears - **Counters.** - ECU powered time and engine hours - starts - trips by cause - watchdog resets - time at each quarter of rated load - **Trip recorder.** For the last two overspeed or crank-signal trips: speed, fuel, rail pressure and state every 20 ms, from 2 s before to 1 s after. Normal ignition-off stops are not recorded, or the daily stop would overwrite the last real trip.

Surviving power loss and flash wear

- **Two copies.** The host keeps two copies of the image and writes them alternately. Each copy carries a sequence number and a CRC-32, and at power-up the newest good copy wins. A power cut in the middle of a write loses that one save, never the history.
- **Sealed copy for the host.** The image the host writes is a sealed copy, so the 1 ms tick can keep updating the live one during the write.
- **Saves only from the main loop.** A save happens on:
- a fault's first occurrence
- every trip (1 s after it, once the recording is complete)
- entering shutdown
- each start
- every hour of running

The rate is capped at one save per 10 s, except a trip. - **Wear.** That is about 30 saves a day, roughly 110 000 per copy over 20 years. The S32K148's storage has to be chosen for that, in the dev-kit spec.

Reading it

Over J1939, which the DSE4522 and standard service tools understand: - **stored faults (DM2)** - **clear stored faults (DM3)** - **fault snapshots (DM4)** - **engine hours**: on request, and broadcast every 10 s until we know whether the DSE asks for them (U4)

The event log and trip recorder need a laptop service tool over CAN (UDS), which belongs in the dev-kit spec. Until then, the simulator reads them straight from the image.

The watchdog needs the main loop now

Until now the 1 ms tick pulsed the watchdog on every pass. A main loop hung while interrupts still ran would have kept the watchdog happy forever, which is exactly the failure a window watchdog exists to catch.

The tick now pulses only if the main loop has checked in within 50 ms. If the loop hangs, the pulses stop, the TPS3850 resets the processor, and the next power-up logs a watchdog reset.

Three more fixes for long running

- **The millisecond counter wraps every hour by design.** It used to wrap after 49.7 days, which would have broken message timing once in the field.
- **A lost crank tooth is logged as a crank-sensor fault.** It used to be logged as overspeed, because the missing tooth makes that edge's speed read high. The decoder also drops its speed history on any fault, so one bad edge can't raise a second, false overspeed.
- **A fault message too long for one CAN frame is sent whole or not at all.** It is never half-sent.

What the tests caught while building

- **Faults read "active" after power-up.** A fault saved while active still read "active" after a power cycle. Load now marks every stored fault as previously active, because nothing is active until it is seen again.
- **Scenario 11 couldn't see the watchdog stop.** With the watchdog newly tied to the main loop, the simulator, which didn't run the main loop yet, stopped pulsing after 50 ms, and scenario 11 still passed. It now also checks that the pulses continue to the end.

What the final review caught

A fresh reviewer, acting as an off-highway ECU firmware specialist, found no critical defects. It did find four that would hurt in months of field running. Each now has a test that failed first: - **A loose sensor connector flooded the event log.** Every flicker logged "fault set" and "fault cleared", and in 10 s that pushed the overspeed record out. Each fault's set and clear is now logged once per ignition cycle, and ignition-off only after 500 ms off, so a bouncing contact logs nothing. - **A service tool reading fault snapshots could silence the DSE's fault list.** Long multi-frame messages queue one at a time, and the tool's requests always won. The active fault list (DM1) now goes first. - **A watchdog reset was never saved on its own.** If the operator power-cycled before anything else triggered a save, the evidence of the hang was lost. A watchdog-reset boot now requests a save, still under the 10 s limit, so a boot loop can't wear the flash. - **A packet of a long message could be dropped** if the CAN queue was full at that moment. It is now retried 1 ms later.

Proof

- **78 unit tests.**
- **15 scenarios.** The four new ones:

#	Scenario	Result
12	Runaway, then power cycle	overspeed fault stored with its snapshot (1650 rpm), shutdown event logged, trip recording kept (last sample before the trip 1633 rpm, 50 after)
13	Power cut halfway through a save	reload takes the previous save; nothing else lost
14	Stored faults over CAN	DM2 returns the stored fault; DM3 clears it and leaves an active one alone
15	One lost crank tooth at 1500 rpm	shutdown, stored as crank sensor (636), not overspeed

Still open (dev-kit spec)

- **Storage choice.** Pick storage for about 110 000 writes per copy.
- **Laptop service tool.** It reads the event log and trip recorder over UDS.
- **DM3 acknowledgement.** Whether the DSE expects one (U4).
- **Interrupt masking.** Mask the tick during the sealed-copy step on the real chip.

Addendum, 14 October: the Protect lamp

The problem: the DSE4522's configuration (`field-evidence-2026-09-18.md`) treats the DM1 **red stop** lamp as a *warning* and shuts down only on the **ECU Protect** lamp. Our firmware set red on its shutdown faults but never Protect. So after our ECU cut the fuel for overspeed, the DSE would have seen only the speed falling and shut down on *under-speed* (1310 rpm), logging the wrong reason.

Chosen: a shutdown fault (overspeed, SPN 190; crank signal lost, SPN 636) now sets **red and Protect** (DM1 byte 1 = 0x11). A sensor fault sets amber only (0x04), so a bad sensor never stops the set through the DSE.

Proof: three unit tests (79 in all): overspeed and crank loss each send 0x11; an open coolant sensor sends 0x04. All 15 scenarios still pass.

Still to confirm on the machine: that the DSE logs "ECU Protect" rather than under-speed when our ECU trips. That's a commissioning check.

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13. The board, prepared for hand routing (11–14 October)

The question: should the PCB be laid out here, or by a person? And since the new ECU must fit somewhere on the engine, where does it go?

Chosen (owner, 11 October): - **I prepare, a person routes.** The board file arrives with every part placed by block and every rule set. A layout engineer routes it by hand in KiCad 10. - **The board goes in the old ECU's die-cast housing.** The housing is also the heat sink. So the outline, the screw holes, the cooling contact areas and the J1001 header all copy the old unit.

What exists now

Item	Where
The board, 379 parts, 373 nets, placed by block	`hw/ecu25kva.kicad_pcb` (KiCad 10)
Stack-up, computed widths, net classes, housing placeholders	`hw/pcb_rules.py` (tested by `hw/test_pcb_rules.py`)
The generator	`hw/gen_pcb.py`. On a routed board it updates only the rules
The check	`hw/check_pcb.py`. Footprints, nets, classes, widths, DRC: all pass, 499 connections unrouted (expected)
Guide for the person routing	`docs/layout-guide.md` (the bible's PCB Layout Guide)
What to measure on the old ECU	inspection brief, Group 6

The decisions inside it

Item	Choice	Why
Stack-up	4 layers, 1.6 mm, Tg ≥ 170 °C; 2 oz outer, 1 oz inner; L2 a solid ground plane	Engine bay; high current on the outer layers; the plane is never cut
Track widths	Computed from current by IPC-2221, for a 20 °C rise. Never typed	The spec named IPC-2152; IPC-2221 has a formula and comes out wider, so it errs safe. 18 A → 5.35 mm, 6.4 A → 1.3 mm on 2 oz
Clearances	IPC-2221B, external, uncoated: 0.6 mm for anything from 31 V up (boost voltage; the EGR gates, which reach 52 V on a load dump); 0.2 mm for CAN and analogue; 0.15 mm for logic and low-voltage power on fine-pitch pins. Battery power at 0.3 mm is a **judgement**, not a table row: the table gives 0.1 mm up to 30 V	Safe even before conformal coating
Net classes	Ten: PWR_HI, RTN, PWR_LOOM, PWR, PWR_LV, HV100, HV100_SIG, CAN, ANALOG, Default	See below for the ones that differ from the plan
The EGR feed (pin 6)	Sized for its 10 A loom fuse, not the 6.4 A load: 135 % of rating (a blade fuse may carry that for up to 30 minutes, ISO 8820-3) = 13.5 A → 3.6 mm	Nothing on the board limits that feed below the loom fuse
Unclassified power nets	The check fails if any net touching a power part (power FET, power diode, inductor, fuse, milliohm shunt) has no class	A new power net can't silently get a thin trace
Pads inside one part	Exempt from class clearance (a custom rule)	A 0.5 mm-pitch chip can't meet a 0.3 mm class rule; that spacing is the maker's. The classes govern the copper the router adds
Single ground join	All high-current returns meet the board ground at J1001's battery-ground pins, and only there	Keeps injector and switching current out of the sensor ground (layout guide §4)

Three changes from the plan, found while doing it: - The injector's 100 V nets carry 18 A, not 1 A. The plan sized the 100 V class for 1 A. But the injector's peak current flows through the boost rail, the bank nodes and the low-side drains. So that class takes the 18 A width. A new class, **HV100_SIG**, gives the small nets that sit at boost voltage (gates,

bootstraps, divider midpoints) the 0.6 mm clearance without a 5 mm default width. - **Ground has its own class (RTN).** Ground was planned under the 18 A class, with its 0.6 mm clearance. That put 0.6 mm between every ground pin and its neighbour, which no fine-pitch chip meets. Ground, with the injector shunt tops, keeps the 18 A width at the default clearance. Anything at 100 V still gets 0.6 mm from its own class. The same reasoning gives 3V3, 5 V and the boost controller's pins the PWR_LV class at 0.15 mm. - **The fused pin-21 path is 6.4 A class, not 18 A.** It sits behind the 5 A fuse F101. The injector's power comes in on pin 4.

Placeholders, and what the measurements change

Item	Placeholder	Settled by
Board outline	180 × 130 mm	Group 6.1–6.2
Mounting holes	4 × M3, corners	6.1–6.2
Housing contact band	top 25 mm	6.1–6.2
J1001 footprint	6 large + 4 × 22 at an assumed 2.5 mm pitch, through-hole, marked "DO NOT FABRICATE"	**6.3**
Battery-ground pins	1 and 2, assumed. The schematic leaves them unconnected	5.1, 6.4
D806 footprint	SMA	the trip-loop pin, 5.2

When the measurements arrive, `HOUSING` and `REGIONS` in `pcb_rules.py` change. Before routing, re-run the generator. After routing starts, move the outline and J1001 by hand.

One finding the owner should know. The spec's placeholder outline was 160 × 120 mm, and **the parts don't fit on one side of it**. Packed, they need about 15,300 mm^2. At 160 × 120, the power-input block alone had no room beside the connector. The placeholder is therefore 180 × 130. If the old housing's board is smaller, our parts go on **both sides**. That is normal for an ECU, but it needs the housing's inside depth on both faces (Group 6.1 asks for it).

Two things the schematic doesn't have yet, which the layout guide reserves room for: - a common-mode choke at the power input and on each CAN pair (spec §6, needed for CISPR 25 Class 5) - test points

One KiCad trap. kicad-cli 10 silently ignores a custom-rules file it can't parse; KiCad 10 needs units on every number (`0mm`, not `0`). So `check_pcb.py` confirms the rules file is the generated one before it trusts the DRC.

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14. Calibrate from the old ECU's own recording (14 October)

The evidence: a 35-minute XCP recording of the original Bosch ECU on a running 3-cylinder engine at 1500 rpm (`refs/field-evidence-2026-10-14.md`).

Asked: should the firmware take its numbers from it? **Owner:** yes, "we need it to work right."

What changed:

Item	Before	Now	From
Metering valve	normally closed (`METER_NC 1`)	**normally open** (`METER_NC 0`): full duty stops flow	current falls as pressure rises; it rises to stop the engine
Injector law	300 µs + q / (0.0009 √p)	**323 µs + q / (0.00191 √p)**	fitted to 20,473 samples, within ±6 µs
Start of injection	5° BTDC fixed	**table on fuel**: 11.5° at 6.5 mg → 15.9° at 17 mg → 14.2° from 26.5 mg	the old ECU's own values, median per mg
Rail pressure	1000 bar fixed	**table on fuel**: 827 → 1171 bar	the same
Droop	4 % only with the droop switch closed	unchanged	the old ECU runs isochronous, and so does ours with the switch open

What it costs, and what is still open: - **The trip module must move.** With a normally-open valve, cutting the valve's supply gives *full* flow, so the trip removes no fuel. The trip has to cut the injector and boost supply instead. This is an **external wiring change, not the PCB**: the trip goes in series with the injector feed, pin 04. It is written up in spec §3 item 6 (14 Oct), with the field survey in inspection brief Group 7. If the module's own contact can't carry the 10 A feed, it switches a run relay that is energised to run. Not yet wired. - **The valve's part number** is the last confirmation. If it turned out normally closed, `METER_NC` goes back to 1; both polarities are tested. - **The tables cover 6.5–26.5 mg at 1500 rpm.** Above that, and at other speeds (cranking, the run-up), they hold their end values. Those parts need the real engine. - **The old ECU dips 9.8 % on a load step,** at G2's limit. Our model shows 3.4 % on a bigger step, so it is probably optimistic. Commissioning has to measure it.

Proof: 80 unit tests, and 15/15 scenarios. The engine model now uses the measured injector and valve too. Scenario 10 checks rail pressure against the firmware's live setpoint (worst 2.3 % at no load, 1.3 % at full load).

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15. A prototype board, buildable now (15–17 October)

The question: the production board waits on the Bosch connector and the old ECU's housing. Nothing has shown the design working. Can a board that sets both aside be built now?

Chosen (owner, 15 October): yes. Everything for it is in `prototype/` (spec `prototype/docs/spec.md`, plan `prototype/docs/plan.md`, bring-up guide `prototype/docs/bring-up.md`).

Question	Owner's answer
What goes on it	**The full production circuit.** Its eight sheets are built by the production generators, unchanged
MCU	**STM32G474VET6**, cheaper and easier to get than the S32K148. The production MCU choice is revisited once the prototype works
Dev board or one board	**One board,** MCU soldered on. A WeAct STM32G474 module is bought separately, for early firmware only
Assembly	**Fab-assembled.** The files are fab-neutral; the fab is chosen at quoting
Routing	**Done here,** by script. The owner does not route
Connector and housing	**Set aside.** Pluggable 5.08 mm screw terminals, labelled with the production pin numbers
High-current copper (16 Oct)	**Pours carry the current.** 585 of the 640 power pads are narrower than their class width, so no track can meet the class at the pad. Each power net's power parts are joined by class-width pours; tracks to small parts on a power net may be pad-width
Board size (16 Oct)	**300 × 200 mm** ("you can use bigger board dimensions"). 220 × 160 and 240 × 180 were too congested to route the injector stage at full width
Pin necks (16 Oct)	**0.15 mm, inside a part's courtyard only.** 0.2 mm (the 2 oz minimum) does not clear between 0.5 mm-pitch pins; 0.2 mm everywhere else
Layers (16 Oct)	**Stay at 4.** L2 stays a solid ground plane, never cut; signals use F.Cu, In2 and B.Cu

What it adds to production, and only to the prototype

- **Test points** on every rail, the boost rail (marked HV), CAN, the crank and cam edges and the four current senses: TP1–TP24.
- **A boost-rail monitor** `BOOST_MON`: 2 × 499 k + 24.9 k, so 116 V reads 2.82 V.
- **A battery monitor** `VBAT_MON` on the MCU sheet: 120 k / 10 k.
- **A 2 × 10 bridge header** for the future simulator board.
- **USB-C, data only.** It does not power the board, so the MCU rail has one source.

How the board is routed

- `prototype/power_router.py` routes the 29 power nets as class-width bars. It is a maze router on a 0.5 mm grid over B.Cu, F.Cu and In2. Nets negotiate for space (the PathFinder method) until no two overlap. Where a bar changes layer it gets a via array at 2 A per via. On In2, which is 1 oz copper, bars are twice as wide.

- **Freerouting 2.4.1** routes everything else. **It treats a KiCad pour as no obstacle** (in its own code, a DSN plane loads with `isObstacle=false`). So a first run routed signals straight across the power bars, which the refill would have cut. In its input each bar is now a fixed wire and each pour via a fixed via; they come out again after import. `test_router_sees_pours` guards this.
- **Ground reaches L2 through its own vias** (`gnd_fanout`), one per ground pad, so Freerouting never sees L2. It routes three layers; its vias come back through all four.
- **Freerouting leaves about 45 connections open** (run 12, 29 passes, 1 h 50 min). `prototype/finish_router.py` routes them, one at a time, with KiCad's DRC as the judge: anything DRC rejects comes out whole. What it needed, each found on a connection that would not route:
- **A start anywhere on the end's copper island**, not only its centre point. Off a 0.5 mm-pitch pin's centre is the only legal start beside a 0.3 mm clearance net.
- **Vias through a pour's interior**, with 1 mm of pour kept round the hole. A deeper bite into a pour's edge is allowed but priced as a 20 mm detour, and `necks()` judges it afterwards.
- **A last pass 50 mm round the ends on a 0.1 mm grid**, for HSB_BAT_GATE: 115 mm across the injector pours.
- **Rip-up**: other nets' tracks round the open ends, then along the line between them, are taken out; the open connection routes first, the rest after. Rounds that leave more open are undone.
- `tidy` removes the vias and stubs rip-up leaves dangling.
- **Result (17 October): nothing unconnected, 0 DRC errors, schematic parity clean** but for D806's placeholder footprint. Freerouting started from 625 nets.
- **Proofs:**
- `test_power_pours_connect`: with every track removed, every power part's pads are still joined, so no track carries load current.
- `gen_board.necks()`: every bar keeps at least 3 of 5 sample points across its width filled, outside its own pads.

Rules the routing needed (17 October)

Rule	Why	Cost if wrong
Pin escape: inside a fine-pitch chip's courtyard, a PWR or PWR_LOOM track keeps the chip's own 0.2 mm pin gap to that chip's pads, not 0.3 mm. The 100 V classes and PWR_HI keep theirs	U203 pin 13 (VSNS) lies between OVC_BST and OVC_GATE on a 0.5 mm pitch: even a 0.15 mm track dead centre is 0.275 mm from them. The chip itself holds those voltages across 0.2 mm. **Production has the same pin**	A creepage review of the 7 chips (U101, U202, U203, U301, U302, U401, U701)
Fine-pitch chips keep 1 mm clear round their courtyard in placement	Run 9 left U401, U203 and U504 walled in by their own satellites	A placement rerun

Rule	Why	Cost if wrong
R503 is turned 180° after packing	U504's gate-source resistor faced gate-down, source-up, against the driver's pin order: EGR_AH_G and EGR_HIGH_59 had to cross at 0.6 mm clearance, and EGR_A_VB was walled in	None: the same box
Finishing tracks on power nets are 0.2 mm	They reach small parts; the current rides the pours	None measurable
A power net with no pour carries its own current on tracks (`TRACK_AMPS` : 1 A on the 5 V buck's switch node and output, 0.05–0.3 A on the rest). Each track is IPC-2221 wide for that current on its layer. A net that would need more than 0.2 mm on In2 stays on the outer layers	Eight power-class nets have at most one power part, so the pour router gives them no bars. The 5 V buck's output ran 0.2 mm on 1 oz In2: 65 °C rise at the buck's 1 A, which a chafed sensor lead can hold (each sensor tracker limits at 301–520 mA). `test_track_power_widths`	A rerun if a load grows: the table is the budget

Found on the way

- **A production defect, fixed in `hw/` .** In the supervisor, the `~RESET` and `~WDO` pull-ups overlapped on the schematic. That shorted the TPS3850's reset and watchdog-fault outputs into one net. Fixed in commit 491f37e. `check_netlist.MUST_SEPARATE` now guards it.
- **That fix broke the watchdog kill (found in review, 19 October).** The TPS3850 asserts `WDO` on a watchdog timeout and never `RESET` (SBVS301B §7.3.6.3). Once the short was gone, a hung MCU no longer reset or killed the gates. **D302, a BAT54Q-7-F Schottky** (anode `~RESET` , cathode `WDT_FAULT`), puts the path back: `WDO` low pulls `RESET` to about 0.3 V. That is below Q301's 1.1 V minimum threshold, so `GATE_KILL` rises. `RESET` low leaves `WDO` alone. In production and on the prototype; `check_netlist` WDO_KILL guards it. **The cost:** the MCU is held in reset for as long as `WDO` is low, so `WDT_FAULT` no longer tells firmware which trip it was. No firmware reads it.
- **U606's library footprint** (Freescale 98ARH99066A) has a malformed courtyard. The prototype redraws it as a rectangle. **Production uses the same footprint** and passes the design-rule check only because of where it sits. Check it before production layout.
- **ST's datasheet DS12288 could not be downloaded** (st.com refused). The power-distribution check keeps production's 0.5 A load step: 70.3 mΩ cold and 78.5 mΩ warm against the 100 mΩ target. The STM32's real step is smaller.

The cost to know before quoting

- **The stack-up is production's:** 4 layers, 2 oz outer copper, 1 oz inner. The pour widths are computed for 2 oz.
- **A 1 oz quote is not a drop-in.** The bars would carry about half the current.
- **300 × 200 mm with 2 oz outer copper** is outside most fabs' cheapest prototype tier. Expect to pay for size and copper weight, not part count.

- **95 pin necks are 0.15 mm on 2 oz outer copper.** Some fabs' 2 oz minimum is 0.15–0.2 mm: ask at quoting.
- **20 small islands of pour fill** are left floating (current-sense pours on In2, three on the ground plane). They carry nothing; a fab may ask about them.

16. A service link: bootloader, end-of-line record, underspeed warning (17–20 October)

Prompted by a friend's screenshot of a comparable production ECU's firmware, a TargetLink model. It has a bootloader, an end-of-line write and an underspeed fix. Ours had none of the three.

The owner's answers (17 October)

Question	Answer
Underspeed action	**Warn; the DSE decides.** The DSE has its own under-frequency shutdown; the ECU does not stop the engine for underspeed
Link	**Both.** USB on the prototype for quick iterations; CAN channel 2 in production
End-of-line scope	**The unit record**, locked once written. A factory output-test mode is later work
Protection	**Seed-key**, and the engine must be stopped

What was built

All portable C, tested on the PC (firmware guide §16). Spec `docs/superpowers/specs/2026-10-17-service-link-design.md`.

Part	Decision	Why
Protocol	A subset of **UDS (ISO 14229)**, the same on both links	A technician's UDS tester can read the IDs; `python-udsoncan` could replace our tool
Underspeed	Below 1350 rpm for 3 s: SPN 190 FMI 18, **amber lamp only**; above 1400 rpm for 1 s clears it	The DSE shuts down on the Protect lamp, which every SPN 190 used to light. A warning must not stop the set
End-of-line record	84 bytes, two copies apart from the logs, trims applied to every input; none good means SPN 628 FMI 13	A unit that skipped the line says so on the DSE's screen
Bootloader	Image header written **last**, magic last of all; never touches its own 32 KiB	A power cut anywhere leaves no valid header: the update is just run again. Tested by cutting the power at every write
Hardware check	The bootloader runs an image only if the record's hardware revision is in the image's range	A good image for another board revision stays in the bootloader
Engine stopped	OFF, WAIT or SHUTDOWN and no tooth for 1 s; an open session ends at the first tooth; an ECU reset is refused while it turns	A laptop left connected, or a generic tester's reset button, must never stop a running set
End-of-line "passed"	SPN 628 stays on until the record carries the passed stamp	A station that writes the serial and then aborts must not ship a unit running silently on default trims

Found on the way

- **The SIL engine's load was constant power**, so any overload stalled it in under 3 s, faster than the warning. A real alternator's regulator rolls the voltage off below about 47 Hz, so a resistive load's power falls with speed. The model now has that roll-off (`AVR_KNEE_RPM` , an assumption); scenario 16 settles near 1190 rpm at 30 kW.
- The spec's record size (88 bytes) was wrong: its own fields pack to 84.
- **The final review (20 October)** found an ECU reset accepted on a running set, a lost CAN flow-control frame that silenced channel 2 until power-off, "stopped" reading "turning" after 36 minutes at rest, a partial record clearing SPN 628, USB served from the preemptible background, a save handshake that could lose a write, and the affine CRC-32 key. All seven fixed, each with a test that failed first.

Risks

Risk	What it costs	Answer
The seed-key secret is in every unit's flash	Anyone who reads one unit can compute keys for all	Acceptable for stopping casual writes. Signed images are the answer to tampering. (The first design's CRC-32 key was worse: it is affine, so one sniffed unlock gave keys for every seed. The final review caught it; the key is now SipHash-2-4)
A CAN update at 500 kbit/s	About 30 KiB/s: a 256 KiB image takes about 10 s plus the erase	USB on the prototype
The DSE's under-frequency settings are unknown (U4)	If the DSE ignores DM1 and has no under-frequency trip, nothing stops a set that sits overloaded	In the U4 questions
No rollback	A bad image that passes its CRC needs the previous one flashed	One application slot, by choice
Still to write: the chip layers	Flash, CAN, USB and reset drivers on the STM32G474 and S32K148	The port contract, firmware guide §14 items 10–13

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17. A simulator anyone can run (20 October)

Asked for: three simulators. (1) The firmware alone. (2) The prototype board fed sensor values and faults over USB, its firmware using them in place of its inputs. (3) The board on a bench rig that plays the engine. Each runs live or from a script, and everything is traced in a browser. This entry is mode 1. The owner's friend uses Windows: "it should be an easy setup for him too".

Question	Answer	Why
Language	**All Rust**: engine model, loop, server	One `.exe`, nothing to install. About 100 times faster than real time, against about 3 times for the Python model, so scripts run in seconds. The same model can later run on the mode 3 bench's microcontroller
The firmware	**Unchanged C**, built in, and loadable from the UI as a `.dll` / `.dylib` / `.so`	The simulator tests the code that ships. A loaded build is refused unless `ecu_abi_version()` and `ecu_io_size()` match
The UI	A web page served by the program itself, on 127.0.0.1 only	Works the same on every OS. `--share` lets a second PC watch, and then refuses firmware uploads, which run code
Scripts	`.seq` text: timed actions, `end`, and `expect` checks	Readable by someone who has never seen the code. A live session is saved in the same syntax, so it replays
The Python SIL	**Kept** as the firmware's gate	It checks what the script language cannot say: the sync edge count, watchdog periods, CAN intervals, the NVM image byte by byte

How it is checked

- The Rust engine model gives the Python model's crank edge times exactly, and speed and rail within 1e-6, over four open-loop profiles.
- The whole loop gives the Python SIL's 5 s start with identical states and speed within 1 rpm.
- All 16 SIL scenarios run as scripts and pass. CI runs them on Windows, macOS and Linux once the branch is pushed.

Found on the way

- The droop scenario claims a 4 % drop, not a speed. Scripts gained one check, `mean change`, for it.
- A DM1 with two faults is sent as a multi-frame BAM. The Python SIL reads only single frames, so a second fault would have gone unseen there. The simulator reassembles them.
- Operating systems hand back an already-loaded library for the same path, so a rebuilt firmware uploaded under the old name would have run the old code. Every load is a fresh copy.

Risks

Risk	Answer
Loaded firmware runs with the user's rights	The UI says so; uploads are refused under `--share`. Only the simulator's own page may send it anything: the final review found that any web site the user visits could POST a library to 127.0.0.1 (no preflight for a plain body), so the Host and Origin headers are checked
The Windows `.exe` needing the VC++ runtime	The C runtime is linked in (`+crt-static`, `/MT`), so nothing is installed
The binary is unsigned	SmartScreen and Gatekeeper warn once; the guide says how to proceed

Risk	Answer
A live session keeps its whole trace in memory	About 300 MB per simulated hour; streaming to disk is the upgrade if long sessions matter
`sin` and `exp` differ in the last bit between OSes	Traces can differ slightly across machines; the checks' bands absorb it

Guide: `simulator/README.md`. Spec and plan: `simulator/docs/`.

18. The Bosch pinout becomes the master (7–8 November)

Asked for: the board had been designed from KOEL's harness diagram (GP3.8703.C4). The owner sent Bosch's terminal diagram for the ECU itself (`T4.435.04.0.PR` , Bosch drawing 0 462 YU4 15D, "MD1CS162 KOEL CPCB4+ 3 cyl"), and the two disagree on power. The owner's answers (7 November): this project replicates the MD1CS162. The Bosch drawing applies to our 3GK550ETA4SR1, because the models it lists share the same sensors and ratings. The "0 281 ..." number on the old ECU is an access code and does not matter. Then: **"use only Bosch pins"**, and **"go with Bosch"**. The pin list is `refs/bosch-pinout-md1cs162.md` , read off a 400 dpi render.

Pin	KOEL diagram (what the board did)	Bosch (what it does now)
21	Permanent battery feed for the electronics	`I_S_WFLS` , the **water-in-fuel** switch, active high
04 / 06	Injector and EGR feeds, plus voltage sense	`V_V_BAT_1R/2R` , the main relay's contact, the board's **only supply**. 04 now also feeds the electronics chain
50	Main relay output, no special role	Closes the board's **own supply** relay
71	Ignition input	Ignition, and the **wake**: it closes the relay in hardware
01 / 02	Unknown, no power ground anywhere	`G_G_BAT1/2` , the **power ground**
09	"Synchronization ground", unassigned	`I_A_SYNC` , an analog input from the GCU
14, 25	Not shown	Diagnosis request; **stop / emergency switch**
CAN	55/77 "channel 1"	87/86 is Bosch's CAN 1 (1000 kbit/s, flashing and UDS); 55/77 is CAN 2 (250 kbit/s, J1939)

Question	Answer	Why
How the board comes up	**Pin 71 turns on a second FET on pin 50** (10k / 22k, 12 V zener): the relay closes in hardware, 04/06 come up, the MCU boots and holds the relay itself	With no permanent feed nothing else can close the relay before the MCU has power. Vgs 5.5 V at 8 V in, 12.0 V on pulse 2a, -0.7 V reversed (`main_relay.cir` A)
Where the electronics feed comes from	**Pin 04**, ahead of its own protection, beside the injector feed	2.95 A of electronics plus 3.8 A RMS of injector stays under the 10 A loom fuse. A blown 04 fuse also stops injection, so losing the ECU with it costs nothing extra

Question	Answer	Why
Pin 50's turn-off clamp	**A TVS to ground** (SMAJ48CA), not a diode to VBAT_PROT	See "Found on the way"
The after-run	Ignition off: the firmware stops the engine, asks for a save, and **releases the relay only once the engine has stopped and the save is written**, at most 60 s	The board drops its own supply. Releasing early would lose the stop's record. The save goes through the 10 s rate limit, and the relay waits for it, so a flapping ignition costs no extra flash writes
The trip loop	**Removed.** The overspeed trip module opens the main relay's coil feed	It had no Bosch pin. Opening the coil drops the ECU, the injectors and the metering unit together, and the GCU sees the CAN go quiet
Pin 25	An operator stop: SHUTDOWN, no DTC, no red lamp, latched until an ignition cycle. **Also in hardware since 10 Nov** (below)	Like the ignition. **Polarity as Bosch draws it** (contact to terminal 31 = stop), the owner's decision on 10 Nov; seen once at first power-up (bring-up step 6)
Pin 21	Amber DTC, SPN 97 FMI 31, after 2 s; clears after 2 s dry	A warning, never a stop. 2 s each way so fuel sloshing past the probe does nothing
Pins 09 and 14	Read and passed to the firmware; **no function yet**	Bosch's drawing names them but does not say what the GCU sends. 09 reads 0–10 V
CAN	Net names kept; the service link moves to **1000 kbit/s**	Renaming nets churns three sheets for nothing. The mapping is written on the connector sheet

Found on the way

- **A sneak path that would have held the relay in forever.** The relay driver had a flyback diode from the drain to VBAT_PROT. That was safe while the coil was fed from the board's own battery feed. Under Bosch's wiring the coil sits on Term. 30, always live. With the relay open, the coil and diode fed the "unpowered" board: 22.8 mA through the coil, 2.7 times its release current, with the board at 9.1 V (`main_relay.cir` B). The relay could never drop out. Bosch marks pin 50 "not equipped with freewheeling diode": the diode belongs in the relay. The netlist check now fails on any diode from pin 50 to a supply.
- **The decoupling check had been passing on nothing.** KiCad 10's `kicad-cli` writes the netlist one element per line. The check's pattern expected KiCad 7's inline form, found no parts, and passed on "0 supply pins". It now parses either form, and fails if it finds none. It checks 50 pins, all decoupled.
- **Two open items closed.** The power ground (since 19 September) and the trip loop's missing pin. The netlist check reports none open.

Proof

- `run_sim.py` : 26/26 blocks. `main_relay` is new; `trip_module_sense` is retired with the loop.
- `check_netlist.py` : connectivity OK, 0 open items. Five new checks: GND on pins 01/02, the 5 A fuse on pin 04, pin 21 an input, no diode from pin 50 to a supply, and two drivers on pin 50 with one woken from pin 71.

- Firmware: the unit tests pass (seven new), as do the boot tests, 16/16 SIL scenarios and the service tool. Scenario 7 now checks that the relay opens 22.6 s after ignition off: after the engine has stopped, and after the stop is saved. `ECU_ABI_VERSION` is 2.
- `check_pcb.py`: footprints, links, nets, classes, widths and DRC pass after `gen_pcb.py`. Schematic parity fails on the Windows machine this was done on, 198 entries, almost all pins with no net. It fails the same on the unchanged board (199), so it is the KiCad build, not this change. It passed on the macOS setup. **Found 9 Nov:** KiCad 10.0.6 drops every pin on an auto-named net (`Net-(...)`, `unconnected-(...)`) from parity, and only those. `check_pcb.py` now checks exactly those pins against KiCad's exported netlist, and parity passes. A pad on the wrong net still fails.

The prototype, brought up to it (8–9 November)

The routed prototype board was updated in place, not placed and routed again (`prototype/gen_board.py`):

- `migrate` records what `sync` cannot decide on its own (`BOSCH_MIGRATE`):
- `/VBAT_IN` becomes `/V_BAT_1R` with its copper. The same rename goes into the committed router inputs `build/power.json` and `build/prototype.dsn`, which the board tests read.
- The trip loop's parts go.
- U301, J1001, J1007 and F101 take the new pad nets. J1007 grows from 6 to 9 positions, with pin 1 kept in place.
- **J1001-1 and -2 are joined by a 2.5 mm F.Cu link.** Both positions are pin 04 now, but their pours had belonged to two nets and did not meet.
- It is safe to run again.
- `sync` **places the 30 new parts.** Grown from the MCU end, the first try put three inputs' front ends 90–150 mm from J1007. Placement now:
- starts from the connector and works inward
- ignores GND and the supply rails when choosing a neighbour
- on a tie, prefers a part placed in the same run

Every input's protection and divider now sits at J1007. - **The three new inputs moved on the MCU:** water in fuel, diagnosis request and the stop switch went from PE7–PE9 to **PD8–PD10** (`stm32_pins.py`). PE7–PE9 are three neighbours mid-way along the bottom row, in the CAN lines' corridor. Routed there, PE8 was walled in, and so were three CAN lines and SENSOR_5V_MON. PD8–PD10 are free pins on the side facing J1007. The firmware reads them through `ecu_io`, so it does not change. - `gnd_stitch` gives each new F.Cu ground pad that reached no via a via to the L2 plane. `pour()` does this for every part, but `sync` did not, so 5 pads lacked one (Q807, Q808, C808, D807, C616). - **D806 is no longer a**

placeholder (`proto_rules.PLACEHOLDER` and `PARITY_OK` are empty). The reference now names the water-in-fuel clamp, a BAV199. - Routing: `finish` and `finish_pass` routed all 42 new connections. A `rip` stage clears a net's tracks before its pins move.

The result: **0 DRC errors, 0 unconnected items, schematic parity clean** (KiCad 10, this machine).

Pin 25 kills the gates in hardware (10 November)

The owner: pin 25 is the operator's force shutdown, pressed when the generator is not behaving properly. So it has to work when the firmware is what is misbehaving. Until now it reached only the firmware.

The change (`hw/gen_sheet_mcu.py`, `stop_kill()`, on both boards): - Q301, the inverter that raises `GATE_KILL`, now takes its gate from `~RESET` through a 100k (R307; R314 on the prototype). - A 2N7002BK (Q302) with its gate on `SW_STOP` pulls that gate low while the button is pressed. `GATE_KILL` rises and every injector, metering and EGR gate drops, with no firmware in the path. - `~RESET` stays at 3.0 V, above the MCU's 2.31 V threshold, so the MCU runs on, sees the stop and latches SHUTDOWN. Releasing the button does not restart injection.

Proven by: - `sim/blocks/stop_kill.cir`, at the worst-high FET threshold: - pressed: 11.9 V on `GATE_KILL` - a 1 kΩ contact still kills; 47 kΩ of harness leakage does not; it trips at 10.8 kΩ - 0.9 ms from press to kill - `check_netlist` STOP_KILL, on both boards.

Polarity: the drawing's (contact to terminal 31 when pressed), chosen by the owner on 10 Nov. If the machine shows a normally-closed contact, this path and the firmware both invert.

Still open

- **The Rust simulator** mirrors `ecu_io`. Its fields and ABI are updated, but there was no Rust toolchain on this machine to build it. CI will build it, and so will a `cargo` on any of the three OSes. The committed Windows `.exe` is the old build until then.
- ~~At the machine: whether the fitted main relay carries its own suppressor.~~ Closed 10 Nov. Pin 50 drives the main relay and 69 the buzzer relay (owner). Neither needs to know about suppressors:
- Pin 50's TVS works with one in the relay or without.
- Pin 69's flyback diode (D811) goes to the board's supply. That is safe because the buzzer coil is fed from the same main-relay line, as Bosch draws it, so there is no sneak path when the relay opens.
- **Pin 25's polarity: as Bosch draws it** (owner, 10 Nov 2026). The switch is the external stop/emergency button, and pressing it connects pin 25 to terminal 31. It is no longer a question for the machine; first power-up shows it (bring-up step 6).

- **Pins 09 and 14 (owner, 10 Nov):** they stay as an analog input (09, 0–10 V into the ADC) and a digital switch input (14, contact to ground), at their current ratings. What they do is configured in firmware later. Until then the firmware reads both and acts on neither, so neither can change how the engine runs.

19. The boost cannot run away; heat and coating (12 November)

Asked for: the owner asked what still bothered us, then for fixes for the 100 V boost and for heat: **"fix this"**.

Found: nothing stopped the boost in hardware. The TPS40210 holds 99.7 V through 2 × 499 k into FB. If either resistor opens (a cracked 1206, a lifted joint), FB reads 0 V and the controller runs at maximum duty. Its DIS/EN pin was tied to "on" through 10 k, and nothing else stopped the rail before a 150 V part failed (FET, diode or the 160 V capacitor stack). The production board doesn't measure the rail at all.

Question	Answer	Why
The cutoff	**A 110 V zener (BZG03C110-HM3) from BOOST_100V, through 470 Ω, into DIS/EN**; the pulldown goes from 10 k to 470 Ω	DIS/EN is an active-high shutdown, 0.7–1.3 V, rated 10 V (TI SLVS861F §6.1, §6.5, §7.3.7). 470/470 puts the zener at 1.5–2.8 mA at the trip, near its 5 mA test current, so its 104–116 V range applies. **Trip 105.0–119.3 V** over both corners; with feedback open the rail is held at 119.3 V, dissipating 0.66 W in the zener (`run_sim.py boost_ovp`)
Above the trip	**A 5.1 V zener (BZX84-C5V1-Q) on DIS/EN**	Holds the pin at 6.2 V with the rail at 160 V, under its 10 V rating
The MCU's stop	**`BOOST_DIS` : PTD15 (production), PD14 (prototype), through 100 Ω and a BAT54Q into DIS/EN**	2.1 V on EN at the MCU's 2.9 V minimum high, 4.5 mA. The Schottky keeps a low or Hi-Z pin from loading the zener's divider. Hi-Z at reset leaves the boost running, as before
Firmware	**Above 106 V for 20 ms: `boost_dis` , SHUTDOWN, red DTC SPN 520192 FMI 0**, latched to the next ignition cycle. The boost is also off in SHUTDOWN	With the boost off, 12V_GATE collapses and nothing can inject, so the stop is made explicit. 520192 is the first proprietary SPN: J1939 has none for an injector boost rail. Production has no rail channel (`v_boost_rail 0`), so there only the zener guards it
Passives at the fault peak	**BOOST peak 104 → 120 V** in `assign_passives.py`	Nothing changes size: the 0805s on the rail are rated 150 V

Heat. The hot parts were sized against a housing the prototype doesn't have: Q104 at 139 °C junction with the EGR valve stalled at 6.4 A, on a 105 °C board. The firmware had no EGR control loop, so it had no stall limit either. **Now it has a stall guard.** Over 5 A for 1.5 s, the bridge coasts (`egr_off` : the valve springs shut) for 5 s and an amber DTC is raised: SPN 2791 (EGR valve control), FMI 6. The third stall holds it off until the next ignition cycle. EGR is never a reason to stop the engine. Bring-up step 7 adds a 10-minute full-load heat check with a 100 °C limit.

Spacing. Every 100 V net keeps 0.6 mm. That is IPC-2221B's B2 row for an uncoated external layer up to 150 V, so it's fine on a bench. Raising it to 1.0 mm would fail hundreds of places, and many of them are inside the FET packages themselves, where layout can't help. So **the board is conformal-coated before it goes on the engine** (acrylic or silicone, connectors and test points masked). The bring-up guide says so, and so does the layout guide for production.

The prototype, brought up to it. `migrate boost_ovp` refits R403 (10 k → 470 Ω) and U301 (PD14 now drives `BOOST_DIS`); `sync` places the five new parts beside U401. Sync dropped R437 inside the ISNS_BOOST pour (Q401's source to the 62 mΩ sense resistor), and its first routing cut that pour in two (`test_power_pours_connect`). R437 was moved to the nearest clear spot, left of the pour, and the four nets were re-routed in one pass. 26/26 prototype tests pass, with 0 DRC errors, nothing unconnected, and the fab files regenerated.

ABI 3. `ecu_io` gains `v_boost_rail`, `i_egr`, `boost_dis` and `egr_off`. The Python SIL and the Rust simulator's copies are updated. The simulator wasn't rebuilt (owner: "no simulator").

20. The application in Simulink and Stateflow, with generated code (13 November)

Asked for: "a complete state flow model in MATLAB Simulink that is able to generate code, and the new generated code we should be able to flash in our ECU". The owner left the how to us ("do the best you can"). Design: `docs/superpowers/specs/2026-11-13-simulink-model-design.md`.

Question	Answer	Why
What the models own	**The application layer**: the engine state machine, governor, rail control, the two calibration tables, the injector law, the trip decision and the five timed guards. Twelve models	That is the logic engineers tune and review. Tooth decoding, J1939, UDS, storage and the bootloader are event- and byte-driven platform code, already tested, and stay C
How they plug in	**Drop-in components** behind today's functions (`gov_step`, `mode_cmd` ...), selected by `ECU_MODEL`	One model for the whole tick would either slow the per-tooth trips (overspeed, sync) to the tick or split one state machine between C and model
The state machine	**A Stateflow chart driven by commands** (IGN_ON, TOOTH, RUN_OK, TRIP, RESET)	The tooth interrupt and the tick both change the state. `ecu.c` had five places that wrote it; all five now ask the chart
The guards	**Five small charts**, each its own model (`mode.c` holds the hand-written reference, moved out of `ecu.c` unchanged)	They reset at different times (water in fuel only at power-up) and run at different points in the tick
Where constants come from	**The C headers and tables**, read by `model/ecu_params.m`	A calibration typed in two places drifts. The `.slx` files are built by scripts, like the schematics

Question	Answer	Why
Code settings	ERT, reusable functions, inlined parameters, single precision, ARM Cortex-M, shared utilities	Runs on the STM32G474 and the S32K148 alike; no `double` on a single-precision FPU
Which is the reference	**Both builds, for now**	The hand-written modules stay until the generated build has run on the engine

The proof. Every unit test and all 16 SIL scenarios pass on the generated build. A back-to-back test (`model/test/run_b2b.sh`) runs each generated model against its hand-written original on millions of random inputs; the six charts are run 3 million random ticks each, reaching every outcome (84 boost trips, 143 EGR stalls, 46 after-runs, every engine state). They agree exactly, apart from one or two units in the last place in the two tables, where Simulink orders its interpolation arithmetic differently. The closed-loop model (`ecu_app` : the real component models with a mean-value engine) starts, holds 1500 rpm, takes 20 kW with a 1444 rpm low and stops on pin 25 in Simulink (`model/test/sim_ecu_app.m`).

Found on the way: - The governor first came out one unit in the last place off. The compiler, not the model, did it: it fused `set + RAMP*dt` into one multiply-add in the hand-written C. With `-ffp-contract=off` both agree to the bit. - Simulating the charts in Simulink caught locals read before they were first written, which the C tests could not see. They start at zero in the generated C either way, but every chart now says so. - `python -m ziglang cc` swallows the compiler's error messages on this machine. Two silent failures (a duplicated helper across two models, a missing source file) came from that; `zig.exe` called directly prints them.

What it does not do yet: run on the board. The generated code is part of the firmware image and is flashed with it, over SWD or the UDS bootloader. But the STM32 chip layer (ADC, timers, CAN, USB, flash) is still unwritten, for either build.

14 November, closing what still bothered us (the owner: "can we fix any of these"): - **The boost's silent clamp, and a dead boost.** The hardware zener can hold the rail at 105 V, below the firmware's 106 V trip, and nothing said so. A dead boost showed only as "won't start". Now: held over 104 V for 2 s raises amber SPN 520192 FMI 16; under 80 V for 0.5 s while cranking or running raises amber FMI 18. Neither stops the engine. Both are in the C and in a new chart, `ecu_boostm` ; the back-to-back test agrees on every tick. - **The state machine owns its conditions.** `ecu.c` used to decide when the ignition counted and when an ignition cycle had finished; it now sends TICK every millisecond and the chart decides both. The Simulink closed loop lost the copy of that logic it carried. - **One command checks everything:** `./verify.sh` . - **A test that passed by luck.** The hour-counter test depended on the edge

generator's leftover fraction of a microsecond from earlier tests: that bias decided whether the governor's integral drifted up or down over a simulated hour. Adding a test before it flipped it. Each test now starts the edge clock clean.

Where these came from

Nothing here was found by reading the schematic. Each one surfaced when a part number was chosen and its datasheet was checked against what the drawing assumed — which is the same shape as every earlier finding in this project: a document and an executable check disagreeing, with nothing comparing them until someone did.

Firmware Guide: how the engine-control software works

Date: 2026-10-11 **Covers:** `firmware/` as merged on master (commit `1149ecc`), plus the service link of 20 October (§16): bootloader, end-of-line record, underspeed warning. **Read with:** decisions log §11 (why it is built this way), §12 (the logs), §16 (the service link), and the specs in `docs/superpowers/specs/2026-10-1*`.

This guide explains what the firmware does and how, in the order a new engineer needs it: - what runs where - how one crank tooth and one millisecond are handled - the control loops - safety - CAN - the logs - every tuning knob - how it is tested - what the S32K148 port must provide - the service link: the laptop tool, the end-of-line record, the bootloader

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1. What it is, and where it runs

The firmware is **plain C (C99) that touches no hardware.** It has: - no operating system - no memory allocation at run time - no waiting loops - no chip headers

Because of that, the same files run in three places:

Where	Who calls the firmware	State
A PC	`firmware/sim/run_sil.py`, through a Python model of the engine, generator and fuel rail	**Working: 15 scenarios pass**
NXP S32K148 evaluation board	interrupts and a main loop on the chip, with an Arduino faking the engine signals	Next: the dev-kit spec
Our own ECU board	the same, on our PCB	After layout

It is **bare-metal, not RTOS-based**, on purpose. The work is one 1 ms loop plus a crank-tooth interrupt. A scheduler would add stacks and priority problems without adding anything this ECU needs. The short, fixed overspeed path is also easier to prove without one. If a customer ever requires AUTOSAR OS, the 1 ms entry point becomes a 1 ms task and nothing inside the core changes.

2. Map of the code

File	Job
`core/ecu.h`	The whole interface: the `ecu_io` struct and the entry points. Start here.
`core/ecu.c`	The state machine, the 1 ms schedule, trip handling, the save policy and the watchdog
`core/crank.c`	Reads the 60-2 crank wheel and the cam: synchronisation, speed, angle

File	Job
`core/inject.c`	Turns fuel and angle into per-cylinder injection commands
`core/govern.c`	Speed governor: PI, setpoint ramp, droop
`core/rail.c`	Rail-pressure loop on the metering valve
`core/sensors.c`	Volts to engineering units, range faults
`core/safety.c`	The trip decision and the table of active faults
`core/mode.c`	The engine state machine and the guards that keep their own time (boost, EGR stall, water in fuel, underspeed, main relay), stepped by `ecu.c` (§17)
`core/model_glue.c`	`ECU_MODEL` builds: the same functions from the code generated out of `model/` (§17)
`gen/`	The C Embedded Coder generated from the Simulink/Stateflow models in `model/`. Committed; do not edit
`core/j1939.c`	Every CAN frame the ECU sends, and request parsing
`core/nvm.c`	The persistent image: layout, checksum, choosing between the two copies
`core/log.c`	Writes into the image: events, counters, fault memory, trip recorder
`core/isotp.c`	ISO-TP framing for the service link on CAN channel 2
`core/link.c`	USB framing for the service link: sync byte, length, CRC-16
`core/service.c`	The UDS core both sides share: sessions, seed-key, tester present, reset
`core/eol.c`	The end-of-line record: defaults, checksum, trim limits, picking a copy
`boot/boot.c`	The bootloader: run the application or stay, erase, download, check
`tools/ecu_service.py`	The laptop service tool (§16)
`sim/test_core.c`	101 unit tests of the C code
`sim/test_boot.c`	7 bootloader tests against a simulated flash
`sim/ecu.py`	Python wrapper around the compiled core; `Unit`, a simulated board for the tool
`sim/engine.py`	The engine model
`sim/run_sil.py`	The 16 scenarios
`sim/test_service.py`	The service tool end to end against `Unit`

All of the core's state lives in `static` variables inside `ecu.c` and `mode.c`. There is one ECU per program, and `ecu_init` resets it.

3. The interface

3.1 The `ecu_io` struct

All the firmware sees of the outside world is this one struct. The platform writes the inputs and reads the outputs.

Group	Fields	Meaning
Inputs	`v_clt`, `v_oil`, `v_rail`, `v_boost`, `v_egr`	Sensor voltages at the ADC pins (V)
	`v_5v`	Measured 5 V sensor supply; the ratiometric sensors are corrected by it
	`v_bat`	Battery voltage, recorded in fault snapshots
	`ignition`, `droop_sw`	Switch inputs, 0 or 1
	`reset_cause`	Set once before loading: 0 power-on, 1 watchdog, 2 other
Outputs	`inj[3]`	Per cylinder: start time, duration (µs) and a sequence number `seq`
	`meter_duty`	Metering-valve duty, 0 to 1
	`wdg`	The watchdog pin level
	`can[16]`, `can_n`, `can_drops`	CAN frames to send; the platform sends `can_n` of them, then zeroes it
	`nvm_save`, `nvm_slot`	"Write the image to this copy now", cleared by the platform when done
Status	`state`, `rpm`, `fuel_mg`, `rail_bar`, `clt_c`, `oil_kpa`, `fault_n`, `synced`	For display and tests
Service link (§16)	`boot_ver`	Input: the bootloader's version, for UDS 0xF180
	`usb_in[64]`, `usb_in_n`	Bytes received on USB; the next `ecu_tick_1ms` consumes them and zeroes the count (at tick priority: they change state the interrupts share)
	`usb_out[512]`, `usb_out_n`	Bytes to send on USB; the platform sends them, then zeroes the count
	`can2[16]`, `can2_n`	Frames to send on CAN channel 2, as `can[]`
	`eol_save`	"Write the end-of-line record to copy a, then copy b", cleared when done. The image does not change while it is set; a later change asks again once it is cleared
	`boot_req`, `reset_req`	Set the stay flag and reset into the bootloader; a plain reset
	`session`	Status: the UDS session (1 default, 2 programming, 3 extended)
Bosch pinout (ABI 2)	`stop_sw`, `wif`, `diag_sw`	Inputs, logical (the platform applies each contact's polarity): pin 25 stop, pin 21 water in fuel, pin 14 diagnosis request
	`v_sync`	Input: pin 09 volts, the GCU's sync. Read, no function yet
	`main_relay`, `afterrun`	Output: hold the main relay (pin 50), the board's own supply; status: holding it after ignition off
Boost and EGR guards (ABI 3)	`v_boost_rail`	Input: BOOST_100V volts. 0 means not measured (production has no channel; the prototype reads `BOOST_MON`)
	`i_egr`	Input: EGR bridge current (A)
	`boost_dis`	Output: 1 = stop the boost converter (`BOOST_DIS`, DIS/EN)
	`egr_off`	Output: 1 = the EGR bridge coasts (`EGR_EN` low)

Injection is given as commands, not pin levels. For each cylinder the command is "start at time T, for D microseconds". The chip's timer output-compare runs that directly. Every change to a command, including a cancel (duration 0), increments `seq`, so the platform only needs to watch `seq`. A cancel stops a command that hasn't started. An injection already open runs to its end.

3.2 Entry points

Function	Called by	How often	Does
`ecu_init`	start-up	once	Resets everything
`ecu_nvm_load`	start-up, after init	once	Loads the newer good copy of the image, logs the power-up
`ecu_crank_edge`	crank-sensor interrupt	every tooth (about 1450/s at 1500 rpm)	Decodes the tooth, **checks overspeed**, schedules injection
`ecu_cam_edge`	cam-sensor interrupt	once per 2 revolutions	Marks which revolution has cylinder 1 firing
`ecu_tick_1ms`	1 ms timer interrupt	1000/s	Everything slow (§5)
`ecu_can_rx`	CAN receive	per frame	Answers J1939 requests (§9)
`ecu_background`	main loop	as often as it likes	Saves, and checks in for the watchdog
`ecu_nvm_image`	main loop	when `nvm_save` is set	Gives the sealed image to write
`ecu_eol_load`	start-up, after init	once	Loads the good copy of the end-of-line record; neither good means SPN 628 FMI 13 and untrimmed inputs
`ecu_eol_image`	main loop	when `eol_save` is set	Gives the record to write
`ecu_can2_rx`	CAN channel 2 receive	per frame	The service link (only ID 0x7E0)
`ecu_abi_version`, `ecu_io_size`	the simulator, on load	once	The interface version and `sizeof(ecu_io)`; `ecusim` refuses a firmware library whose values differ from its own. Bump `ECU_ABI_VERSION` with any change to `ecu_io` or an entry point

One rule for the port: `ecu_crank_edge`, `ecu_cam_edge`, `ecu_tick_1ms`, `ecu_can_rx` and `ecu_can2_rx` share data and must not interrupt each other. Give their interrupts the same priority. `ecu_background` runs at the lowest level and may be interrupted at any time.

4. One crank tooth

This is the fastest and most important path. At 1500 rpm a tooth arrives every 0.67 ms.

1. **Noise filter.** A tooth that arrives in under half the previous tooth's time is ignored. No engine accelerates that fast, so it can only be noise. A tooth arriving after more than 50 ms of silence restarts the decoder's

history.

2. **Gap and sync.** The wheel has 58 teeth and a gap where two are missing, so the gap shows up as a period more than 1.5× the last one. Two gaps exactly 58 teeth apart mean **synchronised**. Once synchronised, a gap in the wrong place, or no gap where one belongs, is **sync loss**.
3. **Phase.** The crank turns twice per engine cycle, and the cam pulse says which turn has cylinder 1 firing. Until a cam pulse has been seen, no fuel is injected. A cam pulse in the wrong turn stops injection until the next one agrees. It does not shut the engine down, because one noise spike shouldn't stop a genset.
4. **Speed.** rpm comes from the last 6 tooth periods, with the gap counted as its true 18°.
5. **Safety, on this very tooth.** If speed is above **1650 rpm**, the ECU shuts down here, inside the interrupt, and cancels every injection that hasn't started. If this tooth itself raised a sync fault (a lost tooth reads as a speed jump), it is diagnosed as a crank sensor fault, not overspeed.
6. **Injection.** Each cylinder starts injecting **11.5–16° before top dead centre**, from a table on fuel copied from the old ECU (11.5° at 6.5 mg, 15.9° at 17 mg, 14.2° from 26.5 mg up). Cylinder 1's TDC is 90° after the gap; cylinders 2 and 3 follow at 240° and 480°. The command is written on the last tooth at least **1° before** that angle, so the start time is never already past. The start time is extrapolated from the current tooth speed.

5. One millisecond

`ecu_tick_1ms` runs, in this order:

1. The ms counter, which wraps every hour (3 600 000) by design.
2. **Watchdog:** toggle the WDI pin every 5 ms (a falling edge every 10 ms), but **only if the main loop has checked in within 50 ms**.
3. Once a second: the counters (§10).
4. Ignition handling: the shutdown latch (§6), and the ignition-off event after 500 ms off.
5. Crank timeout check: 50 ms without a tooth while synchronised is sync loss.
6. Sensors: convert, range-check, update the fault table (§8).
7. Safety check (overspeed, sync loss, ignition off).
8. Governor (in RUN) and rail loop (in CRANK and RUN).
9. CAN messages due this millisecond (§9).
10. A trip-recorder sample every 20 ms.
11. The status fields.

6. The state machine

```
OFF ──ignition on──▶ WAIT ──first tooth──▶ CRANK ──phased and ≥ 400 rpm──▶ RUN
                                                          (any state)
            overspeed > 1650 rpm  ─┐
            sync loss while RUN   ─┼──▶ SHUTDOWN (latched)
            ignition off          ─┘
SHUTDOWN ──ignition off ≥ 500 ms with the engine stopped, then on──▶ OFF ──▶ WAIT
```

- **WAIT:** powered, ignition on, waiting for the engine to turn.
- **CRANK:**
- the starter is turning the engine
- fixed start fuel of 30 mg once phased
- a decoder fault just restarts synchronisation
- **RUN:**
- the governor owns the fuel
- on entry the speed setpoint starts at the current speed and ramps at 500 rpm/s to 1500, which stops the start from overshooting towards the trip
- **SHUTDOWN:**
- no injection
- metering valve closed
- fault codes active
- CAN keeps reporting, so the DSE sees the engine run down

Underspeed is a warning, not a state. In RUN, once the setpoint ramp has finished, speed below **1350 rpm for 3 s** raises SPN 190 FMI 18 in DM1 with the **amber** lamp only. Speed above **1400 rpm for 1 s** clears it. The engine keeps governing: the DSE's own under-frequency settings decide whether the set stops. 1350 rpm is also the load-step scenario's floor, so a normal load step never raises it. (The red and Protect lamps stay with the real shutdowns: overspeed, SPN 190 FMI 0, and sync loss.)

The latch clears only on a real ignition cycle: ignition off for at least 500 ms *with the engine stopped*, then on. A bouncing contact or a flicker while the engine still turns does nothing.

7. The control loops

Governor (`govern.c`): - PI on the speed error, output = fuel per injection, clamped to 0–45 mg. - Isochronous at 1500 rpm. - With the droop switch closed, the setpoint falls 4 % from no load to full load (34 mg, about 20 kW). - The integrator is clamped (anti-windup). - Gains: 0.5 mg/rpm and 2.0 mg/(rpm·s), tuned in simulation, which gives a 3.4 % dip (1500 to 1449 rpm) on a full 20 kW step.

Rail pressure (`rail.c`): - PI drives the metering valve to a **setpoint that follows fuel**: 827 bar at 6.5 mg rising to 1171 bar at 26.5 mg, from the old ECU's own table at 1500 rpm (held flat beyond the data). - The valve is **normally open** (`METER_NC 0`): no power, full flow; more duty, less flow; full duty stops it. The old ECU's recording shows exactly this (decisions log §14). The valve's part number is the final confirmation. Because losing the drive now means full flow, **the trip module has to act on the injector supply**, not the valve's: a hardware change, still open. - In SHUTDOWN the valve is commanded closed. - If the rail sensor fails, the valve is held at a fixed 30 % opening (duty 0.7 for a normally-open valve), so the engine keeps running.

Injection duration (`inject.c`): duration = 323 µs dead time + fuel ÷ (0.00191 × √rail pressure). Both constants are fitted to the old ECU's own energising times (20 473 samples, within ±6 µs). Rail pressure is floored at 100 bar, so a failed or zero reading can't produce an enormous pulse.

8. Sensors

Every input is corrected by the measured 5 V supply. If that supply reads outside 4.5–5.5 V, 5.0 V is used instead.

Sensor	Scaling (placeholder)	Valid volts	Default on fault	J1939 SPN
Coolant temperature	NTC 2.5 kΩ at 20 °C, β 3450, 2.49 kΩ pull-up	0.1–4.8 V	90 °C	110
Oil pressure	0.5–4.5 V = 0–1000 kPa	0.2–4.8 V	300 kPa	100
Rail pressure	0.5–4.5 V = 0–1800 bar	0.2–4.8 V	1000 bar	157
Boost pressure	0.5–4.5 V = 50–400 kPa	0.2–4.8 V	100 kPa	102
EGR position	0.5–4.5 V = 0–100 %	0.2–4.8 V	0 %	27

End-of-line trims. Each input, and the 5 V reference, is corrected as `v × gain + offset` before anything else, with the six gains and six offsets from the unit's end-of-line record (§16). The factory measures them; the limits are gain 0.9–1.1 and offset ±0.1 V. With no good record the trims are 1 and 0, and SPN 628 FMI 13 ("out of calibration") stays on DM1, so a unit that skipped the line says so on the DSE's screen.

Range faults: - Out of range for **100 ms** raises a fault: FMI 3 for too high (an open circuit), FMI 4 for too low (a short). - **From the first moment out of range, the last good value is held.** The 100 ms delay only delays the fault code; it never lets a bad value through. - While a channel is faulted, its default is used for control and "not available" (0xFF) is sent on CAN. - The coolant low limit is 0.1 V, not 0.2 V. 0.2 V is a real 129 °C, and a hot engine must not be mistaken for a shorted sensor.

9. CAN (J1939, source address 234)

Sent all the time:

Message	PGN	Every	Carries
EEC1	61444	20 ms	Engine speed (0.125 rpm/bit)
EFL/P1	65263	500 ms	Oil pressure
ET1	65262	1 s	Coolant temperature
DM1	65226	1 s	Active faults and lamps: red + Protect = shutdown fault, amber = sensor fault. The DSE shuts down on Protect and treats red as a warning, so a shutdown fault must set both
HOURS	65253	10 s	Engine hours (0.05 h/bit)

Sent on request (a J1939 request addressed to 234 or to everyone):

Request	PGN	Answer
DM2	65227	Stored (previously active) faults
DM3	65228	Clears stored faults; active faults are untouched
DM4	65229	Fault snapshots: speed, load, coolant, then rail, oil, battery, state
HOURS	65253	Engine hours

Long messages. A message longer than 8 bytes goes as a broadcast sequence: one announcement, then packets 50 ms apart. Three rules protect it: - Only one sequence is in flight at a time. - The active fault list (DM1) always goes before DM2/DM4, so a service tool can't starve the controller of faults. - A sequence is sent whole or not at all: no announcement, no packets, and a packet that meets a full queue is retried 1 ms later.

The exact set of messages the DSE4522 expects is still open item U4.

10. The logs

Everything below lives in one **persistent image** of 3036 bytes, which fits the S32K148's 4 KB emulated EEPROM.

Record	Holds
Counters	ECU powered seconds, engine hours, starts, trips by cause, watchdog resets, time at each quarter of rated load
Fault memory	16 faults; each with an occurrence count, first and last time seen, and a **snapshot taken the first time**: speed, load, fuel, rail pressure, coolant, oil, battery, state
Event log	The last 64: power-up (with cause), ignition on/off, crank, run, shutdown (with reason), fault set/cleared, stored-fault clear, image reset
Trip recorder	The last 2 overspeed or crank-signal trips: speed, fuel, rail pressure and state every 20 ms, from 2 s before to 1 s after

Saving: - The platform keeps **two copies** and writes them alternately. Each copy carries a sequence number and a CRC-32 over the whole image. At power-up the newer good copy wins, so a power cut during a write costs one save, never the history. - An image from a different layout version is never misread. It is treated as blank, and an "image

reset" event is logged. - At power-up, every stored fault is marked previously active until it is seen again. - Saves are requested only by `ecu_background`, never from an interrupt, on any of these: - a fault's first occurrence - each start - entering shutdown - each hour of running - a watchdog-reset boot - a trip (1 s after it, once the recording is complete) - Saves are capped at one per 10 s, except a trip. That is about 30 a day, roughly 110 000 per copy over 20 years. - A fault's set and clear are logged once per ignition cycle, so a loose connector can't flood the event log.

Reading: stored faults, snapshots and engine hours over J1939 (§9). The event log and trip recorder need the laptop service tool from the dev-kit spec. Until then, the simulator decodes them from the image (`decode()` in `run_sil.py`).

11. Safety, in layers

Layer	Catches	Where
Overspeed check on every tooth	Runaway: trips at 1650 rpm, under the DSE's own 1680/1710 rpm trips	`ecu_crank_edge`
Sync-loss trip	Lost crank signal (50 ms) or a wheel that no longer makes sense	`crank.c` , `ecu.c`
Injection cancel	Any command not yet started when a trip happens	`inject_cancel`
Metering valve closed in SHUTDOWN	Rail pressure falls after a trip	`rail.h` `METER_CLOSED`
Underspeed warning	A set held below 1350 rpm for 3 s (overload, a fuel fault): SPN 190 FMI 18, amber. **No shutdown**: the DSE decides	`ecu.c` `underspeed()`
Boost rail warnings	Held over 104 V for 2 s (the hardware zener clamping, below the 106 V trip): amber SPN 520192 FMI 16. Under 80 V for 0.5 s while cranking or running (a dead boost): amber FMI 18. Both clear after as long the other way	`mode.c` `boostm_step`
Boost overvoltage	Rail over 106 V for 20 ms: stop the boost, SHUTDOWN, red SPN 520192 FMI 0. The boost is also off in SHUTDOWN. A 110 V zener on DIS/EN backs it in hardware (105–119 V)	`ecu.c` `boost_guard()` ; injector sheet
EGR stall	Over 5 A for 1.5 s: the bridge coasts 5 s, amber SPN 2791 FMI 6; the third stall holds it off to the next ignition cycle. **No shutdown**	`ecu.c` `egr_guard()`
Window watchdog (TPS3850)	A hung main loop: the pulses stop and the chip resets the processor; the next power-up logs it	`ecu_tick_1ms` , `ecu_background`
Independent trip module	Everything above failing: it cuts the metering valve's supply from outside the board	Hardware, spec §3

The firmware is the first line, not the only one. The trip module doesn't depend on the firmware at all.

12. Every tuning knob

Each is one `#define`. *Assumed* means a guess to be replaced with the real engine's figure. *Calibration* means tuned in simulation and to be re-tuned on the engine.

Knob	File	Value	Status
`CRANK_TDC1_DEG`	crank.h	90° after the gap	assumed (60-2 wheel assumed; **firing order 1-2-3 confirmed** 13 Oct 2026)
`CRANK_TIMEOUT_US`	crank.h	50 ms	design
`OVERSPEED_RPM`	safety.h	1650	design
`GOV_RPM_SET` , `GOV_DROOP`	govern.h	1500, 4 %	design
`GOV_KP` , `GOV_KI`	govern.h	0.5 mg/rpm, 2.0 mg/(rpm·s)	calibration
`GOV_RAMP_RPM_S`	govern.h	500 rpm/s	calibration
`FUEL_MAX_MG` , `FUEL_RATED_MG`	govern.h	45, 34 mg	assumed
`FUEL_START_MG`	inject.h	30 mg	calibration
SOI table `SOI_Q` / `SOI_D`	inject.c	11.5–16° BTDC on fuel	measured (old ECU, 1500 rpm)
`INJ_DEAD_US` , `INJ_K`	inject.h	323 µs, 0.00191	measured (old ECU)
`METER_NC`	rail.h	0 (normally open)	measured behaviour; confirm with the part number
Rail table `RAIL_Q` / `RAIL_P`	rail.c	827–1171 bar on fuel	measured (old ECU, 1500 rpm)
`RAIL_KP` , `RAIL_KI`	rail.h	0.002, 0.01	calibration
`RAIL_OPEN_LIMP`	rail.h	30 % opening	calibration
Sensor tables	sensors.c	§8	assumed until each front-end is measured
`RUN_RPM`	ecu.c	400	design
`IGN_OFF_MS`	ecu.h	500 ms	design
`BOOST_OV_V` , `BOOST_OV_MS`	ecu.h	106 V for 20 ms	design (rail 99.7 V ±2 %)
`BOOST_HIGH_V` , `BOOST_HIGH_MS`	ecu.h	104 V for 2 s	design (the zener's 105 V low corner)
`BOOST_LOW_V` , `BOOST_LOW_MS`	ecu.h	80 V for 0.5 s, cranking or running	design
`EGR_STALL_A` , `EGR_STALL_MS`	ecu.h	5 A for 1.5 s	assumed (stall 6.4 A; the running current is unmeasured)
`EGR_COOL_MS` , `EGR_STALL_TRIES`	ecu.h	5 s, 3	design
`SAVE_GAP_S`	ecu.h	10 s	design (flash wear)
`UNDERSPEED_RPM` , `UNDERSPEED_MS`	safety.h	1350 rpm for 3 s	design (90 % of 1500)
`UNDERSPEED_CLEAR_RPM` , `UNDERSPEED_CLEAR_MS`	safety.h	1400 rpm for 1 s	design
`SEC_SECRET`	service.h	16 bytes, overridable per build	**set per product before shipping**
`SVC_SESSION_MS` , `SVC_LOCK_MS`	service.h	5 s, 10 s	design
`EOL_GAIN_MIN/MAX` , `EOL_OFFSET_MAX`	eol.h	0.9–1.1, ±0.1 V	design

The engine model in `sim/engine.py` has its own assumed constants, also marked. When real engine data arrives, change the constant, re-run the scenarios, and they say at once whether the control still holds.

13. How it is tested

```
make -C firmware/sim test                        # 101 unit tests and 7 bootloader tests, a few s
.venv/bin/python firmware/sim/run_sil.py         # 16 scenarios, about 2-3 min
.venv/bin/python firmware/sim/run_sil.py rail    # scenarios whose name contains "rail"
.venv/bin/python firmware/sim/test_service.py    # the service tool end to end, about 10 s
```

#	Scenario	Proves
1	Start	in sync within 2 revolutions; 1500 rpm by 5 s; peak under 1600
2	Load step 0 → 20 kW	dip stays above 1350 rpm; back within ±15 rpm in 5 s; no underspeed warning
3	Load rejection 20 → 0 kW	peak stays under the trip
4	Droop switch	speed falls 4 ± 1 % at full load
5	Stuck injector (runaway)	shuts down on the very tooth it first reads > 1650 rpm; no injection after; rail pressure falls
6	Crank signal lost	shutdown within 100 ms, crank-sensor fault
7	Ignition off	shutdown with no fault code
8	Coolant sensor unplugged	fault within 1.2 s; engine keeps running
9	CAN timing and content	every message on time; speed within 0.5 %
10	Rail pressure	within 5 % of the setpoint for the fuel delivered
11	Watchdog	pulses every 10 ms, right to the end of the run
12	Runaway, then power cycle	fault, snapshot, event and trip recording all survive
13	Power cut mid-save	the previous save loads
14	Stored faults over CAN	DM2 reads them; DM3 clears them and leaves active ones
15	One lost crank tooth	logged as a crank-sensor fault, not overspeed
16	Overload, 30 kW	settles near 1190 rpm; SPN 190 FMI 18 within 5 s, amber only; never shuts down

Scenario 16 relies on one model assumption: the alternator's voltage regulator rolls the voltage off below 47 Hz (`AVR_KNEE_RPM`), so the load's power falls with speed and an overloaded set settles instead of stalling. With a pure constant-power load every overload stalls in under 3 s, faster than the warning.

What a pass means, and what it doesn't. The scenarios prove the logic and the safety behaviour. They don't prove the tuning on a real engine. The engine model uses the firmware's own injector and sensor formulas, and it has no electrical noise. Two independent expert reviews, each reproducing its findings, were run against the code (decisions log §11 and §12).

14. What the S32K148 port must provide

The core needs nothing else. The platform layer must:

1. **Crank input:** input capture on the crank sensor; call `ecu_crank_edge` with the capture time in µs (a free-running 32-bit counter). Same for the cam.
2. **1 ms timer:** call `ecu_tick_1ms` with the same µs clock.
3. **Injection:** after each call, if `inj[k].seq` changed, load `start_us` / `dur_us` into that cylinder's output-compare channels. Duration 0 means disarm, unless the injection has already started.
4. **Metering valve:** drive the PWM from `meter_duty`.
5. **CAN:** send `can[0..can_n)` and zero `can_n`. Pass received frames to `ecu_can_rx`.
6. **Watchdog:** copy `wdg` to the TPS3850's WDI pin. Strap its window (SET0/SET1, CWD) to accept a 5–20 ms kick period.
7. **Storage:** keep two copies of the image. When `nvm_save` is set, write `ecu_nvm_image()` to copy `nvm_slot`, **without blocking the main loop for more than 50 ms** (or the watchdog resets the chip), then clear `nvm_save`. At start-up pass both copies to `ecu_nvm_load`, with `reset_cause` from the chip's reset status.
8. **Main loop:** call `ecu_background` continuously.
9. **Priorities:** the edge, tick and CAN-receive interrupts at one priority. Mask them during the image copy inside `ecu_background`.
10. **Service link:** pass CAN channel 2 frames to `ecu_can2_rx` (at the tick's priority); send `can2[0..can2_n)`. On the prototype, copy USB virtual-serial bytes into `usb_in` (up to 64 per ms; the tick takes them) and send `usb_out`.
11. **End-of-line record:** two copies in their own sector (S32K148: a D-flash sector; STM32G474: the last page). Pass both to `ecu_eol_load` at start-up. When `eol_save` is set, write `ecu_eol_image()` to copy a, then copy b, then clear it. Pass the bootloader's version in `boot_ver`.
12. **Resets:** on `reset_req`, a software reset. On `boot_req`, set the stay flag (one RAM word the reset does not clear) and reset.
13. **The bootloader's host:** `flash_erase`, `flash_write` (whole 8-byte words), `flash_read` over the application area only, the stay flag (read and clear), and `jump_to_app` (stack pointer and vector table from the application's base, then branch). §16 has the rest.

15. Known limits and open items

Needs the machine or a supplier: - the metering valve's part number (behaviour already measured: normally open) - crank wheel (firing order 1-2-3 is confirmed) - the sensor tables

Needs the real engine: - governor and rail tuning; the old ECU dips 9.8 % on a load step, so our simulated 3.4 % is likely optimistic - start fuel, and timing and rail pressure while cranking (the recording starts with the engine already running) - timing and rail tables above 26.5 mg, and at speeds other than 1500 rpm - pilot and post injections (the old ECU uses none in this mode)

Needs U4 (the DSE's exact expectations): - the exact J1939 message set - whether DM3 needs an acknowledgement

Dev-kit spec: - flash choice for about 110 000 writes per copy - the service tool's reading of the event log and trip recorder (the record and flashing are done, §16) - the chip layers of the service link: STM32G474 and S32K148 flash, CAN, USB and reset drivers (§14 items 10–13) - MISRA C checks - governor performance graded against ISO 8528-5 - a failure analysis of the overspeed path

Known minor items: - An extra crank tooth from a noise spike still logs as overspeed. The engine stops correctly either way. - A faulted sensor's snapshot stores its default value rather than "not available". - DM1 always reports an occurrence count of 1.

16. The service link

One protocol, a subset of UDS (ISO 14229), on two links: **USB** on the prototype for quick iterations, **CAN channel 2** in production (ECU pins 86/87, off the DSE's J1939 bus). The application and the bootloader share `core/service.c`; only the framing differs.

Link	Framing
CAN channel 2	ISO-TP, 11-bit: request `0x7E0`, response `0x7E8`, 1000 kbit/s (Bosch's application CAN, pins 87/86), frames padded with `0xAA`, block size 0, STmin 0, up to 4095 bytes
USB (virtual serial)	`0x55`, length (2 bytes LE), payload, CRC-16/CCITT of the payload (2 bytes LE). A bad CRC or a length over 4095 drops the message

Services:

Service	Application	Bootloader
`0x10` session	default `01`, extended `03`; programming `02` (unlocked) resets into the bootloader	default, extended, programming
`0x27` security	seed `01`, key `02`; not in the default session	same
`0x3E` tester present	keeps a session alive (`3E 80`: no reply)	same
`0x11` reset	`01`, engine stopped only	`01`: runs the image if it is good
`0x22` read	`F18C` serial, `F191` hardware revision, `F18B` build date, `F195` software version, `F180` boot version, `0100` trims, `0101` station, passed, locked	`F180`, `F195` (from the image header), `FD00` the application area

Service	Application	Bootloader
`0x2E` write	the record's IDs: extended, unlocked, engine stopped, record not locked	—
`0x31` routine	`F000` lock the record, `F001` unlock it (logged)	`FF00` erase (address, length), `FF01` check (CRC, version, hardware range)
`0x34` / `0x36` / `0x37`	—	download, 1024-byte blocks, exit

Refusals are `7F, service, reason`: `22` conditions (the engine turning, or the record locked), `33` security, `35` wrong key, `36` too many wrong keys, `31` out of range, `24` wrong sequence, `13` wrong length, `72` programming failure, `7F` wrong session.

Rules: - **Engine stopped** means OFF, WAIT or SHUTDOWN and no crank tooth for 1 s (a saturating counter, so it holds however long the set rests). Extended and programming sessions, and an ECU reset, are refused otherwise, and an open session ends at the first tooth. - A non-default session ends 5 s after the last message. - **Seed-key:** the key is the low 32 bits of **SipHash-2-4** of the seed (4 bytes LE), keyed with the 16-byte `SEC_SECRET`. Three wrong keys lock security access for 10 s. A sniffed seed and key reveal nothing about other seeds (the first design's CRC-32 key did: it is affine). It still stops casual writes, not a determined attacker: anyone who reads one unit's flash has the secret for all.

The end-of-line record (`core/eol.h`, 84 bytes): magic `0xE01D`, version, serial (16 ASCII), hardware revision, build date (yyyymmdd), six gains and six offsets (coolant, oil, rail, boost, EGR, 5 V reference), station, passed, locked, CRC-32. Two copies apart from the logs; copy a is written first, so when both are good copy a is the newer. **Only a record stamped "passed" counts:** until the station writes that stamp, SPN 628 FMI 13 stays on, so a station that stops after the serial ships no silent unit. Once locked (`31 01 F000`) every write is refused until `31 01 F001`.

The bootloader (`boot/boot.c`, 32 KiB at the bottom of flash): - At reset: if the stay flag is set, stay. Otherwise run the application only if its header is good, its CRC matches, and the end-of-line record's hardware revision is inside the header's range. - Programming: programming session, unlock, erase, download to the application base + 64, transfer, exit, then the check routine. **The header is written last**, after the CRC of what is in flash matches the tool's, and its magic is the very last write. A power cut anywhere before leaves no valid header, so the next reset stays in the bootloader and the update is simply run again (`test_boot.c` cuts the power at every single write). - It never erases or writes outside the application area. While it runs the engine cannot: no

injection, the metering valve at its safe state, the gates held off by their pull-downs; the watchdog is still kicked. - No rollback: a bad image that passes its CRC is fixed by flashing the previous one.

The laptop tool:

```
T=firmware/tools/ecu_service.py; K=--secret key.bin      # key.bin: the 16 bytes of SEC_SECRET
.venv/bin/python $T --link usb:/dev/tty.usbmodem1 $K info
.venv/bin/python $T --link can:socketcan:can0 $K eol-read
.venv/bin/python $T --link usb:/dev/tty.usbmodem1 $K eol-write --serial SN0042 \
    --hw-rev 0x0100 --date 20261020 --trims trims.json --station 3 --passed
.venv/bin/python $T --link usb:/dev/tty.usbmodem1 $K eol-lock          # eol-unlock for a rework
.venv/bin/python $T --link usb:/dev/tty.usbmodem1 $K flash app.bin \
    --sw-ver 0x00010002 --hw-min 0x0100 --hw-max 0x01FF
.venv/bin/python $T --link sim $K info                                 # the simulated board
```

`trims.json` is `{"gain": [6 values], "offset": [6 values]}` in the order above. `pyserial` (USB) and `python-can` with `can-isotp` (CAN) are needed only by their links. A 256 KiB image takes about 10 s over CAN plus the erase; USB is faster.

Not yet: the chip layers (§14 items 10–13). The prototype is first flashed over SWD (J301), as any new board is.

17. The Simulink and Stateflow models (13 November 2026)

The control and state logic also exists as models, in `model/`, and Embedded Coder generates C from them into `gen/`. Building with `ECU_MODEL` defined puts that generated code in place of the hand-written modules, behind the same functions:

Function(s)	Model	Kind
`mode_cmd`	`ecu_mode`	Stateflow: OFF, WAIT, CRANK, RUN, SHUTDOWN
`boostg_*`, `boostm_*`, `egrg_*`, `wifg_*`, `underg_*`, `relay_*`	`ecu_boostg`, `ecu_boostm`, `ecu_egrg`, `ecu_wifg`, `ecu_underg`, `ecu_relay`	Stateflow
`gov_reset`, `gov_step`	`ecu_gov`	Simulink
`rail_reset`, `rail_step`, `rail_set_bar`	`ecu_rail`, `ecu_railset`	Simulink
`inject_soi_btdc`, `inject_dur_us`	`ecu_soi`, `ecu_injdur`	Simulink
`safety_check`	`ecu_safety`	Simulink

The state machine is driven by commands, because both the crank-tooth interrupt and the 1 ms tick change it: `ecu.c` sends TICK (every 1 ms, with the ignition and speed), TOOTH, RUN_OK, TRIP or RESET, and the chart answers with the new state. Since 14 November the chart owns its conditions: the ignition for OFF → WAIT, and the ignition cycle that ends SHUTDOWN (low for `IGN_OFF_MS` with the engine stopped, then on). When the chart lands in OFF that way, `ecu.c` resets

everything else. `ecu.c` keeps everything around it (events, the governor reset on RUN entry, counters, DTCs), and `gov_setpoint` stays hand-written in both builds.

Checking everything at once: `./verify.sh` at the top of the repository runs the circuits, schematics, both firmware builds, the back-to-back test, both sets of scenarios and the service tool, and names whatever failed. `--boards` adds the KiCad board checks, `--matlab` the Simulink closed loop. Run it before every commit that touches `firmware/`, `model/` or `hw/`.

Building it: `make test-model` (unit tests) and `build/libecu_model.*` (the scenarios, with `ECU_LIB=libecu_model`) in `sim/`. On Windows, call `zig.exe` from the venv directly (`.venv/Lib/site-packages/ziglang/zig.exe cc`): `python -m ziglang` swallows the compiler's error messages.

The evidence it is the same firmware: every unit test and all 16 scenarios pass on both builds; `model/test/run_b2b.sh` runs each generated model against its hand-written original on millions of random inputs, and they agree exactly, apart from one or two units in the last place in the two tables (Simulink orders its interpolation arithmetic differently from `table.h`). See `model/README.md`.

For the target: the generated code is single-precision C99 for an ARM Cortex-M. With the whole core it compiles warning-free for the Cortex-M4F (`zig cc -target thumb-linux-musleabihf -mcpu=cortex_m4+vfp4d16sp -mfloat-abi=hard -DECU_MODEL`, compile only), and the governor comes out as native `vadd.f32` / `vmul.f32` / `vdiv.f32` with no double-precision code. It is flashed as part of the firmware image. The chip layers (§14) are still what stands between either build and the board.

Which is the source of truth: for now, both. The hand-written modules stay the reference build until the generated build has run on the engine. A change goes into both: the model's script in `model/` (constants come from the C headers automatically), then `build_all` and `gen_code`, then the C reference, then both test runs.

PCB Layout Guide

Written for: the layout engineer who routes `hw/ecu25kva.kicad_pcb` by hand.

Date: 14 October 2026.

What you get: - a board with all 379 parts placed by block - every net - every net class and rule

What you do: fine placement and routing. This guide says what matters in each block and why. Where it quotes a number, the source is named beside it. A simulation file is `sim/blocks/<name>.cir`. A sheet is the matching `hw/gen_sheet_<name>.py`.

---

1. What is ready, and what is a placeholder

Item	State
379 footprints, 373 nets	From the schematic, exactly. `hw/check_pcb.py` proves it
Net classes, widths, clearances	In `ecu25kva.kicad_pro`. Generated from `hw/pcb_rules.py`
Placement	By block (each schematic sheet is a KiCad group), shelf-packed, no overlaps. Fine placement is yours
Board outline	**PLACEHOLDER:** 180 × 130 mm rectangle
Mounting holes	**PLACEHOLDER:** 4 × M3 at the corners
Housing contact band	**PLACEHOLDER:** top 25 mm (User.Comments layer)
J1001, the 94-way header	**PLACEHOLDER:** pin numbers right, geometry assumed (2.5 mm pitch)
D806	Footprint assumed (SMA). The part waits on the trip-loop pin

> **No board is ordered until J1001 and the outline are measured** from the old ECU (inspection brief, Group 6). Both placeholders say "DO NOT FABRICATE" on the silkscreen. When the measurements arrive, `HOUSING` and `REGIONS` in `pcb_rules.py` change. If you have not started routing, re-run the generator. If you have, move the outline and J1001 by hand.

Why 180 × 130 and not the spec's 160 × 120: the parts need about 15,300 mm² packed on one side. At 160 × 120, the power-input block alone did not fit beside J1001. If the old housing's board turns out smaller, parts go on both sides. That is your call once the outline is known.

2. Tools

- **KiCad 10 is required.** The board file is KiCad 10 format; the schematic stays KiCad 7 format.
- **Rebuild the board:** `$KP hw/gen_pcb.py`, where `KP` is KiCad's own Python (README). On a routed board it rewrites **only the rules** and leaves your copper alone; `--force` throws the routing away.
- **Check the board:** `.venv/bin/python hw/check_pcb.py`. It checks:
- footprints, nets and classes
- widths
- every DRC error (clearance, courtyards, edge and hole clearance, shorts, mask bridges)
- schematic parity: each pad's net, the symbol fields, the footprints. The board's footprints are linked to their symbols, so **Update PCB from Schematic** works once routing has started

Unrouted connections are counted, not failed. - **Custom rule** (`ecu25kva.kicad_dru`): pads inside one footprint are exempt from net-class clearance, because that spacing is the maker's land pattern. Every class still applies to the copper you add.

3. Stack-up and classes

Board: 4 layers, 1.6 mm FR-4, **Tg ≥ 170 °C** (engine bay).

Layer	Copper	Use
L1 top	2 oz	Parts, power copper, short signals
L2	1 oz	**Solid ground plane.** No splits, no slots, unbroken under the switching stages and the MCU
L3	1 oz	Power distribution and slow signals
L4 bottom	2 oz	Power copper; heat spreading into the housing contact areas

Widths are computed (IPC-2221, 20 °C rise) for the outer 2 oz layers. An inner 1 oz layer needs much more: 18 A on an inner layer is over 5× the outer width. **So route high current on L1/L4 or as pours, not on L3.**

Class	What	Current	Width (outer)	Clearance
`PWR_HI`	injector feed: pin 4 → VBAT_INJ, the hold-path diodes	18 A	5.35 mm	0.6 mm
`RTN`	ground, and the injector shunt tops (~0 V, on fine-pitch amplifier pins)	18 A	5.35 mm (pour)	0.15 mm
`PWR_LOOM`	pin 6 → VBAT_EGR → EGR bridge and its shunt. Protected only by the 10 A loom fuse, so sized for 135 % of it (ISO 8820-3)	13.5 A	3.6 mm	0.3 mm
`HV100`	boost rail, bank nodes (pins 3, 5), low-side drains (pins 73, 29, 7), boost switch node	18 A, up to ~116 V	5.35 mm	0.6 mm

Class	What	Current	Width (outer)	Clearance
`HV100_SIG`	injector high-side gates and bootstraps, divider midpoints, PZTA42 collectors, VDD_REF; the EGR high-side gates and bootstraps (VBAT + 12 V: ~27 V running, 52 V on a 40 V load dump)	small, 31–116 V	0.2 mm	0.6 mm
`PWR`	the electronics path behind the 5 A fuse F101 (from pin 04 since the Bosch pinout, 8 Nov), VBAT_PROT, metering valve, buck switch node, 12V_GATE	6.4 A	1.3 mm	0.3 mm
`PWR_LV`	3V3_MCU, 5V_MAIN, the boost controller's VDD and current sense: low voltage on 0.5 mm-pitch pins	6.4 A	1.3 mm	0.15 mm
`CAN`	CAN1/CAN2 H and L	—	0.2 mm, gap 0.25 (set from the fab's stack-up)	0.2 mm
`ANALOG`	sensor inputs, clamps, ADC nets, VR pair	—	0.25 mm	0.2 mm
`Default`	logic	—	0.2 mm	0.15 mm

The 0.3 mm on battery power is a judgement, not a table row. IPC-2221B B2 gives 0.1 mm up to 30 V and 0.6 mm from 31 V; 0.3 mm adds margin over the 12 V system (24 V jump start) for clamped transients.

Peak versus average: the 18 A is the injector peak phase, which is short. The class width is the conservative default. Where a pad forces you to neck down, keep the neck short and add copper on the other outer layer.

4. The ground: one plane, one join

- **L2 is one solid plane.** Power and signal returns are separated by **placement and routing**, not by cutting the plane.
- **The single point:** all high-current returns meet the board ground **at the battery-ground pins of J1001, and only there**. Those returns are:
- the injector shunts R428 and R429
- the boost shunt R402
- the EGR shunt R507
- the metering shunt R501
- the TVS diodes D101, D105 and D108
- the negative clamp Q102
- the freewheel diodes D409 and D411
- the reservoir C420

Run them on L1/L4 copper to that point, then via into L2 there. Don't drop them into L2 next to the part. - **Which pins are battery ground: open.** The schematic has J1001 pins 1 and 2 (`G_G_B_AT1/2`) **unconnected**: the OEM wiring diagram shows no battery negative. Pins 1–2 are the assumption (`HOUSING["ground_pins"]`). The continuity check (inspection brief 5.1) settles it. **Leave room for a wide GND**

connection to two of the six large pins. - **The sensor grounds** (SGND pins 8, 30 and 44, tied to GND on the schematic) join the plane **at their own front ends**, away from the power join. - **SGND_34 and SGND_36 are not ground.** They go to the INA592 difference amplifiers' negative inputs, which is the whole point of `sensor_differential.cir`. Route each with its signal, as a pair.

5. Block by block

Injector stage (`injector`, top band)

The circuit: - Boost converter: L401, Q401, D401, shunt R402, controller U401 (TPS40210). - 100 V reservoir: C401 in series with C420, balanced. - Two high sides per bank: - Q408/Q410 from the boost rail (peak) - Q409/Q411 from the battery via D408/D410 (hold) - Freewheel diodes D409/D411, ground → bank node. - Low sides Q412, Q414, Q416. - Turn-off diodes D412–D414: low side → boost rail. - Shared bank shunts R428 (cylinders 1 and 3) and R429 (cylinder 2).

Loops to keep tight (`injector_boost.cir`, `injector_turnoff.cir`, `boost_converter.cir`):

1. **The turn-off loop, the most important on the board.** Each turn-off returns about **32 mJ** of coil energy (18 A in 200 µH, `injector_turnoff.cir`). The path is: - low-side drain (pins 73, 29, 7) - turn-off diode D412/D413/D414 - BOOST_100V - C401/C420 - ground - shunt - low-side source

Keep the diodes and the reservoir within a few mm of each other, on a wide pour. 2. **The boost switch loop:** Q401 drain → D401 → C401/C420 → R402 → Q401 source. The switch node SW_BOOST is the noisiest copper on the board: make it small, and keep it away from the sensor and VR inputs. 3. **The peak path:** BOOST_100V → Q408/Q410 → INJ_A_03/INJ_B_05 → J1001 pins 3/5. That carries 18 A at up to 100 V: `HV100`, short and wide.

Other rules: - **Shunts:** R428/R429 sit in the low-side source return. Take the sense lines to U406/U407 as a **Kelvin pair** from the shunt pads, not from the pour. - **High-side drivers:** U402–U405 with bootstraps C409/C411/C413/C415. Keep each bootstrap capacitor at its driver's VB/VS pins. VS is the bank node, so the bootstrap loop rides at 100 V. - **Creepage:** the 100 V copper is in `HV100` / `HV100_SIG`, at 0.6 mm (IPC-2221B, external, uncoated, 101–150 V). Don't run Default-class signals between HV pads. - **Conformal coating (12 Nov 2026, decisions log §19):** 0.6 mm is the *uncoated* spacing, which assumes a clean board. In an engine bay, with soot and condensation, it isn't enough on its own. So the assembled board is **conformal-coated** (acrylic or silicone), with the connector, the test points and the debug header masked. Put this on the assembly drawing. - **The boost cutoff (BOOST-OVP):** D417 (110 V zener),

R436 and R403 into U401's DIS/EN, with the clamp and the `BOOST_DIS` Schottky beside them. Keep the zener and R436 next to U401 pin 3. The zener's cathode is on BOOST_100V, so its pad takes that net's 0.6 mm; the rest of the cutoff sits at logic level. - **L401 inductor:** no vias or sensitive traces underneath. - **Heat:** the TO-263 FETs (Q408, Q410, Q412, Q414, Q416) and the SMPD diodes D408–D414 are the board's main heat. Put their tabs on large L1 pours, stitched to L4, over the housing contact area.

Power input (`power_input` , left of J1001)

The circuit: - **Pin 21:** F101 (5 A) → EMI filter L101 / C101–C104 / L102 (`emi_filter.cir`) → TVS D101 → reverse-battery P-FET Q101 → VBAT_PROT. The negative clamp Q102 sits on VBAT_PROT (`negative_pulses.cir`). - **Pins 4 and 6** (the injector and EGR feeds): TVS D105/D108 → reverse FETs Q103/Q104 → filters L103/L104 → VBAT_INJ / VBAT_EGR. Each has bulk HXC capacitors and two R-C dampers (`feed_emi.cir`).

Rules: - Place it in **the stage order**, input to output, with the pin side nearest J1001. A filter only works if its input and output copper don't couple around it. - **Keep the dampers** (R110–R113 with C113/C114/C117/C118) **next to the bulk capacitors.** `feed_emi.cir` needs them for stability against the boost converter's negative input resistance; they are not optional EMI decoration. - **The common-mode choke** that spec §6 calls for **is not on the schematic.** `emi_filter.cir` says it can only be modelled once a layout gives it a chassis return. Leave space at the electronics feed's entry (pin 04, Bosch pinout) for a common-mode choke and its chassis connection. **CISPR 25 Class 5 is the target** (owner, 12 Oct). - **Q101's gate is passive:** R103 to ground with the zener D102. There is no controller (decisions log §5b), and the same goes for Q103/Q104. - **Q102, the negative clamp, has a driven gate:** comparator U101 → gate driver U102 (UCC27517A) → Q102. Keep U102 at Q102's gate, with its return taken straight to Q102's source (ground), so the gate loop is small. - **Heat:** Q103 and Q104 carry the feeds continuously. The sheet note gives Q103 "Tj 117 C at a 105 C board" and Q104 (EGR, 6.4 A stall) "Tj 139 C at a 105 C board, 155 C while cranking". Give both tabs large L1 pours stitched to L4, and put them on the housing contact area if the measured housing allows.

Rails (`rails` , right)

The circuit: - **Buck:** LM5164 (U201), L201, bootstrap C203, switch node `/rails/SW` . - **Overvoltage cut-off:** U203 + Q201. - **Sensor trackers:** U204–U206, TPS7B4253 (DDA package).

Rules: - **Buck loop:** VIN capacitor → U201 → SW → L201 → output capacitor → ground. Keep it small. The datasheet's own layout applies. - **L201** (WE-PD 1050): **no vias or sensitive traces underneath** (datasheet). - **U204–U206 thermal pads:** solder-mask-defined, 2.6 × 3.2 mm opening on 2.95 × 4.9 copper (the drawn footprint). TI's optional thermal vias are your call.

Metering valve and EGR bridge (`metering_egr` , top right, in the contact band)

The circuit: - **Metering valve:** low-side FET Q501 on pin 88's return (`MU_RETURN`), 50 mΩ shunt R501, freewheel D501 to VBAT_PROT (`metering_unit_pwm.cir`). - **EGR:** discrete H-bridge Q503–Q506, drivers U504/U505, 20 mΩ shunt R507, pins 59/81.

Rules: - Kelvin sense from R501 and R507 to U501 and U503. - **Bridge loop:** VBAT_EGR → high FET → motor pins → low FET → R507 → ground. Keep it compact, with bulk capacitance at the high-side drains. - **Heat:** the bridge FETs Q503–Q506 and the metering FET Q501 make heat (the EGR stall is 6.4 A continuous). Keep their tabs over the housing contact band, on pours stitched to L4.

MCU and supervisor (`mcu` , centre)

The circuit: - S32K148 (U301), 144-LQFP. - Decoupling. - Bulk capacitor C313 with its **22 mΩ series resistor R306** (`/mcu/BULK_RTN`). - Supervisor TPS3850 (U302). - Debug header J301.

Rules: - **Every decoupling capacitor at its pin**, with the via to L2 at the capacitor pad. - **R306 is deliberate.** `mcu_pdn.cir` : "a very low-ESR bulk part makes this peak worse, not better." Keep R306 in series with C313. Don't bypass it. - The quiet zone: no switching copper (SW_BOOST, `/rails/SW` , the bank nodes) under or beside the MCU.

Sensors (`sensors_analog` , centre right)

The circuit: - Divider, ratiometric, differential (INA592, U601–U605) and NTC front ends. - Each ends in a 22 nF filter and a 3.3 V clamp at the MCU pin. - U606 is the barometric pressure sensor (MPXHZ6115A).

Rules: - Route each differential input with its SGND return as a pair, from the connector pin to the amplifier. - Put the filter capacitor at the MCU pin, not at the connector. - **U606 needs air:** keep its port clear, and line it up with the housing's vent (the round vent plug on the old housing). Confirm against the housing.

Speed inputs (`speed_inputs` , centre right)

The circuit: - **Crank VR**, pins 52/74: TVS D701 → resistors → clamps → comparator U701. - **Cam Hall**, pin 46.

Rules: - **Route the VR pair as a differential pair.** `vr_conditioner.cir`: "common-mode rejection at the input is doing real work against the injector currents flowing nearby." Keep it away from the injector block and SW_BOOST, over unbroken L2. - At cranking the VR signal can be **under a volt** (`vr_conditioner.cir`), so coupled noise matters most exactly when the engine needs to start.

CAN (`can` , right of J1001)

The circuit: two TCAN1042 transceivers (U901/U902), each with a split termination: R902/R903 + C903, and R905/R906 + C906.

Rules: - **Route H/L as a pair**, matched, with the termination at the transceiver end and the midpoint capacitor to ground with a short via. - **The common-mode choke** (`can_termination.cir` , like the power input) is not on the schematic. Leave space for one per channel between the connector and the transceiver.

Discrete I/O (`discrete_io` , right)

The circuit: switch inputs, the supervised trip loop (`TRIP_SENSE`), and two relay drivers (Q806/Q807, freewheel D807/D808 to VBAT_PROT).

Rules: nothing critical beyond keeping each input's RC filter at the MCU end.

6. Test points

There are no test-point footprints in the schematic yet. Add them as `TP` symbols on the schematic first, so the board and schematic stay in step. The minimum set:

- **Rails:** VBAT_PROT, VBAT_INJ, VBAT_EGR, 12V_GATE, 5V_MAIN, 3V3_MCU, 5V_SENSOR_A/B/C, BOOST_100V (with a warning label)
- **CAN:** CAN1_H, CAN1_L, CAN2_H, CAN2_L
- **Timing:** CRANK_EDGE, CAM_EDGE
- **Injector current:** ISNS_INJ_A, ISNS_INJ_B (the amplifier outputs), plus a ground next to each

7. Before fabrication

Nothing is fabricated until all of these are true:

1. J1001's footprint, position and the outline are **measured** from the old ECU (Group 6) and replace the placeholders.
2. The **battery-ground pins** are confirmed (brief 5.1), and the schematic connects GND to them.
3. D806's part is chosen (trip-loop pin, brief 5.2).

4. The **fab confirms the stack-up**, and the CAN pair width/gap is set from it.
5. `check_pcb.py` passes with **zero** unrouted connections.

Prototype bring-up

For: the person at the bench with the assembled prototype board. **You need:** - a bench supply with an adjustable current limit, 0–20 V - a multimeter - a 100 MHz oscilloscope with a probe rated **300 V CAT II or better** (the boost rail reaches 119 V in a fault) - an ST-LINK V3 or J-Link with the 10-pin 1.27 mm Cortex cable - dummy loads for steps 4 and 5 - a 12 V automotive relay (or a 160 Ω, 2 W resistor) for step 6

Power the board up **one supply at a time**, in the order below. The board follows Bosch's pinout (decisions log §18), which has **no separate logic feed**: - **Pin 04** feeds the electronics *and* the injector stage, so the 100 V boost rail comes up with the logic. - **Pin 06** feeds the EGR bridge.

On the engine both pins come from the main relay, which the board closes itself (step 6). On the bench, feed them straight from the supply. Don't skip a step. If a reading is outside its range, **stop** and find out why before going on.

From step 2 on, TP12 is at about 100 V, because the boost converter starts as soon as pin 04 is powered: its DIS/EN pin is pulled low in copper, and the MCU only stops it by driving `BOOST_DIS` (PD14) high. **The boost capacitors stay charged after the supply is off.** TP12 can hold close to 100 V for minutes. Before you touch the board after any powered step, measure TP12 to GND and wait until it reads **under 5 V**.

Expected values come from the production simulation blocks (`sim/blocks/`) and the sheet notes; each is named.

Where things are

Terminal	Edge	Position: signal (production pin)
J1001	left, top	1, 2: V_BAT_1R (04: electronics and injectors; use both above 5 A) · 3: V_BAT_2R (06, EGR) · 4: GND
J1002	left	1, 2: GND, the power return (Bosch pins 01/02)
J1003	left	injectors: INJ_A_03, INJ_B_05, INJ_LS1_73, INJ_LS2_29, INJ_LS3_07
J1004	top	MU_RETURN (88), EGR_HIGH_59, EGR_LOW_81, RLY_MAIN_OUT (50), RLY_BUZZER_OUT (69)
J1005	right	sensors: 5 excitations, 6 sensor inputs, SGND 34/36, GND (rail sensor return, 08)
J1006	right	speed: CRANK_HI_52, CRANK_LO_74, CAM_FREQ_46, CAM_GND_44, SHIELD_30, GND
J1007	bottom	1–5: switches 43/23/20/24/71 (ignition) · 6: water in fuel (21) · 7: diagnosis request (14) · 8: stop switch (25) · 9: GCU sync, analog (09)
J1008	bottom	CAN1_H, CAN1_L, GND, CAN2_H, CAN2_L, GND
J1009	inside, by J1005	2 × 10 bridge header: crank pair, cam, 6 sensor inputs, ignition on the odd pins; GND on the even pins

Terminal	Edge	Position: signal (production pin)
J301	by the MCU	SWD (Cortex 10-pin). SWO is wired to pin 6
J302	by the MCU	USB-C, **data only**: it does not power the board

Test points (TP1–TP24):

TP	Net	TP	Net
TP1	VBAT_PROT	TP13, TP14	GND (injector)
TP2	VBAT_INJ	TP15, TP16	ISNS_INJ_A, ISNS_INJ_B
TP3	VBAT_EGR	TP17, TP18	ISNS_MU, ISNS_EGR
TP4	12V_GATE	TP19, TP20	CRANK_EDGE, CAM_EDGE
TP5	5V_MAIN	TP21, TP22	CAN1_H, CAN1_L
TP6, TP11	GND	TP23, TP24	CAN2_H, CAN2_L
TP7	3V3_MCU	**TP12**	**BOOST_100V: up to 116 V. HV.**
TP8–TP10	5V_SENSOR_A/B/C		

1. Before any power

1. **Look** at every IC for orientation and solder bridges, under a loupe. Check the LQFP-100 (U301) and the HVSSOP boost controller (U401) closely.
2. **Resistance to GND** with the meter. Nothing on this list should read below 100 Ω. A short here would destroy parts at power-up.

Probe	Expect
TP1, TP2, TP3 (the feeds)	> 1 kΩ
TP5 (5V_MAIN), TP7 (3V3_MCU)	> 100 Ω (charging capacitors: the reading climbs)
TP12 (BOOST_100V)	> 10 kΩ, climbing (two 82 µF capacitors charging)
TP21–TP22, TP23–TP24 (H to L of each bus)	**120 Ω**, the split termination 60R0 + 60R0 (`can` sheet)

2. Pin 04: the electronics and the boost rail, limited to 1 A

Supply **13.5 V on J1001-1**, with GND on J1002, at a **1 A** limit. Leave pin 06 (J1001-3) open. Put the HV probe on TP12 before you switch on.

Check	Expect	Source
Supply current	at the limit for a moment while the boost capacitors charge, then well under it. **If it stays at the limit, switch off**	—
TP1 VBAT_PROT	about 13.5 V, less one FET drop	`reverse_battery.cir`
TP2 VBAT_INJ	about 13.5 V	`power_path.cir`
TP5 5V_MAIN	5.0 V, ripple under 50 mV p-p	`buck_preregulator.cir`
TP7 3V3_MCU	3.30 V. **It must sit inside 3.143–3.459 V**, or the supervisor resets the board. The TLV767 holds 1 %	`rails` sheet; `supervisor.cir`

Check	Expect	Source
TP8–TP10 5V_SENSOR_A/B/C	about 5.0 V, following 5V_MAIN	`sensor_rail.cir`
D303 (3V3 LED)	on	—
TP12 BOOST_100V (HV probe)	**99.7 V nominal**, never above about 102 V	`boost_converter.cir` ; `injector` sheet
`BOOST_MON` (PB11)	**2.43 V at 100 V** (24.9 k / (2 × 499 k + 24.9 k) = 0.0243)	`prototype/additions.py`
`EN_BOOST` (U401 pin 3, DIS/EN)	**under 0.1 V** while the boost runs. It rises only when the overvoltage zener D417 conducts, or `BOOST_DIS` is driven high	`boost_ovp.cir`
TP4 12V_GATE	about 12.3 V (a 13 V zener-referenced follower from the boost rail). The drivers' UVLO is 9.8 V	`injector` sheet

Then connect the debugger to J301. **SWD must connect,** and the debugger should read the device ID (STM32G474, `0x469`). Flash a test build that toggles `LED_STATUS` (PB7): D304 should blink.

The supervisor will reset the MCU every few tens of milliseconds unless the firmware kicks `WDT_KICK` (PC11) at 5–20 ms intervals (decisions log 11). With no firmware, a reset pulse train on NRST is correct behaviour, not a fault.

3. CAN and the sensor inputs

1. **CAN:** connect a USB-CAN adapter to J1008: **250 kbit/s on J1008-1/2** (pins 55/77, J1939) and **1000 kbit/s on J1008-4/5** (pins 87/86, the service link; Bosch's rates). Each bus is split-terminated on the board; terminate the adapter end too. Send and receive a frame on each bus.
2. **Sensors:** put a 10 kΩ potentiometer between each excitation and its SGND on J1005, with the wiper on the matching input. Sweep each one and read the ADC. - **The divided channels read about 0–3.07 V for 0–5 V in.** `sensor_ratiometric.cir` checks 5.0 V at the pad giving 3.072 V.
3. **Battery monitor:** `VBAT_MON` (PB12) should read **1.04 V at 13.5 V** (120 k / 10 k).

4. Pin 06 (EGR), then firing the injectors

Pin 06: connect 13.5 V to J1001-3 at a 1 A limit. - TP3 VBAT_EGR should read about 13.5 V. - Load the EGR outputs (J1004-2/-3) with a 10 Ω, 25 W resistor. - The EGR bridge stays coasting until firmware raises `EGR_EN`.

Injectors: put a dummy load on each injector output in J1003: 0.2 Ω + 200 µH, or a real injector on a bench.

Before firing any injector, raise the pin 04 limit to 5 A. The boost recharge after each shot draws more than 1 A; at a 1 A limit the supply folds back and the readings below are wrong.

With firmware firing one injection at 1 ms: - **TP15/TP16** (ISNS_INJ) should show the current rising to the 18 A peak in about **38 µs**, from the boost rail (`injector_boost.cir`). - After turn-off, the current should fall in about **36 µs** (`injector_turnoff.cir`). - **BOOST_100V** should be back at its setpoint before the next cylinder, 26.67 ms later.

5. The watchdog-kill test

1. With pins 04 and 06 powered and the firmware driving the outputs, **stop kicking** `WDT_KICK` .
2. At the end of the watchdog window, the TPS3850 asserts `WDO` . `WDO` pulls `RESET` low through D302 (a BAT54Q Schottky, about 0.3 V), so the MCU resets and Q301 turns off. `GATE_KILL` rises to about **12 V** (zener-clamped, 12.7 V maximum). Probe it at R305/Q301's drain.
3. **Every gate must drop:** - injector high and low sides - metering FET - EGR bridge

The kill FETs that do this are Q402–Q405, Q413, Q415, Q417, Q502 and Q507. The relays are not on `GATE_KILL` . The buzzer relay drops because the firmware is in reset. **The main relay stays closed while the ignition is on**, because the wake FET holds it, so on the engine the board keeps its supply through a reset.

Check the current on each dummy load, and the gate of Q412 (low side 1) on the scope. 4. On the scope, `WDT_FAULT` (PC12) and NRST go low together and come back high together. `WDT_FAULT` is high again by the time firmware runs, so it does not record which trip it was. **History:** RESET and `WDT_FAULT` were shorted on the production schematic until 15 Oct 2026; that fix removed the only path from `WDO` to the reset, and D302 (19 Oct) puts it back.

6. The ignition wake, the main relay and pin 25 (Bosch wiring)

On the engine the board has no supply until **pin 71 (ignition)** closes the main relay through the wake FET on pin 50. The firmware then holds the relay through the after-run (decisions log §18).

1. Board powered as in step 2, firmware running. Connect the relay coil (or the 160 Ω resistor) from **+13.5 V to J1004-4** (`RLY_MAIN_OUT` , pin 50).
2. **Ignition off (J1007-5 open), firmware not yet holding the relay:** J1004-4 sits at **13.5 V**. No coil current flows, so the relay is open. This is the sneak path the TVS replaced the diode to remove: there must be no current here.
3. **Ignition on (13.5 V on J1007-5):** J1004-4 falls to **under 0.1 V** and the relay pulls in. **Repeat with the MCU held in reset** (NRST to GND through the debugger): it must still pull in, because the wake FET closes it in hardware.

4. **Ignition off with the firmware running:** the relay stays in. The firmware saves the fault memory, waits for the engine to be stopped (on the bench it already is), then releases it, within about 10 s. `afterrun` reads 1 until then.
5. **Optional: a real main relay.** Wire its contact from the bench supply to J1001-1, -2 and -3, the way the engine does, and repeat 3–4. The board should start on ignition, and switch itself off after ignition off.
6. **Inputs:** 13.5 V on J1007-6 (water in fuel) for over 2 s raises an amber DTC, SPN 97. Grounding J1007-8 (stop switch, the force shutdown) shuts the firmware down. J1007-9 (GCU sync) reads 0–10 V as 0–3.125 V on PB2.
7. **Pin 25 in hardware** (decisions log §18): with the firmware driving the outputs, **halt the MCU from the debugger** (it stays out of reset), then ground J1007-8. - `GATE_KILL` (Q301's drain) rises to about 12 V **within 1 ms**. - Every gate drops, as in step 5. - `~RESET` stays at about **3.0 V**: the MCU is not reset. - Release J1007-8: `GATE_KILL` falls again.

7. The boost cutoff and the heat check

The boost has a hardware overvoltage cutoff since 12 November 2026 (decisions log §19). A 110 V zener (D417) feeds the controller's DIS/EN pin, so if the rail climbs to **105–119 V** the converter stops with no firmware involved. The firmware can also stop it through `BOOST_DIS` (PD14), and does so above 106 V, latching a red DTC (SPN 520192).

1. **Firmware stop:** with step 2's supply on, drive `BOOST_DIS` (PD14) high. - `EN_BOOST` (U401 pin 3) reads about **2.1 V**. - TP12 decays through the 22 k bleed, under 5 V in about 3 s. - Release PD14: TP12 returns to 99.7 V.
2. **Optional, the zener trip itself.** This needs a current-limited supply that reaches 150 V. With pin 04 **unpowered**, feed TP12 from that supply at a **20 mA** limit and raise it slowly from 95 V. `EN_BOOST` must stay under 0.1 V up to 102 V and cross **0.7–1.3 V between 105 and 119.3 V**. Discharge TP12 afterwards.
3. **Heat.** The hot parts were designed to be cooled by the ECU housing, which the prototype doesn't have. Run each load at full current for **10 minutes** and measure each part with a thermocouple or an IR thermometer. **Stop if any part goes over 100 °C.** - the feed FETs Q103 and Q104 - the metering FET Q501 - the EGR bridge FETs Q503–Q506 - the injector FETs Q408–Q416 and diodes D408–D414

The worst case is the EGR load stalled: the bridge carries 6.4 A. 4. **EGR stall guard:** hold the EGR load at stall (or fit a 2 Ω load). After **1.5 s** the firmware lets the bridge coast (`EGR_EN` low) and raises an amber DTC, SPN 2791 FMI 6. It retries after 5 s. After the third stall it stays off until the ignition cycles.

Before the board goes on the engine

- **Conformal-coat it.** Use acrylic or silicone, and mask the connectors, the test points and the debug header. The 100 V nets keep 0.6 mm, which is IPC-2221B's spacing for an *uncoated* board up to 150 V. That is fine on a clean bench, but not with diesel soot and condensation.
- **Heat.** If any part ran above 100 °C in step 7, or the engine bay is hotter than the bench, fit stick-on heatsinks to the FETs above, or a small fan.
- **An independent stop.** Before the first start, confirm that something besides this board can stop the engine on overspeed: the DSE controller's overspeed shutdown, or a fuel or air cut-off within reach.

Pass: all seven steps within their ranges, and TP12 under 5 V before you disconnect. Record the readings in `docs/status.md` with the date and board serial.

ECU simulator (`ecusim`)

The genset ECU's real firmware, running on your computer against a model of the engine, generator and fuel rail. You drive it from a browser: start the engine, change the load, break a sensor, and watch speed, rail pressure, the fault codes (DM1) and the CAN traffic react. Scripts (`.seq` files) run whole test scenarios and check the results.

This is mode 1 of three: firmware only. The design is in docs/spec.md.

On Windows: getting started

1. Download the latest `ecusim-windows` from the repository's **Actions** tab: open the newest green **simulator** run, then **Artifacts**. Unzip it.
2. Double-click `ecusim-windows.exe` .
3. Windows may say "Windows protected your PC", because the program isn't signed. Click **More info**, then **Run anyway**.
4. A black window opens and says where it is running, and your browser opens the simulator. Keep the black window open; closing it stops the simulator.

Nothing to install. Run folders are written into a `runs` folder next to where you started it.

On a Mac it is the same, with `ecusim-macos` . The first time, right-click it and choose **Open** (it isn't notarised). On Linux, use `ecusim-linux` .

Using it

Live. Start cranks the engine until the firmware reaches RUN. Then move the **Load** slider, flip the droop switch, or pick a fault and **Apply** it. **Stop (ignition off)** stops the engine. The speed buttons run at 1×, 2× or 10× real time, or as fast as the computer goes (**max**). **Stop** in the top bar ends the run and writes its folder. **Reset** throws it away.

Scripts. Choose one from the library and **Load** it, or **Open...** a `.seq` file, or drag one onto the editor. Then **Run script**. Errors are listed with their line number; click one to jump to the line. **Save as...** downloads what is in the editor. `.seq` files in a `scenarios` folder next to where you started `ecusim` appear in the library too.

When a script ends, the charts show the whole run and the report lists each check as PASS or FAIL with the value found. The **Runs** list opens any earlier run. The page's address then ends in `#run=<folder>` , which you can bookmark.

Firmware. `ecusim` has the firmware built in. To try another build, **Load firmware…** and pick `ecu_fw.dll` (or `.dylib`, `.so`). It is refused if it was built for a different interface (ABI) and the reason is shown. **Only load firmware you trust: it runs on your computer with your rights.** Each run's folder records which firmware made it.

Writing scripts

One line each; `#` starts a comment; times are seconds of simulated time and never go backwards. `end` is required and comes last.

```
# load step 0 → 20 kW
0    start
8    load 20
13   end
expect rpm min 1350.. over 8..13
expect rpm mean 1485..1515 over 12.9..13
expect never dm1 190/18
```

Action	Meaning
`start`	starter on until the firmware reaches RUN, then off
`starter on / off`	the starter by hand
`load <kW>`	electrical load
`ignition on / off`	ignition input
`ignition bounce <ms>`	off for that long, then on
`droop on / off`	droop switch
`sensor <ch> open` / `short` / `short5v` / `stuck` / `clear`	`ch` is `clt oil rail boost egr 5v`. Open reads 5 V on the coolant sensor (pull-up), 0 V on the others
`sensor <ch> volts <V> / offset <V>`	a forced reading, or the model's plus an offset
`crank lost` / `crank restore`	no crank signal / back to normal
`crank drop <n> / crank extra <n>`	the next n teeth missing / n spurious teeth
`cam lost` / `cam restore`	no cam signal / back
`fuel stuck <mg> / fuel stuck <mg> cyl <1-3> / fuel free`	fuel regardless of the firmware (a runaway, or one stuck injector)
`leak <x>`	the rail leaks x times normal
`request dm2 / dm3 / dm4 / hours`	a J1939 request from a service tool
`power cycle`	power off and on: the firmware restarts from its saved memory
`tear next save`	the next memory save is cut halfway, as by a power loss

Check	Passes when
`expect <signal> <mean|min|max> <lo>..<hi> over <t0>..<t1>`	the value over that window is inside the band (`lo..` or `..hi` leave one end open)
`expect <signal> mean change <lo>..<hi> from <a0>..<a1> to <b0>..<b1>`	the mean moves by that much between the windows
`expect dm1 <spn>/<fmi> within <s> of <t>`	a DM1 carrying it is sent within s seconds after t

Check	Passes when
`expect never dm1 <spn>/<fmi>`	it never appears
`expect dm1 clear over <t0>..<t1>`	every DM1 in the window is empty
`expect lamp amber / red+protect / ... with <spn>/<fmi>`	the first DM1 carrying it lights exactly those lamps
`expect trip within <ms> ms of <t>` / `expect no trip`	SHUTDOWN within that time / never
`expect state RUN at end`	the state when the run stops (`OFF WAIT CRANK RUN SHUTDOWN`)
`expect stored <spn>/<fmi>` / `expect not stored ...`	in the fault memory at the end
`expect dm2 <spn>/<fmi> after <t>` / `expect dm2 empty after <t>`	the first DM2 answer after t

Signals: `rpm` (the engine's), `ecu_rpm` (the firmware's reading), `rail`, `rail_err` (rail over the firmware's setpoint, minus 1), `duty`, `fuel`, `clt`, `oil`, `load`. An FMI of `*` matches any.

The library's `sil/` scripts are the firmware's 16 test scenarios. `faults/` has one demonstration per fault.

Run folders

Every run writes `runs/<date>-<time>-<name>/` (the time is UTC):

File	Content
`trace.csv`	one row per simulated millisecond
`can.csv`	every CAN frame, sent and received
`events.txt`	everything done, scripted or by hand, as a script: it replays
`report.txt`	each check and what was found
`firmware.txt`	which firmware: built in, or the file's name and SHA-256

From the command line

```
ecusim                          open the browser UI (port 8025, or the next free one)
ecusim --port 9000 --share      let other computers on the network watch; firmware uploads are r
ecusim run scenarios/sil/*.seq  run scripts headless, as fast as possible
ecusim run my.seq --fw ecu_fw.dll --out results
```

`ecusim run` exits 0 if every check passes, 1 if any fails, and 2 on a script or firmware error.

Building from source

Rust 1.80 or newer (https://rustup.rs). On Windows that needs the Visual Studio C++ Build Tools, which rustup offers to install. They also compile the firmware.

```
cd simulator
cargo build --release -p ecusim     # target/release/ecusim(.exe)
cargo test --workspace              # everything, including the 16 scenarios
cargo run --release -p fwbuild      # the firmware as build/ecu_fw.dll / .dylib / .so
```

`fwbuild --core DIR --out FILE -D NAME=VALUE` builds another copy of the firmware, such as a branch you are trying.

The crates: `plant` is the engine model, ported from `firmware/sim/engine.py` and checked against its traces. `sim` is the firmware binding, the loop, scripts and checks. `ecusim` is the program and its page (`crates/ecusim/ui/`). `fwbuild` builds the firmware library.

Changing `ecu_io` or an entry point in `firmware/core/ecu.h`? Bump `ECU_ABI_VERSION` there and `ABI_VERSION` in `crates/sim/src/fw.rs`, and copy the struct change into `EcuIo`. The `io_size_matches` test catches a missed field.

Research Memo 01: Engine Identity

Task: U11 (engine make/model/displacement/cylinder count/rated speed/emissions tier). Informs U8, U9. **Date:** 2026-09-13 **Author:** Phase-1 research (task 1)

Verdict

> **SUPERSEDED 18 Sep 2026 — READ THIS FIRST**
>
> The engine's rating plate has been photographed. It reads `3GK550ETA 4SR1`, application code `GK3.8703`, 26.5 kW / 36 hp at 1500 rpm, type approval `ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24`, manufactured 12 Sep 2025.
>
> **This memo's model-number conclusion below is wrong.** It identified the engine as `3R550ETA 4G1` and explicitly rejected "GK550" as a garbled recollection of "KG4" + "550", advising that GK550 "should not be used as a search term or part-lookup key going forward". The original designation given to this project was right; this memo's inference was not.
>
> **What survives:** every hardware constraint in the Step 3 cross-check — three cylinders, common rail, turbocharged-aftercooled, cooled EGR with DOC and no SCR, 1500 rpm, CPCB IV+, 26.5 kW — matches the plate exactly. The reasoning from the wiring diagram was sound. What failed was the model-number identification built on top of it, which converged on a real Kirloskar engine that is not this one.
>
> **Also resolved:** this memo recorded as a negative result that "GP3 cannot be resolved to a specific engine model or family". It is a project/application code — the wiring diagram is `GP3.8703.C4.pdf` and the engine's serial number begins `GK3.8703/`. The 8703 matches.
>
> Evidence: `refs/field-evidence-2026-09-18.md`. **U11 is closed**, by plate rather than by inference.

Verdict (as originally written, superseded above)

The engine is almost certainly the **Kirloskar 3R550ETA 4G1** (Kirloskar "R550" family, 3-cylinder, 1.65 L, turbocharged-aftercooled, common-rail, cooled-EGR + DOC), as fitted to the **Kirloskar KG4-25WS1** 25 kVA CPCB IV+ genset. Every hardware constraint from the wiring-diagram cross-check (Step 3) is satisfied, and the genset controller named in the OEM brochure (KG640C) is the exact same controller named in the OEM wiring-diagram extraction. This is treated as **inferred with high confidence** rather than **confirmed**, because no single primary source explicitly states "manual 08-GP3-60-001 / file GP3.8703.C4.pdf belongs to engine 3R550ETA 4G1" — the identification is a convergence of independent documents, not a direct statement. A rating-plate photograph (see "What would close this," below) would upgrade this to confirmed.

"GK550" as reported by the user does not match any Kirloskar Oil Engines (KOEL) designation found in searching. It is most plausibly the user's own conflation of the genset model prefix **KG4** and the engine family number **550** — but this is speculation on my part, not a sourced fact, and is flagged as such.

Step 1: What does "GK550" designate?

Searched: `Kirloskar GK550`, `Kirloskar Green GK 550 25 kVA genset`, `KOEL GK550 25 kVA diesel genset specification`, `"GK550" Kirloskar engine OR genset OR generator`, `Kirloskar "GK" series genset air-cooled OR water-cooled kVA range`.

Result: no primary source uses the string "GK550." No Kirloskar Oil Engines product page, brochure, distributor listing, or spare-parts catalogue found in this search returned that exact designation.

(Negative result — searched via general web search, no domain restriction, September 2026.)

One lead was run down and ruled out: - ManualsLib lists a document titled "KIRLOSKAR GK SERIES INSTRUCTION ON INSTALLATION, OPERATION AND MAINTENANCE" (https://www.manualslib.com/manual/1698613/Kirloskar-Gk-Series.html). Search-result snippets describe this as covering Kirloskar pump type **GK(P)** — a **Kirloskar Brothers Limited** (pumps) product, not a Kirloskar Oil Engines (engines/gensets) product. These are different companies under the Kirloskar Group umbrella. **Ruled out** — different Kirloskar Group entity, wrong product category. *(Confirmed via search-result description; the page itself returned HTTP 403 to direct fetch and could not be read in full.)*

What KOEL genset/engine naming actually looks like, established from primary sources fetched directly: - Genset models use a `KG<n>-<kVA><suffix>` pattern, e.g. **KG4-25WS1** (25 kVA), KG4-30WS1, KG4-40WS1, KG4-58.5WS (all from the same brochure) — source: Kirloskar Oil Engines, "CPCBIV+ COMPLIANT — INDIA'S LARGEST FLEET OF GENSETS 25-58.5 kVA," https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf (downloaded and text-extracted directly; doc ref `20240610/REV01/A4 CPCB IV+25-82.5 kVA`, footer code `IBG.V.045 Rev0`). **Confirmed.** - Engine models use an `<cyl>R<displacement-code><aspiration><generation>` pattern, e.g. **3R550ETA 4G1** (3-cylinder, "550" family, Electronic-governed, Turbocharged-Aftercooled, generation suffix "4G1") for the 25 kVA genset in the same brochure. Source: same PDF, table row "ENGINE / Engine Model." **Confirmed.** - A related, non-genset-specific brochure from Kirloskar Americas (https://www.kirloskaramericas.com/documents/5928996/5929154/Kirloskar+R550+G+Drive+Brochure+25052022+Rev0.pdf — downloaded directly) lists the bare engine-platform naming: **2R550NA, 3R550NA, 3R550TC, 3R550TA** — i.e., "R550" is the engine *platform* name, and the suffix encodes cylinder count and aspiration. **Confirmed.**

Working hypothesis (inferred, not sourced): the user's "GK550" is most likely a garbled or telescoped memory of **"KG4"** (the genset-model prefix) and **"550"** (the engine-family number that actually appears in the engine model string, 3R**550**ETA), possibly by transposing K/G. I found no evidence this is an official SKU, a genset model, or a Kirloskar Green product code in its own right. It should not be used as a search term or part-lookup key going forward — use **KG4-25WS1** (genset) and **3R550ETA 4G1** (engine) instead.

Step 2: Working backwards from the manual number

Manual number `08-GP3-60-001`, file `GP3.8703.C4.pdf`, served from `https://ikonnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram?manualName=08-GP3-60-001` (confirmed present in the PDF's own footer on every page, verified via `pdftotext` extraction of the source file).

Attempts made: - Fetched `https://ikonnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram` (no query string) directly — connection failed (no response body; the endpoint requires the `manualName` parameter and/or an authenticated session). - Fetched `https://ikonnect.kirloskar.com/KOEL` (portal home) — retrieved (22 KB), but it is a login/marketing shell with no manual index; no list of manual-family codes is exposed to an unauthenticated fetch. - Searched `Kirloskar "GP3" engine family diesel`, `Kirloskar "GP3" ECU OR "engine control unit" wiring diagram manual`, `"08-GP3-60-001" Kirloskar`, `Kirloskar "GP3" platform OR series diesel engine common rail EGR CPCB` — **no result anywhere maps "GP3" to a specific engine model or family.** *(Negative result.)* - Read the wiring diagram itself in full (`pdftotext -layout` extraction of `/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf`, all 5 pages) — it does not state an engine model number anywhere in its body text. It is a browser print-to-PDF of the ikonnect portal page (page metadata shows

a print timestamp), consistent with the note already recorded in `ecu-pinout-extracted.md` that the diagram is troubleshooting-oriented and does not carry a full connector/engine cross-reference.

Conclusion: "GP3" cannot be resolved to a specific engine model or family from public sources. *(Negative result — this closes off Step 2 as originally scoped.)* A plausible but **unconfirmed and unsourced** guess: KOEL's CPCB IV+ genset range (7.5–800 kW) uses the same control architecture (common rail, cooled EGR, DOC, KG640C controller) across several engine displacements (R550, R1040/R1190, R810, per the CPCB IV+ 25-58.5 kVA brochure), so "GP3" may denote a shared ECU/wiring *platform generation* spanning multiple engine sizes rather than one engine. This would explain why the diagram itself carries no engine-specific model text — it may be reused, with only the pin count and sensor complement varying by engine. This is **speculation, explicitly not a finding**, and should not be treated as established.

Step 3: Cross-check against the hardware evidence

Per the brief, a valid candidate must match **all five** of: 3 cylinders; common-rail; EGR fitted; catalyst temp sensor present but no SCR/DEF; 12 V electrical system. Evidence for each column below is drawn directly from `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` (hardware) and the KOEL CPCB IV+ 25-58.5 kVA brochure (candidate engine spec, downloaded and extracted directly from `https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf`).

Constraint	Wiring-diagram evidence	3R550ETA 4G1 (candidate)	Match
3 cylinders	Injector bank wiring: cyl 1/2/3, pins 03/05/73/07/29 (pinout doc, "Injectors" section)	"No. of cylinder: 3" (brochure table)	**Match — confirmed**
Common-rail	Rail pressure sensor (pins 32/35/08) + PWM-driven fuel metering unit (pin 88) (pinout doc, "Analog sensors" and "Outputs")	CRDi (Common Rail Direct Injection); brochure narrative: "High pressure common rail system employed on Kirloskar CPCB IV+ Gensets"	**Match — confirmed**
EGR fitted	Dedicated EGR position sensor (pins 15/37/36) and EGR H/L bridge-driven actuator output (pins 59/81) (pinout doc, "Analog sensors" and "Outputs")	"EGR System used to reduce the level of NOx emitted by Engine" (brochure feature callout)	**Match — confirmed**
Catalyst temp sensor present, no SCR/DEF	Catalyst Temperature Sensor connector present in connector appendix (2-pin, one only); pinout doc's "Absent from the entire system" section: "No DEF/urea sensor... No SCR"	Brochure lists **DOC** ("DOC system sets off the reaction to meet the CPCB norms") for this kVA family; **no SCR or DEF mentioned anywhere** in the 25–58.5 kVA brochure	**Match — confirmed**
12 V electrical system	Battery+ feed pins 04/06/21, all via fused red supply; no 24 V evidence anywhere in diagram	"Electrical Battery Start in R Voltage: 12 Volts-DC" (brochure); user-reported 12 V/75 Ah battery	**Match — confirmed**

All five constraints are satisfied simultaneously by only one engine in the brochure's four-model 25–58.5 kVA lineup (the other three — 3R1190ENA, 3R1190ETA, 4R810ETA — are 30/40/58.5 kVA respectively, not 25 kVA, and one is naturally aspirated, so they fail on rated output alone even before considering hardware detail). **This is the strongest evidence in the memo** and is why the identity is called "inferred with high confidence" rather than a bare guess.

Independent corroboration found during this cross-check (not requested by the task, but material): the pinout extraction names the genset control unit as **Kirloskar KG640C** (`ecu-pinout-extracted.md`, "Other system components named"). The CPCB IV+ 25-58.5 kVA brochure independently names its standard controller as the **KG640C Controller** (same PDF, "State of the art Genset Controller" section, pictured with a photo captioned "KG640C Controller"). Two independently-obtained documents (a scraped OEM troubleshooting diagram, and a public marketing brochure) naming the identical, fairly specific controller part number is strong circumstantial confirmation that the wiring diagram belongs to this genset family. **Confirmed** (both documents state the KG640C designation directly; the *inference* is that they describe the same unit).

Step 4: Emissions tier

CPCB IV+ is confirmed as the applicable tier for this class of genset:

- Regulatory basis: **GSR 804(E), dated 3 November 2022**, Ministry of Environment, Forest and Climate Change — "Emission limits for new internal combustion engine up to 800 kW gross power used for power generating set (Genset) application," commonly referred to as the CPCB IV+ regulation. *(Confirmed via independent secondary source: https://www.ricago.com/blog/revised-emission-standards-for-gensets-a-step-towards-cleaner-air, which states the GSR number and date; corroborated by a second secondary source in the same search. Primary source page https://cpcb.gov.in/genset-notifications/ (redirected from `cpcb.nic.in`) confirms the base notification chain "GSR 771(E) dated 11 Dec 2013" and its amendments exist at CPCB, but its listing text as fetched did not expose the GSR 804(E) reference by name — flagged as a fetch limitation, not a contradiction.)*
- Effective/compliance date: 1 July 2023 (secondary sources; not independently verified against gazette text).
- Power-band limits (19–56 kW band, which contains this engine's 26.5 kW prime rating): PM ≤ 0.03 g/kWh, combined NOx+HC ≤ 4.7 g/kWh, CO ≤ 3.5 g/kWh, smoke ≤ 0.7 — per dieselnet.com's published summary table of the 2022/2023 Indian genset standard (https://dieselnet.com/standards/in/genset.php). **Inferred/secondary-sourced** — dieselnet is a respected industry standards clearinghouse but this is not the gazette text itself.
- The genset manufacturer's own literature states this engine/genset combination is CPCB IV+ certified and uses cooled EGR + DOC (no SCR, no DPF mentioned) to meet it — consistent with the 19–56 kW band not requiring the separate, stricter NOx-only limit (0.40 g/kWh) that appears at the 56–560 kW band, where SCR is more commonly needed to hit the tighter NOx number. This is **inferred**: no source states "this band never needs SCR," but the absence of SCR/DEF hardware, plus the combined NOx+HC limit form used below 56 kW, is internally consistent with why KOEL doesn't fit SCR to this genset size.

Implication for aftertreatment scope (informs U8): design should assume a **DOC-only aftertreatment path with cooled EGR** — no SCR, no DEF/urea system, no DPF regeneration logic. This matches the hardware evidence exactly (single catalyst-temperature-sensor connector, no urea pump/quality sensor anywhere in the harness).

Step 5: Summary — engine identity record

Field	Value	Status	Source
Engine make	Kirloskar (Kirloskar Oil Engines Ltd, KOEL)	Confirmed	KG4-25WS1 brochure (below)
Engine model	3R550ETA 4G1 (OCR'd in one source as "3R55OETA 4G1" — "O"/"0" artifact)	Inferred, high confidence (see Step 3)	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure, https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf ; corroborated by https://www.indiamart.com/proddetail/kirloskar-kg4-25ws1-diesel-generator-2854043860488.html
Engine platform/family	"R550" (Kirloskar's own naming)	Confirmed	Kirloskar Americas R550 G-Drive brochure, https://www.kirloskaramericas.com/documents/5928996/5929154/Kirloskar+R550+G+Drive+Brochure+25052022+Rev0.pdf
Genset model	KG4-25WS1	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above); indiamart listing (above)
Displacement	1.65 L (1,650 cc, 3-cylinder)	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above); cross-checked arithmetically: 86 mm bore × 94 mm stroke × 3 cyl ≈ 1,638 cc, consistent with rounded 1.65 L spec
Bore × stroke	86 mm × 94 mm	Confirmed	Kirloskar Americas R550 G-Drive brochure (above)
Cylinder count	3 (inline)	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above); matches wiring-diagram injector wiring (cyl 1/2/3)
Compression ratio	19.7:1	Confirmed	Kirloskar Americas R550 G-Drive brochure (above)
Aspiration	Turbocharged, aftercooled (TA)	Confirmed	Both Kirloskar brochures (above)
Rated speed	1500 rpm	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above) — matches 50 Hz expectation from the brief
Rated output	26.5 kW / 36 HP prime, per ISO 8528-1	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above)

Field	Value	Status	Source
Genset rated output	25 kVA prime, 20 kW, 0.8 PF, 50 Hz, 230 V(1φ)/415 V(3φ)	Confirmed	Same brochure
Fuel injection	Common rail direct injection (CRDi), electronic governor	Confirmed	Same brochure; hardware corroboration via rail pressure sensor + PWM fuel metering unit in pinout doc
EGR	Cooled EGR fitted	Confirmed	Same brochure; hardware corroboration via EGR position sensor + EGR actuator pins in pinout doc
Aftertreatment	DOC only; no SCR, no DEF, no DPF mentioned	Confirmed (absence)	Same brochure narrative; hardware corroboration via single catalyst-temp-sensor connector and complete absence of urea/SCR hardware in pinout doc
Emissions tier	CPCB IV+ (GSR 804(E), 3-Nov-2022; effective 1-Jul-2023)	Confirmed regulation exists; inferred that this specific unit is certified to it (brochure says so but I have not seen a type-approval certificate)	Brochure states CPCB IV+ compliance; GSR number/date per https://www.ricago.com/blog/revised-emission-standards-for-gensets-a-step-towards-cleaner-air ; band limits per https://dieselnet.com/standards/in/genset.php
Genset controller	Kirloskar KG640C	Confirmed	Both the wiring-diagram pinout extraction and the CPCB IV+ brochure name this controller independently
Battery	12 V DC starting	Confirmed	Brochure ("Electrical Battery Start in R Voltage: 12 Volts-DC"); matches user-reported 12 V system
Battery Ah	User reports 75 Ah; generic Kirloskar Americas R550 brochure lists 65 Ah for the bare-engine starter spec; a KOEL-Green-branded 12 V/75 Ah battery (part 02.709.24.0.Pr, model 75D31R-A) is sold as a genuine Kirloskar spare	Inferred (discrepancy noted, not resolved)	Kirloskar Americas R550 G-Drive brochure (65 Ah, generic/global engine-only spec); https://www.safesparesonline.com/product/kirloskar-genuine-02-709-24-0-pr-12v-dc-75ah-lead-acid-koel-green-battery-model-75d31r-a/ (75 Ah, India-market genuine part)
OEM ECU part number	Not found	Not established	See Step 2 — no source ties a Bosch/Continental/etc. ECU part number to this engine or to manual `08-GP3-60-001`
ECU manual/family code	`08-GP3-60-001` / "GP3" — meaning of "GP3" not established	Not established	See Step 2 (negative result)

What would close this — the one photograph that settles it

The engine's own **rating plate** (usually riveted to the rocker cover, flywheel housing, or side of the cylinder block) would resolve every remaining "inferred" item above in one shot. Specifically, photograph it so the following are legible:

1. **Engine model string** — should read something matching `3R550ETA` (or a close variant) if this identification is correct. If it instead reads e.g. `3R1040...` or a different family entirely, this whole identification is wrong and should be discarded rather than patched.
2. **Serial number** — allows a direct KOEL/dealer parts-catalogue or warranty lookup, independent of any web search.
3. **Rated output, rated speed, and frequency** stamped on the plate — cross-check against the 26.5 kW / 1500 rpm / 50 Hz figures above.

4. **Emission certification marking** if present (CPCB IV+ or equivalent BEE/ARAI type-approval sticker) — would directly confirm Step 4 rather than relying on brochure-level marketing claims.
5. **The genset's own nameplate** (usually on the canopy or control panel, separate from the engine plate) — should read `KG4-25WS1` or a close variant if this identification is correct, and will separately carry the genset serial number, which is the cleanest path to an authoritative KOEL parts/spec lookup via an authorized dealer or KOEL Care.

Absent that photograph, treat every "inferred" row above as provisional. The hardware cross-check in Step 3 is strong (five independent constraints all satisfied by exactly one candidate in the only public document that lists 25 kVA CPCB IV+ engines), but it is still an inference from convergence, not a direct statement of identity from a primary source that names manual `08-GP3-60-001` and engine `3R550ETA 4G1` in the same document.

Sources consulted (full list)

- `/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf` — OEM wiring diagram, manual `08-GP3-60-001`, read in full via `pdftotext -layout`.
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — pin-level extraction of the above, read in full.
- Kirloskar Oil Engines, "CPCBIV+ COMPLIANT — INDIA'S LARGEST FLEET OF GENSETS 25-58.5 kVA": https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf — downloaded directly and text-extracted in full (283 lines).
- Kirloskar Americas, "R550 G-Drive" brochure: https://www.kirloskaramericas.com/documents/5928996/5929154/Kirloskar+R550+G+Drive+Brochure+25052022+Rev0.pdf — downloaded directly and text-extracted in full (blocked to the WebFetch tool with HTTP 403; retrieved successfully via direct HTTP GET).
- https://www.kirloskaroilengines.com/products/power-gen/genset/lhp/diesel-generators/25-kva-diesel-genset — fetched, gives KG4-25WS1 and engine model 3R550ETA 4G1.
- https://www.indiamart.com/proddetail/kirloskar-kg4-25ws1-diesel-generator-2854043860488.html — fetched, corroborates engine model, rated speed 1500 rpm, CRDi, EGR+DOC/CPCB IV+.
- https://www.indiamart.com/proddetail/kirloskar-3r550-engine-2850959578848.html — fetched, US-market bare-engine listing corroborating 3-cylinder, ~1.64 L (100 cu in), NA/TC/TA variants.
- https://www.safesparesonline.com/product/kirloskar-genuine-02-709-24-0-pr-12v-dc-75ah-lead-acid-koel-green-battery-model-75d31r-a/ — found via search, corroborates a 12 V/75 Ah Kirloskar Green genuine battery part exists in the India market.
- https://www.manualslib.com/manual/1698613/Kirloskar-Gk-Series.html — found via search, used to rule out "GK Series" as a Kirloskar Brothers pump line unrelated to this engine (page itself returned HTTP 403 on direct fetch; ruled out based on search-result description only — flagged as weaker evidence).
- https://ikonnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram (no query) and https://ikonnect.kirloskar.com/KOEL — fetched directly; no manual-family index exposed without authentication.
- https://dieselnet.com/standards/in/genset.php — fetched, CPCB IV+ power-band emission limits table.
- https://cpcb.gov.in/genset-notifications/ (redirected from https://cpcb.nic.in/genset-notifications/) — fetched, confirms existence of the GSR 771(E)/2013 notification chain and its amendments; did not by itself surface the GSR 804(E)/2022 CPCB IV+ reference.
- https://www.ricago.com/blog/revised-emission-standards-for-gensets-a-step-towards-cleaner-air — fetched, states GSR 804(E) dated 3 November 2022 as the CPCB IV+ notification.
- General web searches (not individually cited above where they returned only negative results or were superseded by direct fetches): `Kirloskar GK550`, `Kirloskar Green GK 550 25 kVA genset`, `KOEL`

`GK550 25 kVA diesel genset specification`, `Kirloskar "GP3" engine family diesel`, `ikonnect.kirloskar.com TroubleShootingWiringDiagram manualName`, `Kirloskar "3R550ETA" engine specifications`, `Kirloskar "R550" OR "3R550" displacement bore stroke cc`, `site:kirloskaramericas.com R550`, `Kirloskar R550 engine "1500 rpm" OR "1800 rpm" kW genset industrial`, `Kirloskar "GP3" ECU OR "engine control unit" wiring diagram manual`, `"08-GP3-60-001" Kirloskar`, `CPCB IV+ norms diesel genset 19-56 kW NOx PM DOC EGR SCR requirement notification date`, `CPCB IV+ GSR notification number "3rd November 2022" gensets gazette`.

Research Memo 02: KG640C Genset Controller (GCU) — J1939 Contract and the MPU Safety Gate

Task: U3 (safety gate), U4. **Task 2, phase-1 research — highest priority in the phase. Date:** 2026-09-13 **Author:** Phase-1 research (task 2)

SUPERSEDED, 18 Sep 2026 — the fitted controller is NOT a KG640C

The controller has been photographed. It is a **Deep Sea Electronics DSE4522 MKII AMF (India SP)**, part `4522-001-01`, serial 11021794, made in the UK. Not a KG640C. A Kirloskar-issued configuration for it was supplied alongside: `GP3.314.01.0.PR – DSE 4522 CONTROLLER PROGRAM FOR 3R550.25KVA, 3PHASE (GK PROJECT)`, naming this engine and this rating.

This memo analyses a KG640C. The OEM wiring diagram in this repo shows a KG640C, and both documents carry `GP3.` numbering, so which controller is actually fitted is now an open question rather than a settled one. Spec section 3's safety position is built on this memo.

Two specific contradictions, neither yet resolved:

- **Output mapping.** This memo assigns output A to the start relay and output B to the ignition relay. The DSE4522 configuration assigns **A = Fuel Relay, B = Start Relay**. A separate reading of the wiring diagram gives `-13RB1` = ignition on B and `-13RB2` = start on A. Three sources, three mappings.
- **Speed source.** This memo's central finding is that the controller senses speed from alternator frequency independently of the ECU. The DSE4522 configuration carries `Module To Use Engine Speed = No`, whose meaning in DSE's own semantics must be established from their documentation before anything is concluded from it.

What the new document adds rather than contradicts: an Emergency Stop input wired fail-safe, `ECU Data Fail → Shutdown`, CAN source addresses 234 and 44, and the protection thresholds the controller applies to values our ECU supplies. Those partly answer U4.

Both contradictions are now resolved against this memo. DSE's own manual (`057-260`) states that with a CAN engine the ECU transmits engine speed to the controller — so the independent alternator-frequency *speed* path this memo describes does not exist on the fitted device. Spec §3 has been rewritten on the DSE4522, and the standalone overspeed trip module reverts to **mandatory**.

What survives and is still worth reading here: the reasoning method, the J1939 background, and the costing of the two trip-module candidates (GAC SSW675, Murphy HD9063), which is device-independent. The memo's conclusion that no magnetic pickup exists also turned out to be right — the DSE4522 has no MPU terminal either — but it was right by coincidence, about a different controller.

Treat everything below as **historical**: an analysis of a device that is not on this machine.

Evidence: `refs/field-evidence-2026-09-18.md`.

Verdict (U3 — read this first)

The KG640C genset controller has no dedicated magnetic pickup (MPU) input terminal. Verdict: No.

Status: inferred, high confidence (not "confirmed" outright — see caveat below) — from two Kirloskar Oil Engines Limited (KOEL) primary documents, not from the wiring diagram's silence:

1. **The OEM wiring diagram itself** (`/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf`, page 1, "GENSET CONTROL UNIT KG640C" block) shows the controller's terminal groups **J1 through J8 in full** — every terminal on the block is labelled: J1 (terminals 1–2, `12/24` battery-sense), J2 (terminals 3–6, digital outputs A–D: Start Relay, Ignition Relay, MC Connect Relay, Common Fault), J3 (terminals 7–10: `D+CHG ALT`, digital outputs/inputs), J4 (terminals 11–16: digital input, **CAN L/H**, RS485 A/B), J5 (terminals 17–26: reserved, digital inputs D/E, analog inputs), J6 (terminals 27–30: genset alternator voltage/frequency sense, fed from the DG alternator's 415 VAC 3-phase output), J7 (mains voltage sense N/B/Y/R), J8 (terminals 35–42: CT current-sense inputs, SCP). **No terminal in J1–J8 is labelled MPU, speed pickup, or magnetic pickup.** Every terminal on the block is accounted for, so this is a complete negative, not silence about an unused pin. *(Confirmed — read directly from the diagram at 400 DPI.)*
2. **A genuine KOEL controller manual for the same product family, "KG640 CONTROLLER MANUAL"** (KOEL letterhead, Laxmanrao Kirloskar Road, Khadki, Pune; internal document number `SED-MAN-KG640-002`), found hosted on Scribd (uploaded by an account named `rajinder.kirloskar`) at https://www.scribd.com/document/878092942/SED-MAN-KG640-002-Koel-Manual-for-KG640-Genset-Controller and retrieved in full (92 pages). This is the **base "KG640" variant** (no CAN bus at all — see §2 below), but its terminal architecture matches the KG640C block on the wiring diagram almost exactly: terminals 1–2 battery, 3–6 digital outputs, 7 `D+ CHARGE ALT`, 27–30 `GEN_V IN` (N,B,Y,R), 35–40 `GEN_CT_IN`, 41 `SCP` — the same terminal numbers doing the same jobs as the KG640C block in the wiring diagram. The manual's **complete 42-terminal table (its Table 15, "Terminal Description")** enumerates every terminal from 1 to 42 with a name; there is no MPU/speed-pickup terminal anywhere in it. Its glossary lists "MPU" as an abbreviation (page iii) but the term is **never used again anywhere in the 92-page body** — consistent with a shared glossary template across KOEL controller documents rather than evidence this unit has one. **Its own configuration parameter for engine speed, "Engine Speed Sense Source," offers only one value:** `Alternator frequency` (§16.8.2, "Speed monitoring engine screen": *"Allows to select electrical signal to be use for measurement of engine speed. Alternator frequency: Engine speed will be sensed from alternator frequency."*). No magnetic-pickup or CAN option is listed for that parameter anywhere in the document. *(Inferred, high confidence, from a primary KOEL manufacturer document — see caveat below on why this isn't "confirmed" outright.)*

Caveat, stated plainly: this manual documents "KG640," not "KG640C" by name, and it predates or excludes the CAN bus that the wiring diagram's KG640C block clearly has (J4 terminals 13/14, CAN L/H — repurposing what the base manual lists as "Reserved" terminals 13/14). The "C" suffix is therefore read as "CAN-added" variant of the same terminal architecture, not a different product — the terminal-for-terminal match on every other function (genset voltage sense, CT sense, digital I/O, D+ charge alt, RS485) is too exact to be coincidental.

But no manual titled specifically "KG640C" was located, so this is an inference from a closely related sibling document, not a direct statement about the exact SKU. This is why the verdict is **"No, inferred with high confidence"** rather than **"No, confirmed."** Per the task's own safety framing, concluding "no MPU" when the true answer might be "yes" is the safe-direction error (it triggers an unnecessary but harmless redundant hardware purchase); concluding "yes" when the answer is "no" is the dangerous-direction error. This memo is deliberately conservative.

What would upgrade this to "confirmed": a manual or datasheet with "KG640C" in the title/header (not just the base KG640), a photograph of a physical KG640C rear terminal block, or a call to KOEL technical support quoting manual number `08-GP3-60-001` (the engine wiring diagram's own manual number, from Task 1) and asking directly.

A materially important side-finding: the GCU already has a speed-sensing path independent of the engine ECU and CAN bus

The base KG640 manual's Table 5 ("Genset Voltages and Frequency") specifies terminals 27–30 sample the **genset's own AC output** at **5 kHz**, deriving frequency over a **3–75 Hz range at 0.1 Hz resolution, 0.25% accuracy**. Because this 4-pole alternator's electrical frequency is rigidly tied to shaft speed (1500 rpm ↔ 50 Hz, no slip), and because the GCU's "Engine Speed Sense Source" defaults to reading this alternator frequency rather than a CAN message from the engine ECU, **the GCU's own over-speed and gross-over-speed shutdown logic (manual §16.8.2, alarms #27/#28 in the base manual's alarm table) does not depend on the engine ECU or the CAN bus between them** — it depends only on the GCU's own AC front-end and its own microcontroller. This is **not** a substitute for a hardware MPU trip (it still shares fate with the GCU's firmware/microcontroller, and a CAN-bus failure alone would not defeat it, but a GCU firmware hang would), and it does **not** change the verdict above. But it is directly relevant to the design decision that follows from the verdict, and should inform how the team weighs the residual risk before deciding how much redundancy the standalone module below needs to add. *(Inferred from the base KG640 manual's specification tables and alarm list; not confirmed for the KG640C variant specifically, since KG640C-specific documentation was not located.)*

Recommended standalone overspeed trip module candidates (since the answer is "No")

Two candidates, both electronic speed switches driven from a magnetic pickup at the flywheel ring gear, independent of any ECU or GCU microcontroller:

#	Part	Function	Key specs (manufacturer-stated)	India availability / price found
1	**Governors America Corp (GAC) SSW675** "2-Element Speed Switch — Overspeed, Crank"	Dedicated dual-setpoint electromechanical-relay trip switch: low setpoint = crank disconnect, high setpoint = overspeed shutdown. Magnetic-pickup input, 5 A relay output, on-board overspeed test circuit.	Manufacturer product page: https://www.governors-america.com/product/ssw675/ (fetch blocked by site, confirmed via search-result excerpt quoting the manufacturer's own description)	**India price not confirmed this session** — GAC/Governors America Corp parts are stocked by Indian industrial resellers (e.g., Spares Bazaar Pvt Ltd, New Delhi, and HT Industrial Automation Ltd, which list other GAC part numbers), but no INR figure for SSW675 specifically was found. A related GAC part family (ESD5100-series analog governors, a **different product — NOT a dedicated overspeed relay**, and GAC's own ESD5500E documentation explicitly states a separate overspeed shutdown device is still required) is listed at **₹9,500/piece** by Spares Bazaar Pvt Ltd, New Delhi (https://www.indiamart.com/proddetail/esd5500e-esd5111-esd5550-esd5221-gac-speed-control-unit-governor-america-corp-21293246791.html) — cited only as a rough order-of-magnitude anchor for this brand/class of device in the Indian market, **not** as pricing for the correct part.
2	**Murphy by Enovation Controls HD9063** dual-setpoint electronic speed switch	Crank disconnect (250–6,000 Hz) + overspeed (1,100–10,000 Hz) on one module. Magnetic-pickup input, 0.35–60 Vrms signal range, 8–30 VDC supply, SPDT relay output 5 A/30 VDC, built-in overspeed test circuit and trip-point LEDs. Explicitly marketed for "generator sets, pumps, and compressors."	Manufacturer product page: https://www.enovationcontrols.com/products/hd9063-os77d-and-ss300/ (fetched directly, specs quoted above verbatim)	Indian distributor confirmed: **"Murphy By Enovation Controls," Pune, Maharashtra**, IndiaMART storefront lists HD9063/OS77D/SS300 (https://www.indiamart.com/proddetail/hd9063-os77d-and-ss300-electronic-speed-switches-15889251691.html) but the listing is quote-only ("Get Latest Price") — **no published INR figure**.

Negative result, recorded honestly: this session did not obtain a firm INR list price for either correctly-matched dedicated overspeed module (SSW675 or HD9063). Both have confirmed Indian distributors who would need to be contacted directly for a quote. A single-function analog GAC governor (a different, not-directly-comparable product) was found at ₹9,500 in India as the only concrete INR figure this search turned up for the GAC brand; it is reported here only for order-of-magnitude context, flagged explicitly as the wrong product class, and should not be used as a quote basis.

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1. Controller identity and sourcing basis

- **Controller:** Kirloskar KG640C, named independently by the OEM wiring diagram (`GP3.8703.C4.pdf` , page 1) and by KOEL's public CPCB IV+ 25–58.5 kVA brochure (Task 1, `01-engine-identity.md`). *(Confirmed.)*
- **Manual used:** `SED-MAN-KG640-002` , "KG640 CONTROLLER MANUAL," KOEL, found via Scribd (https://www.scribd.com/document/878092942/SED-MAN-KG640-002-Koel-Manual-for-KG640-Genset-Controller), retrieved and read in full (92 pages). This is a genuine KOEL-authored document (KOEL letterhead, internal doc number, copyright notice), located via a document-sharing site rather than kirloskar.com directly — the **content is primary** (manufacturer-authored), but its **hosting is secondary** (Scribd, not KOEL's own domain). Flagged for transparency; the manual's content is treated as manufacturer documentation for the purposes of the "no safety-relevant conclusion from a secondary source" rule, because the document itself is KOEL's own, not a reseller's or forum's description of KOEL's product.

- **Not investigated further as a "rebadge" lead:** the brief flagged Deep Sea Electronics, ComAp, Datakom, SmartGen and Lovato as common Indian-market OEM sources. A SmartGen HGM6100N manual was pulled during this investigation as a comparison candidate, but the KOEL-letterhead KG640 manual's near-exact terminal-for-terminal match to the KG640C block in the OEM wiring diagram is strong enough evidence that **KG640C is Kirloskar's own documented product**, not a rebadge requiring third-party manual substitution. This conclusion is **inferred**, not from a statement anywhere that KG640C is Kirloskar's original design, but from the convergence of two independent KOEL-branded documents describing the identical terminal architecture.
- **No manual titled specifically "KG640C" was located** despite searching `Kirloskar KG640C manual`, `"KG640C" Kirloskar controller manual pdf`, `scribd "KG640C" OR "SED-MAN-KG640C"`, and the `ikonnect.kirloskar.com` portal (which exposes no manual index to an unauthenticated fetch, consistent with Task 1's finding). A sibling document, "KG645 Genset Controller User Manual" (https://www.scribd.com/document/407079798/Kirloskar-KG645-User-manual-for-genset-controller-pdf, 123 pages), was found and may be a closer CAN-capable relative, but its full text could not be retrieved this session (Scribd's preview blocked the fetch tool; only metadata was obtained). **Flagged as an unresolved lead**, not pursued further under this task's budget constraint.

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2. J1939 PGN expectations (Step 3) — mostly unconfirmed; documented honestly

This section could not be closed with primary-source confirmation. The only KOEL controller manual obtained in full (the base "KG640," §1 above) has **no CAN bus and no J1939 content whatsoever** — no mention of PGN, SPN, EEC1, ET1, EFL/P1, VEP1, DM1, or TSC1 anywhere in its 92 pages (verified by full-text search of the extracted manual). Its only digital communications are RS485/Modbus (for panel-to-PC configuration, terminals 15/16) and USB (configuration tool). This confirms the base KG640 does not need this section at all — but it also means **the base manual cannot answer Step 3 for the CAN-equipped KG640C variant actually in the field.**

What **is** confirmed directly from the two primary hardware documents:

Fact	Source	Status
KG640C has exactly **one** CAN interface on its terminal block: J4, terminal 13 = CAN L, terminal 14 = CAN H	OEM wiring diagram, page 1 (`kg_bot.png` crop)	Confirmed
The engine ECU has **two** CAN channels: channel 1 on pins 55 (CAN H)/77 (CAN L), channel 2 on pins 87 (CAN H)/86 (CAN L)	`ecu-pinout-extracted.md`, "CAN — two channels"	Confirmed
Which of the ECU's two CAN channels physically connects to the GCU's single CAN port is **not stated** in either document — the wiring diagram's harness connectors (XC1/XC11) were not traced pin-for-pin to this specific link in the pinout extraction	Both documents	**Not established — open item**
Baud rate is not stated on either the wiring diagram or in the base KG640 manual	—	**Not established — open item**
Engine ECU source address (whether 0x00 per SAE J1939 convention) is not stated anywhere in the available documents	—	**Not established — open item**

What is *not* confirmed, and is presented here only as an informed, explicitly-labelled expectation, not a finding: a J1939 link between a proprietary genset controller and a Tier-4/CPCB-IV+ common-rail ECU of this type would, by nearly universal industry convention (SAE J1939-71), be expected to carry at minimum: **EEC1 (PGN 61444)** for engine speed, **ET1 (PGN 65262)** for coolant temperature, **EFL/P1 (PGN 65263)**

for oil pressure, **VEP1 (PGN 65271)** for battery/system voltage, **DM1 (PGN 65226)** for active diagnostic trouble codes, and **TSC1 (PGN 0)** if the GCU (rather than the ECU alone) is meant to command engine speed/torque. Standard J1939 baud rate for this device class is 250 kbit/s, and the engine ECU node would conventionally sit at source address 0x00. **None of this is sourced to a KG640C document — it is a generic industry baseline offered so the design isn't starting from nothing, and it must be treated as inferred/unconfirmed until either a real KG640C CAN specification is found or the bus is logged directly off the existing panel.**

What would close this: (a) locating and fully retrieving the KG645 manual noted above, on the chance it documents the CAN link in detail; (b) a KG640C-specific manual or CAN database (DBC) from KOEL; (c) the most direct route — connecting a CAN logger to the existing genset's harness at the XC1/XC11 connectors and recording live traffic, which would settle baud rate, source addresses, and the actual PGN set in one measurement, superseding any manual-based inference.

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3. Digital I/O contract (Step 4)

Confirmed terminal assignments on the KG640C block (wiring diagram page 1, `kg_top.png` / `kg_bot.png` crops), cross-referenced against the base KG640 manual's terminal table and against `ecu-pinout-extracted.md`'s ECU-side digital I/O list:

KG640C terminal	Group	Function	Direction relative to ECU
J2-3 (terminal 3, "A")	Digital output	Drives **START RELAY** coil	GCU issues the start *request*; see reasoning below on what actually cranks the engine
J2-4 (terminal 4, "B")	Digital output	Drives **IGNITION RELAY** coil (`-13RB1` , 70 A)	GCU-issued; **corrected 14 Sep 2026** — this memo originally named this relay `-13RB2` , which is the START relay on output A; the ignition relay is `-13RB1` . The binding spec always had it right, so the error did not propagate. This relay's contact almost certainly switches battery+ onto ECU pin 71 (`IGNITION` , confirmed active-high/switched-battery per `ecu-pinout-extracted.md`), i.e. **the GCU powers the ECU on**
J2-5 (terminal 5, "C")	Digital output	Drives **MC CONNECT RELAY** (mains contactor connect)	Not ECU-related — genset/mains transfer logic, panel-side only
J2-6 / J3 area	Digital output	**COMMON FAULT** relay, **GC CONTACT RELAY** (genset contactor)	Not ECU-related — panel/contactor logic
J3-7	Combined I/O	**D+ CHARGE ALT** — GCU provides alternator field excitation and reads back charge-alternator voltage for charge-fail detection	**Entirely GCU-and-alternator; the ECU is not involved.** `ecu-pinout-extracted.md` 's connector appendix lists a 2-pin "Charging Alternator" connector present in the harness but never traced to an ECU pin, consistent with this.
J4-13/14	CAN L/H	Single CAN port	Almost certainly the GCU's link to one of the ECU's two CAN channels (see §2) — exact channel not traced
J5-21/22 ("D"/"E")	Digital inputs	Wired to **OVERRIDE SS SW** `-S7`	Matches `ecu-pinout-extracted.md` 's note that ECU pin 24 ("OVERSIDE SWITCH") traces via harness connectors XC1/XC11 to the same `-S7 Override SS SW` component — **this is a shared physical switch, not two different switches**, and per the pinout doc it is explicitly **not** a dedicated overspeed switch: *"OVERSIDE SWITCH (pin 24) and Override SS SW -S7 are the same component. There is no separate overspeed switch."*

What actually executes cranking — inferred, not fully traceable: `ecu-pinout-extracted.md` lists ECU pin 50 (`MAIN RELAY`) as a **low-side output from the ECU itself**, and separately names a starter (`8STR1`) with its own solenoid (`-SOL`) among the panel components. The most consistent reading of the two documents together is: the GCU's Ignition Relay output energizes the ECU (powers it up); the ECU then runs its own internal start sequence (preheat, fuel priming, crank-permission

logic) and asserts its own `MAIN RELAY` output to engage the starter solenoid — i.e., **the GCU requests a start (by powering the ECU on) and the ECU executes it** (drives the actual crank relay), matching the brief's framing exactly. However, **the GCU's own "START RELAY" terminal (J2-3) has no confirmed downstream connection to any ECU pin in either document** — it may feed a separate/legacy manual-crank path, a crank-permissive input the pinout extraction didn't capture, or something not traceable without the full XC1/XC11 harness cross-reference (which page 1 of the wiring diagram, the page examined in most detail for this memo, does not itself contain — that harness detail lives on later pages not fully re-examined under this task's time budget). **Flagged as an open item**, not asserted as resolved.

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4. What remains unknown

1. **No document titled specifically "KG640C" was found** — the MPU verdict and terminal architecture rest on a sibling "KG640" (non-CAN) manual plus the OEM wiring diagram's own KG640C block, not on a KG640C-specific manual. High confidence, not full confirmation.
2. **J1939 baud rate, engine-node source address, and the exact PGN set are not confirmed** — no KG640C CAN specification or DBC was located. Direct CAN bus logging off the existing panel is the fastest way to close this and should be prioritized over further document search.
3. **Which of the ECU's two CAN channels (55/77 or 87/86) physically connects to the GCU's single CAN port (J4, 13/14) is not established.** If the second channel is used for something else (a dealer diagnostic tool, telematics/remote-monitoring per KOEL's own marketing material mentioning a "remote monitoring system"), that has implications for whether diagnostic tooling can coexist with the GCU link without contention.
4. **The GCU's own "START RELAY" output (terminal J2-3) has no confirmed destination** — whether it reaches the ECU, the starter solenoid directly, or a legacy manual-start path is unresolved from the documents in hand.
5. **INR pricing for the two correctly-matched standalone overspeed modules (GAC SSW675, Murphy/Enovation HD9063) was not obtained** — both have confirmed Indian distributors but require a direct quote request.
6. **The "KG645" manual** (a possible CAN-capable sibling found via search, https://www.scribd.com/document/407079798/) was not retrieved in full this session and may resolve some or all of items 1–3 above if obtained.
7. As with Task 1, **a photograph of the physical KG640C rear terminal block would be the single most valuable piece of evidence** this design phase could still obtain — it would upgrade the MPU verdict from "inferred, high confidence" to "confirmed" in one image, and would likely also show CAN port labelling that clarifies item 3.

Sources consulted (full list)

- `/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf` — OEM wiring diagram, page 1, rendered at 400 DPI and read directly (crops examined: KG640C block terminals J1–J8, in this session's scratchpad).
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — ECU-side pin-level extraction of the same diagram, read in full.
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — Task 1 memo, engine and controller identity.

- KOEL, "KG640 CONTROLLER MANUAL" / `SED-MAN-KG640-002`, 92 pages, retrieved via https://www.scribd.com/document/878092942/SED-MAN-KG640-002-Koel-Manual-for-KG640-Genset-Controller — read in full.
- KOEL, "KG645 Genset Controller User Manual," https://www.scribd.com/document/407079798/Kirloskar-KG645-User-manual-for-genset-controller-pdf — located, **not** retrieved in full (fetch returned only page metadata); flagged as an open lead.
- KOEL, "KG934V1 Genset Controller User Manual" (2010), https://5.imimg.com/data5/SELLER/Doc/2021/7/JH/DB/YP/28202182/kirloskar-green-kg934v1-genset-controller.pdf — located as an older/smaller sibling product during search, not used as evidence in this memo (different, smaller controller family).
- SmartGen (Zhengzhou) Technology Co., "HGM6100N Series Genset Controller User Manual" — pulled as a rebadge-comparison candidate; ruled out as the KG640C's underlying product per §1.
- Governors America Corp (GAC), SSW675 product page, https://www.governors-america.com/product/ssw675/ (direct fetch blocked by site; description obtained via search-result excerpt of the manufacturer's own page).
- Governors America Corp (GAC), ESD5500E documentation (via search snippet) — cited only to show GAC's own explicit statement that a separate overspeed shutdown device is required in addition to its analog governors.
- Enovation Controls (Murphy), HD9063/OS77D/SS300 product page, https://www.enovationcontrols.com/products/hd9063-os77d-and-ss300/ — fetched directly, specs quoted verbatim.
- IndiaMART listing, Spares Bazaar Pvt Ltd (New Delhi), GAC ESD5500E/ESD5111/ESD5550/ESD5221 at ₹9,500/piece, https://www.indiamart.com/proddetail/esd5500e-esd5111-esd5550-esd5221-gac-speed-control-unit-governor-america-corp-21293246791.html — cited only as order-of-magnitude brand/market context, not as pricing for the correct part (see caveat in the recommendation table).
- IndiaMART listing, Murphy By Enovation Controls (Pune), HD9063/OS77D/SS300, https://www.indiamart.com/proddetail/hd9063-os77d-and-ss300-electronic-speed-switches-15889251691.html — quote-only, no published price.
- `ikonnect.kirloskar.com` portal — checked for a KG640C-specific manual index; none exposed without authentication (consistent with Task 1's finding).
- General web searches (recorded for completeness, mostly negative or superseded by the direct fetches above): `Kirloskar KG640C manual`, `KG640C genset controller datasheet`, `KOEL KG640C wiring`, `Kirloskar KG640C manual J1939 CAN engine PGN`, `"KG640C" Kirloskar controller manual pdf`, `scribd "KG640C" OR "SED-MAN-KG640C" Kirloskar CAN J1939`, `GAC ESD5500 electronic overspeed shutdown device price India magnetic pickup`, `GAC ESD5111 electronic overspeed shutdown device specification magnetic pickup frequency`, `"Governors America" OR GAC overspeed shutdown device dedicated relay model number`, `GAC SSW675 speed switch price India`, `"SSW675" OR "SSW676" OR "speed switch" GAC indiamart price generator overspeed India`, `Murphy overspeed shutdown module magnetic pickup speed switch price India`, `Murphy OS77D speed switch India IndiaMART price`, `Murphy Enovation OS77D specification frequency input voltage amplitude magnetic pickup`, `Electra Brandt SSW675 price`.

Research Memo 03: ECU Connector Identification

> **U1 CLOSED, 18 Sep 2026 — this memo's leading candidate is confirmed**
>
> The connector has been photographed with its markings legible. It carries the **Bosch** name and armature logo, part numbers `1 928 405 192` and `1 928 405 194`, `code C`, and `2.7`. The face-on photograph shows 94 cavities in two chambers: 8 large power contacts (1–8) plus four fine rows numbered 9–28, 29–50, 51–72, 73–94.
>
> **This memo ranked "Bosch EDC17-style 94-pin" as the leading candidate on inference. Confirmed, with part numbers read off the physical part.**
>
> One detail this memo did not anticipate: `code C` is a **mechanical coding variant**. Bosch supplies these housings in coded versions that will not mate across codes, so any replacement or adapter housing must be code C.
>
> A search for the two part numbers returned no direct catalogue hit, so sourcing should go through a Bosch distributor quoting the numbers rather than a search engine. The shortlist and sourcing analysis below remain useful for that conversation.
>
> Evidence: `refs/field-evidence-2026-09-18.md`.

Task: U1 (ECU connector part number / mating-half sourcing), partial closure — full closure needs a photograph. Consumes Task 1 (engine identity). Feeds the adapter-harness deliverable (spec §7.1). **Date:** 2026-09-13 **Author:** Phase-1 research (task 3) **Engine/genset:** Kirloskar 3R550ETA 4G1 / KG4-25WS1, per `docs/research/01-engine-identity.md` (inferred, high confidence).

Verdict (read this first)

The OEM ECU part number **could not be established** — same negative result already recorded in `01-engine-identity.md`, Step 2, and not re-openable with the sources available here. What this memo adds is a **physical characterisation of the 94-pin ECU connector** from the wiring diagram, and a **ranked shortlist of connector families** that plausibly match it, each with a buyability verdict for India. No part number is

asserted with confidence — per the task's own framing, a line drawing cannot support that, and a confident wrong connector is worse than an honest shortlist.

Rank	Family	Visual match	Buyable in India	Verdict
1	**Bosch EDC17/EDC15/EDC16-style 94-pin ECU connector** (TE-Connectivity-built, Bosch-proprietary tooling)	Strong — pin count exact, mixed large/small contact pattern plausible, widely used on Bosch common-rail diesel ECUs of this class	**Yes, but only through the automotive ECU-repair/chiptuning aftermarket, not mainstream component distributors** — order via specialty ECU-connector shops (international, ships to India)	**Leading candidate — photograph required to confirm**
2	**Aptiv/Delphi GT-series (150-position family) rectangular multi-row connector**	Moderate — rectangular, multi-row, mixed contact sizes are a documented feature of this family; exact 94-position variant not confirmed	**Yes** — stocked by Mouser (mainstream distributor with an India storefront)	Plausible alternate
3	**TE Connectivity AMPSEAL (16/23/35/direct-descendant 90-series)**	Weak — AMPSEAL housings are characteristically oval/rounded single-lever shells; the PDF drawing shows a rectangular, sharp-cornered, multi-row grid that fits this family less well	**Yes** — mainstream Mouser/DigiKey stock	Weakest visual fit of the three buyable candidates
—	Molex CMC / high-density cylindrical automotive headers	Ruled out — CMC-family connectors are round/cylindrical pin-and-socket bundles, not a flat rectangular multi-row grid	N/A	**Ruled out on shape alone**

None of these is asserted as "the" connector. All three buyable candidates are plausible; the photograph brief (final section) is designed to distinguish them in a single site visit.

Step 1: What the wiring diagram actually shows

Source: `refs/GP3.8703.C4.pdf`, page 4, "Engine Control Unit Connector" panel (one of six connector-detail panels on that page), re-rendered at 400 dpi with `pdftoppm` and cropped with PIL for close reading.

Confirmed directly from the drawing: - The connector is drawn as **one single housing outline** (one continuous molded shell in the 2D projection) — not two or three physically separate ganged blocks side by side. Pins 1–94 are called out on this one outline. - The pin field is arranged as **four rows** of small, evenly pitched square contacts, split top/bottom: an upper pair of rows (running from a low pin number

near "7" up to "28" at the top-right corner) and a lower pair of rows (running from "73" at bottom-left to "94" at bottom-right corner). This matches `refs/ecu-pinout-extracted.md`'s statement that the connector is 94-pin. - At the **top-left and bottom-left corners**, outside the dense small-pin grid, sit two separate 2-contact clusters, individually labelled "PIN NO.1" (top) and "PIN NO.5" (bottom). These two clusters are drawn with the **identical stepped-rectangle pin symbol** used elsewhere on the *same PDF page* for the standalone 2-pin **Fuel Pump Connector** and **Charging Alternator Connector** panels — both of which are physically large, single-current-path power/current connectors in this harness. **Inferred** (by direct comparison of drawing conventions within one source document, not an explicit label): pins 1, 2, 5, 6 are drawn as large/keyed contacts, distinct in kind from the ~88 small pins in the main grid. This is consistent with the brief's expectation that a 94-pin automotive ECU header of this class mixes large power/injector contacts with small signal contacts in one shell. - Between the upper and lower row-pairs, **four elongated rectangular features** run across the housing's mid-band. None of them carries a pin number anywhere in the source (every actual contact in this diagram set is numbered). **Inferred:** these are molded polarization/alignment ribs in the plastic shroud (a standard feature on wide, high-pin-count automotive headers to prevent pin damage on blind mating), not additional large contacts. - On the **right-hand (mating/cable) side** of the housing, the outline shows a curved, bowed wall plus two small rectangular tabs protruding further right than the main shell, near the top-right and bottom-right corners. **Inferred, not confirmed:** this is most consistent with pivot bosses and body of a **lever-actuated latch** (used on high pin-count automotive connectors to provide the large mating force ~94 contacts require) — but a cable-entry strain-relief boot is a plausible alternative reading of the same 2D outline, and the drawing does not disambiguate. This is the single largest open question the photograph needs to resolve. - Small rectangular tabs are also present at even intervals along the top and bottom edges of the housing (visible in the full-page crop) — consistent with either mounting/latch-catch features or additional polarization ribs; not resolvable further from a line drawing. - **Not a housing marking:** the "Kirloskar / Oil Engines" watermark visible across the page image is the ikonnect web portal's page watermark (confirmed present on every page of this browser-printed PDF, per `01-engine-identity.md` Step 2), not a molded marking on the physical connector. This is flagged explicitly so it is not mistaken for a part marking later.

What the drawing does not show and cannot be inferred: exact housing dimensions, contact pitch, wire-seal type (individual seals vs. a single mat seal), material/color, or any molded part number/logo. `refs/ecu-pinout-extracted.md` already records that only 44 of the 94

pins are traceable to a signal in the wiring diagram; the connector-detail panel on page 4 is a generic "how to count pins" reference (it exists to let a technician find pin 1/5/28/73/94 as anchor points), not a manufacturer datasheet.

Step 2: Candidate families, ranked, with reasoning

1. Bosch EDC15/EDC16/EDC17-style 94-pin ECU connector (leading candidate)

- **Why it's plausible:** The pin count (94) is an exact match, not just "in the ballpark" — multiple independent listings in the automotive ECU-repair/chiptuning aftermarket specifically describe a "Bosch EDC17 94-pin" (and a companion 60-pin, for two-connector EDC17 variants) ECU connector, sold as bench/pigtail cable kits: e.g. u-obd.com ("BOSCH EDC17 Connector 94 PIN 60 PIN plug cable set full PIN wired," https://www.u-obd.com/product/edc17-connector/, US$99), ecudepot.com (same product, https://ecudepot.com/product/bosch-edc17-connector/), autoecupart.net (multiple listings, e.g. https://www.autoecupart.net/products/original-full-94-pin-connector-for-bosch-edc17-electronic-control-unit-ecu), and eBay listings (e.g. https://www.ebay.com/itm/265128037857 "OEM Full 94 Pin Connector for Bosch ME17.9.11 EDC17"). **Confirmed:** a documented, commercially available "Bosch 94-pin" ECU connector family exists and is actively sold as a discrete product. **Not confirmed:** that this specific family is the one on the Kirloskar genset — no source ties a Bosch part number to manual `08-GP3-60-001` or to this engine.
- **Corroborating context from this project's own prior research (not new evidence, but relevant):** `docs/research/04-injector-drive.md` independently concludes the injector architecture is most likely a Bosch solenoid common-rail family (CRIN2/CRIN3-class), based on wiring-diagram evidence (rail-pressure sensor + PWM metering-unit pattern) and Kirloskar's confirmed practice of re-badging genuine Bosch injectors under Kirloskar part numbers for at least one other engine family (Bosch 0445120646 = Kirloskar F6.248.08.0.pr). If the injectors are Bosch, an ECU built around a Bosch-family connector (rather than a mixed-vendor harness) is the more parsimonious assumption — **inferred, not a separate confirmed fact.**
- **Retention/shape match:** the EDC15/16/17 94-pin housing family is documented in the tuning trade as using a mixed grid of small signal pins with a smaller number of larger power/injector pins in one shell, closely matching the page-4 drawing's large-corner-pair-plus-dense-grid layout described in Step 1. This comparison is based on general familiarity with these connectors in the aftermarket record, not a side-by-side dimensioned drawing — **inferred, moderate confidence.**

- **Buyability (India): Yes, but not through a normal electronic-component distributor.** No listing was found on Mouser India, DigiKey India, or Element14 for a "Bosch EDC17 94-pin" connector under any manufacturer part number — searches for a TE Connectivity part number tied to this housing returned no result (**negative result**). It is however readily orderable from the international automotive ECU-repair trade (u-obd.com, ecudepot.com, autoecupart.net, eBay/AliExpress sellers), all of which ship internationally including to India, typically **US$65–99 per full connector-pair kit** (94-pin + companion 60-pin, wired pigtail) plus international shipping and India customs duty/GST on import (landed cost not independently verified — treat the $65–99 base figure as confirmed-at-source, the India-landed total as **unverified, order-of-magnitude estimate only**). This satisfies "a connector we cannot buy is not a candidate" — it is buyable — but the sourcing channel is the aftermarket repair trade, not a component distributor, which has implications for lead time, quality/counterfeit risk, and lack of a formal datasheet.

2. Aptiv/Delphi GT-series rectangular multi-row connector (150-position family)

- **Why it's plausible:** The GT150 family is a genuine, rectangular, multi-row automotive ECU/PCM connector platform distributed through mainstream channels (Mouser lists it directly: https://www.mouser.com/Delphi-Connection-Systems/Connectors/Automotive-Connectors/GT-150-Series/, and specialist motorsport/ECU-wiring shops such as Corsa Technic and Waytek Wire stock GT150 housings, terminals, and seals across a range of position counts). Its general silhouette — rectangular shell, dense rows of small pins with provision for larger power contacts in the same housing family — is structurally similar to what the page-4 drawing shows. **Confirmed:** this connector family exists, is rectangular/multi-row, and is mainstream-distributor-stocked.
- **What is not confirmed:** whether a 94-position GT-series variant exists and matches the specific layout (four small-pin rows plus two 2-large-pin corner blocks) seen on page 4. This was not verified in this pass — **negative result, flagged as an open item**, not a ruled-out candidate.
- **Buyability (India): Yes** — Mouser operates an India storefront with INR pricing and ships GT-series connectors; DigiKey/Element14 India stock was not separately checked in this pass (**not verified**, time-boxed). Exact INR unit pricing for a 94-position variant was not obtained in this pass — recommend confirming against Mouser India's live catalog once/if this candidate is pursued further.

3. TE Connectivity AMPSEAL (90-series descendants)

- **Why it's a weaker fit:** AMPSEAL is a genuinely mainstream, India-buyable automotive sealed-connector family (Mouser lists the series directly: https://www.mouser.com/en/c/connectors/automotive-connectors/?m=TE+Connectivity&series=AMPSEAL) and is used on some diesel ECUs. But the AMPSEAL housing silhouette is characteristically **oval/rounded with a single center or side lever**, single or double staggered pin rows — not the sharp-rectangular, four-row, dense micro-pin grid with two separate large-pin corner clusters that page 4 shows. **Inferred:** shape mismatch downgrades this candidate relative to #1 and #2, even though it is the easiest of the three to actually buy through normal distribution.
- **Buyability (India): Yes, straightforwardly** — standard Mouser/DigiKey stock, no aftermarket-only caveat. This is the safest fallback if the photograph rules out #1 and #2.

Ruled out: Molex CMC and similar high-density cylindrical automotive headers

Molex's CMC (and comparable high-pin-density automotive families built around round/cylindrical pin-and-socket bundles rather than a flat rectangular grid) do not match the page-4 drawing's flat, rectangular, multi-row silhouette at all. **Ruled out on shape alone** — no further sourcing work done on this family. *(Negative result, recorded per the global instruction that negative results are results.)*

Step 3: Sourcing summary table

Candidate family	Housing/mating-half source	Contacts/seals source	Price (as found)	India availability	Buy verdict
Bosch EDC15/16/17 94-pin (+60-pin companion)	u-obd.com, ecudepot.com, autoecupart.net, eBay sellers (aftermarket ECU-repair trade)	Bundled with the wired pigtail kits above; loose terminals also listed separately on some of the same sites	US$65–99 per kit (source-confirmed); India-landed cost incl. shipping/duty **not verified, estimate only**	Yes, via international aftermarket shipping — **not stocked by Mouser/DigiKey/Element14 India** (checked, negative result)	**Buyable — proceed if photograph confirms Bosch-style housing**
Aptiv/Delphi GT150-series	Mouser (mainstream distributor); Corsa Technic, Waytek Wire (motorsport specialty)	Same distributors, per-position contact/seal kits	Not obtained in this pass for a 94-position variant — **unverified**	Yes — Mouser has an India storefront	**Buyable — needs live-catalog price/position check before ordering**
TE Connectivity AMPSEAL (90-series)	Mouser, DigiKey (mainstream, confirmed stocked)	Same distributors	Not itemised in this pass	Yes — mainstream stock	**Buyable — safest fallback, weakest visual match**

Candidate family	Housing/mating-half source	Contacts/seals source	Price (as found)	India availability	Buy verdict
Molex CMC / cylindrical high-density	—	—	—	—	**Not a candidate — shape mismatch**

Step 4: The photograph brief that closes this

One site visit, done right, resolves which of the three buyable candidates is correct. Photograph the ECU connector **on the engine harness side (the plug, not the ECU-mounted socket)** and the **ECU case itself**, as follows:

1. **Full-face shot of the connector's mating face**, camera square-on (not at an angle) to the pin field, filling the frame. This is the single most important shot: it will immediately show whether the housing is one rectangular shell with a flat multi-row grid (Bosch/Delphi-style — candidates #1/#2) or an oval/rounded single-lever shell (AMPSEAL-style — candidate #3), settling the ranking's top-level question in one image.
2. **A shot of the retention mechanism actuated/engaged and again disengaged** (if it can be safely operated with the engine off and battery disconnected) — specifically to determine whether it is a pivoting lever, a slide latch, a center bolt, or simple friction/CPA-clip retention. This resolves the Step-1 ambiguity about the bowed feature on the connector's right side.
3. **A raking (low-angle, side-lit) shot of every flat surface of the housing** — top, both sides, and the cable-entry boot — specifically hunting for a **molded (raised or recessed) part number, date code, or manufacturer logo**. These are usually small (2–4 mm character height) and shallow, so: use a raking light held at a low angle roughly parallel to the surface (a phone flashlight held to the side works well) rather than the on-axis camera flash, which washes out shallow embossing. Take multiple shots at different light angles if the first pass doesn't reveal legible text. Manufacturer logos to watch for specifically: the Bosch roundel, a Delphi/Aptiv wordmark or triangle logo, or a TE Connectivity ("te" stylized) mark — presence of any one of these directly resolves the family question.
4. **The ECU's own case label/nameplate**, photographed flat-on and well-lit — looking for a Bosch part number of the pattern `0 281 0xx xxx` (EDC-family) or `F 01R 0xx xxx` (Bosch reseller code, sometimes used by Kirloskar per the injector cross-reference in `04-injector-drive.md`), or any Continental/Delphi/Denso equivalent. Finding this would independently resolve the still-open "OEM ECU part number" question from `01-engine-identity.md` Step 2, which is a bigger win than the connector question alone.

5. **A wide shot showing the connector still mated to the harness**, to capture the harness-side (socket) connector's backshell and cable exit angle/strain relief — useful for the adapter-harness design even if it adds nothing to family identification.
6. **A pin-count check shot**: with the connector unplugged (engine off, battery disconnected), count and photograph the pin rows directly, confirming the "four rows, two 2-pin corner blocks" structure this memo infers from the line drawing, and specifically checking whether the two corner-pin contacts are visibly larger in diameter/blade-width than the grid pins — this is the single check that would upgrade Step 1's "inferred" finding about mixed contact sizes to "confirmed."

Lighting/angle summary: natural or diffuse white light for the full-face and pin-count shots (avoid on-axis flash — it flattens the pin field and hides depth); raking side-light specifically for hunting molded text. A macro mode or phone "portrait"/close-focus mode is recommended for the part-number hunt, since the expected character height is small.

Sources consulted

- `/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf` — page 4, "Engine Control Unit Connector" panel, re-rendered at 400 dpi (`pdftoppm -r 400 -f 4 -l 4`) and examined via cropped PIL renders.
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — confirms 94-pin count and that only 44/94 pins are traceable in the wiring diagram.
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — engine/genset identity; Step 2 negative result on OEM ECU part number, carried forward unchanged (not re-openable with sources available here).
- `/home/mohan/workws/ecu25kva/docs/research/04-injector-drive.md` — Bosch-injector-family inference used as corroborating (not standalone) context for candidate #1.
- https://www.u-obd.com/product/edc17-connector/ — Bosch EDC17 94-pin+60-pin connector kit, US$99, fetched directly.
- https://ecudepot.com/product/bosch-edc17-connector/ — same product, found via search.
- https://www.autoecupart.net/products/original-new-full-94-pin-connector-for-bosch-edc17-electronic-control-unit-ecu and related autoecupart.net listings — found via search, corroborate the "Bosch 94-pin EDC17" connector as a distinct, repeatedly-listed aftermarket product.

- https://www.ebay.com/itm/265128037857 and https://www.ebay.com/itm/254545773762 — eBay listings, found via search, further corroboration.
- https://www.mouser.com/Delphi-Connection-Systems/Connectors/Automotive-Connectors/GT-150-Series/ — Aptiv/Delphi GT150 series distributor listing, found via search.
- https://www.corsa-technic.com/category.php?category_id=178 and https://www.waytekwire.com/catalog/connectors/aptiv-gt150-connectors — GT150 specialty-distributor listings, found via search.
- https://www.mouser.com/en/c/connectors/automotive-connectors/?m=TE+Connectivity&series=AMPSEAL — TE Connectivity AMPSEAL series distributor listing, found via search.
- https://bosch-connectors.com/ (PS Connectors) — fetched directly; page returned only navigation/legal boilerplate, no usable technical content (**negative result**, recorded rather than omitted).
- Searches returning negative or inconclusive results (recorded per the global instruction): `Bosch EDC17 94 pin connector India price Delhi distributor OR Amazon.in OR IndiaMART` (no India-specific listing found); `Bosch MD1 MDG1 154 pin ECU connector ampseal delphi GT94 comparison` (no direct technical comparison found); `TE Connectivity AMPSEAL 90 way OR Delphi GT150 150 pin connector Mouser India price` (distribution channel confirmed, exact India pricing not obtained).
- Prior-session scratchpad files reviewed for reusable material (`/tmp/claude-1000/.../scratchpad/`): `drivven.txt` (Drivven DI Driver Module manual — relevant to Task 4, not this task), `bpc960.txt` (Bliss Powercom BPC960 genset-controller manual — a different controller, not the KG640C or the engine ECU, not used in this memo), `apalrd.txt` (generic common-rail-diesel-control-system slide deck — not connector-specific, not used). None of these contained ECU-connector-family evidence directly usable here; reviewed to avoid duplicate fetching, not cited as sources for any claim above.

Addendum, 11 October 2026: online search for the J1001 header drawing

The board needs the **PCB-side header** (J1001) that mates with the harness connector photographed on the machine (Bosch `1 928 405 192` / `194`, code C). A web search found the family, but **no public drawing of the header**:

- **Harness side, confirmed buyable.** Bosch **1 928 405 063** is "Connector, 94-Way, Code C, Black". Chief Enterprises, an authorised Bosch connector distributor in Germany, stocks 593 of them at €18.26 each, falling to €6.50 per 108.

The same family has these accessories: - 1 928 405 069 (cover, exit right) - 1 928 405 071 (cover, exit left) - 1 928 405 074 (secondary lock) - 1 928 405 067 (retaining plate)

Our 192 / 194 numbers are not listed anywhere online. They are most likely variants of this family. **Not yet confirmed.** - **Board side, a family match but no drawing.** TE Connectivity **936524-2** is an "ECU series" right-angle, through-hole, 2.5 mm pitch, 2-row shrouded header with 154 contacts: a 60-way engine module plus a **94-way vehicle module**, in a silicone-sealed housing, rated −40 to 130 °C. RS lists it (stock numbers 719-9696 and 161-4043, including RS India). TE's own page marks it **"restricted product, no documentation available"**: it is built for Bosch, and its drawing is not public. - **No standalone 94-way header was found** with a published drawing. Aftermarket "EDC17 94-pin PC board socket" listings exist, cross-referenced to VAG 7L0906385A. They come from eBay sellers, with no drawing.

Ways to close J1001, cheapest first: 1. **Ask Chief Enterprises (or Bosch Connectors) for the PCB header that mates with 1 928 405 063, code C, and its drawing.** An authorised distributor can request a customer drawing, sometimes under NDA. 2. **Buy one TE 936524-2 from RS and measure it.** Its 94-way vehicle module is very likely the same header, and the pin grid, pitch, row spacing and mounting pegs can be measured to make a land pattern. The risk is that it is the combined 154-way part, so the single-94 outline and pegs may differ. 3. **Measure a scrap ECU board.** A dead ECU of this type from a repair shop has the real header soldered on, so the footprint can be read straight off the PCB.

Until one of these is done, J1001 stays without a footprint. Layout can start everywhere else.

Research Memo 04: Injector Identification and Drive Profile

Task: U5 (injector drive stage — boost rail voltage, peak/hold current, pulse timing). Consumes Task 1 (engine identity). Feeds the injector driver sheet and Phase-2 boost converter design. **Date:** 2026-09-13 **Author:** Phase-1 research (task 4) **Engine:** Kirloskar **3GK550ETA 4SR1** (this memo was written as `3R550ETA 4G1`; corrected 18 Sep 2026 from the rating plate — the family envelope this memo derives is unaffected, since it was never tied to a part number). 3-cyl inline, 3-cyl inline, 1.65 L, common-rail, turbocharged-aftercooled, 1500 rpm, 26.5 kW prime (per `docs/research/01-engine-identity.md`, inferred with high confidence). **Genset:** Kirloskar KG4-25WS1.

Verdict (read this first)

Item	Value	Status
Injection system type	Solenoid common-rail, Bosch-family architecture (not piezo)	Inferred
Injector supplier/part number	**Not established** for this specific engine (3R550ETA)	Negative result — see Step 1
Boost rail voltage envelope	**65–115 V DC**, most solenoid CR designs cluster **90–115 V**; this engine most likely sits near **~100 V ± 15 V**	Inferred envelope
Peak current	**12–24 A**, most designs **18–20 A**	Inferred envelope
Peak current duration	**~0.3–1.0 ms** (boost phase, until current threshold reached)	Inferred envelope
Hold current	**~8–13 A**, most designs **~10–12 A**	Inferred envelope
Minimum controllable pulse width	**~0.1–0.3 ms**	Inferred envelope
Required driver slew rate	**40–100 A/ms** covers the family	Inferred envelope (matches brief's design target)
Bank-sharing timing check (Step 3)	**PASSES**, with large margin (≥240 crank degrees / 26.67 ms minimum separation vs. ≤~3 ms max injection duration)	**Confirmed by calculation** — see Step 3
Calibration boundary (Step 5)	Hardware can be built and bench-verified; **cannot run this engine correctly** without flow-vs-pulsewidth maps not publicly available	Stated plainly — see Step 5

The headline finding for Step 3 is good news, stated loudly because the brief asked for it either way: **the bank-shared wiring (cyl 1+3 on one high side, pin 03) is inherently safe for any plausible 3-cylinder firing order**, because a 3-cylinder 4-stroke engine with even firing intervals has exactly three, evenly-spaced injection events per 720° cycle — so *any* two of the three cylinders are always separated by one full firing interval (240°) in the closer direction, regardless of which firing order (1-2-3 or 1-3-2) is used. This is a property of "3 evenly spaced cylinders," not of the specific order label. See Step 3 for the full derivation and the one assumption that could break it.

Step 1: Injector family identification

Search performed: `Kirloskar R550 common rail injector Bosch part number`, `Kirloskar 3R550 injector Denso`, cross-reference of `crdiparts.com` and `turbotech.in` Kirloskar-branded injector listings.

What was found: - Kirloskar packages Bosch common-rail injectors under its own part-number scheme for at least one other engine family: Bosch injector **0445120646** is sold as Kirloskar part **F6.248.08.0.pr**, "suitable for Kirloskar engine in genset and Liugong loader" (confirmed listing, https://www.crdiparts.com/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/, mirrored at https://turbotech.in/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/). The "F6.248" prefix is consistent with Kirloskar's engine-family/displacement coding (analogous to the "3R550" pattern already established in Task 1) and most plausibly denotes a different, larger (6-cylinder-class) Kirloskar engine, **not** the 3-cylinder R550 family used in the KG4-25WS1. **Confirmed:** Kirloskar sources genuine Bosch solenoid CR injectors and re-badges them with Kirloskar part numbers of the form `<code>.<code>.<code>.<code>.pr`. **Not confirmed:** which Bosch part number corresponds to the 3R550ETA

specifically. - No source found ties a specific Bosch (or Denso) injector part number to "3R550" or "3R550ETA" by name. Searches tried: `Kirloskar 3R550 common rail injector Bosch part number`, `Kirloskar R550 CRDI injector Denso`. **Negative result**, recorded per the global instruction that negative results are results. - The wiring evidence itself (rail-pressure sensor on pins 32/35/08, PWM-driven metering unit on pin 88 per `refs/ecu-pinout-extracted.md`) confirms a solenoid-type high-pressure common-rail architecture (a metering-unit-controlled rail, not a unit-injector or unit-pump system), which is consistent with the Bosch CRSN2/CRIN family design pattern used broadly in this displacement/power class. **Inferred** from architecture, not a part-number match.

Working assumption for this memo (inferred, not sourced): the injector is a **Bosch CRIN-family solenoid injector**, most likely in the CRIN2/CRIN3 generation used across Bosch's small-to-mid displacement common-rail industrial/genset engines circa the CPCB IV+ generation (2022–2023 introduction per Task 1). This is a reasonable design assumption but should not be treated as a confirmed part number. **A rating-plate or injector-body photograph (stamped Bosch part number, e.g. `0445 1xx xxx`) would close this** — the same recommendation Task 1 made for engine identity.

Step 2: Drive profile envelope

Exact peak/hold/boost figures are proprietary to the injector manufacturer and are not published for a specific part number without an NDA or a physical sample. Per the brief, the correct engineering response is **an envelope that covers the family**, not a guess at one number.

The envelope, and why these bounds

Solenoid common-rail injectors (Bosch CRIN, Denso G3/G4, Delphi/Continental DFI) all use the same two-stage drive strategy for the same physical reason: the injector solenoid must open against rail pressure (which can exceed 1600–2000 bar in this class) within a few hundred microseconds, which requires driving far more current into the coil, far faster, than a 12 V battery rail can deliver into a ~0.3–1 mH / <1 Ω coil. The universal solution:

1. **Boost phase:** a boost converter (12 V → tens to ~100+ V) is switched across the coil to slew current up rapidly to a peak threshold.
2. **Peak hold:** current is regulated (chopped) at the peak level for a short duration to fully open the pin/needle against flow forces.
3. **Hold phase:** current is dropped to a lower "hold" level (higher-voltage boost removed, battery rail or PWM-chopped boost used) for the remainder of the injection, to reduce power dissipation and heating.
4. **Fast turn-off:** the low side is opened and freewheel/clamp circuitry rapidly collapses the field to close the injector precisely.

Sourced figures found for this class of driver: - A peak-and-hold strategy using **~100 V boost to reach 20 A peak, then 24 V (battery-derived) to hold ~10 A** is described directly in an automotive-oscilloscope diagnostic reference for Bosch common-rail solenoid injectors (confirmed, https://www.picoauto.com/library/automotive-guided-tests/bosch-current-at-idle). - NXP's own reference design for its dedicated injector-driver IC (MC33816, see Step 4) documents a three-stage waveform: **48 V boost to 19 A, regulated hold at 19 A for the peak duration, then regulated hold at 11 A** for the remainder of injection (confirmed, drawn from NXP application material referenced in general search results for the MC33816 common-rail reference design). - A peer-reviewed overview of common-rail injector driving circuitry states boost voltages in the **65–115 V** range and peak currents around **20 A**, with coil resistance ≤1 Ω and inductance ≤1 mH typical for this injector class (confirmed, "Circuit for driving common rail diesel injectors," ResearchGate, https://www.researchgate.net/publication/350426572_Circuit_for_driving_common_rail_diesel_injectors). - NXP's MC33816 product page states its pre-driver stage operates up to **72 V** (confirmed, https://www.nxp.com/products/analog-and-mixed-

signal/gate-drivers/high-precision-differential-injector-driver:MC33816) — lower than the ~100–115 V boost used by some Bosch-native designs, which means a driver built around this IC's pre-driver alone would need external high-voltage MOSFETs/gate stages if the target boost rail is pushed toward the top of the family envelope.

Envelope adopted for this project (matches the brief's design target and is corroborated by the above):

Parameter	Envelope	Where this engine likely sits, and why
Boost rail voltage	65–115 V	Likely **~90–105 V**. This is a mid-displacement (0.55 L/cyl) industrial/genset engine, not a high-speed passenger-car CR engine chasing sub-300 µs injector response; it does not need the most aggressive (115 V-class) boost used by some high-speed automotive applications, but a genset running at a fixed 1500 rpm still benefits from fast, repeatable injector opening for combustion-timing accuracy, so the low end of the range (65 V, associated with older/smaller injectors) is also unlikely. **Inferred**, no part-number-specific source.
Peak current	12–24 A	Likely **~18–20 A**, matching both cited reference designs (Picoscope: 20 A; NXP MC33816 reference: 19 A) which cluster tightly despite being independent sources. **Inferred.**
Peak duration	0.3–1.0 ms	Set by however long it takes the boost stage to reach threshold current against coil L/R and back-EMF; no engine-specific source. **Inferred envelope only.**
Hold current	8–13 A	Likely **~10–12 A**, again bracketed by the two cited reference designs (10 A, 11 A). **Inferred.**
Min. pulse width	0.1–0.3 ms	Standard for pilot-injection-capable CR solenoid injectors; this engine's CPCB IV+ emissions calibration likely uses pilot injection for NOx/noise control, so the low end of this range should be treated as a hard requirement, not headroom. **Inferred.**
Driver slew rate	40–100 A/ms	Per the brief's own design guidance; consistent with reaching ~20 A peak within under 0.5 ms at a ~90–100 V boost rail against a sub-1 mH coil (a first-order L/di/dt estimate: 100 V / 1 mH ≈ 100 A/ms at turn-on, tapering as back-EMF rises — internally consistent with the cited figures). **Derived, not directly sourced.**

Design conclusion: a driver spanning **65–115 V boost, 12–24 A peak, 8–13 A hold, 40–100 A/ms slew** will drive any solenoid CR injector in this displacement/power class, this engine's specific injector included, without knowing its exact part number. This is the number set to carry into the Phase-2 boost converter and driver-IC selection.

Step 3: Bank arrangement vs. firing order — the safety check

The wiring fact being checked

Per `refs/ecu-pinout-extracted.md`: pin 03 is a single high-side driver output shared, via harness splice V5, between cylinder 1 and cylinder 3. Pin 05 is a dedicated high side for cylinder 2. Low sides are independent per cylinder (73→cyl1, 07→cyl3, 29→cyl2). **The failure mode to rule out:** if cylinder 1 and cylinder 3 ever needed to inject at overlapping times, sharing one high-side node would prevent independent control — both injectors would be forced onto the same boosted supply simultaneously, which the low-side switches alone cannot arbitrate into two independently-timed pulses.

Establishing the firing order

Three-cylinder inline 4-stroke engines fire every **720°/3 = 240°** of crank rotation when cylinders are evenly spaced (the standard configuration — inline-3 crankshafts use 120° crank-throw spacing, which combined with the 720° four-stroke cycle produces exactly this even 240° firing interval). The two firing orders seen across the industry, **1-3-2** and **1-2-3**, both satisfy this even spacing; they differ only in which physical cylinder fires second, not in the interval between any two firing events. **Confirmed general mechanics** (basic four-stroke engine theory; see e.g. https://en.wikipedia.org/wiki/Firing_order for the general formula). No source was found stating the 3R550ETA's specific firing order by name — searches for `Kirloskar 3R550 firing order`, `Kirloskar R550 firing order 1-2-3 OR 1-3-2` returned no Kirloskar-specific document (**negative result**). This does not block the safety check, for the reason below.

The check

At 1500 rpm rated speed (per Task 1, confirmed): one crankshaft revolution = 60,000 ms / 1500 = 40 ms. A four-stroke cycle is 720° = 2 revolutions = **80 ms**, matching the brief exactly.

Three evenly-spaced injection events per 80 ms cycle → one event every 80/3 ≈ **26.67 ms**, i.e. every 240° of crank rotation.

Label the three firing instants within one 720° cycle as 0°, 240°, 480° (order 1-2-3: cyl1@0°, cyl2@240°, cyl3@480°; order 1-3-2: cyl1@0°, cyl3@240°, cyl2@480° — the derivation below is order-independent, done for both explicitly):

- **Order 1-2-3:** cyl1 at 0°, cyl3 at 480°. Forward gap cyl1→cyl3 = 480° (53.3 ms). Backward gap (cyl3 in the *previous* cycle, at 480−720 = −240°, to cyl1 at 0°) = 240° (26.67 ms). **Minimum separation = 240° = 26.67 ms.**
- **Order 1-3-2:** cyl1 at 0°, cyl3 at 240°. Forward gap cyl1→cyl3 = 240° (26.67 ms). **Minimum separation = 240° = 26.67 ms.**

Both orders produce the **identical minimum separation: 240 crank degrees, 26.67 ms.** This is not a coincidence specific to either order — it is a general property of any 3 evenly-spaced points on a 720° cycle: with exactly three points, every pair is separated by exactly one interval (240°) in one rotational direction and two intervals (480°) in the other, so the *minimum* pairwise gap is always one interval, **regardless of firing-order labeling**. This is why Step 3 does not actually hinge on resolving the unconfirmed firing order — the bank-sharing safety property holds for both candidate orders identically. **This derivation is original calculation for this memo (basic four-stroke kinematics), not drawn from an external source; it is presented as confirmed by calculation, not inferred.**

Margin check: injection duration for common-rail solenoid injectors in this engine class runs from under 0.3 ms (pilot injections, light load) up to roughly 1.5–2.5 ms for a single main injection at higher load, per the cited peak/hold reference designs' typical operating envelope (Step 2 sources); even a generous upper bound of 3 ms for a long, high-load main injection leaves:

26.67 ms available − 3 ms max injection duration ≈ 23.67 ms of margin, an ~8.9× safety factor.

Even stacking a pilot + main injection (e.g., 0.3 ms + 2 ms + a few hundred µs dwell between them, well under 5 ms total) on one cylinder leaves over 20 ms of clearance before the shared high side would need to serve the other cylinder on that bank.

Verdict

The bank-sharing arrangement (cyl 1+3 on pin 03) is safe. Cylinders 1 and 3 are never required to inject within less than 240 crank degrees / 26.67 ms of each other under either plausible firing order, and typical common-rail injection durations (even generously bounded) are roughly an order of magnitude shorter than that window. There is no finding to raise loudly here in the negative sense the brief warned about — the check **passes**, and passes with enough margin that even significant transient-response slop in the driver electronics (boost recharge time between pulses, current decay tails) will not threaten it. The one assumption underlying this result — **even 240° firing spacing**, i.e., a conventional 120°-throw inline-3 crankshaft — is standard for this engine class and considered safe to rely on absent contrary evidence; it would be worth a one-line confirmation against a Kirloskar service manual or a firing-order marking on the timing cover if one becomes available, but it is not a blocking risk for the driver design.

Step 4: Candidate driver ICs

Given budget constraints, this section is intentionally thinner, per the brief's own prioritization (Steps 3 and 5 are load-bearing; this one may be lighter).

Candidate	Channels	Boost handling	Current regulation	High-side bank switching	Package	India availability
NXP MC33816 (SMARTMOS precision differential injector/solenoid driver)	5 high-side + 7 low-side external MOSFET pre-drivers (up to ~4 full injector channels typical in reference designs)	Pre-driver operates up to **72 V**; does not itself generate boost — needs an external boost converter stage; current regulation and PWM channel sequencing are on-chip (confirmed, https://www.nxp.com/products/analog-and-mixed-signal/gate-drivers/high-precision-differential-injector-driver:MC33816)	On-chip programmable current-profile sequencer (the part is explicitly documented and used for common-rail diesel injector waveforms: boost-to-peak, peak-hold, hold, per NXP's own AN4849 application note referenced from the product page)	Yes — external high/low-side FETs are user-selected and the pre-driver sequences which pair is active, so a shared-high-side bank topology like this ECU's can be implemented in firmware/config, not hardware-forced	Not confirmed from the fetched page (typical NXP power packages; needs datasheet confirmation)	**Confirmed general-market part** — listed ' with NXP's Product Longevity Program (1 availability commitment), stocked at Digi (https://www.digikey.com/en/products/de usa-inc/MC33816AE/4693868) and Mouser (https://www.mouser.com/ProductDetail/ Semiconductors/MC33816AE) with norma now, ships today" retail terms — **not auto allocation-only**, orderable in small quant
Discrete architecture (boost converter IC + external high-side/low-side power MOSFETs or IGBTs + current-sense resistor/comparator, sequenced by an MCU or FPGA)	N/A — built from discrete blocks	Fully custom; boost voltage set by converter design (any point in the 65–115 V envelope)	Closed-loop current regulation implemented in firmware/analog comparator, not integrated	Fully under designer control — the natural way to implement any custom bank topology, including this one	N/A	All constituent parts (boost controllers, automotive MOSFETs, current-sense amp broadly available in small quantities fron standard distributors

Note on automotive-allocation risk: several dedicated common-rail injector driver ASICs used by Tier-1s (proprietary Bosch/Denso/Continental in-house silicon, and some Infineon/Elmos/ST parts specifically qualified and allocated for OEM automotive programs) are **not available through normal small-quantity distribution** — searches for Infineon and Elmos part numbers plausibly in this space (`TLE8090` , `E526`) did not surface a confirmed, orderable, small-quantity datasheet within this memo's search budget (**negative result** — this does not prove such parts don't exist or aren't orderable, only that this search did not locate and confirm one; it should be treated as an open item, not a closed "allocation-only" finding, for any part not named above). The **MC33816 is the one confirmed, small-quantity-orderable candidate found in this pass** and is recommended as the primary driver-IC direction for Phase 2, paired with an externally designed boost stage sized to the Step 2 envelope.

Step 5: The calibration boundary — stated plainly

This is the most important paragraph in this memo, and it is written to be read on its own.

What this hardware can do without any injector-specific calibration data: the drive stage described above — a boost converter in the 65–115 V range, a peak/hold current-regulated driver spanning 12–24 A peak and 8–13 A hold, sequenced across the bank-shared topology this ECU actually uses — **can be fully designed, built, and verified on the bench.** It can be proven correct against a **known inductive load** (a resistor/inductor dummy load, or a bare injector solenoid off the engine, driven on a bench rail) — confirming boost rise time, peak/hold current accuracy, turn-off decay, minimum pulse width, and the bank-sequencing logic from Step 3, all without ever touching the running engine. This is real, verifiable engineering progress and should proceed.

What it cannot do without more data: run this specific engine correctly. A correctly *built* driver that outputs the right current profile is not the same as a driver that delivers the right *fuel quantity* at the right *crank angle* at the right *rail pressure* — and that mapping (fuel mass injected as a function of pulse width, at each of several rail pressures, separately for pilot and main injections, including the "ballistic region" where pulse width is too short for the injector to reach full lift) is exactly the calibration data that **Bosch and Denso do not**

publish. It is proprietary flow-bench data tied to the specific injector part number's hydraulic geometry (nozzle hole count/size, needle lift, spring preload), and normally reaches an OEM only bundled with the supplier's own ECU and calibration tools. Without it, this hardware would open and close the injectors with electrically correct pulses but with **no reliable way to know how much fuel each pulse actually delivered** — which means no reliable air-fuel ratio control, no reliable torque control, and for this specific engine, no reliable path to the CPCB IV+ NOx/PM limits its EGR-and-DOC aftertreatment strategy was designed around (Task 1). This is not a minor gap to be tuned out empirically on a running genset; incorrect fueling on a compression-ignition engine at 19.7:1 compression risks knock-like combustion damage, turbocharger overspeed/overtemperature, and DOC/EGR fouling well before a trial-and-error tuning process could converge.

The three routes to flow-vs-pulsewidth data, named plainly:

1. **Supplier licence (Bosch or Denso).** Approach the injector's OEM supplier (once the exact part number is identified, per Step 1's open item) for calibration data or a development/licensing agreement. **Cost:** realistically out of reach for a project at this scale — these programs are structured around OEM production volumes (tens of thousands of units), not a single genset retrofit; even where a "development kit" path nominally exists, expect NDA gates, minimum engineering-service fees, and multi-month lead times with no guarantee of access for a non-OEM customer. **Legality:** fully legal — this is the intended, sanctioned route. It is listed first for correctness and last for realism.
2. **Dyno characterization with rate-of-injection measurement.** Buy or borrow injectors of the identified part number, mount them on an injector-rate test bench, and directly measure fuel delivery vs. pulse width vs. rail pressure using an established method — the **Bosch tube method** (pressure-wave measurement in a long fuel-filled tube) or the **Zeuch method** (constant-volume chamber), both well-documented in the injection-research literature (confirmed: e.g. "Experimental Research on the Injection Rate of DME and Diesel Fuel in Common Rail Injection System by Using Bosch and Zeuch Methods," https://doi.org/10.3390/en11020273; "Benchmark between Bosch and Zeuch method–based flowmeters," https://www.researchgate.net/publication/331523136_Benchmark_between_Bosch_and_Zeuch_method-based_flowmeters_for_the_measurement_of_the_fuel_injection_rate). **Cost:** substantial but bounded and controllable — either access to an existing university/industry fuel-injection lab (a rate-of-injection rig is specialized, not off-the-shelf test equipment, and few Indian institutions maintain one) or building a simplified rate tube in-house, plus the engineering time to sweep pulse width × rail pressure × injector unit-to-unit variation into a usable map. **Legality:** entirely legal — this is independent, first-party measurement of purchased hardware, no reverse engineering of anyone's IP.
3. **OEM calibration extraction** (reading the flash/calibration data out of a genuine Kirloskar/Bosch ECU for this engine, by bench-side flash dump or bootloader access, and decoding the injector maps embedded in it). **Cost:** potentially the cheapest in engineering time if a donor ECU and the right tools are available, but requires reverse-engineering an undocumented, likely-encrypted or checksum-protected calibration binary — a nontrivial and uncertain effort in itself. **Legality: do not soften this.** Extracting and using a manufacturer's proprietary calibration data, without a license, sits in genuinely contested legal territory — it implicates the OEM's copyright in the calibration software/data, and depending on jurisdiction and exact method, protections analogous to anti-circumvention rules for access-controlled software. This is squarely the kind of activity that has drawn civil action against aftermarket tuning firms in other jurisdictions (well-documented in the passenger-vehicle ECU-tuning industry). It is **not recommended** as this project's route absent explicit legal clearance, and should not be treated as a routine engineering shortcut.

This is the single largest risk in the project, larger than any hardware selection question in this memo. The drive electronics are a solved, bench-verifiable problem. Getting *this engine* to run correctly, safely, and within its designed emissions strategy is gated on data this project

does not currently have a confirmed, legal, affordable path to. The owners have been told this before; this memo is the durable written record: **build and bench-verify the driver now; do not plan to commission it on the running genset without first closing Route 1 or Route 2 above (or accepting the documented legal exposure of Route 3).**

Sources consulted

- `/home/mohan/workws/ecu25kva/.superpowers/sdd/2026-09-13-phase1-research/task-4-brief.md` — task requirements (read in full).
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — engine identity memo (read in full).
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — ECU pinout extraction, injector bank wiring (read in full).
- https://www.crdiparts.com/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/ — fetched; Kirloskar-branded Bosch injector cross-reference (different engine family than 3R550).
- https://turbotech.in/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/ — mirror listing, found via search.
- https://www.picoauto.com/library/automotive-guided-tests/bosch-current-at-idle — found via search; Bosch CR solenoid injector current profile (~100 V boost to 20 A peak, 24 V to 10 A hold).
- https://www.researchgate.net/publication/350426572_Circuit_for_driving_common_rail_diesel_injectors — found via search; boost voltage 65–115 V, peak ~20 A, coil R/L typical values.
- https://www.nxp.com/products/analog-and-mixed-signal/gate-drivers/high-precision-differential-injector-driver:MC33816 — fetched; MC33816 channel count, pre-driver voltage (up to 72 V), Product Longevity status.
- https://www.digikey.com/en/products/detail/nxp-usa-inc/MC33816AE/4693868 and https://www.mouser.com/ProductDetail/NXP-Semiconductors/MC33816AE — found via search; small-quantity distributor stocking and pricing (~US$5.19–5.61).
- https://en.wikipedia.org/wiki/Firing_order — found via search; general four-stroke firing-interval formula (720°/n cylinders).
- https://doi.org/10.3390/en11020273 ("Experimental Research on the Injection Rate of DME and Diesel Fuel... Bosch and Zeuch Methods") — found via search; rate-of-injection measurement methods.
- https://www.researchgate.net/publication/331523136_Benchmark_between_Bosch_and_Zeuch_method-based_flowmeters_for_the_measurement_of_the_fuel_injection_rate — found via search; corroborates Bosch/Zeuch method equivalence.
- Searches returning negative results (recorded, not omitted): `Kirloskar R550 common rail injector Bosch part number`, `Kirloskar 3R550 3 cylinder diesel firing order 1-2-3`, `Infineon TLE8090 OR ELMOS E526 OR STMicroelectronics common rail injector driver IC datasheet` (no confirmed small-quantity-orderable alternative IC located in this search pass).

Research Memo 05: MCU Selection — S32K148 Confirmation

Verdict: CONFIRMED. NXP S32K148, Arm Cortex-M4F, recommended in the **144-pin LQFP package** (orderable P/N **FS32K148HAT0MLQT** — 80 MHz clock grade, −40…+125 °C, CAN-FD feature set, tape-and-reel), is retained as the MCU for this design. It clears the channel budget in spec §5 with large margin, is in stock at an India-serving distributor today (not a 40-week part), has a workable — if not ideal — KiCad footprint path, and a free toolchain. **One part of the spec's reasoning is weaker than it looks and is replaced below (§6): the "constant speed relaxes timing" argument checks crank-edge rate, which was never the binding constraint. The binding constraint is injector pulse-width/output-compare resolution, and that also clears with large margin — so the conclusion holds, but for a different reason than §5 states.**

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1. Channel budget vs. the S32K148 datasheet

Source: NXP, *S32K1XX — S32K1xx Data Sheet*, **Rev. 15, 5 March 2026**, https://www.nxp.com/docs/en/data-sheet/S32K1xx.pdf (cached copy inspected locally; page numbers below refer to this PDF's own footer pagination, "N / 108").

Budget item (spec §5)	Need	S32K148 datasheet	Cite	Status
CAN	2×	**3× FlexCAN**, all 3 with CAN-FD (ISO/CD 11898-1)	p.6, Fig. 3	Confirmed — 1 spare instance
ADC	~14 ch (8 sensor + 2 batt-rail + 4 current-sense)	**2× 12-bit SAR ADC, 1 Msps each, up to 32 ch/module** = up to 64 analog-capable pins at full-chip level	p.2 (Key Features), p.6 Fig. 3	Confirmed at instance/full-chip level. **Inferred, not yet confirmed:** how many of those 64 channels are actually broken out on the 144-pin package specifically requires the pin-mux table in the *S32K1xx Reference Manual* — the datasheet itself says pinouts/signal descriptions live there, not in this document (p.10, §10.1). Flagged for Phase 2.
Timer	~10 ch (crank, cam, 3 inj-lo, 2 inj-hi, metering PWM, 2 EGR PWM)	**8× independent 16-bit FlexTimer (FTM) modules, up to 64 standard IC/OC/PWM channels total** (8 ch/instance)	p.3, p.6 Fig. 3	Confirmed — >6× the required channel count

Budget item (spec §5)	Need	S32K148 datasheet	Cite	Status
I2C (baro sensor)	1×	**2× LPI2C**	p.6 Fig. 3	Confirmed
Flash	—	up to **2 MB** program flash (up to 1,984 KB when EEPROM emulation/data flash in use)	p.2, p.6 Fig. 3 footnote 2	Confirmed, generous for this application's code size
RAM	—	up to **256 KB** SRAM (incl. FlexRAM)	p.2, p.6 Fig. 3	Confirmed, generous
Core / clock	—	Arm Cortex-M4F, **80 MHz (RUN)** or **112 MHz (HSRUN)**	p.2, p.6 Fig. 3	Confirmed

The spec's own summary of the datasheet (§5: "3× FlexCAN… dual 12-bit ADC… FlexTimer instances well beyond the ~10 timing channels required") is accurate and is now backed by a primary-source page reference rather than recollection.

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2. Package selection

Recommendation: 144-pin LQFP (NXP case outline **SOT486-2**, drawing **98ASS23177W** — confirmed document numbers, per datasheet §9.1 "Obtaining package dimensions," p.108 of the PDF; the dimensional drawing itself was **not** independently fetched in this session — **inferred, not confirmed:** SOT486 is the standard 20×20 mm / 0.5 mm-pitch LQFP144 outline used across NXP's automotive MCU lines, but this specific drawing should be pulled from nxp.com and checked pad-by-pad before footprint lock, per the PCB Roadmap's warning that a hand-sourced footprint is the most likely place to introduce a silent error).

Why 144-pin over the smaller options, and why LQFP over BGA:

- Package options that exist for S32K148: 100-pin LQFP (QuadSPI *not* supported in this package per datasheet footnote 6, p.6), 100-pin MAPBGA, 144-pin LQFP, 176-pin LQFP.
- The confirmed pin map (spec §4) resolves only 44 of 94 OEM ECU pins; U1 in the unknowns register carries the other 50. The MCU-side signal budget from §5 alone (2 CAN ×2 pins + 14 ADC + 10 timer + 1 I2C ×2 pins ≈ 30 dedicated pins) plus SWD debug, oscillator, and power/ground pins for a part this size (typically another 30–40 pins on a 144-pin LQFP) is comfortably inside even the 100-pin package. The 144-pin package is chosen as headroom against U1, not because the confirmed budget requires it.

- LQFP over MAPBGA: a 100-pin MAPBGA buys nothing the 144-pin LQFP doesn't already have (same instance counts — CAN/ADC/FTM/I2C counts are silicon-level, not package-level, per Fig. 3), while a BGA package requires X-ray or destructive inspection to verify solder joints — a real cost for a small team doing early hand/low-volume assembly. 144-pin LQFP is 0.5 mm pitch, which is reflow-soldering-standard and inspectable by eye or AOI.

Recommended orderable part number: FS32K148HAT0MLQT — decoded against the datasheet's own ordering-information key (p.8, Fig. 5): - `H` = 80 MHz speed grade (RUN mode) - `A` / `A1` = CAN-FD + FlexIO (+ Security) feature option - `M` = **−40 °C to +125 °C** temperature grade - `LQ` = 144-pin LQFP package code - `T` = trays/tubes (an `R` suffix variant, e.g. FS32K148HFT0VLQR, exists for tape-and-reel production quantities)

Confirmed the `U` (112 MHz HSRUN) speed-grade parts are explicitly **not valid at the 125 °C (`M`) temperature grade** (datasheet footnote 1 under the ordering-option table, p.8: "Not valid with M temperature/125C"). This matters directly for the recommendation: an engine-bay-adjacent ECU should be specified to the 125 °C grade, and §6 below shows the extra 32 MHz of HSRUN clock isn't needed to close the timing budget — so there is no reason to trade thermal margin for clock headroom the design doesn't use. **This is a concrete correction to the spec's framing in §5**, which cites "a Cortex-M4F at 112 MHz" as the relevant number; the part that should actually go on the BOM runs at 80 MHz and gets the better temperature rating in exchange.

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3. Availability and pricing — India

Distributor pages for the specific chip part numbers (as opposed to the eval board) are behind bot-detection on live fetch (Mouser: DataDome 403; DigiKey: 403) in this session, so the figures below come from each distributor's own indexed page content returned via search, dated **2026-09-13.** This is flagged so a live re-check before BOM lock is understood as still worthwhile — component pricing/stock is volatile — but the figures are not fabricated; they are quoted from the cited distributor URL's own content as indexed.

S32K148, 144-pin LQFP family (FS32K148UJT0VLQT / FS32K148HFT0VLQR / FS32K148HAT0MLQT):

Distributor	Part	Stock / lead time	Price	Cite
DigiKey (India storefront, digikey.in)	FS32K148UJT0VLQT	In stock, ships to India in ~4 days	₹1,900.22 (qty 1)	DigiKey
DigiKey	FS32K148UJT0VMHT (100-BGA variant, for comparison)	In stock	₹2,016.80 (qty 1)	DigiKey

Distributor	Part	Stock / lead time	Price	Cite
DigiKey (global .com listing)	FS32K148HAT0MLQT	"Ships today" per indexed listing	not captured in INR	DigiKey
Third-party stocking distributor (components-store.com)	FS32K148UJT0VLQT	7,948 units	$4.28/pc	components-store.com
Third-party stocking distributor	FS32K148HFT0VLQR	10,530 units	$3.59/pc	(same search)
Element14 India	S32K148EVB-Q176 (eval board, not bare chip)	listed, "pricing unavailable, contact customer service" for chip SKUs	₹17,922.07 (eval board)	in.element14.com

Verdict on S32K148 availability: confirmed pass. Real stock at DigiKey's India-facing storefront in single-digit quantities, at low-hundreds-of-rupees-to-few- thousand-rupee unit pricing, with ordinary shipping — not a lead-time problem. No need to invoke the S32K146 fallback.

S32K146 fallback (checked per brief step 2, not needed but recorded):

Part	Stock	Cite
FS32K146HFT0VLQT (144-LQFP)	Out of stock, backorder at DigiKey	DigiKey
FS32K146HAT0VLQT (144-LQFP)	Out of stock, backorder	DigiKey
FB32K146HAT0VLLT (related SKU, 100-LQFP)	Manufacturer standard lead time **52 weeks**	found via search index
FS32K146HAT0MLLT (related SKU, 100-LQFP)	Manufacturer standard lead time **12 weeks**	found via search index

Inferred: the 144-pin LQFP S32K146 SKUs specifically are on backorder with no confirmed lead time in the indexed snippets; related SKUs in the same family show a wide 12–52 week spread, which would have made S32K146 a materially worse availability bet than S32K148 if it had been needed. This reinforces, rather than undercuts, the S32K148 recommendation — the fallback is in worse shape right now than the primary choice.

Element14 India does not surface public per-unit pricing for the bare MCU chip in either family (only the eval board); it would need a quote request. Not a blocker given DigiKey's confirmed stock, but worth noting Element14 is the weakest of the three distributors for this specific part.

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4. KiCad symbol and footprint provenance

- **KiCad's own stock symbol library does not carry the S32K1xx family.** A direct search of the official `KiCad/kicad-symbols` GitHub repository for NXP MCU symbols returns only legacy Freescale parts

(`MCU_NXP_HC11.lib`, `MCU_NXP_HC12.lib`, `MCU_NXP_NTAG.lib`) — confirmed, no `S32K` entry exists. Cite: https://github.com/KiCad/kicad-symbols

- **NXP's own EVB design files exist but are schematic-level, not a reusable KiCad/parametric MCU symbol.** The S32K148EVB-Q176 has a published schematic (document SCH-29642, dated 2017-07-12, mirrored via Octopart) but this is a board-level PDF schematic, not a native KiCad library part — using it as a footprint source would mean hand-deriving pin assignments, exactly the "hand-sourced footprint" risk the PCB Roadmap warns about. Cite: https://datasheet.octopart.com/S32K148EVB-Q144-NXP-Semiconductors-datasheet-103093927.pdf
- **SnapEDA (now SnapMagic Search) is the practical source.** It lists symbol/footprint/ 3D-model downloads, exportable to KiCad, for S32K148 parts including 144-LQFP variants (e.g. `FS32K148HAT0MLQT`) — confirmed listing exists. Cite: https://www.snapeda.com/parts/FS32K148HAT0MLLT/NXP%20Semiconductors/view-part/, https://octopart.com/part/nxp-semiconductors/FS32K148HAT0MLQT (Octopart surfaces the same "trusted partner" symbol/footprint/3D-model set)

Pad pitch verification: not done in this session — flagged as an open Phase 2 item. The datasheet points to case outline SOT486-2 / drawing 98ASS23177W for the exact mechanical dimensions (§9.1, p.108 of the PDF) but does not inline the pitch value, and the drawing PDF itself was not fetched here. Per the PCB Roadmap's explicit warning, do **not** carry a SnapEDA footprint into layout without diffing its pad pitch/pad size against document 98ASS23177W pulled fresh from nxp.com — this is exactly the class of silent error the Roadmap calls out, and it has not been closed by this memo.

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5. Toolchain and debug

- **Compiler/IDE: NXP S32 Design Studio for ARM**, free, Eclipse-based, bundles the GNU Arm Embedded Toolchain (GCC/GDB) — confirmed, S32DS for ARM 2.2 explicitly lists S32K148 support. NXP is migrating S32K1 support toward "S32 Design Studio for S32 Platform" going forward; both remain usable today. Cite: https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-arm:S32DS-ARM, release notes PDF via NXP Community. Third-party toolchains (IAR, GHS, Arm/Keil, Lauterbach, iSystems) are also listed as supported ecosystem options in the datasheet itself (p.6, Fig. 3, "Ecosystem" row).

- **Evaluation board exists: S32K148EVB-Q176**, Arduino-Uno-pinout-compatible, confirmed available from DigiKey India (₹17,487.61) and Element14 India (₹17,922.07) — the two prices differ by ~2%, ordinary distributor variance, not a discrepancy worth chasing. Cite: https://www.digikey.in/en/products/detail/nxp-usa-inc/S32K148EVB-Q176/7929314, https://in.element14.com/nxp/s32k148evb-q176/eval-board-32bit-arm-cortex-m4f/dp/2917581. **Confirmed** this lets firmware bring-up (HAL, crank/cam capture, CAN/J1939 stack skeleton, injector timer logic) start well before the custom PCB is fabricated, per spec §10 Phase 4 sequencing.
- **Debug probe for the custom board.** The EVB has an onboard OpenSDA debug/serial adapter (confirmed via NXP's own EVB documentation, indexed) usable during EVB-based bring-up at no extra cost. For the custom PCB (bare SWD header, no onboard debug circuit), a standalone probe is needed. **Important gotcha, not called out in the brief:** SEGGER's cheapest J-Link variant, the **J-Link EDU Mini, is licensed for non-commercial/educational use only** — SEGGER's own product description states it is "sold to private persons, colleges, schools, universities and NPOs for educational purposes only and is not sold to companies." This project is a commercial product development effort, so J-Link EDU Mini is **not** a legally usable option here despite being the cheapest one that shows up in a casual search. Commercially-licensed options:
- **SEGGER J-Link BASE**: $798 (8.08.00) or **J-Link BASE Compact**: $598 (8.19.00) — confirmed pricing at DigiKey (USD; India-specific INR not separately captured in this session — inferred to land in the ₹50,000–₹67,000 band at typical GST/duty markup, needs a live India quote before budgeting). Cite: https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-08-00-J-LINK-BASE/2175882, https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-19-00-J-LINK-BASE-COMPACT/7386652
- **PE Micro U-MULTILINK-FX**: $738.27 at DigiKey — comparable price to J-Link BASE, no commercial-use restriction, and PEmicro explicitly supports S32 Design Studio. Cite: https://www.digikey.com/en/products/detail/nxp-usa-inc/U-MULTILINK-FX/3247905 Recommendation: budget for one J-Link BASE Compact or PEmicro U-MULTILINK-FX for lab/ production debug; use the EVB's free onboard OpenSDA for early firmware work.

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6. Testing the constant-speed timing argument

The spec's §5 argument, restated: a genset governs at a fixed 1500 rpm; a crank degree is 111 µs at that speed; a 60-tooth wheel gives a tooth edge every 667 µs; the Cortex-M4F/ FlexTimer combination has "several

orders of magnitude of headroom" against that, so the eTPU2's angle-synchronous hardware isn't needed.

This is checking the wrong variable. Crank-edge *capture* rate was never the hard constraint that motivated eTPU2 use in automotive engines. A tooth arriving every several hundred microseconds to a few milliseconds is trivial for a Cortex-M4F interrupt or a FlexTimer input-capture channel regardless of whether the engine is a genset or a car — 667 µs is ~75,000 CPU cycles at 80 MHz, and NVIC latency on a Cortex-M4F is on the order of 12 cycles (~150 ns at 80 MHz), several thousand times smaller than the interval being measured. This was true for automotive engines too, even at 6000 rpm (crank edge every ~28 µs at 6000 rpm / 60 teeth — still >2000 CPU cycles). **So the comparison the spec makes was never actually close, in either the automotive or the genset case, and citing constant-speed operation as what makes it comfortable is not quite the right causal story.**

The eTPU2's real value in automotive engines is different: it is a co-processor that autonomously schedules and re-schedules many angle-domain output-compare events (start of injection, duration, sometimes multi-pulse rate-shaping) across a wide, *rapidly changing* speed range, without loading the main CPU's interrupt budget, and without the main core's scheduling jitter (cache effects, higher-priority ISRs, RTOS latency) touching the injection timing chain. That is a real constraint when rpm sweeps 800–6000 rpm with tight closed-loop transient response, many simultaneous channels, and multi-pulse strategies (pilot/main/post injection) that all need to land within microseconds of a moving target angle.

The constraint that should actually be checked is injector pulse-width/output-compare resolution — how finely the MCU can control injector on-time, independent of crank speed. This is what determines fuel-quantity metering accuracy for a given injector's flow characteristic, and it does not go away just because rpm is constant. Checking it:

- FTM output-compare channels run off the bus/peripheral clock, up to 112 MHz (HSRUN) or 80 MHz (RUN) per the datasheet (p.2). At 80 MHz, one timer tick = 12.5 ns; at 112 MHz, ~8.9 ns.
- Typical diesel solenoid injector pulse-width control for quantity repeatability needs on the order of 1–4 µs resolution in demanding automotive multi-pulse (pilot/main/post) strategies — this is a widely-cited engineering figure for common-rail injector drive, not a datasheet fact, so it is marked **inferred** rather than confirmed by a specific citation here; it is not sourced from memory of a specific document, and should be checked against the actual injector's datasheet once U5 (injector part number, in the spec's unknowns register) is closed.

- The pin map (spec §4) shows a single main injection pulse per cylinder per cycle (one high-side bank switch + one low-side select per cylinder, "boosted high-side switch" + "current-controlled low-side, peak-and-hold" — no evidence of multi-pulse rate-shaping in scope for this genset). A single-pulse-per-cycle strategy is materially less demanding on timing resolution than automotive pilot/main/post injection.
- Either way, an 8.9–12.5 ns timer tick against a ~1–4 µs (or looser, for single-pulse metering) requirement leaves roughly **100–450× headroom** — comparable in kind to the margin the spec claimed for crank-edge capture, but now measured against the constraint that actually matters.

Conclusion: the spec's recommendation survives, but its stated reasoning should be corrected, not just accepted. Crank-edge capture rate was never the tight constraint, in either an automotive or a genset engine. The tight constraint — injector pulse-width resolution — is also comfortably cleared by the S32K148's FlexTimer output-compare granularity, with margin that would likely hold even if this design later grew a multi-pulse injection strategy (not currently in scope; U5 in the spec's unknowns register should be revisited if it changes). The two PDBs (Programmable Delay Blocks, 2× on K148, confirmed p.6 Fig. 3) are also relevant here and not mentioned in the spec: they allow triggering ADC conversions or timer events at a defined delay from a capture event in hardware, without CPU intervention — useful headroom for keeping the injection-timing chain jitter-free even if firmware complexity grows, and a partial answer to the "what does the eTPU2 buy that we're giving up" question the spec doesn't fully address.

One caveat this memo cannot close: the flow-rate/resolution figure used above is a general diesel-injector engineering figure, not a citation to this project's actual injector. U5 (injector part number and drive profile) is still open per the spec's unknowns register. If the injector's actual metering-quantity-vs-pulse-width sensitivity turns out to need sub-microsecond control (unusual for single-pulse metering, but not impossible for a very small idle-quantity injector), this section's margin claim should be re-run against the injector's actual datasheet before Phase 2 sign-off. This is the one place where "confirmed" in this memo rests on an inferred industry figure rather than a project-specific document, and it is flagged as such rather than smoothed over.

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7. Residual items for Phase 2

1. Pull NXP package drawing **98ASS23177W** (144-pin LQFP, SOT486-2) directly and diff its pad pitch/pad geometry against whatever KiCad footprint (SnapEDA-sourced or hand-built) is actually used in the schematic sheet — not done in this memo.

2. Pull the **S32K1xx Reference Manual** pinout/IO-signal-description tables to confirm which of the 64 ADC channels and 64 FTM channels are actually broken out on the 144-pin LQFP package specifically (the datasheet defers this to the Reference Manual, §10.1) — the instance-level counts are confirmed, the exact pin-level assignment for this package is not.
3. Get a live India quote (not indexed-search pricing) for FS32K148HAT0MLQT and one commercial debug probe (J-Link BASE Compact or PEmicro U-MULTILINK-FX) before BOM lock — Mouser/DigiKey blocked live automated fetch in this session (bot protection); the figures above are real but were retrieved via search-index snippets of the distributors' own pages, not a live page load.
4. Revisit §6's injector timing margin once U5 (injector part number) is closed.

Citation index

- NXP, *S32K1XX S32K1xx Data Sheet*, Rev. 15, 5 March 2026: https://www.nxp.com/docs/en/data-sheet/S32K1xx.pdf
- KiCad stock symbols repo: https://github.com/KiCad/kicad-symbols
- SnapEDA / SnapMagic Search, S32K148 parts: https://www.snapeda.com/parts/FS32K148HAT0MLLT/NXP%20Semiconductors/view-part/, https://octopart.com/part/nxp-semiconductors/FS32K148HAT0MLQT
- S32K148EVB-Q176 schematic (SCH-29642): https://datasheet.octopart.com/S32K148EVB-Q144-NXP-Semiconductors-datasheet-103093927.pdf
- NXP S32 Design Studio for ARM: https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-arm:S32DS-ARM
- DigiKey India: https://www.digikey.in/en; DigiKey FS32K148HAT0MLQT: https://www.digikey.com/en/products/detail/nxp-usa-inc/FS32K148HAT0MLQT/8628226; S32K148EVB-Q176: https://www.digikey.in/en/products/detail/nxp-usa-inc/S32K148EVB-Q176/7929314
- DigiKey S32K146 SKUs: https://www.digikey.com/en/products/detail/nxp-usa-inc/FS32K146HFT0VLQT/10815998, https://www.digikey.co.il/en/products/detail/nxp-usa-inc/FS32K146HAT0VLQT/10816081
- Element14 India, S32K148EVB-Q176: https://in.element14.com/nxp/s32k148evb-q176/eval-board-32bit-arm-cortex-m4f/dp/2917581

- SEGGER J-Link EDU Mini (educational-use-only licensing): https://www.segger.com/news/segger-introduces-j-link-edu-mini-a-low-cost-j-link-for-education/
- SEGGER J-Link BASE / BASE Compact pricing: https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-08-00-J-LINK-BASE/2175882, https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-19-00-J-LINK-BASE-COMPACT/7386652
- PEmicro U-MULTILINK-FX: https://www.digikey.com/en/products/detail/nxp-usa-inc/U-MULTILINK-FX/3247905
- Project spec: `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` §4, §5

Research Memo 06: Standards, Test Requirements, and Certification Scope

> **UPDATED 18 Sep 2026 — the certificate is now a specific number**
>
> The engine's rating plate has been photographed. Two things change for this memo, both in the direction of making its conclusions **more** concrete, not less:
>
> - The engine is **3GK550ETA 4SR1**, application code `GK3.8703`, not the `3R550ETA 4G1` this memo inherited from research memo 01. The genset model `KG4-25WS1` in the header was inferred from the same chain and should be treated as **unconfirmed** — no genset nameplate has been seen.
> - The plate carries the actual type approval: `ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24`. `IGES4` is India Genset Emission Standard IV, confirming the CPCB IV+ tier from the machine itself rather than from a brochure.
>
> **This memo's regulatory conclusions stand, and now attach to a numbered certificate against a specific serial number** rather than to a product family in the abstract. That is worth stating plainly to whoever takes the decision.
>
> One gap this memo did not examine, raised by a later review: the **state pollution board Consent-to-Operate** regime is a separate gate from central type approval and was never looked at. Recorded as unexamined.
>
> The controller references below to the KG640C are also superseded — the fitted unit is a **DSE4522 MKII AMF**. See `refs/field-evidence-2026-09-18.md`.

Task: U6 (J1939 layers, ISO 7637-2 transient immunity, CISPR 25 conducted emissions, certification position). Consumes Task 1's emissions tier. Produces the test/design numbers Phase 3 layout must honour and an honest scope statement on certification. **Date:** 2026-09-13 **Author:** Phase-1 research (task 6) **Genset:** Kirloskar KG4-25WS1, 25 kVA, 415 V 3-phase / 230 V 1-phase, 50 Hz. **Engine:** Kirloskar 3R550ETA 4G1, 26.5 kW prime (19–56 kW CPCB IV+ band). **Emissions tier:** CPCB IV+, cooled EGR + DOC, no SCR/DEF/DPF (per `docs/research/01-engine-identity.md`). 12 V electrical system. In service, in India.

Verdict (read this first)

Item	Finding	Status
J1939 physical/transport/diagnostic layers	Standard 250 kbit/s twisted-pair CAN with 120 Ω termination, 1 m stub limit; BAM for broadcast multi-packet, RTS/CTS for destination-specific; DM1/DM2 for active/previously-active DTCs; address-claim on power-up	Secondary-sourced (SAE texts paywalled); PGN set itself carried over from Task 2, there marked unconfirmed for this specific controller
ISO 7637-2 transient levels (12 V ref.)	Pulse 1: −150 V (Level IV); Pulse 2a: +112 V; Pulse 2b: +10 V; Pulse 3a: −220 V; Pulse 3b: +150 V (2 ms–0.1 ms widths, 2–50 Ω source impedance per pulse); Pulse 5b (load dump, now properly an ISO 16750-2 pulse, still colloquially cited under 7637-2): 40–105 V suppressed, 0.5–8 Ω, 40–400 ms	Secondary source (industry technical note quoting the standard's table), clearly marked — standard text itself not accessed
CISPR 25 conducted emissions	Class 5 (most stringent), 150 kHz–108 MHz, voltage method via 5 µH/50 Ω LISN; toughest bands are VHF/FM (68–108 MHz): 18 dBµV avg / 38 dBµV peak	Secondary source, clearly marked
Certification position	**Confirmed, from the gazette text itself:** CPCB IV+ type approval is granted to an engine/genset family and obtained by the manufacturer/importer/assembler at the point of manufacture or import — it is not a per-unit approval that travels indefinitely with the physical genset regardless of what is later done to it. **A separate CPCB regulatory document (the RECD System & Procedure) states directly that ECU characteristics governing injection timing, air-mass metering, and emission-reduction strategy must not be altered even under a government-sanctioned retrofit programme, and that modifications limited to engine-control changes fall outside the one retrofit-approval pathway CPCB operates.** No pathway was found by which a third-party ECU replacement on this genset could be approved, and General Condition 3 of GSR 804(E) states plainly that no person shall use a genset lacking a valid Type Approval + CoP certificate. **This is a decision for the project owners to take to qualified counsel and/or CPCB/ARAI directly — not an engineering call, and not something this memo resolves.**	Confirmed (primary gazette text and primary CPCB procedure text, both fetched and read in full) for what the rules say; **not legal advice** on what they mean for this specific project

Step 1: SAE J1939 layers that apply

SAE's J1939 series texts are paywalled (sae.org); nothing below was read from the standard itself. Every figure is from secondary industry sources, cited individually, and should be treated as **secondary/summarized**, not as a verbatim standard quote.

- **J1939-11 (physical layer, twisted-pair variant).** Shielded twisted-pair backbone, nominal 120 Ω characteristic impedance, 250 kbit/s data rate for this device class, 120 Ω (±1%, ¼ W typical) termination resistors at both physical ends of the backbone, and stub ("drop line") length limited to ≈1 m per node to control reflections. *(Secondary: https://simmasoftware.com/products/j1939/sae-j1939-11-physical-layer/ ;*

https://copperhilltech.com/content/SAE%20J1939%20Wiring%20Requirements.pdf ; https://obd-cable.com/j1939-backbone-termination-stub-length-guide/.) **Design relevance:** the genset's CAN segment (GCU↔ECU, per Task 2's KG640C memo) should be laid out to this backbone/stub/termination discipline; a single genset with two nodes is a short, simple topology relative to the standard's truck/trailer use case, so termination is the only piece likely to bite if omitted.

- **J1939-21 (transport protocol).** Multi-packet (>8 byte) messages use either **BAM** (Broadcast Announce Message — connectionless, one-to-all, no flow control, used for messages any node may want, e.g. some diagnostic broadcasts) or **RTS/CTS** (Request-to-Send/Clear-to-Send — connection-oriented, destination-specific, flow-controlled, used for point-to-point large transfers). *(Secondary: https://www.spcontroller.com/j1939-transport-protocol-bam-vs-rts-cts/ ; https://copperhilltech.com/blog/sae-j1939-programming-with-arduino-transport-protocol-rtscts-session-sae-j193921/.)* **Design relevance:** if the replacement ECU needs to send anything larger than a single CAN frame (e.g., a multi-DTC DM1 payload) to the GCU, the firmware needs both mechanisms implemented — most stacks default to BAM for DM1 and would only need RTS/CTS if a proprietary large-payload message is added.
- **J1939-71 (application layer / PGN definitions).** Task 2 (`docs/research/02-kg640c-gcu.md`, §2) already established the working PGN set for this specific genset, **explicitly flagged there as an inferred industry-baseline, not confirmed against a KG640C document**: EEC1 (PGN 61444, engine speed), ET1 (PGN 65262, coolant temp), EFL/P1 (PGN 65263, oil pressure), VEP1 (PGN 65271, battery/system voltage), DM1 (PGN 65226, active DTCs), and TSC1 (PGN 0, torque/speed control, if the GCU is to command the engine). This memo does not re-derive that list; it is carried forward as-is, with the same caveat Task 2 attached.
- **J1939-73 (diagnostics, DM1/DM2). DM1 — Active Diagnostic Trouble Codes** (PGN 65226/0xFECA): broadcast by default at 1 Hz and on state change, carries currently-active DTCs. **DM2 — Previously Active Diagnostic Trouble Codes** (PGN 65227): available on request only, carries DTCs that were active but have cleared. *(Secondary: https://www.csselectronics.com/pages/j1939-73-dm1-diagnostic-message-dtc.)* **Design relevance:** the replacement ECU should broadcast DM1 unsolicited at 1 Hz to be a drop-in match for what the KG640C GCU's fault-display logic almost certainly expects (per Task 2's finding that the GCU is a CAN consumer, not just a display).
- **J1939-81 (network management / address claiming).** On power-up, each node transmits an Address Claimed message; every other node records/verifies it; if two nodes claim the same address, priority is resolved by the NAME field (a 64-bit value encoding manufacturer, function, and an arbitrary/self-configurable-address capability bit). *(Secondary: https://copperhilltech.com/blog/sae-j1939-address-claim-procedure-sae-j193981-network-management/.)* **Design relevance:** the

engine ECU node conventionally claims source address 0x00 by industry convention (again per Task 2, unconfirmed for this controller specifically) — the replacement ECU should claim the same address the GCU expects, which direct bus logging (Task 2's recommended next step) would settle definitively.

Net for Phase 3 layout: none of this drives component values the way Steps 2–3 do; it drives firmware behaviour and connector/stub layout. The one open item carried from Task 2 — the real PGN set and source address, unconfirmed from any KG640C-specific document — should be closed by logging the existing bus before the replacement ECU's CAN stack is finalized.

Step 2: ISO 7637-2 transient immunity — the numbers that set the TVS and buck input rating

Source note: the ISO 7637-2 standard text itself is paywalled and was not accessed. The table below is drawn from an ON Semiconductor automotive applications technical note that reproduces the ISO 7637-2:2011(E) severity-level table directly (title "Supplemental Data for ISO7637 Pulses," https://www.onsemi.com/download/design-notes/pdf/tnd6424-d.pdf, fetched and text-extracted in full) — **marked here as secondary**, since it is an industry document quoting the standard, not the standard.

Pulse	Description	Amplitude, 13.5 V nominal system, Level I/II	Level III	Level IV	Source impedance / pulse width	Test duration
1	Supply disconnection (inductive load switch-off, e.g. relay coil)	−75 V	−112 V	**−150 V**	10 Ω, 2 ms	500 pulses, min. 0.5 s repeat
2a	Sudden interruption of current in a device sharing the harness with the DUT (inductive)	+37 V	+55 V	**+112 V**	2 Ω, 0.05 ms	500 pulses, min. 0.2 s repeat
2b	Same mechanism, lower-energy variant	+10 V	+10 V	+10 V	2 Ω, 0.05 ms	10 pulses, min. 0.5 s repeat
3a	Fast transient bursts (switching-type, EFT-like)	−112 V	−165 V	**−220 V**	50 Ω, 0.1 ms	1 hr test time, 90 ms min. repeat
3b	Same mechanism, opposite polarity	+75 V	+112 V	**+150 V**	50 Ω, 0.1 ms	1 hr test time, 90 ms min. repeat

(All figures per ON Semiconductor TND6424-D, secondary source.)

Load dump (Pulse 5b) — a nuance worth stating plainly: the 2011 edition of ISO 7637-2 removed the load-dump pulses (5a/5b); they now live in **ISO 16750-2** (and are referenced in ISO 21848 for some classes), though industry practice — including many component datasheets and this project's own brief — still refers to "ISO 7637-2 pulse 5b" as shorthand. *(Secondary: search-result synthesis, not independently verified against either standard's text — flagged as an open item rather*

than a confirmed edition history. If the design team needs to cite the pulse in a test plan, ISO 16750-2 §4.6.4 is the more defensible current reference.) For a **suppressed** load dump (the realistic case on a genset with an alternator-integrated or centralized suppression diode/clamp, as is now common) — pulse 5b: peak voltage 40–105 V for a 12 V system, source impedance 0.5–8 Ω, pulse width 40–400 ms. *(Secondary, search-result synthesis — not read from either standard's own table.)*

What this sets for Phase 2/3 hardware: - **TVS clamp voltage:** must sit above the normal 12 V (13.5–14.4 V charging) rail with margin, but clamp well below the injector boost-converter's and any other sensitive node's absolute maximum — the working design window is a clamp that lets pulses 2a/2b (+37 to +112 V) and suppressed 5b (up to ~105 V) pass through to a controlled clamp voltage, while pulses 1/3a (down to −220 V) are clamped on the negative side. A single automotive-grade bidirectional TVS rated for these levels (many parts are explicitly binned "meets ISO 7637-2 pulse 1/2a/3a/3b," per component datasheets — not independently verified against this list here) is the usual answer; the −220 V pulse 3a figure (Level IV, 50 Ω/0.1 ms — low energy despite high voltage) is the most demanding negative excursion to clamp. - **Buck converter input rating:** the input stage (from battery rail to the point after transient protection) should be rated for a working input up to the clamped level of the worst transient the TVS lets through (commonly designed to ~40–45 V clamped for a suppressed load-dump system, consistent with the Vishay/EDN load-dump note found in this search pass but not independently verified here) rather than the full unclamped 220 V pulse, which the TVS/clamp stage is responsible for absorbing before it reaches the buck's input pins.

Step 3: CISPR 25 conducted emissions — band limits and what a bench setup can tell you

Source note: the CISPR 25 standard text itself is paywalled and was not accessed. Figures below are from two secondary sources: a Texas Instruments technical article (https://www.ti.com/lit/pdf/sszt673, fetched and read in full) and an EMC-limits compilation site (https://emccalc.com/limits/cispr25-cev/, fetched). **Both are marked secondary — neither is the standard itself.**

- **Method and frequency range:** conducted emissions are measured at the DUT's power terminals via a 5 µH/50 Ω LISN (line impedance stabilization network), voltage method, over **150 kHz to 108 MHz**, using peak and average detectors. *(TI, sszt673, confirmed as stated by that source.)*
- **Class 5** is the most stringent of CISPR 25's five classes and the one automotive OEMs typically mandate for safety-/receiver-adjacent electronics. *(TI, sszt673.)* Selected Class 5 limits, voltage method (secondary, per emccalc.com's compilation, which states the 2016 (Ed.4) and 2021 (Ed.5) voltage-method limits are identical):

Band	Frequency	Peak limit	Average limit
LW (broadcast)	0.15–0.30 MHz	70 dBµV	50 dBµV
MW (broadcast)	0.53–1.8 MHz	54 dBµV	34 dBµV
SW (broadcast)	5.9–6.2 MHz	53 dBµV	33 dBµV
CB	26–28 MHz	44 dBµV	24 dBµV
VHF (low)	30–54 MHz	44 dBµV	24 dBµV
VHF (high)	68–87 MHz	38 dBµV	18 dBµV
FM broadcast	76–108 MHz (overlapping VHF-high per source)	38 dBµV	18 dBµV

The **68–108 MHz VHF/FM region carries the tightest limit (18 dBµV average / 38 dBµV peak)** — explicitly called out by TI as "extremely challenging." *(TI, sszt673, confirmed as stated by that source.)* This is the band a switching converter's harmonics most often violate, and it is where PCB layout (input filter corner frequency, output ripple, switching-edge control) earns its keep.

What a LISN + spectrum analyser bench setup can and cannot tell you, short of a test house: - **Can:** give a reliable relative measurement of your own converter's conducted noise floor and harmonic content across the 150 kHz–108 MHz range, using a 5 µH/50 Ω LISN (buildable or purchasable as a discrete unit) feeding a spectrum analyser with peak-hold — enough to catch a converter that is grossly non-compliant (wrong switching frequency choice, missing input filter, poor grounding) long before a certification test, and to A/B compare layout or filter revisions against each other. - **Cannot:** substitute for accredited peak/average/quasi-peak detector calibration, the specific bonded ground-plane/test-bench geometry CISPR 25 mandates, cable harness length and dress requirements, or the formal measurement uncertainty budget a test house applies — none of which a bench LISN setup replicates exactly. A bench pass is **necessary-but-not-sufficient** evidence; it de-risks the design before a paid test-house slot, it does not replace one. *(This paragraph is general EMC test-practice knowledge, not tied to a specific citation found in this search pass — flagged as such rather than presented as sourced.)*

Step 4: The certification position — stated plainly

This is the section the brief calls the one that matters most, and it is written to stand on its own.

4.1 What CPCB IV+ actually is, and where it comes from — confirmed from the gazette text itself

This memo obtained and read, in full, the actual Gazette of India notification: **G.S.R. 804(E), Ministry of Environment, Forest and Climate Change, New Delhi, 3 November 2022** — the *Environment (Protection) Third Amendment Rules, 2022*, made under **sections 6 and 25 of the Environment (Protection) Act, 1986**, amending **Schedule I, serial number 88** of the Environment (Protection) Rules, 1986. *(Confirmed — primary source, fetched directly at*

https://cpcb.gov.in/openpdffile.php?id=TGF0ZXN0RmlsZS8zNThfMTY2ODE2NjYzNl9tZWRpYXBob3RvMTkwNTgucGRm and text-extracted via `pdftotext -layout` from the saved PDF; this upgrades Task 1's "inferred/secondary" GSR citation to confirmed.) It comes into force **1 July 2023**, exactly as Task 1 recorded.

The band limits are now confirmed directly from the gazette's own Table 1, not from dieselnet's summary as Task 1 had it: for the **19 < P ≤ 56 kW** power category (this engine's 26.5 kW prime rating falls here) — combined **NOx+HC ≤ 4.7 g/kWh**, **CO ≤ 3.5 g/kWh**, **PM ≤ 0.03 g/kWh**, **smoke ≤ 0.7 m^{-1}** (light absorption coefficient), for CI engines, no separate NOx-only limit in this band. This matches dieselnet's figures exactly and is now **confirmed**, not inferred.

4.2 What the certification covers, and who holds the obligation — confirmed from the gazette's General Conditions

- **General Condition 1 (Applicability):** the rules apply to "**new** engines for power generation application and products **manufactured, assembled or imported to India**" — this is a point-of-manufacture/import regime, not a running in-service inspection scheme (there is no equivalent here to a vehicle's periodic PUC check). *(Confirmed, direct gazette quote.)*
- **General Condition 2 (Requirement of certification):** "**Domestic manufacturer, importer or assembler**... shall obtain **Type Approval** from authorised certifying agency and also comply with **Conformity of Production (CoP)** test of their product(s)..., valid for the next Conformity of Production year." *(Confirmed, direct gazette quote.)* The obligation to *hold* a Type Approval and CoP certificate is placed on the **manufacturer/importer/assembler**, not on the end owner or operator.
- **General Condition 3 (Sale/import/use of non-compliant product):** "**No person shall sell, import or use** an engine and genset for power generation application which is not having a valid Type Approval certificate and certificate of Conformity of Production." *(Confirmed, direct gazette quote — this is the clause that reaches beyond the manufacturer to anyone who uses such equipment, which would include a genset owner/operator.)*
- **General Condition 5 (Nodal agency):** the **Central Pollution Control Board (CPCB)** is the nodal agency for implementation. *(Confirmed.)*
- **General Condition 6 (Authorised certifying agencies):** **Automotive Research Association of India (ARAI, Pune)**, **International Centre for Automotive Technology (ICAT, Manesar)**, and **Indian Institute of Petroleum (IIP, Dehradun)** are the agencies authorised to test and certify Type Approval and CoP for engines and gensets. *(Confirmed, direct gazette quote; a related CPCB procedure document for the separate retrofit-device scheme, §4.2.1, also lists **Vehicle Research and Development Establishment (VRDE, Ahmednagar)** and any NABL-accredited agency meeting that scheme's requirements — this fourth agency and the NABL route are confirmed only for the retrofit-device scheme, not shown in the GSR 804(E) text itself for new-engine Type Approval.)*

- **General Condition 11:** CPCB, the Commission for Air Quality Management, State Pollution Control Boards, or Pollution Control Committees may issue **more stringent** local norms. *(Confirmed — a reminder that state-level rules could add obligations beyond the national floor; not researched further here, flagged as an adjacent open item.)*

What "Type Approval" covers, structurally: it is granted to an engine/genset **family or model** — the regulation's own "engine family" concept (defined in the companion CPCB retrofit-device procedure, §1.1.18: "a manufacturer's grouping of engine systems, which, through their design, have similar exhaust emission characteristics; all members of the family must comply with the applicable emission limit values," citing ISO 8178-7) is confirmed only in that companion document but is standard practice across this same regulatory architecture (same nodal agency, same three certifying agencies, same GSR-804(E)-referenced testing regime). Individual units are not independently re-tested against Table 1 one-by-one; the manufacturer's ongoing conformance is checked by periodic Conformity of Production (CoP) sampling of production units, not a certificate that attaches permanently, in isolation, to one physical serial number regardless of what happens to it afterward.

4.3 The direct hit: CPCB's own text on ECU/calibration modification

This memo located and read, in full, **CPCB's "System and Procedure for Emission Compliance Testing of Retro-fit Emission Control Devices (RECD) for Diesel Power Generating Set Engines up to Gross Mechanical Power 800 kW"** (fetched from https://cpcb.gov.in/uploads/RECD_Procedure_Revised.pdf, text-extracted via `pdftotext -layout` from the saved PDF in full, 3258 lines). This is CPCB's own regulatory document governing the *one* pathway CPCB operates for modifying an already-type-approved genset's emission-control configuration after manufacture. Two passages from it are decisive for this project, quoted verbatim:

> **§4.1.2.2(e):** "The characteristics of the electronic engine control unit (as regards, for example, injection timing, air-mass flow metering, or exhaust emissions reduction strategies) **shall not be altered** by the retrofitting. Any modification of the test engine that changes the original emission behaviour (for example alternation of the injection timing) **is not permitted**."
>
> **§1.1.35 (definition of "Retrofit emission control device"):** "...Systems that **only modify the existing engine system controls are not considered to be RECDs**."

(Both confirmed — direct quotes, primary CPCB source, fetched and read in full.)

Read together, these two passages mean: (1) CPCB explicitly identifies **injection timing, air-mass flow metering, and exhaust-emission-reduction strategy** — precisely the calibration parameters this engine's EGR-rate-scheduling-and-injection-timing NOx/PM strategy runs on (per Task 1's finding of EGR+DOC-only aftertreatment, no SCR) — as characteristics that must not be touched, even under a CPCB-sanctioned emissions-improvement retrofit programme; and (2) a modification that is *purely* a change to the engine's own control system, with no added physical after-treatment hardware — which is exactly what an ECU replacement is — **does not qualify as an RECD at all** and therefore has **no approval pathway under this scheme, full stop.** It is neither covered by the new-manufacture Type Approval/CoP scheme (GSR 804(E), which governs manufacturers at point of production/import, §4.2 above), nor by the one modification-approval scheme CPCB operates for in-service units (RECD, which by its own definition excludes pure control-system changes).

Negative result, recorded per the brief's instruction: no CPCB-administered, or any other, approval pathway was found in this search by which replacing this genset's OEM ECU with a third-party unit of different calibration could be formally re-certified or otherwise sanctioned as compliant. This does not prove no such pathway exists anywhere in Indian regulatory practice — only that this search, within its budget, did not locate one. This is exactly the kind of fact pattern that should go to CPCB, ARAI, or qualified counsel directly, not be inferred further by this memo.

4.4 The calibration angle the brief asked to check — confirmed as directly relevant, not merely plausible

The brief's suspicion is correct, and the gazette text itself confirms the mechanism: GSR 804(E) Table 1 sets limits in **g/kWh over a defined test cycle** (a mass-of-pollutant-per-unit-of-work measure), not as a hardware bill-of-materials requirement. Compliance is therefore inherently a function of *how the engine is run* — injection timing, EGR rate scheduling, fuelling — as much as *what parts are bolted to it*. Combined with §4.3's finding that CPCB itself names injection timing, air-mass metering, and emission-reduction strategy as the calibration parameters that determine emission behaviour, and with Task 1's finding that this specific engine meets its 19–56 kW-band NOx+HC limit (≤4.7 g/kWh) through cooled EGR and DOC alone, with **no SCR** as a chemical backstop: **a replacement ECU with different EGR-scheduling or injection-timing calibration can move this engine out of compliance with its certified emission limits without any hardware changing at all.** This is not a remote or theoretical risk — it is the same mechanism by which the engine meets its limits in the first place, run in the wrong direction. It is directly relevant to, and reinforces, the calibration-data gap already flagged as the project's largest single risk in `docs/research/04-injector-drive.md` (Step 5): that memo's finding was about getting the engine to run *correctly and safely* without OEM fuelling maps; this memo's finding is that even a "correctly running"

replacement ECU, if its EGR/injection-timing calibration differs from the certified original, has no assurance of remaining within the CPCB IV+ limits the genset was certified against — an emissions-compliance risk layered on top of the combustion/mechanical risk Task 4 already documented.

4.5 Statutory exposure — flagged, not advised on

Separately from the certification-scheme mechanics above, the **Environment (Protection) Act, 1986, Section 15** provides a general penalty for contravention of the Act or rules made under it. *(Secondary — this memo relied on a legal-database rendering of the statute, https://indiankanoon.org/doc/24023684/, rather than a freshly fetched Gazette copy of the Act itself, and the search result returned text that appears to blend the original 1986 penalty (imprisonment up to 5 years and/or fine up to ₹1 lakh, escalating for continuing contravention) with a later monetary-penalty band (₹10,000–₹15 lakh) that reads as a post-2023 Jan Vishwas (Amendment of Provisions) Act amendment. Which version is currently in force, and whether it would apply to this specific fact pattern, is exactly the kind of question that needs a lawyer, not this memo — flagged, not resolved.)*

4.6 The decision, framed as the brief asked

This memo does not tell the project owners whether to proceed. What it establishes, with the citations above:

1. CPCB IV+ Type Approval is a family/model-level certificate obtained by the manufacturer, not a permanent per-unit status immune to what happens to the physical genset afterward.
2. The obligation to *hold* Type Approval + CoP sits with the manufacturer/importer/assembler; the obligation not to *use* a non-compliant engine/genset (General Condition 3) reaches the owner/operator directly, by the gazette's own words — "no person shall... use."
3. CPCB's own regulatory text, in the one scheme it operates for modifying in-service gensets, states plainly that altering ECU characteristics governing injection timing and emission-reduction strategy is not permitted even under a sanctioned retrofit programme, and that pure engine-control modifications fall outside that scheme's approval process entirely.
4. No approval pathway for a third-party ECU replacement on this genset was found in this search.
5. The mechanism by which this specific engine meets its CPCB IV+ limits (EGR scheduling + injection timing, no SCR backstop) is exactly the mechanism a replacement ECU's calibration would change.

Whether this means the genset's certification is void, whether it creates a compliance or disclosure obligation, and what exposure attaches to the owner/operator versus this project team, is a legal and regulatory question — not an engineering one. It should go to CPCB,

one of the three authorised certifying agencies (ARAI/ICAT/IIP), or qualified counsel, before this project proceeds past the bench-verification stage Task 4 already scoped as safe to continue.

Sources consulted

- `/home/mohan/workws/ecu25kva/.superpowers/sdd/2026-09-13-phase1-research/task-6-brief.md` — task requirements, read in full.
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — engine identity and emissions-tier memo, read in full (Step 4 specifically).
- `/home/mohan/workws/ecu25kva/docs/research/02-kg640c-gcu.md` — KG640C GCU/J1939 memo, grepped for PGN references (Step 2 of that memo).
- `/home/mohan/workws/ecu25kva/docs/research/04-injector-drive.md` — injector drive/calibration-boundary memo, read in full (Step 5 specifically, for the calibration-risk cross-reference in §4.4 above).
- **G.S.R. 804(E), 3 November 2022**, Ministry of Environment, Forest and Climate Change — fetched directly (https://cpcb.gov.in/openpdffile.php?id=TGF0ZXN0RmlsZS8zNThfMTY2ODE2NjYzNl9tZWRpYXBob3RvMTkwNTgucGRm), saved and text-extracted in full via `pdftotext -layout`. **Primary source, read in full.** This is the actual Gazette of India notification text — confirms GSR number, date, ministry, Table 1/Table 2 emission limits, and all General Conditions cited in Step 4.
- **CPCB, "System and Procedure for Emission Compliance Testing of Retro-fit Emission Control Devices (RECD) for Diesel Power Generating Set Engines up to Gross Mechanical Power 800 kW"** — fetched directly (https://cpcb.gov.in/uploads/RECD_Procedure_Revised.pdf, redirected from cpcb.nic.in), saved and text-extracted in full via `pdftotext -layout` (3258 lines). **Primary source, read in full** for the definitions and §4.1.2.2(e) quoted in Step 4.
- https://cpcb.gov.in/genset-notifications/ — fetched, confirms the GSR 804(E)/GSR 436(E) amendment chain exists on CPCB's own notification index.
- https://indiankanoon.org/doc/24023684/ (via search) — Environment (Protection) Act 1986, Section 15 text, secondary rendering of the statute, not independently fetched from a gazette copy of the Act.
- https://www.onsemi.com/download/design-notes/pdf/tnd6424-d.pdf — ON Semiconductor technical note reproducing the ISO 7637-2:2011(E) pulse severity table; fetched, saved, and text-extracted in full via `pdftotext -layout`. Secondary source.
- https://www.ti.com/lit/pdf/sszt673 — Texas Instruments, "A Review of EMI Standards, Part 1 – Conducted Emissions"; fetched, saved, and text-extracted in full via `pdftotext -layout`. Secondary source, for CISPR 25 method and Class 5 VHF/FM limits.

- https://emccalc.com/limits/cispr25-cev/ — CISPR 25 Class 5 voltage-method limit table compilation; fetched. Secondary source.
- https://www.araiindia.com/certification/certification/genset ; https://cpcb.nic.in/type-approval-cop-data/ ; https://cpcb.nic.in/displaypdf.php?id=Tm9pc2VfYW5kX0dlbnNldHMvU3lzX1Byb2RfZm9yX0VtaXNzX0NvbXBfb2ZfREdTZXRzLnBkZg%3D%3D — found via search, corroborate the general Type Approval/CoP scheme description in §4.2 without being individually quoted.
- Searches returning only partial or negative results (recorded, not omitted): `ISO 7637-2 pulse 1 2a 2b 3a 3b 5b amplitude 12V test levels table` (general web search did not surface a full table directly, resolved via the onsemi PDF instead); `CISPR 25 class 5 limits table dBuV band I II III IV V` (resolved via emccalc.com); direct WebFetch of https://keystonecompliance.com/iso-7637-2/ and https://www.electronicdesign.com/.../cispr-25-class-5... returned only qualitative descriptions, no numeric tables — superseded by the sources above.

Research Memo 07: Consolidated BOM and Sourcing

Task: Consolidated, costed bill of materials covering every functional block in spec §4 and §6. Consumes Tasks 3 (connector), 4 (injector driver), 5 (MCU) — the three hard sourcing problems — plus the simulation-fixed component values supplied directly for this task. **Date:** 2026-09-13 **Author:** Phase-1 research (task 7) **Scope note, stated once and binding throughout:** this ECU has no DEF/urea sensor, no LIN bus, no SCR, no magnetic pickup, and one catalyst temperature sensor. Fuel level and water-in-fuel are panel signals, not ECU signals. None of these appear below. Earlier design artifacts that included them were wrong and are not carried forward.

Read this first — the headline

	Qty 1	Qty 10
BOM total (excl. PCB fab, excl. assembly)	**≈ ₹14,200**	**≈ ₹1,34,500** (≈ ₹13,450/unit)

Two findings drive that number and its shape:

1. **The 94-pin OEM-mating connector is ~45–50% of the per-unit BOM cost, and it does not get cheaper with volume** — it is bought as an aftermarket wired-pigtail kit at a fixed per-kit price, not from a distributor with quantity breaks. This is the single biggest lever on total cost and the single biggest sourcing risk in the design.
2. Real quantity discounts on the catalog parts (MCU, injector driver, CAN transceiver, buck, supervisor) only partly show up at qty 10 — DigiKey's own price breaks for these lines mostly land at qty 100, not qty 10 — so the qty-10 total is not proportionally cheaper than 10× the qty-1 total.

Single-source / long-lead lines flagged: 3 confirmed, 1 partially mitigated, 1 open Phase-2 item. See §3.

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1. Method and honesty note

Per the task's own budget instruction, this memo prices the expensive and risky lines from live or near-live distributor data — **MCU, injector driver, connector, CAN transceivers, the buck** — and gives the remaining functional blocks a **reasonable aggregate estimate, marked as such**, built from a named representative automotive-grade part rather than an invented number with no anchor. Every "estimate" line is a candidate part in a real, in-production automotive family, priced by class-typical India distributor pricing, not fabricated from nothing — but it has **not** been individually fetched from a live distributor page this session, and Phase 2 must confirm it before BOM lock.

Distributor pricing fetches in this session hit DigiKey India live pages successfully; Mouser India pages timed out repeatedly (likely bot-mitigation, consistent with what Memo 05 reported for Mouser/DigiKey chip pages). Where a Mouser-only line is cited, the price is from search-indexed content, not a live page load, and is marked accordingly.

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2. BOM by functional block

Legend: **AEC** = automotive qualification status. **C** = confirmed this session (live fetch or direct citation). **I** = inferred / estimated, not independently priced this session. Currency conversion where only USD was found: ₹88/US$1, noted per line.

#	Functional block (spec §4/§6)	Part	Distributor	Stock	Qty 1 (₹)	Qty 100 (₹)	Lead time	AEC-Q	Status	Cite
1	EMI input common-mode choke	Würth WE-CMB automotive CMC (representative, e.g. 744232801-class)	Mouser/DigiKey (class-typical)	—	~90	~55	—	AEC-Q200 (passive)	**I** — aggregate estimate	class prici not fetch

#	Functional block (spec §4/§6)	Part	Distributor	Stock	Qty 1 (₹)	Qty 100 (₹)	Lead time	AEC-Q	Status	Cite
2a	Reverse-battery ideal-diode controller	TI **LM74610-Q1**	DigiKey / Mouser	listed	~320 (est. from $1.25@1k)	~140	standard	AEC-Q100	**I** — price scaled from confirmed $1.25@1000 unit	TI
2b	Reverse-battery P-FET	Infineon **IPB80P03P4L-01** (representative AEC-Q101 P-FET)	Mouser/DigiKey (class-typical)	—	~150	~90	—	AEC-Q101	**I** — aggregate estimate	class prici not fetch
3a	Transient clamp — TVS	**SMBJ33CA** bidirectional TVS (fixed by simulation)	Mouser/DigiKey (widely stocked jellybean)	—	~18	~8	—	Commercial-grade common; automotive-grade (`-13-F` type suffixes) available from Littelfuse/Vishay/onsemi	**I** — aggregate estimate	class prici not fetch
3b	Transient clamp — 10 µF bulk cap	Automotive X7R/electrolytic, 10 µF (fixed by simulation)	Mouser/DigiKey (class-typical)	—	~15	~7	—	AEC-Q200 (ceramic)	**I** — aggregate estimate	class prici not fetch
4	Buck pre-regulator, 6–40 V → 5 V	~~TI TPS54360B-Q1~~ → **SUPERSEDED 20 Sep 2026, see the note under this table: TI LM5164DDAR**	DigiKey (global; India page not confirmed live)	listed	~458 ($5.20 × 88)	~280 (est.)	standard	AEC-Q100	**C** (price), 60 V-rated device chosen over 42 V LM5175-Q1 for load-dump margin — **I** (India-specific stock/lead time not confirmed)	DigiK searc $5.20 foun via searc inde
5a	`5V_SENSOR` LDO	TI **TLV70233-Q1**-class automotive LDO (representative)	Mouser/DigiKey (class-typical)	—	~120	~70	—	AEC-Q100	**I** — aggregate estimate	class prici not fetch
5b	`5V_SENSOR` PTC resettable fuses ×4 (per sensor group)	Bourns **MF-USMF**-class automotive PTC	Mouser/DigiKey (class-typical)	—	~100 (4×~25)	~60	—	AEC-Q200	**I** — aggregate estimate	class prici not fetch
6	`3V3_MCU` LDO	~~TLV1117-33QDCYRQ1-class~~ → **SUPERSEDED 20 Sep 2026: TI TLV76733QWDRBRQ1**	Mouser/DigiKey (class-typical)	—	~70	~35	—	AEC-Q100	**I** — aggregate estimate	class prici not fetch
7	Supervisor (windowed WD + BOR)	TI **TPS3823-33QDBVRQ1**	DigiKey India	4,605 units in stock	**162.44**	**95.60**	Mfr. standard 16 wk (moot — stock covers near-term demand)	AEC-Q100	**C** — live DigiKey India fetch	DigiK in
8	MCU	NXP **FS32K148HAT0MLQT**, 144-LQFP	DigiKey India	In stock, ships ~4 days (sibling SKU confirmed; exact SKU "ships today" per global listing)	**~1,950** (₹1,900–2,000 band)	~1,400 (est., no qty100 break found)	Not a lead-time problem	AEC-Q100	**C** — per Memo 05	Mem 05, DigiK
9	Crystal (MCU osc.)	NDK/Abracon AEC-Q200 automotive crystal, 8 MHz-class (representative)	Mouser/DigiKey (class-typical)	—	~35	~18	—	AEC-Q200	**I** — aggregate estimate	class prici not fetch
10	CAN transceiver ×2 + protection	TI **TCAN1042HGVDRQ1** (×2) + ESD array (representative NUP2105L-class, ×2)	DigiKey India	**Out of stock; 2,500 units expected 21-Sep-2026**	**426.16** (2× 213.08) + ~40 (2× ~20 ESD)	**257.50** (2× 128.75) + ~30	**Mfr. standard 16 weeks**	AEC-Q100	**C** (transceiver price/stock) — **flagged, see §3**	DigiK in
11	VR conditioner (crank sensor)	Melexis **MLX92242**-class or NXP MC33199-class differential VR interface (representative — not yet selected)	Mouser/DigiKey (class-typical)	—	~180	~100	—	AEC-Q100	**I** — aggregate estimate, **part not yet firmly selected (Phase 2)**	class prici not fetch

#	Functional block (spec §4/§6)	Part	Distributor	Stock	Qty 1 (₹)	Qty 100 (₹)	Lead time	AEC-Q	Status	Cite
12	EGR H-bridge	Infineon **TLE9201SG**-class automotive H-bridge (representative)	Mouser/DigiKey (class-typical)	—	~220	~140	—	AEC-Q100	**I** — aggregate estimate	class prici not fetch
13	Metering-unit low-side driver + current sense	Infineon **BTS443P**-class automotive smart low-side switch (representative)	Mouser/DigiKey (class-typical)	—	~250	~160	—	AEC-Q100	**I** — aggregate estimate	class prici not fetch
14a	Injector driver IC	NXP **MC33816AE**	DigiKey India	**868 units**	**1,040.54**	**693.12**	Std.; note: page shows "Last Time Buy Date 06/08/2027" — **flagged, see §3**	Industrial/automotive SMARTMOS, not itself AEC-Q100-badged in fetched listing — **needs Phase-2 confirmation**	**C** — live DigiKey India fetch	DigiI in
14b	Injector boost stage (65–115 V, 12–24 A pk, 8–13 A hold)	Discrete: boost controller (e.g. LM5155-Q1-class) + HV automotive MOSFETs + magnetics + HV caps	Mouser/DigiKey (class-typical, multi-part)	—	~800	~500	—	Mixed AEC-Q100/Q101	**I** — aggregate estimate for the whole discrete stage, **no specific BOM chosen yet (Phase 2)**	Mem enve
15	Relay FETs ×2 (main relay, buzzer relay) — low-side N-FET + flyback diode each	Automotive N-FET (e.g. IPD50N04S4L-13-class) + automotive Schottky (e.g. SS34-class), ×2 pairs	Mouser/DigiKey (class-typical)	—	~80 (2×~40)	~40	—	AEC-Q101	**I** — aggregate estimate	class prici not fetch
16	Barometric sensor (onboard, per spec §7 deviation #2)	NXP **MPXH6115A6U**-class automotive absolute pressure sensor (representative)	Mouser/DigiKey (class-typical)	—	~450	~300	—	Automotive-grade family (NXP MPXH line is automotive-qualified)	**I** — aggregate estimate, **part not yet firmly selected (Phase 2)**	class prici not fetch
17a	94-pin ECU-mating connector (adapter harness side)	**Bosch EDC17-style 94-pin + 60-pin companion, wired pigtail kit** — leading candidate, unconfirmed part number	u-obd.com / ecudepot.com / autoecupart.net (aftermarket ECU-repair trade)	Listed, multiple sellers	**~6,600** (US$65–99 × 88, band)	**~6,600** (no bulk break — aftermarket kit pricing)	Order-dependent, international shipping + India customs, **not independently verified — Phase 2 item**	N/A — not a distributor part number	**C** (price band at source) / **I** (India-landed total) — **flagged, see §3**	Mem 03, u- obd.c
17b	New ECU-side split connectors (spec §7 deviation #1)	TE Connectivity sealed automotive connector family (e.g. MCON 1.2/Superseal-class, representative)	Mouser/DigiKey (mainstream stock)	—	~300 (aggregate, multiple housings)	~180	—	Automotive sealed connector family	**I** — aggregate estimate, exact housing/pin-count split not yet designed	class prici not fetch
18	Fixed-value passives (battery-sense divider ×2, discrete-input front-end ×3 sets, ratiometric front-end ×4 sets, plus generic bypass/pull-up passives)	1% metal-film resistors, MLCC caps, zener diodes per the simulation-fixed values below	Mouser/DigiKey (class-typical)	—	**~300**	**~150**	—	Mixed AEC-Q200 (passive) availability	**I** — aggregate estimate, see breakdown below	class prici not fetch

Passives aggregate breakdown (line 18), values fixed by simulation, not re-derived: - Battery-sense divider, ×2 (pins 04, 06): 120 kΩ / 10 kΩ, 1% metal film — 4 resistors. - Discrete-input front-end, ×3 (pins 20, 24, 71): 47 kΩ / 68 kΩ, 220 nF, 3.0 V zener — 12 components. - Ratiometric sensor front-end, ×4 (boost pressure, EGR position, oil pressure, rail pressure): 10 kΩ / 16 kΩ, 22 nF, 3.3 V zener — 16 components. - Plus an allowance for generic bypass caps, gate resistors, and pull-ups not individually itemized in this memo. - ~32 fixed-value components + ~30 generic ones ≈ 60–70 small passives total, priced at India-typical qty-1

rates of ₹2–4/resistor, ₹3–6/MLCC cap, ₹8–15/zener; qty-100 rates roughly a third lower. This is the class of line the task explicitly authorizes as an honest aggregate rather than 30 invented individual prices.

Superseded 20 September 2026, during schematic capture

Two rows above are struck through. Both were chosen honestly and both have since been overtaken — one by a design change, one by a requirement that did not exist when the table was written.

Row 4, the buck. TPS54360B-Q1 was picked as a 60 V part "for load-dump margin" over a 42 V alternative. Correct at the time. The input-protection work then raised the TVS standoff to 43 V, so the suppressor stops conducting during a *normal* clamped load dump — and a higher standoff clamps higher, putting the ISO 7637-2 pulse 2a clamp at **73.3 V**. `sim/blocks/transient_clamp.cir` now carries a *passing* check whose text reads "60 V-class parts are NO LONGER viable downstream". A 60 V buck on this input fails the first pulse 2a event.

The replacement is **LM5164DDAR**, 100 V input, 1 A, 6–100 V range. It is not a new candidate: it is the part every rating check in the simulation suite has been written against since the negative-pulse work — its −0.3 V / 100 V VIN limits are what `negative_pulses.cir` and `transient_clamp.cir` check. The BOM table had simply never been reconciled with the suite.

That gap is the lesson worth keeping from this row. **The contradiction sat in the repository for days between an executable file and a prose table, and only the executable one knew.** Nothing flagged it, because nothing compares them; it surfaced when someone had to draw the buck and needed a pinout.

Row 6, the 3V3 LDO. TLV1117-33QDCYRQ1 was listed as representative, with its own caveat "class pricing, not fetched". Choosing the supervisor (TPS3850G33, `docs/pinmap.md` §1.8) turned 3V3 accuracy into a hard constraint: outside 3.143–3.459 V the supervisor resets the board. That makes "representative" no longer good enough. **TLV76733QWDRBRQ1** is 1% over load *and* temperature — 3.267–3.333 V, ~125 mV margin at each end.

Pricing for neither replacement has been fetched; the estimates in the table above are stale for these two rows and should be re-costed before BOM lock. Full reasoning and the derived configuration values are in `docs/bom_requirements.md`.

3. Single-source and long-lead flags

Per the task's global instruction: any line with one supplier, or a lead time beyond eight weeks, gets a named second source or an explicit "no alternative identified." These are the lines that stall a build.

3.1 94-pin ECU-mating connector — FLAGGED, no distributor alternative identified

Single-source in the distribution sense that matters: the leading candidate (Bosch EDC17-style 94-pin housing) is sold **only through the international ECU-repair aftermarket** (u-obd.com, ecudepot.com, autoecupart.net, eBay sellers) — confirmed absent from Mouser India, DigiKey India, and Element14 in Memo 03's search. That channel carries real risks a normal distributor doesn't: no formal datasheet, counterfeit/quality risk, order-dependent lead time, and customs/duty exposure not yet quantified.

Named second sources, both weaker matches, both mainstream-distributor-buyable: - **Aptiv/Delphi GT-150 series** — Mouser-stocked, moderate visual match, exact 94-position variant not yet confirmed to exist (Memo 03, negative result on that specific check). - **TE Connectivity AMPSEAL (90-series)** — Mouser/DigiKey-stocked, weakest visual match (oval/rounded shell vs. the drawing's rectangular multi-row grid), but the safest fallback if the Phase-2 photograph rules out the other two.

This line cannot be closed further without the photograph brief in Memo 03 §4. It is the top risk item in this BOM — both for build-stalling potential and for the ~45–50% of per-unit cost it represents.

3.2 Injector driver IC (MC33816AE) — FLAGGED, no alternative identified

Memo 04 searched specifically for allocation-free alternatives (Infineon TLE8090, Elmos E526) and returned a **negative result** — no confirmed, small-quantity-orderable alternative dedicated common-rail injector-driver IC was located. The MC33816AE is **confirmed in stock (868 units) at DigiKey India** today, which is good news, but it is a single-supplier (NXP), single-product-line dependency with no named second source.

New finding this session, not yet resolved: the live DigiKey India page for this part carries a **"Last Time Buy Date: 06/08/2027"** field. Memo 04 separately reports the part as "Active" under NXP's Product Longevity Program (a 10-year-availability commitment), which is not the same thing as an EOL notice — a longevity-program part can still show a programmatic LTB-style date without meaning the product is being discontinued. **This memo cannot resolve which reading is correct with the sources available this session** — it is flagged, not smoothed over, as a Phase-2 item: confirm directly with NXP or DigiKey sales whether MC33816AE has an active EOL notice before committing the injector driver architecture to this part.

No alternative identified. If the LTB field turns out to mean what it says, this becomes the most urgent open item in the entire BOM, ahead of the connector.

3.3 CAN transceiver (TCAN1042HGVDRQ1) — FLAGGED, second source named

Live DigiKey India data this session: **out of stock**, with 2,500 units expected 2026-09-21, and a **manufacturer standard lead time of 16 weeks** — over the task's 8-week threshold on both counts (current stockout, and the manufacturer-quoted lead time for a fresh order).

Named second source: NXP **TJA1051** / **TJA1042** high-speed CAN transceiver family — AEC-Q100, functionally equivalent (ISO 11898-2 high-speed CAN transceiver), confirmed listed at Mouser (Memo-05-style search-index confirmation; India-specific live price not obtained this session, flagged as a Phase-2 confirmation item). Given the restock date is roughly one week out, this may resolve itself before it matters, but the flag stands per the letter of the instruction — it was out of stock and over 8 weeks lead time at time of writing.

3.4 Supervisor IC (TPS3823-33QDBVRQ1) — flagged on paper, low real risk

DigiKey's manufacturer standard lead time field shows 16 weeks (>8-week threshold), but **4,605 units are in stock right now** at DigiKey India, which resolves the practical risk for any order size this project plausibly needs. Named second source anyway, for completeness: Microchip **MCP130**-class or ON Semi automotive supervisor family, both mainstream-distributor-stocked.

3.5 Injector boost stage, VR conditioner, EGR H-bridge driver, barometric sensor — open Phase-2 selections, not yet single-source risks

These four lines carry a representative candidate part, not a firmly selected one. They are not flagged as single-source because no sourcing decision has been locked in yet — but Phase 2 must close each of them with the same rigor Tasks 3/4/5 applied to the connector, injector driver, and MCU, before this memo's estimate lines convert to real BOM lines.

3.6 MCU (FS32K148HAT0MLQT) — mitigated, not flagged

Per Memo 05: real DigiKey India stock today, ordinary shipping, multiple orderable SKUs within the same S32K148 family (UJT0VLQT / HFT0VLQR / HAT0MLQT) across DigiKey and at least one third-party stocking distributor (components-store.com, thousands of units). The checked fallback family (S32K146) is in *worse* supply shape (12–52 week spread, backordered on the 144-LQFP SKUs), so it is not offered as the named second source — the in-family SKU spread already provides one. Not flagged.

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4. Headline cost, worked

Qty 1 (sum of the "Qty 1" column above): **≈ ₹14,175**, rounded to **≈ ₹14,200**.

Of that, the connector kit (line 17a) alone is **≈ ₹6,600** — about 46% of the total. Remove it and the rest of the BOM (electronics + new-side connectors + passives) is **≈ ₹7,575** at qty 1.

Qty 10, worked with an explicit, stated assumption rather than silently guessed: catalog IC lines are assumed to fall to roughly 90% of their qty-1 unit price at qty 10 (a conservative read — DigiKey's own price breaks for the lines fetched live this session mostly land at qty 100, e.g. TPS3823 falls 41% qty1→100 and TCAN1042 falls 40%, so a qty-10 point partway there is a reasonable, stated interpolation, not a measured figure). The connector kit (line 17a) is assumed to hold its per-unit price flat at qty 10 — it is bought per-kit from an aftermarket seller with no confirmed bulk-order mechanism.

- Non-connector BOM at qty 1: ₹14,175 − ₹6,600 ≈ ₹7,575 → ×0.90 ≈ ₹6,820/unit at qty 10 → ×10 ≈ **₹68,200**
- Connector at qty 10: ₹6,600 × 10 = **₹66,000** (flat, per above)
- **Total, 10 units: ≈ ₹1,34,200**, i.e. **≈ ₹13,420/unit** — barely below the qty-1 unit price, because the largest line item does not get cheaper in volume through this channel.

This is the headline finding worth repeating in plain language: a conventional component BOM would show a strong per-unit cost curve from qty 1 to qty 10. This one barely moves, because its largest cost driver is not a catalog part.

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5. PCB fabrication estimate (separate from the BOM total above)

No layout exists yet — this is a Phase-2-pending, explicitly inferred board-area assumption, built from the parts on this BOM: a 144-pin LQFP MCU, a 94-pin connector-shadow footprint plus new-side connectors, a buck pre-regulator, an injector boost stage with HV magnetics and bulk caps, an H-bridge, and assorted discrete front-ends. A board in the **100–150 mm × 120–160 mm** class (≈150–240 cm^2) is a reasonable planning assumption for this component count; this memo adopts **120 mm × 150 mm (180 cm^2)** as the working figure.

- **China-based low-volume fab, shipped to India** (JLCPCB/PCBWay-class service, the common route for small Indian teams): a 4-layer, standard-FR4, no-ENIG prototype run of 5–10 boards at this area typically lands in the **US$40–70 total** band for the panel, i.e. **≈ ₹3,500–6,200 for the run (≈ ₹600–1,200/board)**, plus international shipping (days to ~2 weeks) and possible customs handling. **Inferred**, not a live quote — Phase 2 should submit real Gerbers once layout is complete.
- **India-based fab** (faster turnaround, no import wait, typically higher per-board cost at prototype quantities for 4-layer with plated vias at this class of complexity): **≈ ₹1,000–2,000/board** for a small prototype run. **Inferred**, same caveat.

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6. What this memo did not do, and why

- Did not individually fetch live pricing for ~14 of the 22 BOM lines. The task's own budget instruction says this explicitly is the right trade: price the expensive and risky lines properly, mark the rest as an aggregate estimate. Ten well-sourced lines (MCU, injector driver, connector, both CAN transceiver components, buck, supervisor) plus one clearly-marked aggregate covers everything the task asked for without chasing forty invented numbers.
- Did not resolve the MC33816AE "Last Time Buy Date" field. This is flagged loudly in §3.2 rather than either alarmed over or ignored — it needs a direct NXP/distributor sales conversation, not another web search.

- Did not select final parts for the VR conditioner, EGR H-bridge driver, metering-unit driver, barometric sensor, or injector boost-stage BOM. These carry representative, real, in-production automotive-grade candidates so the cost estimate has an anchor, but none of the five is a locked sourcing decision. Flagged in §3.5, not smoothed over.

Citation index

- Memo 03 — ECU connector
- Memo 04 — Injector drive
- Memo 05 — MCU selection
- Project spec: `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` §4, §6
- DigiKey India, TPS3823-33QDBVRQ1: https://www.digikey.in/en/products/detail/texas-instruments/TPS3823-33QDBVRQ1/1894636 — fetched live 2026-09-13
- DigiKey India, MC33816AE: https://www.digikey.in/en/products/detail/nxp-usa-inc/MC33816AE/4693868 — fetched live 2026-09-13
- DigiKey India, TCAN1042HGVDRQ1: https://www.digikey.in/en/products/detail/texas-instruments/TCAN1042HGVDRQ1/5967638 — fetched live 2026-09-13
- TI LM74610-Q1 product page and $1.25@1k pricing: https://www.ti.com/product/LM74610-Q1, https://www.how2power.com/newsletters/1512/products/H2PToday1512_products_TexasInstruments.pdf
- TI TPS54360B-Q1 / TPS54360BQDDARQ1 pricing (search-indexed, $5.20–5.24): https://www.ti.com/product/TPS54360B-Q1, DigiKey listing found via search
- DigiKey FS32K148HAT0MLQT: https://www.digikey.com/en/products/detail/nxp-usa-inc/FS32K148HAT0MLQT/8628226
- u-obd.com, Bosch EDC17 94-pin connector kit: https://www.u-obd.com/product/edc17-connector/
- Mouser, Delphi GT-150 series: https://www.mouser.com/Delphi-Connection-Systems/Connectors/Automotive-Connectors/GT-150-Series/
- Mouser, TE Connectivity AMPSEAL: https://www.mouser.com/en/c/connectors/automotive-connectors/?m=TE+Connectivity&series=AMPSEAL

Engine Inspection Brief — Field Checklist

STATUS 18 Sep 2026 — part of this brief is done

A first visit has happened and photographs came back. **Four register items closed**, and the tasks that closed them are ticked below. What remains is mostly the multimeter work, which is the part that actually gates the design.

Closed: U11 engine identity (task 3.1 — the plate reads `3GK550ETA 4SR1`, *not* the model memo 01 inferred) · U1 the connector (tasks 3.4–3.8 — Bosch `1 928 405 192 / 194`, **code C**) · U13 which controller is fitted (**DSE4522 MKII AMF**, not a KG640C — this was not a task in this brief, it was raised by the evidence itself) · U3 no speed pickup (from DSE's manual).

UPDATE, later on 18 Sep: U12 is CLOSED, and it closed badly. Reported from the machine: **the fuel relay cuts a signal, it does not remove power.** That is the outcome task 2.1's table told the reader to flag immediately. §3's second line of defence terminates in the failure it was meant to catch, and the standalone trip module is now the only shutdown path on this machine that does not require the cooperation of the ECU being protected against. Task 1.4's contact-type check is still wanted as confirmation — see `refs/field-evidence-2026-09-18.md §4` for why nothing depends on it.

Also reported: the intake throttle and the pre/post-heaters are **battery-fed**. That answers who supplies them, not who commands them — tasks 3.11 and the U14 trace both still stand.

Also still open and cheap: task 1.3 (U2, one resistance reading) and task 3.11 (U8, follow one cable by hand).

SECOND VISIT, 21 Sep 2026 — see Group 5 at the foot of this brief. Seven new tasks, written after the schematics were drawn rather than from documents. **5.1 blocks the board outright: the 94-way connector has no identified battery-negative pin, and every circuit on the new board returns to a net that currently has nowhere to land.** 5.7 and the still-open 3.11 are the only two items that can make the board bigger, so both want answering before layout.

New since this brief was written, worth adding to the next visit: trace the **pre-heat / post-heat** supply — the DSE4522's configuration enables both at 50 °C and this design has no heater output at all (U14).

Purpose: close every open unknown that no amount of desk research can close — only a person with a multimeter and a camera, at the machine, can. Closes or advances **U1, U2, U4, U6, U7, U8, U9, U10, U11, U12**, plus the pin 01/02 ground question raised in the reconciled spec.

Print this. Carry it. Tick boxes as you go — a half-finished visit is still useful if the ticks say which half.

Sources: reconciled spec `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` (§3 safety, §8 unknowns register), `refs/ecu-pinout-extracted.md`, research memos `docs/research/01-engine-identity.md` and `docs/research/03-ecu-connector.md`. Every expected reading below cites one of these. Anything not sourced is marked **inferred**.

SAFETY — read this before touching anything

This is a 25 kVA genset: a diesel engine that can start, and a 415 V three-phase alternator. This checklist is bench/inspection work only.

- **Battery disconnected before any continuity or resistance measurement.** Negative terminal off first, isolated, out of reach of the positive post.
- **Nothing inside the alternator output enclosure.** No task here requires it. If you find yourself reaching toward the 415 V output terminals, stop — that is not this checklist.
- **The engine does not run during any of this.** Every task below is done with the engine off. Ignition-on tasks mean the key/master switch position only, never cranking or running.
- **If a measurement needs the set running** (this includes the full CAN log in Group 4, which may need it), **that is a separate supervised activity**, done by someone qualified to run the set, not a solo checklist item. Note it as such and move on.
- Reconnect the battery only after Group 1 is fully complete and you've checked no tools or leads are left on the terminals.

Materials: digital multimeter (V, Ω, continuity beep), clamp meter if available (safer than breaking a fused circuit for current), phone/camera with flashlight, notebook or voice memo, 10 mm socket for the battery terminal, tape/marker for labelling. **Group 4 additionally needs a USB-CAN adapter and a laptop** — it may warrant its own trip.

Group 1 — Battery disconnected (continuity / resistance)

Do these first, in this order, with the battery negative terminal removed and isolated.

- [ ] **1.1 — Pin 01/02 ground hypothesis.** Continuity from ECU connector pins **01** and **02** to battery negative / chassis ground.
- *Why:* the connector drawing places 01/02/05/06 as large power contacts, but 01/02 are labelled `G_G_B AT1` / `AT2` — switch-return legs, which wouldn't need a large contact. The alternate reading: `G_G_B` = battery ground, and these are the ECU's main power grounds (also the only explanation for why no power-ground pin appears anywhere in the 44-pin extraction). *(spec §4 note; pinout "Digital inputs")*
- **Continuity (near 0 Ω) → hypothesis confirmed:** 01/02 are power grounds, not switch returns. Update the pin map and power/ground plan.
- **Open circuit → hypothesis wrong.** They're isolated switch-return legs as originally labelled; the power-ground pin(s) remain unidentified — keep looking during Group 3 tracing.

- [ ] **1.2 — Pin 09 "synchronization ground."** Continuity from pin **09** to (a) battery negative and (b) a known sensor ground pin (e.g. 34 or 08).

- *Why:* pin 09 is named as a ground but drawn in **red** (switched-battery colour class), which is contradictory on its face. *(spec §4, "Power"; pinout, U6 entry)*
- **Continuous to battery negative** → it likely is a ground point, oddly coloured in the diagram for harness-routing reasons, not a live signal. Note this for §4 but treat as resolved.
- **Open to both** → it is *not* a simple ground. Carry to Group 2 (2.3) for a voltage measurement — the red wire colour then means what it usually means.

- [ ] **1.3 — Boost sensor thermistor topology (closes U2 outright).** Boost P&T sensor unplugged, engine **cold**. Resistance across sensor pins **2 and 1** (ECU pins 79 / 34 per the pinout — temperature signal / sensor ground). *(brief; spec §8 U2; pinout "Analog sensors")*

- **A few kΩ, falling if you re-check once the engine's warmed a little** → NTC thermistor. Design a pull-up divider front-end. **U2 closed.**
- **Open circuit, or a fixed high reading that doesn't move with temperature** → not a bare NTC — a conditioned/ratiometric output. Design the ratiometric front-end (same as the pressure channel). **U2 closed.**
- This is a single measurement. Either outcome closes U2 — there is no third case.

- [ ] **1.4 — Ignition relay contact type.** *(Still wanted 18 Sep as confirmation; task 2.1 has since been answered the other way round — see the status banner.)* **Do this BEFORE task 2.1; it decides how 2.1's result is read.** Locate ignition relay `-13RB1` on the GCU relay board. With the battery disconnected and the relay coil unpowered, measure continuity across its **load contacts** (not the coil).

Why this matters and why 2.1 is ambiguous without it: task 2.1 reads voltage with the ignition switch off, assuming that is the relay's de-energised state. If the relay turns out to be normally-closed, "de-energised" means contacts **made** and power still present — so a battery-voltage reading in 2.1 would mean the test state was wrong, not that the relay only signals. Those two have opposite conclusions and the same reading. This measurement separates them. **Inferred concern, not sourced — raised because the relay's drive logic is not documented anywhere we have.**

Continuity across load contacts, coil unpowered	Meaning for task 2.1
Open (normally-open, energise to run)	Expected case. 2.1's table reads as written.
Closed (normally-closed, energise to cut)	2.1's table is **inverted**. Battery voltage with the switch off is then normal, and U12 must instead be tested by energising the relay and re-measuring. Note it and flag it.

Group 2 — Ignition switch tests (battery connected, engine OFF, not cranking)

Reconnect the battery for this group only. Key/master switch is the only thing that moves. **Do not crank or start.**

- [ ] **2.1 — U12, the priority item. Complete task 1.4 first.** Ignition/master switch **OFF**. This is assumed to be ignition relay `-13RB1`'s de-energised state — **task 1.4 confirms that assumption, and if it fails, the table below inverts.** You do not need to pull or force the relay. If you can see or hear the relay (GCU relay board), confirm it looks/sounds unenergised before measuring; photograph its label to confirm 70 A rating.

With the switch OFF, measure voltage (relative to chassis/battery negative) at: - ECU connector pin **21** (battery+ feed) - ECU connector pin **04** (`V_V_BAT_1R`) - ECU connector pin **06** (`V_V_BAT_2R`) - the **fuel metering unit's own pin 2** (its battery-fed pin, via 20 A fuse `8F1` — not ECU pin 88, which is the ECU's PWM output to metering unit pin 1)

Why this is the whole point of the trip: spec §3's safety argument — that a hung ECU can't keep injecting because the GCU physically removes fuel-system power — depends entirely on this. *(spec §3, §8 U12)*

Reading at all four points	Meaning
0 V (no battery voltage present)	Relay **removes power**. §3's second line of defence is real. U12 closed in the safe direction.
Battery voltage present at any of them	Relay **signals only** — power stays up through a hung ECU. §3's relaxation is void: reverts to "standalone MPU trip module mandatory." **Flag this immediately, do not wait for the writeup.**

- [ ] **2.2 — U7: are pins 04/06 sense-only or load feeds?** Switch **ON** (engine still off). Re-measure voltage at pins 04 and 06 (expect battery voltage present now, per 2.1's contrast — that alone doesn't answer U7).
- **Then measure current**, preferably with a clamp meter on the wire (safer — no circuit break). If only a multimeter is available, break the circuit at the fuse holder for 04/06 and measure DC current in series, briefly.
- *Why:* sense lines draw milliamps; something that's actually feeding a load draws amps, and each fuse is rated 10 A. *(brief; spec §8 U7; spec §4 "Power" — both pins are 10 A fused)*

Current	Meaning
mA range	Sense-only — these just feed the battery-voltage ADC divider. Power sheet needs no extra current rating here.
Amps, approaching the 10 A fuse rating under some load state	Load feed — something downstream draws real current through the ECU. Power sheet must size for it.

- [ ] **2.3 — Pin 09, if 1.2 left it open.** Switch ON. Voltage at pin 09 vs. chassis.
- **~0 V** → confirms a ground reference despite the odd wire colour.
- **Switched battery voltage** → it's a real signal wearing a "ground" name — treat as an undocumented discrete/analog input, not a ground. Note whatever else in the harness moves when this changes (if observable) as a clue to its function.

Group 3 — Photography (engine off; battery may be either state, disconnect

preferred while working around the engine bay)

For each: square-on unless noted, fill the frame, flash off / raking light for part numbers (on-axis flash washes out shallow embossing).

- [ ] ~~3.1 DONE 18 Sep~~ — **Engine rating plate** (rocker cover / flywheel housing / block side). Get legible: model string (expect something matching `3R550ETA` — anything else means the engine-identity inference in memo 01 is wrong and should be discarded, not patched), serial number, rated output/speed/frequency, any emissions certification mark. **Closes U11.** *(memo 01, "What would close this")*

- [ ] **3.2 Genset nameplate** (canopy/control panel). **Do not go looking for a string you expect.** This brief originally said to expect `KG4-25WS1`, inferred from memo 01's engine identification — and that identification turned out to be wrong when the engine plate was read. The genset model inference came from the same chain and is equally unsupported. Photograph whatever is actually on the plate, including the serial number. *(memo 01, and its correction)*
- [ ] **3.3 ECU case label.** Flat, well-lit. Hunting a vendor part number — Bosch EDC pattern `0 281 0xx xxx`, Bosch-reseller pattern `F 01R 0xx xxx`, or any Continental/Delphi/Denso mark. **Now more likely to pay off than when this was written:** the connector is confirmed Bosch (`1 928 405 192` / `194`), so a Bosch part number on the case would be consistent and would identify the OEM ECU outright. Would resolve the still-open OEM-ECU-vendor question. *(memo 03 §4 item 4)*
- [ ] ~~3.4 DONE 18 Sep~~ — **ECU connector — full-face, mating side, connector unplugged.** Determines at a glance: flat rectangular multi-row grid (Bosch/Delphi-style) vs. oval/rounded single-lever shell (AMPSEAL-style) — the top-level branch of the connector shortlist. **Advances U1.** *(memo 03 §4 item 1, ranked candidate table)*
- [ ] ~~3.5 DONE 18 Sep~~ — **Connector retention mechanism**, actuated and disengaged, if it can be operated safely with the connector unplugged. Lever vs. slide latch vs. bolt vs. friction/CPA clip. *(memo 03 §4 item 2)*
- [ ] ~~3.6 DONE 18 Sep~~ — **Raking-light pass over every flat face of the connector housing** — top, both sides, cable boot. Hunting molded/raised part number, date code, or logo: Bosch roundel, Delphi/Aptiv wordmark, TE Connectivity mark. Take several shots at different light angles if the first doesn't show text. **Would settle U1's connector-family ranking outright if a logo turns up.** *(memo 03 §4 item 3)*
- [ ] ~~3.7 DONE 18 Sep~~ — **Connector still mated, wide shot** — backshell, cable exit angle, strain relief. Feeds the adapter-harness design even without a family ID. *(memo 03 §4 item 5)*
- [ ] ~~3.8 DONE 18 Sep~~ — **Pin-count/contact-size check, connector unplugged.** Confirm four rows of small pins plus two 2-pin corner blocks (near pins 1 and 5). **Check specifically whether the corner-block contacts (pins 1/2/5/6) are visibly larger/wider than the small-pin grid.** This is the photographic half of the 1.1 pin 01/02 question — larger contacts support "these are power," normal-sized contacts support "these are ordinary switch returns after all." *(memo 03 §4 item 6; spec §4 note)*
- [ ] **3.9 Boost P&T sensor body** — part number, if legible. Backs up 1.3's electrical answer with a datasheet-checkable part.

- [ ] **3.10 Injector body** — part number, if accessible without disturbing the fuel system. Corroborates the Bosch-family injector inference already used in memo 03/04.
- [ ] **3.11 Intake throttle — 6-pin connector and cable run (closes U8).** Photograph the connector, then follow the cable by hand through the loom.
  - **Terminates at the ECU's 94-pin connector** → in scope. Photograph which physical position on the ECU housing it plugs into (row/corner), so it can be matched to a pin number once U1 is resolved.
  - **Joins the XC11/XC1 panel loom instead** (same path as fuel level/WIF, per spec §2.3) → out of scope for this ECU; a throttle driver sheet is not needed.
- [ ] **3.12 Catalyst temperature sensor — 2-pin connector and harness run (closes U9).** Same method as 3.11: photograph, then trace.
  - **Terminates at ECU** → in scope, an EGT-style front-end is needed.
  - **Joins the panel/GCU loom** → out of scope, same treatment as fuel level/WIF.
- [ ] **3.13 KG640C terminal strip (J1–J8), photographed with terminal labels legible.** Reference material for the MPU-input question (U3) and future GCU interface work; not expected to close anything by itself this visit.
- [ ] **3.14 Wire-colour spot check at each analog sensor connector (boost, coolant, EGR, oil pressure, rail pressure) — flag toward U10.** SENT and CAN H share the same yellow colour code in this harness's legend *(pinout, "Wire colour classes")*, so this can't be resolved by colour alone. Photograph any **single yellow wire that is not paired with a green wire** (a CAN pair) and is not part of a 3-wire AC bundle — that's the SENT candidate. **This will likely only narrow it down, not close it outright** — say so in the notes rather than guessing a pin.

Group 4 — Separate trip: CAN bus log (closes U4 in practice, not from a manual)

Needs a USB-CAN adapter and a laptop. Do this as its own visit unless you're already bringing the hardware.

- [ ] **4.1** With the OEM ECU still fitted and the harness intact, tap CAN channel 1 (pins **55/77**, H/L) and channel 2 (pins **87/86**, H/L) via breakout or fly leads. *(pinout, "CAN — two channels"; spec §4)*
- [ ] **4.2** Log traffic with the ignition on (ECU and GCU powered). Identify PGNs present, source addresses, and update rates.
  - Basic identification/status broadcasts should appear at key-on. **Engine-state PGNs (speed, load) may not appear, or may read zero/stale, until the engine is actually running** — if you need those, that's the "separate

supervised activity" from the safety preamble, not a solo add-on to this trip.

- [ ] **4.3** Compare the captured PGN set against whatever the KG640C is found to expect (from the log itself, and cross-check against the KG640 manual if it surfaces). This is the practical closure the brief couldn't get from documents — **the log is the source, not a spec.** *(spec §8 U4)*

Group 5 — Second visit, added 21 September 2026

Why there is a Group 5. Groups 1–4 were written from documents. These tasks come out of **drawing the schematics**, which is a different kind of question: not "what is this signal" but "where does this wire actually go, because the board cannot be built without it." Two of them block the board outright.

Priority order is the order below. **5.1 is the blocker** — nothing about the board's ground plane, its connector footprint or its power path can be finished without it.

- [ ] **5.1 — WHERE IS THE POWER GROUND? Answered 8 Nov 2026 by Bosch's terminal diagram: pins 01/02 (`G_G_BAT1/2`).** A continuity check from 01/02 to battery negative still confirms the harness. The original task follows. Battery disconnected, connector **unmated**. Working on the **harness side** plug, probe every populated cavity for continuity to **chassis ground**. Write down every cavity number that beeps, and its resistance.

Why: the ECU's 94-way connector has **no identified battery-negative pin.** `GP3.8703.C4.pdf` shows pin 21 as the battery **positive** feed and shows no negative anywhere — which is not a contradiction, because that document says outright it omits "power grounds, unused pins and internal-only pins." But it means the net that every circuit on the new board returns to, and that the injector low sides put **18 A** into, currently has nowhere to land. It cannot be guessed: a wrong ground pin is not a rework, it is a harness that has to be remade.

Do task 1.1 first if it has not been done — pins 01/02 (`G_G_B AT1` / `AT2`) are the leading hypothesis and 1.1 tests them directly. 5.1 is the wider sweep that catches the answer if 1.1 comes back open.

What you find	What it means
Two or more cavities at well under 1 Ω to chassis	Those are the power grounds. Record every one — an 18 A injector stage does not return through a single contact, so expect more than one and record them all, not just the first. **This closes the blocker.**
Exactly one	Record it, and say so explicitly in the notes. One 18 A return through one contact is a design constraint worth knowing about early, not a measurement error.
A few ohms, not near zero	That is a return **through a device** (a sensor ground, a lamp, a coil), not a power ground. Note it separately — do not count it.

What you find	What it means
Nothing beeps anywhere	The ECU is grounded through its **case/mounting**, not through the connector. Photograph the mounting points and any ground strap, and measure continuity from a mounting bolt to chassis. That answer changes the mechanical design, so it matters as much as the other outcomes.

- **5.2 — Cavity census: which are populated, which are empty.** Same session as 5.1, connector unmated. Photograph the mating face square-on with good light, then list: which cavity numbers have a **wire crimped in**, which have a **contact but no wire**, and which are **empty**.

No longer needed, 8 Nov 2026: the trip loop is removed (no Bosch pin). The trip module opens the main relay's coil feed instead (spec §3 item 7).

Why: the wiring diagram accounts for 44 of 94 pins. The new design needs one more signal than the OEM harness carries — `TRIP_LOOP`, the supervised trip-module loop, which is not on the OEM diagram at all because it is new. **The question is not just "is there a spare pin" but "is there a spare WIRE":** an empty cavity means a contact and a conductor have to be added to a loom that is already on the machine.

What you find	What it means
Spare cavities with wires already run	Best case. Trace where one goes (same method as 3.11) — if it reaches the trip module or the panel, `TRIP_LOOP` has a home with no harness work.
Contacts fitted, no wires	Adding a conductor is possible without re-terminating. Note the contact type so the crimp can be matched.
All 50 empty	`TRIP_LOOP` needs a new contact **and** a new wire. Record the cavity you would use, so the board and the harness rework agree.

- **5.3 — Connector contact geometry (for the board-side footprint). Now load-bearing (22 Sep 2026):** the owner decided the ECU carries this OEM connector directly, with no adapter harness, so the board's own header is the mating half of this plug. Connector unmated. Measure with calipers, and photograph with a scale in frame: **contact pitch** within a row, **row-to-row spacing**, and the **overall footprint** of the housing. If a contact can be seen down a cavity, photograph it.

Why: the connector is identified — **Bosch `1 928 405 192` / `1 928 405 194`, code C** — but neither number returns a catalogue hit, so no land pattern can be downloaded. The board-side footprint has to come either from a Bosch distributor quoting those numbers, or from measurement. Measurement will not be precise enough to build from on its own; what it will do is **confirm which family drawing is the right one** when a distributor sends one.

Also re-photograph the `code C` marking and the `2.7` beside the logo. Code C is mechanical keying — a differently-coded housing will not mate, and that is the kind of detail that is invisible until a part arrives

and will not fit.

- [ ] **5.4 — Crank sensor output while cranking, COLD. SUPERVISED ACTIVITY, NOT A SOLO CHECKLIST ITEM** — the engine cranks, and a diesel with fuel available will start. Treat this like Group 4's running tests: someone qualified to operate the set, with the normal start procedure, not a meter held in one hand.

Measure across ECU connector pins **52** and **74** (crank VR high/low) on the harness side, during the first seconds of cranking, engine cold. **Oscilloscope strongly preferred** — a multimeter on AC volts reads an RMS approximation of a non-sinusoidal waveform and will under-read the peak, which is the number that matters. If a meter is all there is, record that it was a meter.

Why: the crank conditioner is a zero-cross comparator with a **±198 mV** hysteresis window, and a signal that never leaves that window cannot flip it at all, whatever its amplitude margin. The design was sized against a **2 V estimate** at cranking and the simulation says it still works down to about **0.25 V** — roughly a decade of headroom. That headroom has never been checked against this sensor.

Peak amplitude, cranking, cold	What it means
Above 1 V	Comfortable. Record the number and move on.
0.25 V to 1 V	The design still works, with less margin than assumed. Record the number — it becomes the real design figure and replaces the estimate.
Below 0.25 V	The hysteresis window has to shrink, which costs noise immunity, or the front end needs gain ahead of the comparator. **This is the outcome that changes a drawn sheet**, so flag it immediately rather than filing it.

- [ ] **5.5 — Injector part number (was 3.10 — now load-bearing, not corroboration).** Same task, higher priority. Photograph the injector body markings.

Why it was upgraded: the boost rail's **100 V target** and the **18 A peak threshold** come from a family-wide envelope in memo 04, not from this injector. The whole injector stage — the 47 µF reservoir, the 150 V part classes, the recirculation diodes — is sized off those two numbers. A part number turns an envelope into a datasheet. **Closes U5.**

- [ ] **5.6 — Alternator rectifier / regulator identification (closes U16).** Photograph the charging alternator's nameplate, and any separate rectifier or regulator module, with part numbers legible under raking light.

Why: whether the alternator is **suppressed** decides which ISO 7637-2 load-dump pulse the input has to survive — the suppressed 5b case the design is currently built around, or the unsuppressed 5a case at 65–87 V. The input-protection topology and `load_dump`'s failure mode both hang

off this. If no part number is readable, the alternative is a scope during a live disconnect under load, which is a supervised activity and a separate trip.

- [ ] **5.7 — Who commands the pre-heat / post-heat? (closes U14).** Trace the heater feed by hand, the same method as 3.11.

Why: the DSE4522's fitted configuration enables pre-heat and post-heat at 50 °C, and this design has **no heater output at all.** It was already reported that the heaters are **battery-fed** — that answers who supplies them, not who **commands** them.

Where the command comes from	What it means
The DSE4522 / panel	Out of scope. Nothing is added to the board.
The ECU's 94-way connector	**A driver channel has to be added** — which changes the board's area and its output count. Flag this one immediately: it is one of only two open items that can make the board bigger.

Still open from the earlier groups, and cheap

These were written for the first visit and have not been done. Two of them are a single measurement each.

- [ ] **1.3** — one resistance reading across the boost P&T sensor, cold. **Closes U2** either way; there is no third case.
- [ ] **3.11** — follow the intake throttle's 6-pin cable by hand. **Closes U8**, and like 5.7 it is one of the two items that can make the board bigger.

Group 6 — Measure the old ECU, added 14 October 2026

Why. The new board goes into the old ECU's die-cast housing (owner's decision, 11 Oct) and plugs into the same harness. So the board's outline, its screw holes, where it touches the housing to shed heat, and the exact header on it must all copy the old unit. The PCB is ready to route now, but **those four things are placeholders, and no board can be ordered until they're measured.**

> **Do not open the only working ECU on a running genset.** Use the spare or dead unit (the one in the 13 Oct photos looked to be off the engine), or wait until ours replaces it. If the unit is sealed with screws under a label or with glue, stop and send photos first.

For every item: a photo **straight on** (not angled), with a **ruler or calipers in the frame** and touching the part, plus the number read off the calipers. If you can only do some items, do 6.3 first: it is the one with no other source.

☐ **6.1 The housing.**

- Inside length, width and depth.
- The connector opening: its position from two edges, and its size.
- The mounting bosses: how many, where (from one corner), and the hole or thread size.
- Anything inside the lid that presses on the board: thermal pads, raised pedestals, gap-filler. Measure their size and position.
- The round vent plug: where it is. The board has an air-pressure sensor that should sit near it.

☐ **6.2 The board.**

- The outline, with every corner and cut-out.
- Thickness, at the edge, with calipers.
- The mounting holes: position and diameter.
- **Where it touches the housing.** These are usually bare copper or a grey thermal pad. Photograph both sides.
- The tallest parts and their heights.
- Are there parts on **both** sides? (This decides whether ours can fit.)

☐ **6.3 The header on the board.** This closes the connector footprint. Count and measure:

- How many pins in each row. On the outside, the 13 Oct photo showed 6 large + 4 rows of 22.
- The pitch: from the **centre of pin 1 to the centre of pin 11** (ten gaps), divided by 10. That's more accurate than one gap.
- The distance between rows, and the gap at the central rib.
- Which pins are the large ones, and where they sit relative to the small ones.
- Pin thickness, if a pin can be reached with calipers.
- How the header is held: pegs, screws or clips. Measure their position.
- Its distance from the board edge.
- Any marking on the header body (a TE or Bosch number), photographed sharp.

☐ **6.4 Only if the unit is open and safe to probe** (battery disconnected, unit off the engine):

- **Continuity, large pins to board ground.** With a meter on continuity, touch each of the six large pins in turn and the board's ground (a mounting hole's copper ring or a big ground area). Write down which pins beep. **This settles which pins are the battery ground (item 5.1).**
- **Which cavities have a contact fitted.** Look into the connector face and note every empty position. Read along with item 5.2, this narrows the trip-loop pin.

Group 7 — The overspeed trip and the injector feed, added 14 October 2026

Why. The old ECU's recording shows the fuel metering valve is **normally open**: cut its power and it lets *more* fuel through. The plan had been to fit an overspeed trip module that stops the engine by cutting the valve's power. **That would not stop a runaway.** The trip now has to cut the **injector supply**, which is ECU **pin 04** (spec §3 item 6). These items gather what the wiring needs. **Nothing is rewired in this group.**

> Engine **off**, ignition **off**, for everything here. Items 7.2 and 7.3 are looking and photographing only. Don't pull fuses or cut wires.

- [ ] **7.1 Confirm the valve.** This is the most important item. Photograph the label on the **fuel metering valve**: the small 2-pin electrical unit on the high-pressure pump, with its connector going to ECU pin 88. We need the number on it (Bosch numbers start `0 928 400 …`). That number settles "normally open" for certain.
- [ ] **7.2 Find pin 04's fuse.** Follow the wire from ECU pin 04 back towards the battery. Find its **10 A fuse**, then photograph:
  - the fuse box with its label
  - the fuse's position
  - the wire colour on both sides of the fuse
- [ ] **7.3 Is there a trip module already?** Look for any small speed-switch box with its own magnetic pickup on the flywheel housing. Brand names to look for: GAC, Murphy. If there is one, photograph:
  - the box and its label
  - its terminals
  - where its output wires go

If there isn't one, say so. That is the expected answer. - [ ] **7.4 Room for a relay.** Near the pin 04 fuse, is there space and a mounting point for one standard automotive relay (about 30 × 30 × 25 mm) with its socket? Take a photo with a ruler.

What the finished wiring will look like (for reference only, not to be done now):

```
battery + ── 10 A loom fuse ── [run relay: normally-open contact, ≥ 30 A] ── ECU pin 04
                                        │ coil, 12 V
ignition-switched + ── trip module "healthy" contact ── coil ── ground
```

The relay is held **closed while the engine may run**. Any of these stops the injectors: - an overspeed trip - the module losing power - a broken coil wire

Pin 21 (ECU logic), pin 06 (EGR) and the metering valve's own feed are **not** touched.

What this closes

#	Unknown	Closed by
U1	Connector family	3.4, 3.5, 3.6, 3.7, 3.8 (photographic — narrows the ranked shortlist in memo 03, may not fully settle it without a logo hit)
U2	Boost air-temp topology	1.3 — closes outright
U4	KG640C PGN set	Group 4 — practical answer via live log
U6	Pin 09 function	1.2, 2.3
U7	Pins 04/06 sense vs. load	2.2
U8	Intake throttle on ECU?	3.11
U9	Catalyst sensor routing	3.12
U10	Which signal is SENT	3.14 — narrows, may not fully close
U11	Engine identity	3.1 — closes outright if plate matches
U12	Ignition relay: power-cut or signal-only	2.1 — **the safety-gate item; closes §3's open question either way**
—	Pin 01/02 ground hypothesis	1.1, 3.8
U5	Injector part number	5.5 — turns memo 04's family envelope into a datasheet
U14	Who commands the heaters	5.7 — **can add a driver channel to the board**
U16	Alternator suppressed or not	5.6 — decides which load-dump pulse the input must survive
—	**The power ground's pin**	**5.1 — blocks the board outright**
—	A pin *and a wire* for `TRIP_LOOP`	5.2
—	Board-side connector footprint	5.3 — confirms which family drawing is right
—	Crank VR amplitude at cranking	5.4 — supervised; the only Group 5 item that can change a drawn sheet
—	**Board outline, holes, contact areas**	**6.1, 6.2 — placeholders on the PCB until measured**
—	**J1001 header geometry**	**6.3 — the footprint; no board is ordered without it**
—	Battery-ground pins (again)	6.4 — continuity on the open unit
—	**Metering valve part number (normally open?)**	**7.1 — decides the trip wiring; firmware already assumes normally open**
—	Pin 04 feed and its fuse; existing trip module	7.2, 7.3, 7.4 — what the trip wiring needs

Ten register entries touched, two (U2, U11) closeable outright on the spot, one (U12) resolves the live safety-argument branch either way, the rest narrowed with a clear next step recorded regardless of outcome.

Group 5 adds four more register entries and three questions that are not register entries at all — they are things the schematics need and no document in this project contains. Of those, **5.1 is the one that stops the board**, and **5.7 and 3.11 are the only two that can make the board bigger**, so they should be answered before layout starts rather than after.

Research Memo 09: Resistive Temperature Front-Ends — DSE and SEDEMAC Prior Art

> **UPDATED 18 Sep 2026 — DSE is not just prior art here, it is the fitted controller**
>
> This memo studied Deep Sea Electronics and SEDEMAC as **prior art** — other vendors' answers to a problem we share. It opened by noting that both are genset-controller class devices, "the same class of device as the KG640C we talk to", and was careful that findings from them might not transfer.
>
> The controller on this machine has since been photographed. It is a **Deep Sea Electronics DSE4522 MKII AMF**. The DSE practice described below is therefore not analogous guidance from a comparable product — **it is the behaviour of the device this ECU actually has to work alongside.**
>
> That strengthens §3.1's ground-offset finding considerably. DSE's own manual for the fitted module's siblings (`057-260`) is emphatic about it: *"It is VERY important that terminal 10 (sensor common) is connected to an earth point on the ENGINE BLOCK, not within the control panel … This connection MUST NOT be used to provide an earth connection for other terminals or devices."*The differential front-end recommendation in §4 has been adopted into spec §4.
>
> §3.4's scaling caveat also held up under its own test: `sim/blocks/ntc_frontend.cir` found that copying DSE's constant-current topology onto a 670:1 automotive NTC runs out of rail compliance at the cold end. The divider survived.
>
> The SEDEMAC material remains what it was — prior art, snippet-sourced, and still not verified against a retrieved manual.

Task: U2 supporting research — how two established genset-controller vendors build resistive temperature sensor inputs. Consumes memo 02 (KG640C is the same class of device). Feeds the sheet 3b analog front-end and a re-examination of the already-simulated `sensor_ratiometric` block. **Date:** 2026-09-14 **Author:** Follow-up research, phase 1 exit

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Verdict (read this first)

Neither vendor uses a plain pull-up divider to a fixed rail — the topology this project had planned for the U2 NTC branch.

- **Deep Sea Electronics** forces a **fixed current** through the sender and measures the voltage across it with a **differential** input, specified for **±2 V of common-mode** (ground offset) rejection. *(Confirmed — read directly from two DSE operator manuals.)*
- **SEDEMAC** uses **ratiometric** resistive sensing with a dedicated **sensor common point** terminal bonded to the engine body. *(Secondary — see the sourcing caveat in §2; the manual itself could not be retrieved.)*

This does not close U2. Both are genset *controllers* — the same class of device as the KG640C we talk to, not the common-rail ECU we are replacing. They read VDO/Datcon-style senders bolted into the block; U2 concerns a charge-air temperature element inside a 4-pin P&T combo on a common-rail engine. This is prior art on how to build the resistive branch **well**, not evidence about which branch this engine is on. Brief task 1.3's resistance measurement remains the only thing that settles U2.

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1. Deep Sea Electronics — constant current, differential

Two generations of the same controller, both read directly from the operator manuals via text extraction. **Confirmed.**

DSE8610, Operator Manual Issue 5, document 057-115

Channel	Excitation current	Full scale	Over range / fail
Oil pressure	15 mA	240 Ω	270 Ω
Coolant temperature	10 mA	480 Ω	540 Ω
Flexible sensor (×2)	10 mA	480 Ω	540 Ω

DSE8610 MKII, document 057-254 Issue 3

Unified every channel on **15 mA, 480 Ω full scale, 600 Ω over range/fail**, and made all four inputs selectable between resistive, 0–10 V and 4–20 mA modes.

The wording that matters

Identical across every channel of both generations:

> **Measurement type:** Resistance measurement by measuring voltage across sensor with a fixed current applied **Arrangement:** Differential resistance measurement input **Max common mode voltage:** ±2 V

> **Accuracy:** ±2 % of full scale resistance, excluding sensor error

Sources (both downloaded and text-extracted, not read from search snippets): - https://www.reactpower.com/wp-content/uploads/2020/07/DSE8610-Control-Panel.pdf - http://www.davidsonsalesshop.com/catalog/files/Products/Deep%20Sea/DSE8610MKII%20Manual.pdf

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2. SEDEMAC — ratiometric, with a dedicated sensor common point

Sourcing caveat, stated plainly: every attempt to retrieve a SEDEMAC manual failed. ManualsLib, Manualzz and Scribd all returned HTTP 403 to direct fetch, and one aggregator link redirected to an unrelated cryptocurrency domain and was not followed. The figures below come from **search-result snippets**, consistent across two independently phrased queries, and are recorded as **secondary, unverified against the document itself.** They should not be used as design numbers without retrieving the manual.

GC1200 series, analog resistive sensor inputs on terminals 24 / 25 / 26 (engine temperature, fuel level, lube oil pressure):

Property	Value
Type	Ratiometric sensing
Range	10 Ω to 1000 Ω
Open-circuit detection	Above 5.5 kΩ
Accuracy	±2 % of FSD up to 1000 Ω

Separately, terminal 41 is a **sensor common point (SCP)**, specified to bond directly to an electrically sound point on the engine body, serving as the shared reference for all analog senders.

Sources (snippet-level only): - https://www.scribd.com/document/558526157/sedmac-controller-manual-1202 - https://www.manualslib.com/manual/1474533/Sedemac-Gc1100.html

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3. What transfers to this design

3.1 The differential input is the load-bearing finding

DSE specifies ±2 V common-mode rejection on every sender channel, and SEDEMAC solves the same problem from the other direction by insisting on a dedicated sensor common point bonded to the engine body. Both are addressing **ground offset between the sensor's return and the controller's reference.**

This project has that exact problem, and it is visible in the OEM wiring: **pin 34 is a shared sensor ground, spliced between the boost P&T sensor and the coolant sensor** (per `refs/ecu-pinout-extracted.md`, splice V3). A shared return carries the other sensor's current, developing an offset across the harness resistance. A differential input rejects that offset; a single-ended input reads it as signal.

`sim/blocks/sensor_ratiometric.cir` is single-ended. It passes its checks, but those checks never modelled a ground offset, so passing does not mean the topology is right for a shared return. This is a gap in the *check set*, not a failure of the simulation — and it is exactly the kind of thing the simulation programme exists to catch. It should be re-examined before sheet 3b is drawn.

Worth noting the OEM's own hierarchy here: the rail-pressure sensor gets a **dedicated** ground (pin 08, not spliced), while boost and coolant share pin 34. Whoever designed that harness ranked the channels by how much ground offset each could tolerate. That ranking is information, and it agrees with SEDEMAC's SCP discipline.

3.2 Fault detection belongs in the range, not bolted on

Both vendors reserve a band above full scale as the fault region — DSE 480 Ω → 600 Ω, SEDEMAC anything above 5.5 kΩ. An open or shorted sender lands somewhere that is unambiguously **not a temperature**, rather than pinning to a rail where it could be read as a plausible cold or hot value. Front-end component values should be chosen so both failure modes land outside the signal band, and firmware should treat that band as a distinct state.

3.3 Constant current linearises the readout

With a pull-up divider, the measured voltage is `Vref · R / (R + Rpu)` — non-linear in R, so resolution varies across the range and the pull-up's tolerance enters the result directly. With a forced current it is `I · R`, linear, with the current source's accuracy as the only gain error. That is why DSE does it, and it costs a current source per channel.

Ratiometric sensing (SEDEMAC's approach, and the one already used on this project's pressure channels) does not linearise, but it does cancel supply drift by making the ADC reference track the excitation. The two approaches solve different halves of the problem.

3.4 The current values do NOT transfer — a scaling caveat

These controllers expect senders in the **tens to hundreds of ohms** (VDO/Datcon range). This project's own inspection brief predicts **"a few kΩ"** for the NTC case at task 1.3. Forcing DSE's 15 mA through a few kΩ would demand tens of volts and is impossible on a 5 V sensor rail.

If the constant-current topology is adopted, excitation must be scaled to roughly the **100 µA class** for a kΩ-range NTC. Copying DSE's numbers directly would be a straightforward and expensive mistake. **Inferred** from the resistance ranges involved, not sourced.

---

4. Recommendation

1. **Upgrade the U2 mitigation.** The register's original hedge was a do-not-populate pull-up so either sensor type would work. That hedge assumed the NTC branch is a divider. Design the front-end so the branch can be either a scaled current source or a divider, and the choice remains a populate option after task 1.3 answers U2.
2. **Add a ground-offset check to** `sensor_ratiometric` before drawing sheet 3b, and reconsider a differential front-end for the two channels that share pin 34. The block's current PASS verdict does not cover this case.
3. **Retrieve a SEDEMAC manual properly** before using any figure in §2 as a design number. Everything there is snippet-level.

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5. Negative results, recorded

- No DSE application note on wiring resistive sensors was retrieved, though search results reference one existing on the DSE site. Its guidance on common wiring mistakes would likely be relevant and was not obtained.
- No SEDEMAC document was retrieved in full. Three separate hosts refused.
- Neither vendor publishes a schematic of the input stage — only its specification. The internal topology in §1 is read from the specification's own wording ("fixed current applied", "differential"), which is explicit enough to be treated as confirmed, but no circuit diagram was seen.

Research Memo: Injector Turn-Off / Recirculation Circuit

Task: Close the gap `injector_boost.cir`'s RESULT NOTE names — turn-off is not modelled, no clamp/recirculation path is drawn anywhere in the design, and interrupting 18 A in 200 µH releases ~32 mJ per event with nowhere defined for it to go. This memo establishes what real automotive injector drivers do about that, before a SPICE block gets built. **Date:** 2026-09-18 **Scope:** Read-only research. No repo files modified, nothing committed.

Verdict (read this first)

Production peak-and-hold injector drivers do **not** use a plain freewheel diode for turn-off — that topology is real (it's the textbook baseline and the cheapest option) but it is also the slowest, and slow turn-off is exactly what this project's boost rail was built to avoid on the *opening* side. The turn-off side has the same problem in reverse. Two topologies dominate production practice for this injector class: **active/Zener clamping** (fast, dissipative, tunable voltage) and **recirculation of the coil's turn-off energy back into the boost reservoir capacitor** (fast, largely non-dissipative, and confirmed as real automotive practice going back to at least a 1988 Magneti Marelli diesel-injector patent, not a textbook idealization). For this design, **recirculation to the existing 100 V / 47 µF boost rail is the recommended topology**, clamped by a diode rated for the boost-rail voltage: it collapses 18 A in 200 µH in about **36 µs** (physics below), and the ~32 mJ recovered per event is the same order of magnitude as the ~7.3 V peak-phase droop `boost_converter.cir` already calculated — this is not a rounding error, it changes that block's charge balance.

Sourcing note, stated as plainly as memo 09 states its own: this pass retrieved one strong, directly on-topic primary source in full (Nexperia AN50003, an application note that surveys exactly the four candidate topologies the task asked about, with a quantified loss/performance comparison table) and one confirming patent (Marelli Autronica, 1988, Google Patents full text) that is diesel-injector-specific and explicitly diesel-electro-injector. Several other leads that looked promising by title (ST L9781, NXP AN4849, the ISO 7637-2 standard text itself) **could not be retrieved** — repeated 403s, timeouts, or a 404 — and are recorded as negative results below rather than filled in from memory or a search snippet dressed up as a document. TI DRV8701 was checked and ruled

out: it is a brushed-DC-motor H-bridge driver, not an injector-class part, despite surface-level keyword overlap ("solenoid control" appears in its marketing copy).

---

1. What actually absorbs the coil energy at turn-off

1.1 Nexperia AN50003, "Driving solenoids in automotive applications" — CONFIRMED, retrieved and read in full

https://www.digikey.com/Site/Global/Layouts/DownloadPdf.ashx?pdfUrl=7547AD78C870436881732FA36D2AAAFA (Nexperia B.V., Rev 1.0, 4 December 2020 — an automotive-solenoid-driver application note, explicitly citing "peak-and-hold" fuel injection current profiles as its motivating example, §3, Fig. 3)

This is the single best-matched source found in this pass: it surveys **exactly** the four candidate topologies named in the task, with SPICE-based loss and performance numbers (their reference case: 12 V battery, 5 mH inductor, 3 A peak, 1.2 A hold — a much smaller solenoid than this injector, but the physics and the relative comparison transfer directly). Quoting Table 1 and Table 2 (§5) in full:

Topology	Total energy loss (mJ)	Cost	Speed	Efficiency	Reliability
Free-wheeling diode	17.0	Low	Low	High	Long term
MOSFET avalanche	20.7	Low	Medium	Low	Long term
Active clamp (Zener, drain-to-gate)	20.7	Low	Medium	Low	**Questionable**
Boost converter (energy recovery)	26.0	High	High	High	Long term

Their own text on the "Questionable" reliability rating for active clamp (§4.3), quoted directly: *"During active clamp there are high energy charge carriers generated in proximity of the MOSFET's gate oxide. These carriers might be injected into the oxide and cause damage. Over many active clamp cycles the gate oxide can wear out and cause parametric shift and ultimately device failure. Currently it is not recommended to use MOSFETs in repetitive active clamp."**This directly rules out the naive Zener-across-the-FET version of "active clamp" for a component that will do this thousands of times a minute for the life of the genset.**

On the boost/recovery topology (§4.4), quoted: *"During the Peak and Hold Phases the nominal battery voltage is switched, as in the previous cases. This allows for fast actuation of the solenoid, but also the energy from the solenoid is regenerated into the DC link capacitor of the boost converter... Although the losses appear to be higher in the Boost topology, the recuperation of the energy means that its efficiency is on par with the free-wheeling topology, despite using higher R_DSon*

components." (§5) — i.e. the boost/recovery topology has the **highest absolute component losses in their table but the best net efficiency**, because most of the 32 mJ-scale energy is returned to the rail rather than burned.

On the free-wheeling diode being the slowest (§4.1), quoted: *"Compared to the other driver topologies, the free-wheeling driver is simple, has a low component count, but it is the slowest due to the inductor voltage being approximately equal to the battery voltage."*

1.2 The industry-standard schematic vocabulary — CONFIRMED across two independent IC datasheets

- **Infineon TLE8242-2** datasheet, retrieved in full (rev 1.0, Feb 2010): https://www.infineon.com/dgdl/Infineon-TLE8242_2-DS-v01_00-en.pdf — this is a **low-side constant-current pre-driver IC**, explicitly stated to be for "Variable Force Solenoids (e.g. automotive transmission solenoids)" and other constant-current solenoids (EGR, idle air control), not a diesel common-rail injector driver — **important scale mismatch**: its current range is 0–1.2 A and its absolute max POSx/NEGx rating is 50 V, roughly an order of magnitude below this injector's 18 A / 100 V. It is cited here only for what it confirms about topology convention, not for numbers. §1.3.1 and §1.3.2, quoted directly: *"Note: An external flyback clamp is required in [Direct PWM] configuration otherwise the IC may be damaged"* and *"Note: An external recirculation diode is required in [constant current mode] configuration otherwise the IC may be damaged."* **The IC itself never integrates the clamp/recirculation element — it is always an external, designer-chosen part**, which matches what this project needs to design rather than buy.
- **ST L9781 — SNIPPET only, not retrieved.** Two direct fetch attempts (st.com PDF, alldatasheet mirror) both failed — one timed out, one returned HTTP 403. What search-result snippets say (unverified against the document itself): the L9781 is a multi-valve pre-driver with an integrated boost step-up converter reported as "V-boost up to 80 V," and its recirculation/clamp behaviour is described in a snippet as *"Fast recirculation happens through clamp activation, with S2 open and clamp on S1 activated"* — language consistent with recirculation back toward the boost-side switch/rail rather than a simple flyback-to-ground diode. This is recorded as a data point corroborating the recirculation-to-boost topology, but it is explicitly **not** elevated to CONFIRMED status because the datasheet itself was never read.

1.3 Energy recovery in real diesel-injector drive circuits — CONFIRMED via patent full text (Google Patents), not a datasheet

US Patent 4,862,866, "Circuit for the piloting of inductive loads, particularly for operating the electro-injectors of a diesel-cycle internal combustion engine," assignee **Marelli Autronica S.p.A.** (Magneti

Marelli's electronics arm — a genuine Tier-1 diesel-ECU supplier, a Bosch competitor), filed 22 August 1988. https://patents.google.com/patent/US4862866A/en

Quoted directly from the retrieved claims/description text: the circuit includes *"energy-recovery circuit means controlled by the unit and adapted to enable part of the reactive energy stored in the load to be recycled towards the supply VB each time a load is deactivated."* This is explicit, diesel-electro-injector-specific, decades-old confirmation that **energy recovery at injector turn-off is real production-line thinking, not a textbook flourish invented for this memo.** No numeric clamp voltage or recovered-energy figure is given in the retrieved text — the patent describes the mechanism (an OR-diode network into a junction with a recovery switch SW4 back to the supply), not performance numbers.

A second patent, EP0622536A2 (Chrysler, priority 1993), was also retrieved via Google Patents and is recorded for completeness but is **not** treated as strong evidence for this design: it is a **gasoline two-stroke direct-injection** patent, not diesel common-rail, and its numbers (flyback clamped to "15–20× battery potential," i.e. ~180–240 V on a 12 V system, against an 8–10× battery boost voltage of ~96–120 V) describe a self-boosted clamp stage deliberately driven *above* the boost rail, a different design choice than recirculating into the existing boost capacitor. Noted as related prior art, not as a number to adopt.

1.4 Bosch EDC17/EDC7 architecture — negative result

No public technical description of Bosch's own injector-driver ASIC internals (clamp topology, voltage, or recirculation) was found. Search results for EDC17 return tuning/remapping community material (map structures, torque-based control philosophy) with no circuit-level disclosure. This is consistent with memo 04's own finding that OEM injector-driver silicon and calibration data are not publicly documented — recorded here as a negative result, not filled in by inference from the adjacent (and more open) NXP/Nexperia/Infineon material.

1.5 TI DRV8701-class — ruled out, not applicable

The DRV8701 datasheet (https://www.ti.com/lit/ds/symlink/drv8701.pdf) is a single H-bridge **gate driver for a 5.9–45 V, bidirectional brushed-DC motor**. It is not a peak-and-hold constant-current injector driver and has no boost-voltage handling in this part's role. It surfaced in search results only because "solenoid control" appears in TI's generic marketing description for the DRV87xx/DRV870x family. **Recorded as a checked-and- rejected candidate, not a silent omission.**

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2. Clamp voltage vs turn-off time — the actual numbers

2.1 The physics, cross-checked against a retrieved source

`di/dt = V_clamp / L` while the coil discharges above zero, so `Δt ≈ L·ΔI / V_clamp`. This is not a new derivation — it is exactly the relation Nexperia's AN50003 uses (§4.1, Eq. 1: `Δt = L·ΔI/V`), and their own worked numbers cross-check cleanly: their free-wheeling case (12 V, 5 mH, 1.2 A hold current) gives `Δt = 5e-3 × 1.2 / 12 = 0.5 ms`, matching their stated "close to 0.5 ms" (§4.1). Their avalanche case, clamped at a measured 68 V (Fig. 7, read directly off their MOSFET avalanche plot) against the same 5 mH/1.2 A step, gives `Δt = 5e-3 × 1.2 / 68 ≈ 88 µs`, and their text states the reduction is "close to... a five-fold reduction" — 12/68 ≈ 0.176, i.e. a 5.7× voltage ratio, consistent with their "five times faster with five times larger voltage" framing (§4.2). **This cross-check is original arithmetic done for this memo against Nexperia's own published numbers, not itself sourced — marked INFERRED, but it validates that the formula and their reported figures agree with each other.**

2.2 Applied to this injector — INFERRED, derived, not sourced for this specific part

Using this project's own numbers (18 A, 200 µH, from `injector_boost.cir`):

Clamp path	Clamp voltage	Turn-off time (18 A → 0)	Note
Freewheel to battery	~13.5–14 V	**~257–267 µs**	Baseline, slowest
Avalanche/Zener active clamp	68 V (Nexperia's example FET rating)	**~53 µs**	~5× faster, matches their ratio
Recirculate to boost rail	100 V (this project's existing rail)	**~36 µs**	Recommended path, see §3
Dedicated higher clamp above boost	150 V (illustrative)	**~24 µs**	Faster still, needs a part rated well above the 100 V rail

At a 1–3 ms injection duration (memo 04's envelope, corroborated by `injector_boost.cir`'s own framing), even the slowest option (267 µs, battery freewheel) is a smaller fraction of the injection window than the 38 µs turn-on ramp is — but turn-off precision matters disproportionately for **small-quantity pilot injections**, where end-of-delivery timing directly sets fuel quantity. No source in this pass gave a quantified "needle closes N µs after current reaches zero" mechanical lag figure for this injector class — that remains an open item, same as memo 04's Step 5 calibration gap. **This whole table is INFERRED (derived arithmetic), not a number pulled from a source for this specific injector** — the only source-backed clamp-voltage figure found in this pass is Nexperia's own 68 V MOSFET avalanche example, which is a generic automotive part rating, not an injector-specific spec.

2.3 Why higher clamp voltage is not free — confirmed qualitatively, not quantitatively

AN50003 states plainly (§4.2, §4.3) that faster (higher-voltage) turn-off concentrates the *same total energy* into a shorter, hotter pulse, and that this is the mechanism behind their "Questionable" reliability flag on active clamp (gate-oxide stress) and their explicit statement that existing MOSFET datasheets give only "scarce data" on repetitive avalanche ratings (§6). **No source in this pass gave a quantitative die-temperature-rise-per-volt figure** — that would require a specific FET's thermal datasheet and is a detailed-design question, not a topology question. Recorded as an open item rather than estimated.

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3. Energy recovery — real practice or textbook complication

Real practice, with a caveat about how much it matters here.

- Confirmed via the Marelli Autronica 1988 patent (§1.3 above): energy recovery at diesel-injector deactivation is documented production-supplier engineering, not a modern textbook idea being retrofitted onto this project.
- Confirmed via Nexperia AN50003 (§1.1 above): the boost-recovery topology has the best net efficiency of the four surveyed, specifically *because* turn-off energy is regenerated into the DC-link capacitor rather than burned.
- SNIPPET-level (ST L9781, not retrieved in full) is consistent with the same idea in a currently-sold automotive injector pre-driver part.

Does it meaningfully change this project's boost converter load? This is where the sourcing runs out and the memo has to say so: **no source found gives a quantitative "% duty cycle reduction" figure for recovery vs. no recovery on a diesel common-rail boost stage.** What follows is original arithmetic against this project's own `boost_converter.cir` numbers, marked **INFERRED**:

- Coil turn-off energy per event: ~32.4 mJ (`0.5 × 200µH × 18A² = 32.4 mJ`, matching `injector_boost.cir`'s RESULT NOTE figure of "about 32 mJ").
- `boost_converter.cir`'s reservoir is 47 µF at a 100 V setpoint, and its own RESULT NOTE calculates a **7.3 V droop** from the peak-phase charge draw (342 µC / 47 µF).
- If that same 32.4 mJ were returned to the same 47 µF/100 V capacitor (energy balance `ΔE = C·V·ΔV`, linearised around 100 V): `ΔV = ΔE / (C·V) = 32.4e-3 / (47e-6 × 100) ≈ 6.9 V`.
- **That recovered-voltage bump (≈6.9 V) is essentially the same size as the droop the peak-phase draw already causes (7.3 V).** This is not a small correction — on this project's own reservoir sizing, full recovery

would come close to cancelling the very droop `boost_converter.cir` exists to characterize. It would change the charger's required average current (`Bchga`'s 50 mA figure) and could push the rail *above* its 100 V setpoint on back-to-back events if the regulator doesn't shed the excess, which is exactly the "climbed past target, reported 122 V" failure mode that block's own header describes for an unregulated charger.

This is the concrete reason recovery is worth building here, not a default recommendation copied from the literature: it isn't a minor efficiency nicety, it is the same order of magnitude as a number this project has already calculated and cared about.

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4. Thermal — events per second and average dissipation

Assumptions stated plainly, because the task brief itself contains a contradiction worth flagging: the task text refers to "a 4-cylinder at 1500 rpm genset speed" in one clause and then names the actual engine as "3GK550ETA 4SR1, 3-cylinder" in the same sentence. Per memo 04 (confirmed, engine identity task), **this engine is a 3-cylinder** (Kirloskar 3GK550ETA 4SR1, 3-cyl inline, KG4-25WS1 genset). This memo uses **3-cylinder**, not 4, and flags the discrepancy rather than silently picking one.

Derivation (original calculation, INFERRED, following memo 04's own Step 3 method exactly): - At 1500 rpm: one crank revolution = 40 ms; a 4-stroke cycle (720°) = 2 revolutions = **80 ms** (matches memo 04 and `boost_converter.cir`'s own 26.67 ms per-cylinder figure: 80/3 = 26.67 ms). - Assumption: pilot + main + post injection per cylinder per cycle = **3 injection events per cylinder per 720° cycle**. This is stated as an assumption because no source in this pass (or in memo 04) gives a confirmed injection-strategy count for this specific engine/calibration — it is a reasonable assumption for a CPCB IV+-class common-rail genset (memo 04 already assumes pilot-injection capability is a hard requirement for its minimum-pulse-width figure), not a measured fact. - Total events per cycle: 3 cylinders × 3 injections = **9 events / 80 ms**. - **Events per second: 9 / 0.080 s = 112.5 Hz.**

Worst-case average dissipation, assuming every one of the 9 events collapses from the full 18 A peak (i.e., assuming a purely dissipative clamp — freewheel-to-battery or avalanche/Zener, not the recovery topology recommended in §3): `P_avg = 32.4 mJ × 112.5 /s ≈ 3.6 W`.

This is a **modest, not alarming, average number** — well within what a TO-220/DPAK-class clamp FET or Zener can shed continuously, even with a fairly restrained heatsink. The reason it still matters is not the average, it's the **peak instantaneous dissipation during each ~30–50 µs clamp interval** (32.4 mJ delivered in ~36–53 µs is a multi-hundred-watt

instantaneous pulse into the clamp element even though the time-averaged figure is small) — this is precisely the stress mode AN50003 flags as "Questionable" for active-clamp reliability (§1.1) and the reason its own avalanche-rated parts are only characterized up to single-digit-mJ per-event energies (their BUK9K35-60RA example: 4.9 mJ/event, ~2.5 billion cycle life at that energy, §6) — **an order of magnitude below this injector's 32.4 mJ per event.** No source in this pass characterizes a repetitive-avalanche or active-clamp part rated for 32 mJ/event at anything like injector-relevant cycle counts; this is recorded as an open item for the eventual FET selection, not answered here. **This is exactly the argument for recirculating the energy into the boost cap rather than dissipating it in a clamp device at all** — it sidesteps the question of whether a suitably-rated dissipative part even exists in this energy class.

If pilot/post events are assumed to turn off from hold current (~10 A) rather than full peak (plausible if they are short enough to be within the hold phase at cutoff, though memo 04 notes the boost ramp to 18A takes only ~38 µs, well under the ~100–300 µs minimum pulse width, so even short pilot pulses likely do reach peak current before cutoff) — the worst case above should be treated as the design figure, not a conservative upper bound to be discounted.

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5. ISO 7637-2 / CISPR 25 implications of the turn-off edge

Partially sourced — the standard's own text could not be retrieved.

- An attempt to fetch the ISO 7637-2:2011 standard text directly (a third-party PDF mirror) returned **HTTP 401 (paywalled/blocked)**. The official standard was not retrieved in this pass. What follows is drawn from search-result summaries describing the standard's pulse definitions — **SNIPPET level**, not confirmed against the standard itself.
- Per those snippets: **ISO 7637-2 pulse 2a** is specifically defined to simulate *"transients due to sudden interruption of currents in a device connected in parallel with the DUT due to the inductances of the wiring harness"* — this is, in general terms, structurally the same event class as an injector solenoid turn-off on a shared harness (an inductive load on the same supply/return network being suddenly interrupted). **This is an inference about applicability, not a confirmed statement from the standard that injector turn-off is pulse 2a's origin** — no source found states that explicitly for this project's harness.
- Pulses 3a/3b are described (snippet level) as driven by "distributed capacitance and inductance of the wiring harness" during switching processes — again structurally relevant to a fast clamped turn-off edge propagating down a wire that runs the length of the engine, but not confirmed against the standard text.

- `emi_filter.cir`'s own RESULT NOTE already establishes this project's relevant framing: it models **differential-mode conducted emissions only** (150 kHz–108 MHz, CISPR 25 Class 5, 18 dBµV average in the FM band per research memo 06), and explicitly defers common-mode analysis (which a fast dV/dt edge referenced to chassis would excite more than a differential edge) as "a specification, not a simulated result, until there is a layout to model it against."

What transfers, stated as INFERRED, not sourced: a clamped turn-off at 100 V in 36 µs is a dV/dt of roughly 100 V / 36 µs ≈ **2.8 V/µs** — two orders of magnitude gentler than the buck stage's ~400 kHz switching edges that `emi_filter.cir` already exists to filter (a 400 kHz square edge with even a few-hundred-ns risetime is tens to hundreds of V/µs). This suggests the turn-off edge is **unlikely to be the dominant conducted-emissions contributor** compared to the buck converter already being filtered, but this is a first-order comparison, not a simulated result, and it says nothing about the common-mode / radiated-emissions picture, which `emi_filter.cir` itself has not modelled for any source yet. **This is the one open item from this section worth carrying into the eventual SPICE block:** once a turn-off netlist exists, its edge should be checked against `emi_filter.cir`'s existing differential-mode model rather than assumed benign.

What transfers to this design

1. **Topology choice is a solved, four-way-classified problem** (Nexperia AN50003), not something to invent from first principles. Free-wheel is cheapest and slowest; avalanche and active-clamp are similar in speed and loss to each other but active-clamp carries a documented long-term reliability caution this project should not casually inherit; boost/recovery is the most complex and highest nominal component cost but the best net efficiency *and* is the one option with genuine diesel-injector production precedent (Marelli, 1988).
2. **The clamp-voltage-vs-speed trade-off is a straightforward L/V relation** that this memo's own arithmetic cross-checks cleanly against Nexperia's published numbers — no separate injector-specific figure was found or is needed to use the relation correctly.
3. **Recirculation to the boost rail is not a marginal optimization for this specific design** — the ~6.9 V recovered bump is the same order of magnitude as the ~7.3 V droop `boost_converter.cir` already treats as the central sizing question for that block. Building the turn-off circuit without considering recovery would leave that block's charge balance silently wrong once turn-off events are added to the picture.

4. **32.4 mJ/event is a real thermal design constraint for any dissipative clamp part**, even though the *average* power (3.6 W worst case) is modest — it is the peak/instantaneous stress during each 30–50 µs clamp event that matters, and no source in this pass identifies an avalanche-rated or Zener-clamp part characterized anywhere near this energy class per event. This is an argument in favor of recovery (minimizing energy actually dissipated in a switch/clamp device) rather than a solved problem to defer.
5. **The turn-off edge should be checked against, not assumed clear of,** the existing `emi_filter.cir` differential-mode model, and eventually against a common-mode model neither this memo nor that block has built.

Recommendation: the circuit to build

Topology: Low-side switch (per injector channel, matching the existing bank-shared high-side / independent low-side wiring memo 04 already verified) with a **recirculation diode from the injector's low side back into the 100 V boost rail capacitor** — the "boost converter" / energy-recovery variant from AN50003 §4.4, adapted from its PWM-ripple-recovery role (used there during the Peak/Hold chop) to also handle full End-of- Injection turn-off, which is the diesel-electro-injector-specific architecture the Marelli 1988 patent describes in general terms. This is **not** a Zener-across-the-FET active clamp (ruled out per §1.1's reliability finding) and **not** a bare freewheel-to-battery diode (too slow relative to the boost rail already built for the opposite reason on turn-on).

Clamp voltage: the existing **100 V boost rail itself** — no separate, higher clamp voltage is recommended. This keeps the clamp element's voltage rating aligned with parts already being selected for the boost stage, avoids introducing a fifth voltage domain, and gives a computed ~36 µs turn-off (§2.2), comfortably fast against the 1–3 ms injection window. A higher dedicated clamp (§2.2's 150 V illustrative row) would turn off faster still, but nothing in this pass's sourcing justifies paying for a second high-voltage rail when the existing one is already close to the useful range AN50003's own avalanche example (68 V) demonstrates.

Part class: a diode (ultrafast recovery, rated for boost-rail voltage and the 18 A peak current, matching the class TLE8242-2's own datasheet calls out generically — "recirculation diode (ultrafast)" — even though that IC itself is the wrong current/voltage class for this injector) in series with the existing low-side switch topology; no new active clamp IC is recommended. Should the boost rail's regulator (`Bchga` in `boost_converter.cir`) need to shed recovered energy that pushes it

above 100 V on back-to-back events, that is a boost-converter-block problem to solve in that block's next revision, not a reason to add a second dissipative clamp path here.

Specific claims the next SPICE block should check: 1. Turn-off time from 18 A to ~0 A when clamped to a 100 V rail: expect **~36 µs** (physics-derived, §2.2) — if the simulated result is far off this, the model's assumptions (diode drop, rail sag under load) need re-examination before trusting it. 2. Boost-rail voltage excursion from one full-energy recovery event on the existing 47 µF/100 V reservoir: expect **~+6.9 V** (§3) — compare against `boost_converter.cir`'s own ~7.3 V peak-phase droop to see whether recovery approximately cancels it, overshoots it, or undershoots it once diode/switch losses are included (this memo's number ignores those losses; AN50003's own data suggests recovery topologies still have real, non-zero losses even though net efficiency is favorable). 3. Whether recovered energy pushes the rail above the 100 V setpoint on back-to-back pilot+main+post events at the 26.67 ms per-cylinder spacing, and whether the existing `Bchga` charge-regulation logic needs a bleed path. 4. The clamp diode's peak instantaneous power/junction stress during the ~36 µs turn-off pulse, not just the ~3.6 W time-averaged figure (§4) — the average number is not the one that will size the part. 5. The turn-off edge's differential-mode contribution against `emi_filter.cir`'s existing model (§5) — first-order comparison here suggests it is gentler than the buck stage's edges, but this should be checked, not assumed.

Negative results, recorded

- **ST L9781 datasheet** — not retrieved. Two fetch attempts (st.com direct PDF, alldatasheet mirror) failed: one timed out (60 s), one returned HTTP 403. Everything attributed to it in this memo is explicitly marked SNIPPET and should not be used as a design number without retrieving the document.
- **NXP AN4849** ("Four Injector and Fuel Pump Drive Featuring the MC33816") — the PDF URL found via search (nxp.com/docs/en/application- note/AN4849.pdf) returned **HTTP 404**. Not retrieved. Search-result summaries describing its "boosting"/"holding" high-side driver pairing per bank are recorded in this memo's working notes but were not used as a cited finding above, since the document itself was never seen.
- **Bosch EDC17/EDC7 injector-driver ASIC architecture** — no public technical/circuit-level source found at all (see §1.4). This is a negative result about public availability, not evidence that Bosch's internal architecture differs from what's described here.

- **ISO 7637-2:2011 standard text** — one direct-PDF fetch attempt returned **HTTP 401.** Not retrieved. §5's discussion of pulses 2a/3a/3b is search-snippet level throughout and is flagged as such at each claim.
- **A quantitative die-temperature-rise-per-clamp-voltage figure** (task item 2's request for "actual numbers real drivers use," beyond the single 68 V generic-automotive avalanche example found) — not found. No injector-specific clamp-voltage number (as opposed to a generic solenoid or a 15–20×-battery gasoline-DI figure) was retrieved from any source in this pass. The 100 V recommendation in this memo's §"Recommendation" is a design choice justified by matching the existing boost rail, not a sourced industry-standard clamp voltage for this injector class.
- **A confirmed injection strategy (pilot/main/post count) for this specific engine/calibration** — not found; memo 04 does not establish this either. §4's "3 events per cylinder per cycle" is stated as an assumption, not a sourced fact.
- **TI DRV8701 relevance** — checked and explicitly ruled out (§1.5), not silently dropped, since the task named it as a candidate to check.

Research Memo: Supervisor Circuit — Watchdog, Brownout, and Output State During Reset

Task: Spec §3's chain of protection collapsed on 18 Sep 2026 (U12, U13): the DSE4522 reads engine speed from this ECU over CAN, and the only relay in the path signals rather than removes power. Nothing external to this ECU currently interrupts fuelling without the ECU's cooperation, and a standalone trip module is specified but not fitted. This memo asks the question that gap exposes: what does this board do when its own MCU stops executing correctly? There is no supervisor circuit anywhere in the current design — no external watchdog, no brownout/reset supervisor, and nothing defining output state during MCU reset. **Date:** 2026-09-18 **Scope:** Read-only research. No repo files modified, nothing committed, no subagents dispatched.

Verdict (read this first)

The S32K148 datasheet and reference manual, retrieved and read directly, confirm the hazard is real and quantify it: **Low Voltage Reset (VLVR, 2.50–2.7 V) is always active and forces a reset in every mode, but Low Voltage Detect (VLVD, 2.8–3.0 V) only forces a reset if firmware sets** `LVDRE`, and the part's own datasheet states the S32K148 is not guaranteed to execute correctly below 2.97 V with the PLL engaged. That leaves a **sourced, non-trivial band — roughly 2.97 V down to 2.58 V typ, ~390 mV wide — where the MCU is out of its own guaranteed operating envelope but nothing internal has reset it yet**, unless `LVDRE` is set (which narrows but does not close the band). Reset-and-boot text in the same manual confirms **non-analog GPIO pins are configured disabled on reset** — high-impedance is the documented behaviour, not a guess. Separately, NXP's own MC33814 automotive injector/ignition driver datasheet documents production practice for exactly this exposure: **the driver IC's own POR forces every output off, independent of what the host MCU's pins are doing** — the fix lives partly in the driver stage, not only in pull-down resistors.

For the watchdog question, four parts were retrieved and read in enough depth to compare architectures, not just marketing copy: **TI TPS3850/TPS3813** are cheap, small, standalone windowed-watchdog supervisors (10-pin VSON / 6-pin SOT-23) that monitor a rail directly and drive an open-drain RESET; **NXP FS26** and **Infineon TLF35584** are full safety SBCs/PMICs that additionally offer **true question/answer**

(challenge-response) watchdogs and **dedicated safe-state output pins (FS0B/FS1B, SS1/SS2)** that the device's own fail-safe state machine can assert independent of any live MCU communication — but adopting either means replacing this design's existing buck pre-regulator and LDO architecture (spec §6), not adding one part beside it.

Recommendation: TPS3850-class windowed supervisor, threshold set above VLVD's 3.0 V max, RESET output gating a discrete "kill" pulldown on every injector and metering-unit gate node, backed by 10 kΩ gate-source pulldowns (Infineon's own stated typical value) as the passive baseline. This is buildable now, without restructuring the power tree, and is independently justified regardless of the mandatory trip module — they protect different failure points (Q6).

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1. Internal vs external watchdog — what the on-die peripherals cannot cover

CONFIRMED, from the S32K1xx Series Reference Manual (Rev. 13, 04/2020), retrieved in full (14 MB PDF, `community.nxp.com` mirror) and the S32K1xx Data Sheet (Rev. 6, 01/2018, via `www1.futureelectronics.com/doc/NXP/FS32K146HAT0VLQT.pdf`):

- The S32K1xx has **two separate on-die watchdog-class peripherals**, and they are not redundant with each other: the **WDOG** (internal watchdog, resets the MCU's own core and peripherals) and the **EWM** (External Watchdog Monitor). Quoting the reference manual directly on EWM: *"The EWM differs from the internal watchdog in that it does not reset the MCU's CPU and peripherals. The EWM provides an independent EWM_OUT_b signal that when asserted resets or places an external circuit into a safe mode."* This means the S32K148 already has an on-die mechanism whose *output* is meant to drive external hardware into a safe state — but it shares the same die, the same power rail, and (absent specific clock-source configuration) can share failure modes with the core it is meant to backstop.
- **WDOG supports windowed mode** (`CS[WIN]`, `WIN` register) and **can be clocked from a source other than the bus clock**: *"The options allow software to select a clock source independent of the bus clock for applications that need to meet more robust safety requirements. Using a clock source other than the bus clock ensures that the watchdog counter continues to run if the bus clock is somehow halted."* There is even a **"backup reset"**: *"a safeguard feature that independently generates a reset in case the main WDOG logic loses its clock... If the watchdog counter overflows twice in succession (without an intervening reset), the backup reset function takes effect."* This is a

materially more capable on-die watchdog architecture than "just a timeout counter" — NXP has already engineered around the plain clock-failure case.

- **What none of this covers, and the reference manual does not claim to cover:** a core that is executing the *wrong* application code but whose scheduler or a low-priority ISR still reaches the watchdog-service call on schedule. Both WDOG and EWM are serviced by writing a value from *somewhere* in the code the core is executing; neither peripheral can distinguish "correct control flow reached this service call" from "any code path reached this service call," because both live on the silicon being monitored and have no visibility into which code path issued the refresh. This is exactly the case a **question/answer watchdog closes** (§2) and windowing alone does not (§2) — INFERRED from combining the confirmed WDOG/EWM mechanics above with the TLF35584 functional-watchdog mechanism read in §2, not stated explicitly as a limitation in NXP's own text.
- **Every one of these peripherals is configured by firmware after reset.** A firmware bug that fails to enable `LVDRE`, fails to configure WDOG windowing, or fails to route EWM_OUT_b usefully leaves the corresponding protection silently absent — and there is no external check on whether that configuration happened correctly. An external supervisor's threshold and window are set by resistors/capacitors or SPI-from-a-second- die, not by the same firmware whose failure it exists to catch.

Why automotive practice still fits an external part, stated plainly: it is not that the on-die peripherals are poorly engineered — the WDOG/EWM/backup-reset chain above is genuinely thorough. It is that **every one of them is the same silicon, the same power domain, and (for WDOG/EWM specifically) configured by the same firmware being protected against.** An external supervisor is a second, physically independent part: different die, different supply-monitoring analog front end, and — for the question/answer class — a challenge the monitored firmware must *compute*, not just schedule a call to.

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2. Windowed watchdogs and question/answer supervisors — what four real parts provide

2.1 TI TPS3850 — CONFIRMED, full datasheet retrieved (`ti.com/lit/ds/symlink/tps3850.pdf`, SBVS301B, Oct 2016 rev. Sep 2021)

Precision voltage supervisor **plus** programmable window watchdog in one 10-pin VSON part. Directly confirmed electrical table (§6.5, at 1.6 V ≤ VDD ≤ 6.5 V, −40 °C to 125 °C):

Parameter	Min	Typ	Max	Unit
VDD supply range	1.6	—	6.5	V
VPOR (power-on reset voltage)	—	—	0.8	V
VUVLO	—	—	1.35	V
Overvoltage/undervoltage SENSE threshold accuracy	—	—	±0.8 %	—

Timing table (§6.6) confirms **factory-selectable window ratios of 1/8, 1/2, and 3/4** (lower boundary : upper boundary), and — with the `CWD` pin left unconnected — **t_WDL spans ~1.85 ms to ~800 ms and t_WDU spans ~2.5 ms to ~1840 ms** depending on the `SET0` / `SET1` pins; an external capacitor on `CWD` (0.1 nF to 1000 nF) extends this further, with the footnote giving 62.74 ms (0.1 nF) to 77.45 s (1000 nF) for t_WDU(typ). `RESET` and `WDO` are both open-drain, pulled up externally. Footnote directly states: *"When VDD falls below VPOR, RESET and WDO are undefined."* — i.e. even this supervisor has its own undefined band below its own POR threshold (0.8 V max), which is far below anything relevant to a 3.3 V rail, but is worth carrying as a general principle: **every supervisor has its own minimum operating voltage below which it, too, is undefined.**

2.2 TI TPS3813 — CONFIRMED, full datasheet retrieved (`ti.com/lit/ds/symlink/tps3813.pdf`)

Same family concept, smaller (6-pin SOT-23), simpler configuration: `WDT` sets the upper timeout limit (GND/VDD/external cap), `WDR` sets the window ratio (GND/VDD only, two ratio classes rather than three). Confirmed electrical table: negative-going input threshold `VIT` for the L30 variant is **2.58 V min / 2.64 V typ / 2.7 V max** (directly usable as an off-the-shelf threshold sitting almost exactly on the S32K148's own VLVR typ of 2.58 V — see §3 on why that is *not* actually the right threshold to pick). Timing table gives watchdog window ratios up to **1:127.7** and a base delay `td` of 20/25/30 ms after VDD exceeds VIT+0.2 V. Both this and TPS3850 are genuinely low-part-count, low-cost additions: a handful of passives plus one SOT-23/VSON IC.

2.3 NXP FS26 — CONFIRMED, short data sheet retrieved and read as images (`nxp.com/docs/en/data-sheet/FS26_SDS.pdf` , Rev 6.1, 30 Jul 2026; full data sheet requires NDA per the document's own cover page — noted, not obtained)

FS26 is a **safety system basis chip (SBC)**, not a standalone supervisor: it integrates a wide-input (VBST 5–18 V, VPRE HVBUCK buck configurable 3.7–6.35 V/1.5 A) pre-regulator, a VCORE regulator (0.8–3.35 V), two LDOs, two trackers, and a voltage reference — i.e. it **replaces**, not supplements, this design's existing buck pre-regulator and LDO (spec §6). Its watchdog block is explicitly dual-mode — the device's own part table distinguishes **"Simple"** vs **"Challenger"** watchdog variants by part suffix, and the block diagram labels it directly: *"WINDOW WATCHDOG*

(SIMPLE/CHALLENGER)." Confirmed via a companion NXP search summary that ASIL D applications require the challenger variant. FS26 also has **two dedicated fail-safe output pins, FS0B and FS1B (plus RSTB)**, driven by an independent "FAIL SAFE OUTPUT DRIVER" block inside the chip's own fail-safe logic domain, explicitly drawn in the block diagram feeding an external **"ACTUATORS"** block — this is the "hard-disable independent of MCU" mechanism the task asks about, built into a full SBC rather than a standalone add-on. Package: LQFP48 with exposed pad.

2.4 Infineon TLF35584 — CONFIRMED, full 4-page-summarized (actually multi-hundred-page) datasheet retrieved and read (`infineon.com/assets/row/public/documents/10/49/infineon-tlf35584-datasheet-en.pdf`, Rev 2.0, 2017-03-16)

Also a full PMIC (step-up/step-down pre-regulator 3–40 V input, multiple LDOs/trackers), **not** a standalone supervisor — same caveat as FS26. But its watchdog architecture is the clearest confirmed answer to "what does question/answer actually mean, mechanically." Quoting §15.1 directly: *"Two independent types of watchdogs are implemented in the TLF35584: A standalone window watchdog (WWD)... A standalone functional or question/answer watchdog (FWD)... The watchdogs have independent timers and error counters, which allows to run both watchdogs in parallel."* And §15.3, quoted directly on the FWD mechanism: *"A question is generated (taken out of table)... The question consists of 4 bits, the expected answer consists of 4 responses of 8 bits each. The four responses shall be sent before the heartbeat period has ended... If the complete answer (all four responses – 32 bits) is correct... it is regarded as 'Valid FWD triggering.'"* **This is the concrete mechanical answer to why question/answer defeats a stuck-loop-that-still-kicks-the-dog: the correct 32-bit answer must be looked up from a table and assembled into four SPI writes in the right sequence — a code path that happens to call a single `WDOG_Refresh()` function cannot produce this by accident.** TLF35584 also has **SS1/SS2 ("Secondary Safety Shutdown") pins**, driven directly by the window-watchdog and functional-watchdog failure counters through dedicated safety logic (Figure 59, §12.5): *"ΣWWO = Number of 'Invalid WWD triggering', after safe state control shall activate the safe state signal SS1 and SS2."* Packages: PG-VQFN-48 or PG-LQFP-64.

2.5 Maxim MAX16997/MAX16998 — SNIPPET only, not retrieved in full

Two full-text fetch attempts (`analog.com` direct, `datasheet4u.com` mirror) both failed — one timed out twice (60 s and 140 s), the other returned non-PDF content mislabeled as a PDF. What search-result snippets state (unverified against the document itself): four independent inputs for reset/watchdog, `SWT` / `SRT` pins set watchdog and reset timeout via external capacitors, active-low `RESET` and active-

low `ENABLE` outputs, 8-pin µMAX, −40 °C to 125 °C automotive-rated. **Not elevated to CONFIRMED.** The `ENABLE` output is notable if true — a dedicated enable-gating pin distinct from `RESET` is exactly the mechanism §4/§6 need — but this project's own discipline (memo 09, memo 10) is explicit that snippet-level claims are not design inputs, so this is recorded as a lead, not a basis for the recommendation.

What each actually costs in parts, stated plainly

- **TPS3850 / TPS3813:** one small IC (VSON-10 / SOT23-6) plus a handful of resistors and optionally one capacitor. Adds to the existing power architecture without changing it. **This is the class that fits a "supervisor bolted onto an otherwise- finished power tree," which is this project's situation.**
- **FS26 / TLF35584:** one 48–64-pin IC that **is** the power tree — buck pre-regulator, core LDO, sensor LDOs, tracker regulators, reference, and the safety logic all in one part. Adopting either means re-deriving spec §6 from scratch, not adding a block beside `buck_preregulator.cir` and `mcu_pdn.cir`. Also gates significant parts of the FS26 full datasheet behind NDA (confirmed from the short data sheet's own cover page), which is a real cost for a small team, separate from silicon price.

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3. Brownout — the threshold, the hazard band, and what the 150 µF cap does to it

CONFIRMED, from the S32K1xx Data Sheet (Rev. 6, 01/2018), Table 2 and Table 5, retrieved and read directly (text-extracted, not paraphrased from a search snippet):

- **Minimum operating voltage, stated in the datasheet's own words** (footnote 3 to Table 2): *"S32K148 will operate from 2.7 V when executing from internal FIRC. When the PLL is engaged S32K148 is guaranteed to operate from 2.97 V."*Table 2 itself: `VDD` min **2.73 V**, and a boxed NOTE above the table: *"Full functionality/ specifications cannot be guaranteed when voltage drops below 2.7 V."*
- **Table 5, "VDD supply LVR, LVD and POR operating requirements"** — the exact table the task asked for:

Symbol	Description	Min	Typ	Max	Unit
VPOR	Rising and falling VDD POR detect voltage	1.1	1.6	2.0	V
VLVR	LVR falling threshold (RUN, HSRUN, STOP)	2.50	2.58	2.7	V
VLVR_LP	LVR falling threshold (VLPS/VLPR)	1.97	2.22	2.44	V
VLVD	Falling low-voltage detect threshold	2.8	2.875	3	V
VLVW	Falling low-voltage warning threshold	4.19	4.305	4.5	V

(VLVW is stated for completeness; at this design's 3.3 V nominal `3V3_MCU` rail, VLVW's 4.19 V min floor is above the rail's own nominal voltage and therefore cannot function as a meaningful warning here — flagged as an **INFERRED implication**, not stated as such in the datasheet, which documents the part across its full 2.7–5.5 V VDD range.)

- **Reference manual, §25.2.2.2, confirms the reset-forcing behaviour directly**: *"Besides LVD operation, the device also supports LVR (Low Voltage Reset) operation. If the supply voltage falls below the reset trip point (VLVR), a system reset will be generated.* ***LVR system is enabled in all modes.*** *LVDRE has effect on LVD operation only."* I.e. **LVR always resets the part; LVD only resets it if firmware has set `PMC_LVDSC1[LVDRE]`.** This is the load-bearing distinction for the "band" question: without `LVDRE` set, the *only* automatic reset floor is VLVR at 2.50–2.7 V, not VLVD at 2.8–3.0 V.

The hazard band, quantified from these confirmed numbers, not invented:

1. **With `LVDRE` unset (a plausible firmware omission, not a hypothetical)**: the MCU's own datasheet stops guaranteeing correct operation at 2.97 V (PLL engaged) but the first automatic reset does not occur until VLVR's typ 2.58 V / worst-case 2.50 V. That is a **~390–470 mV band, executing with outputs still under firmware control, in which the part's own datasheet has already withdrawn its correctness guarantee.**
2. **With `LVDRE` set** (zero-BOM-cost, firmware-only fix, and this memo recommends it regardless of any external supervisor): the automatic floor rises to VLVD's 2.8–3.0 V, which **still sits below the 2.97 V PLL-guarantee floor at the typ corner (2.875 V)** — a narrower but nonzero ~95 mV gap remains at typical parts, wider at worst-case corners (2.97 V guarantee vs. 2.8 V min LVD trip = 170 mV).
3. **Neither of these is zero**, and both exist entirely within NXP's own published numbers — this is not a manufactured hazard.

What the 150 µF bulk cap does to the *duration* of this band — INFERRED, original arithmetic, not sourced to a load-current measurement: `mcu_pdn.cir`'s RESULT NOTE (read in full, §"Power chain stages 6 and 7") records the bulk capacitor on `3V3_MCU` was raised from 47 µF to 150 µF on 18 Sep 2026 for a PDN-impedance reason unrelated to brownout (damping the LDO/bulk anti-resonance). That capacitor still governs how long a *lost* rail takes to sag through the ~390 mV LVR-to-guarantee band once the LDO can no longer supply it (e.g. on an upstream supply loss, not a cranking dip — the buck stage's 6–40 V input range with 6 V covering the cranking dip means a *normal* crank event should not reach this rail at all). Using `dt = C·ΔV/I` with the

now-confirmed 390 mV band and an assumed, unsourced load current in the tens-to-low-hundreds-of-mA class typical for this MCU plus its immediate peripherals:

Assumed load	dt across the 390 mV band
50 mA	1.17 ms
100 mA	585 µs
200 mA	293 µs

These are sub-millisecond to low-millisecond, i.e. shorter than even the fastest window-watchdog configurations surveyed in §2 — for a *lost-supply* event, the on-die LVR should reset the part well before any watchdog would have caught it. The scenario that actually matters is a **slow sag** (not a clean loss): if the buck stage's own regulation degrades gradually rather than losing input entirely, the rail could linger in the band far longer than this RC estimate, which assumes no regulation at all during the sag. No source found quantifies this slower-sag case for this specific buck design; flagged as an open item for `buck_preregulator.cir`'s own next revision rather than guessed here.

What margin a supervisor threshold needs above VLVR/VLVD, stated plainly: an external supervisor exists specifically to fire *before* the MCU's own guarantee lapses, and to do so **regardless of whether firmware ever set `LVDRE`.** Its threshold should sit above VLVD's stated max (3.0 V) with margin for the supervisor's own accuracy (TPS3850's ±0.8% threshold accuracy, confirmed in §2.1) and for the fact that VLVD is a *falling* threshold with separately-specified rising hysteresis — i.e. roughly **3.05–3.10 V** is a reasonable target for the external supervisor's undervoltage trip, giving clearance above both VLVD's worst case and VLVR's worst case, and doing so independent of any internal MCU register configuration. This is a design recommendation for this memo, not a number pulled from a source.

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4. THE QUESTION I MOST WANT ANSWERED — output state during reset

CONFIRMED, GPIO default state. S32K1xx Series Reference Manual, §25.2.2 (System reset sources), quoted directly: *"The on-chip peripheral modules are disabled and the non-analog I/O pins are initially configured as disabled. The pins with analog functions assigned to them are configured for their analog functions after reset."* And §12.5.1 (`PORT_PCRn`), MUX field: *"000 Pin disabled (Alternative 0) (analog)."* This confirms the task's premise directly from the primary source, not from a forum answer: **GPIOs are high-impedance / disabled out of reset, and stay that way until firmware writes the PCR MUX field.** The register's `PE` (pull enable) and `PS` (pull select) bit reset values are explicitly

documented as **"Varies by port. See Signal Multiplexing and Signal Descriptions chapter for reset values per port"** — i.e. **not every pin defaults to the same pull state.** Some dedicated pins have a stated default (JTAG: *"TDI as pullup, TCK as pulldown, TMS as pullup"*, confirmed in the same reset-and-boot section); general-purpose pins do not have a universal default documented at this level. **This is an open item specific to this design, not resolved by this memo**: the S32K148's pin-mux/signal-multiplexing table — which physical `PTxx` pin ECU pins 88, 73, 07, 29, 03 and 05 land on — is explicitly deferred to Phase 2 per spec §5 ("confirming how many ADC channels the 144-pin package actually breaks out (needs the reference manual's pin-mux table")). **The exact PE/PS reset default for those six specific pins cannot be stated until that pin-mux assignment exists.** Do not treat "high-impedance, no pull" as confirmed for those specific pins — treat "high-impedance by default, pull state pin-specific and not yet mapped" as the confirmed statement.

CONFIRMED, production practice for the driver stage itself. NXP MC33814 ("Advanced Fuel Injector and Ignition Coil Driver Controller IC," full datasheet retrieved as a saved PDF via `nxp.com/docs/en/data-sheet/MC33814.pdf` and text-extracted) is a real Bosch-family-class injector/ignition low-side driver IC — the same device class this design's own injector and metering-unit drivers occupy. §5.1.2.2, quoted directly: *"SPI register settings from Power On Reset (POR) are as follows: • All outputs turned off • Off State open load detection enabled (LSD) • Default values in the SPI Configuration, Control and Status registers."*This is the production answer to the task's question, from a real injector-class driver IC: **the driver IC's own POR (its `VCC(POR)` threshold, separately confirmed in the same datasheet as 3.9–4.9 V rising) forces every output off, independent of whatever the host MCU's control pins are doing during the host's own reset.** The driver does not trust the host's GPIO state; it has its own POR and its own default-off register state.

One nuance worth flagging rather than glossing over — a genuine gap in what was retrieved, not filled in: MC33814 §5.3.2 lists watchdog timeout, VCC undervoltage, and VPWR overvoltage as sources of a `RESETB` pulse (used to reset the *host MCU*), but the retrieved text does not state whether a watchdog-timeout-triggered `RESETB` pulse *also* resets the 33814's own output/control registers to their POR default, the way an actual POR (sleep→active transition) does per §5.1.2.2. §5.3.3 separately describes an SPI-commanded "internal reset" that does clear all registers, but that requires the host to still be capable of issuing SPI commands — not a given during the fault this memo is about. **This is recorded as an open question about MC33814 specifically, not resolved**

by inference, and it is exactly the kind of assumption ("the driver IC handles it") that should not be adopted without checking, which is the point of flagging it.

Pin-specific exposure in this design, reasoned from spec §2.2, §4 and the sim blocks read for this task (INFERRED synthesis of confirmed facts, not itself sourced):

- **Metering unit (pin 88, low-side PWM into a fused-battery-fed coil, per spec §2.2):** `metering_unit_pwm.cir`'s own header states the coil's electrical time constant is L/R = 3 ms. If the GPIO driving this gate floats high during a reset/brownout window (§3's band, or the tens-of-ms class of watchdog timeout surveyed in §2), the coil current will visibly ramp toward its fully-on value within a small multiple of that 3 ms constant — this is not a momentary glitch, it is enough time for the metering unit's actual fuel-metering position to move.
- **Injector low-side selects (73/07/29) and bank high-sides (03/05):** per spec §2.1, these switch a shared 100 V boost rail (`injector_boost.cir` / `injector_turnoff.cir`). A floating low-side select gate during reset is lower-consequence alone (no boost-rail path is complete unless the corresponding high-side is also conducting), but the bank-shared high-side switches (03/05) are the ones exposed to the full 100 V rail on their drain — if either floats on during a reset window while any low-side is also ambiguous, current can flow into whichever injector coil is selected.

What standard practice actually is, assembled from the confirmed sources above — be specific, per the task's own instruction:

1. **Passive gate-source pulldown on every gate node, sized from a real application note, not a round number.** Infineon's 2022 application note "Gate drive for power MOSFETs in switching applications" (`infineon.com/assets/row/public/documents/24/42/ infineon-gate-drive-for-power-mosfets-in-switchtin-applications-applicationnotes- en.pdf`, retrieved in full), quoted directly: *"A resistor RGS, in the kΩ range (typically 10 kΩ), is highly recommended between the gate and source so that the MOSFET gate will be discharged if the gate becomes disconnected from the driver circuit. Without this a MOSFET may remain on when it should be off..."* **10 kΩ is this memo's recommended value, and here is why it clears this design's own numbers, not just the AN's round figure:** - *Leakage check*: S32K1xx Data Sheet Table 10 (§5.3, CONFIRMED), input leakage per pin at VDD = 3.3 V is **0.005 µA typ / 0.5 µA max.** Through 10 kΩ that is at most 5 mV — utterly negligible against any gate threshold. - *Discharge-speed check*: RC = 10 kΩ × Ciss. No specific gate driver FET part number is chosen yet for this design (spec §8, U5-adjacent gap), so this uses a representative automotive power-MOSFET class Ciss of 1–4 nF — **INFERRED, not sourced to a chosen part.** RC ≈ 10–40 µs, three to four

orders of magnitude faster than the shortest injection event (1 ms) or the fastest watchdog window considered in §2/§5. The pulldown does not need to be any stronger than 10 kΩ to be fast enough for this application's own timescales. - *dV/dt-coupling check, the case the same Infineon AN specifically warns needs a "strong" pulldown*: quoted directly, *"A strong pull-down is important in hard-switching half-bridge or full-bridge configurations to prevent C.dv/dt-induced turn on."*This design's own already-computed turn-off edge (`injector_turnoff.cir` 's RESULT NOTE / research memo 10 §5) is **~2.8 V/µs** across the 100 V boost rail — the memo's own number, reused here, not re-derived. Using an illustrative Crss of 50–200 pF (again a class figure, INFERRED, not a chosen part), the coupled current is `I = Crss·dV/dt ≈ 140–560 µA` .

> **CORRECTION, 18 Sep 2026 — this paragraph was wrong by a factor of 1000, and the conclusion it reached does not survive the fix.**
>
> It originally read: *"Through 10 kΩ that is at most ~5.6 mV — again far below any realistic gate threshold,"* and concluded that a plain 10 kΩ weak pulldown is adequate without an active clamp.
>
> 560 µA through 10 kΩ is **5.6 V**, not 5.6 mV. `560e-6 × 10e3 = 5.6` .
>
> `sim/blocks/supervisor.cir` was built to test this paragraph's conclusion as claim 3, and it **FAILS**. Simulated peak gate voltage, which includes the gate's own Ciss dividing the step down and so reads somewhat below the bare arithmetic:
>
> | Crss | V(gate) through 10 kΩ alone |
> |---|---|
> | 20 pF | 0.465 V |
> | 50 pF | **1.155 V** |
> | 100 pF | 2.289 V |
> | 200 pF | 4.497 V |
> | 500 pF | 10.647 V |
>
> Against a representative Vgs(th) of 1.0 V it fails at **50 pF — the low end of this memo's own illustrative range**, not at some pessimistic extension of it.
>
> The edge-rate argument above is still correct as far as it goes: 2.8 V/µs really is two orders of magnitude gentler than the hard-switched half-bridge case the "strong pulldown" literature addresses. It just does not license 10 kΩ, and the arithmetic slip is what made it look as though it did.
>
> **The pulldown cannot be a fixed number in this memo.** It is set by the FET actually chosen, through `Rpd < Vgs(th) / (Crss · dV/dt)` . At this design's 2.8 V/µs and a 1.0 V threshold: Crss 200 pF

> needs under 1.79 kΩ, 500 pF under 714 Ω, 1000 pF under 357 Ω. Halve for margin. Bounded from below by the gate driver's source current, since the pulldown fights it continuously during conduction — 300 Ω holding a 10 V gate costs 33 mA per gate, and there are six.
>
> What survives unchanged is everything the kill transistor does. Claim 4 passes across six brownout ramps with margin. This correction is about the passive resistor beside it, not the supervisor path. 2. **A driver IC whose own POR forces outputs off, independent of host GPIO state** — confirmed real practice per MC33814 above. This is the stronger mechanism where applicable, because it does not depend on a passive resistor racing an active driver trying to turn the gate on; it removes the ambiguity at its source. 3. **A supervisor output that hard-disables the drivers independent of the MCU** — this is what FS26's FS0B/FS1B and TLF35584's SS1/SS2 are *for* (§2.3, §2.4), confirmed from their own block diagrams and safe-state-control chapters. This project's recommended part class (§2's TPS3850/TPS3813) does not have a dedicated safe-state pin the way those SBCs do, but its open-drain `RESET` output serves the same role when wired through one inverting stage into a dedicated "kill" pulldown transistor at each protected gate node — see the Recommendation below.

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5. Reset timing against the injection event

INFERRED — original arithmetic against this project's own confirmed engine numbers, not sourced to a watchdog-timeout industry standard, per the task's instruction to state assumptions plainly.

Base figures, already established and confirmed elsewhere in this project (memo 04, memo 10, `boost_converter.cir`): 1500 rpm, 3-cylinder, 4-stroke ⇒ one full engine cycle (720°, 2 revolutions) = **80 ms**; per-cylinder firing interval (cylinders phased 240° apart) = **26.67 ms**. Memo 10's own assumption (explicitly flagged there as an assumption, not a sourced fact) of a pilot+main+post = 3 injection events per cylinder per cycle gives **9 events / 80 ms ≈ 112.5 Hz**, i.e. an average of one injection event every ~8.9 ms, though in practice a cylinder's triplet lands in a short burst (memo 10's own `injector_turnoff.cir` spaces them 300 µs apart, itself unsourced) rather than evenly across the 26.67 ms gap.

Watchdog timeout	Cylinder-intervals covered (÷26.67 ms)	What a hung ECU could do inside it
1 ms	0.037	Shorter than a single injection event (1–3 ms) — could interrupt *one event in progress*, not even complete it

Watchdog timeout	Cylinder-intervals covered (÷26.67 ms)	What a hung ECU could do inside it
5 ms	0.19	Within one cylinder's pilot/main/post triplet only
10 ms	0.375	Still within one triplet's worth of time, comfortably
26.67 ms	1.0	Exactly one additional cylinder gets a full, possibly wrong or stuck, firing decision
50 ms	1.87	~2 more cylinders fire under the hung condition before recovery
100 ms	3.75	Every cylinder gets an extra full engine cycle (all 3 fire again)

The metering unit (pin 88) is the more continuously-exposed output, not the injectors: it is not event-scoped, it is a continuous PWM duty. `metering_unit_pwm.cir` gives the coil's electrical time constant as 3 ms; a stuck duty cycle held for any of the timeouts above except the 1 ms row is long enough for the coil current — and therefore the actual fuel-metering position — to have **settled** at whatever the stuck command was, not merely glitched toward it.

What this implies for choosing a timeout, stated as a design conclusion rather than a sourced number: keeping total hung-ECU exposure **under one cylinder-to-cylinder interval (26.67 ms)** keeps the worst case to "part of one cylinder's injection triplet affected," rather than "one or more entire extra cylinders fire under an uncommanded condition." That argues for a **window watchdog with its closed (upper) boundary well under 26.67 ms** — a closed window in the 15–20 ms class, with an open (lower) boundary comfortably above the 1–3 ms injection duration (so normal interrupt/scheduling jitter around a real injection event does not false-trip the window) is the target this memo recommends qualitatively. **The exact capacitor/register values to hit that window on a chosen supervisor part are explicitly left open here** — TPS3850's own equations (referenced in its datasheet's revision history but not extracted in this pass) would size it precisely once a part and threshold are finalized. Guessing a specific picofarad value into this memo would be exactly the "confident desk reasoning" failure mode this project's own sourcing discipline exists to catch; it is left as a Phase-2 hardware/firmware co-design item instead.

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6. Does the supervisor need to be independent of the trip module, or can they be one thing?

INFERRED — reasoned from spec §3 read in full, not sourced to an external reference, and stated as this memo's considered position rather than hedged.

They are not redundant. They protect different points in the chain, and removing either weakens the design in a way the other does not cover:

- **The trip module (spec §3 item 5) is downstream and symptom-facing.** It must break the metering unit's own battery feed (through fuse `8F1`) — i.e. it acts on the *load side* of the metering unit, after everything upstream (MCU, GPIO state, gate driver, FET) has already done whatever it was going to do. It is triggered by an engine-level fault signal (most plausibly overspeed, per spec §3's own reasoning about the standalone MPU-driven trip module) and its detection latency is bounded by how fast the engine can visibly speed up, not by how fast the MCU misbehaves. It is also the *only* mechanism in this whole discussion that can stop a **hardware fault downstream of the gate** — a low-side FET that has failed shorted, for instance, is completely unaffected by anything a supervisor does to that FET's gate, because the gate is no longer what is holding the FET on. Only removing supply at the relay reaches that failure.
- **A supervisor that hard-disables the drivers is upstream and root-cause-facing.** It watches whether the MCU itself is behaving (via watchdog) and whether its supply is in-spec (via undervoltage threshold), and it acts at the gate, before current ever reaches the injector or metering coil. It catches failure modes that never progress to an engine-level symptom the trip module's overspeed logic would notice — a hung MCU commanding a *wrong but not obviously runaway* fuelling profile (rough running, white smoke, a stall under load, a genset that drops load without ever exceeding 1680/1710 rpm) is invisible to an overspeed-triggered trip module but is exactly what a watchdog-driven gate kill catches, because it does not care what the fuelling profile *looks like* downstream — it only cares whether the MCU is still answering correctly.
- **They also differ in what they depend on.** The trip module (per spec §3 item 5's own requirement) must not depend on the GCU relay chain that U12 just showed fails — it has to be its own mechanical/electrical path to the metering unit's supply. A supervisor on this board depends on nothing outside this board: it does not care whether the DSE4522 is configured correctly, whether the alternator is exciting, or whether the trip module itself has been correctly wired — all conditions spec §3 already flags as real, live uncertainties for the *other* two lines of defence.

Where they do overlap, and why that overlap is a feature, not waste: in the specific scenario of "MCU hangs, metering unit gets stuck on," both mechanisms independently break the fuelling path — the supervisor at the gate, the trip module at the relay. That is genuine defence-in-depth precisely because their independence is real (different physical points, different trigger conditions, different failure dependencies), not because either one is a weaker copy of the other. **A supervisor is not a substitute for the trip module** — it has no path to the alternator-side over-frequency signal and no mechanical contact breaking the battery

feed, so it cannot replace what spec §3 item 5 requires. **The trip module is not a substitute for a supervisor** — it does nothing until the engine has already visibly misbehaved, and does nothing at all for a stuck FET's gate. Both belong in this design.

What transfers to this design

1. **Enable `PMC_LVDSC1[LVDRE]` in firmware startup.** Zero-BOM-cost, moves the internal automatic reset floor from VLVR (2.50–2.7 V) up to VLVD (2.8–3.0 V), narrowing (not closing) the band identified in §3. This is a firmware action, not a hardware one, but it belongs in the same design review that adds the external supervisor, because it is the cheapest partial mitigation available and there is no reason not to take it.
2. **A TPS3850/TPS3813-class supervisor, not an FS26/TLF35584-class SBC.** The latter pair would require re-deriving spec §6's entire power architecture; the former adds one small part beside `buck_preregulator.cir` / `mcu_pdn.cir` without disturbing either.
3. **10 kΩ gate-source pulldowns on every gate this task named** (pin 88's metering FET; the 73/07/29 low-side selects; the 03/05 bank high-sides), justified against this design's own confirmed leakage number (S32K1xx datasheet, ≤0.5 µA) and this project's own already-computed turn-off edge rate (~2.8 V/µs, memo 10), not against a generic round number.
4. **Route the supervisor's RESET through one inverting stage into a dedicated kill pulldown at each protected gate**, so that reset/brownout/watchdog-fault conditions are enforced at the gate independent of GPIO state, matching what FS0B/FS1B and SS1/SS2 do on the parts surveyed in §2, without needing to adopt a full SBC to get it.
5. **The pin-mux table gap (spec §5, deferred to Phase 2) directly blocks finishing this analysis.** Until ECU pins 88/73/07/29/03/05 are mapped to physical `PTxx` pins, this memo cannot state those specific pins' PE/PS reset defaults from the reference manual — only the general GPIO-disabled-on-reset behaviour is confirmed. This should be flagged alongside U2/U8 as a dependency for finishing the supervisor/output-safety sheet, not silently assumed away.
6. **The trip module and a board-level supervisor are both required — see §6.** Neither is optional given the other.

Recommendation: the circuit to build

Topology: A TPS3850-class (or TPS3813-class, if the third window-ratio option is not needed) precision voltage supervisor with programmable window watchdog, monitoring `3V3_MCU` via its `SENSE` pin (fixed-threshold variant) or, if a threshold other than a factory-trimmed

option is needed, the adjustable `H01` variant with an external divider per its own `V_MON = V_IT(ADJ) × (1 + R1/R2)` equation (confirmed present in the datasheet, §7.3.4). Its open-drain `RESET` output, pulled up to `3V3_MCU`, feeds:

- the S32K148's `RESET_B` pin, as conventional; and
- one inverting stage (a single small-signal BJT or logic inverter — not specified further here, a Phase-2 detail) whose output drives the gate of a dedicated N-channel "kill" transistor placed from each protected gate node (pin 88's metering FET gate; 73/07/29 low-side select gates; 03/05 bank high-side gates) to ground, in parallel with the standing 10 kΩ gate-source pulldown already recommended above. The inversion is needed because `RESET` is active-low (asserted during fault) while the kill transistor must turn ON — pull the gate down — precisely when `RESET` is asserted.

Thresholds: - Undervoltage trip: **~3.05–3.10 V** on `3V3_MCU`, chosen to sit above VLVD's confirmed 3.0 V max with margin for the supervisor's own ±0.8% accuracy — independent of whether firmware ever configured `LVDRE`. - Watchdog window: **closed (upper) boundary well under 26.67 ms** (§5), open (lower) boundary comfortably above the 1–3 ms injection duration. Exact CWD/SET0/SET1 programming left to Phase 2 once a specific part variant is chosen — not guessed here. - Gate pulldown: **10 kΩ** at every named gate node (§4).

Part class: TI TPS3850 (VSON-10) or TPS3813 (SOT23-6) for the supervisor; a small-signal N-FET (e.g. SOT-23 class, no current-handling requirement beyond discharging gate capacitance) for each kill transistor; standard 10 kΩ 0603/0805 resistors for the passive pulldowns.

Specific, falsifiable claims the next SPICE block should check:

1. With the supervisor's `RESET` asserted and the kill transistor's gate driven high (via the inverting stage), every protected gate node should settle below the FET class's `Vgs(th)` within the 10–40 µs RC estimate from §4 — not just "eventually," a specific time bound the simulation can assert.
2. With `RESET` deasserted (normal operation) and the MCU actively driving a gate high through its normal drive path, the kill transistor and its 10 kΩ pulldown together should not measurably slow the intended switching edge — i.e. the pulldown's loading effect on turn-on time should be checked against whatever gate-drive current budget the eventual driver stage has, not assumed negligible.
3. A simulated Miller/dV/dt coupling event at this design's own ~2.8 V/µs boost-rail turn-off edge (memo 10, reused here) should show the gate node staying below `Vgs(th)` through the 10 kΩ pulldown alone, across a swept Crss range (the 50–200 pF class used in §4's arithmetic) — this

directly tests whether the "weak pulldown is adequate, no strong/active clamp needed" conclusion in §4 actually holds, or whether it was an artifact of the illustrative Crss assumption.

4. A brownout ramp on `3V3_MCU` sagging through the confirmed VLVR/VLVD band (§3) at a swept rate (from the sub-millisecond RC estimate up through several orders of magnitude slower, to bound the "slow sag" open item from §3) should show the external supervisor's own threshold tripping *before* the MCU's outputs could plausibly be left in an undefined state by internal LVR/LVD alone — this is the block that would either validate or falsify this whole memo's central claim.
5. Total exposure time from watchdog-fault-onset to kill-transistor-conducting should be checked against the §5 table's per-timeout-value cylinder-interval count, once an actual window value is chosen in Phase 2 — confirming the chosen window keeps worst- case exposure under one 26.67 ms cylinder interval as intended.

Negative results, recorded

- **Maxim MAX16997/MAX16998 full datasheet** — not retrieved. Direct fetch from `analog.com` timed out twice (60 s, then 140 s in the background); a `datasheet4u.com` mirror returned content that was not actually a readable PDF. Everything attributed to these parts in §2.5 is explicitly marked SNIPPET and should not be used as a design number without retrieving the document.
- **FS26 full data sheet** — the retrieved document is explicitly a "short data sheet" whose own cover page states *"For detailed and full information, see the relevant FS26 full data sheet... available via the NXP Secure Files content interface to those with non-disclosure agreement (NDA) access."*The full watchdog-timing and safe-state electrical-characteristics tables were therefore not available in this pass. The architectural claims in §2.3 (Simple/Challenger watchdog variants, FS0B/FS1B fail-safe output driver) are read directly from the short data sheet's own block diagram and part-ordering table, which is CONFIRMED as far as it goes, but does not include timing numbers the way TPS3850's public datasheet does.
- **A specific, sourced Crss/Ciss figure for this design's actual injector/metering FET** — not found, because no FET part number has been chosen yet (spec §8 does not carry this as its own numbered unknown, but it is the same category of gap as U5's injector part number). §4's dV/dt-coupling and RC-discharge arithmetic uses illustrative, clearly-flagged INFERRED class values (Crss 50–200 pF, Ciss 1–4 nF) rather than a sourced number, and the "10 kΩ is adequate" conclusion should be re-checked once a specific FET is selected.

- **MC33814's behaviour on a watchdog-triggered (as opposed to POR-triggered) reset, specifically whether it also clears the driver's own output/control registers** — not resolved from the retrieved text (§4's nuance paragraph). The datasheet is explicit about POR behaviour and about SPI-commanded internal reset, but the retrieved §5.3.2 text does not state whether a watchdog-timeout-triggered `RESETB` pulse has the same register-clearing effect. Recorded as open rather than assumed favorably.
- **A quantified "how much longer does the brownout band last under a slow supply sag" figure** — not found and not derivable from any source retrieved in this pass; §3's RC-decay table explicitly assumes a lost supply with no ongoing regulation, and flags the slow-sag case as a `buck_preregulator.cir`-adjacent open item rather than estimating it without a basis.
- **TPS3850's exact `t_WDL` / `t_WDU` equations for a custom `CWD` capacitor value** — the datasheet's revision history references "Equation 4," "Equation 5," and "Equation 11" governing these, but this pass did not extract the equations themselves (only the factory-preset timing table, §2.1/§6.6). §5's window-watchdog recommendation is therefore qualitative (target ranges) rather than a specific capacitor value, by design — filling that in without the equations would be exactly the guessed-into-copper failure mode this project's sourcing discipline exists to prevent.
- **Infineon TLF35584 direct datasheet PDF** — the first attempted URL pattern (`dgdl/... DataSheet-v01_02-EN.pdf`) returned HTTP 404; the working URL (`infineon.com/assets/row/public/documents/10/49/infineon-tlf35584-datasheet-en.pdf` , Rev 2.0) was found via a second search and did succeed. Recorded so the failure isn't silently absent from the trail — the guess at a version-numbered filename was wrong, the actual current document uses a different naming convention.

Research Memo 12: Driver Architecture — Integrated IC vs Discrete FETs, and the Coupled Gate-Kill Question

Task: Two schematic sheets (injector driver, metering-unit driver) are blocked on one decision each, and the two decisions are coupled: **integrated injector-driver IC vs discrete FETs with custom gate drive**, and **passive gate pulldown vs active clamp** for holding every driver gate off during reset/brownout (spec §4's "Every driver gate is held off by hardware" requirement; status.md §6). This memo answers both, together, because `supervisor.cir`'s claim 3/claim 6 result and the spec's own correction already established that the second question cannot be closed without the first. **Date:** 2026-09-19 **Scope:** Read-only research. No repo files modified except this one, nothing committed, no subagents dispatched.

Verdict (read this first)

Discrete FETs with a custom pre-driver/gate-drive stage, keeping the already-simulated 100 V boost rail, for the injector banks (pins 03/05, 73/07/29). An integrated smart low-side switch (already memo 07's direction) for the metering unit (pin 88). An active kill-clamp at every gate — not a passive pulldown alone — built discretely, exactly as `supervisor.cir` already validated it, for both output classes.

The two purpose-built integrated injector-driver ICs close enough to fit this design's channel count and current class — **NXP MC33816** and **ST L9781** — both cap their boost-rail voltage **below** the 100 V this project already chose and already built three passing simulation blocks around: MC33816's `VBOOST` absolute maximum is **72 V** (CONFIRMED, NXP datasheet Rev 10.0), L9781's `VTank` is **up to 80 V** (CONFIRMED 8 Oct 2026 from ST's data brief DB3720, supplied by the owner after the datasheet failed retrieval five times — see §2.4). Neither integrated part eliminates the external power MOSFETs, the external boost inductor/diode, or the external current-sense shunt — both are **pre-drivers**, not integrated power stages, so choosing either does not remove the discrete-FET selection and thermal-design work this decision is nominally about; it only relocates the gate-drive sequencing, current-comparator logic, and (for MC33816 specifically) a real microcode toolchain into a part. **MC33814** and (per memo 10, not re-derived here) **TLE8242-2** are confirmed wrong current/voltage class

outright — MC33814's injector output is rated 1.3 A continuous against an 18 A/8-13 A requirement, and it has no boost stage at all (CONFIRMED, NXP/Freescale datasheet Rev 5.0).

On the coupled question: the integrated route does not remove the passive-vs-active problem — it relocates it, in a form worth copying. MC33816's own datasheet states (§6.3.4, CONFIRMED) that its high-side pre-driver output is **actively forced low** — not merely released to a resistor — whenever reset is asserted or `DrvEn` is low, so long as the driver's bootstrap capacitor still holds charge above 1.1 V; only when that bootstrap charge is exhausted does it fall back to a genuinely weak 500 kΩ–2 MΩ gate-source resistor (CONFIRMED, Table 31). That is the same two-tier structure this project's own `supervisor.cir` already built and validated for a discrete design: an active kill path that dominates (claim 1, claim 4: tens of nanoseconds, wins against an adversarial driver with margin) backed by a passive pulldown that is not, on its own, strong enough against Miller coupling at this design's own edge rate (claim 3's FAIL, claim 6's 470 Ω fix). **Buying the IC does not buy exemption from this problem; it buys someone else's already-shipped version of the same fix**, on a rail voltage this project did not choose.

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1. What the design actually needs from a driver — established from the repo

Read from spec §2.1, §2.2, §4; `injector_boost.cir`; `injector_turnoff.cir`; `boost_converter.cir`; `metering_unit_pwm.cir`; memo 04; memo 07. CONFIRMED (repo's own simulation results and spec text, not re-derived):

Requirement	Value	Source
Injector bank high sides	2 channels (pins 03, 05), bank-shared	spec §2.1, memo 04 §3
Injector low-side selects	3 channels (73, 07, 29), independent, current-controlled peak-and-hold	spec §2.1
Injector peak current	~18 A from the boost rail	`injector_boost.cir`, memo 04 envelope 12–24 A
Injector hold current	8–13 A family envelope, ~10 A used in sim	`boost_converter.cir`, memo 04
Boost rail voltage	**100 V**, chosen mid-envelope (memo 04: 65–115 V) and matched to memo 10's turn-off clamp recommendation	`boost_converter.cir`, `injector_turnoff.cir`
Turn-off topology	Recirculation into the same 100 V/47 µF boost reservoir, not freewheel-to-battery, not Zener active clamp	memo 10 (settled, not reopened here)

Requirement	Value	Source
Metering unit	1 low-side PWM channel, current sense, battery-fed through fuse `8F1`, ~0.6–1.35 A class current, no boost, no bank-sharing	spec §2.2, `metering_unit_pwm.cir`
EGR	H-bridge, pins 59/81, not yet designed	status.md §2 — explicitly out of this decision's scope; none of the candidates evaluated below are H-bridge parts
Fail-safe	Every driver gate (88, 73/07/29, 03/05) held off by hardware during reset/brownout, independent of GPIO state	spec §4, memo 11, `supervisor.cir`

The number that actually decides this memo is the 100 V boost rail, not the 18 A peak current — every candidate below has enough current-handling margin once external FETs are sized for it (the ICs are pre-drivers; the FETs carry the current, not the silicon). Voltage is where the candidates actually fail.

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2. Candidate ICs, evaluated against these numbers

2.1 NXP MC33816 — CONFIRMED, full datasheet retrieved and read directly (NXP, "SD6 programmable solenoid controller," Rev 10.0, 2/2023, `nxp.com/docs/en/data-sheet/MC33816.pdf`)

Architecture, read from the block diagram and pin table (p.5–8): a **pre-driver only**. Five external high-side MOSFET gate drivers (with integrated bootstrap and charge pump per channel) and seven external low-side MOSFET gate drivers (one dedicated to a boost-converter switch), four independent current-measurement comparator/ADC blocks, and four digital microcores running encrypted, user-written microcode (two 1024×16 code RAM banks, two 64×16 data RAM banks) that sequence the pre-drivers and implement the peak-and-hold control law using the current-measurement blocks as feedback. **It does not integrate the power FETs, the boost inductor, the boost diode, or the current-sense shunt** — all of that remains external and identical in kind to what a discrete design would need.

Voltage — the disqualifying number, CONFIRMED (Table 3, Maximum Ratings, p.9):

Parameter	Value
`VBOOSTMAX` (steady-state and unpowered)	**0–72 V**
`VBATT` absolute max	–0.3 to 72 V
`VBATT` normal operating (internal VCCP reg.)	9–16 V typ, up to 36 V (broken-alternator, ≤1 h), up to 48 V (jump start, external VCCP)

This is an absolute maximum rating, not a recommendation. This project's boost rail is already built, simulated, and passing at **100 V** (`boost_converter.cir`, `injector_turnoff.cir`). Running MC33816's `VBOOST` pin at 100 V would exceed its absolute maximum by 28 V — real risk of permanent device damage, not a derating question. This is independently corroborated at SNIPPET level by hobbyist common-rail ECU builders working with this exact part: *"The MC33816 only boosts to 72V, and the [MC33816]/PT2000 can't drive high side transistors over 72v"* (rusefi.com forum, `www.rusefi.com/forum/viewtopic.php?p=28212`, community project notes — not a datasheet, cited only because it independently confirms the same number this memo already read from the datasheet's own table).

Current: channel count and current-measurement architecture are not the limiting factor — 5 HS + 7 LS pre-driver outputs comfortably cover 2 banks + 3 selects + 1 boost switch, with two channels to spare. The actual current the design draws is set by the external FETs and the current-sense shunt, which this IC does not supply.

Peak-and-hold "in hardware": partially true, and the qualification matters. The current-measurement comparators and PWM gate outputs are hardware; the sequencing, thresholds, and boost/peak/hold state machine are **user-written microcode** running on the part's own microcores, protected by encryption and loaded via SPI (p.14–16, CONFIRMED). This is a real toolchain and a real skill the spec's own §5 already flagged as a hiring risk in a different context (rejecting the MPC5744P's eTPU2 microcode for exactly this reason: *"No eTPU2 microcode, which is a specialist skill and a hiring risk on a small team"*). The same argument applies here, on a different part.

Power-on-reset / fail-safe behaviour — CONFIRMED, §6.3.4 and §6.4.1, and this is the part of the datasheet most directly relevant to spec §4's requirement:

> *"In order to guarantee a safe condition while the device is not operating, the G_HSX output is immediately forced to the low level, switching the external MOSFET off when reset (RSTB) is asserted, or DrvEn is low. This behavior is effective as long as the bootstrap capacitor voltage is greater than a typical voltage of 1.1 V. When the bootstrap capacitor voltage is lower than or equal to 1.1 V, the pre-driver output state is undefined, but the pre-driver is not in a high state. In addition, an integrated pull-down resistor R_PD_HSX between G_HSX and S_HSX keeps the external MOSFET in an OFF state."*

Table 31 gives `R_PD_HSX` = **500 kΩ min, 2000 kΩ (2 MΩ) max** — this is the fallback path, and it is markedly weaker than the 470 Ω this project's own `supervisor.cir` claim 6 found necessary against Miller coupling

at this design's own edge rate. The low-side pre-driver description (§6.4.1, CONFIRMED) states the identical structure: *"a gate to source pull-down resistor R_PD_LSX holds the external MOSFETs in the off state, while the device is in a power on reset state (RSTB low)"*, with the truth table (Table 38, footnote 48) confirming the driver **actively** drives the gate low (not just releases it to the resistor) whenever `lsx_command` is commanded low — including the off/reset condition, as long as the digital core and `VCCP` supply are alive. The numeric off-state drive impedance for the low-side output specifically was not located in the pages read this session (the high-side equivalent, Table 33, gives 5.9–138 Ω across slew-rate settings) — **INFERRED by architectural symmetry with Figure 13/16's matching push-pull output structure, not itself read from a table**, that the low-side off-state impedance is in the same class.

What this means, stated plainly: MC33816 does not remove the exposure spec §4 and memo 11 identified — it has the **same** two-tier structure (active clamp dominant, weak resistor as last resort when the local bias rail is gone) this project's own supervisor block already designed and proved. The corner case that breaks the resistor alone in this project's own design (a fully depleted local bias) is analogous to the corner case that breaks MC33816's own resistor alone (a fully depleted bootstrap cap). Buying the part buys a working instance of this structure already built and characterized by NXP; it does not buy a different, exemption-granting mechanism.

Sourcing — CONFIRMED, reused from memo 07, not re-fetched this session: MC33816AE, DigiKey India, 868 units in stock, ₹1,040.54 qty 1 / ₹693.12 qty 100 (`digikey.in/en/products/detail/nxp-usa-inc/MC33816AE/4693868`). **Flagged and still unresolved:** the live page shows a "Last Time Buy Date: 06/08/2027" field that memo 07 could not reconcile against NXP's own Product Longevity Program listing. This memo does not resolve it either — it is restated here because it bears directly on whether MC33816 is even a live option, not just on its cost.

2.2 NXP MC33814 — CONFIRMED, full datasheet retrieved and read directly (Freescale/NXP, "Two Cylinder Small Engine Control IC," Rev 5.0, 6/2013, mirrored at `datasheet.octopart.com/MC33814AER2-NXP-Semiconductors-datasheet-32860591.pdf`)

Wrong current and voltage class outright, confirmed from the datasheet's own Maximum Ratings table (Table 3, p.8), not inferred from its marketing copy:

Parameter	Value	Against this design's requirement
`VPWR` supply voltage, absolute max	−0.3 to **45 V**	No margin for a 100 V boost rail — and there is no boost rail at all in this part's architecture
Injector driver output continuous current (`IOC_INJX`)	**1.3 A** max	**~14× under** this design's 18 A peak requirement

MC33814's own application diagram (p.1) drives injectors **directly off** `VPWR`, with no boost stage anywhere in the part — consistent with its stated market (lawn mowers, motor scooters, small motorcycles, snow blowers, chain saws, outboard motors, gasoline-driven generators, per its own datasheet's Applications list) being low-pressure port injection or carbureted-class solenoids, not common-rail diesel. This is the same shape of mismatch memo 10 already found for TLE8242-2 (a variable-force-solenoid pre-driver rated for 0–1.2 A / 50 V, an order of magnitude below this injector), not repeated here per the task's own instruction, and confirmed independently for a second part rather than assumed to generalize. **MC33814 does not fit and should not be considered further for the injector banks.**

2.3 Infineon TLE8242-2 — not re-evaluated, per instruction. Memo 10 §1.2 already

found this is a variable-force-solenoid pre-driver (0–1.2 A, 50 V absolute max), an order of magnitude below this injector's current/voltage class. That finding stands and is not repeated here.

2.4 ST L9781 (and L9782) — data brief retrieved 8 Oct 2026; full datasheet not.

> **Update, 8 October 2026.** The owner supplied the ST data brief DB3720 Rev 1 (Sept 2018), supplied by the owner on 8 Oct 2026. It confirms the snippet-level claims below from the maker's own document: - the DC/DC controller generates `VTank` **up to 80 V**, so 20 V short of this design's 100 V rail; - it is a **pre-driver** for **11 external N-channel MOSFETs**, controlling up to **5 inductive loads** (two banks of two, plus one extra peak-and-hold channel); - peak and hold are run by **internal configurable finite state machines** set over a 32-bit SPI; - it is AEC-Q100, in an LQFP64 (10 × 10 mm) package.
>
> A data brief has no absolute-maximum or electrical tables, but the deciding number no longer rests on a snippet. The 19 September decision (discrete FETs, keep the 100 V rail) stands, and the open sourcing item is closed. The full datasheet would matter only if the rail were ever lowered to 80 V or below.

Two direct-fetch attempts this session — `st.com/resource/en/data_brief/l9781.pdf` and an `alldatasheet.com` mirror — both failed (60-second timeout on the ST data brief; HTTP 403 on the alldatasheet page). A third attempt at the full datasheet (`st.com/resource/en/datasheet/l9781.pdf`) also timed out. This matches memo 10's own negative result for the same part (§1.2: *"Two direct fetch attempts (st.com PDF, alldatasheet mirror) both failed"*) — **the same document has now failed retrieval on two separate research passes**, which is itself a data point about how usable this part is for a small team without a distributor account or an ST FAE relationship.

What search-result snippets state (unverified against the document itself, SNIPPET level only, not elevated further this session): - Drives **11 external N-channel logic-level MOSFETs**, controlling up to **5 inductive loads** through peak-and-hold current control plus one DC/DC step-up converter (ST data brief, indexed search summary). - The DC/DC converter's `VTank` output is reported as **"up to 80 V"** — like MC33816, this is **below** this design's chosen 100 V boost rail, and by a wider margin. - Peak/hold sequencing is handled by **internal configurable Finite State Machines (FSMs)**, described as needing only start-of-actuation commands from the host MCU — if accurate, this is architecturally **simpler and less of a toolchain commitment** than MC33816's user-programmed microcode, because an FSM configured by registers is not the same engineering surface as a part with its own encrypted instruction set. **This is exactly the kind of claim this project's own discipline says should not be designed against without the datasheet**, so it is recorded as a reason to keep pursuing the document, not as a basis for a decision. - Orderable: DigiKey lists **L9781** at **US$10.92** (cut tape, MOQ 1), 64-pin package, found via search index (`digikey.com/en/products/detail/stmicroelectronics/L9781/...`) — **SNIPPET**, the live page itself returned HTTP 403 to this session's fetch, the same bot-mitigation behaviour memo 07 already reported for Mouser/DigiKey chip pages in general.

No power-on-reset behaviour, no absolute maximum ratings table, and no per-channel current rating for L9781 could be confirmed this session. It cannot be recommended as a primary path on SNIPPET-level numbers — not because the part looks unsuitable (the reported architecture is a closer conceptual match to this design than MC33816's microcode approach), but because this project's own sourcing discipline (stated in the task, and already the subject of two "caught only after they became requirements" failures) rules out designing a schematic sheet around numbers nobody has actually read.

Summary table

Candidate	Channels	Voltage vs 100 V rail	Current vs 18 A/8-13 A	Peak-hold	External FETs needed	POR/fail-safe	Fit
MC33816	5 HS + 7 LS	**72 V abs max — under, CONFIRMED**	Set by external FETs, no IC limit	Hardware comparators + user microcode	Yes, always	Active-forced-low, backed by weak 500 kΩ–2 MΩ resistor when bootstrap depleted	**Voltage mismatch; everything else fits**

Candidate	Channels	Voltage vs 100 V rail	Current vs 18 A/8-13 A	Peak-hold	External FETs needed	POR/fail-safe	Fit
MC33814	2 injector + misc	45 V abs max, no boost stage	**1.3 A max — 14× under, CONFIRMED**	N/A — no boost/peak-hold architecture	Integrated (wrong class)	Not evaluated — moot	**Does not fit at all**
TLE8242-2	—	50 V abs max (memo 10)	0–1.2 A (memo 10)	—	Yes	—	**Does not fit at all (memo 10)**
L9781/2	11 gate outputs / 5 loads	**~80 V reported — under, SNIPPET**	Set by external FETs, no IC limit reported	FSM-based, less toolchain than MC33816 (SNIPPET)	Yes, always	Not confirmed	**Voltage mismatch (if the snippet holds); unconfirmed otherwise**

3. What the integrated route actually saves — Q2

Reading MC33816's own architecture against this design's needs, what moves into the part and what does not:

Moves into the part, CONFIRMED: - Gate-drive timing and slew-rate control for the power FETs (four selectable slew rates, bootstrap/charge-pump management) — this is real engineering the discrete route has to build (a gate-driver IC or discrete driver stage, still needed either way, but the sequencing-across-channels logic is a bespoke firmware/FPGA job in the discrete case). - Current-sense comparison against programmable thresholds — the analog front end for closed-loop peak/hold, which a discrete design would otherwise build from an op-amp/comparator plus MCU ADC/comparator peripherals. - VDS monitoring and open/short diagnostics per channel (five HS + six LS, independently) — a genuinely useful diagnostic capability neither `injector_boost.cir` nor `injector_turnoff.cir` currently model or require, but real value if the design wants it. - Part of spec §4's fail-safe requirement — the active-forced-low behaviour at RSTB/DrvEn (§2.1 above) — **but only conditionally**, and with the same bootstrap-dependent fallback this project's own discrete kill-FET design already had to solve for a different bias rail.

Does not move into the part, confirmed by the same datasheet: - The power MOSFETs themselves (all five HS and seven LS pre-driver outputs are external-gate-only). - The boost converter's inductor, diode, switching FET (only the switching FET's *gate* is driven by the IC; the power components are external, same as `boost_converter.cir`'s model). - The current-sense shunt resistor. - The turn-off recirculation topology memo 10 and `injector_turnoff.cir` already chose and validated — this is a board-level wiring decision the pre-driver is agnostic to, regardless of which pre-driver IC (or discrete gate driver)

drives the FETs. - The peak-and-hold **control law's own numeric thresholds and timing** — those are written as microcode (MC33816) or firmware (discrete), either way a real design artifact this project has not yet written for either route.

Net assessment: the integrated route buys a pre-built gate-drive/diagnostics/ current-comparator front end and a partial answer to the fail-safe requirement, in exchange for a hard voltage ceiling below this project's chosen rail and a real microcode toolchain commitment. It does not touch the power stage, the boost stage, or the turn-off topology, all of which are unaffected by this decision either way.

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4. What it costs — Q3

Factor	MC33816	Discrete
India availability	CONFIRMED in stock at DigiKey India (memo 07), but with an unresolved EOL-date flag	Broadly available — automotive gate-driver ICs and power MOSFETs in this voltage/current class are mainstream-distributor stock (memo 07's own aggregate-estimate lines for comparable parts)
Price	₹1,040.54 qty 1 / ₹693.12 qty 100 (CONFIRMED, memo 07)	Estimated ~₹800 aggregate for the whole discrete boost/driver stage (memo 07 line 14b, **I**-class estimate, not yet itemized) — comparable order of magnitude, not a clear win either way
Package/thermal	64-pin LQFP-EP, R_θJA 24.3–29.7 °C/W (CONFIRMED, Table 4) — the IC itself dissipates little; the external FETs carry the thermal load in both routes	Same external-FET thermal burden, plus a separate gate-driver package (typically SOIC/SOT, lower pin count, simpler layout)
Toolchain	**A real one.** Encrypted microcode, two microcores, SPI-loaded configuration, a 93-instruction ISA (datasheet TOC runs to p.312) — this is not a footnote, it is a second embedded-systems skill set beyond the S32K148 firmware this project is already building	None beyond the firmware current-loop this project would write regardless, using peripherals (FlexTimer, dual ADC, PDB) spec §5 already justified as adequate headroom
End-of-life exposure	**Real and specific.** A single NXP part, single product line, with an unresolved "Last Time Buy Date" flag (memo 07 §3.2) — if genuine EOL, the injector driver sheet would need a second redesign, not a substitution, because the microcode and the pin-out are both part-specific	A discrete stage is rebuildable around a different FET/gate-driver family without redesigning the current-control architecture, because the control law lives in this project's own firmware, not in a vendor's encrypted microcode
Rail-voltage cost, if adopted	**Forces the boost rail down from 100 V to ≤72 V** — re-derives three already-passing simulation blocks (`boost_converter.cir` , `injector_turnoff.cir` , and this file's own claim inputs) and measurably erodes pilot-injection turn-on margin (§5 below)	None — the existing 100 V rail and its three passing sim blocks are unaffected

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5. Re-derived arithmetic: what a 72 V or 80 V rail actually costs, if MC33816 or L9781 were adopted

Turn-on ramp time, using the exact RL step-response formula `injector_boost.cir`'s own RESULT NOTE cites: `t = (L/R)·ln(V/(V - I_th·R))`, with this project's own L = 200 µH, R = 0.5 Ω, I_th = 18 A (re-derived here, not assumed):

Boost voltage	t (turn-on to 18 A)	vs 100 µs minimum pilot pulse (memo 04)
100 V (as built)	L/R·ln(100/91) = 400 µs × 0.0943 = **37.7 µs** (matches `injector_boost.cir`'s simulated 38 µs)	consumes **37.7 %** of the shortest pilot pulse
80 V (L9781's reported ceiling)	400 µs × ln(80/71) = 400 µs × 0.1194 = **47.8 µs**	consumes **47.8 %**
72 V (MC33816's confirmed ceiling)	400 µs × ln(72/63) = 400 µs × 0.1335 = **53.4 µs**	consumes **53.4 %**

Turn-off clamp time, using memo 10's own relation `Δt = L·ΔI/V_clamp` (re-derived, not assumed):

Clamp voltage	Δt (18 A → 0)
100 V (as built, `injector_turnoff.cir` measured ~34 µs)	200 µH×18/100 = **36 µs** (hand figure; sim measured 34 µs, memo 10 §2.2)
80 V	200 µH×18/80 = **45 µs**
72 V	200 µH×18/72 = **50 µs**

Reading this plainly: none of these numbers are disqualifying on their own — even at 72 V, 53.4 µs is still comfortably inside a 100–300 µs minimum pulse and vastly inside a 1–3 ms main injection. But the margin **more than doubles its consumption of the shortest pilot pulse** (37.7 % → 53.4 %) at the exact operating point (short pilot injections, the class memo 04 flagged as "a hard requirement, not headroom" for this engine's CPCB IV+ calibration) where turn-on dead time matters most. This is a real, quantified cost of adopting either integrated part — not a hand-wave "it's probably fine" — and it stacks on top of re-deriving `boost_converter.cir`'s droop/recovery numbers and `injector_turnoff.cir`'s recirculation-bump numbers, both of which are computed at the 100 V rail those two blocks currently pass at.

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6. The coupled question, answered directly — Q4

Does an integrated driver remove the passive/active clamp problem, or relocate it? Relocate it. MC33816's own confirmed behaviour (§2.1 above) is structurally identical to what this project's own `supervisor.cir` already built for a discrete design: an active, low-impedance forced-off path that dominates whenever the relevant local supply (MC33816's bootstrap cap; this design's kill-FET drive, sourced

independently from `3V3_MCU`/supervisor logic) is alive, backed by a passive resistor that is, on its own, weak relative to this design's own established Miller-coupling requirement (470 Ω at Crss = 500 pF, claim 6) — MC33816's fallback resistor is 500 kΩ–2 MΩ, three to four orders of magnitude weaker again. **The IC does not grant exemption from the physics `supervisor.cir` claim 3 found; it ships a version of the same fix, on a rail voltage this project did not choose, with a corner case (bootstrap depletion) this project has no visibility into or control over.**

Does a discrete design with a proper active clamp beat both? For this project's specific fail-safe requirement, **yes, and this project has already built and verified it:** `supervisor.cir` claim 4 proves the discrete kill path (an active kill FET, backed by a properly-sized passive pulldown per claim 6) beats the S32K148's own 2.97 V correctness-guarantee floor with margins from 18 µs (fastest brownout ramp modelled) up to 61 ms (slowest), **using a kill-drive path this project chose to be independent of the boost rail's own state** — the adversarial-driver stimulus in claims 1b/4 is explicitly modelled as running off a rail separate from the one that's failing, per memo 11's own "Phase-2 detail" note. MC33816's equivalent mechanism is tied to its own bootstrap capacitor, which is charged from `VBOOST` / `VBATT` — the same rails this design's brownout scenario is about. **No source in this pass (NXP's datasheet included) makes a timed claim about how fast MC33816's own bootstrap cap decays relative to a brownout event**, the way this project's `supervisor.cir` makes a timed claim about its own kill path relative to the 150 µF `mcu_pdn.cir` bulk-cap decay. That is a real, specific advantage of building the active clamp discretely, on a bias this project controls, rather than trusting a bootstrap cap of unstated decay behaviour under exactly the fault condition being protected against.

Answer to the coupled question, stated once, plainly: neither integrated candidate removes the need for an active clamp — both already use one, internally, for the same reason `supervisor.cir` does. **Build the active clamp discretely, as already designed and validated** (kill FET, Ron-class ≤5 Ω model, backed by an Rgs pulldown sized per claim 6's `Rpd < Vgs(th)/(Crss·dV/dt)` relation once a real FET is chosen), for every gate this decision touches — pins 88, 73/07/29, 03/05 — regardless of which driver architecture wins for the power stage. This part of spec §4's requirement does not change with this decision; it was never actually in question.

---

7. Does the answer differ between injectors and the metering unit? — Q5

Yes, and the split is real, not a hedge.

Injectors (03/05, 73/07/29): the requirement is peak-and-hold at 18 A from a shared 100 V boost rail across five channels with bank-sharing logic — the class of problem MC33816 and L9781 exist for. The recommendation above (discrete, keep the 100 V rail) is driven by the voltage mismatch, not by the current or channel count, which both candidates handle comfortably via external FETs.

Metering unit (88): a single low-side PWM channel, ~0.6–1.35 A, no boost, no bank-sharing, current sense against a rail-pressure control loop (`metering_unit_pwm.cir`). **None of the four evaluated candidates are a sensible fit for this channel at all** — MC33816 and L9781 are five-to-eleven-channel high-voltage peak-and-hold parts, absurdly over-specified for one low-current low-side switch; MC33814 and TLE8242-2 are wrong-class outright per §2.2 and memo 10. Memo 07 already recommends the correct class for this channel — **an integrated automotive smart low-side switch** (BTS443P-class, memo 07 line 13), which is itself an "integrated driver," just in a completely different, better-matched product category (single-channel, current-sensing, typically with its own thermal/short-circuit shutdown), not one of the four candidates this task named. This memo does not re-evaluate that choice — it is not in question here — but it confirms the same fail-safe gate-kill requirement (spec §4, `supervisor.cir`) still has to be built at pin 88's gate node regardless of which smart low-side switch is chosen, because a generic smart low-side switch's own internal safe-state behaviour on power loss should not be assumed, without checking that specific part's datasheet, to satisfy this project's own 2.97–3.10 V reset-band requirement (memo 11 §3). **That check is a Phase-2 item for whichever BTS443P-class part memo 07's Phase-2 sourcing pass selects, not resolved here.**

What transfers to this design

1. **The injector driver sheet is drawn discrete**, keeping the 100 V boost rail exactly as `boost_converter.cir` and `injector_turnoff.cir` already built and passed. Neither MC33816 nor L9781 removes enough engineering work to justify re-deriving those two blocks and eroding pilot-injection turn-on margin by roughly 16 percentage points (§5's 37.7 %→53.4 % figure).
2. **The current-regulation loop for peak-and-hold is firmware, using the ADC/FlexTimer/ PDB headroom spec §5 already justified** (2× 12-bit ADC against ~14 channels of demand, FlexTimer well beyond the ~10 timing channels needed, 2× PDB for jitter isolation) — this project already argued it does not need a dedicated timing co-processor for injection

scheduling; the same argument extends to not needing a dedicated current-regulation ASIC for the same reason (fixed 1500 rpm, no eTPU2-class real-time pressure).

3. **The active-clamp requirement is unchanged by this decision** — it was never a question this decision could resolve differently. `supervisor.cir`'s kill-FET + sized-Rgs structure (§6 above) is what gets drawn, at every one of pins 88, 73/07/29, 03/05, regardless of which power-stage architecture was chosen.
4. **The metering unit stays on memo 07's existing track** (smart low-side switch), with one new Phase-2 checkbox: confirm that specific part's own datasheet against the 2.97–3.10 V reset-band requirement before assuming its internal safe-state behaviour covers what `supervisor.cir` was built to guarantee.
5. **MC33816's "Last Time Buy Date" ambiguity (memo 07 §3.2) is now doubly relevant** — it was already a BOM risk; it is now also the fallback path's own risk if this decision is ever revisited under the "what would change your mind" conditions below.
6. **L9781/L9782 remain undesigned-against.** Two independent sourcing passes (memo 10, this memo) have both failed to retrieve its datasheet. It should not appear in a BOM or a schematic net until someone actually reads it.

Recommendation: the circuit to build

Injector driver sheet (pins 03, 05, 73, 07, 29): discrete N-channel power MOSFETs (two high-side bank switches, three low-side selects), rated with margin above 100 V and 18 A peak — exact part TBD, gated on U5 (injector part number) the same way memo 04 and `boost_converter.cir` already flag — driven by a conventional automotive high-side/ low-side gate-driver IC (general-purpose class, not an injector-specific part; broadly sourced, e.g. the same class of part memo 07 line 12 already prices for the EGR H-bridge). Current sense via a shunt per channel or per bank into the MCU's own ADC, closed-loop peak-and-hold implemented in firmware using PDB-triggered sampling. Turn-off recirculation exactly as memo 10 and `injector_turnoff.cir` already specify — this sheet does not reopen that decision.

At every one of pins 88, 73, 07, 29, 03, 05: the kill-FET + sized-Rgs active clamp from `supervisor.cir`'s Recommendation and claim 6 — an N-channel kill transistor from gate to source, driven (through one inverting stage) by the supervisor's `RESET`, in parallel with a standing Rgs pulldown sized per `Rpd < Vgs(th)/(Crss·dV/dt)` once the actual FET is chosen (claim 6's own worked example: 470 Ω at Crss = 500 pF,

this design's 2.8 V/µs edge, Vgs(th) = 1.0 V — re-derive once a real part is picked, per `supervisor.cir`'s own header caveat that this scales with 1/Crss).

Metering-unit sheet (pin 88): unchanged from memo 07's direction — an integrated smart automotive low-side switch (BTS443P-class), plus the same kill-FET/pulldown at its gate node, plus the Phase-2 datasheet check named in §7.

Part class named for the injector power stage's gate-driver IC: not a specific part number — this is deliberately left as a Phase-2 sourcing item (the same status memo 04's Step 4 already gave the discrete route: "all constituent parts… are broadly available in small quantities from standard distributors"), because the point of this recommendation is that no single-source, voltage-capped, microcode-dependent part is needed to close this sheet.

What would change this recommendation:

1. **If U5 closes and shows this engine's injector actually wants a clamp/boost voltage at or below ~72 V** (the low end of memo 04's 65–115 V envelope, not the ~90–105 V midpoint this project chose), the MC33816 voltage mismatch disappears and its pre-built current-regulation/diagnostics stage becomes a straightforwardly good trade — revisit this memo's §2.1 and §5 arithmetic at the actual chosen voltage.
2. **If the firmware effort for a real-time peak-and-hold current loop turns out to be materially harder than spec §5's ADC/FlexTimer/PDB headroom argument assumed** — discovered only once firmware development starts — the calculus shifts toward MC33816 despite the rail-voltage cost, because buying a working current-loop implementation may be cheaper than debugging one under real injector loads.
3. **If MC33816's DigiKey "Last Time Buy Date" is confirmed as a genuine EOL notice** (memo 07 §3.2, still open), MC33816 is removed from consideration entirely, leaving only L9781/L9782 as an integrated fallback — which cannot be adopted until its datasheet is actually retrieved and read, not before.
4. **If a full L9781/L9782 datasheet is obtained** and shows either (a) a `VTank` capability compatible with a 90–100 V rail, or (b) an FSM-based control architecture confirmed to need materially less firmware/toolchain investment than MC33816's microcode, it should be re-evaluated as the primary integrated candidate rather than dismissed alongside MC33816.

Specific, falsifiable claims a Phase-2 SPICE block should check, once an actual FET and gate-driver IC are chosen for the discrete route:

1. The chosen gate-driver IC's output impedance, combined with the claim-6-sized Rpd, should reproduce `supervisor.cir` claim 2's finding (turn-on edge slowed by ≤3 %, not the "moved destination" effect claim 6's RESULT NOTE found at 470 Ω against a 100 Ω driver) — i.e. confirm `Rdrv ≤ 24.7 Ω`-class headroom against the real driver chosen, not the illustrative 100 Ω `supervisor.cir` used.
2. The actual chosen injector-bank FET's `Crss` / `Ciss` / `Vgs(th)` should be re-run through `supervisor.cir` claim 3's sweep to confirm the 470 Ω figure (or its recalculation) still clears Miller coupling at this design's own 2.8 V/µs edge, per claim 6's own stated caveat that the result scales as 1/Crss.
3. If a firmware current-loop prototype is built before this decision is finalized, its achieved loop bandwidth/jitter against the 1–3 ms injection window and the ~38 µs boost-ramp time should be measured and compared against what MC33816's hardware comparators would have guaranteed — this is the concrete evidence that would either confirm or overturn recommendation §item 2 above.

Negative results, recorded

- **ST L9781/L9782 full datasheet** — not retrieved. Three fetch attempts this session (`st.com/resource/en/data_brief/l9781.pdf` — 60 s timeout; `alldatasheet.com` mirror — HTTP 403; `st.com/resource/en/datasheet/l9781.pdf` — 60 s timeout), on top of memo 10's own two prior failed attempts against the same document. Every claim about this part in §2.4 is explicitly SNIPPET-level and must not be used as a schematic input without the actual datasheet — recommended action for whoever owns Phase 2: request the document directly from an ST distributor or FAE contact rather than another automated fetch, since the pattern across two independent sessions is a consistent block, not a transient failure.
- **ST L9781 DigiKey live product page** — HTTP 403 on direct fetch, consistent with memo 07's own documented DigiKey/Mouser bot-mitigation behaviour. Price and package are SNIPPET-level (search-index only).
- **MC33816's low-side pre-driver off-state drive impedance** (the LS equivalent of Table 33's HS numbers) — not located in the pages read this session. The claim that it is in the same tens-of-ohms class as the high-side table is INFERRED from the matching push-pull output topology shown in Figure 16, not read from a numbered table.
- **MC33816's bootstrap-capacitor decay time under a brownout/VBATT-loss condition** — not stated anywhere in the retrieved datasheet. This is the specific gap that prevents treating MC33816's fail-safe behaviour as

equivalent to `supervisor.cir`'s claim-4 guarantee rather than merely analogous to it (§6 above).

- **A confirmed resolution of MC33816AE's "Last Time Buy Date: 06/08/2027" field** (memo 07 §3.2) — still open. Not re-attempted this session; flagged again because this memo's recommendation depends on it remaining a live part if the fallback conditions in the Recommendation section are ever triggered.
- **A quantitative per-channel current rating for L9781/L9782** — not found in any source retrieved this session (SNIPPET only: "5 inductive loads," no amperage figure).

Phase 1 — Research Implementation Plan

> **For agentic workers:** REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (`- [ ]`) syntax for tracking.

Goal: Close the unknowns that block schematic capture, and produce the sourcing and standards groundwork the later phases depend on.

Architecture: Seven independent research memos, one per question, each written to `docs/research/`. Every memo ends with a verdict that either closes a numbered unknown from the spec's §8 register or states precisely why it cannot be closed from desk research and what physical access would close it. Memos are evidence documents: every factual claim carries a citation, and an uncited claim is a defect.

Tech Stack: WebSearch and WebFetch for primary sources; manufacturer datasheets and distributor stock APIs; Markdown output under version control.

Spec: `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md`

Global Constraints

- **Every factual claim carries a citation** — a URL, a document number, or an explicit "measured on the engine by on ". No claim from memory.
- **Distinguish confirmed from inferred.** Confirmed means a primary source says it. Inferred means we reasoned to it. Memos mark every non-obvious claim as one or the other.
- **A closed unknown states what closed it.** "U5 closed — injector is Bosch 0445110xxx per photograph refs/injector-body.jpg" is a close. "U5 probably Bosch" is not.
- **Negative results are results.** A memo that establishes the KG640C manual is not publicly available has done its job, provided it records where it looked.
- **No safety-relevant conclusion from a secondary source.** Anything feeding the §3 overspeed gate needs a manufacturer document or a measurement, not a forum post.
- Currency for sourcing: **INR**, with availability assessed for **India** (Mouser/DigiKey India, Element14, Robu, local distributors).
- Memos live at `docs/research/<NN>-<topic>.md`, numbered to match task order.

Task 1: Engine identity

Files: - Create: `docs/research/01-engine-identity.md`

Interfaces: - Consumes: nothing. - Produces: engine make/model/displacement/cylinder count/rated speed/emissions tier, used by Tasks 4 and 6 and by the injector and aftertreatment scope decisions.

Closes: U11. Informs U8, U9.

- [ ] **Step 1: Establish what "GK550" designates**

The user reports the engine as "GK550". This does not match a Kirloskar engine designation. Determine whether it is a genset model, an engine model, a Kirloskar Green product code, or a distributor SKU.

Search: `Kirloskar GK550`, `Kirloskar Green GK 550 genset`, `KOEL GK550 25 kVA`.

- [ ] **Step 2: Work backwards from the manual number**

The wiring diagram is `08-GP3-60-001`, file `GP3.8703.C4.pdf`, hosted at `ikonnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram`. Determine what the `GP3` family designates. Try fetching the parent index of that URL path for a manual list that maps families to engine models.

- [ ] **Step 3: Cross-check against the hardware evidence**

The wiring diagram independently constrains the engine. Any candidate must match **all** of: 3 cylinders (three injectors, cylinders 1/2/3); common-rail (rail pressure sensor plus fuel metering unit); EGR fitted; a catalyst temperature sensor present but no SCR or DEF; 12 V electrical system. Reject any candidate that fails one.

- [ ] **Step 4: Determine the emissions tier**

A 25 kVA genset in India with common-rail, EGR and a DOC but no SCR points to CPCB IV+. Confirm against CPCB IV+ requirements for the 19–56 kW band and record what that implies for aftertreatment scope.

- [ ] **Step 5: Write the memo**

Record: engine make, model, displacement, bore/stroke if available, cylinder count, aspiration, rated speed (expected 1500 rpm for 50 Hz), rated output, emissions tier, and the OEM ECU part number if discoverable. Mark each confirmed or inferred. If identity cannot be established, state exactly what photograph of the engine rating plate would close it.

- [ ] **Step 6: Commit**

```
git add docs/research/01-engine-identity.md
git commit -m "research: engine identity and emissions tier"
```

Task 2: KG640C genset controller

Files: - Create: `docs/research/02-kg640c-gcu.md`

Interfaces: - Consumes: Task 1's engine identity (the GCU is sold against engine families). - Produces: the J1939 PGN set the ECU must transmit and receive, and a yes/no on the MPU input that the spec's §3 safety gate turns on.

Closes: U3, U4. **U3 is the safety gate — treat this task as the highest priority in the phase.**

- [ ] **Step 1: Locate the KG640C manual**

Search: `Kirloskar KG640C manual`, `KG640C genset controller datasheet`, `KOEL KG640C wiring`. Also try the `ikonnect.kirloskar.com` path that served the wiring diagram, substituting the controller's manual number if a list is discoverable.

The KG640C may be a rebadged third-party controller — Deep Sea, ComAp, Datakom and SmartGen all OEM for Indian genset makers. If the manual is unavailable under the Kirloskar name, identify the underlying product and use its documentation, recording the identification basis.

- [ ] **Step 2: Answer the MPU question explicitly**

Determine whether the KG640C has a magnetic pickup input terminal. The spec's §3 gate depends on this and nothing else.

Three possible verdicts, all acceptable outputs: - **Yes** — record the terminal designation and the input's electrical spec. The design wires an MPU to it. - **No** — the design needs a standalone overspeed trip module in the panel. Identify two candidate modules with part numbers and prices. - **Cannot determine** — state what would: a photograph of the KG640C terminal strip, or the physical unit.

Do not infer from the wiring diagram's silence. The diagram shows terminals J1–J8 in use; an unused MPU terminal would not appear.

- [ ] **Step 3: Extract the J1939 expectations**

Determine which PGNs the controller expects from an engine node, and which it transmits. At minimum establish whether it uses: EEC1 (61444) for speed, ET1 (65262) for coolant temperature, EFL/P1 (65263) for oil pressure, VEP1 (65271) for battery voltage, DM1 (65226) for active faults, and TSC1 (0) for speed commands to the engine.

Record baud rate (expected 250 kbit/s) and whether it expects the engine at source address 0x00.

- [ ] **Step 4: Record the digital I/O contract**

From the wiring diagram, terminals J1/J2 carry `D+CHG ALT`, `DIG I/P`, `DIG O/P`, `IGNITION RELAY`, `START RELAY`. Establish which of these the ECU interacts with and in which direction, since start/stop sequencing is split between the GCU requesting and the ECU executing.

- [ ] **Step 5: Write the memo**

Lead with the U3 verdict — it is the reason this memo exists. Then the PGN tables, the digital I/O contract, and a list of what remains unknown.

- [ ] **Step 6: Commit**

```
git add docs/research/02-kg640c-gcu.md
git commit -m "research: KG640C controller, J1939 contract and MPU verdict"
```

Task 3: ECU connector identification

Files: - Create: `docs/research/03-ecu-connector.md`

Interfaces: - Consumes: Task 1's engine identity. - Produces: candidate connector part numbers and the mating-half sourcing that the adapter-harness deliverable (spec §7.1) depends on.

Closes: U1, partially — full closure needs the photograph.

- [ ] **Step 1: Characterise the connector from the PDF**

PDF page 4 shows the ECU connector with pins 1, 5, 28, 73 and 94 called out. Re-render that page region at high DPI and examine the housing: number of pin rows, whether it is a single housing or ganged blocks, the lever or bolt retention style.

```
pdftoppm -r 400 -f 4 -l 4 -png refs/GP3.8703.C4.pdf /tmp/p4
```

- [ ] **Step 2: Match against standard 94-pin automotive ECU headers**

94-pin is a common automotive ECU size. Check the usual families: TE Connectivity AMPSEAL, Bosch's own ECU headers (the Bosch EDC17/MD1 series uses 94-pin variants), Molex CMC, Aptiv/Delphi GT. Record which are 94-pin and how their housings compare to the PDF image.

- [ ] **Step 3: Establish mating-half availability and price in India**

For each candidate, find the harness-side connector, contacts and seals, with distributor stock and INR pricing. A connector we cannot buy is not a candidate.

- [ ] **Step 4: Specify the photograph that closes U1**

Write explicit instructions for what to photograph on the engine: the ECU connector housing from the mating face, any moulded part number or logo on the housing and on the ECU case, and the ECU's own label. Include what lighting and angle make moulded numbers legible.

- [ ] **Step 5: Write the memo**

Ranked candidate list with the evidence for each, sourcing table, and the photograph brief.

- [ ] **Step 6: Commit**

```
git add docs/research/03-ecu-connector.md
git commit -m "research: ECU connector candidates and sourcing"
```

Task 4: Injector identification and drive profile

Files: - Create: `docs/research/04-injector-drive.md`

Interfaces: - Consumes: Task 1's engine identity. - Produces: boost rail voltage, peak and hold current targets, and pulse timing — the inputs to the injector driver sheet and the boost converter design in Phase 2.

Closes: U5.

- [ ] **Step 1: Identify the injector family**

From the engine identity, determine the common-rail system supplier. Indian 3-cylinder gensets in this class typically use Bosch CRS or Denso. Record the injector part number if discoverable.

- [ ] **Step 2: Establish the drive profile envelope**

Solenoid common-rail injectors share a characteristic profile: a boosted opening pulse, then a lower hold current. Establish typical values for the identified family — boost rail voltage, peak current and its duration, hold current, and minimum controllable pulse width.

Where exact figures are unavailable, record the **envelope** across the family and design the hardware to cover it. A driver that spans 40–100 A/ms slew and 12–24 A peak will drive any of them; the calibration narrows it later.

- [ ] **Step 3: Confirm the bank arrangement against firing order**

The spec establishes bank A = cylinders 1+3, bank B = cylinder 2. Verify this is consistent with a 3-cylinder firing order (typically 1-3-2 or 1-2-3) — cylinders sharing a high-side bank must never need simultaneous injection. Record the firing order and show the check.

- [ ] **Step 4: Select candidate driver ICs**

Identify integrated common-rail injector drivers with integrated or companion boost control. Record for each: channel count, boost capability, current regulation method, whether it handles the high-side bank switching, package, availability in India, and price. Note explicitly whether each is available in small quantities or is automotive-tier allocation only.

- [ ] **Step 5: State the calibration boundary plainly**

Record what this hardware can and cannot do without injector characterisation data. The memo must say, unambiguously, that the driver can be built and bench-verified against a known load, and that making it run *this* engine correctly needs flow-vs-pulsewidth data that is not publicly available. Name the three routes to that data — supplier licence, dyno with rate-of-injection measurement, or OEM calibration extraction — with an honest note on the cost and legality of each.

- [ ] **Step 6: Write the memo and commit**

```
git add docs/research/04-injector-drive.md
git commit -m "research: injector identification, drive envelope and calibration boundary"
```

Task 5: MCU confirmation and availability

Files: - Create: `docs/research/05-mcu-selection.md`

Interfaces: - Consumes: the channel budget in spec §5 — 2 CAN, ~14 ADC, ~10 timer channels. - Produces: the confirmed MCU part number, package and KiCad symbol source for Phase 2.

Confirms: spec §5. Does not close a numbered unknown.

- [ ] **Step 1: Verify the channel budget against the S32K148 datasheet**

Fetch the S32K148 datasheet and confirm: FlexCAN instance count, ADC channel count and resolution, FlexTimer instances and channels per instance, LPI2C for the barometric sensor, flash and RAM. Record the specific package under consideration and its pin count.

- [ ] **Step 2: Check real availability and price in India**

Check Mouser India, DigiKey India and Element14 for S32K148 stock in the chosen package, in single-digit and hundred-unit quantities, with INR pricing and lead time. Repeat for the S32K146 fallback named in

the spec.

An MCU with a 40-week lead time fails this task regardless of how good the datasheet is. If both S32K parts fail on availability, evaluate alternatives against the same channel budget and recommend one — do not silently proceed with an unobtainable part.

- [ ] **Step 3: Confirm KiCad symbol and footprint provenance**

Determine whether a symbol and footprint exist in KiCad's stock libraries, in NXP's published EVB design files, or on SnapEDA/Ultra Librarian. Record the source and whether pad pitch was verified against the datasheet. Per the PCB Roadmap's warning, a hand-sourced footprint is the single most likely place to introduce a silent error.

- [ ] **Step 4: Confirm the toolchain**

Record: compiler (S32 Design Studio / GCC ARM), debug probe compatible with the chosen part and its price in India, and whether an evaluation board exists that would let firmware work start before the PCB is fabricated.

- [ ] **Step 5: Write the memo and commit**

State the verdict in the first line: confirmed, or changed with reasoning.

```
git add docs/research/05-mcu-selection.md
git commit -m "research: MCU confirmation, availability and toolchain"
```

Task 6: Standards and compliance scope

Files: - Create: `docs/research/06-standards.md`

Interfaces: - Consumes: Task 1's emissions tier. - Produces: the test and design requirements Phase 3 layout must honour, and an honest scope statement on certification.

- [ ] **Step 1: SAE J1939 layers that apply**

Summarise what each relevant part requires of this design: J1939-11 (twisted-pair physical layer, termination, stub lengths), J1939-21 (transport protocol, BAM vs RTS/CTS), J1939-71 (the PGN definitions Task 2 identified), J1939-73 (diagnostics, DM1/DM2), J1939-81 (address claiming).

- [ ] **Step 2: ISO 7637-2 transient immunity**

Record the pulse definitions that apply to a 12 V system — 1, 2a, 2b, 3a, 3b and load dump 5b — with their amplitudes and the severity levels typically specified. This sets the TVS clamp voltage and the buck's input rating.

- [ ] **Step 3: CISPR 25 conducted emissions**

Record the frequency bands and class limits. Note what bench equivalent (LISN plus spectrum analyser) gives useful indication short of a test house, and what it cannot tell you.

- [ ] **Step 4: State the certification position honestly**

This is a replacement controller on an emissions-certified engine. Record plainly what that means: modifying the engine controller on a CPCB-IV+ certified genset affects its certification status, and a replacement ECU is not covered by the original certificate. State the regulatory position for a genset in service in India and flag it as a decision for the project owners, not an engineering detail.

- [ ] **Step 5: Write the memo and commit**

```
git add docs/research/06-standards.md
git commit -m "research: J1939, ISO 7637-2, CISPR 25 and certification scope"
```

Task 7: Consolidated BOM and sourcing

Files: - Create: `docs/research/07-bom-sourcing.md`

Interfaces: - Consumes: Tasks 3, 4 and 5 — connector, injector driver and MCU are the three hard sourcing problems. - Produces: a costed BOM skeleton for Phase 2 capture, with every line either sourced or flagged.

- [ ] **Step 1: Build the BOM skeleton from the spec**

One line per functional block in spec §4 and §6: EMI filter, reverse-battery controller and P-FET, TVS, buck, sensor-rail LDO and PTCs, 3V3 LDO, supervisor, MCU, crystal, 2× CAN transceiver and protection, VR conditioner, EGR H-bridge, metering-unit low-side driver, injector driver and boost stage, relay-drive FETs, barometric sensor, connectors.

- [ ] **Step 2: Price and check stock for every line**

For each: manufacturer part number, distributor, stock quantity, INR unit price at qty 1 and qty 100, lead time. Prefer automotive-qualified (AEC-Q100/Q101) parts and record the qualification status per line.

- [ ] **Step 3: Flag every single-source and long-lead line**

Any line with one supplier, or a lead time beyond eight weeks, gets flagged with a named second source or an explicit "no alternative identified". These are the lines that stall a build.

- [ ] **Step 4: Produce a headline cost**

Total BOM cost at quantity 1 and quantity 10, excluding PCB fabrication and assembly. Note separately the estimated 4-layer PCB fab cost for the expected board area.

- [ ] **Step 5: Write the memo and commit**

```
git add docs/research/07-bom-sourcing.md
git commit -m "research: consolidated BOM with India sourcing and cost"
```

Task 8: Engine inspection brief

Files: - Create: `docs/research/08-engine-inspection-brief.md`

Interfaces: - Consumes: Tasks 1, 3 and 4 — each contributes an item that only physical access resolves. - Produces: a single checklist someone takes to the machine, closing six unknowns in one visit.

Closes: U2, U6, U7, U8, U9, U10 — none of which desk research can close.

- [ ] **Step 1: Collect every unknown that needs the physical engine**

From spec §8: U2 (boost air-temp topology), U6 (pin 09 synchronisation ground), U7 (whether pins 04/06 are sense-only or load feeds), U8 (6-pin intake throttle — on the ECU at all?), U9 (catalyst temperature sensor routing), U10 (which signal uses SENT). Add U1's connector photograph from Task 3 and U11's rating plate from Task 1.

- [ ] **Step 2: Write the photograph list**

For each: what to photograph, from what angle, and what detail must be legible. Engine rating plate. ECU case label and connector mating face. Boost P&T sensor body — the part number on it settles U2 outright. Injector body. Intake throttle connector and its cable run. Catalyst temperature sensor and where its harness goes. KG640C terminal strip.

- [ ] **Step 3: Write the measurement list**

Measurements need the harness live or disconnected — say which for each, and give the expected reading so an anomaly is recognisable on the spot.

For U2: resistance across boost sensor pins 2 and 1 with the connector unplugged and the engine cold. A few kΩ that falls as it warms means NTC; an open or a fixed high reading suggests a conditioned ratiometric output. This single measurement closes U2.

For U7: voltage on ECU pins 04 and 06 with ignition on, then current draw if it can be measured safely at the fuses. Sense-only pins draw milliamps; load feeds draw amps.

For U6: continuity from pin 09 to battery negative and to sensor ground, engine off, battery disconnected.

- [ ] **Step 4: Write the safety preamble**

The brief is for someone working on a genset. Lead with: battery disconnected before any continuity measurement; no measurement inside the alternator output enclosure; the engine does not run during any of this. State that if the set must run for a measurement, that is a separate supervised activity and not part of this checklist.

- [ ] **Step 5: Write the memo and commit**

Format it as a checklist that works printed and carried, not as a report.

```
git add docs/research/08-engine-inspection-brief.md
git commit -m "research: engine inspection brief closing six physical unknowns"
```

Phase exit criteria

Phase 1 is complete when:

- [ ] U3 has a verdict. The safety gate in spec §3 is either satisfiable or has a named trip module.
- [ ] U5 has a drive envelope sufficient to design the injector stage, and the calibration boundary is stated in writing.
- [ ] U11 is closed, or the rating-plate photograph is specified and requested.
- [ ] U1 has a ranked candidate list and a photograph brief.
- [ ] The MCU is confirmed or replaced, with stock verified.
- [ ] The engine inspection brief exists and has been handed to whoever has access to the machine.
- [ ] Every unknown that remains open is recorded in an updated spec §8 with what would close it.

Final step: update spec §8 in place so the register reflects reality, and commit that edit with the phase.

What this plan does not cover

Phases 2, 3 and 4 get their own plans, written once Phase 1 closes the unknowns they depend on:

- **Phase 2 — schematic capture.** Can begin on power, CAN, MCU support, speed inputs and relay outputs before Phase 1 finishes. Blocked on U1 for connectors, U2 for the boost air-temp front-end, U5 for the injector stage, U8 for the throttle.

- **Phase 3 — PCB layout.** Blocked on Phase 2.
- **Phase 4 — firmware skeleton.** Unblocked by Phase 1 except for the J1939 layer, which needs Task 2. Could run in parallel.